lyche-numerical-linear-algebra — formula report

4638 inline formulas (section omitted; all rendered), 1056 display equations, 7 tables, 58 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_EQ0706 **[conf 0.001]**	253	■ 0.001 ● C|+16	`\begin{array}{rl} \boldsymbol{T} \boldsymbol{V}+ \boldsymbol{V} \boldsymbol{T} & =h^{2} \boldsymbol{F} \\ \boldsymbol{V}^{\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}+ \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} & =h^{2} \boldsymbol{F} \\ \stackrel{ \boldsymbol{S}() \boldsymbol{S}}{\Longleftrightarrow} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^{2}+\boldsymbol{S}^{2} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} & =h^{2} \boldsymbol{S} \boldsymbol{F} \boldsymbol{S}=h^{2} \boldsymbol{G} \\ \stackrel{ \boldsymbol{T}}{\Leftrightarrow} \boldsymbol{S}^{ \mathbf{2}} \boldsymbol{D} & \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^{2}+\boldsymbol{S}^{2} \boldsymbol{X} \boldsymbol{S}^{2} \boldsymbol{D} \\ \boldsymbol{S}^{2} & =h^{2} \boldsymbol{G} \\ \Longleftrightarrow & \boldsymbol{I} /(2 h) \end{array}`	$\begin{array}{rl} \boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T} & =h^{2} \boldsymbol{F} \\ \boldsymbol{V}^{\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}+\boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} & =h^{2} \boldsymbol{F} \\ \stackrel{\boldsymbol{S}() \boldsymbol{S}}{\Longleftrightarrow} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^{2}+\boldsymbol{S}^{2} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} & =h^{2} \boldsymbol{S} \boldsymbol{F} \boldsymbol{S}=h^{2} \boldsymbol{G} \\ \stackrel{\boldsymbol{T}}{\Leftrightarrow} \boldsymbol{S}^{\mathbf{2}} \boldsymbol{D} & \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^{2}+\boldsymbol{S}^{2} \boldsymbol{X} \boldsymbol{S}^{2} \boldsymbol{D} \\ \boldsymbol{S}^{2} & =h^{2} \boldsymbol{G} \\ \Longleftrightarrow & \boldsymbol{I} /(2 h) \end{array}$	$\begin{array}{l} \boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}=h^{2}\boldsymbol{F} \\ \stackrel{\boldsymbol{V}=\boldsymbol{S}\boldsymbol{X}\boldsymbol{S}}{\Longleftrightarrow} \boldsymbol{T}\boldsymbol{S}\boldsymbol{X}\boldsymbol{S}+\boldsymbol{S}\boldsymbol{X}\boldsymbol{S}\boldsymbol{T}=h^{2}\boldsymbol{F} \\ \stackrel{\boldsymbol{S}()\boldsymbol{S}}{\Longleftrightarrow} \boldsymbol{S}\boldsymbol{T}\boldsymbol{S}\boldsymbol{X}\boldsymbol{S}^{2}+\boldsymbol{S}^{2}\boldsymbol{X}\boldsymbol{S}\boldsymbol{T}\boldsymbol{S}=h^{2}\boldsymbol{S}\boldsymbol{F}\boldsymbol{S}=h^{2}\boldsymbol{G} \\ \stackrel{\boldsymbol{T}\boldsymbol{S}=\boldsymbol{S}\boldsymbol{D}}{\Longleftrightarrow} \boldsymbol{S}^{2}\boldsymbol{D}\boldsymbol{X}\boldsymbol{S}^{2}+\boldsymbol{S}^{2}\boldsymbol{X}\boldsymbol{S}^{2}\boldsymbol{D}=h^{2}\boldsymbol{G} \\ \stackrel{\boldsymbol{S}^{2}=\boldsymbol{I}/(2h)}{\Longleftrightarrow} \boldsymbol{D}\boldsymbol{X}+\boldsymbol{X}\boldsymbol{D}=4h^{4}\boldsymbol{G}. \end{array}$
lyche-numerical-linear-algebra_EQ0729 **[conf 0.002]**	258	■ 0.002 ● C|+19	`\begin{array}{lll} \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p, q} & =\omega_{2 m}^{j(2 k)} & =\omega_{m}^{j k}, \\ \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p+m, q} & =\omega_{2 m}^{(j+m)(2 k)} & =\omega_{m}^{j+m) k} \\ \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p, q+m} & =\omega_{2 m}^{j(2 k+1)} & =\omega_{2 m}^{j} \omega_{m}^{j k}, \\ \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p+m, q+m} & \left.=\omega_{2 m}^{j+m}\right)(2 k+1) & \left.=\omega_{2 m}^{j+m} \omega_{m}^{j+m}\right) k \\ & \\ (j+m & & \end{array}`	$\begin{array}{lll} \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p, q} & =\omega_{2 m}^{j(2 k)} & =\omega_{m}^{j k}, \\ \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p+m, q} & =\omega_{2 m}^{(j+m)(2 k)} & =\omega_{m}^{j+m) k} \\ \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p, q+m} & =\omega_{2 m}^{j(2 k+1)} & =\omega_{2 m}^{j} \omega_{m}^{j k}, \\ \left(\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}\right)_{p+m, q+m} & \left.=\omega_{2 m}^{j+m}\right)(2 k+1) & \left.=\omega_{2 m}^{j+m} \omega_{m}^{j+m}\right) k \\ & \\ (j+m & & \end{array}$	$\begin{array}{llll} (\boldsymbol{F}_{2m}\boldsymbol{P}_{2m})_{p,q} & =\omega_{2m}^{j(2k)} & =\omega_{m}^{jk}, & \\ (\boldsymbol{F}_{2m}\boldsymbol{P}_{2m})_{p+m,q} & =\omega_{2m}^{(j+m)(2k)} & =\omega_{m}^{(j+m)k} & =\omega_{m}^{jk}, \\ (\boldsymbol{F}_{2m}\boldsymbol{P}_{2m})_{p,q+m} & =\omega_{2m}^{j(2k+1)} & =\omega_{2m}^{j}\omega_{m}^{jk}, & \\ (\boldsymbol{F}_{2m}\boldsymbol{P}_{2m})_{p+m,q+m} & =\omega_{2m}^{(j+m)(2k+1)} & =\omega_{2m}^{j+m}\omega_{m}^{(j+m)k} & =-\omega_{2m}^{j}\omega_{m}^{jk}. \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr>
<td>lyche-numerical-linear-algebra_EQ0326 (5.18)</td>
<td>133</td>
<td>■ 0.118
● C|+15</td>
<td><code>\begin{aligned} & \boldsymbol{Q}_{1}:=\left[\begin{array}{llll} \boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{n} \end{array}\right], \\ & \boldsymbol{R}_{1}:= \\ & \boldsymbol{q}_{j}:=\frac{ \boldsymbol{v}_{j}}{\left\|\boldsymbol{v}_{j}\right \|_{2}}, \quad j=1, \ldots, n \text { and } \\ & {\left[\begin{array}{ccccc} \left\|\boldsymbol{v}_{1}\right \|_{2} & \boldsymbol{a}_{2}^{T} \boldsymbol{q}_{1} & \boldsymbol{a}_{3}^{T} \boldsymbol{q}_{1} & \cdots & \boldsymbol{a}_{n-1}^{T} \boldsymbol{q}_{1} \\ 0 & \left \|\boldsymbol{v}_{2}\right\|_{2} & \boldsymbol{a}_{3}^{T} \boldsymbol{q}_{2} & \cdots & \boldsymbol{a}_{n-1}^{T} \boldsymbol{q}_{2} \\ & 0 & \left\|\boldsymbol{v}_{3} \right\|_{2} & \cdots & \boldsymbol{a}_{n-1}^{T} \boldsymbol{q}_{2} \\ & & \ddots & \ddots & \boldsymbol{a}_{n}^{T} \boldsymbol{q}_{3} \\ & & & \ddots & \vdots \\ & & & \left\|\boldsymbol{v}_{n-1}\right\|_{2} & \boldsymbol{a}_{n}^{T} \boldsymbol{q}_{n-1} \\ & & & 0 & \left\|\boldsymbol{v}_{n}\right\|_{2} \end{array} \right] .} \end{aligned}</code></td>
<td>$$\begin{aligned} & \boldsymbol{Q}_{1}:=\left[\begin{array}{llll} \boldsymbol{q}_{1}, \ldots, \boldsymbol{q}_{n} \end{array}\right], \\ & \boldsymbol{R}_{1}:= \\ & \boldsymbol{q}_{j}:=\frac{ \boldsymbol{v}_{j}}{\left\|\boldsymbol{v}_{j}\right \|_{2}}, \quad j=1, \ldots, n \text { and } \\ & {\left[\begin{array}{ccccc} \left\|\boldsymbol{v}_{1}\right \|_{2} & \boldsymbol{a}_{2}^{T} \boldsymbol{q}_{1} & \boldsymbol{a}_{3}^{T} \boldsymbol{q}_{1} & \cdots & \boldsymbol{a}_{n-1}^{T} \boldsymbol{q}_{1} \\ 0 & \left \|\boldsymbol{v}_{2}\right\|_{2} & \boldsymbol{a}_{3}^{T} \boldsymbol{q}_{2} & \cdots & \boldsymbol{a}_{n-1}^{T} \boldsymbol{q}_{2} \\ & 0 & \left\|\boldsymbol{v}_{3} \right\|_{2} & \cdots & \boldsymbol{a}_{n-1}^{T} \boldsymbol{q}_{2} \\ & & \ddots & \ddots & \boldsymbol{a}_{n}^{T} \boldsymbol{q}_{3} \\ & & & \ddots & \vdots \\ & & & \left\|\boldsymbol{v}_{n-1}\right\|_{2} & \boldsymbol{a}_{n}^{T} \boldsymbol{q}_{n-1} \\ & & & 0 & \left\|\boldsymbol{v}_{n}\right\|_{2} \end{array} \right] .} \end{aligned}$$</td>
<td>$$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\|\boldsymbol{v}_j\|_2}, \quad j = 1, \ldots, n \text{ and}$$ $$\boldsymbol{R}_1 := \begin{bmatrix} \|\boldsymbol{v}_1\|_2 & \boldsymbol{a}_2^T\boldsymbol{q}_1 & \boldsymbol{a}_3^T\boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T\boldsymbol{q}_1 & \boldsymbol{a}_n^T\boldsymbol{q}_1 \\ 0 & \|\boldsymbol{v}_2\|_2 & \boldsymbol{a}_3^T\boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T\boldsymbol{q}_2 & \boldsymbol{a}_n^T\boldsymbol{q}_2 \\ & 0 & \|\boldsymbol{v}_3\|_2 & \cdots & \boldsymbol{a}_{n-1}^T\boldsymbol{q}_3 & \boldsymbol{a}_n^T\boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \|\boldsymbol{v}_{n-1}\|_2 & \boldsymbol{a}_n^T\boldsymbol{q}_{n-1} \\ & & & & 0 & \|\boldsymbol{v}_n\|_2 \end{bmatrix}.$$</td>
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<td>lyche-numerical-linear-algebra_EQ0029</td>
<td>031</td>
<td>■ 0.138
● S|+0</td>
<td><code>\begin{array}{ccc} a_{11} x_{1}+a_{12} x_{2}+\cdots+ & a_{1 n} x_{n}=b_{1} \\ a_{21} x_{1}+a_{22} x_{2}+\cdots+ & a_{2 n} x_{n}= & b_{2} \\ \vdots & \vdots & \vdots \\ a_{m 1} x_{1}+a_{m 2} x_{2}+\cdots+ & \vdots \\ a_{m n} x_{n} & =b_{m} \end{array}</code></td>
<td>$$\begin{array}{ccc} a_{11} x_{1}+a_{12} x_{2}+\cdots+ & a_{1 n} x_{n}=b_{1} \\ a_{21} x_{1}+a_{22} x_{2}+\cdots+ & a_{2 n} x_{n}= & b_{2} \\ \vdots & \vdots & \vdots \\ a_{m 1} x_{1}+a_{m 2} x_{2}+\cdots+ & \vdots \\ a_{m n} x_{n} & =b_{m} \end{array}$$</td>
<td>$$\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots \qquad \vdots \qquad\qquad \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m \end{array}$$</td>
</tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_EQ0630	229	0.167 C|+3	\begin{aligned} {\left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \boldsymbol{0} \end{array}\right]\left[\begin{array}{l} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{array} \right] } & =\left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_{i} \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} \end{array}\right]=\left[\begin{array}{l} \alpha_{i} \boldsymbol{u}_{i} \\ \alpha_{i} \boldsymbol{v}_{i} \end{array}\right]= \alpha_{i}\left[\begin{array}{l} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{array}\right], \quad i=1, \ldots, r, \\ {\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right]\left[\begin{array}{c} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{array}\right] } & =\left[\begin{array}{c} -\boldsymbol{A} \boldsymbol{v}_{i} \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} \end{array}\right]=\left[\begin{array}{c} -\alpha_{i} \boldsymbol{u}_{i} \\ \alpha_{i} \boldsymbol{v}_{i} \end{array}\right]=- \alpha_{i}\left[\begin{array}{c} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{array}\right], \quad i=1, \ldots, r, \\ {\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right]\left[\begin{array}{c} \boldsymbol{u}_{i} \\ \mathbf{0} \end{array}\right] } & = \left[\begin{array}{c} \mathbf{0} \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} \end{array}\right]=\left[\begin{array}{l} \mathbf{0} \\ \mathbf{0} \end{array} \right]=0\left[\begin{array}{c} \boldsymbol{u}_{i} \\ \mathbf{0} \end{array}\right], \quad i=r+1, \ldots, m, \\ {\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right]\left[\begin{array}{c} \mathbf{0} \\ \boldsymbol{v}_{i} \end{array}\right] } & =\left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_{i} \\ \mathbf{0} \end{array}\right]=\left[\begin{array}{l} \mathbf{0} \\ \mathbf{0} \end{array}\right]=0\left[\begin{array}{c} \mathbf{0} \\ \boldsymbol{v}_{i} \end{array}\right], \quad i=r+1, \ldots, n \end{aligned}	$\begin{aligned} {\left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \boldsymbol{0} \end{array}\right]\left[\begin{array}{l} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{array} \right] } & =\left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_{i} \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} \end{array}\right]=\left[\begin{array}{l} \alpha_{i} \boldsymbol{u}_{i} \\ \alpha_{i} \boldsymbol{v}_{i} \end{array}\right]= \alpha_{i}\left[\begin{array}{l} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{array}\right], \quad i=1, \ldots, r, \\ {\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right]\left[\begin{array}{c} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{array}\right] } & =\left[\begin{array}{c} -\boldsymbol{A} \boldsymbol{v}_{i} \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} \end{array}\right]=\left[\begin{array}{c} -\alpha_{i} \boldsymbol{u}_{i} \\ \alpha_{i} \boldsymbol{v}_{i} \end{array}\right]=- \alpha_{i}\left[\begin{array}{c} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{array}\right], \quad i=1, \ldots, r, \\ {\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right]\left[\begin{array}{c} \boldsymbol{u}_{i} \\ \mathbf{0} \end{array}\right] } & = \left[\begin{array}{c} \mathbf{0} \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} \end{array}\right]=\left[\begin{array}{l} \mathbf{0} \\ \mathbf{0} \end{array} \right]=0\left[\begin{array}{c} \boldsymbol{u}_{i} \\ \mathbf{0} \end{array}\right], \quad i=r+1, \ldots, m, \\ {\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right]\left[\begin{array}{c} \mathbf{0} \\ \boldsymbol{v}_{i} \end{array}\right] } & =\left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_{i} \\ \mathbf{0} \end{array}\right]=\left[\begin{array}{l} \mathbf{0} \\ \mathbf{0} \end{array}\right]=0\left[\begin{array}{c} \mathbf{0} \\ \boldsymbol{v}_{i} \end{array}\right], \quad i=r+1, \ldots, n \end{aligned}$	$\begin{aligned} \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix} &= \begin{bmatrix} \boldsymbol{A}\boldsymbol{v}_i \\ \boldsymbol{A}^*\boldsymbol{u}_i \end{bmatrix} = \begin{bmatrix} \alpha_i\boldsymbol{u}_i \\ \alpha_i\boldsymbol{v}_i \end{bmatrix} = \alpha_i \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad i=1,\ldots,r, \\ \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix} &= \begin{bmatrix} -\boldsymbol{A}\boldsymbol{v}_i \\ \boldsymbol{A}^*\boldsymbol{u}_i \end{bmatrix} = \begin{bmatrix} -\alpha_i\boldsymbol{u}_i \\ \alpha_i\boldsymbol{v}_i \end{bmatrix} = -\alpha_i \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad i=1,\ldots,r, \\ \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_i \\ \mathbf{0} \end{bmatrix} &= \begin{bmatrix} \mathbf{0} \\ \boldsymbol{A}^*\boldsymbol{u}_i \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \boldsymbol{u}_i \\ \mathbf{0} \end{bmatrix}, \quad i=r+1,\ldots,m, \\ \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \boldsymbol{v}_i \end{bmatrix} &= \begin{bmatrix} \boldsymbol{A}\boldsymbol{v}_i \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \mathbf{0} \\ \boldsymbol{v}_i \end{bmatrix}, \quad i=r+1,\ldots,n. \end{aligned}$
lyche-numerical-linear-algebra_EQ0371	151	0.194 N|+1	\boldsymbol{J}:=\left[\begin{array}{ccccccc} 3 & 1 & & & & & \\ 0 & 3 & & & & & \\ & & 2 & 1 & & & \\ & & 0 & 2 & & & \\ & & & & 2 & & \\ & & & & 0 & 1 & 0 \\ & & & & 0 & 2 & 1 \\ & & & & 0 & 2 \end{array}\right]	$\boldsymbol{J}:=\left[\begin{array}{ccccccc} 3 & 1 & & & & & \\ 0 & 3 & & & & & \\ & & 2 & 1 & & & \\ & & 0 & 2 & & & \\ & & & & 2 & & \\ & & & & 0 & 1 & 0 \\ & & & & 0 & 2 & 1 \\ & & & & 0 & 2 \end{array}\right]$	$\boldsymbol{J} := \left[\begin{array}{cccccccc} 3 & 1 & & & & & & \\ 0 & 3 & & & & & & \\ & & 2 & 1 & & & & \\ & & 0 & 2 & & & & \\ & & & & 2 & & & \\ & & & & & 2 & 1 & 0 \\ & & & & & 0 & 2 & 1 \\ & & & & & 0 & 0 & 2 \end{array}\right]$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>lyche-numerical-linear-algebra_
EQ0492 (8.20)</td><td>193</td><td>■ 0.232
● W|+1</td><td><code>\begin{aligned} & \|\boldsymbol{A}\|_{2}=\sigma_{1} \text { and }\left\|\boldsymbol{A}^{-1}\right\|_{2}=\frac{1}{ \sigma_{n}} \\ & \|\boldsymbol{A}\|_{2}=\lambda_{1} \text { and }\left\|\boldsymbol{A}^{-1}\right\|_{2}= \frac{1}{\lambda_{n}}, \text { if } \boldsymbol{A} \text { is positive definite, } \\ & \|\boldsymbol{A} \|_{2}=\left|\lambda_{1}\right| \text { and }\left \|\boldsymbol{A}^{-1}\right\|_{2}=\frac{1}{\left| \lambda_{n}\right|}, \text { if } \boldsymbol{A} \text { is normal. } \end{aligned}</code></td><td>$$\begin{aligned} & \|\boldsymbol{A}\|_{2}=\sigma_{1} \text { and }\left\|\boldsymbol{A}^{-1}\right\|_{2}=\frac{1}{\sigma_{n}} \\ & \|\boldsymbol{A}\|_{2}=\lambda_{1} \text { and }\left\|\boldsymbol{A}^{-1}\right\|_{2}=\frac{1}{\lambda_{n}}, \text { if } \boldsymbol{A} \text { is positive definite, } \\ & \|\boldsymbol{A}\|_{2}=\left|\lambda_{1}\right| \text { and }\left\|\boldsymbol{A}^{-1}\right\|_{2}=\frac{1}{\left|\lambda_{n}\right|}, \text { if } \boldsymbol{A} \text { is normal. } \end{aligned}$$</td><td>$$\begin{aligned} &\|\boldsymbol{A}\|_2 = \sigma_1 \textit{ and } \|\boldsymbol{A}^{-1}\|_2 = \frac{1}{\sigma_n}, \\ &\|\boldsymbol{A}\|_2 = \lambda_1 \textit{ and } \|\boldsymbol{A}^{-1}\|_2 = \frac{1}{\lambda_n}, \quad \textit{if } \boldsymbol{A} \textit{ is positive definite,} \\ &\|\boldsymbol{A}\|_2 = |\lambda_1| \textit{ and } \|\boldsymbol{A}^{-1}\|_2 = \frac{1}{|\lambda_n|}, \quad \textit{if } \boldsymbol{A} \textit{ is normal.} \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0010</td><td>024</td><td>■ 0.244
● C|-11</td><td><code>\begin{gathered} \operatorname{diag}\left(d_{i}\right)= \operatorname{diag}\left(d_{1}, \ldots, d_{n}\right):= \left[\begin{array}{cccc} d_{1} & 0 & \cdots & 0 \\ 0 & d_{2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{n} \end{array}\right]= \left[\begin{array}{lll} d_{1} & & \\ & \ddots & \ & & d_{n} \end{array}\right], \\ \boldsymbol{B}= \operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right)= \operatorname{tridiag}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{c}):=\left[\begin{array}{cccc} d_{1} & c_{1} & & \\ a_{1} & d_{2} & c_{2} & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} \\ & & & a_{n-1} \\ & & d_{n} \end{array}\right] . \end{gathered}</code></td><td>$$\begin{gathered} \operatorname{diag}\left(d_{i}\right)=\operatorname{diag}\left(d_{1}, \ldots, d_{n}\right):=\left[\begin{array}{cccc} d_{1} & 0 & \cdots & 0 \\ 0 & d_{2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{n} \end{array}\right]=\left[\begin{array}{lll} d_{1} & & \\ & \ddots & \\ & & d_{n} \end{array}\right], \\ \boldsymbol{B}=\operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right)=\operatorname{tridiag}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{c}):=\left[\begin{array}{cccc} d_{1} & c_{1} & & \\ a_{1} & d_{2} & c_{2} & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} \\ & & & a_{n-1} \\ & & d_{n} & \end{array}\right] . \end{gathered}$$</td><td>$$\operatorname{diag}(d_i) = \operatorname{diag}(d_1, \ldots, d_n) := \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix},$$ $$\boldsymbol{B} = \operatorname{tridiag}(a_i, d_i, c_i) = \operatorname{tridiag}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{c}) := \begin{bmatrix} d_1 & c_1 & & & \\ a_1 & d_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix}.$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0924 (13.53)</td><td>314</td><td>■ 0.288
● W|+1</td><td><code>\begin{array}{rlr} \left(L_{h} v\right)_{j, k} & := \frac{(d v)_{j, k}}{h^{2}}=f_{j, k}, & (j, k) \in I_{m}, \\ v_{j, k} & =0, & (j, k) \in \partial I_{m}, \end{array}</code></td><td>$$\begin{array}{rlr} \left(L_{h} v\right)_{j, k} & := \frac{(d v)_{j, k}}{h^{2}}=f_{j, k}, & (j, k) \in I_{m}, \\ v_{j, k} & =0, & (j, k) \in \partial I_{m}, \end{array}$$</td><td>$$\begin{aligned} (L_h v)_{j,k} &:= \frac{(dv)_{j,k}}{h^2} = f_{j,k}, && (j,k) \in I_m, \\ v_{j,k} &= 0, && (j,k) \in \partial I_m, \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0369 (6.7)</td><td>150</td><td>■ 0.298
● N|+1</td><td><code>\boldsymbol{J}:=\operatorname{diag}\left(\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right)=\left[\begin{array}{ccccccc} 2 & 1 & 0 & & & & \\ 0 & 2 & 1 & & & & \\ 0 & 0 & 2 & & 1 & & \\ 0 & 0 & 2 & & 1 & & \\ & & 0 & 1 & & \\ & & 0 & 2 & & \\ & & & & & 0 & \\ & & & & & 0 & 1 \end{array} \right] \in \mathbb{R}^{8 \times 8} .</code></td><td>$$\boldsymbol{J}:=\operatorname{diag}\left(\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right)=\left[\begin{array}{ccccccc} 2 & 1 & 0 & & & & \\ 0 & 2 & 1 & & & & \\ 0 & 0 & 2 & & 1 & & \\ & & 0 & 1 & & & \\ & & 0 & 2 & & & \\ & & & & & 0 & \\ & & & & & 0 & 1 \end{array}\right] \in \mathbb{R}^{8 \times 8}.$$</td><td>$$\boldsymbol{J} := \operatorname{diag}(\boldsymbol{U}_1, \boldsymbol{U}_2) = \begin{bmatrix} \begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{smallmatrix} & & & \\ & \begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix} & & \\ & & 2 & \\ & & & \begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix} \end{bmatrix} \in \mathbb{R}^{8\times 8}.$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0375</td><td>152</td><td>0.305 W|+0</td><td>\begin{aligned} \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{W}^{*}\left(\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}\right) \boldsymbol{W} & = \left[\begin{array}{l|l} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1}^{*} \end{array}\right] \left[\begin{array}{l|l} \lambda_{1} & \boldsymbol{x}^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right]\left[\begin{array}{l|l} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1} \end{array} \right] \\ & =\left[\begin{array}{l|l} \lambda_{1} & \boldsymbol{x}^{*} \boldsymbol{W}_{1} \\ \hline \mathbf{0} & \boldsymbol{W}_{1}^{*} \boldsymbol{M} \boldsymbol{W}_{1} \end{array}\right] \end{aligned}</td><td>$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{W}^{*}\left(\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}\right) \boldsymbol{W} = \left[\begin{array}{l|l} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1}^{*} \end{array}\right] \left[\begin{array}{l|l} \lambda_{1} & \boldsymbol{x}^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right]\left[\begin{array}{l|l} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1} \end{array} \right] =\left[\begin{array}{l|l} \lambda_{1} & \boldsymbol{x}^{*} \boldsymbol{W}_{1} \\ \hline \mathbf{0} & \boldsymbol{W}_{1}^{*} \boldsymbol{M} \boldsymbol{W}_{1} \end{array}\right]$</td><td>$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{W}^{*}(\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}) \boldsymbol{W} = \left[\begin{array}{l|l} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1}^{*} \end{array}\right] \left[\begin{array}{l|l} \lambda_{1} & \boldsymbol{x}^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right]\left[\begin{array}{l|l} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1} \end{array} \right] =\left[\begin{array}{l|l} \lambda_{1} & \boldsymbol{x}^{*} \boldsymbol{W}_{1} \\ \hline \mathbf{0} & \boldsymbol{W}_{1}^{*} \boldsymbol{M} \boldsymbol{W}_{1} \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0366 (6.4)</td><td>149</td><td>0.315 W|+0</td><td>\boldsymbol{J}_{m}(\lambda):=\left[\begin{array}{cccccc} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array} \right]=\lambda \boldsymbol{I}_{m}+\boldsymbol{E}_{m}, \quad \boldsymbol{E}_{m}:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right] .</td><td>$\boldsymbol{J}_{m}(\lambda):=\left[\begin{array}{cccccc} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array} \right]=\lambda \boldsymbol{I}_{m}+\boldsymbol{E}_{m}, \quad \boldsymbol{E}_{m}:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right] .$</td><td>$\boldsymbol{J}_{m}(\lambda):=\left[\begin{array}{cccccc} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array} \right]=\lambda \boldsymbol{I}_{m}+\boldsymbol{E}_{m}, \quad \boldsymbol{E}_{m}:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0579</td><td>215</td><td>0.322 W|+1</td><td>\begin{aligned} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x} & =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_{1} & \cdots & t_{m} \end{array}\right]\left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ 1 \\ m \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array} \right]=\left[\begin{array}{cc} m & \sum t_{k} \\ \sum t_{k} & \sum t_{k}^{2} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right], \\ & =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_{1} & \cdots & t_{m} \end{array}\right]\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array}\right]=\left[\begin{array}{c} \sum y_{k} \\ \sum t_{k} y_{k} \end{array}\right]=\boldsymbol{A}^{*} \boldsymbol{b}, \end{aligned}</td><td>$\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x} =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_{1} & \cdots & t_{m} \end{array}\right]\left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ 1 \\ m \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array} \right]=\left[\begin{array}{cc} m & \sum t_{k} \\ \sum t_{k} & \sum t_{k}^{2} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right], =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_{1} & \cdots & t_{m} \end{array}\right]\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array}\right]=\left[\begin{array}{c} \sum y_{k} \\ \sum t_{k} y_{k} \end{array}\right]=\boldsymbol{A}^{*} \boldsymbol{b},$</td><td>$\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x} =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_{1} & \cdots & t_{m} \end{array}\right]\left[\begin{array}{cc} 1 & t_1 \\ \vdots & \\ 1 & t_m \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array} \right]=\left[\begin{array}{cc} m & \sum t_{k} \\ \sum t_{k} & \sum t_{k}^{2} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right], =\left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_{1} & \cdots & t_{m} \end{array}\right]\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array}\right]=\left[\begin{array}{c} \sum y_{k} \\ \sum t_{k} y_{k} \end{array}\right]=\boldsymbol{A}^{*} \boldsymbol{b},$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0335 (5.19)	135	■ 0.329 ● N|+0	`\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right]=\left[\begin{array}{rc} c & s \\ -s c & \end{array}\right] \left[\begin{array}{l} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right],[\boldsymbol{C}(:, i) \boldsymbol{C}(:, j)]=[\boldsymbol{A}(:, i) \boldsymbol{A}(:, j)]\left[\begin{array}{rc} c & s \\ -s & c \end{array}\right] .`	$\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right]=\left[\begin{array}{rc} c & s \\ -s c & \end{array}\right] \left[\begin{array}{l} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right],[\boldsymbol{C}(:, i) \boldsymbol{C}(:, j)]=[\boldsymbol{A}(:, i) \boldsymbol{A}(:, j)]\left[\begin{array}{rc} c & s \\ -s & c \end{array}\right] .$	$\begin{bmatrix} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{bmatrix}, \ [\boldsymbol{C}(:,i)\ \boldsymbol{C}(:,j)] = [\boldsymbol{A}(:,i)\ \boldsymbol{A}(:,j)] \begin{bmatrix} c & s \\ -s & c \end{bmatrix}.$
lyche-numerical-linear-algebra_ EQ0917	312	■ 0.371 ● W|+2	`\frac{\left|\boldsymbol{x}-\boldsymbol{x}_{k}\right|_{\boldsymbol{A}}}{\left|\boldsymbol{x}-\boldsymbol{x}_{0}\right|_{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k} \quad \text{ for } \quad k \geq 0`	$\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0}\right\|_{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k} \quad \text{ for } \quad k \geq 0$	$\frac{\|\boldsymbol{x}-\boldsymbol{x}_k\|_A}{\|\boldsymbol{x}-\boldsymbol{x}_0\|_A} \le 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k, \quad \textit{for} \quad k \ge 0,$
lyche-numerical-linear-algebra_ EQ0547	207	■ 0.373 ● N|+0	`c_{i, k+1}= \begin{cases}0 & \text { if } \sigma_{i}= \lambda_{k+1}, \\ c_{i, k}\left(1-\frac{\sigma_{i}}{ \lambda_{k+1}}\right) & \text { otherwise } .\end{cases}`	$c_{i, k+1}= \begin{cases}0 & \text { if } \sigma_{i}= \lambda_{k+1}, \\ c_{i, k}\left(1-\frac{\sigma_{i}}{ \lambda_{k+1}}\right) & \text { otherwise } .\end{cases}$	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k}\left(1-\frac{\sigma_i}{\lambda_{k+1}}\right) & \text{otherwise.} \end{cases}$
lyche-numerical-linear-algebra_ EQ0855	296	■ 0.386 ● C|-49	`\begin{array}{ll} \boldsymbol{t}_{0}=\left[\begin{array}{c} 3 \\ -3 / 2 \end{array}\right], \quad \alpha_{0}=1 / 2, \quad \boldsymbol{x}_{1}=\left[\begin{array}{l} -1 / 4 \\ -1 / 2 \end{array}\right], \quad \boldsymbol{r}_{1}=\left[\begin{array}{c} 0 \\ 3 / 4 \end{array}\right], \quad \beta_{0}=1 / 4, \\ \boldsymbol{p}_{1}=\left[\begin{array}{l} 3 / 8 \\ 3 / 4 \end{array}\right], \quad \boldsymbol{t}_{1}=\left[\begin{array}{c} 0 \\ 9 / 8 \end{array}\right], \quad \alpha_{1}=2 / 3, \quad \boldsymbol{x}_{2}=\mathbf{0}, \quad \boldsymbol{r}_{2}=\mathbf{0} . \end{array}`	$\begin{array}{ll} \boldsymbol{t}_{0}=\left[\begin{array}{c} 3 \\ -3 / 2 \end{array}\right], \quad \alpha_{0}=1 / 2, \quad \boldsymbol{x}_{1}=\left[\begin{array}{l} -1 / 4 \\ -1 / 2 \end{array}\right], \quad \boldsymbol{r}_{1}=\left[\begin{array}{c} 0 \\ 3 / 4 \end{array}\right], \quad \beta_{0}=1 / 4, \\ \boldsymbol{p}_{1}=\left[\begin{array}{l} 3 / 8 \\ 3 / 4 \end{array}\right], \quad \boldsymbol{t}_{1}=\left[\begin{array}{c} 0 \\ 9 / 8 \end{array}\right], \quad \alpha_{1}=2 / 3, \quad \boldsymbol{x}_{2}=\mathbf{0}, \quad \boldsymbol{r}_{2}=\mathbf{0} . \end{array}$	$\boldsymbol{t}_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4,$ $\boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}.$
lyche-numerical-linear-algebra_ EQ0670 (10.10)	241	■ 0.390 ● W|+2	`\begin{aligned} a_{i i} & =2 d, i=1, \ldots, n, \\ a_{i, i+1}=a_{i+1, i} & =a, \quad i=1, \ldots, n-1, \quad i \neq m, 2 m, \ldots,(m-1) m, \\ a_{i, i+m}=a_{i+m, i} & =a, \quad i=1, \ldots, n-m, \\ a_{i j} & =0, \text { otherwise } \end{aligned}`	$\begin{aligned} a_{i i} & =2 d, i=1, \ldots, n, \\ a_{i, i+1}=a_{i+1, i} & =a, \quad i=1, \ldots, n-1, \quad i \neq m, 2 m, \ldots,(m-1) m, \\ a_{i, i+m}=a_{i+m, i} & =a, \quad i=1, \ldots, n-m, \\ a_{i j} & =0, \text { otherwise } \end{aligned}$	$\begin{aligned} a_{ii} &= 2d,\ i = 1,\dots,n, \\ a_{i,i+1} = a_{i+1,i} &= a, \quad i = 1,\dots,n-1, \quad i \ne m, 2m, \dots, (m-1)m, \\ a_{i,i+m} = a_{i+m,i} &= a, \quad i = 1,\dots,n-m, \\ a_{ij} &= 0, \quad \text{otherwise,} \end{aligned}$
lyche-numerical-linear-algebra_ EQ0763 (12.14)	271	■ 0.402 ● N|+0	`\boldsymbol{G}_{\omega}=\left[\begin{array}{ll} l & 0 \\ 0 & 1 \end{array}\right]-\left[\begin{array}{cc} \omega / 2 & 0 \\ \omega^{2} / 4 & \omega / 2 \end{array}\right] \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1-\omega & \omega / 2 \\ \omega(1-\omega) / 2 & 1-\omega+\omega^{2} / 4 \end{array}\right] .`	$\boldsymbol{G}_{\omega}=\left[\begin{array}{ll} l & 0 \\ 0 & 1 \end{array}\right]-\left[\begin{array}{cc} \omega / 2 & 0 \\ \omega^{2} / 4 & \omega / 2 \end{array}\right] \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1-\omega & \omega / 2 \\ \omega(1-\omega) / 2 & 1-\omega+\omega^{2} / 4 \end{array}\right] .$	$\boldsymbol{G}_\omega = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{bmatrix}.$
lyche-numerical-linear-algebra_ EQ0498	194	■ 0.420 ● C|+4	`\begin{aligned} & x_{1}+\left(1-10^{-16}\right) x_{2}=20 \\ & x_{1}+\left(1-10^{-15}\right.\end{aligned}`	$\begin{aligned} & x_{1}+\left(1-10^{-16}\right) x_{2}=20 \\ & x_{1}+\left(1-10^{-15}\right.\end{aligned}$	$\begin{aligned} x_1 + \qquad\qquad x_2 &= 20 \\ x_1 + (1-10^{-16})x_2 &= 20 - 10^{-15} \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0262	107	0.421 N|-1	`\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=z^{*} \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} \boldsymbol{z}=z^{*} \operatorname{diag}\left( \lambda_{1}, \ldots, \lambda_{n}\right) \boldsymbol{z}= \sum_{j=1}^{n} \lambda_{j}\left|z_{j}\right|^{2} \geq 0 .`	$\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x}=z^{*}\boldsymbol{U}^{*}\boldsymbol{A}\boldsymbol{U}\boldsymbol{z}=z^{*}\operatorname{diag}\left(\lambda_{1},\ldots,\lambda_{n}\right)\boldsymbol{z}=\sum_{j=1}^{n}\lambda_{j}\left\vert z_{j}\right\vert^{2}\geq 0.$	$\boldsymbol{x}^{*}\boldsymbol{A}\boldsymbol{x}=\boldsymbol{z}^{*}\boldsymbol{U}^{*}\boldsymbol{A}\boldsymbol{U}\boldsymbol{z}=\boldsymbol{z}^{*}\operatorname{diag}(\lambda_{1},\ldots,\lambda_{n})\boldsymbol{z}=\sum_{j=1}^{n}\lambda_{j}\vert z_{j}\vert^{2}\geq 0.$
lyche-numerical-linear-algebra_ EQ0404	161	0.448 C|+10	`\left[\begin{array}{llllllll} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{8,8} ?`	$\left[\begin{array}{llllllll} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{8,8} ?$	$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix} \in \mathbb{R}^{8,8}?$
lyche-numerical-linear-algebra_ EQ0914 (13.48)	311	0.453 N|+0	`\begin{aligned} & \boldsymbol{y}_{k+1}= \boldsymbol{y}_{k}+\alpha_{k} \boldsymbol{q}_{k}, \quad \alpha_{k}=z_{k}^{T} z_{k} / \boldsymbol{q}_{k}^{T}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{q}_{k}, \\ & \boldsymbol{z}_{k+1}=z_{k}-\alpha_{k}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{q}_{k}, \\ & \boldsymbol{q}_{k+1}=z_{k+1}+\beta_{k} \boldsymbol{q}_{k}, \quad \beta_{k}=z_{k+1}^{T} z_{k+1} / z_{k}^{T} z_{k} . \end{aligned}`	$\begin{aligned} & \boldsymbol{y}_{k+1}=\boldsymbol{y}_{k}+\alpha_{k}\boldsymbol{q}_{k}, \quad \alpha_{k}=z_{k}^{T}z_{k}/\boldsymbol{q}_{k}^{T}\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T}\right)\boldsymbol{q}_{k}, \\ & \boldsymbol{z}_{k+1}=z_{k}-\alpha_{k}\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T}\right)\boldsymbol{q}_{k}, \\ & \boldsymbol{q}_{k+1}=z_{k+1}+\beta_{k}\boldsymbol{q}_{k}, \quad \beta_{k}=z_{k+1}^{T}z_{k+1}/z_{k}^{T}z_{k}. \end{aligned}$	$\begin{aligned} & \boldsymbol{y}_{k+1}=\boldsymbol{y}_{k}+\alpha_{k}\boldsymbol{q}_{k}, \quad \alpha_{k}=\boldsymbol{z}_{k}^{T}\boldsymbol{z}_{k}/\boldsymbol{q}_{k}^{T}(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T})\boldsymbol{q}_{k}, \\ & \boldsymbol{z}_{k+1}=\boldsymbol{z}_{k}-\alpha_{k}(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T})\boldsymbol{q}_{k}, \\ & \boldsymbol{q}_{k+1}=\boldsymbol{z}_{k+1}+\beta_{k}\boldsymbol{q}_{k}, \quad \beta_{k}=\boldsymbol{z}_{k+1}^{T}\boldsymbol{z}_{k+1}/\boldsymbol{z}_{k}^{T}\boldsymbol{z}_{k}. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0245	100	0.458 N|-1	`\left[\begin{array}{ll} a & \bar{b} \\ b & d \end{array} \right]=\left[\begin{array}{ll} 1 & 0 \\ l_{1} & 1 \end{array}\right]\left[\begin{array}{cc} d_{1} & 0 \\ 0 & d_{2} \end{array}\right]\left[\begin{array}{cc} 1 & \overline{l_{1}} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} d_{1} & d_{1} \overline{l_{1}} \\ d_{1} l_{1} & d_{1}\left|l_{1}\right|^{2}+d_{2} \end{array} \right]`	$\left[\begin{array}{ll} a & \bar{b} \\ b & d \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ l_{1} & 1 \end{array}\right]\left[\begin{array}{cc} d_{1} & 0 \\ 0 & d_{2} \end{array}\right]\left[\begin{array}{cc} 1 & \overline{l_{1}} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} d_{1} & d_{1}\overline{l_{1}} \\ d_{1}l_{1} & d_{1}\left\vert l_{1}\right\vert^{2}+d_{2} \end{array}\right]$	$\begin{bmatrix} a & \bar{b} \\ b & d \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ l_{1} & 1 \end{bmatrix}\begin{bmatrix} d_{1} & 0 \\ 0 & d_{2} \end{bmatrix}\begin{bmatrix} 1 & \bar{l}_{1} \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} d_{1} & d_{1}\bar{l}_{1} \\ d_{1}l_{1} & d_{1}\vert l_{1}\vert^{2}+d_{2} \end{bmatrix}$
lyche-numerical-linear-algebra_ EQ0370	150	0.466 N|+1	`\begin{aligned} & \boldsymbol{A} \boldsymbol{s}_{1}=2 \boldsymbol{s}_{1}, \quad \boldsymbol{A} \boldsymbol{s}_{2}=\boldsymbol{s}_{1}+2 \boldsymbol{s}_{2}, \quad \boldsymbol{A} \boldsymbol{s}_{3}=\boldsymbol{s}_{2}+2 \boldsymbol{s}_{3}, \\ & \boldsymbol{A} \boldsymbol{s}_{4}=2 \boldsymbol{s}_{4}, \quad \boldsymbol{A} \boldsymbol{s}_{5}=\boldsymbol{s}_{4}+2 \boldsymbol{s}_{5} \\ & \boldsymbol{A} \boldsymbol{s}_{6}=2 \boldsymbol{s}_{6}, \\ & \boldsymbol{A} \boldsymbol{s}_{7}=3 \boldsymbol{s}_{7}, \quad \boldsymbol{A} \boldsymbol{s}_{8}= \boldsymbol{s}_{7}+3 \boldsymbol{s}_{8} . \end{aligned}`	$\begin{aligned} & \boldsymbol{A}\boldsymbol{s}_{1}=2\boldsymbol{s}_{1}, \quad \boldsymbol{A}\boldsymbol{s}_{2}=\boldsymbol{s}_{1}+2\boldsymbol{s}_{2}, \quad \boldsymbol{A}\boldsymbol{s}_{3}=\boldsymbol{s}_{2}+2\boldsymbol{s}_{3}, \\ & \boldsymbol{A}\boldsymbol{s}_{4}=2\boldsymbol{s}_{4}, \quad \boldsymbol{A}\boldsymbol{s}_{5}=\boldsymbol{s}_{4}+2\boldsymbol{s}_{5} \\ & \boldsymbol{A}\boldsymbol{s}_{6}=2\boldsymbol{s}_{6}, \\ & \boldsymbol{A}\boldsymbol{s}_{7}=3\boldsymbol{s}_{7}, \quad \boldsymbol{A}\boldsymbol{s}_{8}=\boldsymbol{s}_{7}+3\boldsymbol{s}_{8}. \end{aligned}$	$\begin{aligned} & \boldsymbol{A}\boldsymbol{s}_{1}=2\boldsymbol{s}_{1}, \quad \boldsymbol{A}\boldsymbol{s}_{2}=\boldsymbol{s}_{1}+2\boldsymbol{s}_{2}, \quad \boldsymbol{A}\boldsymbol{s}_{3}=\boldsymbol{s}_{2}+2\boldsymbol{s}_{3}, \\ & \boldsymbol{A}\boldsymbol{s}_{4}=2\boldsymbol{s}_{4}, \quad \boldsymbol{A}\boldsymbol{s}_{5}=\boldsymbol{s}_{4}+2\boldsymbol{s}_{5}, \\ & \boldsymbol{A}\boldsymbol{s}_{6}=2\boldsymbol{s}_{6}, \\ & \boldsymbol{A}\boldsymbol{s}_{7}=3\boldsymbol{s}_{7}, \quad \boldsymbol{A}\boldsymbol{s}_{8}=\boldsymbol{s}_{7}+3\boldsymbol{s}_{8}. \end{aligned}$

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0396</td><td>158</td><td>■ 0.467 ● C|-3</td><td>\begin{aligned} \beta_{k} & \leq \max _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{B}}(\boldsymbol{x}) \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{A}}(\boldsymbol{x})+\max _{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{E}}(\boldsymbol{x}) \\ & \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{A}}(\boldsymbol{x})+\max _{\substack{ \boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{E}}(\boldsymbol{x})= \alpha_{k}+\varepsilon_{1}, \end{aligned}</td><td>$\begin{aligned} \beta_{k} & \leq \max _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{B}}(\boldsymbol{x}) \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{A}}(\boldsymbol{x})+\max _{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{E}}(\boldsymbol{x}) \\ & \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{A}}(\boldsymbol{x})+\max _{\substack{ \boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{E}}(\boldsymbol{x})= \alpha_{k}+\varepsilon_{1}, \end{aligned}$</td><td>$\begin{aligned} \beta_{k} & \leq \max _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{B}}(\boldsymbol{x}) \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{A}}(\boldsymbol{x})+\max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{E}}(\boldsymbol{x}) \\ & \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{A}}(\boldsymbol{x})+\max _{\substack{ \boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} R_{\boldsymbol{E}}(\boldsymbol{x})= \alpha_{k}+\varepsilon_{1}, \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0308</td><td>125</td><td>■ 0.469 ● N|+0</td><td>\boldsymbol{P} \boldsymbol{x}=\boldsymbol{x}-\frac{ \boldsymbol{v}^{*} \boldsymbol{x}}{\boldsymbol{v}^{*} \boldsymbol{v}} \boldsymbol{v} \stackrel{(5.13)}{=} \boldsymbol{x}-\frac{1}{2} \boldsymbol{v}=\frac{1}{2}(\boldsymbol{x}+\boldsymbol{y}) .</td><td>$\boldsymbol{P} \boldsymbol{x}=\boldsymbol{x}-\frac{ \boldsymbol{v}^{*} \boldsymbol{x}}{\boldsymbol{v}^{*} \boldsymbol{v}} \boldsymbol{v} \stackrel{(5.13)}{=} \boldsymbol{x}-\frac{1}{2} \boldsymbol{v}=\frac{1}{2}(\boldsymbol{x}+\boldsymbol{y}) .$</td><td>$\boldsymbol{P} \boldsymbol{x}=\boldsymbol{x}-\frac{ \boldsymbol{v}^{*} \boldsymbol{x}}{\boldsymbol{v}^{*} \boldsymbol{v}} \boldsymbol{v} \stackrel{(5.13)}{=} \boldsymbol{x}-\frac{1}{2} \boldsymbol{v}=\frac{1}{2}(\boldsymbol{x}+\boldsymbol{y}) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0600 (0)</td><td>221</td><td>■ 0.474 ● N|+0</td><td>\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrcr} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right]= \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right],</td><td>$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrcr} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right]= \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right],$</td><td>$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right]= \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0028 (1.11)</td><td>031</td><td>■ 0.479 ● N|+0</td><td>\|x\|_{2}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{2} \right)^{1 / 2}=\sqrt{x^{*} x} .</td><td>$\|x\|_{2}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{2} \right)^{1 / 2}=\sqrt{x^{*} x} .$</td><td>$\|\boldsymbol{x}\|_{2}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{2} \right)^{1 / 2}=\sqrt{\boldsymbol{x}^{*} \boldsymbol{x}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0241 (3.33)</td><td>097</td><td>■ 0.492 ● C|+5</td><td>\boldsymbol{B}:=\left[\begin{array}{cc} \left(1+\boldsymbol{u}_{1}^{T}\right. & \left. \boldsymbol{B}_{2,2} \boldsymbol{\ell}_{1} \right) / u_{1,1} \\ -\boldsymbol{u}_{2,2}^{T} & \boldsymbol{B}_{2,2} / u_{1,1} \\ & \boldsymbol{B}_{2,2} \end{array}\right] .</td><td>$\boldsymbol{B}:=\left[\begin{array}{cc} \left(1+\boldsymbol{u}_{1}^{T}\right. & \left. \boldsymbol{B}_{2,2} \boldsymbol{\ell}_{1} \right) / u_{1,1} \\ -\boldsymbol{u}_{2,2}^{T} & \boldsymbol{B}_{2,2} / u_{1,1} \\ & \boldsymbol{B}_{2,2} \end{array}\right] .$</td><td>$\boldsymbol{B}:=\left[\begin{array}{cc} (1+\boldsymbol{u}_{1}^{T} \boldsymbol{B}_{2,2} \boldsymbol{\ell}_{1}) / u_{1,1} & -\boldsymbol{u}_{1}^{T} \boldsymbol{B}_{2,2} / u_{1,1} \\ -\boldsymbol{B}_{2,2} \boldsymbol{\ell}_{1} & \boldsymbol{B}_{2,2} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0757 (12.11)</td><td>270</td><td>■ 0.498 ● C|+6</td><td>\boldsymbol{A}_{\boldsymbol{L}}:=\left[\begin{array}{ccc} 0 & & \\ -a_{2,1} & 0 & \\ \vdots & \ddots & \ddots \\ -a_{n, 1} & \cdots & -a_{n, n-1} \end{array} \right], \quad \boldsymbol{A}_{\boldsymbol{R}}:=\left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ & 0 & -a_{n-1, n} \\ & & 0 \end{array} \right] .</td><td>$\boldsymbol{A}_{\boldsymbol{L}}:=\left[\begin{array}{ccc} 0 & & \\ -a_{2,1} & 0 & \\ \vdots & \ddots & \ddots \\ -a_{n, 1} & \cdots & -a_{n, n-1} \end{array} \right], \quad \boldsymbol{A}_{\boldsymbol{R}}:=\left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ & 0 & -a_{n-1, n} \\ & & 0 \end{array} \right] .$</td><td>$\boldsymbol{A}_{\boldsymbol{L}}:=\left[\begin{array}{cccc} 0 & & & \\ -a_{2,1} & 0 & & \\ \vdots & \ddots & \ddots & \\ -a_{n, 1} & \cdots & -a_{n, n-1} & 0 \end{array} \right], \quad \boldsymbol{A}_{\boldsymbol{R}}:=\left[\begin{array}{cccc} 0 & -a_{1,2} & \cdots & -a_{1,n} \\ & \ddots & \ddots & \vdots \\ & & 0 & -a_{n-1, n} \\ & & & 0 \end{array} \right] .$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0138</td><td>064</td><td>0.503 W|+1</td><td>\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right]\left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right]=\left[\begin{array}{ll} A_{11} B_{11}+A_{12} B_{21} & A_{11} B_{12}+A_{12} B_{22} \\ A_{21} B_{11}+A_{22} B_{21} & A_{21} B_{12}+A_{22} B_{22} \end{array}\right],</td><td>$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right]\left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right]=\left[\begin{array}{ll} A_{11} B_{11}+A_{12} B_{21} & A_{11} B_{12}+A_{12} B_{22} \\ A_{21} B_{11}+A_{22} B_{21} & A_{21} B_{12}+A_{22} B_{22} \end{array}\right],$</td><td>$\begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix}\begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{bmatrix}=\begin{bmatrix} \boldsymbol{A}_{11}\boldsymbol{B}_{11}+\boldsymbol{A}_{12}\boldsymbol{B}_{21} & \boldsymbol{A}_{11}\boldsymbol{B}_{12}+\boldsymbol{A}_{12}\boldsymbol{B}_{22} \\ \boldsymbol{A}_{21}\boldsymbol{B}_{11}+\boldsymbol{A}_{22}\boldsymbol{B}_{21} & \boldsymbol{A}_{21}\boldsymbol{B}_{12}+\boldsymbol{A}_{22}\boldsymbol{B}_{22} \end{bmatrix},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0465 (8.1)</td><td>186</td><td>0.507 N|+0</td><td>\begin{aligned} \|\boldsymbol{x}\|_{p} & :=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{p}\right)^{1 / p}, \\ \|\boldsymbol{x}\|_{\infty} & :=\max _{1 \leq j \leq n}\left|x_{j}\right| . \end{aligned}</td><td>$\begin{aligned} \|\boldsymbol{x}\|_{p} & :=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{p}\right)^{1 / p}, \\ \|\boldsymbol{x}\|_{\infty} & :=\max _{1 \leq j \leq n}\left|x_{j}\right| . \end{aligned}$</td><td>$\|\boldsymbol{x}\|_p := \Big(\sum_{j=1}^{n}|x_j|^p\Big)^{1/p},$
$\|\boldsymbol{x}\|_\infty := \max_{1\le j\le n}|x_j|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0486</td><td>191</td><td>0.508 S|-1</td><td>\|x\|_{p}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{p} \right)^{1 / p}, p \geq 1, \quad\|x\|_{\infty}:=\max _{1 \leq j \leq n}\left|x_{j}\right| .</td><td>$\|x\|_{p}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{p} \right)^{1 / p}, p \geq 1, \quad\|x\|_{\infty}:=\max _{1 \leq j \leq n}\left|x_{j}\right| .$</td><td>$\|\boldsymbol{x}\|_p := \big(\sum_{j=1}^{n}|x_j|^p\big)^{1/p},\ p\ge 1, \quad \|\boldsymbol{x}\|_\infty := \max_{1\le j\le n}|x_j|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0766</td><td>272</td><td>0.512 N|+0</td><td>x_{k+1}(2)=\frac{1}{2}\left(\frac{1}{2} x_{k}(2)+ \frac{1}{2}\right)+\frac{1}{2}=\frac{1}{4} x_{k}(2)+ \frac{3}{4} .</td><td>$x_{k+1}(2)=\frac{1}{2}\left(\frac{1}{2} x_{k}(2)+ \frac{1}{2}\right)+\frac{1}{2}=\frac{1}{4} x_{k}(2)+ \frac{3}{4} .$</td><td>$\boldsymbol{x}_{k+1}(2) = \frac{1}{2}(\frac{1}{2}\boldsymbol{x}_k(2)+\frac{1}{2})+\frac{1}{2} = \frac{1}{4}\boldsymbol{x}_k(2)+\frac{3}{4}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1017</td><td>345</td><td>0.514 K|+0</td><td>z_{0}=\frac{1}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right]+\frac{1}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]=c_{1} \boldsymbol{v}_{1}+c_{2} \boldsymbol{v}_{2} .</td><td>$z_{0}=\frac{1}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right]+\frac{1}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]=c_{1} \boldsymbol{v}_{1}+c_{2} \boldsymbol{v}_{2} .$</td><td>$\boldsymbol{z}_0 = \frac{1}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0798</td><td>280</td><td>0.516 W|+1</td><td>\begin{aligned} \left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\| & =\left\|\boldsymbol{G}\left(\boldsymbol{x}_{k-1}-\boldsymbol{x}\right)\right\| \leq\|\boldsymbol{G}\|\left\|\boldsymbol{x}_{k-1}-\boldsymbol{x}\right\| \\ & =\|\boldsymbol{G}\| \left\|\boldsymbol{x}_{k-1}-\boldsymbol{x}_{k}+\boldsymbol{x}_{k}-\boldsymbol{x}\right\| \leq\|\boldsymbol{G}\|\left(\left\|\boldsymbol{x}_{k-1}-\boldsymbol{x}_{k}\right\|+\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|\right) \end{aligned}</td><td>$\begin{aligned} \left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\| & =\left\|\boldsymbol{G}\left(\boldsymbol{x}_{k-1}-\boldsymbol{x}\right)\right\| \leq\|\boldsymbol{G}\|\left\|\boldsymbol{x}_{k-1}-\boldsymbol{x}\right\| \\ & =\|\boldsymbol{G}\| \left\|\boldsymbol{x}_{k-1}-\boldsymbol{x}_{k}+\boldsymbol{x}_{k}-\boldsymbol{x}\right\| \leq\|\boldsymbol{G}\|\left(\left\|\boldsymbol{x}_{k-1}-\boldsymbol{x}_{k}\right\|+\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|\right) \end{aligned}$</td><td>$\begin{aligned} \|\boldsymbol{x}_k-\boldsymbol{x}\| &= \|\boldsymbol{G}(\boldsymbol{x}_{k-1}-\boldsymbol{x})\| \le \|\boldsymbol{G}\|\|\boldsymbol{x}_{k-1}-\boldsymbol{x}\| \\ &= \|\boldsymbol{G}\|\|\boldsymbol{x}_{k-1}-\boldsymbol{x}_k+\boldsymbol{x}_k-\boldsymbol{x}\| \le \|\boldsymbol{G}\|\big(\|\boldsymbol{x}_{k-1}-\boldsymbol{x}_k\|+\|\boldsymbol{x}_k-\boldsymbol{x}\|\big). \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0837 (13.3)</td><td>291</td><td>0.516 N|+0</td><td>Q(\boldsymbol{y}):=\frac{1}{2} \boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}^{T} \boldsymbol{y}+c .</td><td>$Q(\boldsymbol{y}):=\frac{1}{2} \boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}^{T} \boldsymbol{y}+c .$</td><td>$Q(\boldsymbol{y}) := \frac{1}{2}\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}^T\boldsymbol{y} + c.$</td></tr>
</table>

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0854 (13.18)	296	■ 0.521 ● W|+0	\begin{aligned} \boldsymbol{t}_{k} & =\boldsymbol{A} \boldsymbol{p}_{k}, \\ \alpha_{k} & =\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right) /\left(\boldsymbol{p}_{k}^{T} \boldsymbol{t}_{k} \right), \\ \boldsymbol{x}_{k+1} & =\boldsymbol{x}_{k}+ \alpha_{k} \boldsymbol{p}_{k}, \\ \boldsymbol{r}_{k+1} & =\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{t}_{k}, \\ \beta_{k} & =\left(\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}\right) /\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right), \\ \boldsymbol{p}_{k+1} & := \boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k} . \end{aligned}	$\begin{aligned} \boldsymbol{t}_{k} & =\boldsymbol{A} \boldsymbol{p}_{k}, \\ \alpha_{k} & =\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right) /\left(\boldsymbol{p}_{k}^{T} \boldsymbol{t}_{k} \right), \\ \boldsymbol{x}_{k+1} & =\boldsymbol{x}_{k}+ \alpha_{k} \boldsymbol{p}_{k}, \\ \boldsymbol{r}_{k+1} & =\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{t}_{k}, \\ \beta_{k} & =\left(\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}\right) /\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right), \\ \boldsymbol{p}_{k+1} & := \boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k} . \end{aligned}$	$\begin{aligned} \boldsymbol{t}_{k} & =\boldsymbol{A} \boldsymbol{p}_{k}, \\ \alpha_{k} & =(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k})/(\boldsymbol{p}_{k}^{T} \boldsymbol{t}_{k}), \\ \boldsymbol{x}_{k+1} & =\boldsymbol{x}_{k}+ \alpha_{k} \boldsymbol{p}_{k}, \\ \boldsymbol{r}_{k+1} & =\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{t}_{k}, \\ \beta_{k} & =(\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1})/(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}), \\ \boldsymbol{p}_{k+1} & := \boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{p}_{k} . \end{aligned}$
lyche-numerical-linear-algebra_ EQ0329	134	■ 0.523 ● K|+0	\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] .	$\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0862	300	■ 0.523 ● S|-1	\kappa_{P}=\frac{\lambda_{\max }}{\lambda_{\min }}=\frac{ \cos ^{2}(\pi h / 2)}{\sin ^{2}(\pi h / 2)} \text { and } \sqrt{\kappa_{P}}=\frac{\cos (\pi h / 2)}{\sin (\pi h / 2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n} \text {. }	$\kappa_{P}=\frac{\lambda_{\max }}{\lambda_{\min }}=\frac{ \cos ^{2}(\pi h / 2)}{\sin ^{2}(\pi h / 2)} \text { and } \sqrt{\kappa_{P}}=\frac{\cos (\pi h / 2)}{\sin (\pi h / 2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n} \text {. }$	$\kappa_{P}=\frac{\lambda_{max}}{\lambda_{min}}=\frac{\cos^{2}(\pi h/2)}{\sin^{2}(\pi h/2)} \text{ and } \sqrt{\kappa_{P}}=\frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi}\sqrt{n}.$
lyche-numerical-linear-algebra_ EQ0504 (8.26)	197	■ 0.531 ● W|+0	K_{2}(\boldsymbol{A})= \begin{cases}\lambda_{1} / \lambda_{n}, & \text { if } \boldsymbol{A} \text { is positive definite, } \\ \left|\lambda_{1}\right| /\left|\lambda_{n}\right|, & \text { if } \boldsymbol{A} \text { is normal, } \\ \sigma_{1} / \sigma_{n}, & \text { in general. }\end{cases}	$K_{2}(\boldsymbol{A})= \begin{cases}\lambda_{1} / \lambda_{n}, & \text { if } \boldsymbol{A} \text { is positive definite, } \\ \left\|\lambda_{1}\right\| /\left\|\lambda_{n}\right\|, & \text { if } \boldsymbol{A} \text { is normal, } \\ \sigma_{1} / \sigma_{n}, & \text { in general. }\end{cases}$	$K_{2}(\boldsymbol{A})= \begin{cases}\lambda_{1}/\lambda_{n}, & \textit{if } \boldsymbol{A} \textit{ is positive definite,} \\ \|\lambda_{1}\|/\|\lambda_{n}\|, & \textit{if } \boldsymbol{A} \textit{ is normal,} \\ \sigma_{1}/\sigma_{n}, & \textit{in general.}\end{cases}$
lyche-numerical-linear-algebra_ EQ0830 (12.46)	287	■ 0.531 ● N|+0	\boldsymbol{A}=\boldsymbol{M}-\boldsymbol{N}, \quad \boldsymbol{M}=(\boldsymbol{I}-\boldsymbol{L})\left(\boldsymbol{I}-\boldsymbol{L}^{T}\right), \quad \boldsymbol{N}=\boldsymbol{L} \boldsymbol{L}^{T} .	$\boldsymbol{A}=\boldsymbol{M}-\boldsymbol{N}, \quad \boldsymbol{M}=(\boldsymbol{I}-\boldsymbol{L})\left(\boldsymbol{I}-\boldsymbol{L}^{T}\right), \quad \boldsymbol{N}=\boldsymbol{L} \boldsymbol{L}^{T} .$	$\boldsymbol{A}=\boldsymbol{M}-\boldsymbol{N}, \quad \boldsymbol{M}=(\boldsymbol{I}-\boldsymbol{L})(\boldsymbol{I}-\boldsymbol{L}^{T}), \quad \boldsymbol{N}=\boldsymbol{L}\boldsymbol{L}^{T}.$
lyche-numerical-linear-algebra_ EQ0098	054	■ 0.532 ● C|+3	\begin{aligned} & u_{1}=d_{1}, \quad l_{1}=a_{1} / u_{1}, \quad u_{2}=d_{2}-l_{1} c_{1}, \quad l_{2}=a_{2} / u_{2}, \quad u_{3}=d_{3}-l_{2} c_{2} \\ & z_{1}=b_{1}, \quad z_{2}=b_{2}-l_{1} z_{1}, \quad z_{3}=b_{3}-l_{2} z_{2} \\ & x_{3}=z_{3} / u_{3}, \quad x_{2}=\left(z_{2}-c_{2} x_{3}\right) / u_{2}, \quad x_{1}=\left(z_{1}-c_{1} x_{2} \right) / u_{1} \end{aligned}	$\begin{aligned} & u_{1}=d_{1}, \quad l_{1}=a_{1} / u_{1}, \quad u_{2}=d_{2}-l_{1} c_{1}, \quad l_{2}=a_{2} / u_{2}, \quad u_{3}=d_{3}-l_{2} c_{2} \\ & z_{1}=b_{1}, \quad z_{2}=b_{2}-l_{1} z_{1}, \quad z_{3}=b_{3}-l_{2} z_{2} \\ & x_{3}=z_{3} / u_{3}, \quad x_{2}=\left(z_{2}-c_{2} x_{3}\right) / u_{2}, \quad x_{1}=\left(z_{1}-c_{1} x_{2} \right) / u_{1} \end{aligned}$	$\begin{aligned} & u_{1}=d_{1}, \quad l_{1}=a_{1}/u_{1}, \quad u_{2}=d_{2}-l_{1}c_{1}, \quad l_{2}=a_{2}/u_{2}, \quad u_{3}=d_{3}-l_{2}c_{2}, \\ & z_{1}=b_{1}, \quad z_{2}=b_{2}-l_{1}z_{1}, \quad z_{3}=b_{3}-l_{2}z_{2}, \\ & x_{3}=z_{3}/u_{3}, \quad x_{2}=(z_{2}-c_{2}x_{3})/u_{2}, \quad x_{1}=(z_{1}-c_{1}x_{2})/u_{1}. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0836 (13.2)</td><td>291</td><td>■ 0.546 ● N|+0</td><td><code>\left|\boldsymbol{x}^{T} \boldsymbol{A y}\right|^{2} \leq\left(\boldsymbol{x}^{T} \boldsymbol{A x}\right) \left(\boldsymbol{y}^{T} \boldsymbol{A y}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n} .</code></td><td>$\left|\boldsymbol{x}^{T} \boldsymbol{A y}\right|^{2} \leq\left(\boldsymbol{x}^{T} \boldsymbol{A x}\right) \left(\boldsymbol{y}^{T} \boldsymbol{A y}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n} .$</td><td>$|\boldsymbol{x}^{T}\boldsymbol{A}\boldsymbol{y}|^{2} \leq (\boldsymbol{x}^{T}\boldsymbol{A}\boldsymbol{x})(\boldsymbol{y}^{T}\boldsymbol{A}\boldsymbol{y}), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0868</td><td>301</td><td>■ 0.546 ● N|+0</td><td><code>\left\|\boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\right\|_{\boldsymbol{A}}=\left\|\boldsymbol{x}-\boldsymbol{x}_{k}+\boldsymbol{u}\right\|_{\boldsymbol{A}} \geq\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}},</code></td><td>$\left\|\boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\right\|_{\boldsymbol{A}}=\left\|\boldsymbol{x}-\boldsymbol{x}_{k}+\boldsymbol{u}\right\|_{\boldsymbol{A}} \geq\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}},$</td><td>$\|\boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\|_{A}=\|\boldsymbol{x}-\boldsymbol{x}_{k}+\boldsymbol{u}\|_{A} \geq\|\boldsymbol{x}-\boldsymbol{x}_{k}\|_{A},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0918</td><td>312</td><td>■ 0.547 ● W|+0</td><td><code>\frac{\left\|\boldsymbol{y}-\boldsymbol{y}_{k}\right\|_{\boldsymbol{C A C}^{T}}}{\left\|\boldsymbol{y}-\boldsymbol{y}_{0}\right\|_{\boldsymbol{C A C}^{T}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \right)^{k}, \quad \text { for } \quad k \geq 0,</code></td><td>$\frac{\left\|\boldsymbol{y}-\boldsymbol{y}_{k}\right\|_{\boldsymbol{C A C}^{T}}}{\left\|\boldsymbol{y}-\boldsymbol{y}_{0}\right\|_{\boldsymbol{C A C}^{T}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \right)^{k}, \quad \text { for } \quad k \geq 0,$</td><td>$\frac{||\boldsymbol{y}-\boldsymbol{y}_{k}||_{\boldsymbol{CAC}^{T}}}{||\boldsymbol{y}-\boldsymbol{y}_{0}||_{\boldsymbol{CAC}^{T}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}, \quad \text{for} \quad k \geq 0,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0386</td><td>156</td><td>■ 0.548 ● N|+0</td><td><code>\boldsymbol{R}:=\left[\begin{array}{ccc} \boldsymbol{D}_{1} & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \mathbf{0} & \boldsymbol{D}_{2} & \boldsymbol{R}_{2,3} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{D}_{3} \end{array}\right], \boldsymbol{D}_{1}:=\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array}\right], \boldsymbol{D}_{2}:=[1], \boldsymbol{D}_{3}:=\left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array}\right] .</code></td><td>$\boldsymbol{R}:=\left[\begin{array}{ccc} \boldsymbol{D}_{1} & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \mathbf{0} & \boldsymbol{D}_{2} & \boldsymbol{R}_{2,3} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{D}_{3} \end{array}\right], \boldsymbol{D}_{1}:=\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array}\right], \boldsymbol{D}_{2}:=[1], \boldsymbol{D}_{3}:=\left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array}\right] .$</td><td>$\boldsymbol{R} := \begin{bmatrix} \boldsymbol{D}_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \mathbf{0} & \boldsymbol{D}_2 & \boldsymbol{R}_{2,3} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{D}_3 \end{bmatrix}, \ \boldsymbol{D}_1 := \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}, \ \boldsymbol{D}_2 := \begin{bmatrix} 1 \end{bmatrix}, \ \boldsymbol{D}_3 := \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0185</td><td>079</td><td>■ 0.548 ● W|+0</td><td><code>l_{i k}^{(k)} a_{k j}^{(k)}=a_{i j}^{(k)}-a_{i j}^{(k+1)} \text { for } k<\min (i, j) \text {, and } l_{i j}^{(k)} a_{j j}^{(j)}=a_{i j}^{(j)} \text { for } i>j .</code></td><td>$l_{i k}^{(k)} a_{k j}^{(k)}=a_{i j}^{(k)}-a_{i j}^{(k+1)} \text { for } k<\min (i, j) \text {, and } l_{i j}^{(k)} a_{j j}^{(j)}=a_{i j}^{(j)} \text { for } i>j .$</td><td>$l_{ik}^{(k)} a_{kj}^{(k)} = a_{ij}^{(k)} - a_{ij}^{(k+1)}$ for $k < \min(i, j)$, and $l_{ij}^{(k)} a_{jj}^{(j)} = a_{ij}^{(j)}$ for $i > j$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0313</td><td>126</td><td>■ 0.549 ● N|+1</td><td><code>\begin{aligned} \boldsymbol{H} \boldsymbol{x} & = \boldsymbol{x}-\left(\boldsymbol{u}^{*} \boldsymbol{x}\right) \boldsymbol{u}=\rho\|\boldsymbol{x}\|_{2}\left(\boldsymbol{z}-\left(\boldsymbol{u}^{*} \boldsymbol{z}\right) \boldsymbol{u}\right)=\rho\|\boldsymbol{x}\|_{2} \left(z-\frac{\left(z^{*}+\boldsymbol{e}_{1}^{*}\right) \boldsymbol{z}}{1+z_{1}}\left(z+\boldsymbol{e}_{1}\right) \right) \\ & =\rho\|\boldsymbol{x}\|_{2}\left(z-\left(z+ \boldsymbol{e}_{1}\right)\right)=-\rho\|\boldsymbol{x} \|_{2} \boldsymbol{e}_{1}=a \boldsymbol{e}_{1} \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{H} \boldsymbol{x} & = \boldsymbol{x}-\left(\boldsymbol{u}^{*} \boldsymbol{x}\right) \boldsymbol{u}=\rho\|\boldsymbol{x}\|_{2}\left(\boldsymbol{z}-\left(\boldsymbol{u}^{*} \boldsymbol{z}\right) \boldsymbol{u}\right)=\rho\|\boldsymbol{x}\|_{2} \left(z-\frac{\left(z^{*}+\boldsymbol{e}_{1}^{*}\right) \boldsymbol{z}}{1+z_{1}}\left(z+\boldsymbol{e}_{1}\right) \right) \\ & =\rho\|\boldsymbol{x}\|_{2}\left(z-\left(z+ \boldsymbol{e}_{1}\right)\right)=-\rho\|\boldsymbol{x} \|_{2} \boldsymbol{e}_{1}=a \boldsymbol{e}_{1} \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{H}\boldsymbol{x} &= \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u} = \rho\|\boldsymbol{x}\|_2(\boldsymbol{z} - (\boldsymbol{u}^*\boldsymbol{z})\boldsymbol{u}) = \rho\|\boldsymbol{x}\|_2\Big(\boldsymbol{z} - \frac{(\boldsymbol{z}^* + \boldsymbol{e}_1^*)\boldsymbol{z}}{1+z_1}(\boldsymbol{z}+\boldsymbol{e}_1)\Big) \\ &= \rho\|\boldsymbol{x}\|_2(\boldsymbol{z} - (\boldsymbol{z}+\boldsymbol{e}_1)) = -\rho\|\boldsymbol{x}\|_2\boldsymbol{e}_1 = a\boldsymbol{e}_1. \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0129 (89)</td><td>063</td><td>■ 0.550 ● N|+0</td><td><code>\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3], \quad \boldsymbol{A}_{21}=\left[\begin{array}{l} 4 \\ 7 \end{array}\right], \quad \text { and } \boldsymbol{A}_{22}=\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right],</code></td><td>$\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3], \quad \boldsymbol{A}_{21}=\left[\begin{array}{l} 4 \\ 7 \end{array}\right], \quad \text { and } \boldsymbol{A}_{22}=\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right],$</td><td>$\boldsymbol{A}_{11} = \begin{bmatrix} 1 \end{bmatrix}, \ \boldsymbol{A}_{12} = \begin{bmatrix} 2, 3 \end{bmatrix}, \ \boldsymbol{A}_{21} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}, \ \text{and } \boldsymbol{A}_{22} = \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix},$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0117 (2.27)	060	0.551 • C|-5	`\boldsymbol{T}=\boldsymbol{T}_{m}:= \operatorname{tridiag}_{m}(-1,2,-1)-=\left[ \begin{array}{rrrrr} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \ \ 0 & \ddots & \ddots & \ddots & \\ & & & 0 \\ & & & -1 & 2-1 \\ & & & 0-1 & 2 \end{array}\right] \in \mathbb{R}^{m \times m} .`	$\boldsymbol{T}=\boldsymbol{T}_{m}:= \operatorname{tridiag}_{m}(-1,2,-1)-=\left[\begin{array}{rrrrr} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & & 0 \\ & & & -1 & 2-1 \\ & & & 0-1 & 2 \end{array}\right] \in \mathbb{R}^{m \times m} .$	$\boldsymbol{T}=\boldsymbol{T}_{m}:= \operatorname{tridiag}_{m}(-1,2,-1)-=\left[\begin{array}{rrrrr} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots & \\ & & & & 0 \\ & & & -1 & 2\ -1 \\ & & & 0\ -1 & 2 \end{array}\right] \in \mathbb{R}^{m \times m} .$
lyche-numerical-linear-algebra_ EQ0577	214	0.552 • N|+0	`\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p \left(t_{1}\right) \\ \vdots \\ p\left(t_{m}\right) \end{array}\right]=\left[\begin{array}{ll} 1 & t_{1} \\ \vdots \\ 1 & t_{m} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array} \right]=\boldsymbol{b} .`	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p \left(t_{1}\right) \\ \vdots \\ p\left(t_{m}\right) \end{array}\right]=\left[\begin{array}{ll} 1 & t_{1} \\ \vdots \\ 1 & t_{m} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array} \right]=\boldsymbol{b} .$	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p(t_{1}) \\ \vdots \\ p(t_{m}) \end{array}\right]=\left[\begin{array}{ll} 1 & t_{1} \\ \vdots \\ 1 & t_{m} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array} \right]=\boldsymbol{b}.$
lyche-numerical-linear-algebra_ EQ0569	211	0.554 • K|+0	`\boldsymbol{A} \hat{\boldsymbol{s}}=\boldsymbol{b}+ \boldsymbol{e} .`	$\boldsymbol{A} \hat{\boldsymbol{s}}=\boldsymbol{b}+ \boldsymbol{e} .$	$\boldsymbol{A} \hat{\boldsymbol{s}}=\boldsymbol{b}+ \boldsymbol{e}.$
lyche-numerical-linear-algebra_ EQ0336	136	0.557 • C|-8	`\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{12}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{ \boldsymbol{P}_{23}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{34}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array} \right] .`	$\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{12}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{ \boldsymbol{P}_{23}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{34}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array} \right] .$	$\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{12}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{ \boldsymbol{P}_{23}}\left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{34}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array} \right].$
lyche-numerical-linear-algebra_ EQ1015	344	0.561 • N|+0	`\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{z}_{0}:=\left[ \begin{array}{l} 1 \\ 0 \end{array}\right] .`	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{z}_{0}:=\left[\begin{array}{l} 1 \\ 0 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{z}_{0}:=\left[\begin{array}{l} 1 \\ 0 \end{array}\right].$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0919</td><td>312</td><td>■ 0.563
● N|+0</td><td>\begin{aligned} \left\|\boldsymbol{y}-\boldsymbol{y}_{k} \right\|_{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}}^{2} & =\left(\boldsymbol{y}-\boldsymbol{y}_{k}\right)^{T}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right)\left(\boldsymbol{y}-\boldsymbol{y}_{k}\right) \\ & =\left(\boldsymbol{C}^{T}\left(\boldsymbol{y}-\boldsymbol{y}_{k} \right)\right)^{T} \boldsymbol{A}\left(\boldsymbol{C}^{T} \left(\boldsymbol{y}-\boldsymbol{y}_{k}\right)\right)= \left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{ \boldsymbol{A}}^{2}, \end{aligned}</td><td>$\begin{aligned} \left\|\boldsymbol{y}-\boldsymbol{y}_{k}\right\|_{\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T}}^{2} & =\left(\boldsymbol{y}-\boldsymbol{y}_{k}\right)^{T}\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T}\right)\left(\boldsymbol{y}-\boldsymbol{y}_{k}\right) \\ & =\left(\boldsymbol{C}^{T}\left(\boldsymbol{y}-\boldsymbol{y}_{k}\right)\right)^{T}\boldsymbol{A}\left(\boldsymbol{C}^{T}\left(\boldsymbol{y}-\boldsymbol{y}_{k}\right)\right)=\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}^{2}, \end{aligned}$</td><td>$\begin{aligned} \|\boldsymbol{y}-\boldsymbol{y}_{k}\|_{\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T}}^{2} & =(\boldsymbol{y}-\boldsymbol{y}_{k})^{T}(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^{T})(\boldsymbol{y}-\boldsymbol{y}_{k}) \\ & =(\boldsymbol{C}^{T}(\boldsymbol{y}-\boldsymbol{y}_{k}))^{T}\boldsymbol{A}(\boldsymbol{C}^{T}(\boldsymbol{y}-\boldsymbol{y}_{k}))=\|\boldsymbol{x}-\boldsymbol{x}_{k}\|_{\boldsymbol{A}}^{2}, \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0132</td><td>063</td><td>■ 0.564
● N|+0</td><td>\boldsymbol{A} \boldsymbol{B}=\left[\begin{array}{c} a_{1:}^{T} \\ a_{2:}^{T} \\ \vdots \\ a_{m:}^{T} \end{array}\right] \boldsymbol{B}=\left[\begin{array}{c} a_{1:}^{T} \boldsymbol{B} \\ a_{2:}^{T} \boldsymbol{B} \\ \vdots \\ a_{m:}^{T} \boldsymbol{B} \end{array}\right],</td><td>$\boldsymbol{A}\boldsymbol{B}=\left[\begin{array}{c} a_{1:}^{T} \\ a_{2:}^{T} \\ \vdots \\ a_{m:}^{T} \end{array}\right]\boldsymbol{B}=\left[\begin{array}{c} a_{1:}^{T}\boldsymbol{B} \\ a_{2:}^{T}\boldsymbol{B} \\ \vdots \\ a_{m:}^{T}\boldsymbol{B} \end{array}\right],$</td><td>$\boldsymbol{A}\boldsymbol{B}=\begin{bmatrix} a_{1:}^{T} \\ a_{2:}^{T} \\ \vdots \\ a_{m:}^{T} \end{bmatrix}\boldsymbol{B}=\begin{bmatrix} a_{1:}^{T}\boldsymbol{B} \\ a_{2:}^{T}\boldsymbol{B} \\ \vdots \\ a_{m:}^{T}\boldsymbol{B} \end{bmatrix},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0425 (7.4)</td><td>171</td><td>■ 0.564
● W|-1</td><td>\begin{aligned} & \boldsymbol{U}=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_{1}:=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}\right], \quad \boldsymbol{U}_{2}:=\left[\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}\right], \\ & \boldsymbol{V}= \left[\boldsymbol{V}_{1}, \boldsymbol{V}_{2}\right] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_{1}:=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r}\right], \quad \boldsymbol{V}_{2}:=\left[\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right], \\ & \boldsymbol{ \Sigma}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & \mathbf{0}_{r, n-r} \\ \mathbf{0}_{m-r, r} & \mathbf{0}_{m-r, n-r} \end{array}\right] \in \mathbb{R}^{m \times n}, \text { where } \boldsymbol{ \Sigma}_{1}:=\operatorname{diag}\left(\sigma_{1}, \ldots, \sigma_{r}\right) . \end{aligned}</td><td>$\begin{aligned} & \boldsymbol{U}=\left[\boldsymbol{U}_{1},\boldsymbol{U}_{2}\right]\in\mathbb{C}^{m\times m}, \quad \boldsymbol{U}_{1}:=\left[\boldsymbol{u}_{1},\ldots,\boldsymbol{u}_{r}\right], \quad \boldsymbol{U}_{2}:=\left[\boldsymbol{u}_{r+1},\ldots,\boldsymbol{u}_{m}\right], \\ & \boldsymbol{V}=\left[\boldsymbol{V}_{1},\boldsymbol{V}_{2}\right]\in\mathbb{C}^{n\times n}, \quad \boldsymbol{V}_{1}:=\left[\boldsymbol{v}_{1},\ldots,\boldsymbol{v}_{r}\right], \quad \boldsymbol{V}_{2}:=\left[\boldsymbol{v}_{r+1},\ldots,\boldsymbol{v}_{n}\right], \\ & \boldsymbol{\Sigma}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & \mathbf{0}_{r,n-r} \\ \mathbf{0}_{m-r,r} & \mathbf{0}_{m-r,n-r} \end{array}\right]\in\mathbb{R}^{m\times n}, \text{ where } \boldsymbol{\Sigma}_{1}:=\operatorname{diag}\left(\sigma_{1},\ldots,\sigma_{r}\right). \end{aligned}$</td><td>$\begin{aligned} & \boldsymbol{U}=[\boldsymbol{U}_{1},\boldsymbol{U}_{2}]\in\mathbb{C}^{m\times m}, \quad \boldsymbol{U}_{1}:=[\boldsymbol{u}_{1},\ldots,\boldsymbol{u}_{r}], \quad \boldsymbol{U}_{2}:=[\boldsymbol{u}_{r+1},\ldots,\boldsymbol{u}_{m}], \\ & \boldsymbol{V}=[\boldsymbol{V}_{1},\boldsymbol{V}_{2}]\in\mathbb{C}^{n\times n}, \quad \boldsymbol{V}_{1}:=[\boldsymbol{v}_{1},\ldots,\boldsymbol{v}_{r}], \quad \boldsymbol{V}_{2}:=[\boldsymbol{v}_{r+1},\ldots,\boldsymbol{v}_{n}], \\ & \boldsymbol{\Sigma}=\begin{bmatrix} \boldsymbol{\Sigma}_{1} & \mathbf{0}_{r,n-r} \\ \mathbf{0}_{m-r,r} & \mathbf{0}_{m-r,n-r} \end{bmatrix}\in\mathbb{R}^{m\times n}, \text{ where } \boldsymbol{\Sigma}_{1}:=\operatorname{diag}(\sigma_{1},\ldots,\sigma_{r}). \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ1016</td><td>345</td><td>■ 0.571
● N|+0</td><td>z_{1}=\boldsymbol{A} z_{0}=\left[\begin{array}{c} 2 \\ -1 \end{array}\right], \quad z_{2}=\boldsymbol{A} z_{1}=\left[\begin{array}{c} 5 \\ -4 \end{array}\right], \cdots, z_{k}=\frac{1}{2}\left[\begin{array}{l} 1+3^{k} \\ 1-3^{k} \end{array}\right], \cdots .</td><td>$z_{1}=\boldsymbol{A}z_{0}=\left[\begin{array}{c} 2 \\ -1 \end{array}\right], \quad z_{2}=\boldsymbol{A}z_{1}=\left[\begin{array}{c} 5 \\ -4 \end{array}\right], \cdots, z_{k}=\frac{1}{2}\left[\begin{array}{l} 1+3^{k} \\ 1-3^{k} \end{array}\right], \cdots.$</td><td>$z_{1}=\boldsymbol{A}z_{0}=\begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad z_{2}=\boldsymbol{A}z_{1}=\begin{bmatrix} 5 \\ -4 \end{bmatrix}, \cdots, z_{k}=\frac{1}{2}\begin{bmatrix} 1+3^{k} \\ 1-3^{k} \end{bmatrix}, \cdots.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0834</td><td>289</td><td>■ 0.578 ● W|+1</td><td>\boldsymbol{A}:=\left[\begin{array}{cccccc} \lambda & 0 & \cdots & 0 & 0 \\ 0 & \lambda & a & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & & \vdots \ \ 0 & 0 & 0 & \cdots & \lambda & a \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array}\right] \in \mathbb{R}^{n \times n} .</td><td>$\boldsymbol{A}:=\left[\begin{array}{cccccc} \lambda & 0 & \cdots & 0 & 0 \\ 0 & \lambda & a & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & a \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array}\right] \in \mathbb{R}^{n \times n} .$</td><td>$\boldsymbol{A}:=\left[\begin{array}{cccccc} \lambda & a & 0 & \cdots & 0 & 0 \\ 0 & \lambda & a & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & a \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{array}\right] \in \mathbb{R}^{n \times n}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0831 (12.47)</td><td>287</td><td>■ 0.581 ● N|+0</td><td>\lambda=\frac{\boldsymbol{x}^{T} \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^{T} \boldsymbol{A x}+ \boldsymbol{x}^{T} \boldsymbol{N} \boldsymbol{x}} .</td><td>$\lambda=\frac{\boldsymbol{x}^{T} \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^{T} \boldsymbol{A x}+ \boldsymbol{x}^{T} \boldsymbol{N} \boldsymbol{x}} .$</td><td>$\lambda=\frac{\boldsymbol{x}^{T} \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^{T} \boldsymbol{A x}+ \boldsymbol{x}^{T} \boldsymbol{N} \boldsymbol{x}}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0420</td><td>169</td><td>■ 0.585 ● K|+0</td><td>\lambda^{m} \pi_{\boldsymbol{A}^{*} \boldsymbol{A}}(\lambda)=\lambda^{n} \pi_{\boldsymbol{A} \boldsymbol{A}^{*}}(\lambda), \quad \lambda \in \mathbb{C},</td><td>$\lambda^{m} \pi_{\boldsymbol{A}^{*} \boldsymbol{A}}(\lambda)=\lambda^{n} \pi_{\boldsymbol{A} \boldsymbol{A}^{*}}(\lambda), \quad \lambda \in \mathbb{C},$</td><td>$\lambda^{m} \pi_{\boldsymbol{A}^{*} \boldsymbol{A}}(\lambda)=\lambda^{n} \pi_{\boldsymbol{A} \boldsymbol{A}^{*}}(\lambda), \quad \lambda \in \mathbb{C},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0709</td><td>254</td><td>■ 0.587 ● N|+0</td><td>\begin{aligned} & \boldsymbol{X}=h^{4} \boldsymbol{G} / \boldsymbol{M}, \text { where } \boldsymbol{M}:=\left[\begin{array}{c} \sigma_{1} \\ \vdots \\ \sigma_{m} \end{array}\right][1, \ldots, 1]+\left[\begin{array}{c} 1 \\ \vdots \\ i \end{array}\right]\left[\sigma_{1} \ldots . \sigma_{m}\right], \\ & \boldsymbol{S}=\sin \left(\pi h \left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} \right][12 \ldots m]\right), \quad \boldsymbol{\sigma}= \sin \left(\frac{\pi h}{2}\left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array}\right]\right) \wedge 2 . \end{aligned}</td><td>$\begin{aligned} & \boldsymbol{X}=h^{4} \boldsymbol{G} / \boldsymbol{M}, \text { where } \boldsymbol{M}:=\left[\begin{array}{c} \sigma_{1} \\ \vdots \\ \sigma_{m} \end{array}\right][1, \ldots, 1]+\left[\begin{array}{c} 1 \\ \vdots \\ i \end{array}\right]\left[\sigma_{1} \ldots . \sigma_{m}\right], \\ & \boldsymbol{S}=\sin \left(\pi h \left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} \right][12 \ldots m]\right), \quad \boldsymbol{\sigma}= \sin \left(\frac{\pi h}{2}\left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array}\right]\right) \wedge 2 . \end{aligned}$</td><td>$\boldsymbol{X}=h^{4} \boldsymbol{G} / \boldsymbol{M}, \text{ where } \boldsymbol{M}:=\left[\begin{array}{c} \sigma_{1} \\ \vdots \\ \sigma_{m} \end{array}\right][1, \ldots, 1]+\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array}\right][\sigma_{1} . \ldots . \sigma_{m}],$ $\boldsymbol{S}=\sin \left(\pi h\left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array}\right][1\ 2 \ldots m]\right), \quad \boldsymbol{\sigma}=\sin \left(\frac{\pi h}{2}\left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array}\right]\right) \wedge 2.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0644</td><td>232</td><td>■ 0.587 ● W|+0</td><td>\begin{array}{cl} \boldsymbol{A B A}=\boldsymbol{A} & (1) \quad \boldsymbol{A C A}=\boldsymbol{A}, \\ \boldsymbol{B A B}=\boldsymbol{B} & (2) \quad \boldsymbol{C A C}= \boldsymbol{C}, \\ (\boldsymbol{A B})^{*}=\boldsymbol{A B} & (3) \quad(\boldsymbol{A C})^{*}=\boldsymbol{A C}, \\ (\boldsymbol{B A})^{*}=\boldsymbol{B A} & (4) \quad(\boldsymbol{C A})^{*}=\boldsymbol{C A} . \end{array}</td><td>$\begin{array}{cl} \boldsymbol{A B A}=\boldsymbol{A} & (1) \quad \boldsymbol{A C A}=\boldsymbol{A}, \\ \boldsymbol{B A B}=\boldsymbol{B} & (2) \quad \boldsymbol{C A C}= \boldsymbol{C}, \\ (\boldsymbol{A B})^{*}=\boldsymbol{A B} & (3) \quad(\boldsymbol{A C})^{*}=\boldsymbol{A C}, \\ (\boldsymbol{B A})^{*}=\boldsymbol{B A} & (4) \quad(\boldsymbol{C A})^{*}=\boldsymbol{C A} . \end{array}$</td><td>$\begin{array}{cccc} \boldsymbol{A B A}=\boldsymbol{A} & (1) & \boldsymbol{A C A}=\boldsymbol{A}, \\ \boldsymbol{B A B}=\boldsymbol{B} & (2) & \boldsymbol{C A C}=\boldsymbol{C}, \\ (\boldsymbol{A B})^{*}=\boldsymbol{A B} & (3) & (\boldsymbol{A C})^{*}=\boldsymbol{A C}, \\ (\boldsymbol{B A})^{*}=\boldsymbol{B A} & (4) & (\boldsymbol{C A})^{*}=\boldsymbol{C A}. \end{array}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0171</td><td>076</td><td>■ 0.588 ● W|+0</td><td>\begin{aligned} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{22}^{(2)} x_{2}+a_{23}^{(2)} x_{3}=b_{2}^{(2)}, & \mathrm{II}^{\prime} \\ a_{32}^{(2)} x_{2}+a_{33}^{(2)} x_{3}=b_{3}^{(2)}, & \mathrm{III}^{\prime}, \end{aligned}</td><td>$\begin{aligned} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{22}^{(2)} x_{2}+a_{23}^{(2)} x_{3}=b_{2}^{(2)}, & \mathrm{II}^{\prime} \\ a_{32}^{(2)} x_{2}+a_{33}^{(2)} x_{3}=b_{3}^{(2)}, & \mathrm{III}^{\prime}, \end{aligned}$</td><td>$\begin{aligned} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, &\quad \mathrm{I} \\ a_{22}^{(2)} x_{2}+a_{23}^{(2)} x_{3}=b_{2}^{(2)}, &\quad \mathrm{II}^{\prime} \\ a_{32}^{(2)} x_{2}+a_{33}^{(2)} x_{3}=b_{3}^{(2)}, &\quad \mathrm{III}^{\prime}, \end{aligned}$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0467 (8.4)</td><td>186</td><td>■ 0.589 ● N|-1</td><td>\lim _{p \rightarrow \infty}\|\boldsymbol{x}\|_{p}= \|\boldsymbol{x}\|_{\infty} \text { for all } \boldsymbol{x} \in \mathbb{C}^{n} .</td><td>$\lim_{p \rightarrow \infty}\|\boldsymbol{x}\|_{p}=\|\boldsymbol{x}\|_{\infty}$ for all $\boldsymbol{x} \in \mathbb{C}^{n}$.</td><td>$\lim_{p \rightarrow \infty}\|\boldsymbol{x}\|_{p}=\|\boldsymbol{x}\|_{\infty}$ for all $\boldsymbol{x} \in \mathbb{C}^{n}$.</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0793</td><td>279</td><td>■ 0.592 ● N|+0</td><td>k=-\frac{s \log (10)}{\log (1-\eta)} \approx \frac{s \log (10)}{\eta}=\tilde{k} .</td><td>$k=-\frac{s \log (10)}{\log (1-\eta)} \approx \frac{s \log (10)}{\eta}=\tilde{k}.$</td><td>$k=-\frac{s \log (10)}{\log (1-\eta)} \approx \frac{s \log (10)}{\eta}=\tilde{k}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1023 (15.6)</td><td>346</td><td>■ 0.600 ● W|+0</td><td>\begin{array}{ll} \text { (i) } & \boldsymbol{y}_{k}= \boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text { (ii) } & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} /\left\| \boldsymbol{y}_{k}\right\| . \end{array}</td><td>$\begin{array}{ll} \text{(i)} & \boldsymbol{y}_{k}=\boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text{(ii)} & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} /\left\|\boldsymbol{y}_{k}\right\| . \end{array}$</td><td>$\begin{array}{ll} (i) & \boldsymbol{y}_{k}=\boldsymbol{A} \boldsymbol{x}_{k-1} \\ (ii) & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} /\|\boldsymbol{y}_{k}\|. \end{array}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0935</td><td>317</td><td>■ 0.600 ● K|+0</td><td>\lambda=\frac{x^{T} \boldsymbol{A x}}{x^{T} \boldsymbol{A}_{p} x} .</td><td>$\lambda=\frac{x^{T} \boldsymbol{A x}}{x^{T} \boldsymbol{A}_{p} x}.$</td><td>$\lambda=\frac{x^{T} \boldsymbol{A x}}{x^{T} \boldsymbol{A}_{p} x}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0901</td><td>308</td><td>■ 0.602 ● W|+2</td><td>\begin{aligned} \left\|\boldsymbol{\epsilon}_{k} \right\|_{\boldsymbol{A}}^{2} & =\boldsymbol{ \epsilon}_{k}^{T} \boldsymbol{A} \boldsymbol{ \epsilon}_{k}=\boldsymbol{r}_{k}^{T} \boldsymbol{A}^{-1} \boldsymbol{r}_{k}, \\ \left\|\boldsymbol{\epsilon}_{k+1} \right\|_{\boldsymbol{A}}^{2} & =\left(\boldsymbol{ \epsilon}_{k}-\alpha_{k} \boldsymbol{r}_{k}\right)^{T} \boldsymbol{A}\left(\boldsymbol{\epsilon}_{k}- \alpha_{k} \boldsymbol{r}_{k}\right) \\ & =\boldsymbol{ \epsilon}_{k}^{T} \boldsymbol{A} \boldsymbol{ \epsilon}_{k}-2 \alpha_{k} \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}+\alpha_{k}^{2} \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}= \left\|\boldsymbol{\epsilon}_{k}\right\|_{ \boldsymbol{A}}^{2}-\frac{\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right)^{2}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}} \end{aligned}</td><td>$\begin{aligned} \left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}}^{2} & =\boldsymbol{\epsilon}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}=\boldsymbol{r}_{k}^{T} \boldsymbol{A}^{-1} \boldsymbol{r}_{k}, \\ \left\|\boldsymbol{\epsilon}_{k+1}\right\|_{\boldsymbol{A}}^{2} & =\left(\boldsymbol{\epsilon}_{k}-\alpha_{k} \boldsymbol{r}_{k}\right)^{T} \boldsymbol{A}\left(\boldsymbol{\epsilon}_{k}-\alpha_{k} \boldsymbol{r}_{k}\right) \\ & =\boldsymbol{\epsilon}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}-2 \alpha_{k} \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}+\alpha_{k}^{2} \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}=\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}}^{2}-\frac{\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right)^{2}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}} \end{aligned}$</td><td>$\begin{aligned} \|\boldsymbol{\epsilon}_{k}\|_{\boldsymbol{A}}^{2} & =\boldsymbol{\epsilon}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}=\boldsymbol{r}_{k}^{T} \boldsymbol{A}^{-1} \boldsymbol{r}_{k}, \\ \|\boldsymbol{\epsilon}_{k+1}\|_{\boldsymbol{A}}^{2} & =(\boldsymbol{\epsilon}_{k}-\alpha_{k} \boldsymbol{r}_{k})^{T} \boldsymbol{A}(\boldsymbol{\epsilon}_{k}-\alpha_{k} \boldsymbol{r}_{k}) \\ & =\boldsymbol{\epsilon}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}-2 \alpha_{k} \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{\epsilon}_{k}+\alpha_{k}^{2} \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}=\|\boldsymbol{\epsilon}_{k}\|_{\boldsymbol{A}}^{2}-\frac{(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k})^{2}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}}. \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0892 (13.37)</td><td>307</td><td>■ 0.606 ● W|+0</td><td>a:=\frac{\boldsymbol{y}^{T} \boldsymbol{A y}}{ \boldsymbol{y}^{T} \boldsymbol{y}}=\sum_{i=1}^{n} t_{i} \lambda_{i}, \quad b:=\frac{\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^{T} \boldsymbol{y}}=\sum_{i=1}^{n} \frac{t_{i}}{\lambda_{i}},</td><td>$a:=\frac{\boldsymbol{y}^{T} \boldsymbol{A y}}{\boldsymbol{y}^{T} \boldsymbol{y}}=\sum_{i=1}^{n} t_{i} \lambda_{i}, \quad b:=\frac{\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^{T} \boldsymbol{y}}=\sum_{i=1}^{n} \frac{t_{i}}{\lambda_{i}},$</td><td>$a:=\frac{\boldsymbol{y}^{T} \boldsymbol{A y}}{\boldsymbol{y}^{T} \boldsymbol{y}}=\sum_{i=1}^{n} t_{i} \lambda_{i}, \quad b:=\frac{\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^{T} \boldsymbol{y}}=\sum_{i=1}^{n} \frac{t_{i}}{\lambda_{i}},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1029</td><td>348</td><td>■ 0.608 ● K|+0</td><td>\boldsymbol{A}_{1}:=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right], \quad \boldsymbol{A}_{2}:=\left[\begin{array}{cc} 1.7 & -0.4 \\ 0.15 & 2.2 \end{array} \right], \quad \text { and } \boldsymbol{A}_{3}=\left[\begin{array}{cc} 1 & 2 \\ -3 & 4 \end{array}\right] .</td><td>$\boldsymbol{A}_{1}:=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right], \quad \boldsymbol{A}_{2}:=\left[\begin{array}{cc} 1.7 & -0.4 \\ 0.15 & 2.2 \end{array}\right], \quad \text{and } \boldsymbol{A}_{3}=\left[\begin{array}{cc} 1 & 2 \\ -3 & 4 \end{array}\right].$</td><td>$\boldsymbol{A}_{1}:=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right], \quad \boldsymbol{A}_{2}:=\left[\begin{array}{cc} 1.7 & -0.4 \\ 0.15 & 2.2 \end{array}\right], \text{ and } \boldsymbol{A}_{3}=\left[\begin{array}{cc} 1 & 2 \\ -3 & 4 \end{array}\right].$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0379	154	■ 0.611 ● W|+0	`\begin{aligned} \boldsymbol{A} \boldsymbol{A}^{*} & = \left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{*} \right)\left(\boldsymbol{U} \boldsymbol{B}^{*} \boldsymbol{U}^{*}\right)=\boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^{*} \boldsymbol{U}^{*} \text { and } \\ \boldsymbol{A}^{*} \boldsymbol{A} & =\left(\boldsymbol{U} \boldsymbol{B}^{*} \boldsymbol{U}^{*}\right)\left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{*}\right)= \boldsymbol{U} \boldsymbol{B}^{*} \boldsymbol{B} \boldsymbol{U}^{*} . \end{aligned}`	$\boldsymbol{A} \boldsymbol{A}^{*} = \left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{*} \right)\left(\boldsymbol{U} \boldsymbol{B}^{*} \boldsymbol{U}^{*}\right)=\boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^{*} \boldsymbol{U}^{*} \text { and }$ $\boldsymbol{A}^{*} \boldsymbol{A} =\left(\boldsymbol{U} \boldsymbol{B}^{*} \boldsymbol{U}^{*}\right)\left(\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{*}\right)= \boldsymbol{U} \boldsymbol{B}^{*} \boldsymbol{B} \boldsymbol{U}^{*} .$	$\boldsymbol{A}\boldsymbol{A}^* = (\boldsymbol{U}\boldsymbol{B}\boldsymbol{U}^*)(\boldsymbol{U}\boldsymbol{B}^*\boldsymbol{U}^*) = \boldsymbol{U}\boldsymbol{B}\boldsymbol{B}^*\boldsymbol{U}^*$ and $\boldsymbol{A}^*\boldsymbol{A} = (\boldsymbol{U}\boldsymbol{B}^*\boldsymbol{U}^*)(\boldsymbol{U}\boldsymbol{B}\boldsymbol{U}^*) = \boldsymbol{U}\boldsymbol{B}^*\boldsymbol{B}\boldsymbol{U}^*.$
lyche-numerical-linear-algebra_ EQ1031 (15.7)	349	■ 0.611 ● W|+0	`\begin{array}{ll} \text { (i) } & (\boldsymbol{A}-s \boldsymbol{I}) \boldsymbol{y}_{k}=\boldsymbol{x}_{k-1} \\ \text { (ii) } & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} /\left|\boldsymbol{y}_{k}\right| . \end{array}`	$\begin{array}{ll} \text { (i) } & (\boldsymbol{A}-s \boldsymbol{I}) \boldsymbol{y}_{k}=\boldsymbol{x}_{k-1} \\ \text { (ii) } & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} /\left\|\boldsymbol{y}_{k}\right\| . \end{array}$	$(i) \quad (\boldsymbol{A} - s\boldsymbol{I})\boldsymbol{y}_k = \boldsymbol{x}_{k-1}$ $(ii) \quad \boldsymbol{x}_k = \boldsymbol{y}_k/\|\boldsymbol{y}_k\|.$
lyche-numerical-linear-algebra_ EQ1011	341	■ 0.614 ● K|+0	`\boldsymbol{A}:=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array}\right], \quad \boldsymbol{B}:=\left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right] .`	$\boldsymbol{A}:=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array}\right], \quad \boldsymbol{B}:=\left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right] .$	$\boldsymbol{A} := \begin{bmatrix} 0\,1\,0\,0 \\ 0\,0\,1\,0 \\ 0\,0\,0\,1 \\ 0\,0\,0\,0 \end{bmatrix}, \quad \boldsymbol{E} := \begin{bmatrix} 0\,0\,0\,0 \\ 0\,0\,0\,0 \\ 0\,0\,0\,0 \\ \epsilon\;0\,0\,0 \end{bmatrix}, \quad \boldsymbol{B} := \begin{bmatrix} 0\,1\,0\,0 \\ 0\,0\,1\,0 \\ 0\,0\,0\,1 \\ \epsilon\;0\,0\,0 \end{bmatrix}.$
lyche-numerical-linear-algebra_ EQ0011	024	■ 0.618 ● N|+0	`\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}):= \boldsymbol{A}\left(\begin{array}{cccc} i_{1} & i_{2} & \cdots & i_{r} \\ j_{1} & j_{2} & \cdots & j_{c} \end{array}\right)=\left[\begin{array}{cccc} a_{i_{1}, j_{1}} & a_{i_{1}, j_{2}} & \cdots & a_{i_{1}, j_{c}} \\ a_{i_{2}, j_{1}} & a_{i_{2}, j_{2}} & \cdots & a_{i_{2}, j_{c}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_{r}, j_{1}} & a_{i_{r}, j_{2}} & \cdots & a_{i_{r}, j_{c}} \end{array}\right] .`	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}):= \boldsymbol{A}\left(\begin{array}{cccc} i_{1} & i_{2} & \cdots & i_{r} \\ j_{1} & j_{2} & \cdots & j_{c} \end{array}\right)=\left[\begin{array}{cccc} a_{i_{1}, j_{1}} & a_{i_{1}, j_{2}} & \cdots & a_{i_{1}, j_{c}} \\ a_{i_{2}, j_{1}} & a_{i_{2}, j_{2}} & \cdots & a_{i_{2}, j_{c}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_{r}, j_{1}} & a_{i_{r}, j_{2}} & \cdots & a_{i_{r}, j_{c}} \end{array}\right] .$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A}\begin{pmatrix} i_1\; i_2 \cdots i_r \\ j_1\; j_2 \cdots j_c \end{pmatrix} = \begin{bmatrix} a_{i_1,j_1} & a_{i_1,j_2} & \cdots & a_{i_1,j_c} \\ a_{i_2,j_1} & a_{i_2,j_2} & \cdots & a_{i_2,j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r,j_1} & a_{i_r,j_2} & \cdots & a_{i_r,j_c} \end{bmatrix}.$
lyche-numerical-linear-algebra_ EQ0249 (4.4)	102	■ 0.629 ● S|+0	`\boldsymbol{a}_{n}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}, \quad a_{n n}=\boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}+d_{n n} .`	$\boldsymbol{a}_{n}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}, \quad a_{n n}=\boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}+d_{n n} .$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1}\boldsymbol{D}_{n-1}\boldsymbol{l}_n, \quad a_{nn} = \boldsymbol{l}_n^*\boldsymbol{D}_{n-1}\boldsymbol{l}_n + d_{nn}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0849</td><td>295</td><td>0.636 W|+0</td><td>\begin{aligned} \boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{j} & \stackrel{(13.13)}{=}\left(\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}\right)^{T} \boldsymbol{r}_{j} \ \ & \stackrel{(13.11)}{=} \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{j}-\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A}\left(\boldsymbol{p}_{j}+\sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}\right) \boldsymbol{p}_{i}\right) \\ & \stackrel{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=0}{=} \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{j}-\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{j}=0, \quad j=0,1, \ldots, k . \end{aligned}</td><td>$$\begin{aligned} \boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{j} & \stackrel{(13.13)}{=}\left(\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}\right)^{T} \boldsymbol{r}_{j} \\ & \stackrel{(13.11)}{=} \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{j}-\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A}\left(\boldsymbol{p}_{j}+\sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}\right) \boldsymbol{p}_{i}\right) \\ & \stackrel{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=0}{=} \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{j}-\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{j}=0, \quad j=0,1, \ldots, k . \end{aligned}$$</td><td>$$\begin{aligned} \boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{j} & \stackrel{(13.13)}{=} (\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k})^{T} \boldsymbol{r}_{j} \\ & \stackrel{(13.11)}{=} \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{j}-\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A}\big(\boldsymbol{p}_{j}+\sum_{i=0}^{j-1} (\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}) \boldsymbol{p}_{i}\big) \\ & \stackrel{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=0}{=} \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{j}-\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{j}=0, \quad j=0,1, \ldots, k . \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0592 (9.4)</td><td>218</td><td>0.637 K|+0</td><td>\mathbb{C}^{m}=\mathcal{R}(\boldsymbol{A}) \stackrel{ \perp}{\oplus} \mathcal{N}\left(\boldsymbol{A}^{*}\right) .</td><td>$$\mathbb{C}^{m}=\mathcal{R}(\boldsymbol{A}) \stackrel{ \perp}{\oplus} \mathcal{N}\left(\boldsymbol{A}^{*}\right) .$$</td><td>$$\mathbb{C}^{m}=\mathcal{R}(\boldsymbol{A}) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A}^{*}).$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0553 (8.42)</td><td>208</td><td>0.638 N|+1</td><td>\operatorname{cond}_{1}(\boldsymbol{T})= \operatorname{cond}_{\infty}(\boldsymbol{T})=\frac{1}{2} \begin{cases}h^{-2}, & m \text { odd } \\ h^{-2}-1, & m \text { even }\end{cases}</td><td>$$\operatorname{cond}_{1}(\boldsymbol{T})= \operatorname{cond}_{\infty}(\boldsymbol{T})=\frac{1}{2} \begin{cases}h^{-2}, & m \text { odd } \\ h^{-2}-1, & m \text { even }\end{cases}$$</td><td>$$\operatorname{cond}_{1}(\boldsymbol{T})= \operatorname{cond}_{\infty}(\boldsymbol{T})=\frac{1}{2} \begin{cases}h^{-2}, & m \text{ odd}, \\ h^{-2}-1, & m \text{ even}.\end{cases}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1032</td><td>349</td><td>0.642 W|+0</td><td>\begin{aligned} \text { (i) } & \left(\boldsymbol{A}-s_{k-1} \boldsymbol{I}\right) \boldsymbol{y}_{k}=\boldsymbol{x}_{k-1}, \\ \text { (ii) } & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} / \left\|\boldsymbol{y}_{k}\right\|, \\ \text { (iii) } & \boldsymbol{s}_{k}=\boldsymbol{x}_{k}^{*} \boldsymbol{A} \boldsymbol{x}_{k}, \\ \text { (iv) } & \boldsymbol{r}_{k}=\boldsymbol{A} \boldsymbol{x}_{k}-s_{k} \boldsymbol{x}_{k} . \end{aligned}</td><td>$$\begin{aligned} \text { (i) } & \left(\boldsymbol{A}-s_{k-1} \boldsymbol{I}\right) \boldsymbol{y}_{k}=\boldsymbol{x}_{k-1}, \\ \text { (ii) } & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} / \left\|\boldsymbol{y}_{k}\right\|, \\ \text { (iii) } & \boldsymbol{s}_{k}=\boldsymbol{x}_{k}^{*} \boldsymbol{A} \boldsymbol{x}_{k}, \\ \text { (iv) } & \boldsymbol{r}_{k}=\boldsymbol{A} \boldsymbol{x}_{k}-s_{k} \boldsymbol{x}_{k} . \end{aligned}$$</td><td>$$\begin{aligned} (i) \quad & (\boldsymbol{A}-s_{k-1} \boldsymbol{I}) \boldsymbol{y}_{k}=\boldsymbol{x}_{k-1}, \\ (ii) \quad & \boldsymbol{x}_{k}=\boldsymbol{y}_{k} / \|\boldsymbol{y}_{k}\|, \\ (iii) \quad & \boldsymbol{s}_{k}=\boldsymbol{x}_{k}^{*} \boldsymbol{A} \boldsymbol{x}_{k}, \\ (iv) \quad & \boldsymbol{r}_{k}=\boldsymbol{A} \boldsymbol{x}_{k}-s_{k} \boldsymbol{x}_{k}. \end{aligned}$$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0698</td><td>250</td><td>■ 0.645
● S|+0</td><td><code>\begin{aligned} \boldsymbol{A} & =\left[\begin{array}{rrr|rrr|rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline-1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0-1 & 4 \end{array}\right] \\ & =\left[\begin{array}{ccc} \boldsymbol{T}+2 \boldsymbol{I} & -\boldsymbol{I} & \mathbf{0} \\ -\boldsymbol{I} & \boldsymbol{T}+2 \boldsymbol{I} & -\boldsymbol{I} \\ \mathbf{0} & -\boldsymbol{I} & \boldsymbol{T}+2 \boldsymbol{I} \end{array}\right], \end{aligned}</code></td><td>$$\begin{aligned} \boldsymbol{A} & =\left[\begin{array}{rrr|rrr|rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline-1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0-1 & 4 \end{array}\right] \\ & =\left[\begin{array}{ccc} \boldsymbol{T}+2 \boldsymbol{I} & -\boldsymbol{I} & \mathbf{0} \\ -\boldsymbol{I} & \boldsymbol{T}+2 \boldsymbol{I} & -\boldsymbol{I} \\ \mathbf{0} & -\boldsymbol{I} & \boldsymbol{T}+2 \boldsymbol{I} \end{array}\right], \end{aligned}$$</td><td>$$\begin{aligned} \boldsymbol{A} & =\left[\begin{array}{rrr|rrr|rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array}\right] \\ & =\left[\begin{array}{ccc} \boldsymbol{T}+2 \boldsymbol{I} & -\boldsymbol{I} & \mathbf{0} \\ -\boldsymbol{I} & \boldsymbol{T}+2 \boldsymbol{I} & -\boldsymbol{I} \\ \mathbf{0} & -\boldsymbol{I} & \boldsymbol{T}+2 \boldsymbol{I} \end{array}\right], \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0641</td><td>232</td><td>■ 0.645
● N|+0</td><td><code>r=r(x)=b-A x .</code></td><td>$r=r(x)=b-A x .$</td><td>$\boldsymbol{r}=\boldsymbol{r}(\boldsymbol{x})=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0601 (9.7)</td><td>222</td><td>■ 0.650
● N|-1</td><td><code>\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right]\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]\left[\begin{array}{l} \boldsymbol{V}_{1}^{*} \\ \boldsymbol{V}_{2}^{*} \end{array}\right]=\boldsymbol{U}_{1} \boldsymbol{ \Sigma}_{1} \boldsymbol{V}_{1}^{*}, \quad \boldsymbol{ \Sigma}_{1}=\operatorname{diag}\left(\sigma_{1}, \ldots, \sigma_{r}\right),</code></td><td>$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\right]\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]\left[\begin{array}{l} \boldsymbol{V}_{1}^{*} \\ \boldsymbol{V}_{2}^{*} \end{array}\right]=\boldsymbol{U}_{1} \boldsymbol{ \Sigma}_{1} \boldsymbol{V}_{1}^{*}, \quad \boldsymbol{ \Sigma}_{1}=\operatorname{diag}\left(\sigma_{1}, \ldots, \sigma_{r}\right),$</td><td>$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{*}=[\boldsymbol{U}_1,\boldsymbol{U}_2]\begin{bmatrix} \Sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}\begin{bmatrix} \boldsymbol{V}_1^{*} \\ \boldsymbol{V}_2^{*} \end{bmatrix}=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^{*}, \quad \Sigma_1=\operatorname{diag}(\sigma_1,\ldots,\sigma_r),$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0201 (3.11)</td><td>085</td><td>■ 0.651
● N|-1</td><td><code>\boldsymbol{p}:=\boldsymbol{p}_{n}, \text { where } \boldsymbol{p}_{1}:=[1,2, \ldots, n]^{T}, \text { and } \boldsymbol{p}_{k+1}:=\boldsymbol{I}_{r_{k}, k} \boldsymbol{p}_{k} \text { for } k=1, \ldots, n-1 .</code></td><td>$\boldsymbol{p}:=\boldsymbol{p}_{n}, \text { where } \boldsymbol{p}_{1}:=[1,2, \ldots, n]^{T}, \text { and } \boldsymbol{p}_{k+1}:=\boldsymbol{I}_{r_{k}, k} \boldsymbol{p}_{k} \text { for } k=1, \ldots, n-1 .$</td><td>$\boldsymbol{p}:=\boldsymbol{p}_{n}, \text { where } \boldsymbol{p}_{1}:=[1,2, \ldots, n]^{T}, \text { and } \boldsymbol{p}_{k+1}:=\boldsymbol{I}_{r_{k}, k} \boldsymbol{p}_{k} \text { for } k=1, \ldots, n-1 .$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0947	320	■ 0.654 ● N|+0	\begin{aligned} & {\left[\boldsymbol{x}_{0}, \boldsymbol{x}_{1}, \boldsymbol{x}_{2}, \boldsymbol{x}_{3}\right]=\left[\begin{array}{rrrr} 0 & 2 & 8 / 3 & 3 \\ 0 & 0 & 4 / 3 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right],} \\ & {\left[\boldsymbol{r}_{0}, \boldsymbol{r}_{1}, \boldsymbol{r}_{2}, \boldsymbol{r}_{3}\right]=\left[\begin{array}{rrrr} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 / 3 & 0 \end{array}\right],} \end{aligned}	$\begin{aligned} & {\left[\boldsymbol{x}_{0}, \boldsymbol{x}_{1}, \boldsymbol{x}_{2}, \boldsymbol{x}_{3}\right]=\left[\begin{array}{rrrr} 0 & 2 & 8 / 3 & 3 \\ 0 & 0 & 4 / 3 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right],} \\ & {\left[\boldsymbol{r}_{0}, \boldsymbol{r}_{1}, \boldsymbol{r}_{2}, \boldsymbol{r}_{3}\right]=\left[\begin{array}{rrrr} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 / 3 & 0 \end{array}\right],} \end{aligned}$	$[\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3] = \begin{bmatrix} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & & 0 & 1 \end{bmatrix},$ $[\boldsymbol{r}_0, \boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3] = \begin{bmatrix} 4 & 0 & & 0 & 0 \\ 0 & 2 & & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{bmatrix},$
lyche-numerical-linear-algebra_ EQ0415	166	■ 0.666 ● N|+0	R(\boldsymbol{x}-t \boldsymbol{y})=R(\boldsymbol{x})-2 t(\boldsymbol{A x}-R(\boldsymbol{x}) \boldsymbol{x})^{T} \boldsymbol{y}+\mathcal{O}\left(t^{2}\right),	$R(\boldsymbol{x}-t \boldsymbol{y})=R(\boldsymbol{x})-2 t(\boldsymbol{A x}-R(\boldsymbol{x}) \boldsymbol{x})^{T} \boldsymbol{y}+\mathcal{O}\left(t^{2}\right),$	$R(\boldsymbol{x} - t\,\boldsymbol{y}) = R(\boldsymbol{x}) - 2t\big(\boldsymbol{A}\boldsymbol{x} - R(\boldsymbol{x})\boldsymbol{x}\big)^T \boldsymbol{y} + \mathcal{O}(t^2),$
lyche-numerical-linear-algebra_ EQ0602	222	■ 0.667 ● C|+16	\begin{array}{ll} \boldsymbol{U}_{1}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}\right], & \boldsymbol{U}_{2}=\left[\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}\right], \\ \boldsymbol{V}_{1}=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r}^{*} \right], & \boldsymbol{V}_{2}=\left[\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right], \end{array} \quad \begin{aligned} & \boldsymbol{V}_{1}^{*} \boldsymbol{V}_{1}=\boldsymbol{I}, \quad \boldsymbol{V}_{2}^{*} \boldsymbol{V}_{2}=\boldsymbol{I}, \end{aligned}	$\begin{array}{ll} \boldsymbol{U}_{1}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{r}\right], & \boldsymbol{U}_{2}=\left[\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_{m}\right], \\ \boldsymbol{V}_{1}=\left[\boldsymbol{v}_{1}, \ldots, \boldsymbol{v}_{r}^{*} \right], & \boldsymbol{V}_{2}=\left[\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_{n}\right], \end{array} \quad \begin{aligned} & \boldsymbol{V}_{1}^{*} \boldsymbol{V}_{1}=\boldsymbol{I}, \quad \boldsymbol{V}_{2}^{*} \boldsymbol{V}_{2}=\boldsymbol{I}, \end{aligned}$	$\boldsymbol{U}_1 = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r],\ \boldsymbol{U}_2 = [\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m], \quad \boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I},\ \boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I},$ $\boldsymbol{V}_1 = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r],\ \boldsymbol{V}_2 = [\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n], \quad \boldsymbol{V}_1^* \boldsymbol{V}_1 = \boldsymbol{I},\ \boldsymbol{V}_2^* \boldsymbol{V}_2 = \boldsymbol{I}$
lyche-numerical-linear-algebra_ EQ0751	267	■ 0.670 ● W|+0	\begin{array}{r} J: \boldsymbol{v}_{k+1}(i, j)=\left(\boldsymbol{v}_{k}(i-1, j)+\boldsymbol{v}_{k}(i, j-1)+\boldsymbol{v}_{k}(i+1, j)+\boldsymbol{v}_{k}(i, j+1)\right. \\ +\boldsymbol{e}(i, j)) / 4 \\ G S: \boldsymbol{v}_{k+1}(i, j)=\left(\boldsymbol{v}_{k+1}(i-1, j)+\boldsymbol{v}_{k+1}(i, j-1)+\boldsymbol{v}_{k}(i+1, j)+\boldsymbol{v}_{k}(i, j+1)\right. \\ +\boldsymbol{e}(i, j)) / 4 \end{array}	$\begin{array}{r} J: \boldsymbol{v}_{k+1}(i, j)=\left(\boldsymbol{v}_{k}(i-1, j)+\boldsymbol{v}_{k}(i, j-1)+\boldsymbol{v}_{k}(i+1, j)+\boldsymbol{v}_{k}(i, j+1)\right. \\ +\boldsymbol{e}(i, j)) / 4 \\ G S: \boldsymbol{v}_{k+1}(i, j)=\left(\boldsymbol{v}_{k+1}(i-1, j)+\boldsymbol{v}_{k+1}(i, j-1)+\boldsymbol{v}_{k}(i+1, j)+\boldsymbol{v}_{k}(i, j+1)\right. \\ +\boldsymbol{e}(i, j)) / 4 \end{array}$	$J: \boldsymbol{v}_{k+1}(i,j) = \big(\boldsymbol{v}_k(i-1,j) + \boldsymbol{v}_k(i,j-1) + \boldsymbol{v}_k(i+1,j) + \boldsymbol{v}_k(i,j+1) + \boldsymbol{e}(i,j)\big)/4$ $GS: \boldsymbol{v}_{k+1}(i,j) = \big(\boldsymbol{v}_{k+1}(i-1,j) + \boldsymbol{v}_{k+1}(i,j-1) + \boldsymbol{v}_k(i+1,j) + \boldsymbol{v}_k(i,j+1) + \boldsymbol{e}(i,j)\big)/4$
lyche-numerical-linear-algebra_ EQ0190 (3.8)	080	■ 0.672 ● N|+0	x_{k}=\left(b_{k}-\sum_{j=l_{k}}^{k-1} a_{k, j} x_{j}\right) / a_{k k}, \quad k=1,2, \ldots, n .	$x_{k}=\left(b_{k}-\sum_{j=l_{k}}^{k-1} a_{k, j} x_{j}\right) / a_{k k}, \quad k=1,2, \ldots, n .$	$x_k = \big(b_k - \sum_{j=l_k}^{k-1} a_{k,j} x_j\big)/a_{kk}, \quad k = 1, 2, \ldots, n.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>lyche-numerical-linear-algebra_ EQ0719</td><td>257</td><td>■ 0.674 • W|+0</td><td><code>\begin{aligned} \left(\boldsymbol{F}_{2 m+2} \boldsymbol{z}\right)_{k+1} & =\sum_{j=0}^{2 m-1} \omega^{j k} z_{j+1}=\sum_{j=1}^{m} x_{j} \omega^{j k}-\sum_{j=1}^{m} x_{j} \omega^{(2 m+2-j) k} \\ & = \sum_{j=1}^{m} x_{j}\left(e^{-\pi i j k /(m+1)}-e^{ \pi i j k /(m+1)}\right) \\ & =-2 i \sum_{j=1}^{m} x_{j} \sin \left(\frac{j k \pi}{m+1}\right)=-2 i \left(\boldsymbol{S}_{m} \boldsymbol{x}\right)_{k} . \end{aligned}</code></td><td>$$\begin{aligned} \left(\boldsymbol{F}_{2 m+2} \boldsymbol{z}\right)_{k+1} & =\sum_{j=0}^{2 m-1} \omega^{j k} z_{j+1}=\sum_{j=1}^{m} x_{j} \omega^{j k}-\sum_{j=1}^{m} x_{j} \omega^{(2 m+2-j) k} \\ & = \sum_{j=1}^{m} x_{j}\left(e^{-\pi i j k /(m+1)}-e^{ \pi i j k /(m+1)}\right) \\ & =-2 i \sum_{j=1}^{m} x_{j} \sin \left(\frac{j k \pi}{m+1}\right)=-2 i \left(\boldsymbol{S}_{m} \boldsymbol{x}\right)_{k} . \end{aligned}$$</td><td>$$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^{m} x_j \omega^{jk} - \sum_{j=1}^{m} x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^{m} x_j \big(e^{-\pi ijk/(m+1)} - e^{\pi ijk/(m+1)}\big) \\ &= -2i \sum_{j=1}^{m} x_j \sin\left(\frac{jk\pi}{m+1}\right) = -2i(\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0599</td><td>221</td><td>■ 0.676 • N|-1</td><td><code>\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text { and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right] .</code></td><td>$$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text { and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right] .$$</td><td>$$\boldsymbol{A} = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{bmatrix} \text{ and } \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0726 (11.7)</td><td>258</td><td>■ 0.678 • W|+0</td><td><code>\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}= \left[\begin{array}{c|c} \boldsymbol{F}_{m} & \boldsymbol{D}_{m} \boldsymbol{F}_{m} \\ \hline \boldsymbol{F}_{m} & -\boldsymbol{D}_{m} \boldsymbol{F}_{m} \end{array}\right],</code></td><td>$$\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}= \left[\begin{array}{c|c} \boldsymbol{F}_{m} & \boldsymbol{D}_{m} \boldsymbol{F}_{m} \\ \hline \boldsymbol{F}_{m} & -\boldsymbol{D}_{m} \boldsymbol{F}_{m} \end{array}\right],$$</td><td>$$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} = \left[\begin{array}{c|c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right],$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0513</td><td>200</td><td>■ 0.679 • N|+1</td><td><code>\begin{aligned} f(x) & =\frac{x_{2}-x}{x_{2}-x_{1}} f\left(x_{1}\right)+\frac{x-x_{1}}{x_{2}-x_{1}} f \left(x_{2}\right)+\left(x-x_{1}\right)\left(x-x_{2} \right) f^{\prime \prime}(c) / 2 \\ & =(1- \lambda) f\left(x_{1}\right)+\lambda f\left(x_{2} \right)+\left(x_{2}-x_{1}\right)^{2} \lambda(\lambda-1) f^{\prime \prime}(c) / 2, \quad \lambda:= \frac{x-x_{1}}{x_{2}-x_{1}} \end{aligned}</code></td><td>$$\begin{aligned} f(x) & =\frac{x_{2}-x}{x_{2}-x_{1}} f\left(x_{1}\right)+\frac{x-x_{1}}{x_{2}-x_{1}} f \left(x_{2}\right)+\left(x-x_{1}\right)\left(x-x_{2} \right) f^{\prime \prime}(c) / 2 \\ & =(1- \lambda) f\left(x_{1}\right)+\lambda f\left(x_{2} \right)+\left(x_{2}-x_{1}\right)^{2} \lambda(\lambda-1) f^{\prime \prime}(c) / 2, \quad \lambda:= \frac{x-x_{1}}{x_{2}-x_{1}} \end{aligned}$$</td><td>$$\begin{aligned} f(x) &= \frac{x_2 - x}{x_2 - x_1} f(x_1) + \frac{x - x_1}{x_2 - x_1} f(x_2) + (x - x_1)(x - x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2 - x_1)^2 \lambda(\lambda - 1) f''(c)/2, \quad \lambda := \frac{x - x_1}{x_2 - x_1}. \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0607</td><td>222</td><td>■ 0.679 • W|+1</td><td><code>\begin{aligned} A A^{\dagger} A & =U_{1} \Sigma_{1} V_{1}^{*} V_{1} V_{1}^{*}=U_{1} \Sigma_{1} V_{1}^{*}=A \\ A^{\dagger} A A^{\dagger} & =V_{1} \Sigma_{1}^{-1} U_{1}^{*} U_{1} U_{1}^{*}=V_{1} \Sigma^{-1} U_{1}^{*}=A^{ \dagger} \end{aligned}</code></td><td>$$\begin{aligned} A A^{\dagger} A & =U_{1} \Sigma_{1} V_{1}^{*} V_{1} V_{1}^{*}=U_{1} \Sigma_{1} V_{1}^{*}=A \\ A^{\dagger} A A^{\dagger} & =V_{1} \Sigma_{1}^{-1} U_{1}^{*} U_{1} U_{1}^{*}=V_{1} \Sigma^{-1} U_{1}^{*}=A^{ \dagger} \end{aligned}$$</td><td>$$\begin{aligned} \boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^{\dagger}\boldsymbol{A}\boldsymbol{A}^{\dagger} &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^{\dagger}. \end{aligned}$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0383</td><td>155</td><td>0.685 K|+0</td><td>R(x)=R_{A}(x):=\frac{x^{*} A x}{x^{*} x}</td><td>$R(x)=R_{A}(x):=\frac{x^{*} A x}{x^{*} x}$</td><td>$R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0578</td><td>214</td><td>0.686 N|+0</td><td>c_{1}\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array} \right]+c_{2}\left[\begin{array}{c} t_{1} \\ \vdots \\ t_{m} \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right] \Longrightarrow\left[\begin{array}{cc} 1 & t_{i} \\ 1 & t_{j} \end{array} \right]\left[\begin{array}{c} c_{1} \\ c_{2} \end{array} \right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right] \Longrightarrow c_{1}=c_{2}=0 .</td><td>$c_{1}\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array} \right]+c_{2}\left[\begin{array}{c} t_{1} \\ \vdots \\ t_{m} \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right] \Longrightarrow\left[\begin{array}{cc} 1 & t_{i} \\ 1 & t_{j} \end{array} \right]\left[\begin{array}{c} c_{1} \\ c_{2} \end{array} \right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right] \Longrightarrow c_{1}=c_{2}=0 .$</td><td>$c_{1}\left[\begin{array}{c} 1 \\ \vdots \\ 1 \end{array} \right]+c_{2}\left[\begin{array}{c} t_{1} \\ \vdots \\ t_{m} \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right] \Longrightarrow\left[\begin{array}{cc} 1 & t_{i} \\ 1 & t_{j} \end{array} \right]\left[\begin{array}{c} c_{1} \\ c_{2} \end{array} \right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right] \Longrightarrow c_{1}=c_{2}=0 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0535</td><td>203</td><td>0.693 N|+0</td><td>a_{n}\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\left \langle a_{n} \boldsymbol{x}, \boldsymbol{y}\right\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\left\|a_{n} \boldsymbol{x}+\boldsymbol{y}\right\|^{2}-\left\|a_{n} \boldsymbol{x}-\boldsymbol{y}\right\|^{2}\right) .</td><td>$a_{n}\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\left \langle a_{n} \boldsymbol{x}, \boldsymbol{y}\right\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\left\|a_{n} \boldsymbol{x}+\boldsymbol{y}\right\|^{2}-\left\|a_{n} \boldsymbol{x}-\boldsymbol{y}\right\|^{2}\right) .$</td><td>$a_{n}\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle a_{n} \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\|a_{n} \boldsymbol{x}+\boldsymbol{y}\|^{2}-\|a_{n} \boldsymbol{x}-\boldsymbol{y}\|^{2}\right).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1026</td><td>347</td><td>0.696 N|+0</td><td>\left[\begin{array}{c} \boldsymbol{u}^{*} \\ \boldsymbol{U}^{*} \end{array}\right](\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}^{*} \boldsymbol{u} \\ \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u}- \lambda \boldsymbol{U}^{*} \boldsymbol{u} \end{array} \right]=\left[\begin{array}{c} \boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda \\ \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u} \end{array}\right] .</td><td>$\left[\begin{array}{c} \boldsymbol{u}^{*} \\ \boldsymbol{U}^{*} \end{array}\right](\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}^{*} \boldsymbol{u} \\ \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u}- \lambda \boldsymbol{U}^{*} \boldsymbol{u} \end{array} \right]=\left[\begin{array}{c} \boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda \\ \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u} \end{array}\right] .$</td><td>$\left[\begin{array}{c} \boldsymbol{u}^{*} \\ \boldsymbol{U}^{*} \end{array}\right](\boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda \boldsymbol{u}^{*} \boldsymbol{u} \\ \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u}- \lambda \boldsymbol{U}^{*} \boldsymbol{u} \end{array} \right]=\left[\begin{array}{c} \boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda \\ \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u} \end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0410 (6.19)</td><td>164</td><td>0.702 W|+0</td><td>\mu_{\boldsymbol{A}}(\lambda):=\prod_{i=1}^{k}\left(\lambda_{i}-\lambda\right)^{m_{i}} \text { where } m_{i}:=\max _{1 \leq j \leq g_{i}} m_{i, j} \text {, }</td><td>$\mu_{\boldsymbol{A}}(\lambda):=\prod_{i=1}^{k}\left(\lambda_{i}-\lambda\right)^{m_{i}} \text { where } m_{i}:=\max _{1 \leq j \leq g_{i}} m_{i, j} \text {, }$</td><td>$\mu_{\boldsymbol{A}}(\lambda):=\prod_{i=1}^{k}(\lambda_{i}-\lambda)^{m_{i}} \text { where } m_{i}:=\max _{1 \leq j \leq g_{i}} m_{i, j},$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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EQ0699 (11.1)</td><td>251</td><td>■ 0.706
● C|+4</td><td><code>\left[\begin{array}{cccc} \boldsymbol{D}_{1} & \boldsymbol{C}_{1} & & \\ \boldsymbol{A}_{1} & \boldsymbol{D}_{2} & \boldsymbol{C}_{2} & \\ & \ddots & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_{m} \end{array}\right]=\left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_{1} & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cccc} \boldsymbol{U}_{1} & \boldsymbol{C}_{1} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_{m} \end{array}\right] .</code></td><td>$$\left[\begin{array}{cccc} \boldsymbol{D}_{1} & \boldsymbol{C}_{1} & & \\ \boldsymbol{A}_{1} & \boldsymbol{D}_{2} & \boldsymbol{C}_{2} & \\ & \ddots & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_{m} \end{array}\right]=\left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_{1} & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cccc} \boldsymbol{U}_{1} & \boldsymbol{C}_{1} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_{m} \end{array}\right] .$$</td><td>$$\left[\begin{array}{ccccc} \boldsymbol{D}_{1} & \boldsymbol{C}_{1} & & & \\ \boldsymbol{A}_{1} & \boldsymbol{D}_{2} & \boldsymbol{C}_{2} & & \\ & \ddots & \ddots & \ddots & \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_{m} \end{array}\right]=\left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_{1} & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cccc} \boldsymbol{U}_{1} & \boldsymbol{C}_{1} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_{m} \end{array}\right].$$</td></tr>
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EQ0717</td><td>256</td><td>■ 0.709
● N|+0</td><td><code>(\boldsymbol{S} \boldsymbol{x})_{k}=\frac{i}{2}\left(\boldsymbol{F}_{2 m+2} \boldsymbol{z}\right)_{k+1}, \quad k=1, \ldots, m .</code></td><td>$$(\boldsymbol{S} \boldsymbol{x})_{k}=\frac{i}{2}\left(\boldsymbol{F}_{2 m+2} \boldsymbol{z}\right)_{k+1}, \quad k=1, \ldots, m .$$</td><td>$$(\boldsymbol{S x})_{k}=\frac{i}{2}\left(\boldsymbol{F}_{2m+2} \boldsymbol{z}\right)_{k+1}, \quad k=1, \ldots, m .$$</td></tr>
<tr><td>lyche-numerical-linear-al…
EQ0593</td><td>218</td><td>■ 0.712
● N|+0</td><td><code>\|A x-b\|_{2}^{2}=\left\|\left(A x-b_{1}\right)-b_{2} \right\|_{2}^{2}=\left\|A x-b_{1}\right\|_{2}^{2}+\left \|b_{2}\right\|_{2}^{2} \geq\left\|b_{2}\right\|_{2}^{2}</code></td><td>$$\|A x-b\|_{2}^{2}=\left\|\left(A x-b_{1}\right)-b_{2} \right\|_{2}^{2}=\left\|A x-b_{1}\right\|_{2}^{2}+\left \|b_{2}\right\|_{2}^{2} \geq\left\|b_{2}\right\|_{2}^{2}$$</td><td>$$\|\boldsymbol{A x}-\boldsymbol{b}\|_{2}^{2}=\|(\boldsymbol{A x}-\boldsymbol{b}_{1})-\boldsymbol{b}_{2}\|_{2}^{2}=\|\boldsymbol{A x}-\boldsymbol{b}_{1}\|_{2}^{2}+\|\boldsymbol{b}_{2}\|_{2}^{2} \geq\|\boldsymbol{b}_{2}\|_{2}^{2}$$</td></tr>
<tr><td>lyche-numerical-linear-al…
EQ0530 (8.38)</td><td>203</td><td>■ 0.713
● W|+1</td><td><code>\begin{aligned} & 4\langle x, z\rangle+4\langle y, z \rangle \stackrel{(8.36)}{=}\|x+z\|^{2}-\|x-z\|^{2}+\|y+z \|^{2}-\|y-z\|^{2} \\ & =\left\|\left(z+\frac{x+y}{2} \right)+\frac{x-y}{2}\right\|^{2}-\left\|\left(z- \frac{x+y}{2}\right)+\frac{y-x}{2}\right\|^{2} \\ & + \left\|\left(z+\frac{x+y}{2}\right)-\frac{x-y}{2}\right \|^{2}-\left\|\left(z-\frac{x+y}{2}\right)-\frac{y-x}{2} \right\|^{2} \\ & \stackrel{(8.35)}{=} 2\left\|z+ \frac{x+y}{2}\right\|^{2}+2\left\|\frac{x-y}{2}\right \|^{2}-2\left\|z-\frac{x+y}{2}\right\|^{2}-2\left\| \frac{y-x}{2}\right\|^{2} \\ & \stackrel{(8.36)}{=} 8 \left\langle\frac{x+y}{2}, z\right\rangle \end{aligned}</code></td><td>$$\begin{aligned} & 4\langle x, z\rangle+4\langle y, z \rangle \stackrel{(8.36)}{=}\|x+z\|^{2}-\|x-z\|^{2}+\|y+z \|^{2}-\|y-z\|^{2} \\ & =\left\|\left(z+\frac{x+y}{2} \right)+\frac{x-y}{2}\right\|^{2}-\left\|\left(z- \frac{x+y}{2}\right)+\frac{y-x}{2}\right\|^{2} \\ & + \left\|\left(z+\frac{x+y}{2}\right)-\frac{x-y}{2}\right \|^{2}-\left\|\left(z-\frac{x+y}{2}\right)-\frac{y-x}{2} \right\|^{2} \\ & \stackrel{(8.35)}{=} 2\left\|z+ \frac{x+y}{2}\right\|^{2}+2\left\|\frac{x-y}{2}\right \|^{2}-2\left\|z-\frac{x+y}{2}\right\|^{2}-2\left\| \frac{y-x}{2}\right\|^{2} \\ & \stackrel{(8.36)}{=} 8 \left\langle\frac{x+y}{2}, z\right\rangle \end{aligned}$$</td><td>$$\begin{aligned} & 4\langle\boldsymbol{x}, \boldsymbol{z}\rangle+4\langle\boldsymbol{y}, \boldsymbol{z}\rangle \stackrel{(8.36)}{=}\|\boldsymbol{x}+\boldsymbol{z}\|^{2}-\|\boldsymbol{x}-\boldsymbol{z}\|^{2}+\|\boldsymbol{y}+\boldsymbol{z}\|^{2}-\|\boldsymbol{y}-\boldsymbol{z}\|^{2} \\ & =\|(\boldsymbol{z}+\frac{\boldsymbol{x}+\boldsymbol{y}}{2})+\frac{\boldsymbol{x}-\boldsymbol{y}}{2}\|^{2}-\|(\boldsymbol{z}-\frac{\boldsymbol{x}+\boldsymbol{y}}{2})+\frac{\boldsymbol{y}-\boldsymbol{x}}{2}\|^{2} \\ & +\|(\boldsymbol{z}+\frac{\boldsymbol{x}+\boldsymbol{y}}{2})-\frac{\boldsymbol{x}-\boldsymbol{y}}{2}\|^{2}-\|(\boldsymbol{z}-\frac{\boldsymbol{x}+\boldsymbol{y}}{2})-\frac{\boldsymbol{y}-\boldsymbol{x}}{2}\|^{2} \\ & \stackrel{(8.35)}{=} 2\|\boldsymbol{z}+\frac{\boldsymbol{x}+\boldsymbol{y}}{2}\|^{2}+2\|\frac{\boldsymbol{x}-\boldsymbol{y}}{2}\|^{2}-2\|\boldsymbol{z}-\frac{\boldsymbol{x}+\boldsymbol{y}}{2}\|^{2}-2\|\frac{\boldsymbol{y}-\boldsymbol{x}}{2}\|^{2} \\ & \stackrel{(8.36)}{=} 8\langle\frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z}\rangle, \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-al…
EQ0615</td><td>226</td><td>■ 0.721
● N|+0</td><td><code>\boldsymbol{A}^{*} \boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 2 \end{array}\right],\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array}\right], \boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*}=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 1 & 0 \end{array} \right] .</code></td><td>$$\boldsymbol{A}^{*} \boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 2 \end{array}\right],\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array}\right], \boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*}=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 1 & 0 \end{array} \right] .$$</td><td>$$\boldsymbol{A}^{*} \boldsymbol{A}=\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \ (\boldsymbol{A}^{*} \boldsymbol{A})^{-1}=\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \ \boldsymbol{A}^{\dagger}=(\boldsymbol{A}^{*} \boldsymbol{A})^{-1} \boldsymbol{A}^{*}=\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0460 (2)</td><td>182</td><td>■ 0.722
● C|+10</td><td><code>\boldsymbol{A}:=\left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array}\right]=\left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right]\left[\begin{array}{lll} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ \ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{rrr} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{array}\right]</code></td><td>$\boldsymbol{A}:=\left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array}\right]=\left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right]\left[\begin{array}{lll} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{rrr} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{array}\right]$</td><td>$\boldsymbol{A}:=\left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array}\right]=\left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right]\left[\begin{array}{lll} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]\left[\begin{array}{rrr} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0736</td><td>261</td><td>■ 0.724
● N|-1</td><td><code>\boldsymbol{T} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=\boldsymbol{C}, \text { where } \boldsymbol{X}=\boldsymbol{V} \boldsymbol{S}, \text { and } \boldsymbol{C}=h^{2} \boldsymbol{F} \boldsymbol{S} .</code></td><td>$\boldsymbol{T} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=\boldsymbol{C}, \text { where } \boldsymbol{X}=\boldsymbol{V} \boldsymbol{S}, \text { and } \boldsymbol{C}=h^{2} \boldsymbol{F} \boldsymbol{S} .$</td><td>$\boldsymbol{T} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}=\boldsymbol{C}, \text { where } \boldsymbol{X}=\boldsymbol{V} \boldsymbol{S}, \text { and } \boldsymbol{C}=h^{2} \boldsymbol{F} \boldsymbol{S} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0668</td><td>241</td><td>■ 0.727
● C|+11</td><td><code>\boldsymbol{A}=\left[\begin{array}{rrrrrrrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \end{array}\right] .</code></td><td>$\boldsymbol{A}=\left[\begin{array}{rrrrrrrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \end{array}\right] .$</td><td>$\boldsymbol{A}=\left[\begin{array}{rrrrrrrrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0976 (14.1)</td><td>330</td><td>■ 0.729
● N|+0</td><td><code>|\mu-\lambda| \leq K\|E\|_{2}^{1 / n}, \quad K=\left(\|A\|_{2}+\|A+E\|_{2}\right)^{1-1 / n} .</code></td><td>$|\mu-\lambda| \leq K\|E\|_{2}^{1 / n}, \quad K=\left(\|A\|_{2}+\|A+E\|_{2}\right)^{1-1 / n} .$</td><td>$|\mu-\lambda| \leq K\|\boldsymbol{E}\|_{2}^{1 / n}, \quad K=\big(\|\boldsymbol{A}\|_{2}+\|\boldsymbol{A}+\boldsymbol{E}\|_{2}\big)^{1-1 / n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0676 (10.13)</td><td>243</td><td>■ 0.737
● W|+2</td><td><code>\begin{aligned} (\lambda \boldsymbol{A}) \otimes(\mu \boldsymbol{B}) & =\lambda \mu(\boldsymbol{A} \otimes \boldsymbol{B}) \\ \left(\boldsymbol{A}_{1}+ \boldsymbol{A}_{2}\right) \otimes \boldsymbol{B} & =\boldsymbol{A}_{1} \otimes \boldsymbol{B}+ \boldsymbol{A}_{2} \otimes \boldsymbol{B}, \\ \boldsymbol{A} \otimes\left(\boldsymbol{B}_{1}+ \boldsymbol{B}_{2}\right) & =\boldsymbol{A} \otimes \boldsymbol{B}_{1}+\boldsymbol{A} \otimes \boldsymbol{B}_{2} \\ (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} & =\boldsymbol{A} \otimes(\boldsymbol{B} \otimes \boldsymbol{C}) . \end{aligned}</code></td><td>$\begin{aligned} (\lambda \boldsymbol{A}) \otimes(\mu \boldsymbol{B}) & =\lambda \mu(\boldsymbol{A} \otimes \boldsymbol{B}) \\ \left(\boldsymbol{A}_{1}+ \boldsymbol{A}_{2}\right) \otimes \boldsymbol{B} & =\boldsymbol{A}_{1} \otimes \boldsymbol{B}+ \boldsymbol{A}_{2} \otimes \boldsymbol{B}, \\ \boldsymbol{A} \otimes\left(\boldsymbol{B}_{1}+ \boldsymbol{B}_{2}\right) & =\boldsymbol{A} \otimes \boldsymbol{B}_{1}+\boldsymbol{A} \otimes \boldsymbol{B}_{2} \\ (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} & =\boldsymbol{A} \otimes(\boldsymbol{B} \otimes \boldsymbol{C}) . \end{aligned}$</td><td>$\begin{aligned} \big(\lambda \boldsymbol{A}\big) \otimes \big(\mu \boldsymbol{B}\big) & =\lambda \mu\big(\boldsymbol{A} \otimes \boldsymbol{B}\big), \\ \big(\boldsymbol{A}_{1}+ \boldsymbol{A}_{2}\big) \otimes \boldsymbol{B} & =\boldsymbol{A}_{1} \otimes \boldsymbol{B}+ \boldsymbol{A}_{2} \otimes \boldsymbol{B}, \\ \boldsymbol{A} \otimes\big(\boldsymbol{B}_{1}+ \boldsymbol{B}_{2}\big) & =\boldsymbol{A} \otimes \boldsymbol{B}_{1}+\boldsymbol{A} \otimes \boldsymbol{B}_{2}, \\ (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} & =\boldsymbol{A} \otimes(\boldsymbol{B} \otimes \boldsymbol{C}) . \end{aligned}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
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EQ0466 (8.3)</td><td>186</td><td>0.737
N|+0</td><td>\sum_{j=1}^{n}\left|x_{j} y_{j}\right| \leq\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}, \quad \frac{1}{p}+\frac{1}{q}=1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n} .</td><td>$\sum_{j=1}^{n}\left|x_{j} y_{j}\right| \leq\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}, \quad \frac{1}{p}+\frac{1}{q}=1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n} .$</td><td>$\sum_{j=1}^{n}|x_j y_j| \leq \|\boldsymbol{x}\|_p \|\boldsymbol{y}\|_q, \quad \frac{1}{p}+\frac{1}{q}=1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0897</td><td>308</td><td>0.743
W|+0</td><td>\frac{\left(\boldsymbol{y}^{T} \boldsymbol{A y} \right)\left(\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}\right)}{\left(\boldsymbol{y}^{T} \boldsymbol{y}\right)^{2}}=a b \leq \frac{(M+m)^{2}}{4 M m},</td><td>$\frac{\left(\boldsymbol{y}^{T} \boldsymbol{A y} \right)\left(\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}\right)}{\left(\boldsymbol{y}^{T} \boldsymbol{y}\right)^{2}}=a b \leq \frac{(M+m)^{2}}{4 M m},$</td><td>$\frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} = ab \leq \frac{(M+m)^2}{4Mm},$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0582</td><td>216</td><td>0.744
W|+0</td><td>\begin{array}{c|c|c|c|c} \boldsymbol{u} & \boldsymbol{u}_{1} & \boldsymbol{u}_{2} & \cdots & \boldsymbol{u}_{m} \\ \hline \hline & y_{1} & y_{2} & \cdots & y_{m} \end{array} .</td><td>$\begin{array}{c|c|c|c|c} \boldsymbol{u} & \boldsymbol{u}_{1} & \boldsymbol{u}_{2} & \cdots & \boldsymbol{u}_{m} \\ \hline \hline & y_{1} & y_{2} & \cdots & y_{m} \end{array} .$</td><td>$\begin{array}{c|c|c|c|c} \boldsymbol{u} & \boldsymbol{u}_{1} & \boldsymbol{u}_{2} & \cdots & \boldsymbol{u}_{m} \\ \hline y & y_{1} & y_{2} & \cdots & y_{m} \end{array} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0204</td><td>085</td><td>0.749
W|+1</td><td>\begin{aligned} & \text { for } i=k+1: n \\ & \qquad \begin{array}{l} a_{p_{i}, k}^{(k)}=a_{p_{i}, k}^{(k)} / a_{p_{k}, k}^{(k)} \\ \text { for } j=k: n \\ \qquad a_{p_{i}, j}^{(k+1)}=a_{p_{i}, j}^{(k)}-a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)} \end{array} \end{aligned}</td><td>$\begin{aligned} & \text { for } i=k+1: n \\ & \qquad \begin{array}{l} a_{p_{i}, k}^{(k)}=a_{p_{i}, k}^{(k)} / a_{p_{k}, k}^{(k)} \\ \text { for } j=k: n \\ \qquad a_{p_{i}, j}^{(k+1)}=a_{p_{i}, j}^{(k)}-a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)} \end{array} \end{aligned}$</td><td>$\begin{aligned} & \text{for } i=k+1:n \\ & \quad a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)}/a_{p_k,k}^{(k)} \\ & \quad \text{for } j=k:n \\ & \qquad a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0142</td><td>065</td><td>0.749
K|+0</td><td>\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]$</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0707</td><td>253</td><td>0.749
N|+0</td><td>(\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D})_{j, k}=\sum_{\ell=1}^{m} d_{j, \ell} x_{\ell, k}+\sum_{\ell=1}^{m} x_{j, \ell} d_{\ell, k}= \lambda_{j} x_{j, k}+\lambda_{k} x_{j, k}</td><td>$(\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D})_{j, k}=\sum_{\ell=1}^{m} d_{j, \ell} x_{\ell, k}+\sum_{\ell=1}^{m} x_{j, \ell} d_{\ell, k}= \lambda_{j} x_{j, k}+\lambda_{k} x_{j, k}$</td><td>$(\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D})_{j, k}=\sum_{\ell=1}^{m} d_{j, \ell} x_{\ell, k}+\sum_{\ell=1}^{m} x_{j, \ell} d_{\ell, k}= \lambda_{j} x_{j, k}+\lambda_{k} x_{j, k}$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0959 (13.68)</td><td>322</td><td>0.749
N|+0</td><td>\omega_{k}:= \begin{cases}1, & \text { if } k=0 \\ \left(1+\omega_{k-1}^{-1} \rho_{k} / \rho_{k-1}\right)^{-1}, & \text { otherwise. }\end{cases}</td><td>$\omega_{k}:= \begin{cases}1, & \text { if } k=0 \\ \left(1+\omega_{k-1}^{-1} \rho_{k} / \rho_{k-1}\right)^{-1}, & \text { otherwise. }\end{cases}$</td><td>$\omega_{k}:= \begin{cases}1, & \text { if } k=0 \\ (1+\omega_{k-1}^{-1} \rho_{k} / \rho_{k-1})^{-1}, & \text { otherwise. }\end{cases}$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0575</td><td>213</td><td>0.751
N|+1</td><td>\begin{aligned} & x_{1}=1 \\ & x_{1}=1, \quad \boldsymbol{A}=\left[\begin{array}{l} 1 \\ 1 \\ x_{1}=2 \end{array}\right], \quad \boldsymbol{x}=\left[x_{1} \right], \quad \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \end{array}\right] . \end{aligned}</td><td>$\begin{aligned} & x_{1}=1 \\ & x_{1}=1, \quad \boldsymbol{A}=\left[\begin{array}{l} 1 \\ 1 \\ x_{1}=2 \end{array}\right], \quad \boldsymbol{x}=\left[x_{1} \right], \quad \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \end{array}\right] . \end{aligned}$</td><td>$\begin{array}{l} x_1 = 1 \\ x_1 = 1, \\ x_1 = 2 \end{array} \quad \boldsymbol{A}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \end{array}\right], \quad \boldsymbol{x}=[x_1], \quad \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 2 \end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0649</td><td>234</td><td>0.756
K|+0</td><td>U^{*} A V=\left[\begin{array}{cc} \Sigma_{1} & 0 \\ \mathbf{0} & \mathbf{0} \end{array}\right]</td><td>$U^{*} A V=\left[\begin{array}{cc} \Sigma_{1} & 0 \\ \mathbf{0} & \mathbf{0} \end{array}\right]$</td><td>$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0667 (10.8)	240	■ 0.758 ● W|-2	\begin{aligned} a_{i i}=4, & i=1, \ldots, n, \\ a_{i+1, i}=a_{i, i+1}=-1, & i=1, \ldots, n-1, \quad i \neq m, 2 m, \ldots,(m-1) m, \\ a_{i+m, i}=a_{i, i+m}=-1, & i=1, \ldots, n-m, \\ a_{i j}=0, & \text { otherwise } . \end{aligned}	$\begin{aligned} a_{ii}=4, & i=1, \ldots, n, \\ a_{i+1,i}=a_{i,i+1}=-1, & i=1, \ldots, n-1, \quad i \neq m, 2m, \ldots,(m-1)m, \\ a_{i+m,i}=a_{i,i+m}=-1, & i=1, \ldots, n-m, \\ a_{ij}=0, & \text{ otherwise } . \end{aligned}$	$\begin{aligned} a_{ii}=4, &\quad i=1, \ldots, n, \\ a_{i+1,i}=a_{i,i+1}=-1, &\quad i=1, \ldots, n-1,\ i \neq m, 2m, \ldots,(m-1)m, \\ a_{i+m,i}=a_{i,i+m}=-1, &\quad i=1, \ldots, n-m, \\ a_{ij}=0, &\quad \text{otherwise.} \end{aligned}$
lyche-numerical-linear-algebra_ EQ1018	345	■ 0.760 ● K|+0	z_{k}=c_{1} \lambda_{1}^{k} v_{1}+c_{2} \lambda_{2}^{k} v_{2}=c_{1} 3^{k} v_{1}+c_{2} 1^{k} v_{2} .	$z_{k}=c_{1} \lambda_{1}^{k} v_{1}+c_{2} \lambda_{2}^{k} v_{2}=c_{1} 3^{k} v_{1}+c_{2} 1^{k} v_{2} .$	$\boldsymbol{z}_{k}=c_{1} \lambda_{1}^{k} \boldsymbol{v}_{1}+c_{2} \lambda_{2}^{k} \boldsymbol{v}_{2}=c_{1} 3^{k} \boldsymbol{v}_{1}+c_{2} 1^{k} \boldsymbol{v}_{2}.$
lyche-numerical-linear-algebra_ EQ0408	163	■ 0.761 ● K|+0	\boldsymbol{A}=\frac{1}{6}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]	$\boldsymbol{A}=\frac{1}{6}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$	$\boldsymbol{A}=\frac{1}{6}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0651	234	■ 0.763 ● N|+0	\text { (1) } \quad \boldsymbol{A} \boldsymbol{x}= \boldsymbol{b}, \quad \text { (2) } \quad \boldsymbol{A}^{*} \boldsymbol{y}=\mathbf{0}, \boldsymbol{y}^{*} \boldsymbol{b} \neq 0 \text {. }	$\text{ (1) } \quad \boldsymbol{A} \boldsymbol{x}= \boldsymbol{b}, \quad \text{ (2) } \quad \boldsymbol{A}^{*} \boldsymbol{y}=\mathbf{0}, \boldsymbol{y}^{*} \boldsymbol{b} \neq 0 \text{. }$	$(1) \quad \boldsymbol{A} \boldsymbol{x}= \boldsymbol{b}, \qquad (2) \quad \boldsymbol{A}^{*} \boldsymbol{y}=\mathbf{0},\ \boldsymbol{y}^{*} \boldsymbol{b} \neq 0.$
lyche-numerical-linear-algebra_ EQ0176	077	■ 0.770 ● W|+0	\text { for } \begin{aligned} i & =k+1: n \\ l_{i k}^{(k)} & =a_{i k}^{(k)} / a_{k k}^{(k)} \end{aligned}	$\text{ for } \begin{aligned} i & =k+1: n \\ l_{i k}^{(k)} & =a_{i k}^{(k)} / a_{k k}^{(k)} \end{aligned}$	for $i=k+1:n$ $l_{ik}^{(k)}=a_{ik}^{(k)}/a_{kk}^{(k)}$
lyche-numerical-linear-algebra_ EQ0772 (12.19)	273	■ 0.772 ● N|+0	\begin{gathered} \rho(\boldsymbol{I}-\alpha \boldsymbol{A})>\rho\left(\boldsymbol{I}-\alpha_{o} \boldsymbol{A}\right)=\frac{\kappa-1}{\kappa+1}, \quad \alpha \in \mathbb{R} \backslash\left\{\alpha_{o} \right\}, \\ \kappa:=\frac{\lambda_{1}}{\lambda_{n}}, \quad \alpha_{o}:=\frac{2}{\lambda_{1}+\lambda_{n}} . \end{gathered}	$\begin{gathered} \rho(\boldsymbol{I}-\alpha \boldsymbol{A})>\rho\left(\boldsymbol{I}-\alpha_{o} \boldsymbol{A}\right)=\frac{\kappa-1}{\kappa+1}, \quad \alpha \in \mathbb{R} \backslash\left\{\alpha_{o} \right\}, \\ \kappa:=\frac{\lambda_{1}}{\lambda_{n}}, \quad \alpha_{o}:=\frac{2}{\lambda_{1}+\lambda_{n}} . \end{gathered}$	$\begin{gathered} \rho(\boldsymbol{I}-\alpha \boldsymbol{A})>\rho(\boldsymbol{I}-\alpha_{o} \boldsymbol{A})=\frac{\kappa-1}{\kappa+1}, \quad \alpha \in \mathbb{R} \setminus \{\alpha_{o}\}, \\ \kappa:=\frac{\lambda_{1}}{\lambda_{n}}, \quad \alpha_{o}:=\frac{2}{\lambda_{1}+\lambda_{n}}. \end{gathered}$
lyche-numerical-linear-algebra_ EQ0915 (13.47)	311	■ 0.779 ● K|+0	\boldsymbol{x}_{k}:=\boldsymbol{C}^{T} \boldsymbol{y}_{k}, \quad \boldsymbol{p}_{k}:= \boldsymbol{C}^{T} \boldsymbol{q}_{k}, \quad \boldsymbol{s}_{k}:=\boldsymbol{C}^{T} z_{k}, \quad \boldsymbol{r}_{k}:=\boldsymbol{C}^{-1} z_{k}	$\boldsymbol{x}_{k}:=\boldsymbol{C}^{T} \boldsymbol{y}_{k}, \quad \boldsymbol{p}_{k}:= \boldsymbol{C}^{T} \boldsymbol{q}_{k}, \quad \boldsymbol{s}_{k}:=\boldsymbol{C}^{T} z_{k}, \quad \boldsymbol{r}_{k}:=\boldsymbol{C}^{-1} z_{k}$	$\boldsymbol{x}_{k}:=\boldsymbol{C}^{T} \boldsymbol{y}_{k}, \quad \boldsymbol{p}_{k}:= \boldsymbol{C}^{T} \boldsymbol{q}_{k}, \quad \boldsymbol{s}_{k}:=\boldsymbol{C}^{T} \boldsymbol{z}_{k}, \quad \boldsymbol{r}_{k}:=\boldsymbol{C}^{-1} \boldsymbol{z}_{k}$
lyche-numerical-linear-algebra_ EQ0610	223	■ 0.780 ● K|+0	\boldsymbol{v}=\boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{v}=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\left[\begin{array}{l} \boldsymbol{U}_{1}^{*} \\ \boldsymbol{U}_{2}^{*} \end{array}\right] \boldsymbol{v}=\boldsymbol{s}+ \boldsymbol{t},\right.	$\boldsymbol{v}=\boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{v}=\left[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\left[\begin{array}{l} \boldsymbol{U}_{1}^{*} \\ \boldsymbol{U}_{2}^{*} \end{array}\right] \boldsymbol{v}=\boldsymbol{s}+ \boldsymbol{t},\right.$	$\boldsymbol{v}=\boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{v}=[\boldsymbol{U}_{1}, \boldsymbol{U}_{2}\begin{bmatrix} \boldsymbol{U}_{1}^{*} \\ \boldsymbol{U}_{2}^{*} \end{bmatrix} \boldsymbol{v}=\boldsymbol{s}+ \boldsymbol{t},$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_EQ0203 (3.13)	085	■0.781 ● W|+0	`\begin{aligned} & \boldsymbol{p}=[1, \ldots, n]^{T} ; \ \ & \text { for } k=1: n-1 \\ & \text { choose } r_{k} \geq k \text { so that } a_{p_{r_{k}}, k}^{(k)} \neq 0 \text {. } \\ & \qquad \boldsymbol{p}=I_{r_{k}, k} \boldsymbol{p} \end{aligned}`	$\begin{aligned} & \boldsymbol{p}=[1, \ldots, n]^{T} ; \\ & \text { for } k=1: n-1 \\ & \text { choose } r_{k} \geq k \text { so that } a_{p_{r_{k}}, k}^{(k)} \neq 0 . \\ & \qquad \boldsymbol{p}=I_{r_{k}, k} \boldsymbol{p} \end{aligned}$	$\begin{aligned} & \boldsymbol{p}=[1, \ldots, n]^{T} ; \\ & \text{for } k=1: n-1 \\ & \quad \text{choose } r_{k} \geq k \text{ so that } a_{p_{r_{k}}, k}^{(k)} \neq 0 . \\ & \quad \boldsymbol{p}=I_{r_{k}, k} \boldsymbol{p} \end{aligned}$
lyche-numerical-linear-algebra_EQ0852 (13.15)	295	■0.782 ● W|+0	`\begin{aligned} & \text { For } k=0,1,2, \ldots \\ & \qquad \boldsymbol{x}_{k+1}:= \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \quad \alpha_{k}:=\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}, \\ & \boldsymbol{r}_{k+1}:= \boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}, \end{aligned}`	$\begin{aligned} & \text { For } k=0,1,2, \ldots \\ & \qquad \boldsymbol{x}_{k+1}:= \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \quad \alpha_{k}:=\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}, \\ & \boldsymbol{r}_{k+1}:= \boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}, \end{aligned}$	$\begin{aligned} & \text{For } k=0,1,2, \ldots \\ & \qquad \boldsymbol{x}_{k+1}:= \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \quad \alpha_{k}:=\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}, \\ & \qquad \boldsymbol{r}_{k+1}:= \boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}, \end{aligned}$
lyche-numerical-linear-algebra_EQ0531	203	■0.783 ● N|+0	`\langle\boldsymbol{x}, \boldsymbol{z}\rangle+\langle \boldsymbol{y}, \boldsymbol{z}\rangle=2\left\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V} .`	$\langle\boldsymbol{x}, \boldsymbol{z}\rangle+\langle \boldsymbol{y}, \boldsymbol{z}\rangle=2\left\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V} .$	$\langle\boldsymbol{x}, \boldsymbol{z}\rangle+\langle \boldsymbol{y}, \boldsymbol{z}\rangle=2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V} .$
lyche-numerical-linear-algebra_EQ0009 (1.1)	023	■0.784 ● N|+0	`\delta_{i j}:= \begin{cases}1 & \text { if } i=j \\ 0 & \text { otherwise }\end{cases}`	$\delta_{i j}:= \begin{cases}1 & \text { if } i=j \\ 0 & \text { otherwise }\end{cases}$	$\delta_{i j}:= \begin{cases}1 & \text { if } i=j, \\ 0 & \text { otherwise, }\end{cases}$
lyche-numerical-linear-algebra_EQ0108 (2.21)	057	■0.788 ● N|-1	`\boldsymbol{T} \boldsymbol{v}=\left[\begin{array}{rrrrr} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots \\ & & & & 0 \\ & & & -1 & 2-1 \\ & & & 0-1 & 2 \end{array}\right]\left[\begin{array}{c} v_{1} \\ v_{2} \\ \vdots \\ v_{m-1} \\ v_{m} \end{array} \right]=h^{2}\left[\begin{array}{c} f(h) \\ f(2 h) \\ \vdots \\ f((m-1) h) \\ f(m h) \end{array}\right]=: \boldsymbol{b} .`	$\boldsymbol{T} \boldsymbol{v}=\left[\begin{array}{rrrrr} 2 & -1 & 0 & & \\ -1 & 2 & -1 & & \\ 0 & \ddots & \ddots & \ddots \\ & & & & 0 \\ & & & -1 & 2-1 \\ & & & 0-1 & 2 \end{array}\right]\left[\begin{array}{c} v_{1} \\ v_{2} \\ \vdots \\ v_{m-1} \\ v_{m} \end{array} \right]=h^{2}\left[\begin{array}{c} f(h) \\ f(2 h) \\ \vdots \\ f((m-1) h) \\ f(m h) \end{array}\right]=: \boldsymbol{b} .$	$\boldsymbol{T} \boldsymbol{v}=\left[\begin{array}{cccccc} 2 & -1 & 0 & & & \\ -1 & 2 & -1 & & & \\ 0 & \ddots & \ddots & \ddots & & \\ & & & & & 0 \\ & & & & -1 & 2 \;\; -1 \\ & & & & 0 & -1 \;\; 2 \end{array}\right]\left[\begin{array}{c} v_{1} \\ v_{2} \\ \vdots \\ v_{m-1} \\ v_{m} \end{array} \right]=h^{2}\left[\begin{array}{c} f(h) \\ f(2 h) \\ \vdots \\ f\big((m-1) h\big) \\ f(m h) \end{array}\right]=: \boldsymbol{b} .$
lyche-numerical-linear-algebra_EQ0401	160	■0.791 ● N|+0	`\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}=\left[ \begin{array}{c|c} \lambda & z^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right],`	$\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c\|c} \lambda & z^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right],$	$\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c\|c} \lambda & z^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right],$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0590</td><td>218</td><td>■ 0.792
● W|+1</td><td><code>\begin{aligned} \frac{1}{m} b_{i, j} & =\frac{1}{m} \sum_{k=1}^{m} t_{k}^{i+j-2}=\frac{1}{m} \sum_{k=1}^{m} \left(\frac{k-1}{m-1}\right)^{i+j-2} \\ & \approx \int_{0}^{1} x^{i+j-2} d x=\frac{1}{i+j-1}=h_{i, j} \end{aligned}</code></td><td>$$\begin{aligned} \frac{1}{m} b_{i, j} & =\frac{1}{m} \sum_{k=1}^{m} t_{k}^{i+j-2}=\frac{1}{m} \sum_{k=1}^{m} \left(\frac{k-1}{m-1}\right)^{i+j-2} \\ & \approx \int_{0}^{1} x^{i+j-2} d x=\frac{1}{i+j-1}=h_{i, j} \end{aligned}$$</td><td>$$\begin{aligned} \frac{1}{m} b_{i, j} & =\frac{1}{m} \sum_{k=1}^{m} t_{k}^{i+j-2}=\frac{1}{m} \sum_{k=1}^{m} \left(\frac{k-1}{m-1}\right)^{i+j-2} \\ & \approx \int_{0}^{1} x^{i+j-2} d x=\frac{1}{i+j-1}=h_{i, j}. \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0096 (0)</td><td>054</td><td>■ 0.793
● W|+0</td><td><code>\left[\begin{array}{lll} d_{1} & c_{1} & 0 \\ a_{1} & d_{2} & c_{2} \\ 0 & a_{2} & d_{3} \end{array}\right]=\left[\begin{array}{lll} 1 & 0 & 0 \\ l_{1} & 1 & 0 \\ 0 & l_{2} & 1 \end{array}\right]\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ 0 & u_{2} & c_{2} \\ 0 & 0 & u_{3} \end{array}\right]=\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ l_{1} u_{1} & l_{1} c_{1}+u_{2} & c_{2} \\ 0 & l_{2} u_{2} & l_{2} c_{2}+u_{3} \end{array}\right],</code></td><td>$$\left[\begin{array}{lll} d_{1} & c_{1} & 0 \\ a_{1} & d_{2} & c_{2} \\ 0 & a_{2} & d_{3} \end{array}\right]=\left[\begin{array}{lll} 1 & 0 & 0 \\ l_{1} & 1 & 0 \\ 0 & l_{2} & 1 \end{array}\right]\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ 0 & u_{2} & c_{2} \\ 0 & 0 & u_{3} \end{array}\right]=\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ l_{1} u_{1} & l_{1} c_{1}+u_{2} & c_{2} \\ 0 & l_{2} u_{2} & l_{2} c_{2}+u_{3} \end{array}\right],$$</td><td>$$\left[\begin{array}{lll} d_{1} & c_{1} & 0 \\ a_{1} & d_{2} & c_{2} \\ 0 & a_{2} & d_{3} \end{array}\right]=\left[\begin{array}{lll} 1 & 0 & 0 \\ l_{1} & 1 & 0 \\ 0 & l_{2} & 1 \end{array}\right]\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ 0 & u_{2} & c_{2} \\ 0 & 0 & u_{3} \end{array}\right]=\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ l_{1} u_{1} & l_{1} c_{1}+u_{2} & c_{2} \\ 0 & l_{2} u_{2} & l_{2} c_{2}+u_{3} \end{array}\right],$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0752 (12.7)</td><td>268</td><td>■ 0.800
● W|+0</td><td><code>\begin{array}{r} S O R: \boldsymbol{v}_{k+1}(i, j)=\omega\left(\boldsymbol{v}_{k+1}(i-1, j)+ \boldsymbol{v}_{k+1}(i, j-1)+\boldsymbol{v}_{k}(i+1, j)\right. \\ \left.+\boldsymbol{v}_{k}(i, j+1)+ \boldsymbol{e}(i, j)\right) / 4+(1-\omega) \boldsymbol{v}_{k}(i, j) . \end{array}</code></td><td>$$\begin{array}{r} S O R: \boldsymbol{v}_{k+1}(i, j)=\omega\left(\boldsymbol{v}_{k+1}(i-1, j)+ \boldsymbol{v}_{k+1}(i, j-1)+\boldsymbol{v}_{k}(i+1, j)\right. \\ \left.+\boldsymbol{v}_{k}(i, j+1)+ \boldsymbol{e}(i, j)\right) / 4+(1-\omega) \boldsymbol{v}_{k}(i, j) . \end{array}$$</td><td>$$\begin{aligned} SOR: \boldsymbol{v}_{k+1}(i, j) = \omega\big(&\boldsymbol{v}_{k+1}(i-1, j)+ \boldsymbol{v}_{k+1}(i, j-1)+\boldsymbol{v}_{k}(i+1, j) \\ &+\boldsymbol{v}_{k}(i, j+1)+ \boldsymbol{e}(i, j)\big)/4+(1-\omega) \boldsymbol{v}_{k}(i, j). \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0403</td><td>161</td><td>■ 0.801
● N|+0</td><td><code>\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V} \boldsymbol{u}=\left[\begin{array}{c|c} \bar{ \lambda} & \mathbf{0}^{*} \\ \hline \boldsymbol{z} & \boldsymbol{M}^{*} \end{array}\right]\left[\begin{array}{l} 0 \\ \boldsymbol{v} \end{array} \right]=\left[\begin{array}{c} 0 \\ \boldsymbol{M}^{*} \boldsymbol{v} \end{array}\right]=\bar{\lambda}\left[\begin{array}{l} 0 \\ \boldsymbol{v} \end{array}\right]</code></td><td>$$\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V} \boldsymbol{u}=\left[\begin{array}{c|c} \bar{ \lambda} & \mathbf{0}^{*} \\ \hline \boldsymbol{z} & \boldsymbol{M}^{*} \end{array}\right]\left[\begin{array}{l} 0 \\ \boldsymbol{v} \end{array} \right]=\left[\begin{array}{c} 0 \\ \boldsymbol{M}^{*} \boldsymbol{v} \end{array}\right]=\bar{\lambda}\left[\begin{array}{l} 0 \\ \boldsymbol{v} \end{array}\right]$$</td><td>$$\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V} \boldsymbol{u}=\left[\begin{array}{c|c} \bar{ \lambda} & \mathbf{0}^{*} \\ \hline \boldsymbol{z} & \boldsymbol{M}^{*} \end{array}\right]\left[\begin{array}{c} 0 \\ \boldsymbol{v} \end{array} \right]=\left[\begin{array}{c} 0 \\ \boldsymbol{M}^{*} \boldsymbol{v} \end{array}\right]=\bar{\lambda}\left[\begin{array}{c} 0 \\ \boldsymbol{v} \end{array}\right]$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0588</td><td>217</td><td>■ 0.812
● W|+0</td><td><code>\boldsymbol{B}_{3} \boldsymbol{x}:=\left[\begin{array}{ccc} m & \sum t_{k} & \sum t_{k}^{2} \ \ \sum t_{k} & \sum t_{k}^{2} & \sum t_{k}^{3} \\ \sum t_{k}^{2} & \sum t_{k}^{3} & \sum t_{k}^{4} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{c} \sum y_{k} \ \ \sum t_{k} y_{k} \\ \sum t_{k}^{2} y_{k} \end{array} \right] .</code></td><td>$$\boldsymbol{B}_{3} \boldsymbol{x}:=\left[\begin{array}{ccc} m & \sum t_{k} & \sum t_{k}^{2} \\ \sum t_{k} & \sum t_{k}^{2} & \sum t_{k}^{3} \\ \sum t_{k}^{2} & \sum t_{k}^{3} & \sum t_{k}^{4} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{c} \sum y_{k} \\ \sum t_{k} y_{k} \\ \sum t_{k}^{2} y_{k} \end{array} \right] .$$</td><td>$$\boldsymbol{B}_{3} \boldsymbol{x}:=\left[\begin{array}{ccc} m & \sum t_{k} & \sum t_{k}^{2} \\ \sum t_{k} & \sum t_{k}^{2} & \sum t_{k}^{3} \\ \sum t_{k}^{2} & \sum t_{k}^{3} & \sum t_{k}^{4} \end{array} \right]\left[\begin{array}{c} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{c} \sum y_{k} \\ \sum t_{k} y_{k} \\ \sum t_{k}^{2} y_{k} \end{array} \right] .$$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0529 (8.37)</td><td>203</td><td>■ 0.818 ● W|+1</td><td><code>\begin{gathered} \langle\boldsymbol{x}, \boldsymbol{z} \rangle+\langle\boldsymbol{y}, \boldsymbol{z}\rangle= \langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V} \\ \langle a \boldsymbol{x}, \boldsymbol{y}\rangle=a\langle \boldsymbol{x}, \boldsymbol{y}\rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V} . \end{gathered}</code></td><td>$$\begin{gathered} \langle\boldsymbol{x}, \boldsymbol{z} \rangle+\langle\boldsymbol{y}, \boldsymbol{z}\rangle= \langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V} \\ \langle a \boldsymbol{x}, \boldsymbol{y}\rangle=a\langle \boldsymbol{x}, \boldsymbol{y}\rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V} . \end{gathered}$$</td><td>$$\begin{gathered} \langle\boldsymbol{x}, \boldsymbol{z} \rangle+\langle\boldsymbol{y}, \boldsymbol{z}\rangle= \langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}, \\ \langle a \boldsymbol{x}, \boldsymbol{y}\rangle=a\langle \boldsymbol{x}, \boldsymbol{y}\rangle, \quad a \in \mathbb{R}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V} . \end{gathered}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0089 (2.11)</td><td>051</td><td>■ 0.822 ● C|-6</td><td><code>\left[\begin{array}{ccccc} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array}\right]\left[\begin{array}{c} \mu_{2} \\ \mu_{3} \\ \vdots \\ \mu_{n-1} \\ \mu_{n} \end{array} \right]=\frac{6}{h^{2}}\left[\begin{array}{c} \delta^{2} y_{2}-\mu_{1} \\ \delta^{2} y_{3} \\ \vdots \\ \delta^{2} y_{n-1} \\ \delta^{2} y_{n}-\mu_{n+1} \end{array}\right], \delta^{2} y_{i}:=y_{i+1}-2 y_{i}+y_{i-1} .</code></td><td>$$\left[\begin{array}{ccccc} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array}\right]\left[\begin{array}{c} \mu_{2} \\ \mu_{3} \\ \vdots \\ \mu_{n-1} \\ \mu_{n} \end{array} \right]=\frac{6}{h^{2}}\left[\begin{array}{c} \delta^{2} y_{2}-\mu_{1} \\ \delta^{2} y_{3} \\ \vdots \\ \delta^{2} y_{n-1} \\ \delta^{2} y_{n}-\mu_{n+1} \end{array}\right], \delta^{2} y_{i}:=y_{i+1}-2 y_{i}+y_{i-1} .$$</td><td>$$\left[\begin{array}{ccccc} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 4 \end{array}\right]\left[\begin{array}{c} \mu_{2} \\ \mu_{3} \\ \vdots \\ \mu_{n-1} \\ \mu_{n} \end{array} \right]=\frac{6}{h^{2}}\left[\begin{array}{c} \delta^{2} y_{2}-\mu_{1} \\ \delta^{2} y_{3} \\ \vdots \\ \delta^{2} y_{n-1} \\ \delta^{2} y_{n}-\mu_{n+1} \end{array}\right], \delta^{2} y_{i}:=y_{i+1}-2 y_{i}+y_{i-1} .$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0563 (8.44)</td><td>210</td><td>■ 0.822 ● K|+0</td><td><code>\boldsymbol{A} \boldsymbol{s}=\boldsymbol{b},</code></td><td>$\boldsymbol{A} \boldsymbol{s}=\boldsymbol{b},$</td><td>$\boldsymbol{A} \boldsymbol{s}=\boldsymbol{b},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0354</td><td>142</td><td>■ 0.825 ● N|+0</td><td><code>\left[\begin{array}{c} \boldsymbol{L}^{*} \\ \boldsymbol{z}^{*} \end{array}\right]=\boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],</code></td><td>$$\left[\begin{array}{c} \boldsymbol{L}^{*} \\ \boldsymbol{z}^{*} \end{array}\right]=\boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],$$</td><td>$$\left[\begin{array}{c} \boldsymbol{L}^{*} \\ z^{*} \end{array}\right]=\boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array} \right],$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0040 (1.14)</td><td>036</td><td>■ 0.826 ● N|+0</td><td><code>\begin{aligned} \operatorname{det}(\boldsymbol{A}) & = \sum_{j=1}^{n}(-1)^{i+j} a_{i j} \operatorname{det}\left(\boldsymbol{A}_{i j}\right) \text { for } i=1, \ldots, n, \text { row } \\ \operatorname{det}(\boldsymbol{A}) & = \sum_{i=1}^{n}(-1)^{i+j} a_{i j} \operatorname{det}\left(\boldsymbol{A}_{i j}\right) \text { for } j=1, \ldots, n, \text { column. } \end{aligned}</code></td><td>$$\begin{aligned} \operatorname{det}(\boldsymbol{A}) & = \sum_{j=1}^{n}(-1)^{i+j} a_{i j} \operatorname{det}\left(\boldsymbol{A}_{i j}\right) \text { for } i=1, \ldots, n, \text { row } \\ \operatorname{det}(\boldsymbol{A}) & = \sum_{i=1}^{n}(-1)^{i+j} a_{i j} \operatorname{det}\left(\boldsymbol{A}_{i j}\right) \text { for } j=1, \ldots, n, \text { column. } \end{aligned}$$</td><td>$$\begin{aligned} \operatorname{det}(\boldsymbol{A}) & = \sum_{j=1}^{n}(-1)^{i+j} a_{i j} \operatorname{det}(\boldsymbol{A}_{i j}) \text { for } i=1, \ldots, n, \text { row } \\ \operatorname{det}(\boldsymbol{A}) & = \sum_{i=1}^{n}(-1)^{i+j} a_{i j} \operatorname{det}(\boldsymbol{A}_{i j}) \text { for } j=1, \ldots, n, \text { column. } \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0338</td><td>136</td><td>■ 0.827 ● N|+0</td><td><code>\left|\boldsymbol{x}^{T} \boldsymbol{A y}\right|^{2} \leq \boldsymbol{x}^{T} \boldsymbol{A x} \boldsymbol{y}^{T} \boldsymbol{A y}</code></td><td>$\left|\boldsymbol{x}^{T} \boldsymbol{A y}\right|^{2} \leq \boldsymbol{x}^{T} \boldsymbol{A x} \boldsymbol{y}^{T} \boldsymbol{A y}$</td><td>$|\boldsymbol{x}^{T} \boldsymbol{A y}|^{2} \leq \boldsymbol{x}^{T} \boldsymbol{A x}\, \boldsymbol{y}^{T} \boldsymbol{A y}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1004</td><td>338</td><td>■ 0.827 ● N|+0</td><td><code>\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{y}= \boldsymbol{y}^{*} \boldsymbol{E}^{*} \boldsymbol{B} \boldsymbol{E} \boldsymbol{y}=z^{*} \boldsymbol{B} z= \sum_{i=m+1}^{n} \mu_{i}\left|z_{i}\right|^{2} \leq 0,</code></td><td>$$\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{y}= \boldsymbol{y}^{*} \boldsymbol{E}^{*} \boldsymbol{B} \boldsymbol{E} \boldsymbol{y}=z^{*} \boldsymbol{B} z= \sum_{i=m+1}^{n} \mu_{i}\left|z_{i}\right|^{2} \leq 0,$$</td><td>$$\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{y}= \boldsymbol{y}^{*} \boldsymbol{E}^{*} \boldsymbol{B} \boldsymbol{E} \boldsymbol{y}=\boldsymbol{z}^{*} \boldsymbol{B} \boldsymbol{z}= \sum_{i=m+1}^{n} \mu_{i}|z_{i}|^{2} \leq 0,$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0298	123	■ 0.828 ● W|+0	`\langle\boldsymbol{x}, \overline{\boldsymbol{A}} \boldsymbol{y}\rangle=(\overline{\boldsymbol{A}} \boldsymbol{y})^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \overline{\boldsymbol{A}}^{*} \boldsymbol{x}= \boldsymbol{y}^{*} \boldsymbol{A}^{T} \boldsymbol{x}= \boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}=\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle .`	$\langle\boldsymbol{x}, \overline{\boldsymbol{A}} \boldsymbol{y}\rangle=(\overline{\boldsymbol{A}} \boldsymbol{y})^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \overline{\boldsymbol{A}}^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \boldsymbol{A}^{T} \boldsymbol{x}=\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}=\langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle.$	$\langle\boldsymbol{x}, \overline{\boldsymbol{A}} \boldsymbol{y}\rangle=(\overline{\boldsymbol{A}} \boldsymbol{y})^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \overline{\boldsymbol{A}}^{*} \boldsymbol{x}=\boldsymbol{y}^{*} \boldsymbol{A}^{T} \boldsymbol{x}=\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}=\langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{y}\rangle.$
lyche-numerical-linear-algebra_ EQ0263	108	■ 0.830 ● N|+0	`\boldsymbol{A}=\boldsymbol{B B}^{*}, \quad \boldsymbol{B}=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} \sqrt{3} & 0 \\ 0 & \sqrt{8 / 3} \end{array}\right] .`	$\boldsymbol{A}=\boldsymbol{B B}^{*}, \quad \boldsymbol{B}=\left[\begin{array}{cc} 1 & 0 \\ 1/3 & 1 \end{array}\right]\left[\begin{array}{cc} \sqrt{3} & 0 \\ 0 & \sqrt{8/3} \end{array}\right].$	$\boldsymbol{A}=\boldsymbol{B B}^{*}, \quad \boldsymbol{B}=\left[\begin{array}{cc} 1 & 0 \\ 1/3 & 1 \end{array}\right]\left[\begin{array}{cc} \sqrt{3} & 0 \\ 0 & \sqrt{8/3} \end{array}\right].$
lyche-numerical-linear-algebra_ EQ0267	109	■ 0.831 ● N|+0	`\boldsymbol{L} \boldsymbol{L}^{*}=\left[\begin{array}{cc} \beta & \mathbf{0} \\ \boldsymbol{v} / \beta & \boldsymbol{L}_{1} \end{array}\right]\left[\begin{array}{cc} \beta & \boldsymbol{v}^{*} / \beta \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right]= \left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \ \ \boldsymbol{v} & \boldsymbol{B} \end{array}\right]= \boldsymbol{A}`	$\boldsymbol{L} \boldsymbol{L}^{*}=\left[\begin{array}{cc} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_{1} \end{array}\right]\left[\begin{array}{cc} \beta & \boldsymbol{v}^{*}/\beta \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right]=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right]=\boldsymbol{A}$	$\boldsymbol{L} \boldsymbol{L}^{*}=\left[\begin{array}{cc} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_{1} \end{array}\right]\left[\begin{array}{cc} \beta & \boldsymbol{v}^{*}/\beta \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right]=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right]=\boldsymbol{A}$
lyche-numerical-linear-algebra_ EQ0189 (3.7)	080	■ 0.831 ● N|+0	`x_{k}=\left(b_{k}-\sum_{j=1}^{k-1} a_{k, j} x_{j}\right) / a_{k k}, \quad k=1,2, \ldots, n .`	$x_{k}=\left(b_{k}-\sum_{j=1}^{k-1} a_{k, j} x_{j}\right) / a_{k k}, \quad k=1,2, \ldots, n.$	$x_{k}=\big(b_{k}-\sum_{j=1}^{k-1} a_{k, j} x_{j}\big) / a_{k k}, \quad k=1,2, \ldots, n.$
lyche-numerical-linear-algebra_ EQ0122 (2.30)	061	■ 0.833 ● S|+0	`\begin{aligned} & \boldsymbol{s}_{j}=[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T}, \\ & \lambda_{j}=d+2 a \cos (j \pi h) . \end{aligned}`	$\begin{aligned} & \boldsymbol{s}_{j}=[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T}, \\ & \lambda_{j}=d+2 a \cos (j \pi h). \end{aligned}$	$\begin{aligned} & \boldsymbol{s}_{j}=[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T}, \\ & \lambda_{j}=d+2 a \cos (j \pi h). \end{aligned}$
lyche-numerical-linear-algebra_ EQ0950 (13.64)	321	■ 0.838 ● N|+0	`\begin{aligned} & \langle\boldsymbol{B} \boldsymbol{x}, \boldsymbol{x}\rangle=0, \quad \boldsymbol{x} \in \mathbb{R}^{n}, \\ & \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{x}\rangle=\langle \boldsymbol{x}, \boldsymbol{x}\rangle=\\|\boldsymbol{x} \\|_{2}^{2} . \end{aligned}`	$\begin{aligned} & \langle\boldsymbol{B} \boldsymbol{x}, \boldsymbol{x}\rangle=0, \quad \boldsymbol{x} \in \mathbb{R}^{n}, \\ & \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{x}\rangle=\langle\boldsymbol{x}, \boldsymbol{x}\rangle=\\|\boldsymbol{x}\\|_{2}^{2}. \end{aligned}$	$\begin{aligned} & \langle\boldsymbol{B} \boldsymbol{x}, \boldsymbol{x}\rangle=0, \quad \boldsymbol{x} \in \mathbb{R}^{n}, \\ & \langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{x}\rangle=\langle\boldsymbol{x}, \boldsymbol{x}\rangle=\\|\boldsymbol{x}\\|_{2}^{2}. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0321 (5.16)	130	■ 0.839 ● N|+0	`\int_{0}^{n} 4(m-k)(n+r-k) d k=2 m(n+r)^{2}- \frac{2}{3}(n+r)^{3} .`	$\int_{0}^{n} 4(m-k)(n+r-k) d k=2 m(n+r)^{2}-\frac{2}{3}(n+r)^{3}.$	$\int_{0}^{n} 4(m-k)(n+r-k) d k=2 m(n+r)^{2}-\frac{2}{3}(n+r)^{3}.$
lyche-numerical-linear-algebra_ EQ0693 (10.23)	248	■ 0.839 ● C|+9	`\begin{array}{ll} \Delta^{2} u(s, t):=\Delta(\Delta u(s, t))=f(s, t) & (s, t) \in \Omega, \\ u(s, t)=0, & \Delta u(s, t)=0 \end{array}`	$\begin{array}{ll} \Delta^{2} u(s, t):=\Delta(\Delta u(s, t))=f(s, t) & (s, t) \in \Omega, \\ u(s, t)=0, & \Delta u(s, t)=0 \end{array}$	$\begin{array}{lll} \Delta^{2} u(s, t):=\Delta\big(\Delta u(s, t)\big)=f(s, t) & & (s, t) \in \Omega, \\ u(s, t)=0, & \Delta u(s, t)=0 & (s, t) \in \partial \Omega. \end{array}$
lyche-numerical-linear-algebra_ EQ0828	287	■ 0.840 ● K|+0	`\boldsymbol{x}_{k}(1)=\boldsymbol{x}_{k}(2)=1-a^{k}, \quad k \geq 0,`	$\boldsymbol{x}_{k}(1)=\boldsymbol{x}_{k}(2)=1-a^{k}, \quad k \geq 0,$	$\boldsymbol{x}_{k}(1)=\boldsymbol{x}_{k}(2)=1-a^{k}, \quad k \geq 0,$
lyche-numerical-linear-algebra_ EQ0282	118	■ 0.842 ● W|+0	`\boldsymbol{z}:=\boldsymbol{x}-a \boldsymbol{y}, \quad a:=\frac{\langle\boldsymbol{x}, \boldsymbol{y}\rangle}{ \langle\boldsymbol{y}, \boldsymbol{y}\rangle} .`	$\boldsymbol{z}:=\boldsymbol{x}-a \boldsymbol{y}, \quad a:=\frac{\langle\boldsymbol{x}, \boldsymbol{y}\rangle}{\langle\boldsymbol{y}, \boldsymbol{y}\rangle}.$	$\boldsymbol{z}:=\boldsymbol{x}-a \boldsymbol{y}, \quad a:=\frac{\langle\boldsymbol{x}, \boldsymbol{y}\rangle}{\langle\boldsymbol{y}, \boldsymbol{y}\rangle}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0624</td><td>228</td><td>■ 0.842 ● W|+0</td><td>\begin{aligned} \alpha_{j} & \leq \max _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|(\boldsymbol{B}+(\boldsymbol{A}-\boldsymbol{B})) \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{B} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}+\max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|(\boldsymbol{A}-\boldsymbol{B}) \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} \\ & \leq \beta_{j}+\|\boldsymbol{A}-\boldsymbol{B}\|_{2} . \end{aligned}</td><td>$\begin{aligned} \alpha_{j} & \leq \max _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|(\boldsymbol{B}+(\boldsymbol{A}-\boldsymbol{B})) \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{B} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}+\max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|(\boldsymbol{A}-\boldsymbol{B}) \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} \\ & \leq \beta_{j}+\|\boldsymbol{A}-\boldsymbol{B}\|_{2} . \end{aligned}$</td><td>$\begin{aligned} \alpha_{j} & \leq \max _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|(\boldsymbol{B}+(\boldsymbol{A}-\boldsymbol{B})) \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{B} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}+\max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|(\boldsymbol{A}-\boldsymbol{B}) \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} \\ & \leq \beta_{j}+\|\boldsymbol{A}-\boldsymbol{B}\|_{2} . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0049</td><td>038</td><td>■ 0.844 ● W|+1</td><td>\begin{aligned} \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) & =\left(a_{11}-\lambda\right) \left|\begin{array}{cc} a_{22}-\lambda & a_{23} \\ a_{32} & a_{33}-\lambda \end{array}\right|-a_{21} \left|\begin{array}{cc} a_{12} & a_{13} \\ a_{32} & a_{33}-\lambda \end{array}\right| \\ & +a_{31}\left| \begin{array}{cc} a_{12} & a_{13} \\ a_{22}-\lambda & a_{23} \end{array}\right|=\left(a_{11}-\lambda\right) \left(a_{22}-\lambda\right)\left(a_{33}-\lambda\right)+r(\lambda) \end{aligned}</td><td>$\begin{aligned} \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) & =\left(a_{11}-\lambda\right) \left|\begin{array}{cc} a_{22}-\lambda & a_{23} \\ a_{32} & a_{33}-\lambda \end{array}\right|-a_{21} \left|\begin{array}{cc} a_{12} & a_{13} \\ a_{32} & a_{33}-\lambda \end{array}\right| \\ & +a_{31}\left| \begin{array}{cc} a_{12} & a_{13} \\ a_{22}-\lambda & a_{23} \end{array}\right|=\left(a_{11}-\lambda\right) \left(a_{22}-\lambda\right)\left(a_{33}-\lambda\right)+r(\lambda) \end{aligned}$</td><td>$\begin{aligned} \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) & =(a_{11}-\lambda) \left|\begin{array}{cc} a_{22}-\lambda & a_{23} \\ a_{32} & a_{33}-\lambda \end{array}\right|-a_{21} \left|\begin{array}{cc} a_{12} & a_{13} \\ a_{32} & a_{33}-\lambda \end{array}\right| \\ & +a_{31}\left| \begin{array}{cc} a_{12} & a_{13} \\ a_{22}-\lambda & a_{23} \end{array}\right|=(a_{11}-\lambda)(a_{22}-\lambda)(a_{33}-\lambda)+r(\lambda) \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0851 (13.14)</td><td>295</td><td>■ 0.846 ● W|+0</td><td>\beta_{k}:=-\frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \stackrel{\text { (13.13) }}{=} \frac{\boldsymbol{r}_{k+1}^{T}\left(\boldsymbol{r}_{k+1}-\boldsymbol{r}_{k}\right)}{ \alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \stackrel{\text { (13.12) }}{=} \frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} .</td><td>$\beta_{k}:=-\frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \stackrel{\text { (13.13) }}{=} \frac{\boldsymbol{r}_{k+1}^{T}\left(\boldsymbol{r}_{k+1}-\boldsymbol{r}_{k}\right)}{ \alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \stackrel{\text { (13.12) }}{=} \frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} .$</td><td>$\beta_{k}:=-\frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \stackrel{(13.13)}{=} \frac{\boldsymbol{r}_{k+1}^{T}(\boldsymbol{r}_{k+1}-\boldsymbol{r}_{k})}{ \alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \stackrel{(13.12)}{=} \frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0387</td><td>156</td><td>■ 0.846 ● N|+1</td><td>\pi_{D_{1}}(\lambda)=\pi_{D_{3}}(\lambda)=\lambda^{2}-4 \lambda+5, \quad \pi_{D_{2}}(\lambda)=\lambda-1,</td><td>$\pi_{D_{1}}(\lambda)=\pi_{D_{3}}(\lambda)=\lambda^{2}-4 \lambda+5, \quad \pi_{D_{2}}(\lambda)=\lambda-1,$</td><td>$\pi_{\boldsymbol{D}_{1}}(\lambda)=\pi_{\boldsymbol{D}_{3}}(\lambda)=\lambda^{2}-4 \lambda+5, \quad \pi_{\boldsymbol{D}_{2}}(\lambda)=\lambda-1,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0650</td><td>234</td><td>■ 0.847 ● N|+1</td><td>\left[\begin{array}{cc} \Sigma_{1} & 0 \\ 0 & 0 \end{array}\right] y=c .</td><td>$\left[\begin{array}{cc} \Sigma_{1} & 0 \\ 0 & 0 \end{array}\right] y=c .$</td><td>$\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right] \boldsymbol{y}=\boldsymbol{c} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0034</td><td>033</td><td>■ 0.848 ● K|+0</td><td>C=C I=C(A B)=(C A) B=I B=B</td><td>$C=C I=C(A B)=(C A) B=I B=B$</td><td>$\boldsymbol{C}=\boldsymbol{C I}=\boldsymbol{C}(\boldsymbol{A B})=(\boldsymbol{C A}) \boldsymbol{B}=\boldsymbol{I B}=\boldsymbol{B}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0084 (2.6)</td><td>050</td><td>■ 0.850 ● K|+0</td><td>p_{i}\left(x_{i}\right)=y_{i}, \quad p_{i}\left(x_{i+1} \right)=y_{i+1}, \quad p_{i}^{\prime \prime}\left(x_{i} \right)=\mu_{i}, \quad p_{i}^{\prime \prime}\left(x_{i+1} \right)=\mu_{i+1} .</td><td>$p_{i}\left(x_{i}\right)=y_{i}, \quad p_{i}\left(x_{i+1} \right)=y_{i+1}, \quad p_{i}^{\prime \prime}\left(x_{i} \right)=\mu_{i}, \quad p_{i}^{\prime \prime}\left(x_{i+1} \right)=\mu_{i+1} .$</td><td>$p_{i}(x_{i})=y_{i},\ p_{i}(x_{i+1})=y_{i+1}, \quad p_{i}^{\prime \prime}(x_{i})=\mu_{i},\ p_{i}^{\prime \prime}(x_{i+1})=\mu_{i+1} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0409</td><td>163</td><td>■ 0.850 ● K|+0</td><td>\boldsymbol{A}=\left[\begin{array}{cc} a & 1 \\ 0 & a \end{array}\right]</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} a & 1 \\ 0 & a \end{array}\right]$</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} a & 1 \\ 0 & a \end{array}\right]$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0144</td><td>065</td><td>■ 0.853
● K|+0</td><td><code>\boldsymbol{B} \boldsymbol{A}=\left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array} \right]\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \end{array} \right]=\boldsymbol{I}</code></td><td>$\boldsymbol{B} \boldsymbol{A}=\left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array} \right]\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \end{array} \right]=\boldsymbol{I}$</td><td>$\boldsymbol{B} \boldsymbol{A}=\begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{bmatrix}\begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}=\begin{bmatrix} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \end{bmatrix}=\boldsymbol{I}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0264 (4.8)</td><td>109</td><td>■ 0.854
● N|+0</td><td><code>\boldsymbol{A}=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right], \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times(n-1)} .</code></td><td>$\boldsymbol{A}=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right], \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times(n-1)} .$</td><td>$\boldsymbol{A}=\begin{bmatrix} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix}, \quad \alpha \in \mathbb{C}, \ \boldsymbol{v} \in \mathbb{C}^{n-1}, \ \boldsymbol{B} \in \mathbb{C}^{(n-1) \times(n-1)} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0392</td><td>157</td><td>■ 0.857
● N|+0</td><td><code>\max _{\substack{x \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y})= \sum_{j=1}^{n} \lambda_{j}\left|c_{j}\right|^{2}= \sum_{j=1}^{k} \lambda_{j}\left|c_{j}\right|^{2} \geq \sum_{j=1}^{k} \lambda_{k}\left|c_{j}\right|^{2}= \lambda_{k},</code></td><td>$\max _{\substack{x \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y})= \sum_{j=1}^{n} \lambda_{j}\left|c_{j}\right|^{2}= \sum_{j=1}^{k} \lambda_{j}\left|c_{j}\right|^{2} \geq \sum_{j=1}^{k} \lambda_{k}\left|c_{j}\right|^{2}= \lambda_{k},$</td><td>$\max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y})= \sum_{j=1}^{n} \lambda_{j}|c_{j}|^{2}= \sum_{j=1}^{k} \lambda_{j}|c_{j}|^{2} \geq \sum_{j=1}^{k} \lambda_{k}|c_{j}|^{2}= \lambda_{k},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0032</td><td>032</td><td>■ 0.859
● N|+0</td><td><code>\boldsymbol{B} z=\left[\begin{array}{ll} \boldsymbol{A} & \boldsymbol{b} \end{array}\right]\left[\begin{array}{c} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array}\right]= \boldsymbol{A} \tilde{\boldsymbol{z}}+z_{n+1} \boldsymbol{b}=\mathbf{0} .</code></td><td>$\boldsymbol{B} z=\left[\begin{array}{ll} \boldsymbol{A} & \boldsymbol{b} \end{array}\right]\left[\begin{array}{c} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array}\right]= \boldsymbol{A} \tilde{\boldsymbol{z}}+z_{n+1} \boldsymbol{b}=\mathbf{0} .$</td><td>$\boldsymbol{B} \boldsymbol{z}=[\boldsymbol{A}\ \boldsymbol{b}]\begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}= \boldsymbol{A} \tilde{\boldsymbol{z}}+z_{n+1} \boldsymbol{b}=\mathbf{0} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0891 (13.36)</td><td>307</td><td>■ 0.859
● W|+0</td><td><code>1 \leq \frac{\left(\boldsymbol{y}^{T} \boldsymbol{A y} \right)\left(\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}\right)}{\left(\boldsymbol{y}^{T} \boldsymbol{y}\right)^{2}} \leq \frac{(M+m)^{2}}{4 M m} \quad \boldsymbol{y} \neq \mathbf{0}, \boldsymbol{y} \in \mathbb{R}^{n},</code></td><td>$1 \leq \frac{\left(\boldsymbol{y}^{T} \boldsymbol{A y} \right)\left(\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y}\right)}{\left(\boldsymbol{y}^{T} \boldsymbol{y}\right)^{2}} \leq \frac{(M+m)^{2}}{4 M m} \quad \boldsymbol{y} \neq \mathbf{0}, \boldsymbol{y} \in \mathbb{R}^{n},$</td><td>$1 \leq \frac{(\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y})(\boldsymbol{y}^{T} \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^{T} \boldsymbol{y})^{2}} \leq \frac{(M+m)^{2}}{4 M m} \quad \boldsymbol{y} \neq \mathbf{0},\ \boldsymbol{y} \in \mathbb{R}^{n},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0033</td><td>033</td><td>■ 0.860
● S|+0</td><td><code>\boldsymbol{A} \boldsymbol{x}=-\boldsymbol{A}\left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}}\right)=- \frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}}=- \frac{1}{z_{n+1}}\left(-z_{n+1} \boldsymbol{b}\right)= \boldsymbol{b},</code></td><td>$\boldsymbol{A} \boldsymbol{x}=-\boldsymbol{A}\left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}}\right)=- \frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}}=- \frac{1}{z_{n+1}}\left(-z_{n+1} \boldsymbol{b}\right)= \boldsymbol{b},$</td><td>$\boldsymbol{A} \boldsymbol{x}=-\boldsymbol{A}\left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}}\right)=- \frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}}=- \frac{1}{z_{n+1}}\left(-z_{n+1} \boldsymbol{b}\right)= \boldsymbol{b},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0451</td><td>180</td><td>■ 0.869
● W|+0</td><td><code>\boldsymbol{p}_{i}=\left[\begin{array}{c} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{array} \right], \quad \boldsymbol{q}_{i}=\left[\begin{array}{c} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{array} \right], \quad \boldsymbol{r}_{j}=\left[\begin{array}{c} \boldsymbol{u}_{j} \\ \mathbf{0} \end{array}\right], \text { for } i=1, \ldots, n, j=n+1, \ldots, m .</code></td><td>$\boldsymbol{p}_{i}=\left[\begin{array}{c} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{array} \right], \quad \boldsymbol{q}_{i}=\left[\begin{array}{c} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{array} \right], \quad \boldsymbol{r}_{j}=\left[\begin{array}{c} \boldsymbol{u}_{j} \\ \mathbf{0} \end{array}\right], \text { for } i=1, \ldots, n, j=n+1, \ldots, m .$</td><td>$\boldsymbol{p}_{i}=\begin{bmatrix} \boldsymbol{u}_{i} \\ \boldsymbol{v}_{i} \end{bmatrix}, \quad \boldsymbol{q}_{i}=\begin{bmatrix} \boldsymbol{u}_{i} \\ -\boldsymbol{v}_{i} \end{bmatrix}, \quad \boldsymbol{r}_{j}=\begin{bmatrix} \boldsymbol{u}_{j} \\ \mathbf{0} \end{bmatrix}, \text { for } i=1, \ldots, n,\ j=n+1, \ldots, m .$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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EQ0835 (13.1)</td><td>290</td><td>■ 0.869
● W|+0</td><td><code>\langle\boldsymbol{x}, \boldsymbol{y}\rangle_{\boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}, \quad\|\boldsymbol{y}\|_{\boldsymbol{A}}:=\sqrt{\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n} .</code></td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle_{\boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}, \quad\|\boldsymbol{y}\|_{\boldsymbol{A}}:=\sqrt{\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n} .$</td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle_{\boldsymbol{A}}:=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}, \quad\|\boldsymbol{y}\|_{\boldsymbol{A}}:=\sqrt{\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0910 (13.46)</td><td>310</td><td>■ 0.869
● W|+1</td><td><code>\begin{aligned} & \left\|\boldsymbol{\epsilon}_{k}\right\|_{2}^{2}-\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{2}^{2}=\alpha_{k}\left(2 \boldsymbol{p}_{k}^{T} \boldsymbol{\epsilon}_{k+1}+\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\right) \\ & \quad \stackrel{\text { (13.41) }}{=} \alpha_{k}\left(2 \sum_{i=k+1}^{m} \alpha_{i} \boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}+\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\right)=\left(\sum_{i=k}^{m}+ \sum_{i=k+1}^{m}\right) \alpha_{k} \alpha_{i} \boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k} \\ & \quad \stackrel{\text { (13.43) }}{=}\left(\sum_{i=k}^{m}+ \sum_{i=k+1}^{m}\right) \frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}} \frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k} \\ & \quad \stackrel{\text { (13.42) }}{=} \frac{\left\|\boldsymbol{p}_{k}\right\|_{2}^{2}}{\left\|\boldsymbol{p}_{k}\right\|_{\boldsymbol{A}}^{2}} \left(\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}}^{2}+\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{\boldsymbol{A}}^{2}\right) . \end{aligned}</code></td><td>$\begin{aligned} & \left\|\boldsymbol{\epsilon}_{k}\right\|_{2}^{2}-\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{2}^{2}=\alpha_{k}\left(2 \boldsymbol{p}_{k}^{T} \boldsymbol{\epsilon}_{k+1}+\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\right) \\ & \quad \stackrel{\text { (13.41) }}{=} \alpha_{k}\left(2 \sum_{i=k+1}^{m} \alpha_{i} \boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}+\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\right)=\left(\sum_{i=k}^{m}+ \sum_{i=k+1}^{m}\right) \alpha_{k} \alpha_{i} \boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k} \\ & \quad \stackrel{\text { (13.43) }}{=}\left(\sum_{i=k}^{m}+ \sum_{i=k+1}^{m}\right) \frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}} \frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k} \\ & \quad \stackrel{\text { (13.42) }}{=} \frac{\left\|\boldsymbol{p}_{k}\right\|_{2}^{2}}{\left\|\boldsymbol{p}_{k}\right\|_{\boldsymbol{A}}^{2}} \left(\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}}^{2}+\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{\boldsymbol{A}}^{2}\right) . \end{aligned}$</td><td>$\begin{aligned} \|\boldsymbol{\epsilon}_{k}\|_{2}^{2}-\|\boldsymbol{\epsilon}_{k+1}\|_{2}^{2} &=\alpha_{k}\big(2 \boldsymbol{p}_{k}^{T} \boldsymbol{\epsilon}_{k+1}+\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\big) \\ & \stackrel{(13.41)}{=} \alpha_{k}\big(2 \sum_{i=k+1}^{m} \alpha_{i} \boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}+\alpha_{k} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\big)=\big(\sum_{i=k}^{m}+ \sum_{i=k+1}^{m}\big) \alpha_{k} \alpha_{i} \boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k} \\ & \stackrel{(13.43)}{=}\big(\sum_{i=k}^{m}+ \sum_{i=k+1}^{m}\big) \frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} \frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}} \frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k} \\ & \stackrel{(13.42)}{=} \frac{\|\boldsymbol{p}_{k}\|_{2}^{2}}{\|\boldsymbol{p}_{k}\|_{\boldsymbol{A}}^{2}} \big(\|\boldsymbol{\epsilon}_{k}\|_{\boldsymbol{A}}^{2}+\|\boldsymbol{\epsilon}_{k+1}\|_{\boldsymbol{A}}^{2}\big) . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ1014 (15.1)</td><td>344</td><td>■ 0.870
● K|+0</td><td><code>\boldsymbol{z}_{k}:=\boldsymbol{A}^{k} \boldsymbol{z}_{0}=\boldsymbol{A} \boldsymbol{z}_{k-1}, \quad k=1,2, \ldots .</code></td><td>$\boldsymbol{z}_{k}:=\boldsymbol{A}^{k} \boldsymbol{z}_{0}=\boldsymbol{A} \boldsymbol{z}_{k-1}, \quad k=1,2, \ldots .$</td><td>$\boldsymbol{z}_{k}:=\boldsymbol{A}^{k} \boldsymbol{z}_{0}=\boldsymbol{A} \boldsymbol{z}_{k-1}, \quad k=1,2, \ldots .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0591</td><td>218</td><td>■ 0.877
● N|+0</td><td><code>E(\boldsymbol{x}):=\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\|_{2}^{2}=(\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b})^{*}(\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b})=\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}-2 \boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{b}+\boldsymbol{b}^{*} \boldsymbol{b} .</code></td><td>$E(\boldsymbol{x}):=\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\|_{2}^{2}=(\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b})^{*}(\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b})=\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}-2 \boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{b}+\boldsymbol{b}^{*} \boldsymbol{b} .$</td><td>$E(\boldsymbol{x}):=\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\|_{2}^{2}=(\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b})^{*}(\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b})=\boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}-2 \boldsymbol{x}^{*} \boldsymbol{A}^{*} \boldsymbol{b}+\boldsymbol{b}^{*} \boldsymbol{b} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0181</td><td>078</td><td>■ 0.880
● N|+0</td><td><code>[1],[5],[9],\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array}\right],\left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array}\right],\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right], \boldsymbol{A} .</code></td><td>$[1],[5],[9],\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array}\right],\left[\begin{array}{ll} 1 & 3 \\ 7 & 9 \end{array}\right],\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right], \boldsymbol{A} .$</td><td>$[1],\ [5],\ [9],\ \left[\begin{smallmatrix} 1 & 2 \\ 4 & 5 \end{smallmatrix}\right],\ \left[\begin{smallmatrix} 1 & 3 \\ 7 & 9 \end{smallmatrix}\right],\ \left[\begin{smallmatrix} 5 & 6 \\ 8 & 9 \end{smallmatrix}\right],\ \boldsymbol{A} .$</td></tr>
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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0367 (6.5)</td><td>150</td><td>■0.884 ● N|+0</td><td><code>\boldsymbol{J}:=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\operatorname{diag}\left(\boldsymbol{U}_{1}, \ldots, \boldsymbol{U}_{k}\right), \text { with } \boldsymbol{U}_{i} \in \mathbb{C}^{a_{i} \times a_{i}},</code></td><td>$\boldsymbol{J}:=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\operatorname{diag}\left(\boldsymbol{U}_{1}, \ldots, \boldsymbol{U}_{k}\right), \text { with } \boldsymbol{U}_{i} \in \mathbb{C}^{a_{i} \times a_{i}},$</td><td>$\boldsymbol{J}:=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\operatorname{diag}(\boldsymbol{U}_{1}, \ldots, \boldsymbol{U}_{k}),\ \textit{with } \boldsymbol{U}_{i} \in \mathbb{C}^{a_{i} \times a_{i}},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0594 (9.5)</td><td>219</td><td>■0.885 ● N|+0</td><td><code>\left\{\boldsymbol{x} \in \mathbb{C}^{n}: \boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}_{1}\right\} .</code></td><td>$\left\{\boldsymbol{x} \in \mathbb{C}^{n}: \boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}_{1}\right\} .$</td><td>$\{\boldsymbol{x} \in \mathbb{C}^{n}: \boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}_{1}\}.$</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0977</td><td>330</td><td>■0.892 ● N|+0</td><td><code>\begin{gathered} \left\|(\mu \boldsymbol{I}- \boldsymbol{A}) \boldsymbol{u}_{1}\right\|_{2}=\left \|(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{u}_{1}- \boldsymbol{A} \boldsymbol{u}_{1}\right\|_{2}=\left \|\boldsymbol{E} \boldsymbol{u}_{1}\right\|_{2} \leq \|\boldsymbol{E}\|_{2}, \\ \prod_{j=2}^{n}\left\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{j} \right\|_{2} \leq \prod_{j=2}^{n}\left(|\mu|+\left\| \boldsymbol{A} \boldsymbol{u}_{j}\right\|_{2}\right) \leq\left(\|(\boldsymbol{A}+\boldsymbol{E})\|_{2}+\| \boldsymbol{A}\|_{2}\right)^{n-1} . \end{gathered}</code></td><td>$\begin{gathered} \left\|(\mu \boldsymbol{I}- \boldsymbol{A}) \boldsymbol{u}_{1}\right\|_{2}=\left \|(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{u}_{1}- \boldsymbol{A} \boldsymbol{u}_{1}\right\|_{2}=\left \|\boldsymbol{E} \boldsymbol{u}_{1}\right\|_{2} \leq \|\boldsymbol{E}\|_{2}, \\ \prod_{j=2}^{n}\left\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{j} \right\|_{2} \leq \prod_{j=2}^{n}\left(|\mu|+\left\| \boldsymbol{A} \boldsymbol{u}_{j}\right\|_{2}\right) \leq\left(\|(\boldsymbol{A}+\boldsymbol{E})\|_{2}+\| \boldsymbol{A}\|_{2}\right)^{n-1} . \end{gathered}$</td><td>$\begin{gathered} \|(\mu \boldsymbol{I}- \boldsymbol{A}) \boldsymbol{u}_{1}\|_{2}=\|(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{u}_{1}- \boldsymbol{A} \boldsymbol{u}_{1}\|_{2}=\|\boldsymbol{E} \boldsymbol{u}_{1}\|_{2} \leq \|\boldsymbol{E}\|_{2}, \\ \prod_{j=2}^{n}\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{j}\|_{2} \leq \prod_{j=2}^{n}(|\mu|+\| \boldsymbol{A} \boldsymbol{u}_{j}\|_{2}) \leq\left(\|(\boldsymbol{A}+\boldsymbol{E})\|_{2}+\| \boldsymbol{A}\|_{2}\right)^{n-1}. \end{gathered}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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EQ0147</td><td>066</td><td>■ 0.896
● N|+0</td><td><code>\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \mathbf{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{ll} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \end{array} \right]=\boldsymbol{I}</code></td><td>$$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \mathbf{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{ll} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \end{array} \right]=\boldsymbol{I}$$</td><td>$$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \mathbf{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{ll} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} \end{array} \right]=\boldsymbol{I}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
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● N|+0</td><td><code>\rho\left(\boldsymbol{G}_{J}\right)=\sqrt{\left|a_{21} a_{12} / a_{11} a_{22}\right|} .</code></td><td>$$\rho\left(\boldsymbol{G}_{J}\right)=\sqrt{\left|a_{21} a_{12} / a_{11} a_{22}\right|} .$$</td><td>$$\rho(\boldsymbol{G}_{J})=\sqrt{|a_{21} a_{12} / a_{11} a_{22}|}.$$</td></tr>
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EQ0989</td><td>333</td><td>■ 0.899
● N|+0</td><td><code>\boldsymbol{A}_{k}=\left[\begin{array}{ll} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{array} \right] .</code></td><td>$$\boldsymbol{A}_{k}=\left[\begin{array}{ll} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{array} \right] .$$</td><td>$$\boldsymbol{A}_{k}=\left[\begin{array}{ll} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{array} \right].$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0024 (1.9)</td><td>029</td><td>■ 0.902
● N|+0</td><td><code>\boldsymbol{s}_{j}=\sum_{i=1}^{m} a_{i j} \boldsymbol{v}_{i} \text { for } j=1, \ldots, n .</code></td><td>$$\boldsymbol{s}_{j}=\sum_{i=1}^{m} a_{i j} \boldsymbol{v}_{i} \text { for } j=1, \ldots, n .$$</td><td>$$\boldsymbol{s}_{j}=\sum_{i=1}^{m} a_{i j} \boldsymbol{v}_{i}\, for\ j=1, \ldots, n.$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0614</td><td>225</td><td>■ 0.902
● K|+0</td><td><code>\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right], \quad \boldsymbol{e}=\left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right] .</code></td><td>$$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right], \quad \boldsymbol{e}=\left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right] .$$</td><td>$$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right], \quad \boldsymbol{e}=\left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right].$$</td></tr>
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EQ0168</td><td>074</td><td>■ 0.904
● N|+0</td><td><code>\boldsymbol{A}:=\left[\begin{array}{ll} \lambda & \boldsymbol{a}^{T} \\ \mathbf{0} & \boldsymbol{A}_{1} \end{array}\right], \quad \boldsymbol{B}:=\left[\begin{array}{ll} 1 & \mathbf{0}^{T} \\ \mathbf{0} & \boldsymbol{B}_{1} \end{array}\right], \quad \boldsymbol{C}:=\left[\begin{array}{ll} 1 & \mathbf{0}^{T} \\ \mathbf{0} & \boldsymbol{C}_{1} \end{array}\right],</code></td><td>$$\boldsymbol{A}:=\left[\begin{array}{ll} \lambda & \boldsymbol{a}^{T} \\ \mathbf{0} & \boldsymbol{A}_{1} \end{array}\right], \quad \boldsymbol{B}:=\left[\begin{array}{ll} 1 & \mathbf{0}^{T} \\ \mathbf{0} & \boldsymbol{B}_{1} \end{array}\right], \quad \boldsymbol{C}:=\left[\begin{array}{ll} 1 & \mathbf{0}^{T} \\ \mathbf{0} & \boldsymbol{C}_{1} \end{array}\right],$$</td><td>$$\boldsymbol{A}:=\left[\begin{array}{ll} \lambda & \boldsymbol{a}^{T} \\ \mathbf{0} & \boldsymbol{A}_{1} \end{array}\right], \quad \boldsymbol{B}:=\left[\begin{array}{ll} 1 & \mathbf{0}^{T} \\ \mathbf{0} & \boldsymbol{B}_{1} \end{array}\right], \quad \boldsymbol{C}:=\left[\begin{array}{ll} 1 & \mathbf{0}^{T} \\ \mathbf{0} & \boldsymbol{C}_{1} \end{array}\right],$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0925 (13.55)</td><td>314</td><td>■ 0.907
● W|-1</td><td><code>\begin{aligned} & \left(d_{1} v\right)_{j, k}:=c_{j-\frac{1}{2}, k}\left(v_{j, k}-v_{j-1, k}\right)-c_{j+\frac{1}{2}, k}\left(v_{j+1, k}-v_{j, k}\right) \approx-h^{2} \frac{\partial}{\partial x}\left(c \frac{\partial u}{\partial x}\right)_{j, k}, \\ & \left(d_{2} v \right)_{j, k}:=c_{j, k-\frac{1}{2}}\left(v_{j, k}-v_{j, k-1}\right)-c_{j, k+\frac{1}{2}}\left(v_{j, k+1}-v_{j, k}\right) \approx-h^{2} \frac{\partial}{\partial y} \left(c \frac{\partial u}{\partial y}\right)_{j, k}, \end{aligned}</code></td><td>$$\begin{aligned} & \left(d_{1} v\right)_{j, k}:=c_{j-\frac{1}{2}, k}\left(v_{j, k}-v_{j-1, k}\right)-c_{j+\frac{1}{2}, k}\left(v_{j+1, k}-v_{j, k}\right) \approx-h^{2} \frac{\partial}{\partial x}\left(c \frac{\partial u}{\partial x}\right)_{j, k}, \\ & \left(d_{2} v \right)_{j, k}:=c_{j, k-\frac{1}{2}}\left(v_{j, k}-v_{j, k-1}\right)-c_{j, k+\frac{1}{2}}\left(v_{j, k+1}-v_{j, k}\right) \approx-h^{2} \frac{\partial}{\partial y} \left(c \frac{\partial u}{\partial y}\right)_{j, k}, \end{aligned}$$</td><td>$$\begin{aligned} & (d_{1} v)_{j, k}:=c_{j-\frac{1}{2}, k}(v_{j, k}-v_{j-1, k})-c_{j+\frac{1}{2}, k}(v_{j+1, k}-v_{j, k}) \approx-h^{2} \frac{\partial}{\partial x}(c \frac{\partial u}{\partial x})_{j, k}, \\ & (d_{2} v)_{j, k}:=c_{j, k-\frac{1}{2}}(v_{j, k}-v_{j, k-1})-c_{j, k+\frac{1}{2}}(v_{j, k+1}-v_{j, k}) \approx-h^{2} \frac{\partial}{\partial y}(c \frac{\partial u}{\partial y})_{j, k}, \end{aligned}$$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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● W|+0</td><td>\boldsymbol{F}_{4}=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & \omega_{4} & \omega_{4}^{2} & \omega_{4}^{3} \\ 1 & \omega_{4}^{2} & \omega_{4}^{4} & \omega_{4}^{6} \\ 1 & \omega_{4}^{3} & \omega_{4}^{6} & \omega_{4}^{9} \end{array}\right]=\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{array}\right] .</td><td>$\boldsymbol{F}_{4}=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & \omega_{4} & \omega_{4}^{2} & \omega_{4}^{3} \\ 1 & \omega_{4}^{2} & \omega_{4}^{4} & \omega_{4}^{6} \\ 1 & \omega_{4}^{3} & \omega_{4}^{6} & \omega_{4}^{9} \end{array}\right]=\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{array}\right] .$</td><td>$\boldsymbol{F}_{4}=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & \omega_{4} & \omega_{4}^{2} & \omega_{4}^{3} \\ 1 & \omega_{4}^{2} & \omega_{4}^{4} & \omega_{4}^{6} \\ 1 & \omega_{4}^{3} & \omega_{4}^{6} & \omega_{4}^{9} \end{array}\right]=\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0488 (8.18)</td><td>191</td><td>■0.909
● C|-3</td><td>\begin{aligned} & \|\boldsymbol{A}\|_{1}:=\max _{1 \leq j \leq n}\left\|\boldsymbol{A} \boldsymbol{e}_{j}\right\|_{1}=\max _{1 \leq j \leq n} \sum_{k=1}^{m}\left|a_{k, j}\right|, \quad \text { (max column sum) } \\ & \|\boldsymbol{A}\|_{2}:=\sigma_{1}, \quad \text { (largest singular value of } \boldsymbol{A} \text {) } \\ & \|\boldsymbol{A}\|_{\infty}=\max _{1 \leq k \leq m}\left\|\boldsymbol{e}_{k}^{T} \boldsymbol{A}\right\|_{1}=\max _{1 \leq k \leq m} \sum_{j=1}^{n}\left|a_{k, j}\right| . \quad \text { (max row sum) } \end{aligned}</td><td>$\begin{aligned} & \|\boldsymbol{A}\|_{1}:=\max _{1 \leq j \leq n}\left\|\boldsymbol{A} \boldsymbol{e}_{j}\right\|_{1}=\max _{1 \leq j \leq n} \sum_{k=1}^{m}\left|a_{k, j}\right|, \quad \text { (max column sum) } \\ & \|\boldsymbol{A}\|_{2}:=\sigma_{1}, \quad \text { (largest singular value of } \boldsymbol{A} \text {) } \\ & \|\boldsymbol{A}\|_{\infty}=\max _{1 \leq k \leq m}\left\|\boldsymbol{e}_{k}^{T} \boldsymbol{A}\right\|_{1}=\max _{1 \leq k \leq m} \sum_{j=1}^{n}\left|a_{k, j}\right| . \quad \text { (max row sum) } \end{aligned}$</td><td>$\begin{aligned} & \|\boldsymbol{A}\|_{1}:=\max _{1 \leq j \leq n}\|\boldsymbol{A} \boldsymbol{e}_{j}\|_{1}=\max _{1 \leq j \leq n} \sum_{k=1}^{m}|a_{k, j}|, \quad \textit{(max column sum)} \\ & \|\boldsymbol{A}\|_{2}:=\sigma_{1}, \quad \textit{(largest singular value of } \boldsymbol{A} \textit{)} \\ & \|\boldsymbol{A}\|_{\infty}=\max _{1 \leq k \leq m}\|\boldsymbol{e}_{k}^{T} \boldsymbol{A}\|_{1}=\max _{1 \leq k \leq m} \sum_{j=1}^{n}|a_{k, j}| . \quad \textit{(max row sum)} \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0303</td><td>124</td><td>■0.910
● N|+0</td><td>\boldsymbol{H}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]-\left[\begin{array}{l} u_{1} \\ u_{2} \end{array}\right]\left[\begin{array}{ll} u_{1} & u_{2} \end{array}\right]=\left[\begin{array}{cc} 1-u_{1}^{2} & -u_{1} u_{2} \\ -u_{2} u_{1} & 1-u_{2}^{2} \end{array} \right] .</td><td>$\boldsymbol{H}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]-\left[\begin{array}{l} u_{1} \\ u_{2} \end{array}\right]\left[\begin{array}{ll} u_{1} & u_{2} \end{array}\right]=\left[\begin{array}{cc} 1-u_{1}^{2} & -u_{1} u_{2} \\ -u_{2} u_{1} & 1-u_{2}^{2} \end{array} \right] .$</td><td>$\boldsymbol{H}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]-\left[\begin{array}{l} u_{1} \\ u_{2} \end{array}\right]\left[\begin{array}{ll} u_{1} & u_{2} \end{array}\right]=\left[\begin{array}{cc} 1-u_{1}^{2} & -u_{1} u_{2} \\ -u_{2} u_{1} & 1-u_{2}^{2} \end{array} \right] .$</td></tr>
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● W|+0</td><td>\begin{aligned} z^{*} A z & =(x-i y)^{T} A(x+i y)=x^{T} A x-i y^{T} A x+i x^{T} A y-i^{2} y^{T} A y \\ & =x^{T} A x+y^{T} A y, \end{aligned}</td><td>$\begin{aligned} z^{*} A z & =(x-i y)^{T} A(x+i y)=x^{T} A x-i y^{T} A x+i x^{T} A y-i^{2} y^{T} A y \\ & =x^{T} A x+y^{T} A y, \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{z}^{*} \boldsymbol{A} \boldsymbol{z} & =(\boldsymbol{x}-i \boldsymbol{y})^{T} \boldsymbol{A}(\boldsymbol{x}+i \boldsymbol{y})=\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}-i \boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{x}+i \boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{y}-i^{2} \boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y} \\ & =\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}+\boldsymbol{y}^{T} \boldsymbol{A} \boldsymbol{y}, \end{aligned}$</td></tr>
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● W|+0</td><td>\begin{array}{ll} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{21}^{(1)} x_{1}+a_{22}^{(1)} x_{2}+a_{23}^{(1)} x_{3}=b_{2}^{(1)}, & \mathrm{II} \\ a_{31}^{(1)} x_{1}+a_{32}^{(1)} x_{2}+a_{33}^{(1)} x_{3}=b_{3}^{(1)} . & \mathrm{III} . \end{array}</td><td>$\begin{array}{ll} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{21}^{(1)} x_{1}+a_{22}^{(1)} x_{2}+a_{23}^{(1)} x_{3}=b_{2}^{(1)}, & \mathrm{II} \\ a_{31}^{(1)} x_{1}+a_{32}^{(1)} x_{2}+a_{33}^{(1)} x_{3}=b_{3}^{(1)} . & \mathrm{III} . \end{array}$</td><td>$\begin{array}{ll} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{21}^{(1)} x_{1}+a_{22}^{(1)} x_{2}+a_{23}^{(1)} x_{3}=b_{2}^{(1)}, & \mathrm{II} \\ a_{31}^{(1)} x_{1}+a_{32}^{(1)} x_{2}+a_{33}^{(1)} x_{3}=b_{3}^{(1)} . & \mathrm{III} . \end{array}$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0494</td><td>194</td><td>■0.915
● K|+0</td><td>\left\|\boldsymbol{A}^{*}\right\|_{F}=\|\boldsymbol{A}\|_{F} \text { and }\left\|\boldsymbol{A}^{*}\right\|_{2}=\|\boldsymbol{A}\|_{2} .</td><td>$\left\|\boldsymbol{A}^{*}\right\|_{F}=\|\boldsymbol{A}\|_{F} \text { and }\left\|\boldsymbol{A}^{*}\right\|_{2}=\|\boldsymbol{A}\|_{2} .$</td><td>$\|\boldsymbol{A}^{*}\|_{F}=\|\boldsymbol{A}\|_{F} \textit{ and } \|\boldsymbol{A}^{*}\|_{2}=\|\boldsymbol{A}\|_{2} .$</td></tr>
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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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● C|+32</td><td>\begin{aligned} & \|\boldsymbol{A} \boldsymbol{B}\|=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{B} \boldsymbol{x}\|}\frac{\|\boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|} \\ & \quad \leq \max _{\boldsymbol{y} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{y}\|}{\|\boldsymbol{y}\|} \max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\|\boldsymbol{A}\|\|\boldsymbol{B}\| \end{aligned}</td><td>$\begin{aligned} & \|\boldsymbol{A} \boldsymbol{B}\|=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{B} \boldsymbol{x}\|}\frac{\|\boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|} \\ & \quad \leq \max _{\boldsymbol{y} \neq \mathbf{0}} \frac{\|\boldsymbol{A} \boldsymbol{y}\|}{\|\boldsymbol{y}\|} \max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{B} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\|\boldsymbol{A}\|\|\boldsymbol{B}\| \end{aligned}$</td><td>$\|\boldsymbol{AB}\| = \max_{\boldsymbol{x}\neq\mathbf{0}} \frac{\|\boldsymbol{ABx}\|}{\|\boldsymbol{x}\|} = \max_{\boldsymbol{Bx}\neq\mathbf{0}} \frac{\|\boldsymbol{ABx}\|}{\|\boldsymbol{x}\|} = \max_{\boldsymbol{Bx}\neq\mathbf{0}} \frac{\|\boldsymbol{ABx}\|}{\|\boldsymbol{Bx}\|}\frac{\|\boldsymbol{Bx}\|}{\|\boldsymbol{x}\|}$
$\leq \max_{\boldsymbol{y}\neq\mathbf{0}} \frac{\|\boldsymbol{Ay}\|}{\|\boldsymbol{y}\|} \max_{\boldsymbol{x}\neq\mathbf{0}} \frac{\|\boldsymbol{Bx}\|}{\|\boldsymbol{x}\|} = \|\boldsymbol{A}\|\,\|\boldsymbol{B}\|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1022 (15.5)</td><td>346</td><td>■ 0.918
● N|+0</td><td>\lim _{k \rightarrow \infty} \frac{z_{k}}{ \lambda_{1}^{k}}=c_{1} \boldsymbol{v}_{1},</td><td>$\lim _{k \rightarrow \infty} \frac{z_{k}}{\lambda_{1}^{k}}=c_{1} \boldsymbol{v}_{1},$</td><td>$\lim_{k\to\infty} \frac{z_k}{\lambda_1^k} = c_1\boldsymbol{v}_1,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0580</td><td>215</td><td>■ 0.924
● N|+0</td><td>\begin{array}{c|c|c|c|c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}</td><td>$\begin{array}{c|c|c|c|c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$</td><td>$\begin{array}{c|c|c|c|c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0912 (13.45)</td><td>310</td><td>■ 0.931
● W|+0</td><td>\begin{aligned} A x & =b \\ B A x & =B b \\ \left(C A C^{T}\right) y & =C b, \& x=C^{T} y . \end{aligned}</td><td>$\begin{aligned} A x & =b \\ B A x & =B b \\ \left(C A C^{T}\right) y & =C b, \& x=C^{T} y . \end{aligned}$</td><td>$\boldsymbol{Ax} = \boldsymbol{b}$
$\boldsymbol{BAx} = \boldsymbol{Bb}$
$(\boldsymbol{CAC}^T)\boldsymbol{y} = \boldsymbol{Cb},\ \&\ \boldsymbol{x} = \boldsymbol{C}^T\boldsymbol{y}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0229 (3.20)</td><td>092</td><td>■ 0.933
● N|+0</td><td>\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}= \left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m 1} & \cdots & \boldsymbol{L}_{m, m-1} & \boldsymbol{I} \end{array} \right]\left[\begin{array}{cccc} \boldsymbol{U}_{11} & & \cdots & \boldsymbol{U}_{1 m} \\ & \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2 m} \\ & & \ddots & \vdots \\ & & & \boldsymbol{U}_{m m} \end{array}\right]</td><td>$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}= \left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m 1} & \cdots & \boldsymbol{L}_{m, m-1} & \boldsymbol{I} \end{array} \right]\left[\begin{array}{cccc} \boldsymbol{U}_{11} & & \cdots & \boldsymbol{U}_{1 m} \\ & \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2 m} \\ & & \ddots & \vdots \\ & & & \boldsymbol{U}_{m m} \end{array}\right]$</td><td>$\boldsymbol{A} = \boldsymbol{LU} = \begin{bmatrix} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & & \cdots & \boldsymbol{U}_{1m} \\ & \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ & & \ddots & \vdots \\ & & & \boldsymbol{U}_{mm} \end{bmatrix}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0253</td><td>103</td><td>■ 0.934
● C|-3</td><td>\nabla f(\boldsymbol{x})=\left[\begin{array}{c} \frac{\partial f(\boldsymbol{x})}{\partial x_{1}} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_{n}} \end{array}\right] \in \mathbb{R}^{n}, \quad H f(\boldsymbol{x})=\left[\begin{array}{ccc} \frac{ \partial^{2} f(\boldsymbol{x})}{\partial x_{1} \partial x_{1}} & \ldots & \frac{\partial^{2} f(\boldsymbol{x})}{ \partial x_{1} \partial x_{n}} \\ \vdots & & \vdots \\ \frac{\partial^{2} f(\boldsymbol{x})}{\partial x_{n} \partial x_{1}} & \ldots & \frac{\partial^{2} f(\boldsymbol{x})}{\partial x_{n} \partial x_{n}} \end{array}\right] \in \mathbb{R}^{n \times n} .</td><td>$\nabla f(\boldsymbol{x})=\left[\begin{array}{c} \frac{\partial f(\boldsymbol{x})}{\partial x_{1}} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_{n}} \end{array}\right] \in \mathbb{R}^{n}, \quad H f(\boldsymbol{x})=\left[\begin{array}{ccc} \frac{ \partial^{2} f(\boldsymbol{x})}{\partial x_{1} \partial x_{1}} & \ldots & \frac{\partial^{2} f(\boldsymbol{x})}{ \partial x_{1} \partial x_{n}} \\ \vdots & & \vdots \\ \frac{\partial^{2} f(\boldsymbol{x})}{\partial x_{n} \partial x_{1}} & \ldots & \frac{\partial^{2} f(\boldsymbol{x})}{\partial x_{n} \partial x_{n}} \end{array}\right] \in \mathbb{R}^{n \times n} .$</td><td>$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n, \quad Hf(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_n} \end{bmatrix} \in \mathbb{R}^{n\times n}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra EQ0659 (10.3)</td><td>239</td><td>0.934 W|+0</td><td><code>\begin{aligned} -\Delta_{h} v_{j, k} & =f_{j, k}, \quad(j h, k h) \in \Omega_{h}, \\ v_{j, k} & =0, \quad(j h, k h) \in \partial \Omega_{h}, \end{aligned}</code></td><td>$$\begin{aligned} -\Delta_{h} v_{j, k} & =f_{j, k}, \quad(j h, k h) \in \Omega_{h}, \\ v_{j, k} & =0, \quad(j h, k h) \in \partial \Omega_{h}, \end{aligned}$$</td><td>$$\begin{aligned} -\Delta_{h} v_{j, k} & =f_{j, k}, \quad(j h, k h) \in \Omega_{h}, \\ v_{j, k} & =0, \quad(j h, k h) \in \partial \Omega_{h}, \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0873</td><td>303</td><td>0.936 W|-1</td><td><code>\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}^{2} \leq\left\|\boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\right\|_{\boldsymbol{A}}^{2} \leq \max _{a \leq x \leq b}|Q(x)|^{2} \sum_{j=1}^{n} \frac{\sigma_{j}^{2}}{\lambda_{j}}=\max _{a \leq x \leq b}|Q(x)|^{2} \left\|\boldsymbol{x}-\boldsymbol{x}_{0}\right\|_{\boldsymbol{A}}^{2},</code></td><td>$$\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}^{2} \leq\left\|\boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}\right\|_{\boldsymbol{A}}^{2} \leq \max _{a \leq x \leq b}|Q(x)|^{2} \sum_{j=1}^{n} \frac{\sigma_{j}^{2}}{\lambda_{j}}=\max _{a \leq x \leq b}|Q(x)|^{2} \left\|\boldsymbol{x}-\boldsymbol{x}_{0}\right\|_{\boldsymbol{A}}^{2},$$</td><td>$$||\boldsymbol{x}-\boldsymbol{x}_{k}||_{\boldsymbol{A}}^{2} \leq ||\boldsymbol{x}-\boldsymbol{x}_{0}-\boldsymbol{w}||_{\boldsymbol{A}}^{2} \leq \max _{a \leq x \leq b}|Q(x)|^{2} \sum_{j=1}^{n} \frac{\sigma_{j}^{2}}{\lambda_{j}}=\max _{a \leq x \leq b}|Q(x)|^{2} ||\boldsymbol{x}-\boldsymbol{x}_{0}||_{\boldsymbol{A}}^{2},$$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ1052</td><td>359</td><td>0.936 W|+0</td><td><code>\begin{aligned} & \nabla f(x, y)=\left[\begin{array}{c} 2 x-y \\ -x+2 y \end{array}\right], \quad \nabla^{T} \boldsymbol{g}(x, y)=\left[\begin{array}{cc} 2 x-y-x+2 y \\ 1 & -1 \end{array}\right], \\ & \boldsymbol{H} f(x, y)=\left[\begin{array}{cc} \frac{\partial^{2} f}{\partial x^{2}} & \frac{\partial^{2} f}{\partial x \partial y} \\ \frac{\partial^{2} f}{\partial y \partial x} & \frac{ \partial^{2} f}{\partial y^{2}} \end{array}\right]=\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array}\right] . \end{aligned}</code></td><td>$$\begin{aligned} & \nabla f(x, y)=\left[\begin{array}{c} 2 x-y \\ -x+2 y \end{array}\right], \quad \nabla^{T} \boldsymbol{g}(x, y)=\left[\begin{array}{cc} 2 x-y-x+2 y \\ 1 & -1 \end{array}\right], \\ & \boldsymbol{H} f(x, y)=\left[\begin{array}{cc} \frac{\partial^{2} f}{\partial x^{2}} & \frac{\partial^{2} f}{\partial x \partial y} \\ \frac{\partial^{2} f}{\partial y \partial x} & \frac{ \partial^{2} f}{\partial y^{2}} \end{array}\right]=\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array}\right] . \end{aligned}$$</td><td>$$\begin{aligned} & \nabla f(x, y)=\left[\begin{array}{c} 2 x-y \\ -x+2 y \end{array}\right], \quad \nabla^{T} \boldsymbol{g}(x, y)=\left[\begin{array}{cc} 2 x-y & -x+2 y \\ 1 & -1 \end{array}\right], \\ & \boldsymbol{H} f(x, y)=\left[\begin{array}{cc} \frac{\partial^{2} f}{\partial x^{2}} & \frac{\partial^{2} f}{\partial x \partial y} \\ \frac{\partial^{2} f}{\partial y \partial x} & \frac{ \partial^{2} f}{\partial y^{2}} \end{array}\right]=\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array}\right] . \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0782</td><td>276</td><td>0.938 W|+0</td><td><code>\begin{aligned} (\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}+\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x}) & =\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{R}}\right) \boldsymbol{y}+\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}\right) \boldsymbol{y} \\ & = \boldsymbol{y}^{*}\left(2 \omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{y}=\omega^{-1}(2-\omega)|1-\lambda|^{2} \boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x} . \end{aligned}</code></td><td>$$\begin{aligned} (\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}+\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x}) & =\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{R}}\right) \boldsymbol{y}+\boldsymbol{y}^{*}\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}\right) \boldsymbol{y} \\ & = \boldsymbol{y}^{*}\left(2 \omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{y}=\omega^{-1}(2-\omega)|1-\lambda|^{2} \boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x} . \end{aligned}$$</td><td>$$\begin{aligned} (\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}+\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x}) & =\boldsymbol{y}^{*}(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{R}}) \boldsymbol{y}+\boldsymbol{y}^{*}(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}) \boldsymbol{y} \\ & = \boldsymbol{y}^{*}(2 \omega^{-1}-1) \boldsymbol{D} \boldsymbol{y}=\omega^{-1}(2-\omega)|1-\lambda|^{2} \boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x} . \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0330</td><td>134</td><td>0.938 K|+0</td><td><code>\boldsymbol{P}:=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^{2}+s^{2}=1 .</code></td><td>$\boldsymbol{P}:=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right]$, where $c^{2}+s^{2}=1$.</td><td>$\boldsymbol{P}:=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right]$, where $c^{2}+s^{2}=1$.</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0770 (12.17)</td><td>273</td><td>0.939 W|-1</td><td><code>\rho(\boldsymbol{A}):=\max _{\lambda \in \sigma(\boldsymbol{A})}|\lambda| .</code></td><td>$\rho(\boldsymbol{A}):=\max _{\lambda \in \sigma(\boldsymbol{A})}|\lambda|.$</td><td>$\rho(\boldsymbol{A}):=\max _{\lambda \in \sigma(\boldsymbol{A})}|\lambda|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0482</td><td>190</td><td>0.940 N|-1</td><td><code>\max _{\boldsymbol{x} \neq 0} \frac{\|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{ \boldsymbol{x} \neq 0}\left\|\boldsymbol{A}\left(\frac{ \boldsymbol{x}}{\|\boldsymbol{x}\|}\right)\right\|=\max _{\|\boldsymbol{y}\|=1}\|\boldsymbol{A} \boldsymbol{y}\| .</code></td><td>$$\max _{\boldsymbol{x} \neq 0} \frac{\|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{ \boldsymbol{x} \neq 0}\left\|\boldsymbol{A}\left(\frac{ \boldsymbol{x}}{\|\boldsymbol{x}\|}\right)\right\|=\max _{\|\boldsymbol{y}\|=1}\|\boldsymbol{A} \boldsymbol{y}\| .$$</td><td>$$\max _{\boldsymbol{x} \neq 0} \frac{\|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\|}=\max _{ \boldsymbol{x} \neq 0}\left\|\boldsymbol{A}\left(\frac{ \boldsymbol{x}}{\|\boldsymbol{x}\|}\right)\right\|=\max _{\|\boldsymbol{y}\|=1}\|\boldsymbol{A} \boldsymbol{y}\| .$$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra
EQ0604 (9.9)</td><td>222</td><td>■ 0.940
● K|+0</td><td><code>\boldsymbol{A}^{\dagger}=\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{U}_{1}^{*} .</code></td><td>$\boldsymbol{A}^{\dagger}=\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{U}_{1}^{*} .$</td><td>$\boldsymbol{A}^{\dagger}=\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{U}_{1}^{*} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0806 (12.35)</td><td>282</td><td>■ 0.941
● K|+0</td><td><code>\left\|(\boldsymbol{I}-\boldsymbol{B})^{-1}\right\| \leq \frac{1}{1-\|\boldsymbol{B}\|} .</code></td><td>$\left\|(\boldsymbol{I}-\boldsymbol{B})^{-1}\right\| \leq \frac{1}{1-\|\boldsymbol{B}\|} .$</td><td>$\|(\boldsymbol{I}-\boldsymbol{B})^{-1}\| \leq \frac{1}{1-\|\boldsymbol{B}\|} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ1012</td><td>342</td><td>■ 0.942
● N|+0</td><td><code>A_{n}=\left[\begin{array}{ccccc} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{array}\right] \in \mathbb{R}^{n \times n} .</code></td><td>$A_{n}=\left[\begin{array}{ccccc} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{array}\right] \in \mathbb{R}^{n \times n} .$</td><td>$\boldsymbol{A}_{n}=\left[\begin{array}{ccccc} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{array}\right] \in \mathbb{R}^{n \times n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0174</td><td>076</td><td>■ 0.942
● C|-9</td><td><code>\begin{aligned} \boldsymbol{L} \boldsymbol{U} & := \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right]\left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right] \\ = & {\left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)}+a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)}+a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)}+l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)}+l_{32}^{(2)} a_{23}^{(2)}+a_{33}^{(3)} \end{array}\right]=\left[\begin{array}{lll} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array}\right]=\boldsymbol{A} . } \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{L} \boldsymbol{U} & := \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right]\left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right] \\ = & {\left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)}+a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)}+a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)}+l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)}+l_{32}^{(2)} a_{23}^{(2)}+a_{33}^{(3)} \end{array}\right]=\left[\begin{array}{lll} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array}\right]=\boldsymbol{A} . } \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{L} \boldsymbol{U} & := \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right]\left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right] \\ & = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)}+a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)}+a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)}+l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)}+l_{32}^{(2)} a_{23}^{(2)}+a_{33}^{(3)} \end{array}\right]=\left[\begin{array}{lll} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array}\right]=\boldsymbol{A} . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0628</td><td>229</td><td>■ 0.943
● N|-1</td><td><code>\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right] \text { and } \boldsymbol{D}:= \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^{*} & \mathbf{0} \end{array}\right] \in \mathbb{C}^{(m+n) \times(m+n)} .</code></td><td>$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right] \text { and } \boldsymbol{D}:= \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^{*} & \mathbf{0} \end{array}\right] \in \mathbb{C}^{(m+n) \times(m+n)} .$</td><td>$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right] \text { and } \boldsymbol{D}:= \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^{*} & \mathbf{0} \end{array}\right] \in \mathbb{C}^{(m+n) \times(m+n)} .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ1054	359	■ 0.944 ● W|+1	\begin{aligned} g(0) & =f(\boldsymbol{x}) \quad g(1)=f(\boldsymbol{x}+\boldsymbol{h}), \\ g^{\prime}(t) & =\sum_{i=1}^{n} h_{i} \frac{\partial f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_{i}}=\boldsymbol{h}^{T} \nabla f(\boldsymbol{x}+t \boldsymbol{h}), \\ g^{\prime \prime}(t) & =\sum_{i=1}^{n} \sum_{j=1}^{n} h_{i} h_{j} \frac{\partial^{2} f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_{i} \partial x_{j}}=\boldsymbol{h}^{T} \nabla \nabla^{T} f(\boldsymbol{x}+t \boldsymbol{h}) \boldsymbol{h} . \end{aligned}	$\begin{aligned} g(0) & =f(\boldsymbol{x}) \quad g(1)=f(\boldsymbol{x}+\boldsymbol{h}), \\ g^{\prime}(t) & =\sum_{i=1}^{n} h_{i} \frac{\partial f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_{i}}=\boldsymbol{h}^{T} \nabla f(\boldsymbol{x}+t \boldsymbol{h}), \\ g^{\prime \prime}(t) & =\sum_{i=1}^{n} \sum_{j=1}^{n} h_{i} h_{j} \frac{\partial^{2} f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_{i} \partial x_{j}}=\boldsymbol{h}^{T} \nabla \nabla^{T} f(\boldsymbol{x}+t \boldsymbol{h}) \boldsymbol{h} . \end{aligned}$	$\begin{aligned} g(0) & =f(\boldsymbol{x}) \quad g(1)=f(\boldsymbol{x}+\boldsymbol{h}), \\ g^{\prime}(t) & =\sum_{i=1}^{n} h_{i} \frac{\partial f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_{i}}=\boldsymbol{h}^{T} \nabla f(\boldsymbol{x}+t \boldsymbol{h}), \\ g^{\prime \prime}(t) & =\sum_{i=1}^{n} \sum_{j=1}^{n} h_{i} h_{j} \frac{\partial^{2} f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_{i} \partial x_{j}}=\boldsymbol{h}^{T} \nabla \nabla^{T} f(\boldsymbol{x}+t \boldsymbol{h}) \boldsymbol{h} . \end{aligned}$
lyche-numerical-linear-algebra_ EQ0373 (6.8)	152	■ 0.945 ● W|-1	\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c|c} \lambda_{1} & \boldsymbol{x}^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right], \text { for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text { and } \boldsymbol{x} \in \mathbb{C}^{k} .	$\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c\|c} \lambda_{1} & \boldsymbol{x}^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right], \text { for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text { and } \boldsymbol{x} \in \mathbb{C}^{k} .$	$\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c\|c} \lambda_{1} & \boldsymbol{x}^{*} \\ \hline \mathbf{0} & \boldsymbol{M} \end{array}\right], \text { for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text { and } \boldsymbol{x} \in \mathbb{C}^{k} .$
lyche-numerical-linear-algebra_ EQ0533	203	■ 0.946 ● W|+1	m^{2}\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y}\right\rangle \stackrel{\text { (8.39) }}{=} m\langle n \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{\text { (8.39) }}{=} m n\langle\boldsymbol{x}, \boldsymbol{y}\rangle,	$m^{2}\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y}\right\rangle \stackrel{\text { (8.39) }}{=} m\langle n \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{\text { (8.39) }}{=} m n\langle\boldsymbol{x}, \boldsymbol{y}\rangle,$	$m^{2}\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.39)}{=} m\langle n \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.39)}{=} m n\langle\boldsymbol{x}, \boldsymbol{y}\rangle,$
lyche-numerical-linear-algebra_ EQ0686 (10.19)	246	■ 0.946 ● N|+0	\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k} \right)^{T}\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)=\frac{1}{4 h^{2}} \delta_{j, p} \delta_{k, q}, \quad j, k, p, q=1, \ldots, m,	$\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k} \right)^{T}\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)=\frac{1}{4 h^{2}} \delta_{j, p} \delta_{k, q}, \quad j, k, p, q=1, \ldots, m,$	$\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k} \right)^{T}\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)=\frac{1}{4 h^{2}} \delta_{j, p} \delta_{k, q}, \quad j, k, p, q=1, \ldots, m,$
lyche-numerical-linear-algebra_ EQ0324	132	■ 0.946 ● N|-2	\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}_{1} \boldsymbol{R}_{1} .	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}_{1} \boldsymbol{R}_{1} .$	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}_{1} \boldsymbol{R}_{1} .$
lyche-numerical-linear-algebra_ EQ0331	135	■ 0.947 ● N|+0	\boldsymbol{x}=\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right] \neq \mathbf{0}, \quad c:=\frac{x_{1}}{r}, \quad s:=\frac{x_{2}}{r}, \quad r:=\\|\boldsymbol{x}\\|_{2} .	$\boldsymbol{x}=\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right] \neq \mathbf{0}, \quad c:=\frac{x_{1}}{r}, \quad s:=\frac{x_{2}}{r}, \quad r:=\Vert\boldsymbol{x}\Vert_{2} .$	$\boldsymbol{x}=\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right] \neq \mathbf{0}, \quad c:=\frac{x_{1}}{r}, \quad s:=\frac{x_{2}}{r}, \quad r:=\Vert\boldsymbol{x}\Vert_{2} .$
lyche-numerical-linear-algebra_ EQ1000	337	■ 0.947 ● N|+0	y_{0}<y_{1}<z_{1}<y_{2}<z_{2} \cdots<z_{k-1}<y_{k}<y_{k+1} .	$y_{0}<y_{1}<z_{1}<y_{2}<z_{2} \cdots<z_{k-1}<y_{k}<y_{k+1} .$	$y_{0}<y_{1}<z_{1}<y_{2}<z_{2} \cdots<z_{k-1}<y_{k}<y_{k+1} .$
lyche-numerical-linear-algebra_ EQ0999	337	■ 0.950 ● N|+0	p_{3}(-M)<0, \quad p_{3}\left(y_{1}\right)>0, \quad p_{3}\left(y_{2}\right)<0, \quad p_{3}(+M)>0 .	$p_{3}(-M)<0, \quad p_{3}\left(y_{1}\right)>0, \quad p_{3}\left(y_{2}\right)<0, \quad p_{3}(+M)>0 .$	$p_{3}(-M)<0, \quad p_{3}(y_{1})>0, \quad p_{3}(y_{2})<0, \quad p_{3}(+M)>0 .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0948 (13.61)</td><td>321</td><td>0.950 W|+0</td><td><code>\begin{aligned} {\left[\boldsymbol{A} \boldsymbol{p}_{0}, \boldsymbol{A} \boldsymbol{p}_{1}, \boldsymbol{A} \boldsymbol{p}_{2}\right] } & =\left[\begin{array}{rcr} 8 & 0 & 0 \\ -4 & 3 & 0 \\ 0 & -2 & 16 / 9 \end{array}\right], \\ {\left[\boldsymbol{p}_{0}, \boldsymbol{p}_{1}, \boldsymbol{p}_{2}, \boldsymbol{p}_{3}\right] } & =\left[\begin{array}{rrrr} 4 & 1 & 4 / 9 & 0 \\ 0 & 2 & 8 / 9 & 0 \\ 0 & 0 & 12 / 9 & 0 \end{array}\right], \end{aligned}</code></td><td>$\begin{aligned} {\left[\boldsymbol{A} \boldsymbol{p}_{0}, \boldsymbol{A} \boldsymbol{p}_{1}, \boldsymbol{A} \boldsymbol{p}_{2}\right] } & =\left[\begin{array}{rcr} 8 & 0 & 0 \\ -4 & 3 & 0 \\ 0 & -2 & 16 / 9 \end{array}\right], \\ {\left[\boldsymbol{p}_{0}, \boldsymbol{p}_{1}, \boldsymbol{p}_{2}, \boldsymbol{p}_{3}\right] } & =\left[\begin{array}{rrrr} 4 & 1 & 4 / 9 & 0 \\ 0 & 2 & 8 / 9 & 0 \\ 0 & 0 & 12 / 9 & 0 \end{array}\right], \end{aligned}$</td><td>$\begin{aligned} [\boldsymbol{A}\boldsymbol{p}_0, \boldsymbol{A}\boldsymbol{p}_1, \boldsymbol{A}\boldsymbol{p}_2] &= \left[\begin{array}{rrr} 8 & 0 & 0 \\ -4 & 3 & 0 \\ 0 & -2 & 16/9 \end{array}\right], \\ [\boldsymbol{p}_0, \boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3] &= \left[\begin{array}{cccc} 4 & 1 & 4/9 & 0 \\ 0 & 2 & 8/9 & 0 \\ 0 & 0 & 12/9 & 0 \end{array}\right], \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0058</td><td>041</td><td>0.951 N|+1</td><td><code>\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]^{-1}=\alpha\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right], \quad \alpha=\frac{1}{a d-b c}</code></td><td>$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]^{-1}=\alpha\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right], \quad \alpha=\frac{1}{a d-b c}$</td><td>$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]^{-1} = \alpha\left[\begin{array}{rr} d & -b \\ -c & a \end{array}\right], \quad \alpha = \frac{1}{ad-bc},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0220</td><td>089</td><td>0.951 N|+0</td><td><code>\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]=\left[\begin{array}{ll} 1 & 0 \\ l_{1} & 1 \end{array}\right]\left[\begin{array}{ll} u_{1} & u_{2} \\ 0 & u_{3} \end{array}\right]=\left[\begin{array}{cc} u_{1} & u_{2} \\ u_{1} l_{1} & u_{2} l_{1}+u_{3} \end{array}\right]</code></td><td>$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]=\left[\begin{array}{ll} 1 & 0 \\ l_{1} & 1 \end{array}\right]\left[\begin{array}{ll} u_{1} & u_{2} \\ 0 & u_{3} \end{array}\right]=\left[\begin{array}{cc} u_{1} & u_{2} \\ u_{1} l_{1} & u_{2} l_{1}+u_{3} \end{array}\right]$</td><td>$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right] = \left[\begin{array}{ll} 1 & 0 \\ l_1 & 1 \end{array}\right]\left[\begin{array}{ll} u_1 & u_2 \\ 0 & u_3 \end{array}\right] = \left[\begin{array}{cc} u_1 & u_2 \\ u_1 l_1 & u_2 l_1 + u_3 \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0110</td><td>058</td><td>0.951 K|+0</td><td><code>\boldsymbol{A}_{1}=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{2}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_{3}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right] .</code></td><td>$\boldsymbol{A}_{1}=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{2}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_{3}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right] .$</td><td>$\boldsymbol{A}_1 = \left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2 = \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_3 = \left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0278</td><td>117</td><td>0.953 N|-2</td><td><code>\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle \boldsymbol{y}, \boldsymbol{x}\rangle \quad \text { (symmetry). }</code></td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle=\langle \boldsymbol{y}, \boldsymbol{x}\rangle \quad \text { (symmetry). }$</td><td>$\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \langle \boldsymbol{y}, \boldsymbol{x}\rangle \quad \text{(symmetry).}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0570</td><td>211</td><td>0.953 K|+0</td><td><code>\frac{\|\hat{\boldsymbol{s}}-\boldsymbol{s}\|_{\infty}}{ \|\boldsymbol{s}\|_{\infty}} \quad \text { and } \quad \frac{\|\hat{\boldsymbol{s}}-\boldsymbol{s}\|_{1}}{\| \boldsymbol{s}\|_{1}} .</code></td><td>$\frac{\|\hat{\boldsymbol{s}}-\boldsymbol{s}\|_{\infty}}{ \|\boldsymbol{s}\|_{\infty}} \quad \text { and } \quad \frac{\|\hat{\boldsymbol{s}}-\boldsymbol{s}\|_{1}}{\| \boldsymbol{s}\|_{1}} .$</td><td>$\frac{\|\hat{\boldsymbol{s}}-\boldsymbol{s}\|_\infty}{\|\boldsymbol{s}\|_\infty} \quad \text{and} \quad \frac{\|\hat{\boldsymbol{s}}-\boldsymbol{s}\|_1}{\|\boldsymbol{s}\|_1}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0620 (9.16)</td><td>227</td><td>0.953 C|-5</td><td><code>\sigma_{k}=\min _{\operatorname{dim}(\mathcal{S})=n-k+1} \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}=\max _{ \operatorname{dim}(\mathcal{S})=k} \min _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x} \|_{2}}{\|\boldsymbol{x}\|_{2}} .</code></td><td>$\sigma_{k}=\min _{\operatorname{dim}(\mathcal{S})=n-k+1} \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}=\max _{ \operatorname{dim}(\mathcal{S})=k} \min _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x} \|_{2}}{\|\boldsymbol{x}\|_{2}} .$</td><td>$\sigma_k = \min_{\dim(\mathcal{S})=n-k+1} \max_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \boldsymbol{x}\neq\mathbf{0}}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} = \max_{\dim(\mathcal{S})=k} \min_{\substack{\boldsymbol{x}\in\mathcal{S} \\ \boldsymbol{x}\neq\mathbf{0}}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}.$</td></tr>
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Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0722</td><td>257</td><td>■ 0.954
● C|-8</td><td><code>\boldsymbol{P}_{4}=\left[\begin{array}{lll} \boldsymbol{e}_{1} & \boldsymbol{e}_{3} & \boldsymbol{e}_{2} \\ \boldsymbol{e}_{4} \end{array} \right]=\left[\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \boldsymbol{F}_{4} \boldsymbol{P}_{4}= \left[\begin{array}{rr|rr} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array}\right],</code></td><td>$\boldsymbol{P}_{4}=\left[\begin{array}{lll} \boldsymbol{e}_{1} & \boldsymbol{e}_{3} & \boldsymbol{e}_{2} \\ \boldsymbol{e}_{4} \end{array} \right]=\left[\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \boldsymbol{F}_{4} \boldsymbol{P}_{4}= \left[\begin{array}{rr|rr} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array}\right],$</td><td>$\boldsymbol{P}_4=[\boldsymbol{e}_1\ \boldsymbol{e}_3\ \boldsymbol{e}_2\ \boldsymbol{e}_4]=\left[\begin{array}{llll} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \quad \boldsymbol{F}_4\boldsymbol{P}_4=\left[\begin{array}{rr|rr} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0609 (9.12)</td><td>223</td><td>■ 0.955
● N|+1</td><td><code>\begin{gathered} \boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})}=\boldsymbol{A} \boldsymbol{A}^{ \dagger}=\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}= \sum_{j=1}^{r} \boldsymbol{u}_{j} \boldsymbol{u}_{j}^{*} \in \mathbb{C}^{m \times m}, \\ \boldsymbol{P}_{ \mathcal{N}\left(\boldsymbol{A}^{*}\right)}= \boldsymbol{I}-\boldsymbol{A} \boldsymbol{A}^{ \dagger}=\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}= \sum_{j=r+1}^{m} \boldsymbol{u}_{j} \boldsymbol{u}_{j}^{*} \in \mathbb{C}^{m \times m} . \end{gathered}</code></td><td>$$\begin{gathered} \boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})}=\boldsymbol{A} \boldsymbol{A}^{ \dagger}=\boldsymbol{U}_{1} \boldsymbol{U}_{1}^{*}= \sum_{j=1}^{r} \boldsymbol{u}_{j} \boldsymbol{u}_{j}^{*} \in \mathbb{C}^{m \times m}, \\ \boldsymbol{P}_{ \mathcal{N}\left(\boldsymbol{A}^{*}\right)}= \boldsymbol{I}-\boldsymbol{A} \boldsymbol{A}^{ \dagger}=\boldsymbol{U}_{2} \boldsymbol{U}_{2}^{*}= \sum_{j=r+1}^{m} \boldsymbol{u}_{j} \boldsymbol{u}_{j}^{*} \in \mathbb{C}^{m \times m} . \end{gathered}$$</td><td>$$\begin{gathered} \boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})}=\boldsymbol{A}\boldsymbol{A}^{\dagger}=\boldsymbol{U}_1\boldsymbol{U}_1^*=\sum_{j=1}^{r}\boldsymbol{u}_j\boldsymbol{u}_j^*\in\mathbb{C}^{m\times m}, \\ \boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)}=\boldsymbol{I}-\boldsymbol{A}\boldsymbol{A}^{\dagger}=\boldsymbol{U}_2\boldsymbol{U}_2^*=\sum_{j=r+1}^{m}\boldsymbol{u}_j\boldsymbol{u}_j^*\in\mathbb{C}^{m\times m}. \end{gathered}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0753 (12.8)</td><td>270</td><td>■ 0.958
● N|+0</td><td><code>\boldsymbol{M} \boldsymbol{x}=(\boldsymbol{M}- \boldsymbol{A}) \boldsymbol{x}+\boldsymbol{b} .</code></td><td>$\boldsymbol{M} \boldsymbol{x}=(\boldsymbol{M}- \boldsymbol{A}) \boldsymbol{x}+\boldsymbol{b} .$</td><td>$\boldsymbol{M}\boldsymbol{x}=(\boldsymbol{M}-\boldsymbol{A})\boldsymbol{x}+\boldsymbol{b}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0654</td><td>235</td><td>■ 0.959
● N|+0</td><td><code>\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}= \left[\begin{array}{l} 2 \\ 3 \\ 2 \end{array}\right], \quad \in \in \mathbb{R} .</code></td><td>$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}= \left[\begin{array}{l} 2 \\ 3 \\ 2 \end{array}\right], \quad \in \in \mathbb{R} .$</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{l} 2 \\ 3 \\ 2 \end{array}\right], \quad \epsilon\in\mathbb{R}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0443</td><td>177</td><td>■ 0.959
● N|+0</td><td><code>\|\boldsymbol{A}\|_{F} \stackrel{\text { 3. }}{=}\left\| \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{V}\right \|_{F}=\|\boldsymbol{\Sigma}\|_{F}=\sqrt{\sigma_{1}^{2}+ \cdots+\sigma_{n}^{2}} .</code></td><td>$\|\boldsymbol{A}\|_{F} \stackrel{\text { 3. }}{=}\left\| \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{V}\right \|_{F}=\|\boldsymbol{\Sigma}\|_{F}=\sqrt{\sigma_{1}^{2}+ \cdots+\sigma_{n}^{2}} .$</td><td>$\|\boldsymbol{A}\|_F\stackrel{3.}{=}\|\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{V}\|_F=\|\boldsymbol{\Sigma}\|_F=\sqrt{\sigma_1^2+\cdots+\sigma_n^2}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0469 (8.5)</td><td>186</td><td>■ 0.960
● N|+0</td><td><code>\|x\|_{\infty} \leq\|x\|_{p} \leq n^{1 / p}\|x\|_{ \infty}, \quad p \geq 1 .</code></td><td>$\|x\|_{\infty} \leq\|x\|_{p} \leq n^{1 / p}\|x\|_{ \infty}, \quad p \geq 1 .$</td><td>$\|\boldsymbol{x}\|_\infty\leq\|\boldsymbol{x}\|_p\leq n^{1/p}\|\boldsymbol{x}\|_\infty, \quad p\geq 1.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0606 (9.10)</td><td>222</td><td>■ 0.961
● S|+2</td><td><code>\begin{aligned} & A^{\dagger} A=V_{1} \Sigma_{1}^{-1} U_{1}^{*} U_{1} \Sigma_{1} V_{1}^{*}=V_{1} V_{1}^{*} \\ & A A^{\dagger}=U_{1} \Sigma_{1} V_{1}^{*} V_{1} \Sigma_{1}^{-1} U_{1}^{*}=U_{1} U_{1}^{*} \end{aligned}</code></td><td>$$\begin{aligned} & A^{\dagger} A=V_{1} \Sigma_{1}^{-1} U_{1}^{*} U_{1} \Sigma_{1} V_{1}^{*}=V_{1} V_{1}^{*} \\ & A A^{\dagger}=U_{1} \Sigma_{1} V_{1}^{*} V_{1} \Sigma_{1}^{-1} U_{1}^{*}=U_{1} U_{1}^{*} \end{aligned}$$</td><td>$$\begin{aligned} & \boldsymbol{A}^{\dagger}\boldsymbol{A}=\boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{U}_1^*\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*=\boldsymbol{V}_1\boldsymbol{V}_1^*, \\ & \boldsymbol{A}\boldsymbol{A}^{\dagger}=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*\boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{U}_1^*=\boldsymbol{U}_1\boldsymbol{U}_1^*. \end{aligned}$$</td></tr>
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Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● N|+0</td><td><code>\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2, m-1} \boldsymbol{P}_{m-1, m} \boldsymbol{w}=\left[\begin{array}{c} \alpha \\ 0 \\ \vdots \\ 0 \end{array}\right],</code></td><td>$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2, m-1} \boldsymbol{P}_{m-1, m} \boldsymbol{w}=\left[\begin{array}{c} \alpha \\ 0 \\ \vdots \\ 0 \end{array}\right],$</td><td>$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2, m-1} \boldsymbol{P}_{m-1, m} \boldsymbol{w}=\left[\begin{array}{c} \alpha \\ 0 \\ \vdots \\ 0 \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0961 (13.71)</td><td>323</td><td>■ 0.962
● N|+0</td><td><code>\boldsymbol{B} \boldsymbol{r}_{k}=\alpha_{k} \boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{r}_{k-1},</code></td><td>$\boldsymbol{B} \boldsymbol{r}_{k}=\alpha_{k} \boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{r}_{k-1},$</td><td>$\boldsymbol{B} \boldsymbol{r}_{k}=\alpha_{k} \boldsymbol{r}_{k+1}+\beta_{k} \boldsymbol{r}_{k-1},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0332</td><td>135</td><td>■ 0.963
● N|+0</td><td><code>\boldsymbol{P} \boldsymbol{x}=\frac{1}{r}\left[\begin{array}{rr} x_{1} & x_{2} \\ -x_{2} & x_{1} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\frac{1}{r}\left[\begin{array}{c} x_{1}^{2}+x_{2}^{2} \\ 0 \end{array}\right]=\left[\begin{array}{l} r \\ 0 \end{array}\right],</code></td><td>$\boldsymbol{P} \boldsymbol{x}=\frac{1}{r}\left[\begin{array}{rr} x_{1} & x_{2} \\ -x_{2} & x_{1} \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\frac{1}{r}\left[\begin{array}{c} x_{1}^{2}+x_{2}^{2} \\ 0 \end{array}\right]=\left[\begin{array}{l} r \\ 0 \end{array}\right],$</td><td>$\boldsymbol{P} \boldsymbol{x}=\frac{1}{r}\begin{bmatrix} x_{1} & x_{2} \\ -x_{2} & x_{1} \end{bmatrix}\begin{bmatrix} x_{1} \\ x_{2} \end{bmatrix}=\frac{1}{r}\begin{bmatrix} x_{1}^{2}+x_{2}^{2} \\ 0 \end{bmatrix}=\begin{bmatrix} r \\ 0 \end{bmatrix},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0057</td><td>040</td><td>■ 0.969
● W|+0</td><td><code>\begin{array}{ll} \mathbf{P}_{1}=(\boldsymbol{A}+\boldsymbol{D})(\boldsymbol{E}+\boldsymbol{H}), & \\ \mathbf{P}_{2}=(\boldsymbol{C}+\boldsymbol{D}) \boldsymbol{E}, & \mathbf{P}_{5}=(\boldsymbol{A}+\boldsymbol{B}) \boldsymbol{H}, \\ \mathbf{P}_{3}=\boldsymbol{A}(\boldsymbol{F}-\boldsymbol{H}), & \mathbf{P}_{6}=(\boldsymbol{C}-\boldsymbol{A})(\boldsymbol{E}+\boldsymbol{F}), \\ \mathbf{P}_{4}=\boldsymbol{D}(\boldsymbol{G}-\boldsymbol{E}), & \mathbf{P}_{7}=(\boldsymbol{B}-\boldsymbol{D})(\boldsymbol{G}+\boldsymbol{H}), \\ \boldsymbol{W}=\mathbf{P}_{1}+\mathbf{P}_{4}-\mathbf{P}_{5}+\mathbf{P}_{7}, & \boldsymbol{X}=\mathbf{P}_{3}+\mathbf{P}_{5}, \\ \boldsymbol{Y}=\mathbf{P}_{2}+\mathbf{P}_{4}, & \boldsymbol{Z}=\mathbf{P}_{1}+\mathbf{P}_{3}-\mathbf{P}_{2}+\mathbf{P}_{6} . \end{array}</code></td><td>$\begin{array}{ll} \mathbf{P}_{1}=(\boldsymbol{A}+\boldsymbol{D})(\boldsymbol{E}+\boldsymbol{H}), & \\ \mathbf{P}_{2}=(\boldsymbol{C}+\boldsymbol{D}) \boldsymbol{E}, & \mathbf{P}_{5}=(\boldsymbol{A}+\boldsymbol{B}) \boldsymbol{H}, \\ \mathbf{P}_{3}=\boldsymbol{A}(\boldsymbol{F}-\boldsymbol{H}), & \mathbf{P}_{6}=(\boldsymbol{C}-\boldsymbol{A})(\boldsymbol{E}+\boldsymbol{F}), \\ \mathbf{P}_{4}=\boldsymbol{D}(\boldsymbol{G}-\boldsymbol{E}), & \mathbf{P}_{7}=(\boldsymbol{B}-\boldsymbol{D})(\boldsymbol{G}+\boldsymbol{H}), \\ \boldsymbol{W}=\mathbf{P}_{1}+\mathbf{P}_{4}-\mathbf{P}_{5}+\mathbf{P}_{7}, & \boldsymbol{X}=\mathbf{P}_{3}+\mathbf{P}_{5}, \\ \boldsymbol{Y}=\mathbf{P}_{2}+\mathbf{P}_{4}, & \boldsymbol{Z}=\mathbf{P}_{1}+\mathbf{P}_{3}-\mathbf{P}_{2}+\mathbf{P}_{6} . \end{array}$</td><td>$\begin{array}{ll} \mathbf{P}_{1}=(\boldsymbol{A}+\boldsymbol{D})\,(\boldsymbol{E}+\boldsymbol{H})\,, & \\ \mathbf{P}_{2}=(\boldsymbol{C}+\boldsymbol{D})\,\boldsymbol{E}, & \mathbf{P}_{5}=(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{H}, \\ \mathbf{P}_{3}=\boldsymbol{A}\,(\boldsymbol{F}-\boldsymbol{H})\,, & \mathbf{P}_{6}=(\boldsymbol{C}-\boldsymbol{A})\,(\boldsymbol{E}+\boldsymbol{F})\,, \\ \mathbf{P}_{4}=\boldsymbol{D}\,(\boldsymbol{G}-\boldsymbol{E})\,, & \mathbf{P}_{7}=(\boldsymbol{B}-\boldsymbol{D})\,(\boldsymbol{G}+\boldsymbol{H})\,, \\ \boldsymbol{W}=\mathbf{P}_{1}+\mathbf{P}_{4}-\mathbf{P}_{5}+\mathbf{P}_{7}, & \boldsymbol{X}=\mathbf{P}_{3}+\mathbf{P}_{5}, \\ \boldsymbol{Y}=\mathbf{P}_{2}+\mathbf{P}_{4}, & \boldsymbol{Z}=\mathbf{P}_{1}+\mathbf{P}_{3}-\mathbf{P}_{2}+\mathbf{P}_{6}. \end{array}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0224 (3.16)</td><td>090</td><td>■ 0.969
● W|+0</td><td><code>\left[\begin{array}{cc} \boldsymbol{A}_{[k]} & \boldsymbol{B}_{k} \\ \boldsymbol{C}_{k} & \boldsymbol{F}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{[k]} & \mathbf{0} \\ \boldsymbol{M}_{k} & \boldsymbol{N}_{k} \end{array} \right]\left[\begin{array}{cc} \boldsymbol{U}_{[k]} & \boldsymbol{S}_{k} \\ \mathbf{0} & \boldsymbol{T}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]} & \boldsymbol{L}_{[k]} \boldsymbol{S}_{k} \\ \boldsymbol{M}_{k} \boldsymbol{U}_{[k]} & \boldsymbol{M}_{k} \boldsymbol{S}_{k}+\boldsymbol{N}_{k} \boldsymbol{T}_{k} \end{array}\right],</code></td><td>$\left[\begin{array}{cc} \boldsymbol{A}_{[k]} & \boldsymbol{B}_{k} \\ \boldsymbol{C}_{k} & \boldsymbol{F}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{[k]} & \mathbf{0} \\ \boldsymbol{M}_{k} & \boldsymbol{N}_{k} \end{array} \right]\left[\begin{array}{cc} \boldsymbol{U}_{[k]} & \boldsymbol{S}_{k} \\ \mathbf{0} & \boldsymbol{T}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]} & \boldsymbol{L}_{[k]} \boldsymbol{S}_{k} \\ \boldsymbol{M}_{k} \boldsymbol{U}_{[k]} & \boldsymbol{M}_{k} \boldsymbol{S}_{k}+\boldsymbol{N}_{k} \boldsymbol{T}_{k} \end{array}\right],$</td><td>$\begin{bmatrix} \boldsymbol{A}_{[k]} & \boldsymbol{B}_{k} \\ \boldsymbol{C}_{k} & \boldsymbol{F}_{k} \end{bmatrix}=\begin{bmatrix} \boldsymbol{L}_{[k]} & \mathbf{0} \\ \boldsymbol{M}_{k} & \boldsymbol{N}_{k} \end{bmatrix}\begin{bmatrix} \boldsymbol{U}_{[k]} & \boldsymbol{S}_{k} \\ \mathbf{0} & \boldsymbol{T}_{k} \end{bmatrix}=\begin{bmatrix} \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]} & \boldsymbol{L}_{[k]} \boldsymbol{S}_{k} \\ \boldsymbol{M}_{k} \boldsymbol{U}_{[k]} & \boldsymbol{M}_{k} \boldsymbol{S}_{k}+\boldsymbol{N}_{k} \boldsymbol{T}_{k} \end{bmatrix},$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0106	057	■ 0.970 ● W|-1	`\begin{aligned} u^{\prime \prime}(x) & =\lim _{h \rightarrow 0} \frac{u^{\prime}\left(x+\frac{h}{2}\right)-u^{\prime}\left(x-\frac{h}{2}\right)}{h}= \lim _{h \rightarrow 0} \frac{\frac{u(x+h)-u(x)}{h}-\frac{u(x)-u(x-h)}{h}}{h} \\ & =\lim _{h \rightarrow 0} \frac{u(x+h)-2 u(x)+u(x-h)}{h^{2}} . \end{aligned}`	$\begin{aligned} u^{\prime \prime}(x) & =\lim _{h \rightarrow 0} \frac{u^{\prime}\left(x+\frac{h}{2}\right)-u^{\prime}\left(x-\frac{h}{2}\right)}{h}= \lim _{h \rightarrow 0} \frac{\frac{u(x+h)-u(x)}{h}-\frac{u(x)-u(x-h)}{h}}{h} \\ & =\lim _{h \rightarrow 0} \frac{u(x+h)-2 u(x)+u(x-h)}{h^{2}} . \end{aligned}$	$\begin{aligned} u''(x) &= \lim_{h\rightarrow 0} \frac{u'(x+\frac{h}{2})-u'(x-\frac{h}{2})}{h} = \lim_{h\rightarrow 0} \frac{\frac{u(x+h)-u(x)}{h}-\frac{u(x)-u(x-h)}{h}}{h} \\ &= \lim_{h\rightarrow 0} \frac{u(x+h)-2u(x)+u(x-h)}{h^2}. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0680	245	■ 0.970 ● W|+0	`\begin{aligned} (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) & =\operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow\left[\begin{array}{ccc} \boldsymbol{A} b_{11} & \cdots & \boldsymbol{A} b_{1 s} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{s 1} & \cdots & \boldsymbol{A} b_{s s} \end{array}\right]\left[\begin{array}{c} \boldsymbol{v}_{1} \\ \vdots \\ \boldsymbol{v}_{s} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{f}_{1} \\ \vdots \\ \boldsymbol{f}_{s} \end{array}\right] \\ & \Leftrightarrow \boldsymbol{A} \left[\sum_{j} b_{1 j} \boldsymbol{v}_{j}, \ldots, \sum_{j} b_{s j} \boldsymbol{v}_{j}\right]=\left[\boldsymbol{f}_{1}, \ldots, \boldsymbol{f}_{s} \right] \\ & \Leftrightarrow \boldsymbol{A}\left[\boldsymbol{V} \boldsymbol{b}_{1}, \ldots, \boldsymbol{V} \boldsymbol{b}_{s}\right]=\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^{T}=\boldsymbol{F} . \end{aligned}`	$\begin{aligned} (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) & =\operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow\left[\begin{array}{ccc} \boldsymbol{A} b_{11} & \cdots & \boldsymbol{A} b_{1 s} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{s 1} & \cdots & \boldsymbol{A} b_{s s} \end{array}\right]\left[\begin{array}{c} \boldsymbol{v}_{1} \\ \vdots \\ \boldsymbol{v}_{s} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{f}_{1} \\ \vdots \\ \boldsymbol{f}_{s} \end{array}\right] \\ & \Leftrightarrow \boldsymbol{A} \left[\sum_{j} b_{1 j} \boldsymbol{v}_{j}, \ldots, \sum_{j} b_{s j} \boldsymbol{v}_{j}\right]=\left[\boldsymbol{f}_{1}, \ldots, \boldsymbol{f}_{s} \right] \\ & \Leftrightarrow \boldsymbol{A}\left[\boldsymbol{V} \boldsymbol{b}_{1}, \ldots, \boldsymbol{V} \boldsymbol{b}_{s}\right]=\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^{T}=\boldsymbol{F} . \end{aligned}$	$\begin{aligned} &(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ &\Leftrightarrow \begin{bmatrix} \boldsymbol{A}b_{11} \cdots \boldsymbol{A}b_{1s} \\ \vdots \quad \vdots \\ \boldsymbol{A}b_{s1} \cdots \boldsymbol{A}b_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix} \\ &\Leftrightarrow \boldsymbol{A}\Big[\sum_j b_{1j}\boldsymbol{v}_j, \ldots, \sum_j b_{sj}\boldsymbol{v}_j\Big] = [\boldsymbol{f}_1, \ldots, \boldsymbol{f}_s] \\ &\Leftrightarrow \boldsymbol{A}[\boldsymbol{V}\boldsymbol{b}_1, \ldots, \boldsymbol{V}\boldsymbol{b}_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A}\boldsymbol{V}\boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0161	071	■ 0.971 ● N|+0	`a_{j}=-1-\frac{h}{2} r\left(x_{j}\right), c_{j}=-1+ \frac{h}{2} r\left(x_{j}\right), d_{j}=2+h^{2} q \left(x_{j}\right) \text {, }`	$a_{j}=-1-\frac{h}{2} r\left(x_{j}\right), c_{j}=-1+ \frac{h}{2} r\left(x_{j}\right), d_{j}=2+h^{2} q \left(x_{j}\right) \text {, }$	$a_j = -1 - \frac{h}{2}r(x_j),\ c_j = -1 + \frac{h}{2}r(x_j),\ d_j = 2 + h^2 q(x_j),$
lyche-numerical-linear-algebra_ EQ0312 (5.14)	126	■ 0.971 ● N|+0	`\boldsymbol{u}:=\frac{\boldsymbol{z}+\boldsymbol{e}_{1}}{ \sqrt{1+z_{1}}}, \text { where } \boldsymbol{z}:=\bar{ \rho} \boldsymbol{x} /|\boldsymbol{x}|_{2} .`	$\boldsymbol{u}:=\frac{\boldsymbol{z}+\boldsymbol{e}_{1}}{ \sqrt{1+z_{1}}}, \text { where } \boldsymbol{z}:=\bar{ \rho} \boldsymbol{x} /\Vert\boldsymbol{x}\Vert_{2} .$	$\boldsymbol{u} := \frac{\boldsymbol{z}+\boldsymbol{e}_1}{\sqrt{1+z_1}}, \text{ where } \boldsymbol{z} := \overline{\rho}\boldsymbol{x}/\Vert\boldsymbol{x}\Vert_2.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1043</td><td>353</td><td>■ 0.971 ● U|+78</td><td>\begin{figure} \[\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{\boldsymbol{P}_{12}^{*}}\left[\begin{array}{llll} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{\boldsymbol{P}_{23}^{*}}\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{\boldsymbol{P}_{34}^{*}}\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{array}\right] . \] \captionsetup{labelformat=empty} \caption{Fig. 15.1 Post multiplication in a QR step} \end{figure}</td><td>(not rendered)</td><td>$\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{12}^{*}} \left[\begin{array}{llll} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{23}^{*}} \left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{34}^{*}} \left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0407</td><td>162</td><td>■ 0.971 ● N|+0</td><td>\boldsymbol{B}=\left[\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_{0} \\ 1 & 0 & \cdots & 0 & -q_{1} \\ 0 & 1 & \cdots & 0 & -q_{2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{array} \right] .</td><td>$\boldsymbol{B}=\left[\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_{0} \\ 1 & 0 & \cdots & 0 & -q_{1} \\ 0 & 1 & \cdots & 0 & -q_{2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{array}\right] .$</td><td>$\boldsymbol{B}=\left[\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_{0} \\ 1 & 0 & \cdots & 0 & -q_{1} \\ 0 & 1 & \cdots & 0 & -q_{2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0289 (5.8)</td><td>120</td><td>■ 0.971 ● W|+0</td><td>\boldsymbol{v}_{1}:=\boldsymbol{s}_{1}, \quad \boldsymbol{v}_{j}:=\boldsymbol{s}_{j}-\sum_{i=1}^{j-1} \frac{\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{i} \right\rangle}{\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle} \boldsymbol{v}_{i}, \quad j=2, \ldots, k .</td><td>$\boldsymbol{v}_{1}:=\boldsymbol{s}_{1}, \quad \boldsymbol{v}_{j}:=\boldsymbol{s}_{j}-\sum_{i=1}^{j-1} \frac{\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{i}\right\rangle}{\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle} \boldsymbol{v}_{i}, \quad j=2, \ldots, k .$</td><td>$\boldsymbol{v}_{1}:=\boldsymbol{s}_{1}, \quad \boldsymbol{v}_{j}:=\boldsymbol{s}_{j}-\sum_{i=1}^{j-1} \frac{\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{i}\right\rangle}{\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle} \boldsymbol{v}_{i}, \quad j=2, \ldots, k .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0783</td><td>276</td><td>■ 0.973 ● K|+0</td><td>(\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}+ \boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x})=(1-\lambda) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}+\lambda(1-\bar{\lambda}) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}= \left(1-|\lambda|^{2}\right) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x},</td><td>$(\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}+\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x})=(1-\lambda) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}+\lambda(1-\bar{\lambda}) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\left(1-|\lambda|^{2}\right) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x},$</td><td>$(\boldsymbol{A} \boldsymbol{x})^{*} \boldsymbol{y}+\boldsymbol{y}^{*}(\lambda \boldsymbol{A} \boldsymbol{x})=(1-\lambda) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}+\lambda(1-\bar{\lambda}) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=(1-|\lambda|^{2}) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0677 (10.14)</td><td>243</td><td>■ 0.974 ● K|+0</td><td>(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})=(\boldsymbol{A} \boldsymbol{C}) \otimes(\boldsymbol{B} \boldsymbol{D}) .</td><td>$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})=(\boldsymbol{A} \boldsymbol{C}) \otimes(\boldsymbol{B} \boldsymbol{D}) .$</td><td>$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})=(\boldsymbol{A} \boldsymbol{C}) \otimes(\boldsymbol{B} \boldsymbol{D}) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0527</td><td>202</td><td>■ 0.975 ● K|+0</td><td>\|\boldsymbol{x}+\boldsymbol{y}\|^{2}+\|\boldsymbol{x}- \boldsymbol{y}\|^{2}=\langle\boldsymbol{x}+ \boldsymbol{y}, \boldsymbol{x}+\boldsymbol{y}\rangle+ \langle\boldsymbol{x}-\boldsymbol{y}, \boldsymbol{x}- \boldsymbol{y}\rangle=2\|\boldsymbol{x}\|^{2}+2\| \boldsymbol{y}\|^{2}</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2}+\|\boldsymbol{x}-\boldsymbol{y}\|^{2}=\langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{x}+\boldsymbol{y}\rangle+\langle\boldsymbol{x}-\boldsymbol{y}, \boldsymbol{x}-\boldsymbol{y}\rangle=2\|\boldsymbol{x}\|^{2}+2\|\boldsymbol{y}\|^{2}$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2}+\|\boldsymbol{x}-\boldsymbol{y}\|^{2}=\langle\boldsymbol{x}+\boldsymbol{y}, \boldsymbol{x}+\boldsymbol{y}\rangle+\langle\boldsymbol{x}-\boldsymbol{y}, \boldsymbol{x}-\boldsymbol{y}\rangle=2\|\boldsymbol{x}\|^{2}+2\|\boldsymbol{y}\|^{2}$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0081 (2.4)</td><td>047</td><td>0.975 W|+0</td><td>g(x):= \begin{cases}p_{1}(x), & \text { if } x_{1} \leq x<x_{2}, \\ p_{2}(x), & \text { if } x_{2} \leq x<x_{3}, \\ \vdots & \\ p_{n-1}(x), & \text { if } x_{n-1} \leq x<x_{n}, \\ p_{n}(x), & \text { if } x_{n} \leq x \leq x_{n+1},\end{cases}</td><td>$g(x):= \begin{cases}p_{1}(x), & \text { if } x_{1} \leq x<x_{2}, \\ p_{2}(x), & \text { if } x_{2} \leq x<x_{3}, \\ \vdots & \\ p_{n-1}(x), & \text { if } x_{n-1} \leq x<x_{n}, \\ p_{n}(x), & \text { if } x_{n} \leq x \leq x_{n+1},\end{cases}$</td><td>$g(x):= \begin{cases}p_{1}(x), & \text { if } x_{1} \leq x<x_{2}, \\ p_{2}(x), & \text { if } x_{2} \leq x<x_{3}, \\ \vdots & \\ p_{n-1}(x), & \text { if } x_{n-1} \leq x<x_{n}, \\ p_{n}(x), & \text { if } x_{n} \leq x \leq x_{n+1},\end{cases}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0998</td><td>336</td><td>0.976 N|+0</td><td>p_{3}(x)=\left(x-d_{3}\right) p_{2}(x)-c_{2}^{2} p_{1}(x)=\left(x-d_{3}\right)\left(x-y_{1}\right)\left(x-y_{2}\right)-c_{2}^{2}\left(x-d_{1}\right) .</td><td>$p_{3}(x)=\left(x-d_{3}\right) p_{2}(x)-c_{2}^{2} p_{1}(x)=\left(x-d_{3}\right)\left(x-y_{1}\right)\left(x-y_{2}\right)-c_{2}^{2}\left(x-d_{1}\right) .$</td><td>$p_{3}(x)=(x-d_{3}) p_{2}(x)-c_{2}^{2} p_{1}(x)=(x-d_{3})(x-y_{1})(x-y_{2})-c_{2}^{2}(x-d_{1}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0731</td><td>259</td><td>0.976 W|+0</td><td>\begin{aligned} \boldsymbol{F}_{2 m} \boldsymbol{y} & =\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^{T} \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w} \\ & = \left[\begin{array}{r|r} \boldsymbol{F}_{m} & \boldsymbol{D}_{m} \boldsymbol{F}_{m} \\ \hline \boldsymbol{F}_{m} & -\boldsymbol{D}_{m} \boldsymbol{F}_{m} \end{array}\right]\left[\begin{array}{l} \boldsymbol{w}_{1} \\ \boldsymbol{w}_{2} \end{array}\right]=\left[\begin{array}{l} \boldsymbol{q}_{1}+\boldsymbol{q}_{2} \\ \boldsymbol{q}_{1}-\boldsymbol{q}_{2} \end{array}\right], \end{aligned}</td><td>$\begin{aligned} \boldsymbol{F}_{2 m} \boldsymbol{y} & =\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^{T} \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w} \\ & = \left[\begin{array}{r|r} \boldsymbol{F}_{m} & \boldsymbol{D}_{m} \boldsymbol{F}_{m} \\ \hline \boldsymbol{F}_{m} & -\boldsymbol{D}_{m} \boldsymbol{F}_{m} \end{array}\right]\left[\begin{array}{l} \boldsymbol{w}_{1} \\ \boldsymbol{w}_{2} \end{array}\right]=\left[\begin{array}{l} \boldsymbol{q}_{1}+\boldsymbol{q}_{2} \\ \boldsymbol{q}_{1}-\boldsymbol{q}_{2} \end{array}\right], \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{F}_{2 m} \boldsymbol{y} & =\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^{T} \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w} \\ & = \left[\begin{array}{r|r} \boldsymbol{F}_{m} & \boldsymbol{D}_{m} \boldsymbol{F}_{m} \\ \boldsymbol{F}_{m} & -\boldsymbol{D}_{m} \boldsymbol{F}_{m} \end{array}\right]\left[\begin{array}{l} \boldsymbol{w}_{1} \\ \boldsymbol{w}_{2} \end{array}\right]=\left[\begin{array}{l} \boldsymbol{q}_{1}+\boldsymbol{q}_{2} \\ \boldsymbol{q}_{1}-\boldsymbol{q}_{2} \end{array}\right], \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0046</td><td>037</td><td>0.976 C|-6</td><td>\left[\begin{array}{ccc} c_{i_{1}, j_{1}} & \cdots & c_{i_{1}, j_{r}} \\ \vdots & & \vdots \\ c_{i_{r}, j_{1}} & \cdots & c_{i_{r}, j_{r}} \end{array}\right]=\sum_{ \boldsymbol{k}}\left[\begin{array}{ccc} a_{i_{1}, k_{1}} & \cdots & a_{i_{1}, k_{r}} \\ \vdots & & \vdots \\ a_{i_{r} k_{1}} & \cdots & a_{i_{r}, k_{r}} \end{array} \right]\left[\begin{array}{ccc} b_{k_{1}, j_{1}} & \cdots & b_{k_{1}, j_{r}} \\ \vdots & & \vdots \\ b_{k_{r}, j_{1}} & \cdots & b_{k_{r}, j_{r}} \end{array}\right],</td><td>$\left[\begin{array}{ccc} c_{i_{1}, j_{1}} & \cdots & c_{i_{1}, j_{r}} \\ \vdots & & \vdots \\ c_{i_{r}, j_{1}} & \cdots & c_{i_{r}, j_{r}} \end{array}\right]=\sum_{ \boldsymbol{k}}\left[\begin{array}{ccc} a_{i_{1}, k_{1}} & \cdots & a_{i_{1}, k_{r}} \\ \vdots & & \vdots \\ a_{i_{r} k_{1}} & \cdots & a_{i_{r}, k_{r}} \end{array} \right]\left[\begin{array}{ccc} b_{k_{1}, j_{1}} & \cdots & b_{k_{1}, j_{r}} \\ \vdots & & \vdots \\ b_{k_{r}, j_{1}} & \cdots & b_{k_{r}, j_{r}} \end{array}\right],$</td><td>$\left[\begin{array}{ccc} c_{i_{1}, j_{1}} & \cdots & c_{i_{1}, j_{r}} \\ \vdots & & \vdots \\ c_{i_{r}, j_{1}} & \cdots & c_{i_{r}, j_{r}} \end{array}\right]=\sum_{ \boldsymbol{k}}\left[\begin{array}{ccc} a_{i_{1}, k_{1}} & \cdots & a_{i_{1}, k_{r}} \\ \vdots & & \vdots \\ a_{i_{r} k_{1}} & \cdots & a_{i_{r}, k_{r}} \end{array} \right]\left[\begin{array}{ccc} b_{k_{1}, j_{1}} & \cdots & b_{k_{1}, j_{r}} \\ \vdots & & \vdots \\ b_{k_{r}, j_{1}} & \cdots & b_{k_{r}, j_{r}} \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0470 (8.6)</td><td>187</td><td>0.977 N|+0</td><td>\|\boldsymbol{x}\|_{p^{\prime}} \leq\|\boldsymbol{x}\|_{p} \leq n^{1 / p-1 / p^{\prime}}\|\boldsymbol{x}\|_{p^{\prime}}, \quad \boldsymbol{x} \in \mathbb{C}^{n}, \quad 1 \leq p \leq p^{\prime} \leq \infty .</td><td>$\|\boldsymbol{x}\|_{p^{\prime}} \leq\|\boldsymbol{x}\|_{p} \leq n^{1 / p-1 / p^{\prime}}\|\boldsymbol{x}\|_{p^{\prime}}, \quad \boldsymbol{x} \in \mathbb{C}^{n}, \quad 1 \leq p \leq p^{\prime} \leq \infty .$</td><td>$\|\boldsymbol{x}\|_{p^{\prime}} \leq\|\boldsymbol{x}\|_{p} \leq n^{1 / p-1 / p^{\prime}}\|\boldsymbol{x}\|_{p^{\prime}}, \quad \boldsymbol{x} \in \mathbb{C}^{n}, \quad 1 \leq p \leq p^{\prime} \leq \infty .$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0969</td><td>327</td><td>■ 0.977 • N|+1</td><td>\begin{gathered} R_{i}=\left\{z \in \mathbb{C}: \left|z-a_{i i}\right| \leq r_{i}\right\}, \quad r_{i}:= \sum_{\substack{j=1 \\ j \neq i}}^{n}\left|a_{i j} \right|, \\ C_{j}=\left\{z \in \mathbb{C}:\left|z-a_{j j}\right| \leq c_{j}\right\}, \quad c_{j}:=\sum_{ \substack{i=1 \\ i \neq j}}^{n}\left|a_{i j}\right| . \end{gathered}</td><td>$$\begin{gathered} R_{i}=\left\{z \in \mathbb{C}: \left|z-a_{i i}\right| \leq r_{i}\right\}, \quad r_{i}:= \sum_{\substack{j=1 \\ j \neq i}}^{n}\left|a_{i j} \right|, \\ C_{j}=\left\{z \in \mathbb{C}:\left|z-a_{j j}\right| \leq c_{j}\right\}, \quad c_{j}:=\sum_{ \substack{i=1 \\ i \neq j}}^{n}\left|a_{i j}\right| . \end{gathered}$$</td><td>$$\begin{gathered} R_{i}=\{z \in \mathbb{C}: |z-a_{i i}| \leq r_{i}\}, \quad r_{i}:= \sum_{\substack{j=1 \\ j \neq i}}^{n}|a_{i j}|, \\ C_{j}=\{z \in \mathbb{C}:|z-a_{j j}| \leq c_{j}\}, \quad c_{j}:=\sum_{ \substack{i=1 \\ i \neq j}}^{n}|a_{i j}| . \end{gathered}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0342</td><td>137</td><td>■ 0.977 • S|+0</td><td>\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 1 & 2 \\ 1 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{Q}= \frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{array}\right], \quad \boldsymbol{R}=\left[\begin{array}{ll} 2 & 2 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{array}\right] .</td><td>$$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 1 & 2 \\ 1 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{Q}= \frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{array}\right], \quad \boldsymbol{R}=\left[\begin{array}{ll} 2 & 2 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{array}\right] .$$</td><td>$$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 1 & 2 \\ 1 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{Q}= \frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{array}\right], \quad \boldsymbol{R}=\left[\begin{array}{ll} 2 & 2 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{array}\right] .$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0418 (7.2)</td><td>168</td><td>■ 0.977 • S|+0</td><td>\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]=\frac{1}{3} \left[\begin{array}{rrr} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{array}\right] \frac{1}{5}\left[\begin{array}{rr} 3 & 4 \\ 4 & -3 \end{array}\right]= \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} .</td><td>$$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]=\frac{1}{3} \left[\begin{array}{rrr} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{array}\right] \frac{1}{5}\left[\begin{array}{rr} 3 & 4 \\ 4 & -3 \end{array}\right]= \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} .$$</td><td>$$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]=\frac{1}{3} \left[\begin{array}{rrr} 1 & 2 & 2 \\ 2 & -2 & 1 \\ 2 & 1 & -2 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{array}\right] \frac{1}{5}\left[\begin{array}{rr} 3 & 4 \\ 4 & -3 \end{array}\right]= \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*} .$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0536</td><td>204</td><td>■ 0.978 • N|+0</td><td>(-a)\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \langle(-a) \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\|-a \boldsymbol{x}+\boldsymbol{y}\|^{2}-\|-a \boldsymbol{x}- \boldsymbol{y}\|^{2}\right)=-\langle a \boldsymbol{x}, \boldsymbol{y}\rangle,</td><td>$$(-a)\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \langle(-a) \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\|-a \boldsymbol{x}+\boldsymbol{y}\|^{2}-\|-a \boldsymbol{x}- \boldsymbol{y}\|^{2}\right)=-\langle a \boldsymbol{x}, \boldsymbol{y}\rangle,$$</td><td>$$(-a)\langle\boldsymbol{x}, \boldsymbol{y}\rangle= \langle(-a) \boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\|-a \boldsymbol{x}+\boldsymbol{y}\|^{2}-\|-a \boldsymbol{x}- \boldsymbol{y}\|^{2}\right)=-\langle a \boldsymbol{x}, \boldsymbol{y}\rangle,$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0767</td><td>272</td><td>■ 0.979 • N|+0</td><td>\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1) \\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{ll} 0 & 1 / 2 \\ 0 & 1 / 4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_{k}(1) \\ \boldsymbol{x}_{k}(2) \end{array}\right]+\left[\begin{array}{l} 1 / 2 \\ 3 / 4 \end{array}\right]=\boldsymbol{G}_{1} \boldsymbol{x}_{k}+ \boldsymbol{c} .</td><td>$$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1) \\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{ll} 0 & 1 / 2 \\ 0 & 1 / 4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_{k}(1) \\ \boldsymbol{x}_{k}(2) \end{array}\right]+\left[\begin{array}{l} 1 / 2 \\ 3 / 4 \end{array}\right]=\boldsymbol{G}_{1} \boldsymbol{x}_{k}+ \boldsymbol{c} .$$</td><td>$$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1) \\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{ll} 0 & 1 / 2 \\ 0 & 1 / 4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_{k}(1) \\ \boldsymbol{x}_{k}(2) \end{array}\right]+\left[\begin{array}{l} 1 / 2 \\ 3 / 4 \end{array}\right]=\boldsymbol{G}_{1} \boldsymbol{x}_{k}+ \boldsymbol{c} .$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0125</td><td>061</td><td>■ 0.979 ● W|+1</td><td><code>\begin{aligned} \boldsymbol{s}_{j}^{T} \boldsymbol{s}_{j} & =\sum_{k=1}^{m} \sin ^{2}(k j \pi h)=\sum_{k=0}^{m} \sin ^{2}(k j \pi h)=\frac{1}{2} \sum_{k=0}^{m}(1-\cos (2 k j \pi h)) \\ & =\frac{m+1}{2}-\frac{1}{2} \sum_{k=0}^{m} \cos (2 k j \pi h)=\frac{m+1}{2}, \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{s}_{j}^{T} \boldsymbol{s}_{j} & =\sum_{k=1}^{m} \sin ^{2}(k j \pi h)=\sum_{k=0}^{m} \sin ^{2}(k j \pi h)=\frac{1}{2} \sum_{k=0}^{m}(1-\cos (2 k j \pi h)) \\ & =\frac{m+1}{2}-\frac{1}{2} \sum_{k=0}^{m} \cos (2 k j \pi h)=\frac{m+1}{2}, \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{s}_{j}^{T} \boldsymbol{s}_{j} & =\sum_{k=1}^{m} \sin ^{2}(k j \pi h)=\sum_{k=0}^{m} \sin ^{2}(k j \pi h)=\frac{1}{2} \sum_{k=0}^{m}\big(1-\cos (2 k j \pi h)\big) \\ & =\frac{m+1}{2}-\frac{1}{2} \sum_{k=0}^{m} \cos (2 k j \pi h)=\frac{m+1}{2}, \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0172</td><td>076</td><td>■ 0.980 ● W|-2</td><td><code>\begin{aligned} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{22}^{(2)} x_{2}+a_{23}^{(2)} x_{3}=b_{2}^{(2)}, & \mathrm{II}^{\prime} \\ a_{33}^{(3)} x_{3}=b_{3}^{(3)}, & \mathrm{III}^{\prime \prime}, \end{aligned}</code></td><td>$\begin{aligned} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}=b_{1}^{(1)}, & \mathrm{I} \\ a_{22}^{(2)} x_{2}+a_{23}^{(2)} x_{3}=b_{2}^{(2)}, & \mathrm{II}^{\prime} \\ a_{33}^{(3)} x_{3}=b_{3}^{(3)}, & \mathrm{III}^{\prime \prime}, \end{aligned}$</td><td>$\begin{aligned} a_{11}^{(1)} x_{1}+a_{12}^{(1)} x_{2}+a_{13}^{(1)} x_{3}&=b_{1}^{(1)}, \quad \mathrm{I} \\ a_{22}^{(2)} x_{2}+a_{23}^{(2)} x_{3}&=b_{2}^{(2)}, \quad \mathrm{II}^{\prime} \\ a_{33}^{(3)} x_{3}&=b_{3}^{(3)}, \quad \mathrm{III}^{\prime \prime}, \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0703</td><td>253</td><td>■ 0.981 ● W|+0</td><td><code>\begin{aligned} \boldsymbol{T s}_{j} & =\lambda_{j} \boldsymbol{s}_{j}, \quad j=1, \ldots, m, \\ \boldsymbol{s}_{j} & =[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T}, \\ \lambda_{j} & =2-2 \cos (j \pi h)=4 \sin ^{2}(j \pi h / 2), \quad h=1 /(m+1), \\ \boldsymbol{s}_{j}^{T} \boldsymbol{s}_{k} & =\delta_{j, k} /(2 h) \text { for all } j, k . \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{T s}_{j} & =\lambda_{j} \boldsymbol{s}_{j}, \quad j=1, \ldots, m, \\ \boldsymbol{s}_{j} & =[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T}, \\ \lambda_{j} & =2-2 \cos (j \pi h)=4 \sin ^{2}(j \pi h / 2), \quad h=1 /(m+1), \\ \boldsymbol{s}_{j}^{T} \boldsymbol{s}_{k} & =\delta_{j, k} /(2 h) \text { for all } j, k . \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{T s}_{j} & =\lambda_{j} \boldsymbol{s}_{j}, \quad j=1, \ldots, m, \\ \boldsymbol{s}_{j} & =[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T}, \\ \lambda_{j} & =2-2 \cos (j \pi h)=4 \sin ^{2}(j \pi h / 2), \quad h=1 /(m+1), \\ \boldsymbol{s}_{j}^{T} \boldsymbol{s}_{k} & =\delta_{j, k} /(2 h) \text { for all } j, k . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0674</td><td>243</td><td>■ 0.981 ● W|+0</td><td><code>\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}=\left[\begin{array}{ccc} \boldsymbol{T}_{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_{1} \end{array} \right]+\left[\begin{array}{ccc} d \boldsymbol{I} & a \boldsymbol{I} & \mathbf{0} \\ a \boldsymbol{I} & d \boldsymbol{I} & a \boldsymbol{I} \\ \mathbf{0} & a \boldsymbol{I} & d \boldsymbol{I} \end{array}\right]= \boldsymbol{T}_{2}</code></td><td>$\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}=\left[\begin{array}{ccc} \boldsymbol{T}_{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_{1} \end{array} \right]+\left[\begin{array}{ccc} d \boldsymbol{I} & a \boldsymbol{I} & \mathbf{0} \\ a \boldsymbol{I} & d \boldsymbol{I} & a \boldsymbol{I} \\ \mathbf{0} & a \boldsymbol{I} & d \boldsymbol{I} \end{array}\right]= \boldsymbol{T}_{2}$</td><td>$\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}=\left[\begin{array}{ccc} \boldsymbol{T}_{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_{1} \end{array} \right]+\left[\begin{array}{ccc} d \boldsymbol{I} & a \boldsymbol{I} & \mathbf{0} \\ a \boldsymbol{I} & d \boldsymbol{I} & a \boldsymbol{I} \\ \mathbf{0} & a \boldsymbol{I} & d \boldsymbol{I} \end{array}\right]= \boldsymbol{T}_{2}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0613 (9.13)</td><td>225</td><td>■ 0.981 ● N|+0</td><td><code>\frac{1}{K(\boldsymbol{A})} \frac{\left\|\boldsymbol{e}_{1}\right\|}{\left\|\boldsymbol{b}_{1} \right\|} \leq \frac{\|\boldsymbol{y}-\boldsymbol{x}\|}{ \|\boldsymbol{x}\|} \leq K(\boldsymbol{A}) \frac{\left \|\boldsymbol{e}_{1}\right\|}{\left\|\boldsymbol{b}_{1} \right\|}, \quad K(\boldsymbol{A})=\|\boldsymbol{A}\| \left\|\boldsymbol{A}^{\dagger}\right\| .</code></td><td>$\frac{1}{K(\boldsymbol{A})} \frac{\left\|\boldsymbol{e}_{1}\right\|}{\left\|\boldsymbol{b}_{1} \right\|} \leq \frac{\|\boldsymbol{y}-\boldsymbol{x}\|}{ \|\boldsymbol{x}\|} \leq K(\boldsymbol{A}) \frac{\left \|\boldsymbol{e}_{1}\right\|}{\left\|\boldsymbol{b}_{1} \right\|}, \quad K(\boldsymbol{A})=\|\boldsymbol{A}\| \left\|\boldsymbol{A}^{\dagger}\right\| .$</td><td>$\frac{1}{K(\boldsymbol{A})} \frac{\|\boldsymbol{e}_{1}\|}{\|\boldsymbol{b}_{1}\|} \leq \frac{\|\boldsymbol{y}-\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{e}_{1}\|}{\|\boldsymbol{b}_{1}\|}, \quad K(\boldsymbol{A})=\|\boldsymbol{A}\|\|\boldsymbol{A}^{\dagger}\|.$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ1020 (15.3)</td><td>345</td><td>■ 0.981 ● N|+0</td><td><code>\boldsymbol{z}_{k}=c_{1} \lambda_{1}^{k} \boldsymbol{v}_{1}+c_{2} \lambda_{2}^{k} \boldsymbol{v}_{2}+\cdots+c_{n} \lambda_{n}^{k} \boldsymbol{v}_{n}, \quad k=0,1,2, \ldots .</code></td><td>$\boldsymbol{z}_{k}=c_{1} \lambda_{1}^{k} \boldsymbol{v}_{1}+c_{2} \lambda_{2}^{k} \boldsymbol{v}_{2}+\cdots+c_{n} \lambda_{n}^{k} \boldsymbol{v}_{n}, \quad k=0,1,2, \ldots .$</td><td>$\boldsymbol{z}_{k}=c_{1} \lambda_{1}^{k} \boldsymbol{v}_{1}+c_{2} \lambda_{2}^{k} \boldsymbol{v}_{2}+\cdots+c_{n} \lambda_{n}^{k} \boldsymbol{v}_{n}, \quad k=0,1,2, \ldots .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1021 (15.4)</td><td>345</td><td>■ 0.982 ● N|+1</td><td><code>\frac{z_{k}}{\lambda_{1}^{k}}=c_{1} \boldsymbol{v}_{1}+c_{2}\left(\frac{\lambda_{2}}{ \lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+c_{n} \left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}, \quad k=0,1,2, \ldots</code></td><td>$\frac{z_{k}}{\lambda_{1}^{k}}=c_{1} \boldsymbol{v}_{1}+c_{2}\left(\frac{\lambda_{2}}{ \lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+c_{n} \left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}, \quad k=0,1,2, \ldots$</td><td>$\frac{\boldsymbol{z}_{k}}{\lambda_{1}^{k}}=c_{1} \boldsymbol{v}_{1}+c_{2}\left(\frac{\lambda_{2}}{ \lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+c_{n} \left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}, \quad k=0,1,2, \ldots .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0497</td><td>194</td><td>■ 0.982 ● N|+0</td><td><code>\left|x_{i}\right| \leq\left|y_{i}\right|, i=1, \ldots, n \Longrightarrow\|\boldsymbol{x}\| \leq\|\boldsymbol{y}\| \text {, for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n} .</code></td><td>$\left|x_{i}\right| \leq\left|y_{i}\right|, i=1, \ldots, n \Longrightarrow\|\boldsymbol{x}\| \leq\|\boldsymbol{y}\| \text {, for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n} .$</td><td>$|x_{i}| \leq |y_{i}|, \ i=1, \ldots, n \Longrightarrow\|\boldsymbol{x}\| \leq\|\boldsymbol{y}\|, \text { for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^{n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0537</td><td>204</td><td>■ 0.983 ● N|+0</td><td><code>\|\boldsymbol{x}\|_{\boldsymbol{A}} \leq\|\boldsymbol{y}\|_{\boldsymbol{A}} \Longrightarrow\|\boldsymbol{x}\|_{2} \leq \sqrt{\frac{\lambda_{1}}{\lambda_{n}}}\|\boldsymbol{y}\|_{2},</code></td><td>$\|\boldsymbol{x}\|_{\boldsymbol{A}} \leq\|\boldsymbol{y}\|_{\boldsymbol{A}} \Longrightarrow\|\boldsymbol{x}\|_{2} \leq \sqrt{\frac{\lambda_{1}}{\lambda_{n}}}\|\boldsymbol{y}\|_{2},$</td><td>$\|\boldsymbol{x}\|_{\boldsymbol{A}} \leq\|\boldsymbol{y}\|_{\boldsymbol{A}} \quad\Longrightarrow\quad \|\boldsymbol{x}\|_{2} \leq \sqrt{\frac{\lambda_{1}}{\lambda_{n}}}\|\boldsymbol{y}\|_{2},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0573</td><td>212</td><td>■ 0.984 ● N|+0</td><td><code>\|\boldsymbol{x}\|_{p} \leq\|\boldsymbol{x}\|_{q} \leq n^{1 / q-1 / p}\|\boldsymbol{x}\|_{p}, \quad \boldsymbol{x} \in \mathbb{C}^{n} .</code></td><td>$\|\boldsymbol{x}\|_{p} \leq\|\boldsymbol{x}\|_{q} \leq n^{1 / q-1 / p}\|\boldsymbol{x}\|_{p}, \quad \boldsymbol{x} \in \mathbb{C}^{n} .$</td><td>$\|\boldsymbol{x}\|_{p} \leq\|\boldsymbol{x}\|_{q} \leq n^{1 / q-1 / p}\|\boldsymbol{x}\|_{p}, \quad \boldsymbol{x} \in \mathbb{C}^{n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0184 (3.4)</td><td>079</td><td>■ 0.984 ● W|+0</td><td><code>\boldsymbol{L}=\left[\begin{array}{cccc} 1 & & & \\ l_{21}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ l_{n 1}^{(1)} & l_{n 2}^{(2)} & \cdots & 1 \end{array} \right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)} & \cdots & a_{1 n}^{(1)} \\ & \ddots & \vdots \\ & & a_{n n}^{(n)} \end{array}\right],</code></td><td>$\boldsymbol{L}=\left[\begin{array}{cccc} 1 & & & \\ l_{21}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ l_{n 1}^{(1)} & l_{n 2}^{(2)} & \cdots & 1 \end{array} \right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)} & \cdots & a_{1 n}^{(1)} \\ & \ddots & \vdots \\ & & a_{n n}^{(n)} \end{array}\right],$</td><td>$\boldsymbol{L}=\left[\begin{array}{cccc} 1 & & & \\ l_{21}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ l_{n 1}^{(1)} & l_{n 2}^{(2)} & \cdots & 1 \end{array} \right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{11}^{(1)} & \cdots & a_{1 n}^{(1)} \\ & \ddots & \vdots \\ & & a_{n n}^{(n)} \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0019 (1.7)</td><td>029</td><td>■ 0.984 ● N|+0</td><td><code>\operatorname{dim}(\mathcal{S}+\mathcal{T})= \operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S} \cap \mathcal{T}) .</code></td><td>$\operatorname{dim}(\mathcal{S}+\mathcal{T})= \operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S} \cap \mathcal{T}) .$</td><td>$\operatorname{dim}(\mathcal{S}+\mathcal{T})= \operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S} \cap \mathcal{T}) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0502 (8.23)</td><td>196</td><td>■ 0.984 ● N|+0</td><td><code>\frac{1}{K(\boldsymbol{A})} \frac{\|\boldsymbol{r}(\boldsymbol{y})\|}{\|\boldsymbol{b}\|} \leq \frac{\| \boldsymbol{y}-\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{r}(\boldsymbol{y}) \|}{\|\boldsymbol{b}\|} .</code></td><td>$\frac{1}{K(\boldsymbol{A})} \frac{\|\boldsymbol{r}(\boldsymbol{y})\|}{\|\boldsymbol{b}\|} \leq \frac{\| \boldsymbol{y}-\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{r}(\boldsymbol{y}) \|}{\|\boldsymbol{b}\|} .$</td><td>$\frac{1}{K(\boldsymbol{A})} \frac{\|\boldsymbol{r}(\boldsymbol{y})\|}{\|\boldsymbol{b}\|} \leq \frac{\| \boldsymbol{y}-\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{r}(\boldsymbol{y}) \|}{\|\boldsymbol{b}\|} .$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0740	262	■ 0.984 ● W|+1	`x_{j, k}=\frac{h^{4} g_{j, k}}{\sigma_{j}+\sigma_{k}-\frac{2}{3} \sigma_{j} \sigma_{k}}, \text { where } \sigma_{j}=\sin ^{2}((j \pi h) / 2) \text { for } j, k=1,2, \ldots, m .`	$x_{j, k}=\frac{h^{4} g_{j, k}}{\sigma_{j}+\sigma_{k}-\frac{2}{3} \sigma_{j} \sigma_{k}}, \text { where } \sigma_{j}=\sin ^{2}((j \pi h) / 2) \text { for } j, k=1,2, \ldots, m .$	$x_{j,k}=\frac{h^4 g_{j,k}}{\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k}, \text{ where } \sigma_j=\sin^2\big((j\pi h)/2\big) \text{ for } j,k=1,2,\ldots,m.$
lyche-numerical-linear-algebra_ EQ0114	059	■ 0.985 ● N|+0	`u^{\prime \prime}(j h) \approx \frac{u((j+1) h)-2 u(j h)+u((j-1) h)}{h^{2}}, \quad j=1, \ldots, m,`	$u^{\prime \prime}(j h) \approx \frac{u((j+1) h)-2 u(j h)+u((j-1) h)}{h^{2}}, \quad j=1, \ldots, m,$	$u''(jh)\approx\frac{u((j+1)h)-2u(jh)+u((j-1)h)}{h^2}, \quad j=1,\ldots,m,$
lyche-numerical-linear-algebra_ EQ0564	210	■ 0.985 ● N|+0	`A:=\left[\begin{array}{cccccc} 2 & \mu_{1} & 0 & \cdots & 0 & \lambda_{1} \\ \lambda_{2} & 2 & \mu_{2} & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_{n} & 0 & \cdots & 0 & \lambda_{n} & 2 \end{array}\right],`	$A:=\left[\begin{array}{cccccc} 2 & \mu_{1} & 0 & \cdots & 0 & \lambda_{1} \\ \lambda_{2} & 2 & \mu_{2} & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_{n} & 0 & \cdots & 0 & \lambda_{n} & 2 \end{array}\right],$	$\boldsymbol{A}:=\left[\begin{array}{cccccc} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0701 (11.3)	252	■ 0.985 ● W|+0	`\begin{aligned} \boldsymbol{y}_{1} & =\boldsymbol{b}_{1}, \quad \boldsymbol{y}_{k}=\boldsymbol{b}_{k}-\boldsymbol{L}_{k-1} \boldsymbol{y}_{k-1}, \quad k=2,3, \ldots, m, \\ \boldsymbol{x}_{m} & =\boldsymbol{U}_{m}^{-1} \boldsymbol{y}_{m}, \quad \boldsymbol{x}_{k}=\boldsymbol{U}_{k}^{-1} \left(\boldsymbol{y}_{k}-\boldsymbol{C}_{k} \boldsymbol{x}_{k+1}\right), \quad k=m-1, \ldots, 2,1 . \end{aligned}`	$\begin{aligned} \boldsymbol{y}_{1} & =\boldsymbol{b}_{1}, \quad \boldsymbol{y}_{k}=\boldsymbol{b}_{k}-\boldsymbol{L}_{k-1} \boldsymbol{y}_{k-1}, \quad k=2,3, \ldots, m, \\ \boldsymbol{x}_{m} & =\boldsymbol{U}_{m}^{-1} \boldsymbol{y}_{m}, \quad \boldsymbol{x}_{k}=\boldsymbol{U}_{k}^{-1} \left(\boldsymbol{y}_{k}-\boldsymbol{C}_{k} \boldsymbol{x}_{k+1}\right), \quad k=m-1, \ldots, 2,1 . \end{aligned}$	$\begin{aligned} \boldsymbol{y}_1&=\boldsymbol{b}_1, \quad \boldsymbol{y}_k=\boldsymbol{b}_k-\boldsymbol{L}_{k-1}\boldsymbol{y}_{k-1}, \quad k=2,3,\ldots,m, \\ \boldsymbol{x}_m&=\boldsymbol{U}_m^{-1}\boldsymbol{y}_m, \quad \boldsymbol{x}_k=\boldsymbol{U}_k^{-1}(\boldsymbol{y}_k-\boldsymbol{C}_k\boldsymbol{x}_{k+1}), \quad k=m-1,\ldots,2,1. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0848 (13.12)	294	■ 0.985 ● W|+0	`\begin{aligned} \boldsymbol{p}_{j} & := \boldsymbol{r}_{j}-\sum_{i=0}^{j-1}\left(\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}\right) \boldsymbol{p}_{i}, \\ \boldsymbol{x}_{j+1} & := \boldsymbol{x}_{j}+\alpha_{j} \boldsymbol{p}_{j} \quad \alpha_{j}:=\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{r}_{j}}{\boldsymbol{p}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{j}}, \\ \boldsymbol{r}_{j+1} & = \boldsymbol{r}_{j}-\alpha_{j} \boldsymbol{A} \boldsymbol{p}_{j} . \end{aligned}`	$\begin{aligned} \boldsymbol{p}_{j} & := \boldsymbol{r}_{j}-\sum_{i=0}^{j-1}\left(\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}\right) \boldsymbol{p}_{i}, \\ \boldsymbol{x}_{j+1} & := \boldsymbol{x}_{j}+\alpha_{j} \boldsymbol{p}_{j} \quad \alpha_{j}:=\frac{\boldsymbol{r}_{j}^{T} \boldsymbol{r}_{j}}{\boldsymbol{p}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{j}}, \\ \boldsymbol{r}_{j+1} & = \boldsymbol{r}_{j}-\alpha_{j} \boldsymbol{A} \boldsymbol{p}_{j} . \end{aligned}$	$\begin{aligned} \boldsymbol{p}_j&:=\boldsymbol{r}_j-\sum_{i=0}^{j-1}(\frac{\boldsymbol{r}_j^T\boldsymbol{A}\boldsymbol{p}_i}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i})\boldsymbol{p}_i, \\ \boldsymbol{x}_{j+1}&:=\boldsymbol{x}_j+\alpha_j\boldsymbol{p}_j \quad \alpha_j:=\frac{\boldsymbol{r}_j^T\boldsymbol{r}_j}{\boldsymbol{p}_j^T\boldsymbol{A}\boldsymbol{p}_j}, \\ \boldsymbol{r}_{j+1}&=\boldsymbol{r}_j-\alpha_j\boldsymbol{A}\boldsymbol{p}_j. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0210	087	■ 0.985 ● N|+0	`\begin{array}{r} 10^{-4} x_{1}+2 x_{2}=4 \\ x_{1}+x_{2}=3 \end{array}`	$\begin{array}{r} 10^{-4} x_{1}+2 x_{2}=4 \\ x_{1}+x_{2}=3 \end{array}$	$\begin{array}{r} 10^{-4}x_1+2x_2=4 \\ x_1+x_2=3 \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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EQ0429</td><td>172</td><td>■ 0.985
● K|+0</td><td>\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{array}\right] .</td><td>$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{array}\right] .$</td><td>$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{array}\right] .$</td></tr>
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EQ0145</td><td>065</td><td>■ 0.985
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EQ0622 (9.17)</td><td>227</td><td>■ 0.985
● W|-1</td><td>\sigma_{1}=\max _{\substack{\boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\| \boldsymbol{x}\|_{2}}, \quad \sigma_{n}=\min _{\substack{ \boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x} \|_{2}}{\|\boldsymbol{x}\|_{2}} .</td><td>$\sigma_{1}=\max _{\substack{\boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}, \quad \sigma_{n}=\min _{\substack{\boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} .$</td><td>$\sigma_{1}=\max _{\substack{\boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}}, \quad \sigma_{n}=\min _{\substack{\boldsymbol{x} \in \mathbb{C}^{n} \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\|\boldsymbol{A} \boldsymbol{x}\|_{2}}{\|\boldsymbol{x}\|_{2}} .$</td></tr>
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● N|+0</td><td>\boldsymbol{L} \boldsymbol{L}^{*}=\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1} \end{array}\right]\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right]=\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{B} \end{array} \right]=\boldsymbol{A} .</td><td>$\boldsymbol{L} \boldsymbol{L}^{*}=\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1} \end{array}\right]\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right]=\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{B} \end{array}\right]=\boldsymbol{A} .$</td><td>$\boldsymbol{L} \boldsymbol{L}^{*}=\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1} \end{array}\right]\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right]=\left[\begin{array}{ll} 0 & \mathbf{0}^{*} \\ \mathbf{0} & \boldsymbol{B} \end{array}\right]=\boldsymbol{A} .$</td></tr>
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EQ0205 (3.14)</td><td>086</td><td>■ 0.986
● N|+0</td><td>\boldsymbol{L}=\left[\begin{array}{ccc} 1 & & \\ \ a_{p_{2}, 1}^{(1)} & 1 & \\ \vdots & & \ddots \\ \ a_{p_{n}, 1}^{(1)} & a_{p_{n}, 2}^{(2)} & \cdots 1 \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{p_{1}, 1}^{(1)} & \cdots & a_{p_{1}, n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_{n}, n}^{(n)} \end{array}\right] .</td><td>$\boldsymbol{L}=\left[\begin{array}{ccc} 1 & & \\ a_{p_{2}, 1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_{n}, 1}^{(1)} & a_{p_{n}, 2}^{(2)} & \cdots 1 \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{p_{1}, 1}^{(1)} & \cdots & a_{p_{1}, n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_{n}, n}^{(n)} \end{array}\right] .$</td><td>$\boldsymbol{L}=\left[\begin{array}{ccc} 1 & & \\ a_{p_{2}, 1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_{n}, 1}^{(1)} & a_{p_{n}, 2}^{(2)} & \cdots 1 \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{p_{1}, 1}^{(1)} & \cdots & a_{p_{1}, n}^{(1)} \\ & \ddots & \vdots \\ & & a_{p_{n}, n}^{(n)} \end{array}\right] .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra EQ0995</td><td>336</td><td>0.986 N|-1</td><td>\boldsymbol{A}_{1}=\left[\begin{array}{ccccc} d_{1} & c_{1} & & & \\ c_{1} & d_{2} & c_{2} & & \\ & \ddots & \ddots & \ddots & \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_{i} \end{array}\right] \text { and } \boldsymbol{A}_{2}=\left[\begin{array}{ccccc} d_{i+1} & c_{i+1} & & & \\ c_{i+1} & d_{i+2} & c_{i+2} & & \\ & \ddots & \ddots & \ddots & \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_{n} \end{array}\right] .</td><td>$\boldsymbol{A}_{1}=\left[\begin{array}{ccccc} d_{1} & c_{1} & & & \\ c_{1} & d_{2} & c_{2} & & \\ & \ddots & \ddots & \ddots & \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_{i} \end{array}\right] \text { and } \boldsymbol{A}_{2}=\left[\begin{array}{ccccc} d_{i+1} & c_{i+1} & & & \\ c_{i+1} & d_{i+2} & c_{i+2} & & \\ & \ddots & \ddots & \ddots & \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_{n} \end{array}\right] .$</td><td>$\boldsymbol{A}_{1}=\left[\begin{array}{ccccc} d_{1} & c_{1} & & & \\ c_{1} & d_{2} & c_{2} & & \\ & \ddots & \ddots & \ddots & \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_{i} \end{array}\right] \text { and } \boldsymbol{A}_{2}=\left[\begin{array}{ccccc} d_{i+1} & c_{i+1} & & & \\ c_{i+1} & d_{i+2} & c_{i+2} & & \\ & \ddots & \ddots & \ddots & \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_{n} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0300</td><td>124</td><td>0.987 N|+0</td><td>\langle U x, U y\rangle=(U y)^{*}(U x)=y^{*} U^{*} U x=y^{*} x=\langle x, y\rangle .</td><td>$\langle U x, U y\rangle=(U y)^{*}(U x)=y^{*} U^{*} U x=y^{*} x=\langle x, y\rangle .$</td><td>$\langle \boldsymbol{U x}, \boldsymbol{U y}\rangle=(\boldsymbol{U y})^{*}(\boldsymbol{U x})=\boldsymbol{y}^{*} \boldsymbol{U}^{*} \boldsymbol{U x}=\boldsymbol{y}^{*} \boldsymbol{x}=\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0426 (7.5)</td><td>172</td><td>0.987 K|+0</td><td>\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}=\boldsymbol{U}_{1} \boldsymbol{ \Sigma}_{1} \boldsymbol{V}_{1}^{*} .</td><td>$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}=\boldsymbol{U}_{1} \boldsymbol{ \Sigma}_{1} \boldsymbol{V}_{1}^{*} .$</td><td>$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{*}=\boldsymbol{U}_{1} \boldsymbol{\Sigma}_{1} \boldsymbol{V}_{1}^{*} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0397 (6.15)</td><td>159</td><td>0.987 N|+1</td><td>\sum_{j=1}^{n}\left|\mu_{i_{j}}-\lambda_{j}\right|^{2} \leq \sum_{i=1}^{n} \sum_{j=1}^{n}\left|a_{i j}-b_{i j}\right|^{2}</td><td>$\sum_{j=1}^{n}\left|\mu_{i_{j}}-\lambda_{j}\right|^{2} \leq \sum_{i=1}^{n} \sum_{j=1}^{n}\left|a_{i j}-b_{i j}\right|^{2}$</td><td>$\sum_{j=1}^{n}\left|\mu_{i_{j}}-\lambda_{j}\right|^{2} \leq \sum_{i=1}^{n} \sum_{j=1}^{n}\left|a_{i j}-b_{i j}\right|^{2}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0511</td><td>199</td><td>0.987 N|+1</td><td>\|\boldsymbol{x}\|_{p}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{p}\right)^{1 / p}, \quad\|\boldsymbol{x}\|_{\infty}:=\max _{1 \leq j \leq n}\left|x_{j}\right| .</td><td>$\|\boldsymbol{x}\|_{p}:=\left(\sum_{j=1}^{n}\left|x_{j}\right|^{p}\right)^{1 / p}, \quad\|\boldsymbol{x}\|_{\infty}:=\max _{1 \leq j \leq n}\left|x_{j}\right| .$</td><td>$\|\boldsymbol{x}\|_{p}:=\big(\sum_{j=1}^{n}|x_{j}|^{p}\big)^{1 / p}, \quad\|\boldsymbol{x}\|_{\infty}:=\max _{1 \leq j \leq n}|x_{j}| .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0139</td><td>064</td><td>0.987 W|+0</td><td>\boldsymbol{A} \boldsymbol{B}=\left[\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right] \left[\begin{array}{l} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{array}\right], \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{l} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{array}\right]\right] .</td><td>$\boldsymbol{A} \boldsymbol{B}=\left[\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right] \left[\begin{array}{l} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{array}\right], \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{l} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{array}\right]\right] .$</td><td>$\boldsymbol{A} \boldsymbol{B}=\left[\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right] \left[\begin{array}{l} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{array}\right], \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]\left[\begin{array}{l} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{array}\right]\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0164 (2.38)</td><td>073</td><td>0.987 C|-4</td><td>\boldsymbol{L}=\left[\begin{array}{ccccc} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{array}\right], \boldsymbol{U}=\left[\begin{array}{ccccc} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \frac{m}{m-1} & -1 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{array} \right] .</td><td>$\boldsymbol{L}=\left[\begin{array}{ccccc} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{array}\right], \boldsymbol{U}=\left[\begin{array}{ccccc} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \frac{m}{m-1} & -1 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{array} \right] .$</td><td>$\boldsymbol{L}=\left[\begin{array}{ccccc} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{array}\right], \boldsymbol{U}=\left[\begin{array}{ccccc} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \frac{m}{m-1} & -1 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{array} \right] .$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0572</td><td>212</td><td>■ 0.988
● K|+0</td><td>S_{p}=\left\{\boldsymbol{x} \in \mathbb{R}^{n}:\| \boldsymbol{x}\|_{p}=1\right\}</td><td>$S_{p}=\left\{\boldsymbol{x} \in \mathbb{R}^{n}:\| \boldsymbol{x}\|_{p}=1\right\}$</td><td>$S_p = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_p = 1\}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0949 (13.62)</td><td>321</td><td>■ 0.988
● W|+0</td><td>\begin{aligned} \langle\boldsymbol{x}, \boldsymbol{y} \rangle & =\langle\boldsymbol{y}, \boldsymbol{x}\rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \\ \langle\boldsymbol{C} \boldsymbol{x}, \boldsymbol{y} \rangle & =\left\langle\boldsymbol{y}, \boldsymbol{C}^{T} \boldsymbol{x}\right\rangle \stackrel{(13.61)}{=}\left\langle\boldsymbol{C}^{T} \boldsymbol{x}, \boldsymbol{y} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \boldsymbol{C} \in \mathbb{R}^{n \times n} . \end{aligned}</td><td>$$\begin{aligned} \langle\boldsymbol{x}, \boldsymbol{y} \rangle & =\langle\boldsymbol{y}, \boldsymbol{x}\rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \\ \langle\boldsymbol{C} \boldsymbol{x}, \boldsymbol{y} \rangle & =\left\langle\boldsymbol{y}, \boldsymbol{C}^{T} \boldsymbol{x}\right\rangle \stackrel{(13.61)}{=}\left\langle\boldsymbol{C}^{T} \boldsymbol{x}, \boldsymbol{y} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \boldsymbol{C} \in \mathbb{R}^{n \times n} . \end{aligned}$$</td><td>$$\begin{aligned} \langle\boldsymbol{x}, \boldsymbol{y}\rangle &= \langle\boldsymbol{y}, \boldsymbol{x}\rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \\ \langle\boldsymbol{C}\boldsymbol{x}, \boldsymbol{y}\rangle &= \langle\boldsymbol{y}, \boldsymbol{C}^T\boldsymbol{x}\rangle \stackrel{(13.61)}{=} \langle\boldsymbol{C}^T\boldsymbol{x}, \boldsymbol{y}\rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n\times n}. \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0166</td><td>073</td><td>■ 0.988
● N|+0</td><td>\boldsymbol{A} \boldsymbol{B}^{T}=\boldsymbol{a}_{: 1} \boldsymbol{b}_{: 1}^{T}+\boldsymbol{a}_{: 2} \boldsymbol{b}_{: 2}^{T}+\cdots+\boldsymbol{a}_{: n} \boldsymbol{b}_{: n}^{T} .</td><td>$\boldsymbol{A} \boldsymbol{B}^{T}=\boldsymbol{a}_{: 1} \boldsymbol{b}_{: 1}^{T}+\boldsymbol{a}_{: 2} \boldsymbol{b}_{: 2}^{T}+\cdots+\boldsymbol{a}_{: n} \boldsymbol{b}_{: n}^{T} .$</td><td>$\boldsymbol{A}\boldsymbol{B}^T = \boldsymbol{a}_{:1}\boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2}\boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n}\boldsymbol{b}_{:n}^T.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0589</td><td>217</td><td>■ 0.989
● W|+2</td><td>\boldsymbol{H}_{3}=\left[\begin{array}{lll} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array}\right] .</td><td>$\boldsymbol{H}_{3}=\left[\begin{array}{lll} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{array}\right] .$</td><td>$\boldsymbol{H}_3 = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0911 (13.44)</td><td>310</td><td>■ 0.989
● N|+0</td><td>\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^{T}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{C}^{-T} \boldsymbol{x}=\boldsymbol{C}^{T} \boldsymbol{C} \boldsymbol{b} \Leftrightarrow\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{y}= \boldsymbol{C} \boldsymbol{b},</td><td>$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^{T}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{C}^{-T} \boldsymbol{x}=\boldsymbol{C}^{T} \boldsymbol{C} \boldsymbol{b} \Leftrightarrow\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{y}= \boldsymbol{C} \boldsymbol{b},$</td><td>$\boldsymbol{B}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{B}\boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T)\boldsymbol{C}^{-T}\boldsymbol{x} = \boldsymbol{C}^T\boldsymbol{C}\boldsymbol{b} \Leftrightarrow (\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T)\boldsymbol{y} = \boldsymbol{C}\boldsymbol{b},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0500</td><td>195</td><td>■ 0.989
● K|+0</td><td>\|\boldsymbol{A} \boldsymbol{B}\| \leq\|\boldsymbol{A} \|\|\boldsymbol{B}\| \text { and }\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\|\|\boldsymbol{x}\| .</td><td>$\|\boldsymbol{A} \boldsymbol{B}\| \leq\|\boldsymbol{A} \|\|\boldsymbol{B}\| \text { and }\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\|\|\boldsymbol{x}\| .$</td><td>$\|\boldsymbol{A}\boldsymbol{B}\| \le \|\boldsymbol{A}\|\,\|\boldsymbol{B}\| \text{ and } \|\boldsymbol{A}\boldsymbol{x}\| \le \|\boldsymbol{A}\|\,\|\boldsymbol{x}\|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0137</td><td>064</td><td>■ 0.989
● W|+1</td><td>\begin{aligned} (\boldsymbol{A} \boldsymbol{B})_{i j} & =\sum_{k=1}^{p} a_{i k} b_{k j}=\sum_{k=1}^{s} a_{i k} b_{k j}+\sum_{k=s+1}^{p} a_{i k} b_{k j} \\ & =\left(\boldsymbol{A}_{1} \boldsymbol{B}_{1}\right)_{i j}+\left(\boldsymbol{A}_{2} \boldsymbol{B}_{2}\right)_{i j}=\left(\boldsymbol{A}_{1} \boldsymbol{B}_{1}+\boldsymbol{A}_{2} \boldsymbol{B}_{2}\right)_{i j} . \end{aligned}</td><td>$$\begin{aligned} (\boldsymbol{A} \boldsymbol{B})_{i j} & =\sum_{k=1}^{p} a_{i k} b_{k j}=\sum_{k=1}^{s} a_{i k} b_{k j}+\sum_{k=s+1}^{p} a_{i k} b_{k j} \\ & =\left(\boldsymbol{A}_{1} \boldsymbol{B}_{1}\right)_{i j}+\left(\boldsymbol{A}_{2} \boldsymbol{B}_{2}\right)_{i j}=\left(\boldsymbol{A}_{1} \boldsymbol{B}_{1}+\boldsymbol{A}_{2} \boldsymbol{B}_{2}\right)_{i j} . \end{aligned}$$</td><td>$$\begin{aligned} (\boldsymbol{A}\boldsymbol{B})_{ij} &= \sum_{k=1}^p a_{ik}b_{kj} = \sum_{k=1}^s a_{ik}b_{kj} + \sum_{k=s+1}^p a_{ik}b_{kj} \\ &= (\boldsymbol{A}_1\boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2\boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1\boldsymbol{B}_1 + \boldsymbol{A}_2\boldsymbol{B}_2)_{ij}. \end{aligned}$$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0436 (7.7)	174	■0.989 • S|+0	`\begin{aligned} \boldsymbol{A} \boldsymbol{v}_{i} & =\sigma_{i} \boldsymbol{u}_{i}, i=1, \ldots, r, \quad \boldsymbol{A} \boldsymbol{v}_{i}=0, i=r+1, \ldots, n, \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} & =\sigma_{i} \boldsymbol{v}_{i}, i=1, \ldots, r, \quad \boldsymbol{A}^{*} \boldsymbol{u}_{i}=0, i=r+1, \ldots, m . \end{aligned}`	$\begin{aligned} \boldsymbol{A} \boldsymbol{v}_{i} & =\sigma_{i} \boldsymbol{u}_{i}, i=1, \ldots, r, \quad \boldsymbol{A} \boldsymbol{v}_{i}=0, i=r+1, \ldots, n, \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} & =\sigma_{i} \boldsymbol{v}_{i}, i=1, \ldots, r, \quad \boldsymbol{A}^{*} \boldsymbol{u}_{i}=0, i=r+1, \ldots, m . \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{v}_{i} & =\sigma_{i} \boldsymbol{u}_{i},\ i=1, \ldots, r, \quad \boldsymbol{A} \boldsymbol{v}_{i}=0,\ i=r+1, \ldots, n, \\ \boldsymbol{A}^{*} \boldsymbol{u}_{i} & =\sigma_{i} \boldsymbol{v}_{i},\ i=1, \ldots, r, \quad \boldsymbol{A}^{*} \boldsymbol{u}_{i}=0,\ i=r+1, \ldots, m . \end{aligned}$
lyche-numerical-linear-algebra EQ0001	021	■0.990 • K|+0	`\boldsymbol{x}=\left[\begin{array}{c} x_{1} \\ x_{2} \\ \vdots \\ x_{n} \end{array}\right],`	$\boldsymbol{x}=\left[\begin{array}{c} x_{1} \\ x_{2} \\ \vdots \\ x_{n} \end{array}\right],$	$\boldsymbol{x}=\left[\begin{array}{c} x_{1} \\ x_{2} \\ \vdots \\ x_{n} \end{array}\right],$
lyche-numerical-linear-algebra EQ0060	041	■0.990 • K|+0	`\left(\boldsymbol{A}+\boldsymbol{B} \boldsymbol{C}^{T} \right)^{-1}=\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1} \boldsymbol{B}\left(\boldsymbol{I}+\boldsymbol{C}^{T} \boldsymbol{A}^{-1} \boldsymbol{B}\right)^{-1} \boldsymbol{C}^{T} \boldsymbol{A}^{-1} .`	$\left(\boldsymbol{A}+\boldsymbol{B} \boldsymbol{C}^{T} \right)^{-1}=\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1} \boldsymbol{B}\left(\boldsymbol{I}+\boldsymbol{C}^{T} \boldsymbol{A}^{-1} \boldsymbol{B}\right)^{-1} \boldsymbol{C}^{T} \boldsymbol{A}^{-1} .$	$(\boldsymbol{A}+\boldsymbol{B} \boldsymbol{C}^{T})^{-1}=\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1} \boldsymbol{B}(\boldsymbol{I}+\boldsymbol{C}^{T} \boldsymbol{A}^{-1} \boldsymbol{B})^{-1} \boldsymbol{C}^{T} \boldsymbol{A}^{-1} .$
lyche-numerical-linear-algebra EQ0231 (3.21)	092	■0.990 • S|+1	`\begin{aligned} \boldsymbol{A} & =\left[\begin{array}{cc} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^{T} & \boldsymbol{A}_{m m} \end{array} \right] \\ & =\left[\begin{array}{cc} \boldsymbol{L}_{\{m-1\}} & \mathbf{0} \\ \boldsymbol{C}^{T} \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1 \}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{m m}- \boldsymbol{C}^{T} \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{array} \right], \end{aligned}`	$\begin{aligned} \boldsymbol{A} & =\left[\begin{array}{cc} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^{T} & \boldsymbol{A}_{m m} \end{array} \right] \\ & =\left[\begin{array}{cc} \boldsymbol{L}_{\{m-1\}} & \mathbf{0} \\ \boldsymbol{C}^{T} \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1 \}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{m m}- \boldsymbol{C}^{T} \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{array} \right], \end{aligned}$	$\begin{aligned} \boldsymbol{A} & =\left[\begin{array}{cc} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^{T} & \boldsymbol{A}_{m m} \end{array} \right] \\ & =\left[\begin{array}{cc} \boldsymbol{L}_{\{m-1\}} & \mathbf{0} \\ \boldsymbol{C}^{T} \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1 \}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{m m}- \boldsymbol{C}^{T} \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{array} \right], \end{aligned}$
lyche-numerical-linear-algebra EQ0293 (5.9)	122	■0.990 • N|+1	`\left\langle\boldsymbol{s}_{0}, \boldsymbol{s}\right \rangle=\langle\boldsymbol{v}, \boldsymbol{s}\rangle, \quad \text { for all } \boldsymbol{s} \in \mathcal{S}, \quad\left\langle\boldsymbol{t}_{0}, \boldsymbol{t}\right \rangle=\langle\boldsymbol{v}, \boldsymbol{t}\rangle, \quad \text { for all } \boldsymbol{t} \in \mathcal{T} .`	$\left\langle\boldsymbol{s}_{0}, \boldsymbol{s}\right \rangle=\langle\boldsymbol{v}, \boldsymbol{s}\rangle, \quad \text { for all } \boldsymbol{s} \in \mathcal{S}, \quad\left\langle\boldsymbol{t}_{0}, \boldsymbol{t}\right \rangle=\langle\boldsymbol{v}, \boldsymbol{t}\rangle, \quad \text { for all } \boldsymbol{t} \in \mathcal{T} .$	$\langle\boldsymbol{s}_{0}, \boldsymbol{s}\rangle=\langle\boldsymbol{v}, \boldsymbol{s}\rangle, \quad \textit{for all } \boldsymbol{s} \in \mathcal{S}, \quad\langle\boldsymbol{t}_{0}, \boldsymbol{t}\rangle=\langle\boldsymbol{v}, \boldsymbol{t}\rangle, \quad \textit{for all } \boldsymbol{t} \in \mathcal{T} .$
lyche-numerical-linear-algebra EQ0543	206	■0.990 • N|+0	`\left|\boldsymbol{A}^{-1}\right|_{2}=\max _{ \boldsymbol{x} \neq \mathbf{0}} \frac{|\boldsymbol{x} |_{2}}{|\boldsymbol{A} \boldsymbol{x}|_{2}} .`	$\left\|\boldsymbol{A}^{-1}\right\|_{2}=\max _{ \boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{x} \|_{2}}{\|\boldsymbol{A} \boldsymbol{x}\|_{2}} .$	$\|\boldsymbol{A}^{-1}\|_{2}=\max _{ \boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{x} \|_{2}}{\|\boldsymbol{A} \boldsymbol{x}\|_{2}} .$
lyche-numerical-linear-algebra EQ0920 (13.52)	313	■0.990 • W|+0	`\begin{aligned} -\frac{\partial}{\partial x}\left(c(x, y) \frac{\partial u}{\partial x}\right)-\frac{\partial}{ \partial y}\left(c(x, y) \frac{\partial u}{\partial y} \right) & =f(x, y), & & (x, y) \in \Omega=(0,1)^{2}, \\ u(x, y) & =0, & & (x, y) \in \partial \Omega . \end{aligned}`	$\begin{aligned} -\frac{\partial}{\partial x}\left(c(x, y) \frac{\partial u}{\partial x}\right)-\frac{\partial}{ \partial y}\left(c(x, y) \frac{\partial u}{\partial y} \right) & =f(x, y), & & (x, y) \in \Omega=(0,1)^{2}, \\ u(x, y) & =0, & & (x, y) \in \partial \Omega . \end{aligned}$	$\begin{aligned} -\frac{\partial}{\partial x}\left(c(x, y) \frac{\partial u}{\partial x}\right)-\frac{\partial}{ \partial y}\left(c(x, y) \frac{\partial u}{\partial y} \right) & =f(x, y), & & (x, y) \in \mathit{\Omega}=(0,1)^{2}, \\ u(x, y) & =0, & & (x, y) \in \partial \mathit{\Omega} . \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00679</td><td>244</td><td>■ 0.990 ● N|+0</td><td>((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i, j}=\boldsymbol{A} \boldsymbol{C} \sum_{k=1}^{t} b_{i, k} d_{k, j}=(\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i, j}=((\boldsymbol{A} \boldsymbol{C}) \otimes(\boldsymbol{B} \boldsymbol{D}))_{i, j} .</td><td>$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i, j}=\boldsymbol{A} \boldsymbol{C} \sum_{k=1}^{t} b_{i, k} d_{k, j}=(\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i, j}=((\boldsymbol{A} \boldsymbol{C}) \otimes(\boldsymbol{B} \boldsymbol{D}))_{i, j} .$</td><td>$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i, j}=\boldsymbol{A} \boldsymbol{C} \sum_{k=1}^{t} b_{i, k} d_{k, j}=(\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i, j}=((\boldsymbol{A} \boldsymbol{C}) \otimes(\boldsymbol{B} \boldsymbol{D}))_{i, j} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0042</td><td>036</td><td>■ 0.991 ● W|+1</td><td>\begin{aligned} \left|\begin{array}{ccc} 1 & x & y \\ 1 & x_{1} & y_{1} \\ 1 & x_{2} & y_{2} \end{array} \right| & =\left|\begin{array}{ccc} 1 & x & y \\ 0 & x_{1}-x & y_{1}-y \\ 0 & x_{2}-x_{1} & y_{2}-y_{1} \end{array}\right| \\ & =\left|\begin{array}{cc} x_{1}-x & y_{1}-y \\ x_{2}-x_{1} & y_{2}-y_{1} \end{array}\right|=\left(x_{1}-x\right)\left(y_{2}-y_{1} \right)-\left(y_{1}-y\right)\left(x_{2}-x_{1}\right) \end{aligned}</td><td>$\begin{aligned} \left|\begin{array}{ccc} 1 & x & y \\ 1 & x_{1} & y_{1} \\ 1 & x_{2} & y_{2} \end{array} \right| & =\left|\begin{array}{ccc} 1 & x & y \\ 0 & x_{1}-x & y_{1}-y \\ 0 & x_{2}-x_{1} & y_{2}-y_{1} \end{array}\right| \\ & =\left|\begin{array}{cc} x_{1}-x & y_{1}-y \\ x_{2}-x_{1} & y_{2}-y_{1} \end{array}\right|=\left(x_{1}-x\right)\left(y_{2}-y_{1} \right)-\left(y_{1}-y\right)\left(x_{2}-x_{1}\right) \end{aligned}$</td><td>$\begin{aligned} \left|\begin{array}{ccc} 1 & x & y \\ 1 & x_{1} & y_{1} \\ 1 & x_{2} & y_{2} \end{array} \right| & =\left|\begin{array}{ccc} 1 & x & y \\ 0 & x_{1}-x & y_{1}-y \\ 0 & x_{2}-x_{1} & y_{2}-y_{1} \end{array}\right| \\ & =\left|\begin{array}{cc} x_{1}-x & y_{1}-y \\ x_{2}-x_{1} & y_{2}-y_{1} \end{array}\right|=(x_{1}-x)(y_{2}-y_{1})-(y_{1}-y)(x_{2}-x_{1}). \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0213</td><td>087</td><td>■ 0.991 ● N|+0</td><td>\mathrm{fl}\left(x_{2}\right)=2, \quad \mathrm{fl} \left(x_{1}\right)=0 .</td><td>$\mathrm{fl}\left(x_{2}\right)=2, \quad \mathrm{fl} \left(x_{1}\right)=0 .$</td><td>$\mathrm{fl}(x_{2})=2, \quad \mathrm{fl}(x_{1})=0 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0234 (3.25)</td><td>094</td><td>■ 0.991 ● S|+0</td><td>\begin{aligned} 1+2+\cdots+m & =\frac{1}{2} m(m+1), \\ 1^{2}+2^{2}+\cdots+m^{2} & =\frac{1}{3} m\left(m+ \frac{1}{2}\right)(m+1), \\ 1+3+5+\cdots+2 m-1 & =m^{2}, \\ 1 * 2+2 * 3+3 * 4+\cdots+(m-1) m & =\frac{1}{3}(m-1) m(m+1) . \end{aligned}</td><td>$\begin{aligned} 1+2+\cdots+m & =\frac{1}{2} m(m+1), \\ 1^{2}+2^{2}+\cdots+m^{2} & =\frac{1}{3} m\left(m+ \frac{1}{2}\right)(m+1), \\ 1+3+5+\cdots+2 m-1 & =m^{2}, \\ 1 * 2+2 * 3+3 * 4+\cdots+(m-1) m & =\frac{1}{3}(m-1) m(m+1) . \end{aligned}$</td><td>$\begin{aligned} 1+2+\cdots+m & =\frac{1}{2} m(m+1), \\ 1^{2}+2^{2}+\cdots+m^{2} & =\frac{1}{3} m(m+ \frac{1}{2})(m+1), \\ 1+3+5+\cdots+2 m-1 & =m^{2}, \\ 1 * 2+2 * 3+3 * 4+\cdots+(m-1) m & =\frac{1}{3}(m-1) m(m+1) . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0468 (8.2)</td><td>186</td><td>■ 0.991 ● N|+0</td><td>\|\boldsymbol{x}\|_{p}:=\|\boldsymbol{x}\|_{\infty} \left(\sum_{j=1}^{n}\left(\frac{\left|x_{j}\right|}{\| \boldsymbol{x}\|_{\infty}}\right)^{p}\right)^{1 / p} .</td><td>$\|\boldsymbol{x}\|_{p}:=\|\boldsymbol{x}\|_{\infty} \left(\sum_{j=1}^{n}\left(\frac{\left|x_{j}\right|}{\| \boldsymbol{x}\|_{\infty}}\right)^{p}\right)^{1 / p} .$</td><td>$\|\boldsymbol{x}\|_{p}:=\|\boldsymbol{x}\|_{\infty} \left(\sum_{j=1}^{n}(\frac{|x_{j}|}{\| \boldsymbol{x}\|_{\infty}})^{p}\right)^{1 / p} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0584 (9.1)</td><td>216</td><td>■ 0.991 ● W|+0</td><td>\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p \left(t_{1}\right) \\ \vdots \\ p\left(t_{m}\right) \end{array}\right]=\left[\begin{array}{ccc} \phi_{1} \left(t_{1}\right) & \cdots & \phi_{n}\left(t_{1} \right) \\ \vdots & & \vdots \\ \phi_{1}\left(t_{m} \right) & \cdots & \phi_{n}\left(t_{m}\right) \end{array} \right]\left[\begin{array}{c} x_{1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array}\right]=: \boldsymbol{b} .</td><td>$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p \left(t_{1}\right) \\ \vdots \\ p\left(t_{m}\right) \end{array}\right]=\left[\begin{array}{ccc} \phi_{1} \left(t_{1}\right) & \cdots & \phi_{n}\left(t_{1} \right) \\ \vdots & & \vdots \\ \phi_{1}\left(t_{m} \right) & \cdots & \phi_{n}\left(t_{m}\right) \end{array} \right]\left[\begin{array}{c} x_{1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array}\right]=: \boldsymbol{b} .$</td><td>$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} p(t_{1}) \\ \vdots \\ p(t_{m}) \end{array}\right]=\left[\begin{array}{ccc} \phi_{1}(t_{1}) & \cdots & \phi_{n}(t_{1}) \\ \vdots & & \vdots \\ \phi_{1}(t_{m}) & \cdots & \phi_{n}(t_{m}) \end{array} \right]\left[\begin{array}{c} x_{1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} y_{1} \\ \vdots \\ y_{m} \end{array}\right]=: \boldsymbol{b} .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0725</td><td>258</td><td>■ 0.991
● K|+0</td><td><code>\boldsymbol{F}_{4} \boldsymbol{P}_{4}=\left[\begin{array}{r|r} \boldsymbol{F}_{2} & \boldsymbol{D}_{2} \boldsymbol{F}_{2} \\ \hline \boldsymbol{F}_{2} & -\boldsymbol{D}_{2} \boldsymbol{F}_{2} \end{array}\right] .</code></td><td>$\boldsymbol{F}_{4} \boldsymbol{P}_{4}=\left[\begin{array}{r|r} \boldsymbol{F}_{2} & \boldsymbol{D}_{2} \boldsymbol{F}_{2} \\ \hline \boldsymbol{F}_{2} & -\boldsymbol{D}_{2} \boldsymbol{F}_{2} \end{array}\right] .$</td><td>$\boldsymbol{F}_4\boldsymbol{P}_4 = \left[\begin{array}{c|c} \boldsymbol{F}_2 & \boldsymbol{D}_2\boldsymbol{F}_2 \\ \hline \boldsymbol{F}_2 & -\boldsymbol{D}_2\boldsymbol{F}_2\end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0169</td><td>074</td><td>■ 0.991
● N|+0</td><td><code>\boldsymbol{C} \boldsymbol{A} \boldsymbol{B}=\left[\begin{array}{cc} \lambda & \boldsymbol{a}^{T} \boldsymbol{B}_{1} \\ \mathbf{0} & \boldsymbol{C}_{1} \boldsymbol{A}_{1} \boldsymbol{B}_{1} \end{array}\right] .</code></td><td>$\boldsymbol{C} \boldsymbol{A} \boldsymbol{B}=\left[\begin{array}{cc} \lambda & \boldsymbol{a}^{T} \boldsymbol{B}_{1} \\ \mathbf{0} & \boldsymbol{C}_{1} \boldsymbol{A}_{1} \boldsymbol{B}_{1} \end{array}\right] .$</td><td>$\boldsymbol{CAB} = \begin{bmatrix}\lambda & \boldsymbol{a}^T\boldsymbol{B}_1\\ \mathbf{0} & \boldsymbol{C}_1\boldsymbol{A}_1\boldsymbol{B}_1\end{bmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0047 (1.15)</td><td>037</td><td>■ 0.991
● K|+0</td><td><code>\operatorname{det}(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j}))=\sum_{\boldsymbol{k}} \operatorname{det}(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \operatorname{det}(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),</code></td><td>$\operatorname{det}(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j}))=\sum_{\boldsymbol{k}} \operatorname{det}(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \operatorname{det}(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$</td><td>$\det\big(\boldsymbol{C}(\boldsymbol{i},\boldsymbol{j})\big) = \sum_{\boldsymbol{k}} \det\big(\boldsymbol{A}(\boldsymbol{i},\boldsymbol{k})\big)\det\big(\boldsymbol{B}(\boldsymbol{k},\boldsymbol{j})\big),$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0422</td><td>170</td><td>■ 0.992
● N|+0</td><td><code>\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} & =\boldsymbol{U}\left[\sigma_{1} \boldsymbol{e}_{1}, \ldots, \sigma_{r} \boldsymbol{e}_{r}, 0, \ldots, 0 \right] \\ & =\left[\sigma_{1} \boldsymbol{u}_{1}, \ldots, \sigma_{r} \boldsymbol{u}_{r}, 0, \ldots, 0 \right] \\ & =\left[\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{n}\right] . \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} & =\boldsymbol{U}\left[\sigma_{1} \boldsymbol{e}_{1}, \ldots, \sigma_{r} \boldsymbol{e}_{r}, 0, \ldots, 0 \right] \\ & =\left[\sigma_{1} \boldsymbol{u}_{1}, \ldots, \sigma_{r} \boldsymbol{u}_{r}, 0, \ldots, 0 \right] \\ & =\left[\boldsymbol{A} \boldsymbol{v}_{1}, \ldots, \boldsymbol{A} \boldsymbol{v}_{n}\right] . \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{U}\boldsymbol{\Sigma} &= \boldsymbol{U}[\sigma_1\boldsymbol{e}_1,\ldots,\sigma_r\boldsymbol{e}_r,0,\ldots,0]\\ &= [\sigma_1\boldsymbol{u}_1,\ldots,\sigma_r\boldsymbol{u}_r,0,\ldots,0]\\ &= [\boldsymbol{A}\boldsymbol{v}_1,\ldots,\boldsymbol{A}\boldsymbol{v}_n].\end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0702</td><td>252</td><td>■ 0.992
● N|-1</td><td><code>\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F} \quad \text { with } \quad h=1 /(m+1),</code></td><td>$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F} \quad \text { with } \quad h=1 /(m+1),$</td><td>$\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}=h^2\boldsymbol{F} \quad \text{with} \quad h=1/(m+1),$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0857 (13.19)</td><td>299</td><td>■ 0.992
● N|+0</td><td><code>\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{ \boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0} \right\|_{\boldsymbol{A}}} \leq\left(\frac{\kappa-1}{ \kappa+1}\right)^{k}<e^{-\frac{2}{\kappa} k}, \quad, k>0,</code></td><td>$\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{ \boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0} \right\|_{\boldsymbol{A}}} \leq\left(\frac{\kappa-1}{ \kappa+1}\right)^{k}<e^{-\frac{2}{\kappa} k}, \quad, k>0,$</td><td>$\frac{\|\boldsymbol{x}-\boldsymbol{x}_k\|_{\boldsymbol{A}}}{\|\boldsymbol{x}-\boldsymbol{x}_0\|_{\boldsymbol{A}}} \le \left(\frac{\kappa-1}{\kappa+1}\right)^k < e^{-\frac{2}{\kappa}k}, \quad , k>0,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0687 (4h)</td><td>246</td><td>■ 0.992
● N|+0</td><td><code>\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k} \right)^{T}\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)=\left(\boldsymbol{s}_{j}^{T} \otimes \boldsymbol{s}_{k}^{T}\right)\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)= \left(\boldsymbol{s}_{j}^{T} \boldsymbol{s}_{p}\right) \otimes\left(\boldsymbol{s}_{k}^{T} \boldsymbol{s}_{q} \right)=\frac{1}{4 h^{2}} \delta_{j, p} \delta_{k, q}</code></td><td>$\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k} \right)^{T}\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)=\left(\boldsymbol{s}_{j}^{T} \otimes \boldsymbol{s}_{k}^{T}\right)\left(\boldsymbol{s}_{p} \otimes \boldsymbol{s}_{q}\right)= \left(\boldsymbol{s}_{j}^{T} \boldsymbol{s}_{p}\right) \otimes\left(\boldsymbol{s}_{k}^{T} \boldsymbol{s}_{q} \right)=\frac{1}{4 h^{2}} \delta_{j, p} \delta_{k, q}$</td><td>$\big(\boldsymbol{s}_j\otimes\boldsymbol{s}_k\big)^T\big(\boldsymbol{s}_p\otimes\boldsymbol{s}_q\big) = \big(\boldsymbol{s}_j^T\otimes\boldsymbol{s}_k^T\big)\big(\boldsymbol{s}_p\otimes\boldsymbol{s}_q\big) = \big(\boldsymbol{s}_j^T\boldsymbol{s}_p\big)\otimes\big(\boldsymbol{s}_k^T\boldsymbol{s}_q\big) = \frac{1}{4h^2}\delta_{j,p}\delta_{k,q}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0775 (12.20)</td><td>274</td><td>■ 0.992
● N|+0</td><td><code>\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|_{2} \leq\left(\frac{\kappa-1}{\kappa+1}\right)^{k}\left\| \boldsymbol{x}_{0}-\boldsymbol{x}\right\|_{2}, \quad k=0,1,2, \ldots</code></td><td>$\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|_{2} \leq\left(\frac{\kappa-1}{\kappa+1}\right)^{k}\left\| \boldsymbol{x}_{0}-\boldsymbol{x}\right\|_{2}, \quad k=0,1,2, \ldots$</td><td>$\|\boldsymbol{x}_k-\boldsymbol{x}\|_2 \le \big(\frac{\kappa-1}{\kappa+1}\big)^k\|\boldsymbol{x}_0-\boldsymbol{x}\|_2, \quad k=0,1,2,\ldots$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0904	309	0.992 N|+0	\boldsymbol{\epsilon}_{j}=\boldsymbol{x}_{m+1}-\boldsymbol{x}_{j}=\boldsymbol{x}_{m}-\boldsymbol{x}_{j}+\alpha_{m} \boldsymbol{p}_{m}=\boldsymbol{x}_{m-1}-\boldsymbol{x}_{j}+\alpha_{m-1} \boldsymbol{p}_{m-1}+\alpha_{m} \boldsymbol{p}_{m}=\ldots	$\boldsymbol{\epsilon}_{j}=\boldsymbol{x}_{m+1}-\boldsymbol{x}_{j}=\boldsymbol{x}_{m}-\boldsymbol{x}_{j}+\alpha_{m} \boldsymbol{p}_{m}=\boldsymbol{x}_{m-1}-\boldsymbol{x}_{j}+\alpha_{m-1} \boldsymbol{p}_{m-1}+\alpha_{m} \boldsymbol{p}_{m}=\ldots$	$\boldsymbol{\epsilon}_{j}=\boldsymbol{x}_{m+1}-\boldsymbol{x}_{j}=\boldsymbol{x}_{m}-\boldsymbol{x}_{j}+\alpha_{m} \boldsymbol{p}_{m}=\boldsymbol{x}_{m-1}-\boldsymbol{x}_{j}+\alpha_{m-1} \boldsymbol{p}_{m-1}+\alpha_{m} \boldsymbol{p}_{m}=\ldots$
lyche-numerical-linear-algebra EQ0462	183	0.992 K|+0	\boldsymbol{A}=\left[\begin{array}{cc} 3 & 1 \\\\ 2 & 3 \\\\ -1 & 5 \end{array}\right] .	$\boldsymbol{A}=\left[\begin{array}{cc} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{cc} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{array}\right] .$
lyche-numerical-linear-algebra EQ0015 (1.3)	026	0.993 N|+0	c_{1} \boldsymbol{x}_{1}+\cdots+c_{n} \boldsymbol{x}_{n}=\mathbf{0} \quad \Longrightarrow \quad c_{1}=\cdots=c_{n}=0 .	$c_{1} \boldsymbol{x}_{1}+\cdots+c_{n} \boldsymbol{x}_{n}=\mathbf{0} \quad \Longrightarrow \quad c_{1}=\cdots=c_{n}=0 .$	$c_{1} \boldsymbol{x}_{1}+\cdots+c_{n} \boldsymbol{x}_{n}=\mathbf{0} \quad \Longrightarrow \quad c_{1}=\cdots=c_{n}=0 .$
lyche-numerical-linear-algebra EQ0248 (4.3)	101	0.993 W|+0	\begin{aligned} \boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{a}_{n} \\\\ \boldsymbol{a}_{n}^{*} & a_{n n} \end{array}\right] & = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\\\ \boldsymbol{l}_{n}^{*} & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{n-1} & \mathbf{0} \\\\ 0 & d_{n n} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{l}_{n} \\ \ \mathbf{0}^{*} & 1 \end{array}\right] \\\\ & = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n} \\\\ \boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}+d_{n n} \end{array}\right] \end{aligned}	$\begin{aligned} \boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{a}_{n} \\ \boldsymbol{a}_{n}^{*} & a_{n n} \end{array}\right] & = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_{n}^{*} & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{n n} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{l}_{n} \\ \mathbf{0}^{*} & 1 \end{array}\right] \\ & = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n} \\ \boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}+d_{n n} \end{array}\right] \end{aligned}$	$\begin{aligned} \boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{a}_{n} \\ \boldsymbol{a}_{n}^{*} & a_{n n} \end{array}\right] & = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_{n}^{*} & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{n n} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{l}_{n} \\ \mathbf{0}^{*} & 1 \end{array}\right] \\ & = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n} \\ \boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^{*} & \boldsymbol{l}_{n}^{*} \boldsymbol{D}_{n-1} \boldsymbol{l}_{n}+d_{n n} \end{array}\right] \end{aligned}$
lyche-numerical-linear-algebra EQ0583	216	0.993 N|+0	\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{u}_{1}^{*} \\\\ \boldsymbol{u}_{2}^{*} \\\\ \vdots \\\\ \boldsymbol{u}_{m}^{*} \end{array}\right] \boldsymbol{x}=\left[\begin{array}{c} y_{1} \\\\ y_{2} \\\\ \vdots \\\\ y_{m} \end{array}\right]=\boldsymbol{b} .	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{u}_{1}^{*} \\ \boldsymbol{u}_{2}^{*} \\ \vdots \\ \boldsymbol{u}_{m}^{*} \end{array}\right] \boldsymbol{x}=\left[\begin{array}{c} y_{1} \\ y_{2} \\ \vdots \\ y_{m} \end{array}\right]=\boldsymbol{b} .$	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{u}_{1}^{*} \\ \boldsymbol{u}_{2}^{*} \\ \vdots \\ \boldsymbol{u}_{m}^{*} \end{array}\right] \boldsymbol{x}=\left[\begin{array}{c} y_{1} \\ y_{2} \\ \vdots \\ y_{m} \end{array}\right]=\boldsymbol{b} .$
lyche-numerical-linear-algebra EQ0432	173	0.993 N|+0	\boldsymbol{\Sigma}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & 0 \\\\ 0 & 0 \\\\ 0 & 0 \end{array}\right], \quad \boldsymbol{\Sigma}_{1}=[2], \quad \boldsymbol{V}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\\\ 1 & -1 \end{array}\right] .	$\boldsymbol{\Sigma}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{\Sigma}_{1}=[2], \quad \boldsymbol{V}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] .$	$\boldsymbol{\Sigma}=\left[\begin{array}{cc} \boldsymbol{\Sigma}_{1} & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{\Sigma}_{1}=[2], \quad \boldsymbol{V}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0769</td><td>273</td><td>■ 0.993 ● N|+0</td><td><code>\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|=\left\| \boldsymbol{G}^{k}\left(\boldsymbol{x}_{0}-\boldsymbol{x} \right)\right\| \leq\left\|\boldsymbol{G}^{k}\right \|\left\|\boldsymbol{x}_{0}-\boldsymbol{x}\right\| \leq\|\boldsymbol{G}\|^{k}\left\|\boldsymbol{x}_{0}- \boldsymbol{x}\right\| \rightarrow \mathbf{0}, \quad k \rightarrow \infty .</code></td><td>$\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|=\left\|\boldsymbol{G}^{k}\left(\boldsymbol{x}_{0}-\boldsymbol{x}\right)\right\| \leq\left\|\boldsymbol{G}^{k}\right\|\left\|\boldsymbol{x}_{0}-\boldsymbol{x}\right\| \leq\|\boldsymbol{G}\|^{k}\left\|\boldsymbol{x}_{0}-\boldsymbol{x}\right\| \rightarrow \mathbf{0}, \quad k \rightarrow \infty .$</td><td>$\|\boldsymbol{x}_{k}-\boldsymbol{x}\|=\|\boldsymbol{G}^{k}(\boldsymbol{x}_{0}-\boldsymbol{x})\| \leq\|\boldsymbol{G}^{k}\|\|\boldsymbol{x}_{0}-\boldsymbol{x}\| \leq\|\boldsymbol{G}\|^{k}\|\boldsymbol{x}_{0}-\boldsymbol{x}\| \rightarrow \mathbf{0}, \quad k \rightarrow \infty .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0863</td><td>300</td><td>■ 0.993 ● N|-1</td><td><code>\mathbb{W}_{k}=\operatorname{span}\left(\boldsymbol{r}_{0}, \boldsymbol{A} \boldsymbol{r}_{0}, \boldsymbol{A}^{2} \boldsymbol{r}_{0}, \ldots, \boldsymbol{A}^{k-1} \boldsymbol{r}_{0}\right), \quad k=1,2,3, \cdots .</code></td><td>$\mathbb{W}_{k}=\operatorname{span}\left(\boldsymbol{r}_{0}, \boldsymbol{A} \boldsymbol{r}_{0}, \boldsymbol{A}^{2} \boldsymbol{r}_{0}, \ldots, \boldsymbol{A}^{k-1} \boldsymbol{r}_{0}\right), \quad k=1,2,3, \cdots .$</td><td>$\mathbb{W}_{k}=\operatorname{span}(\boldsymbol{r}_{0}, \boldsymbol{A} \boldsymbol{r}_{0}, \boldsymbol{A}^{2} \boldsymbol{r}_{0}, \ldots, \boldsymbol{A}^{k-1} \boldsymbol{r}_{0}), \quad k=1,2,3, \cdots .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0279</td><td>117</td><td>■ 0.993 ● N|+0</td><td><code>\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{x}^{T} \overline{\boldsymbol{y}}=\sum_{j=1}^{n} x_{j} \overline{y_{j}} .</code></td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{x}^{T} \overline{\boldsymbol{y}}=\sum_{j=1}^{n} x_{j} \overline{y_{j}} .$</td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{x}^{T} \overline{\boldsymbol{y}}=\sum_{j=1}^{n} x_{j} \overline{y_{j}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0414</td><td>165</td><td>■ 0.994 ● N|+1</td><td><code>R(x):=\frac{x^{T} A x}{x^{T} x} .</code></td><td>$R(x):=\frac{x^{T} A x}{x^{T} x} .$</td><td>$R(\boldsymbol{x}):=\frac{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{T} \boldsymbol{x}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0802</td><td>281</td><td>■ 0.994 ● W|-1</td><td><code>\begin{aligned} \|\boldsymbol{A}\| & =\left\| \boldsymbol{D}_{t} \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_{t}^{-1}\right \|_{1}=\left\|\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}\right\|_{1}=\max _{1 \leq j \leq n} \sum_{i=1}^{n}\left|\left(\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}\right)_{i j} \right| \\ & \leq \max _{1 \leq j \leq n}\left(\left| \lambda_{j}\right|+\epsilon\right)=\rho(\boldsymbol{A})+ \epsilon . \end{aligned}</code></td><td>$\begin{aligned} \|\boldsymbol{A}\| & =\left\|\boldsymbol{D}_{t} \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_{t}^{-1}\right\|_{1}=\left\|\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}\right\|_{1}=\max _{1 \leq j \leq n} \sum_{i=1}^{n}\left|\left(\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}\right)_{i j}\right| \\ & \leq \max _{1 \leq j \leq n}\left(\left|\lambda_{j}\right|+\epsilon\right)=\rho(\boldsymbol{A})+\epsilon . \end{aligned}$</td><td>$\begin{aligned} \|\boldsymbol{A}\| & =\|\boldsymbol{D}_{t} \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_{t}^{-1}\|_{1}=\|\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}\|_{1}=\max _{1 \leq j \leq n} \sum_{i=1}^{n}|(\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1})_{i j}| \\ & \leq \max _{1 \leq j \leq n}(|\lambda_{j}|+\epsilon)=\rho(\boldsymbol{A})+\epsilon . \end{aligned}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0175 (3.1)</td><td>077</td><td>■ 0.994
● W|+1</td><td><code>\boldsymbol{A}^{(k)}=\left[\begin{array}{ccc|ccccc} a_{1,1}^{(1)} & \cdots & a_{1, k-1}^{(1)} & a_{1, k}^{(1)} & \cdots & a_{1, j}^{(1)} & \cdots & a_{1, n}^{(1)} \\ & \ddots & \vdots & \vdots & & \vdots & & \vdots \\ & & a_{k-1, k-1}^{(k-1)} & a_{k-1, k}^{(k-1)} & \cdots & a_{k-1, j}^{(k-1)} & \cdots & a_{k-1, n}^{(k-1)} \\ \hline & & & a_{k, k}^{(k)} & \cdots & a_{k, j}^{(k)} & \cdots & a_{k, n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{i, k}^{(k)} & \cdots & a_{i, j}^{(k)} & \cdots & a_{i, n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{n, k}^{(k)} & \cdots & a_{n, j}^{(k)} & \cdots & a_{n, n}^{(k)} \end{array}\right] .</code></td><td>$\boldsymbol{A}^{(k)}=\left[\begin{array}{ccc|ccccc} a_{1,1}^{(1)} & \cdots & a_{1, k-1}^{(1)} & a_{1, k}^{(1)} & \cdots & a_{1, j}^{(1)} & \cdots & a_{1, n}^{(1)} \\ & \ddots & \vdots & \vdots & & \vdots & & \vdots \\ & & a_{k-1, k-1}^{(k-1)} & a_{k-1, k}^{(k-1)} & \cdots & a_{k-1, j}^{(k-1)} & \cdots & a_{k-1, n}^{(k-1)} \\ \hline & & & a_{k, k}^{(k)} & \cdots & a_{k, j}^{(k)} & \cdots & a_{k, n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{i, k}^{(k)} & \cdots & a_{i, j}^{(k)} & \cdots & a_{i, n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{n, k}^{(k)} & \cdots & a_{n, j}^{(k)} & \cdots & a_{n, n}^{(k)} \end{array}\right] .$</td><td>$\boldsymbol{A}^{(k)}=\left[\begin{array}{ccc|ccccc} a_{1,1}^{(1)} & \cdots & a_{1, k-1}^{(1)} & a_{1, k}^{(1)} & \cdots & a_{1, j}^{(1)} & \cdots & a_{1, n}^{(1)} \\ & \ddots & \vdots & \vdots & & \vdots & & \vdots \\ & & a_{k-1, k-1}^{(k-1)} & a_{k-1, k}^{(k-1)} & \cdots & a_{k-1, j}^{(k-1)} & \cdots & a_{k-1, n}^{(k-1)} \\ \hline & & & a_{k, k}^{(k)} & \cdots & a_{k, j}^{(k)} & \cdots & a_{k, n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{i, k}^{(k)} & \cdots & a_{i, j}^{(k)} & \cdots & a_{i, n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{n, k}^{(k)} & \cdots & a_{n, j}^{(k)} & \cdots & a_{n, n}^{(k)} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0833</td><td>288</td><td>■ 0.994
● K|+0</td><td><code>A:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]</code></td><td>$A:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$</td><td>$\boldsymbol{A}:=\left[\begin{array}{lll} 3 & 0 & 1 \\ 0 & 7 & 2 \\ 1 & 2 & 4 \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0992</td><td>335</td><td>■ 0.994
● K|+0</td><td><code>\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text { and } \boldsymbol{Q}_{k}=\boldsymbol{H}_{k} \boldsymbol{Q}_{k+1} \text { for } k=n-2, n-3, \ldots, 1 .</code></td><td>$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text { and } \boldsymbol{Q}_{k}=\boldsymbol{H}_{k} \boldsymbol{Q}_{k+1} \text { for } k=n-2, n-3, \ldots, 1 .$</td><td>$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text { and } \boldsymbol{Q}_{k}=\boldsymbol{H}_{k} \boldsymbol{Q}_{k+1} \text { for } k=n-2, n-3, \ldots, 1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ1019 (15.2)</td><td>345</td><td>■ 0.994
● W|+0</td><td><code>\begin{aligned} & \text { (i) }\left|\lambda_{1}\right|>\left|\lambda_{2}\right| \geq\left|\lambda_{3}\right| \geq \cdots \geq\left|\lambda_{n}\right| \text {, } \\ & \text { (ii) } z_{0}^{T} v_{1} \neq 0 \end{aligned}</code></td><td>$\begin{aligned} & \text { (i) }\left|\lambda_{1}\right|>\left|\lambda_{2}\right| \geq\left|\lambda_{3}\right| \geq \cdots \geq\left|\lambda_{n}\right| \text {, } \\ & \text { (ii) } z_{0}^{T} v_{1} \neq 0 \end{aligned}$</td><td>$\begin{aligned} & (i) \quad \left|\lambda_{1}\right|>\left|\lambda_{2}\right| \geq\left|\lambda_{3}\right| \geq \cdots \geq\left|\lambda_{n}\right|, \\ & (ii) \quad \boldsymbol{z}_{0}^{T} \boldsymbol{v}_{1} \neq 0 \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0124</td><td>061</td><td>■ 0.995
● W|+0</td><td><code>\begin{aligned} \left(\boldsymbol{T}_{1} \boldsymbol{s}_{j}\right)_{k} & =\sum_{l=1}^{m} t_{k, l} \sin (l j \pi h) \\ & =a[\sin ((k-1) j \pi h)+\sin ((k+1) j \pi h)]+d \sin (k j \pi h) \\ & =2 a \cos (j \pi h) \sin (k j \pi h)+d \sin (k j \pi h)=\lambda_{j} s_{k, j} . \end{aligned}</code></td><td>$\begin{aligned} \left(\boldsymbol{T}_{1} \boldsymbol{s}_{j}\right)_{k} & =\sum_{l=1}^{m} t_{k, l} \sin (l j \pi h) \\ & =a[\sin ((k-1) j \pi h)+\sin ((k+1) j \pi h)]+d \sin (k j \pi h) \\ & =2 a \cos (j \pi h) \sin (k j \pi h)+d \sin (k j \pi h)=\lambda_{j} s_{k, j} . \end{aligned}$</td><td>$\begin{aligned} \left(\boldsymbol{T}_{1} \boldsymbol{s}_{j}\right)_{k} & =\sum_{l=1}^{m} t_{k, l} \sin (l j \pi h) \\ & =a\big[\sin \big((k-1) j \pi h\big)+\sin \big((k+1) j \pi h\big)\big]+d \sin (k j \pi h) \\ & =2 a \cos (j \pi h) \sin (k j \pi h)+d \sin (k j \pi h)=\lambda_{j} s_{k, j} . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0384 (6.9)</td><td>155</td><td>■ 0.995
● N|+0</td><td><code>R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^{n} \lambda_{i}\left|c_{i}\right|^{2}}{\sum_{j=1}^{n}\left|c_{j}\right|^{2}}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{u}_{j} .</code></td><td>$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^{n} \lambda_{i}\left|c_{i}\right|^{2}}{\sum_{j=1}^{n}\left|c_{j}\right|^{2}}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{u}_{j} .$</td><td>$R_{\boldsymbol{A}}(\boldsymbol{x})=\frac{\sum_{i=1}^{n} \lambda_{i}\left|c_{i}\right|^{2}}{\sum_{j=1}^{n}\left|c_{j}\right|^{2}}, \quad \boldsymbol{x} \neq \mathbf{0}, \quad \boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{u}_{j} .$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0632</td><td>229</td><td>■0.995 ● W|+0</td><td>\begin{gathered} \lambda_{1} \geq \cdots \geq \lambda_{m+n}=\alpha_{1} \geq \cdots \geq \alpha_{t} \geq 0=\cdots=0 \geq-\alpha_{t} \geq \cdots \geq-\alpha_{1}, \\ \mu_{1} \geq \cdots \geq \mu_{m+n}=\beta_{1} \geq \cdots \geq \beta_{t} \geq 0=\cdots=0 \geq-\beta_{t} \geq \cdots \geq-\beta_{1} . \end{gathered}</td><td>$\lambda_{1} \geq \cdots \geq \lambda_{m+n}=\alpha_{1} \geq \cdots \geq \alpha_{t} \geq 0=\cdots=0 \geq-\alpha_{t} \geq \cdots \geq-\alpha_{1},$
$\mu_{1} \geq \cdots \geq \mu_{m+n}=\beta_{1} \geq \cdots \geq \beta_{t} \geq 0=\cdots=0 \geq-\beta_{t} \geq \cdots \geq-\beta_{1}.$</td><td>$\lambda_{1} \geq \cdots \geq \lambda_{m+n}=\alpha_{1} \geq \cdots \geq \alpha_{t} \geq 0=\cdots=0 \geq-\alpha_{t} \geq \cdots \geq-\alpha_{1},$
$\mu_{1} \geq \cdots \geq \mu_{m+n}=\beta_{1} \geq \cdots \geq \beta_{t} \geq 0=\cdots=0 \geq-\beta_{t} \geq \cdots \geq-\beta_{1}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0647</td><td>233</td><td>■0.995 ● K|+0</td><td>\boldsymbol{A}^{\dagger}=\frac{1}{\alpha} \boldsymbol{A}^{*}, \quad \alpha=\|\boldsymbol{u}\|_{2}^{2}\|\boldsymbol{v}\|_{2}^{2} .</td><td>$\boldsymbol{A}^{\dagger}=\frac{1}{\alpha} \boldsymbol{A}^{*}, \quad \alpha=\|\boldsymbol{u}\|_{2}^{2}\|\boldsymbol{v}\|_{2}^{2}.$</td><td>$\boldsymbol{A}^{\dagger}=\frac{1}{\alpha} \boldsymbol{A}^{*}, \quad \alpha=\|\boldsymbol{u}\|_{2}^{2}\|\boldsymbol{v}\|_{2}^{2}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0881 (13.31)</td><td>304</td><td>■0.995 ● N|+0</td><td>x_{n+1}-2 t x_{n}+x_{n-1}=0 \text { for } n \geq 1, \text { with } x_{0}=1, x_{1}=t .</td><td>$x_{n+1}-2 t x_{n}+x_{n-1}=0 \text { for } n \geq 1, \text { with } x_{0}=1, x_{1}=t.$</td><td>$x_{n+1}-2 t x_{n}+x_{n-1}=0 \text { for } n \geq 1, \text { with } x_{0}=1, x_{1}=t.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0301</td><td>124</td><td>■0.995 ● N|+0</td><td>\left(\boldsymbol{U}^{*} \boldsymbol{U}\right)_{i, j}=\boldsymbol{e}_{i}^{*} \boldsymbol{U}^{*} \boldsymbol{U} \boldsymbol{e}_{j}=\left(\boldsymbol{U} \boldsymbol{e}_{i}\right)^{*}\left(\boldsymbol{U} \boldsymbol{e}_{j}\right)=\left\langle\boldsymbol{U} \boldsymbol{e}_{j}, \boldsymbol{U} \boldsymbol{e}_{i} \right\rangle=\left\langle\boldsymbol{e}_{j}, \boldsymbol{e}_{i}\right\rangle=\boldsymbol{e}_{i}^{*} \boldsymbol{e}_{j},</td><td>$\left(\boldsymbol{U}^{*} \boldsymbol{U}\right)_{i, j}=\boldsymbol{e}_{i}^{*} \boldsymbol{U}^{*} \boldsymbol{U} \boldsymbol{e}_{j}=\left(\boldsymbol{U} \boldsymbol{e}_{i}\right)^{*}\left(\boldsymbol{U} \boldsymbol{e}_{j}\right)=\left\langle\boldsymbol{U} \boldsymbol{e}_{j}, \boldsymbol{U} \boldsymbol{e}_{i} \right\rangle=\left\langle\boldsymbol{e}_{j}, \boldsymbol{e}_{i}\right\rangle=\boldsymbol{e}_{i}^{*} \boldsymbol{e}_{j},$</td><td>$(\boldsymbol{U}^{*} \boldsymbol{U})_{i, j}=\boldsymbol{e}_{i}^{*} \boldsymbol{U}^{*} \boldsymbol{U} \boldsymbol{e}_{j}=(\boldsymbol{U} \boldsymbol{e}_{i})^{*}(\boldsymbol{U} \boldsymbol{e}_{j})=\langle\boldsymbol{U} \boldsymbol{e}_{j}, \boldsymbol{U} \boldsymbol{e}_{i} \rangle=\langle\boldsymbol{e}_{j}, \boldsymbol{e}_{i}\rangle=\boldsymbol{e}_{i}^{*} \boldsymbol{e}_{j},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0311</td><td>126</td><td>■0.995 ● S|-1</td><td>\boldsymbol{H} \boldsymbol{x}=a \boldsymbol{e}_{1}, \quad a=-\rho\|\boldsymbol{x}\|_{2}, \quad \rho:= \begin{cases}x_{1} /\left|x_{1}\right|, & \text { if } x_{1} \neq 0, \\ 1, & \text { otherwise } .\end{cases}</td><td>$\boldsymbol{H} \boldsymbol{x}=a \boldsymbol{e}_{1}, \quad a=-\rho\|\boldsymbol{x}\|_{2}, \quad \rho:= \begin{cases}x_{1} /\left|x_{1}\right|, & \text { if } x_{1} \neq 0, \\ 1, & \text { otherwise } .\end{cases}$</td><td>$\boldsymbol{H} \boldsymbol{x}=a \boldsymbol{e}_{1}, \quad a=-\rho\|\boldsymbol{x}\|_{2}, \quad \rho:= \begin{cases}x_{1} /|x_{1}|, & \textit{if } x_{1} \neq 0, \\ 1, & \textit{otherwise}.\end{cases}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0013</td><td>025</td><td>■0.995 ● N|+0</td><td>(\boldsymbol{f}+\boldsymbol{g})(t):=\boldsymbol{f}(t)+\boldsymbol{g}(t), \quad(c \boldsymbol{f})(t):=c \boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R} .</td><td>$(\boldsymbol{f}+\boldsymbol{g})(t):=\boldsymbol{f}(t)+\boldsymbol{g}(t), \quad(c \boldsymbol{f})(t):=c \boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$</td><td>$(\boldsymbol{f}+\boldsymbol{g})(t):=\boldsymbol{f}(t)+\boldsymbol{g}(t), \quad(c \boldsymbol{f})(t):=c \boldsymbol{f}(t), \quad t \in \mathcal{D}, \quad c \in \mathbb{R}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0061 (1.20)</td><td>041</td><td>■0.995 ● K|+0</td><td>\lambda:=\boldsymbol{a}^{T} \boldsymbol{C} \boldsymbol{e}_{i} \neq 0,</td><td>$\lambda:=\boldsymbol{a}^{T} \boldsymbol{C} \boldsymbol{e}_{i} \neq 0,$</td><td>$\lambda:=\boldsymbol{a}^{T} \boldsymbol{C} \boldsymbol{e}_{i} \neq 0,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0216</td><td>088</td><td>■0.995 ● W|+2</td><td>x_{2}=\frac{3.9997}{1.9999} \approx 2, \quad x_{1}=3-x_{2} \approx 1 .</td><td>$x_{2}=\frac{3.9997}{1.9999} \approx 2, \quad x_{1}=3-x_{2} \approx 1.$</td><td>$x_{2}=\frac{3.9997}{1.9999} \approx 2, \quad x_{1}=3-x_{2} \approx 1.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0284</td><td>118</td><td>■0.995 ● W|+0</td><td>\begin{aligned} \|\boldsymbol{x}\|^{2} & =\langle \boldsymbol{x}, \boldsymbol{x}\rangle=\langle \boldsymbol{z}+a \boldsymbol{y}, \boldsymbol{z}+a \boldsymbol{y}\rangle \\ & \stackrel{(5.4)}{=} \langle\boldsymbol{z}, \boldsymbol{z}\rangle+ \langle a \boldsymbol{y}, a \boldsymbol{y}\rangle \stackrel{(5.1)}{=}\|\boldsymbol{z}\|^{2}+|a|^{2}\| \boldsymbol{y}\|^{2} \\ & \geq|a|^{2}\|\boldsymbol{y}\|^{2}=\frac{|\langle\boldsymbol{x}, \boldsymbol{y} \rangle|^{2}}{\|\boldsymbol{y}\|^{2}} . \end{aligned}</td><td>$\begin{aligned} \|\boldsymbol{x}\|^{2} & =\langle \boldsymbol{x}, \boldsymbol{x}\rangle=\langle \boldsymbol{z}+a \boldsymbol{y}, \boldsymbol{z}+a \boldsymbol{y}\rangle \\ & \stackrel{(5.4)}{=} \langle\boldsymbol{z}, \boldsymbol{z}\rangle+ \langle a \boldsymbol{y}, a \boldsymbol{y}\rangle \stackrel{(5.1)}{=}\|\boldsymbol{z}\|^{2}+|a|^{2}\| \boldsymbol{y}\|^{2} \\ & \geq|a|^{2}\|\boldsymbol{y}\|^{2}=\frac{|\langle\boldsymbol{x}, \boldsymbol{y} \rangle|^{2}}{\|\boldsymbol{y}\|^{2}} . \end{aligned}$</td><td>$\begin{aligned} \|\boldsymbol{x}\|^{2} & =\langle \boldsymbol{x}, \boldsymbol{x}\rangle=\langle \boldsymbol{z}+a \boldsymbol{y}, \boldsymbol{z}+a \boldsymbol{y}\rangle \\ & \stackrel{(5.4)}{=} \langle\boldsymbol{z}, \boldsymbol{z}\rangle+ \langle a \boldsymbol{y}, a \boldsymbol{y}\rangle \stackrel{(5.1)}{=}\|\boldsymbol{z}\|^{2}+|a|^{2}\| \boldsymbol{y}\|^{2} \\ & \geq|a|^{2}\|\boldsymbol{y}\|^{2}=\frac{|\langle\boldsymbol{x}, \boldsymbol{y} \rangle|^{2}}{\|\boldsymbol{y}\|^{2}} . \end{aligned}$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra EQ0374</td><td>152</td><td>0.995 N|-1</td><td><code>\boldsymbol{W}=\left[\begin{array}{c|c} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1} \end{array}\right] \text { and } \boldsymbol{U}= \boldsymbol{V} \boldsymbol{W} .</code></td><td>$\boldsymbol{W}=\left[\begin{array}{c|c} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1} \end{array}\right] \text { and } \boldsymbol{U}= \boldsymbol{V} \boldsymbol{W} .$</td><td>$\boldsymbol{W}=\left[\begin{array}{c|c} 1 & \mathbf{0}^{*} \\ \hline \mathbf{0} & \boldsymbol{W}_{1} \end{array}\right] \text { and } \boldsymbol{U}= \boldsymbol{V} \boldsymbol{W} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0858 (13.20)</td><td>299</td><td>0.995 N|+0</td><td><code>\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0}\right\|_{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}<2 e^{-\frac{2}{\sqrt{\kappa}} k}, \quad k \geq 0 .</code></td><td>$\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0}\right\|_{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}<2 e^{-\frac{2}{\sqrt{\kappa}} k}, \quad k \geq 0 .$</td><td>$\frac{\|\boldsymbol{x}-\boldsymbol{x}_{k}\|_{\boldsymbol{A}}}{\|\boldsymbol{x}-\boldsymbol{x}_{0}\|_{\boldsymbol{A}}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}<2 e^{-\frac{2}{\sqrt{\kappa}} k}, \quad k \geq 0 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0459</td><td>182</td><td>0.995 K|+0</td><td><code>\begin{aligned} & x_{1}-x_{3}=4, \\ & x_{1}+x_{2}+x_{3}=12 . \end{aligned}</code></td><td>$\begin{aligned} & x_{1}-x_{3}=4, \\ & x_{1}+x_{2}+x_{3}=12 . \end{aligned}$</td><td>$\begin{aligned} & x_{1}-x_{3}=4, \\ & x_{1}+x_{2}+x_{3}=12 . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0982</td><td>331</td><td>0.995 N|-1</td><td><code>1=\|\boldsymbol{x}\|_{p}=\left\|\boldsymbol{X} \boldsymbol{D}_{1}^{-1} \boldsymbol{X}^{-1} \boldsymbol{r}\right\|_{p} \leq\left\|\boldsymbol{D}_{1}^{-1}\right\|_{p} K_{p}(\boldsymbol{X}) \|\boldsymbol{r}\|_{p}=\frac{K_{p}(\boldsymbol{X})\|\boldsymbol{r}\|_{p}}{\min _{j}\left|\lambda_{j}-\mu \right|} .</code></td><td>$1=\|\boldsymbol{x}\|_{p}=\left\|\boldsymbol{X} \boldsymbol{D}_{1}^{-1} \boldsymbol{X}^{-1} \boldsymbol{r}\right\|_{p} \leq\left\|\boldsymbol{D}_{1}^{-1}\right\|_{p} K_{p}(\boldsymbol{X}) \|\boldsymbol{r}\|_{p}=\frac{K_{p}(\boldsymbol{X})\|\boldsymbol{r}\|_{p}}{\min _{j}\left|\lambda_{j}-\mu \right|} .$</td><td>$1=\|\boldsymbol{x}\|_{p}=\|\boldsymbol{X} \boldsymbol{D}_{1}^{-1} \boldsymbol{X}^{-1} \boldsymbol{r}\|_{p} \leq\|\boldsymbol{D}_{1}^{-1}\|_{p} K_{p}(\boldsymbol{X}) \|\boldsymbol{r}\|_{p}=\frac{K_{p}(\boldsymbol{X})\|\boldsymbol{r}\|_{p}}{\min _{j}|\lambda_{j}-\mu|} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0951 (13.63)</td><td>321</td><td>0.995 N|-1</td><td><code>\left\|\boldsymbol{A}^{-1}\right\|_{2}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{x}\|_{2}}{\|\boldsymbol{A} \boldsymbol{x}\|_{2}},</code></td><td>$\left\|\boldsymbol{A}^{-1}\right\|_{2}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{x}\|_{2}}{\|\boldsymbol{A} \boldsymbol{x}\|_{2}},$</td><td>$\|\boldsymbol{A}^{-1}\|_{2}=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\|\boldsymbol{x}\|_{2}}{\|\boldsymbol{A} \boldsymbol{x}\|_{2}},$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0532 (8.39)</td><td>203</td><td>0.995 N|+0</td><td><code>\langle n \boldsymbol{x}, \boldsymbol{y}\rangle= \langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{\text { (8.37) }}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+ \langle\boldsymbol{x}, \boldsymbol{y}\rangle=n\langle \boldsymbol{x}, \boldsymbol{y}\rangle .</code></td><td>$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle= \langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{\text { (8.37) }}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+ \langle\boldsymbol{x}, \boldsymbol{y}\rangle=n\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$</td><td>$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle= \langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+ \langle\boldsymbol{x}, \boldsymbol{y}\rangle=n\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0648</td><td>233</td><td>0.995 K|+0</td><td><code>\lambda_{i}^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_{i}, & \lambda_{i} \neq 0 \\ 0 & \lambda_{i}=0 . \end{array}\right.</code></td><td>$\lambda_{i}^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_{i}, & \lambda_{i} \neq 0 \\ 0 & \lambda_{i}=0 . \end{array}\right.$</td><td>$\lambda_{i}^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_{i}, & \lambda_{i} \neq 0 \\ 0 & \lambda_{i}=0 . \end{array}\right.$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0975</td><td>330</td><td>0.995 N|+0</td><td><code>\boldsymbol{A}_{1}:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right]=\epsilon \boldsymbol{e}_{n} \boldsymbol{e}_{1}^{T} .</code></td><td>$\boldsymbol{A}_{1}:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right]=\epsilon \boldsymbol{e}_{n} \boldsymbol{e}_{1}^{T} .$</td><td>$\boldsymbol{A}_{1}:=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right], \quad \boldsymbol{E}:=\left[\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right]=\epsilon \boldsymbol{e}_{n} \boldsymbol{e}_{1}^{T} .$</td></tr>
</table>

Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0926 (13.56)</td><td>314</td><td>■ 0.996 ● W|+1</td><td>L_{h} v=\boldsymbol{F}, \quad L_{h} v:=\frac{1}{h^{2}} \left[\begin{array}{ccc} (d v)_{1,1} & \ldots & (d v)_{1, m} \\ \vdots & & \vdots \\ (d v)_{m, 1} & \ldots & (d v)_{m, m} \end{array}\right], \quad \boldsymbol{F}:= \left[\begin{array}{ccc} f_{1,1} & \ldots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \ldots & f_{m, m} \end{array}\right] .</td><td>$L_{h} v=\boldsymbol{F}, \quad L_{h} v:=\frac{1}{h^{2}} \left[\begin{array}{ccc} (d v)_{1,1} & \ldots & (d v)_{1, m} \\ \vdots & & \vdots \\ (d v)_{m, 1} & \ldots & (d v)_{m, m} \end{array}\right], \quad \boldsymbol{F}:= \left[\begin{array}{ccc} f_{1,1} & \ldots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \ldots & f_{m, m} \end{array}\right] .$</td><td>$L_h v = \boldsymbol{F}, \quad L_h v := \frac{1}{h^2}\left[\begin{array}{ccc} (dv)_{1,1} & \ldots & (dv)_{1,m} \\ \vdots & & \vdots \\ (dv)_{m,1} & \ldots & (dv)_{m,m}\end{array}\right], \quad \boldsymbol{F} := \left[\begin{array}{ccc} f_{1,1} & \ldots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \ldots & f_{m,m}\end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1024</td><td>346</td><td>■ 0.996 ● N|+0</td><td>\lim _{k \rightarrow \infty}\left(\frac{\left|\lambda_{1}\right|}{\lambda_{1}}\right)^{k} \boldsymbol{x}_{k}=\frac{c_{1}}{\left|c_{1}\right|} \frac{\boldsymbol{v}_{1}}{\left\|\boldsymbol{v}_{1}\right\|} .</td><td>$\lim _{k \rightarrow \infty}\left(\frac{\left|\lambda_{1}\right|}{\lambda_{1}}\right)^{k} \boldsymbol{x}_{k}=\frac{c_{1}}{\left|c_{1}\right|} \frac{\boldsymbol{v}_{1}}{\left\|\boldsymbol{v}_{1}\right\|} .$</td><td>$\lim_{k\to\infty}\left(\frac{|\lambda_1|}{\lambda_1}\right)^k \boldsymbol{x}_k = \frac{c_1}{|c_1|}\frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0031</td><td>032</td><td>■ 0.996 ● W|+0</td><td>\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m 1} & a_{m 2} & \cdots & a_{m n} \end{array}\right] \left[\begin{array}{c} x_{1} \\ x_{2} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{1} \\ b_{2} \\ \vdots \\ b_{m} \end{array}\right]=\boldsymbol{b} .</td><td>$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m 1} & a_{m 2} & \cdots & a_{m n} \end{array}\right] \left[\begin{array}{c} x_{1} \\ x_{2} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{1} \\ b_{2} \\ \vdots \\ b_{m} \end{array}\right]=\boldsymbol{b} .$</td><td>$\boldsymbol{A}\boldsymbol{x} = \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn}\end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n\end{array}\right] = \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m\end{array}\right] = \boldsymbol{b}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0207</td><td>086</td><td>■ 0.996 ● C|+3</td><td>\begin{aligned} (\boldsymbol{L} \boldsymbol{U})_{i j} & =\sum_{k=1}^{n} l_{i, k} u_{k j}=\sum_{k=1}^{i-1} a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}+a_{p_{i}, j}^{(i)} \\ & =\sum_{k=1}^{i-1}\left(a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)}\right)+a_{p_{i}, j}^{(i)}=a_{p_{i}, j}^{(1)}=a_{p_{i}, j}=\left(\boldsymbol{P}^{T} \boldsymbol{A}\right)_{i j}, \end{aligned}</td><td>$\begin{aligned} (\boldsymbol{L} \boldsymbol{U})_{i j} & =\sum_{k=1}^{n} l_{i, k} u_{k j}=\sum_{k=1}^{i-1} a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}+a_{p_{i}, j}^{(i)} \\ & =\sum_{k=1}^{i-1}\left(a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)}\right)+a_{p_{i}, j}^{(i)}=a_{p_{i}, j}^{(1)}=a_{p_{i}, j}=\left(\boldsymbol{P}^{T} \boldsymbol{A}\right)_{i j}, \end{aligned}$</td><td>$\begin{aligned}(\boldsymbol{L}\boldsymbol{U})_{ij} &= \sum_{k=1}^{n} l_{i,k}u_{kj} = \sum_{k=1}^{i-1} a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} + a_{p_i,j}^{(i)} \\ &= \sum_{k=1}^{i-1}\left(a_{p_i,j}^{(k)} - a_{p_i,j}^{(k+1)}\right) + a_{p_i,j}^{(i)} = a_{p_i,j}^{(1)} = a_{p_i,j} = \left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij},\end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0739</td><td>261</td><td>■ 0.996 ● N|+0</td><td>\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}-\frac{1}{6} \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}=4 h^{4} \boldsymbol{G}, \text { where } \boldsymbol{G}=\boldsymbol{S} \mu \boldsymbol{F} \boldsymbol{S},</td><td>$\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}-\frac{1}{6} \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}=4 h^{4} \boldsymbol{G}, \text { where } \boldsymbol{G}=\boldsymbol{S} \mu \boldsymbol{F} \boldsymbol{S},$</td><td>$\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D} - \frac{1}{6}\boldsymbol{D}\boldsymbol{X}\boldsymbol{D} = 4h^4\boldsymbol{G}, \text{ where } \boldsymbol{G} = \boldsymbol{S}\mu\boldsymbol{F}\boldsymbol{S},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0787 (12.27)</td><td>277</td><td>■ 0.996 ● N|+0</td><td>\rho\left(\boldsymbol{G}_{\omega}\right)= \begin{cases} \frac{1}{4}\left(\omega \beta+\sqrt{(\omega \beta)^{2}-4(\omega-1)}\right)^{2}, & \text { for } 0<\omega \leq \omega^{*}, \\ \omega-1, & \text { for } \omega^{*}< \omega<2,\end{cases}</td><td>$\rho\left(\boldsymbol{G}_{\omega}\right)= \begin{cases} \frac{1}{4}\left(\omega \beta+\sqrt{(\omega \beta)^{2}-4(\omega-1)}\right)^{2}, & \text { for } 0<\omega \leq \omega^{*}, \\ \omega-1, & \text { for } \omega^{*}< \omega<2,\end{cases}$</td><td>$\rho(\boldsymbol{G}_\omega) = \begin{cases} \frac{1}{4}\left(\omega\beta + \sqrt{(\omega\beta)^2 - 4(\omega-1)}\right)^2, & \textit{for } 0 < \omega \le \omega^*, \\ \omega - 1, & \textit{for } \omega^* < \omega < 2,\end{cases}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0819</td><td>285</td><td>■ 0.996 ● K|+0</td><td>d(\omega):=(\omega \mu)^{2}-4(\omega-1) .</td><td>$d(\omega):=(\omega \mu)^{2}-4(\omega-1) .$</td><td>$d(\omega) := (\omega\mu)^2 - 4(\omega - 1).$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0850 (13.16)	295	■ 0.996 • N|+0	`\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i} \stackrel{(13.13)}{=} \boldsymbol{r}_{j}^{T}\left(\frac{ \boldsymbol{r}_{i}-\boldsymbol{r}_{i+1}}{\alpha_{i}} \right)=0, \quad i=0,1, \ldots, j-2 .`	$\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i} \stackrel{(13.13)}{=} \boldsymbol{r}_{j}^{T}\left(\frac{\boldsymbol{r}_{i}-\boldsymbol{r}_{i+1}}{\alpha_{i}}\right)=0, \quad i=0,1, \ldots, j-2 .$	$\boldsymbol{r}_{j}^{T} \boldsymbol{A} \boldsymbol{p}_{i} \stackrel{(13.13)}{=} \boldsymbol{r}_{j}^{T}\big(\frac{\boldsymbol{r}_{i}-\boldsymbol{r}_{i+1}}{\alpha_{i}}\big)=0, \quad i=0,1, \ldots, j-2 .$
lyche-numerical-linear-algebra EQ0991	334	■ 0.996 • W|+0	`\begin{aligned} \boldsymbol{A}_{k+1} & = \boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}= \left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{array} \right]\left[\begin{array}{ll} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{array}\right] \\ & =\left[\begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \boldsymbol{V}_{k} \\ \boldsymbol{V}_{k} \boldsymbol{D}_{k} & \boldsymbol{V}_{k} \boldsymbol{E}_{k} \boldsymbol{V}_{k} \end{array}\right] . \end{aligned}`	$\begin{aligned} \boldsymbol{A}_{k+1} & = \boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}= \left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{array} \right]\left[\begin{array}{ll} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{array}\right] \\ & =\left[\begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \boldsymbol{V}_{k} \\ \boldsymbol{V}_{k} \boldsymbol{D}_{k} & \boldsymbol{V}_{k} \boldsymbol{E}_{k} \boldsymbol{V}_{k} \end{array}\right] . \end{aligned}$	$\begin{aligned} \boldsymbol{A}_{k+1} = \boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k} &= \begin{bmatrix} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{bmatrix}\begin{bmatrix} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{bmatrix}\begin{bmatrix} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \boldsymbol{V}_{k} \\ \boldsymbol{V}_{k} \boldsymbol{D}_{k} & \boldsymbol{V}_{k} \boldsymbol{E}_{k} \boldsymbol{V}_{k} \end{bmatrix}. \end{aligned}$
lyche-numerical-linear-algebra EQ0454 (7.14)	181	■ 0.997 • K|+0	`\boldsymbol{X}_{k+1}=\frac{1}{2}\left(\boldsymbol{X}_{k}+ \boldsymbol{X}_{k}^{-T}\right), k=0,1,2, \ldots \text { with } \boldsymbol{X}_{0}=\boldsymbol{A},`	$\boldsymbol{X}_{k+1}=\frac{1}{2}\left(\boldsymbol{X}_{k}+\boldsymbol{X}_{k}^{-T}\right), k=0,1,2, \ldots \text { with } \boldsymbol{X}_{0}=\boldsymbol{A},$	$\boldsymbol{X}_{k+1}=\frac{1}{2}(\boldsymbol{X}_{k}+\boldsymbol{X}_{k}^{-T}),\ k=0,1,2, \ldots \text { with } \boldsymbol{X}_{0}=\boldsymbol{A},$
lyche-numerical-linear-algebra EQ0135	064	■ 0.997 • K|+0	`\left[\begin{array}{l} A_{1} \\ A_{2} \end{array}\right] B=\left[\begin{array}{l} A_{1} B \\ A_{2} B \end{array} \right] .`	$\left[\begin{array}{l} A_{1} \\ A_{2} \end{array}\right] B=\left[\begin{array}{l} A_{1} B \\ A_{2} B \end{array}\right] .$	$\begin{bmatrix} \boldsymbol{A}_{1} \\ \boldsymbol{A}_{2} \end{bmatrix} \boldsymbol{B}=\begin{bmatrix} \boldsymbol{A}_{1} \boldsymbol{B} \\ \boldsymbol{A}_{2} \boldsymbol{B} \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0269 (4.11)	110	■ 0.997 • W|+2	`\begin{aligned} & \text { if } A(1,1)>0 \\ & A(1,1)= \sqrt{A(1,1)} \\ & A(2: n, 1)=A(2: n, 1) / A(1,1) \\ & \text { for } j=2: n \\ & \quad A(j: n, j)=A(j: n, j)-A(j, 1) * A(j: n, 1) \end{aligned}`	$\begin{aligned} & \text { if } A(1,1)>0 \\ & A(1,1)=\sqrt{A(1,1)} \\ & A(2: n, 1)=A(2: n, 1) / A(1,1) \\ & \text { for } j=2: n \\ & \quad A(j: n, j)=A(j: n, j)-A(j, 1) * A(j: n, 1) \end{aligned}$	$\begin{aligned} & \text{if } A(1,1)>0 \\ & \quad A(1,1)=\sqrt{A(1,1)} \\ & \quad A(2: n, 1)=A(2: n, 1) / A(1,1) \\ & \quad \text{for } j=2: n \\ & \qquad A(j: n, j)=A(j: n, j)-A(j, 1) * A(j: n, 1) \end{aligned}$
lyche-numerical-linear-algebra EQ0431	173	■ 0.997 • N|+0	`\boldsymbol{B}\left[\begin{array}{l} 1 \\ 1 \end{array} \right]=4\left[\begin{array}{l} 1 \\ 1 \end{array} \right], \quad \boldsymbol{B}\left[\begin{array}{r} 1 \\ -1 \end{array}\right]=0\left[\begin{array}{r} 1 \\ -1 \end{array}\right],`	$\boldsymbol{B}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]=4\left[\begin{array}{l} 1 \\ 1 \end{array}\right], \quad \boldsymbol{B}\left[\begin{array}{r} 1 \\ -1 \end{array}\right]=0\left[\begin{array}{r} 1 \\ -1 \end{array}\right],$	$\boldsymbol{B}\begin{bmatrix} 1 \\ 1 \end{bmatrix}=4\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{B}\begin{bmatrix} 1 \\ -1 \end{bmatrix}=0\begin{bmatrix} 1 \\ -1 \end{bmatrix},$
lyche-numerical-linear-algebra EQ0656 (10.1)	238	■ 0.997 • N|+0	`\begin{gathered} -\Delta u:=-\frac{\partial^{2} u}{ \partial x^{2}}-\frac{\partial^{2} u}{\partial y^{2}}=f \text { on } \Omega, \\ u:=0 \text { on } \partial \Omega . \end{gathered}`	$\begin{gathered} -\Delta u:=-\frac{\partial^{2} u}{\partial x^{2}}-\frac{\partial^{2} u}{\partial y^{2}}=f \text { on } \Omega, \\ u:=0 \text { on } \partial \Omega . \end{gathered}$	$\begin{gathered} -\Delta u:=-\frac{\partial^{2} u}{\partial x^{2}}-\frac{\partial^{2} u}{\partial y^{2}}=f \text { on } \Omega, \\ u:=0 \text { on } \partial \Omega . \end{gathered}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0666 (10.7)</td><td>240</td><td>0.997 N|+0</td><td>\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad b \in \mathbb{R}^{n}, \quad n=m^{2},</td><td>$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad b \in \mathbb{R}^{n}, \quad n=m^{2},$</td><td>$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \quad \boldsymbol{A} \in \mathbb{R}^{n \times n}, \quad b \in \mathbb{R}^{n}, \quad n=m^{2},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0671 (10.11)</td><td>242</td><td>0.997 W|+2</td><td>\boldsymbol{T}_{2}=\left[\begin{array}{rrr|rrr|rrr} 2 d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2 d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2 d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2 d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2 d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2 d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2 d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2 d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2 d \end{array}\right] .</td><td>$\boldsymbol{T}_{2}=\left[\begin{array}{rrr|rrr|rrr} 2 d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2 d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2 d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2 d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2 d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2 d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2 d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2 d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2 d \end{array}\right] .$</td><td>$\boldsymbol{T}_{2}=\left[\begin{array}{rrr|rrr|rrr} 2 d & a & 0 & a & 0 & 0 & 0 & 0 & 0 \\ a & 2 d & a & 0 & a & 0 & 0 & 0 & 0 \\ 0 & a & 2 d & 0 & 0 & a & 0 & 0 & 0 \\ \hline a & 0 & 0 & 2 d & a & 0 & a & 0 & 0 \\ 0 & a & 0 & a & 2 d & a & 0 & a & 0 \\ 0 & 0 & a & 0 & a & 2 d & 0 & 0 & a \\ \hline 0 & 0 & 0 & a & 0 & 0 & 2 d & a & 0 \\ 0 & 0 & 0 & 0 & a & 0 & a & 2 d & a \\ 0 & 0 & 0 & 0 & 0 & a & 0 & a & 2 d \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1010 (14.10)</td><td>340</td><td>0.997 W|+1</td><td>a_{k j}=\left\{\begin{array}{ll} 1, & j=k+1, \\ 0, & \text { otherwise }, \end{array} \quad e_{k j}= \begin{cases}\epsilon, & k=n, j=1, \\ 0, & \text { otherwise },\end{cases}\right.</td><td>$a_{k j}=\left\{\begin{array}{ll} 1, & j=k+1, \\ 0, & \text { otherwise }, \end{array} \quad e_{k j}= \begin{cases}\epsilon, & k=n, j=1, \\ 0, & \text { otherwise },\end{cases}\right.$</td><td>$a_{k j}=\begin{cases}1, & j=k+1, \\ 0, & \text{otherwise,}\end{cases} \quad e_{k j}=\begin{cases}\epsilon, & k=n, j=1, \\ 0, & \text{otherwise,}\end{cases}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0294 (5.10)</td><td>122</td><td>0.997 N|+0</td><td>\boldsymbol{s}_{0}=\sum_{i=1}^{k} \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_{i}\right\rangle}{\left \langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right \rangle} \boldsymbol{v}_{i} .</td><td>$\boldsymbol{s}_{0}=\sum_{i=1}^{k} \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_{i}\right\rangle}{\left \langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right \rangle} \boldsymbol{v}_{i} .$</td><td>$\boldsymbol{s}_{0}=\sum_{i=1}^{k} \frac{\langle \boldsymbol{v}, \boldsymbol{v}_{i}\rangle}{\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\rangle} \boldsymbol{v}_{i} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0416</td><td>166</td><td>0.997 N|+0</td><td>B_{1}:=\left\{\boldsymbol{x} \in \mathbb{R}^{n} \mid\|\boldsymbol{x}\|_{2}=1\right\} .</td><td>$B_{1}:=\left\{\boldsymbol{x} \in \mathbb{R}^{n} \mid\|\boldsymbol{x}\|_{2}=1\right\} .$</td><td>$B_{1}:=\{\boldsymbol{x} \in \mathbb{R}^{n} \mid \|\boldsymbol{x}\|_{2}=1\}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0694 (10.24)</td><td>248</td><td>0.997 W|+1</td><td>\begin{array}{rlrl} -\Delta v(s, t) & =f(s, t) & & (s, t) \in \Omega \\ -\Delta u(s, t) & =v(s, t) & & (s, t) \in \Omega \\ u(s, t) & =v(s, t)=0 & (s, t) \in \partial \Omega \end{array}</td><td>$\begin{array}{rlrl} -\Delta v(s, t) & =f(s, t) & & (s, t) \in \Omega \\ -\Delta u(s, t) & =v(s, t) & & (s, t) \in \Omega \\ u(s, t) & =v(s, t)=0 & (s, t) \in \partial \Omega \end{array}$</td><td>$\begin{array}{rll} -\Delta v(s, t) & =f(s, t) & (s, t) \in \Omega \\ -\Delta u(s, t) & =v(s, t) & (s, t) \in \Omega \\ u(s, t) & =v(s, t)=0 & (s, t) \in \partial \Omega . \end{array}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0768 (12.16)</td><td>272</td><td>0.997 K|+0</td><td>\boldsymbol{x}_{k}-\boldsymbol{x}=\boldsymbol{G}^{k} \left(\boldsymbol{x}_{0}-\boldsymbol{x}\right), \quad k=0,1,2, \ldots</td><td>$\boldsymbol{x}_{k}-\boldsymbol{x}=\boldsymbol{G}^{k} \left(\boldsymbol{x}_{0}-\boldsymbol{x}\right), \quad k=0,1,2, \ldots$</td><td>$\boldsymbol{x}_{k}-\boldsymbol{x}=\boldsymbol{G}^{k}(\boldsymbol{x}_{0}-\boldsymbol{x}), \quad k=0,1,2, \ldots$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0815</td><td>284</td><td>0.997 N|+0</td><td>\boldsymbol{G}_{J} \boldsymbol{v}=(\boldsymbol{L}+\boldsymbol{R}) \boldsymbol{v}=\frac{\lambda+\omega-1}{\omega \lambda^{1 / 2}} \boldsymbol{v}=\mu \boldsymbol{v} .</td><td>$\boldsymbol{G}_{J} \boldsymbol{v}=(\boldsymbol{L}+\boldsymbol{R}) \boldsymbol{v}=\frac{\lambda+\omega-1}{\omega \lambda^{1 / 2}} \boldsymbol{v}=\mu \boldsymbol{v} .$</td><td>$\boldsymbol{G}_{J} \boldsymbol{v}=(\boldsymbol{L}+\boldsymbol{R}) \boldsymbol{v}=\frac{\lambda+\omega-1}{\omega \lambda^{1 / 2}} \boldsymbol{v}=\mu \boldsymbol{v} .$</td></tr>
</table>

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_EQ0847 (13.11)	294	■0.997 ● S|+0	`\begin{aligned} & \boldsymbol{t}_{2 k-2}=3 * 4^{1-k} \left[\begin{array}{c} 1 \\ -1 / 2 \end{array} \right], \quad \boldsymbol{x}_{2 k-1}=-4^{-k}\left[\begin{array}{l} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{2 k-1}=3 * 4^{-k}\left[\begin{array}{l} 0 \\ 1 \end{array}\right] \\ & \boldsymbol{t}_{2 k-1}=3 * 4^{-k}\left[\begin{array}{c} -1 \\ 2 \end{array} \right], \quad \boldsymbol{x}_{2 k}=-4^{-k}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{2 k}=3 * 4^{-k}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right] . \end{aligned}`	$\begin{aligned} & \boldsymbol{t}_{2 k-2}=3 * 4^{1-k} \left[\begin{array}{c} 1 \\ -1 / 2 \end{array} \right], \quad \boldsymbol{x}_{2 k-1}=-4^{-k}\left[\begin{array}{l} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{2 k-1}=3 * 4^{-k}\left[\begin{array}{l} 0 \\ 1 \end{array}\right] \\ & \boldsymbol{t}_{2 k-1}=3 * 4^{-k}\left[\begin{array}{c} -1 \\ 2 \end{array} \right], \quad \boldsymbol{x}_{2 k}=-4^{-k}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{2 k}=3 * 4^{-k}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right] . \end{aligned}$	$\begin{aligned} & \boldsymbol{t}_{2 k-2}=3 * 4^{1-k} \left[\begin{smallmatrix} 1 \\ -1 / 2 \end{smallmatrix} \right], \quad \boldsymbol{x}_{2 k-1}=-4^{-k}\left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right], \quad \boldsymbol{r}_{2 k-1}=3 * 4^{-k}\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right] \\ & \boldsymbol{t}_{2 k-1}=3 * 4^{-k}\left[\begin{smallmatrix} -1 \\ 2 \end{smallmatrix} \right], \quad \boldsymbol{x}_{2 k}=-4^{-k}\left[\begin{smallmatrix} 1 \\ 1 / 2 \end{smallmatrix}\right], \quad \boldsymbol{r}_{2 k}=3 * 4^{-k}\left[\begin{smallmatrix} 1 / 2 \\ 0 \end{smallmatrix}\right] . \end{aligned}$
lyche-numerical-linear-algebra_EQ0355 (5.22)	142	■0.998 ● N|+0	`P_{i_{1}, n+1} P_{i_{2}, n+1} \cdots P_{i_{n}, n+1}\left[\begin{array}{c} \boldsymbol{L}^{*} \\ \boldsymbol{z}^{*} \end{array}\right]=\left[ \begin{array}{c} \boldsymbol{R}^{\prime} \\ 0 \end{array} \right],`	$P_{i_{1}, n+1} P_{i_{2}, n+1} \cdots P_{i_{n}, n+1}\left[\begin{array}{c} \boldsymbol{L}^{*} \\ \boldsymbol{z}^{*} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{R}^{\prime} \\ 0 \end{array} \right],$	$P_{i_{1}, n+1} P_{i_{2}, n+1} \cdots P_{i_{n}, n+1}\left[\begin{array}{c} \boldsymbol{L}^{*} \\ \boldsymbol{z}^{*} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{R}^{\prime} \\ 0 \end{array} \right],$
lyche-numerical-linear-algebra_EQ0390 (6.11)	157	■0.998 ● K|+0	`\lambda_{k} \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R( \boldsymbol{x}),`	$\lambda_{k} \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$	$\lambda_{k} \leq \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$
lyche-numerical-linear-algebra_EQ0448	179	■0.998 ● W|+0	`A:=\frac{1}{15}\left[\begin{array}{ccc} 14 & 4 & 16 \\ 2 & 22 & 13 \end{array}\right] \in \mathbb{R}^{2 \times 3} .`	$A:=\frac{1}{15}\left[\begin{array}{ccc} 14 & 4 & 16 \\ 2 & 22 & 13 \end{array}\right] \in \mathbb{R}^{2 \times 3} .$	$\boldsymbol{A}:=\frac{1}{15}\left[\begin{array}{ccc} 14 & 4 & 16 \\ 2 & 22 & 13 \end{array}\right] \in \mathbb{R}^{2 \times 3} .$
lyche-numerical-linear-algebra_EQ0765	272	■0.998 ● N|+1	`x_{k+1}(1)=\frac{1}{2} x_{k}(2)+\frac{1}{2}, \quad x_{k+1}(2)=\frac{1}{2} x_{k+1}(1)+\frac{1}{2} .`	$x_{k+1}(1)=\frac{1}{2} x_{k}(2)+\frac{1}{2}, \quad x_{k+1}(2)=\frac{1}{2} x_{k+1}(1)+\frac{1}{2} .$	$\boldsymbol{x}_{k+1}(1)=\frac{1}{2} \boldsymbol{x}_{k}(2)+\frac{1}{2}, \quad \boldsymbol{x}_{k+1}(2)=\frac{1}{2} \boldsymbol{x}_{k+1}(1)+\frac{1}{2} .$
lyche-numerical-linear-algebra_EQ0228 (3.19)	092	■0.998 ● N|+0	`\boldsymbol{A}:=\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m 1} & \cdots & \boldsymbol{A}_{m m} \end{array}\right],`	$\boldsymbol{A}:=\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m 1} & \cdots & \boldsymbol{A}_{m m} \end{array}\right],$	$\boldsymbol{A}:=\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 m} \\ \vdots & & \vdots \\ \boldsymbol{A}_{m 1} & \cdots & \boldsymbol{A}_{m m} \end{array}\right],$
lyche-numerical-linear-algebra_EQ0603 (9.8)	222	■0.998 ● N|+0	`\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{A}= \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{A} \boldsymbol{A}^{\dagger}=\boldsymbol{A}^{\dagger},\left(\boldsymbol{A}^{\dagger} \boldsymbol{A}\right)^{*}= \boldsymbol{A}^{\dagger} \boldsymbol{A},\left(\boldsymbol{A} \boldsymbol{A}^{\dagger}\right)^{*}= \boldsymbol{A} \boldsymbol{A}^{\dagger} .`	$\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{A}= \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{A} \boldsymbol{A}^{\dagger}=\boldsymbol{A}^{\dagger},\left(\boldsymbol{A}^{\dagger} \boldsymbol{A}\right)^{*}= \boldsymbol{A}^{\dagger} \boldsymbol{A},\left(\boldsymbol{A} \boldsymbol{A}^{\dagger}\right)^{*}= \boldsymbol{A} \boldsymbol{A}^{\dagger} .$	$\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{A}= \boldsymbol{A},\ \boldsymbol{A}^{\dagger} \boldsymbol{A} \boldsymbol{A}^{\dagger}=\boldsymbol{A}^{\dagger},\ (\boldsymbol{A}^{\dagger} \boldsymbol{A})^{*}= \boldsymbol{A}^{\dagger} \boldsymbol{A},\ (\boldsymbol{A} \boldsymbol{A}^{\dagger})^{*}= \boldsymbol{A} \boldsymbol{A}^{\dagger} .$
lyche-numerical-linear-algebra_EQ0691	247	■0.998 ● N|+0	`\boldsymbol{A}=\left[\begin{array}{rrrr} 2 d & a & a & 0 \\ a & 2 d & 0 & a \\ a & 0 & 2 d & a \\ 0 & a & a & 2 d \end{array}\right] .`	$\boldsymbol{A}=\left[\begin{array}{rrrr} 2 d & a & a & 0 \\ a & 2 d & 0 & a \\ a & 0 & 2 d & a \\ 0 & a & a & 2 d \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{rrrr} 2 d & a & a & 0 \\ a & 2 d & 0 & a \\ a & 0 & 2 d & a \\ 0 & a & a & 2 d \end{array}\right] .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0981	331	■ 0.998 ● K|+0	\boldsymbol{X} \boldsymbol{D}_{1}^{-1} \boldsymbol{X}^{-1} \boldsymbol{r}=\left(\boldsymbol{X}(\boldsymbol{D}-\mu \boldsymbol{I}) \boldsymbol{X}^{-1} \right)^{-1} \boldsymbol{r}=(\boldsymbol{A}-\mu \boldsymbol{I})^{-1}(\boldsymbol{A}-\mu \boldsymbol{I}) \boldsymbol{x}=\boldsymbol{x} .	$\boldsymbol{X} \boldsymbol{D}_{1}^{-1} \boldsymbol{X}^{-1} \boldsymbol{r}=\left(\boldsymbol{X}(\boldsymbol{D}-\mu \boldsymbol{I}) \boldsymbol{X}^{-1}\right)^{-1} \boldsymbol{r}=(\boldsymbol{A}-\mu \boldsymbol{I})^{-1}(\boldsymbol{A}-\mu \boldsymbol{I}) \boldsymbol{x}=\boldsymbol{x} .$	$\boldsymbol{X} \boldsymbol{D}_{1}^{-1} \boldsymbol{X}^{-1} \boldsymbol{r}=\left(\boldsymbol{X}(\boldsymbol{D}-\mu \boldsymbol{I}) \boldsymbol{X}^{-1}\right)^{-1} \boldsymbol{r}=(\boldsymbol{A}-\mu \boldsymbol{I})^{-1}(\boldsymbol{A}-\mu \boldsymbol{I}) \boldsymbol{x}=\boldsymbol{x} .$
lyche-numerical-linear-algebra_ EQ1035 (15.8)	351	■ 0.998 ● W|-1	\begin{aligned} & \boldsymbol{A}_{1}=\boldsymbol{A} \\ & \text { for } k=1,2, \ldots \\ & \quad \boldsymbol{Q}_{k} \boldsymbol{R}_{k}=\boldsymbol{A}_{k} \quad\left(\text { QR factorization of } \boldsymbol{A}_{k} \right) \\ & \boldsymbol{A}_{k+1}=\boldsymbol{R}_{k} \boldsymbol{Q}_{k} . \end{aligned}	$\begin{aligned} & \boldsymbol{A}_{1}=\boldsymbol{A} \\ & \text { for } k=1,2, \ldots \\ & \quad \boldsymbol{Q}_{k} \boldsymbol{R}_{k}=\boldsymbol{A}_{k} \quad\left(\text { QR factorization of } \boldsymbol{A}_{k}\right) \\ & \boldsymbol{A}_{k+1}=\boldsymbol{R}_{k} \boldsymbol{Q}_{k} . \end{aligned}$	$\begin{aligned} & \boldsymbol{A}_{1}=\boldsymbol{A} \\ & \text{for } k=1,2, \ldots \\ & \quad \boldsymbol{Q}_{k} \boldsymbol{R}_{k}=\boldsymbol{A}_{k} \qquad (\text{QR factorization of } \boldsymbol{A}_{k}) \\ & \quad \boldsymbol{A}_{k+1}=\boldsymbol{R}_{k} \boldsymbol{Q}_{k} . \end{aligned}$
lyche-numerical-linear-algebra_ EQ0012	024	■ 0.998 ● N|+1	\boldsymbol{A}\left(r_{1}: r_{2}, c_{1}: c_{2}\right):= \left[\begin{array}{rrrr} a_{r_{1}, c_{1}} & a_{r_{1}, c_{1}+1} & \cdots & a_{r_{1}, c_{2}} \\ a_{r_{1}+1, c_{1}} & a_{r_{1}+1, c_{1}+1} & \cdots & a_{r_{1}+1, c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2}, c_{1}} & a_{r_{2}, c_{1}+1} & \cdots & a_{r_{2}, c_{2}} \end{array}\right] .	$\boldsymbol{A}\left(r_{1}: r_{2}, c_{1}: c_{2}\right):=\left[\begin{array}{rrrr} a_{r_{1}, c_{1}} & a_{r_{1}, c_{1}+1} & \cdots & a_{r_{1}, c_{2}} \\ a_{r_{1}+1, c_{1}} & a_{r_{1}+1, c_{1}+1} & \cdots & a_{r_{1}+1, c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2}, c_{1}} & a_{r_{2}, c_{1}+1} & \cdots & a_{r_{2}, c_{2}} \end{array}\right] .$	$\boldsymbol{A}(r_{1}: r_{2}, c_{1}: c_{2}):=\left[\begin{array}{cccc} a_{r_{1}, c_{1}} & a_{r_{1}, c_{1}+1} & \cdots & a_{r_{1}, c_{2}} \\ a_{r_{1}+1, c_{1}} & a_{r_{1}+1, c_{1}+1} & \cdots & a_{r_{1}+1, c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2}, c_{1}} & a_{r_{2}, c_{1}+1} & \cdots & a_{r_{2}, c_{2}} \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0247 (4.2)	101	■ 0.998 ● N|+0	\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[k]} & \boldsymbol{B}_{k}^{*} \\ \boldsymbol{B}_{k} & \boldsymbol{F}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{[k]} & \mathbf{0} \\ \boldsymbol{M}_{k} & \boldsymbol{N}_{k} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{[k]} & \mathbf{0} \ \ \mathbf{0} & \boldsymbol{E}_{k} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{[k]}^{*} & \boldsymbol{M}_{k}^{*} \\ \mathbf{0} & \boldsymbol{N}_{k}^{*} \end{array}\right]=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U},	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[k]} & \boldsymbol{B}_{k}^{*} \\ \boldsymbol{B}_{k} & \boldsymbol{F}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{[k]} & \mathbf{0} \\ \boldsymbol{M}_{k} & \boldsymbol{N}_{k} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{D}_{[k]} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{E}_{k} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{L}_{[k]}^{*} & \boldsymbol{M}_{k}^{*} \\ \mathbf{0} & \boldsymbol{N}_{k}^{*} \end{array}\right]=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U},$	$\boldsymbol{A}=\begin{bmatrix} \boldsymbol{A}_{[k]} & \boldsymbol{B}_{k}^{*} \\ \boldsymbol{B}_{k} & \boldsymbol{F}_{k} \end{bmatrix}=\begin{bmatrix} \boldsymbol{L}_{[k]} & \mathbf{0} \\ \boldsymbol{M}_{k} & \boldsymbol{N}_{k} \end{bmatrix}\begin{bmatrix} \boldsymbol{D}_{[k]} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{E}_{k} \end{bmatrix}\begin{bmatrix} \boldsymbol{L}_{[k]}^{*} & \boldsymbol{M}_{k}^{*} \\ \mathbf{0} & \boldsymbol{N}_{k}^{*} \end{bmatrix}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U},$
lyche-numerical-linear-algebra_ EQ0306 (5.13)	125	■ 0.998 ● N|+0	\boldsymbol{v}^{*} \boldsymbol{v}=(\boldsymbol{x}- \boldsymbol{y})^{*}(\boldsymbol{x}-\boldsymbol{y})=2 \boldsymbol{x}^{*} \boldsymbol{x}-2 \operatorname{Re} \left(\boldsymbol{y}^{*} \boldsymbol{x}\right)=2 \boldsymbol{v}^{*} \boldsymbol{x} .	$\boldsymbol{v}^{*} \boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^{*}(\boldsymbol{x}-\boldsymbol{y})=2 \boldsymbol{x}^{*} \boldsymbol{x}-2 \operatorname{Re}\left(\boldsymbol{y}^{*} \boldsymbol{x}\right)=2 \boldsymbol{v}^{*} \boldsymbol{x} .$	$\boldsymbol{v}^{*} \boldsymbol{v}=(\boldsymbol{x}-\boldsymbol{y})^{*}(\boldsymbol{x}-\boldsymbol{y})=2 \boldsymbol{x}^{*} \boldsymbol{x}-2 \mathrm{Re}(\boldsymbol{y}^{*} \boldsymbol{x})=2 \boldsymbol{v}^{*} \boldsymbol{x} .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0320	129	0.998 C|-4	\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text { or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] .	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text { or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text { or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0438	175	0.998 W|+0	1=\\|z\\|_{2}^{2}=\left\\|\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}\right\\|_{2}^{2}=\left\\|\boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}\right\\|_{2}^{2}=\frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+\frac{y_{n}^{2}}{\sigma_{n}^{2}} .	$1=\lVert z\rVert_{2}^{2}=\left\lVert\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}\right\rVert_{2}^{2}=\left\lVert\boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}\right\rVert_{2}^{2}=\frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+\frac{y_{n}^{2}}{\sigma_{n}^{2}} .$	$1=\lVert z\rVert_{2}^{2}=\lVert\boldsymbol{V}_{1} \boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}\rVert_{2}^{2}=\lVert\boldsymbol{\Sigma}_{1}^{-1} \boldsymbol{y}\rVert_{2}^{2}=\frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+\frac{y_{n}^{2}}{\sigma_{n}^{2}} .$
lyche-numerical-linear-algebra_ EQ0477 (8.10)	189	0.998 N|+0	\left\\|\boldsymbol{A}^{k}\right\\| \leq\\|\boldsymbol{A}\\|^{k} \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N} \text {. }	$\left\lVert\boldsymbol{A}^{k}\right\rVert \leq\lVert\boldsymbol{A}\rVert^{k} \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N} \text {. }$	$\lVert\boldsymbol{A}^{k}\rVert \leq\lVert\boldsymbol{A}\rVert^{k} \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N} \text {. }$
lyche-numerical-linear-algebra_ EQ0487 (8.17)	191	0.998 W|-2	\\|\boldsymbol{A}\\|_{p}:=\max _{\boldsymbol{x} \neq 0} \frac{\\|\boldsymbol{A} \boldsymbol{x}\\|_{p}}{\\|\boldsymbol{x}\\|_{p}}=\max _{\\|\boldsymbol{y}\\|_{p}=1}\\|\boldsymbol{A} \boldsymbol{y}\\|_{p} .	$\lVert\boldsymbol{A}\rVert_{p}:=\max _{\boldsymbol{x} \neq 0} \frac{\lVert\boldsymbol{A} \boldsymbol{x}\rVert_{p}}{\lVert\boldsymbol{x}\rVert_{p}}=\max _{\lVert\boldsymbol{y}\rVert_{p}=1}\lVert\boldsymbol{A} \boldsymbol{y}\rVert_{p} .$	$\lVert\boldsymbol{A}\rVert_{p}:=\max _{\boldsymbol{x} \neq 0} \frac{\lVert\boldsymbol{A} \boldsymbol{x}\rVert_{p}}{\lVert\boldsymbol{x}\rVert_{p}}=\max _{\lVert\boldsymbol{y}\rVert_{p}=1}\lVert\boldsymbol{A} \boldsymbol{y}\rVert_{p} .$
lyche-numerical-linear-algebra_ EQ0517 (8.32)	201	0.998 N|+0	a_{1}^{\lambda_{1}} a_{2}^{\lambda_{2}} \cdots a_{n}^{\lambda_{n}} \leq \sum_{j=1}^{n} \lambda_{j} a_{j},	$a_{1}^{\lambda_{1}} a_{2}^{\lambda_{2}} \cdots a_{n}^{\lambda_{n}} \leq \sum_{j=1}^{n} \lambda_{j} a_{j},$	$a_{1}^{\lambda_{1}} a_{2}^{\lambda_{2}} \cdots a_{n}^{\lambda_{n}} \leq \sum_{j=1}^{n} \lambda_{j} a_{j},$
lyche-numerical-linear-algebra_ EQ0544	206	0.998 N|+0	\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .$	$\boldsymbol{A}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0997	336	0.998 N|+0	p_{2}(-M)>0, \quad p_{2}\left(d_{1}\right)<0, \quad p_{2}(+M)>0 .	$p_{2}(-M)>0, \quad p_{2}\left(d_{1}\right)<0, \quad p_{2}(+M)>0 .$	$p_{2}(-M)>0, \quad p_{2}(d_{1})<0, \quad p_{2}(+M)>0 .$
lyche-numerical-linear-algebra_ EQ1049	358	0.998 W|+0	D_{i} f(\boldsymbol{x}):=\frac{\partial f(\boldsymbol{x})}{\partial x_{i}}:=\lim _{h \rightarrow \mathbf{0}} \frac{f\left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)-f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^{n},	$D_{i} f(\boldsymbol{x}):=\frac{\partial f(\boldsymbol{x})}{\partial x_{i}}:=\lim _{h \rightarrow \mathbf{0}} \frac{f\left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)-f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^{n},$	$D_{i} f(\boldsymbol{x}):=\frac{\partial f(\boldsymbol{x})}{\partial x_{i}}:=\lim _{h \rightarrow \mathbf{0}} \frac{f(\boldsymbol{x}+h \boldsymbol{e}_{i})-f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^{n},$
lyche-numerical-linear-algebra_ EQ0027 (1.10)	031	0.998 N|+0	\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{x}^{T} \overline{\boldsymbol{y}}=\sum_{j=1}^{n} x_{j} \overline{y_{j}} .	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{x}^{T} \overline{\boldsymbol{y}}=\sum_{j=1}^{n} x_{j} \overline{y_{j}} .$	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\boldsymbol{y}^{*} \boldsymbol{x}=\boldsymbol{x}^{T} \overline{\boldsymbol{y}}=\sum_{j=1}^{n} x_{j} \overline{y_{j}} .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>lyche-numerical-linear-algebra_ EQ0362 (6.1)</td><td>147</td><td>■ 0.998 ● N|+0</td><td><code>\pi_{A C}(\lambda)=\lambda^{m-n} \pi_{C A}(\lambda), \quad \lambda \in \mathbb{C} .</code></td><td>$\pi_{A C}(\lambda)=\lambda^{m-n} \pi_{C A}(\lambda), \quad \lambda \in \mathbb{C}.$</td><td>$\pi_{\boldsymbol{AC}}(\lambda)=\lambda^{m-n} \pi_{\boldsymbol{CA}}(\lambda), \quad \lambda \in \mathbb{C}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0611</td><td>224</td><td>■ 0.998 ● N|+0</td><td><code>\boldsymbol{A}:=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right][2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \end{array}\right], \boldsymbol{A}^{\dagger}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]\left[\frac{1}{2}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right]=\frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{b}= \left[\begin{array}{l} 1 / 2 \\ 1 / 2 \end{array}\right] .</code></td><td>$\boldsymbol{A}:=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right][2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \end{array}\right], \boldsymbol{A}^{\dagger}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]\left[\frac{1}{2}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right]=\frac{1}{4} \boldsymbol{A}, \boldsymbol{A}^{\dagger} \boldsymbol{b}= \left[\begin{array}{l} 1 / 2 \\ 1 / 2 \end{array}\right] .$</td><td>$\boldsymbol{A}:=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}[2] \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}, \ \boldsymbol{A}^{\dagger}=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\begin{bmatrix} \frac{1}{2}\end{bmatrix} \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \end{bmatrix}=\frac{1}{4} \boldsymbol{A}, \ \boldsymbol{A}^{\dagger} \boldsymbol{b}= \begin{bmatrix} 1 / 2 \\ 1 / 2 \end{bmatrix} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0872 (13.28)</td><td>303</td><td>■ 0.998 ● N|-1</td><td><code>\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0} \right\|_{\boldsymbol{A}}} \leq \min _{\substack{Q \in \Pi_{k} \\ Q(0)=1}} \max _{a \leq x \leq b}|Q(x)|,</code></td><td>$\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{\boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0} \right\|_{\boldsymbol{A}}} \leq \min _{\substack{Q \in \Pi_{k} \\ Q(0)=1}} \max _{a \leq x \leq b}|Q(x)|,$</td><td>$\frac{\|\boldsymbol{x}-\boldsymbol{x}_{k}\|_{\boldsymbol{A}}}{\|\boldsymbol{x}-\boldsymbol{x}_{0}\|_{\boldsymbol{A}}} \leq \min _{\substack{Q \in \mathit{\Pi}_{k} \\ Q(0)=1}} \max _{a \leq x \leq b}|Q(x)|,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0016 (1.4)</td><td>028</td><td>■ 0.999 ● N|-1</td><td><code>\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S} \text { and } \boldsymbol{t} \in \mathcal{T}\} .</code></td><td>$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S} \text { and } \boldsymbol{t} \in \mathcal{T}\} .$</td><td>$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S} \text { and } \boldsymbol{t} \in \mathcal{T}\} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0073</td><td>044</td><td>■ 0.999 ● N|+0</td><td><code>b_{j, k}=\left(\alpha_{k}+\beta_{j}\right) A_{k}\left(-\beta_{j}\right) B_{j}\left(-\alpha_{k}\right),</code></td><td>$b_{j, k}=\left(\alpha_{k}+\beta_{j}\right) A_{k}\left(-\beta_{j}\right) B_{j}\left(-\alpha_{k}\right),$</td><td>$b_{j, k}=(\alpha_{k}+\beta_{j}) A_{k}(-\beta_{j}) B_{j}(-\alpha_{k}),$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0101 (2.18)	055	■ 0.999 ● W|+1	`\begin{aligned} \left|u_{k+1}\right| & = \left|d_{k+1}-l_{k} c_{k}\right|=\left|d_{k+1}-\frac{a_{k} c_{k}}{u_{k}}\right| \geq\left|d_{k+1}\right|-\frac{\left|a_{k}\right|\left|c_{k}\right|}{\left|u_{k}\right|} \\ & \geq\left|d_{k+1}\right|-\left|a_{k}\right|=\sigma_{k+1}+\left|c_{k+1}\right| \end{aligned}`	$\begin{aligned} \left\|u_{k+1}\right\| & = \left\|d_{k+1}-l_{k} c_{k}\right\|=\left\|d_{k+1}-\frac{a_{k} c_{k}}{u_{k}}\right\| \geq\left\|d_{k+1}\right\|-\frac{\left\|a_{k}\right\|\left\|c_{k}\right\|}{\left\|u_{k}\right\|} \\ & \geq\left\|d_{k+1}\right\|-\left\|a_{k}\right\|=\sigma_{k+1}+\left\|c_{k+1}\right\| \end{aligned}$	$\begin{aligned} \|u_{k+1}\| &= \|d_{k+1}-l_k c_k\| = \|d_{k+1}-\frac{a_k c_k}{u_k}\| \geq \|d_{k+1}\| - \frac{\|a_k\|\|c_k\|}{\|u_k\|} \\ &\geq \|d_{k+1}\| - \|a_k\| = \sigma_{k+1} + \|c_{k+1}\|. \end{aligned}$
lyche-numerical-linear-algebra_ EQ0156	071	■ 0.999 ● N|+0	`\delta^{2} f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^{2}}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .`	$\delta^{2} f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^{2}}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	$\delta^2 f(x) := \frac{f(x+h)-2f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h,x+h] \rightarrow \mathbb{R}.$
lyche-numerical-linear-algebra_ EQ0250	102	■ 0.999 ● N|+1	`\left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array} \right]=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{cc} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]`	$\left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array} \right]=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{cc} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$	$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 8/3 \end{bmatrix} \begin{bmatrix} 1 & 1/3 \\ 0 & 1 \end{bmatrix}$
lyche-numerical-linear-algebra_ EQ0296 (5.11)	123	■ 0.999 ● N|+0	`\left\\|\boldsymbol{v}-\boldsymbol{s}_{0}\right\\|<\\|\boldsymbol{v}-\boldsymbol{s}\\|, \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_{0} .`	$\left\\|\boldsymbol{v}-\boldsymbol{s}_{0}\right\\|<\\|\boldsymbol{v}-\boldsymbol{s}\\|, \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_{0} .$	$\\|\boldsymbol{v}-\boldsymbol{s}_0\\| < \\|\boldsymbol{v}-\boldsymbol{s}\\|, \textit{ for all } \boldsymbol{s} \in \mathcal{S}, \ \boldsymbol{s} \neq \boldsymbol{s}_0.$
lyche-numerical-linear-algebra_ EQ0361	147	■ 0.999 ● N|+0	`\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) & =\operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I} \right)=\operatorname{det}\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S} \right) \\ & =\operatorname{det}\left(\boldsymbol{S}^{-1} \right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) \operatorname{det}(\boldsymbol{S})= \operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \operatorname{det}(\boldsymbol{A}- \lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda), \end{aligned}`	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) & =\operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I} \right)=\operatorname{det}\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S} \right) \\ & =\operatorname{det}\left(\boldsymbol{S}^{-1} \right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) \operatorname{det}(\boldsymbol{S})= \operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \operatorname{det}(\boldsymbol{A}- \lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda), \end{aligned}$	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) &= \det(\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}-\lambda\boldsymbol{I}) = \det\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{S}\right) \\ &= \det(\boldsymbol{S}^{-1})\det(\boldsymbol{A}-\lambda\boldsymbol{I})\det(\boldsymbol{S}) = \det(\boldsymbol{S}^{-1}\boldsymbol{S})\det(\boldsymbol{A}-\lambda\boldsymbol{I}) = \pi_{\boldsymbol{A}}(\lambda), \end{aligned}$
lyche-numerical-linear-algebra_ EQ0381	154	■ 0.999 ● W|-1	`e_{i i}=\sum_{k=1}^{n} \bar{b}_{k i} b_{k i}= \sum_{k=1}^{i}\left|b_{k i}\right|^{2} \text { and } f_{i i}=\sum_{k=1}^{n} b_{i k} \bar{b}_{i k}=\sum_{k=i}^{n} \left|b_{i k}\right|^{2} .`	$e_{i i}=\sum_{k=1}^{n} \bar{b}_{k i} b_{k i}= \sum_{k=1}^{i}\left\|b_{k i}\right\|^{2} \text { and } f_{i i}=\sum_{k=1}^{n} b_{i k} \bar{b}_{i k}=\sum_{k=i}^{n} \left\|b_{i k}\right\|^{2} .$	$e_{ii} = \sum_{k=1}^{n} \overline{b}_{ki} b_{ki} = \sum_{k=1}^{i} \|b_{ki}\|^2 \text{ and } f_{ii} = \sum_{k=1}^{n} b_{ik}\overline{b}_{ik} = \sum_{k=i}^{n} \|b_{ik}\|^2.$
lyche-numerical-linear-algebra_ EQ0406	162	■ 0.999 ● N|+0	`\boldsymbol{A}=\left[\begin{array}{ccccc} -q_{n-1} & -q_{n-2} & \cdots & -q_{1} & -q_{0} \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{array} \right] .`	$\boldsymbol{A}=\left[\begin{array}{ccccc} -q_{n-1} & -q_{n-2} & \cdots & -q_{1} & -q_{0} \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{array} \right] .$	$\boldsymbol{A} = \begin{bmatrix} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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● N|+0</td><td><code>\boldsymbol{A}:=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}</code></td><td>$\boldsymbol{A}:=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}$</td><td>$\boldsymbol{A}:=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{T}$</td></tr>
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EQ0471 (8.7)</td><td>187</td><td>■ 0.999
● N|+0</td><td><code>m\|\boldsymbol{x}\|^{\prime} \leq\|\boldsymbol{x}\| \leq M\|\boldsymbol{x}\|^{\prime} .</code></td><td>$m\|\boldsymbol{x}\|^{\prime} \leq\|\boldsymbol{x}\| \leq M\|\boldsymbol{x}\|^{\prime} .$</td><td>$m\|\boldsymbol{x}\|^{\prime} \leq\|\boldsymbol{x}\| \leq M\|\boldsymbol{x}\|^{\prime} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0481 (8.14)</td><td>190</td><td>■ 0.999
● N|+0</td><td><code>\mathcal{S}:=\left\{x \in \mathbb{C}^{n}:\|\boldsymbol{x}\|=1\right\}</code></td><td>$\mathcal{S}:=\left\{x \in \mathbb{C}^{n}:\|\boldsymbol{x}\|=1\right\}$</td><td>$\mathcal{S}:=\left\{x \in \mathbb{C}^{n}:\|\boldsymbol{x}\|=1\right\}$</td></tr>
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● N|+0</td><td><code>\|\boldsymbol{x}\|_{\boldsymbol{A}}:=\sqrt{ \boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}, \quad \boldsymbol{x} \in \mathbb{R}^{n} .</code></td><td>$\|\boldsymbol{x}\|_{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}, \quad \boldsymbol{x} \in \mathbb{R}^{n} .$</td><td>$\|\boldsymbol{x}\|_{A}:=\sqrt{\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}}, \quad \boldsymbol{x} \in \mathbb{R}^{n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0634</td><td>230</td><td>■ 0.999
● N|+0</td><td><code>\|\boldsymbol{C}-\boldsymbol{D}\|_{F}^{2}=\left\| \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A}-\boldsymbol{B} \\ \boldsymbol{A}^{*}-\boldsymbol{B}^{*} & \mathbf{0} \end{array}\right]\right\|_{F}^{2}=\| \boldsymbol{B}-\boldsymbol{A}\|_{F}^{2}+\left\|(\boldsymbol{B}-\boldsymbol{A})^{*}\right\|_{F}^{2}=2 \|\boldsymbol{B}-\boldsymbol{A}\|_{F}^{2} .</code></td><td>$\|\boldsymbol{C}-\boldsymbol{D}\|_{F}^{2}=\left\| \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A}-\boldsymbol{B} \\ \boldsymbol{A}^{*}-\boldsymbol{B}^{*} & \mathbf{0} \end{array}\right]\right\|_{F}^{2}=\| \boldsymbol{B}-\boldsymbol{A}\|_{F}^{2}+\left\|(\boldsymbol{B}-\boldsymbol{A})^{*}\right\|_{F}^{2}=2 \|\boldsymbol{B}-\boldsymbol{A}\|_{F}^{2} .$</td><td>$\|\boldsymbol{C}-\boldsymbol{D}\|_{F}^{2}=\| \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A}-\boldsymbol{B} \\ \boldsymbol{A}^{*}-\boldsymbol{B}^{*} & \mathbf{0} \end{array}\right]\|_{F}^{2}=\| \boldsymbol{B}-\boldsymbol{A}\|_{F}^{2}+\|(\boldsymbol{B}-\boldsymbol{A})^{*}\|_{F}^{2}=2 \|\boldsymbol{B}-\boldsymbol{A}\|_{F}^{2} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0816</td><td>284</td><td>■ 0.999
● K|+0</td><td><code>\frac{1}{4}\left(\lambda^{1 / 2} w_{i-1, j}+\lambda^{1 / 2} w_{i, j-1}+\lambda^{-1 / 2} w_{i+1, j}+\lambda^{-1 / 2} w_{i, j+1}\right)=\mu w_{i, j} .</code></td><td>$\frac{1}{4}\left(\lambda^{1 / 2} w_{i-1, j}+\lambda^{1 / 2} w_{i, j-1}+\lambda^{-1 / 2} w_{i+1, j}+\lambda^{-1 / 2} w_{i, j+1}\right)=\mu w_{i, j} .$</td><td>$\frac{1}{4}(\lambda^{1 / 2} w_{i-1, j}+\lambda^{1 / 2} w_{i, j-1}+\lambda^{-1 / 2} w_{i+1, j}+\lambda^{-1 / 2} w_{i, j+1})=\mu w_{i, j} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0861</td><td>300</td><td>■ 0.999
● K|+0</td><td><code>\kappa_{A}=\frac{5+4 \cos (\pi h)}{5-4 \cos (\pi h)} \leq 9,</code></td><td>$\kappa_{A}=\frac{5+4 \cos (\pi h)}{5-4 \cos (\pi h)} \leq 9,$</td><td>$\kappa_{A}=\frac{5+4 \cos (\pi h)}{5-4 \cos (\pi h)} \leq 9,$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0954 (13.67)</td><td>322</td><td>■ 0.999
● N|-1</td><td><code>\left\|\boldsymbol{x}^{*}-\boldsymbol{x}\right\|_{2} \leq\|\boldsymbol{A}\|_{2} \min _{\boldsymbol{w} \in \mathcal{W}}\left\|\boldsymbol{x}^{*}-\boldsymbol{w} \right\|_{2} .</code></td><td>$\left\|\boldsymbol{x}^{*}-\boldsymbol{x}\right\|_{2} \leq\|\boldsymbol{A}\|_{2} \min _{\boldsymbol{w} \in \mathcal{W}}\left\|\boldsymbol{x}^{*}-\boldsymbol{w} \right\|_{2} .$</td><td>$\|\boldsymbol{x}^{*}-\boldsymbol{x}\|_{2} \leq\|\boldsymbol{A}\|_{2} \min _{\boldsymbol{w} \in \mathcal{W}}\|\boldsymbol{x}^{*}-\boldsymbol{w}\|_{2} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra
EQ0208</td><td>086</td><td>■ 0.999
● W|+0</td><td><code>\begin{aligned} (\boldsymbol{L} \boldsymbol{U})_{i j} & =\sum_{k=1}^{n} l_{i k}^{(k)} u_{k j}=\sum_{k=1}^{j-1} a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}+a_{p_{i}, j}^{(k)} a_{p_{j}, j}^{(j)} \\ & =\sum_{k=1}^{j-1}\left(a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)}\right)+a_{p_{i}, j}^{(j)}=a_{p_{i}, j}^{(1)}=a_{p_{i}, j}=\left(\boldsymbol{P}^{T} \boldsymbol{A}\right)_{i j} . \end{aligned}</code></td><td>$\begin{aligned} (\boldsymbol{L} \boldsymbol{U})_{i j} & =\sum_{k=1}^{n} l_{i k}^{(k)} u_{k j}=\sum_{k=1}^{j-1} a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}+a_{p_{i}, j}^{(k)} a_{p_{j}, j}^{(j)} \\ & =\sum_{k=1}^{j-1}\left(a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)}\right)+a_{p_{i}, j}^{(j)}=a_{p_{i}, j}^{(1)}=a_{p_{i}, j}=\left(\boldsymbol{P}^{T} \boldsymbol{A}\right)_{i j} . \end{aligned}$</td><td>$\begin{aligned} (\boldsymbol{L} \boldsymbol{U})_{i j} & =\sum_{k=1}^{n} l_{i k}^{(k)} u_{k j}=\sum_{k=1}^{j-1} a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}+a_{p_{i}, j}^{(k)} a_{p_{j}, j}^{(j)} \\ & =\sum_{k=1}^{j-1}(a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)})+a_{p_{i}, j}^{(j)}=a_{p_{i}, j}^{(1)}=a_{p_{i}, j}=(\boldsymbol{P}^{T} \boldsymbol{A})_{i j} . \end{aligned}$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0265 (4.9)	109	0.999 W|+0	\begin{aligned} 0 \leq \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} & =\left[-\boldsymbol{y}^{*} \boldsymbol{v} / \alpha, \boldsymbol{y}^{*}\right] \left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right]\left[\begin{array}{c} -\boldsymbol{v}^{*} \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{array} \right] \\ & =\left[0,-\left(\boldsymbol{y}^{*} \boldsymbol{v}\right) \boldsymbol{v}^{*} / \alpha+ \boldsymbol{y}^{*} \boldsymbol{B}\right]\left[\begin{array}{c} -\boldsymbol{v}^{*} \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{array}\right] \\ & =- \boldsymbol{y}^{*} \boldsymbol{v} \boldsymbol{v}^{*} \boldsymbol{y} / \alpha+\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}=\boldsymbol{y}^{*} \boldsymbol{C} \boldsymbol{y} . \end{aligned}	$\begin{aligned} 0 \leq \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} & =\left[-\boldsymbol{y}^{*} \boldsymbol{v} / \alpha, \boldsymbol{y}^{*}\right] \left[\begin{array}{ll} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right]\left[\begin{array}{c} -\boldsymbol{v}^{*} \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{array} \right] \\ & =\left[0,-\left(\boldsymbol{y}^{*} \boldsymbol{v}\right) \boldsymbol{v}^{*} / \alpha+ \boldsymbol{y}^{*} \boldsymbol{B}\right]\left[\begin{array}{c} -\boldsymbol{v}^{*} \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{array}\right] \\ & =- \boldsymbol{y}^{*} \boldsymbol{v} \boldsymbol{v}^{*} \boldsymbol{y} / \alpha+\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}=\boldsymbol{y}^{*} \boldsymbol{C} \boldsymbol{y} . \end{aligned}$	$\begin{aligned} 0 \leq \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} & =[-\boldsymbol{y}^{*} \boldsymbol{v} / \alpha, \boldsymbol{y}^{*}] \begin{bmatrix} \alpha & \boldsymbol{v}^{*} \\ \boldsymbol{v} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} -\boldsymbol{v}^{*} \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix} \\ & =[0,-(\boldsymbol{y}^{*} \boldsymbol{v}) \boldsymbol{v}^{*} / \alpha+ \boldsymbol{y}^{*} \boldsymbol{B}] \begin{bmatrix} -\boldsymbol{v}^{*} \boldsymbol{y} / \alpha \\ \boldsymbol{y} \end{bmatrix} \\ & =- \boldsymbol{y}^{*} \boldsymbol{v} \boldsymbol{v}^{*} \boldsymbol{y} / \alpha+\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}=\boldsymbol{y}^{*} \boldsymbol{C} \boldsymbol{y} . \end{aligned}$
lyche-numerical-linear-algebra_ EQ0309	126	0.999 C|-12	\begin{gathered} \boldsymbol{H}:=\boldsymbol{I}-\frac{2 \boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]- \frac{2}{4}\left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right]\left[\begin{array}{lll} 2 & 0 & 0 \end{array}\right]=\left[\begin{array}{ccc} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right], \\ \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}= \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]-\frac{1}{4}\left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right]\left[\begin{array}{lll} 2 & 0 & 0 \end{array}\right]=\left[\begin{array}{lll} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] . \end{gathered}	$\begin{gathered} \boldsymbol{H}:=\boldsymbol{I}-\frac{2 \boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]- \frac{2}{4}\left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right]\left[\begin{array}{lll} 2 & 0 & 0 \end{array}\right]=\left[\begin{array}{ccc} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right], \\ \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}= \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]-\frac{1}{4}\left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right]\left[\begin{array}{lll} 2 & 0 & 0 \end{array}\right]=\left[\begin{array}{lll} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] . \end{gathered}$	$\begin{gathered} \boldsymbol{H}:=\boldsymbol{I}-\frac{2 \boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}- \frac{2}{4}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \end{bmatrix}=\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}}= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}-\frac{1}{4}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \end{bmatrix}=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \end{gathered}$
lyche-numerical-linear-algebra_ EQ0645	232	0.999 W|+1	\begin{aligned} B & =(B A) B=\left(A^{*}\right) B^{*} B= \left(A^{*} C^{*}\right) A^{*} B^{*} B=C A\left(A^{*} B^{*}\right) B \\ & =C A(B A B)=(C) A B=C(A C) A B=C C^{*} A^{*}(A B) \\ & =C C^{*}\left(A^{*} B^{*} A^{*}\right)=C\left(C^{*} A^{*}\right)=C A C=C . \end{aligned}	$\begin{aligned} B & =(B A) B=\left(A^{*}\right) B^{*} B= \left(A^{*} C^{*}\right) A^{*} B^{*} B=C A\left(A^{*} B^{*}\right) B \\ & =C A(B A B)=(C) A B=C(A C) A B=C C^{*} A^{*}(A B) \\ & =C C^{*}\left(A^{*} B^{*} A^{*}\right)=C\left(C^{*} A^{*}\right)=C A C=C . \end{aligned}$	$\begin{aligned} \boldsymbol{B} & =(\boldsymbol{B}\boldsymbol{A})\boldsymbol{B}=(\boldsymbol{A}^{*})\boldsymbol{B}^{*}\boldsymbol{B}=(\boldsymbol{A}^{*}\boldsymbol{C}^{*})\boldsymbol{A}^{*}\boldsymbol{B}^{*}\boldsymbol{B}=\boldsymbol{C}\boldsymbol{A}(\boldsymbol{A}^{*}\boldsymbol{B}^{*})\boldsymbol{B} \\ & =\boldsymbol{C}\boldsymbol{A}(\boldsymbol{B}\boldsymbol{A}\boldsymbol{B})=(\boldsymbol{C})\boldsymbol{A}\boldsymbol{B}=\boldsymbol{C}(\boldsymbol{A}\boldsymbol{C})\boldsymbol{A}\boldsymbol{B}=\boldsymbol{C}\boldsymbol{C}^{*}\boldsymbol{A}^{*}(\boldsymbol{A}\boldsymbol{B}) \\ & =\boldsymbol{C}\boldsymbol{C}^{*}(\boldsymbol{A}^{*}\boldsymbol{B}^{*}\boldsymbol{A}^{*})=\boldsymbol{C}(\boldsymbol{C}^{*}\boldsymbol{A}^{*})=\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}=\boldsymbol{C}. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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EQ0827</td><td>287</td><td>■ 0.999
● N|+0</td><td><code>\boldsymbol{x}:=\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{ll} 0 & a \\ a & 0 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]+\left[\begin{array}{l} 1-a \\ 1-a \end{array}\right]=: \boldsymbol{G} \boldsymbol{x}+ \boldsymbol{c} .</code></td><td>$\boldsymbol{x}:=\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{ll} 0 & a \\ a & 0 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]+\left[\begin{array}{l} 1-a \\ 1-a \end{array}\right]=: \boldsymbol{G} \boldsymbol{x}+ \boldsymbol{c} .$</td><td>$\boldsymbol{x}:=\begin{bmatrix} x_{1} \\ x_{2} \end{bmatrix}=\begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix}\begin{bmatrix} x_{1} \\ x_{2} \end{bmatrix}+\begin{bmatrix} 1-a \\ 1-a \end{bmatrix}=: \boldsymbol{G} \boldsymbol{x}+ \boldsymbol{c}.$</td></tr>
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● K|+0</td><td><code>\overline{\boldsymbol{C}}=\boldsymbol{C}\left(\boldsymbol{I}+\frac{1}{\lambda} \boldsymbol{e}_{i}\left(\boldsymbol{e}_{i}^{T}-\boldsymbol{a}^{T} \boldsymbol{C} \right)\right)</code></td><td>$\overline{\boldsymbol{C}}=\boldsymbol{C}\left(\boldsymbol{I}+\frac{1}{\lambda} \boldsymbol{e}_{i}\left(\boldsymbol{e}_{i}^{T}-\boldsymbol{a}^{T} \boldsymbol{C} \right)\right)$</td><td>$\overline{\boldsymbol{C}}=\boldsymbol{C}\big(\boldsymbol{I}+\frac{1}{\lambda} \boldsymbol{e}_{i}(\boldsymbol{e}_{i}^{T}-\boldsymbol{a}^{T} \boldsymbol{C})\big)$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0202 (3.12)</td><td>085</td><td>■ 0.999
● N|+0</td><td><code>\boldsymbol{P}^{T}=\boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_{1}=\boldsymbol{I}(\boldsymbol{p},:), \quad \boldsymbol{P}=\boldsymbol{P}_{1} \boldsymbol{P}_{2} \cdots \boldsymbol{P}_{n-1}= \boldsymbol{I}(:, \boldsymbol{p}) .</code></td><td>$\boldsymbol{P}^{T}=\boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_{1}=\boldsymbol{I}(\boldsymbol{p},:), \quad \boldsymbol{P}=\boldsymbol{P}_{1} \boldsymbol{P}_{2} \cdots \boldsymbol{P}_{n-1}= \boldsymbol{I}(:, \boldsymbol{p}) .$</td><td>$\boldsymbol{P}^{T}=\boldsymbol{P}_{n-1} \cdots \boldsymbol{P}_{1}=\boldsymbol{I}(\boldsymbol{p},:), \quad \boldsymbol{P}=\boldsymbol{P}_{1} \boldsymbol{P}_{2} \cdots \boldsymbol{P}_{n-1}= \boldsymbol{I}(:, \boldsymbol{p}).$</td></tr>
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EQ0254</td><td>104</td><td>■ 0.999
● W|+0</td><td><code>\begin{aligned} \boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x} & =2 \sum_{i=1}^{n} x_{i}^{2}-\sum_{i=1}^{n-1} x_{i} x_{i+1}-\sum_{i=2}^{n} x_{i-1} x_{i} \\ & =\sum_{i=1}^{n-1} x_{i}^{2}-2 \sum_{i=1}^{n-1} x_{i} x_{i+1}+\sum_{i=1}^{n-1} x_{i+1}^{2}+x_{1}^{2}+x_{n}^{2} \\ & =x_{1}^{2}+x_{n}^{2}+\sum_{i=1}^{n-1}\left(x_{i+1}-x_{i} \right)^{2} . \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x} & =2 \sum_{i=1}^{n} x_{i}^{2}-\sum_{i=1}^{n-1} x_{i} x_{i+1}-\sum_{i=2}^{n} x_{i-1} x_{i} \\ & =\sum_{i=1}^{n-1} x_{i}^{2}-2 \sum_{i=1}^{n-1} x_{i} x_{i+1}+\sum_{i=1}^{n-1} x_{i+1}^{2}+x_{1}^{2}+x_{n}^{2} \\ & =x_{1}^{2}+x_{n}^{2}+\sum_{i=1}^{n-1}\left(x_{i+1}-x_{i} \right)^{2} . \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{x}^{T} \boldsymbol{T} \boldsymbol{x} & =2 \sum_{i=1}^{n} x_{i}^{2}-\sum_{i=1}^{n-1} x_{i} x_{i+1}-\sum_{i=2}^{n} x_{i-1} x_{i} \\ & =\sum_{i=1}^{n-1} x_{i}^{2}-2 \sum_{i=1}^{n-1} x_{i} x_{i+1}+\sum_{i=1}^{n-1} x_{i+1}^{2}+x_{1}^{2}+x_{n}^{2} \\ & =x_{1}^{2}+x_{n}^{2}+\sum_{i=1}^{n-1}(x_{i+1}-x_{i})^{2}. \end{aligned}$</td></tr>
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● N|+0</td><td><code>\|\boldsymbol{x}+\boldsymbol{y}\|^{2} \leq\| \boldsymbol{x}\|^{2}+2\|\boldsymbol{x}\|\|\boldsymbol{y} \|+\|\boldsymbol{y}\|^{2}=(\|\boldsymbol{x}\|+\| \boldsymbol{y}\|)^{2} .</code></td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2} \leq\| \boldsymbol{x}\|^{2}+2\|\boldsymbol{x}\|\|\boldsymbol{y} \|+\|\boldsymbol{y}\|^{2}=(\|\boldsymbol{x}\|+\| \boldsymbol{y}\|)^{2} .$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2} \leq\| \boldsymbol{x}\|^{2}+2\|\boldsymbol{x}\|\|\boldsymbol{y} \|+\|\boldsymbol{y}\|^{2}=(\|\boldsymbol{x}\|+\| \boldsymbol{y}\|)^{2}.$</td></tr>
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EQ0310</td><td>126</td><td>■ 0.999
● W|+0</td><td><code>\mathcal{M}:=\left\{\boldsymbol{w} \in \mathbb{R}^{3}: \boldsymbol{w}^{T} \boldsymbol{v}=0\right\}=\left\{\left[\begin{array}{l} w_{1} \\ w_{2} \\ w_{3} \end{array} \right]: 2 w_{1}=0\right\}</code></td><td>$\mathcal{M}:=\left\{\boldsymbol{w} \in \mathbb{R}^{3}: \boldsymbol{w}^{T} \boldsymbol{v}=0\right\}=\left\{\left[\begin{array}{l} w_{1} \\ w_{2} \\ w_{3} \end{array} \right]: 2 w_{1}=0\right\}$</td><td>$\mathcal{M}:=\{\boldsymbol{w} \in \mathbb{R}^{3}: \boldsymbol{w}^{T} \boldsymbol{v}=0\}=\{\left[\begin{smallmatrix} w_{1} \\ w_{2} \\ w_{3} \end{smallmatrix} \right]: 2 w_{1}=0\}$</td></tr>
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EQ0348</td><td>141</td><td>■ 0.999
● W|+0</td><td><code>\left[\begin{array}{ll} p_{i i} & p_{i j} \\ p_{j i} & p_{j j} \end{array}\right]=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right],</code></td><td>$\left[\begin{array}{ll} p_{i i} & p_{i j} \\ p_{j i} & p_{j j} \end{array}\right]=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right],$</td><td>$\begin{bmatrix} p_{i i} & p_{i j} \\ p_{j i} & p_{j j} \end{bmatrix}=\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix},$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra EQ0440</td><td>176</td><td>■ 0.999 • C|-4</td><td><code>\frac{\left(\frac{3}{5} x_{1}+\frac{4}{5} x_{2} \right)^{2}}{9}+\frac{\left(-\frac{4}{5} x_{1}+ \frac{3}{5} x_{2}\right)^{2}}{1}=1,</code></td><td>$\frac{\left(\frac{3}{5} x_{1}+\frac{4}{5} x_{2} \right)^{2}}{9}+\frac{\left(-\frac{4}{5} x_{1}+ \frac{3}{5} x_{2}\right)^{2}}{1}=1,$</td><td>$\frac{(\frac{3}{5} x_{1}+\frac{4}{5} x_{2})^{2}}{9}+\frac{(-\frac{4}{5} x_{1}+ \frac{3}{5} x_{2})^{2}}{1}=1,$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0499</td><td>195</td><td>■ 0.999 • K|+0</td><td><code>x_{1}+\left(1+10^{-16}\right) x_{2}=20-10^{-15},</code></td><td>$x_{1}+\left(1+10^{-16}\right) x_{2}=20-10^{-15},$</td><td>$x_{1}+(1+10^{-16}) x_{2}=20-10^{-15},$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0510</td><td>198</td><td>■ 0.999 • K|+0</td><td><code>\left\|\boldsymbol{B}^{-1}\right\| \leq\left\| \boldsymbol{A}^{-1}\right\|+\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right\| \leq\left\| \boldsymbol{A}^{-1}\right\|+r\left\|\boldsymbol{B}^{-1} \right\| .</code></td><td>$\left\|\boldsymbol{B}^{-1}\right\| \leq\left\| \boldsymbol{A}^{-1}\right\|+\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right\| \leq\left\| \boldsymbol{A}^{-1}\right\|+r\left\|\boldsymbol{B}^{-1} \right\| .$</td><td>$\|\boldsymbol{B}^{-1}\| \leq \|\boldsymbol{A}^{-1}\|+\|\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\| \leq \|\boldsymbol{A}^{-1}\|+r\|\boldsymbol{B}^{-1}\|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0559</td><td>209</td><td>■ 0.999 • N|+0</td><td><code>K(\boldsymbol{B})^{-1} \frac{\|\boldsymbol{E}\|}{\| \boldsymbol{A}\|} \leq \frac{\left\|\boldsymbol{B}^{-1}- \boldsymbol{A}^{-1}\right\|}{\left\|\boldsymbol{B}^{-1} \right\|} .</code></td><td>$K(\boldsymbol{B})^{-1} \frac{\|\boldsymbol{E}\|}{\| \boldsymbol{A}\|} \leq \frac{\left\|\boldsymbol{B}^{-1}- \boldsymbol{A}^{-1}\right\|}{\left\|\boldsymbol{B}^{-1} \right\|} .$</td><td>$K(\boldsymbol{B})^{-1} \frac{\|\boldsymbol{E}\|}{\| \boldsymbol{A}\|} \leq \frac{\|\boldsymbol{B}^{-1}- \boldsymbol{A}^{-1}\|}{\|\boldsymbol{B}^{-1}\|}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0617 (9.14)</td><td>226</td><td>■ 0.999 • N|+0</td><td><code>\frac{1}{4} 10^{-2} \leq \frac{\|\boldsymbol{y}- \boldsymbol{x}\|_{\infty}}{\|\boldsymbol{x}\|_{\infty}} \leq 4 \cdot 10^{-2} .</code></td><td>$\frac{1}{4} 10^{-2} \leq \frac{\|\boldsymbol{y}- \boldsymbol{x}\|_{\infty}}{\|\boldsymbol{x}\|_{\infty}} \leq 4 \cdot 10^{-2} .$</td><td>$\frac{1}{4} 10^{-2} \leq \frac{\|\boldsymbol{y}- \boldsymbol{x}\|_{\infty}}{\|\boldsymbol{x}\|_{\infty}} \leq 4 \cdot 10^{-2}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0661 (10.5)</td><td>239</td><td>■ 0.999 • N|+0</td><td><code>\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F} \quad \text { with } \quad h=1 /(m+1),</code></td><td>$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F} \quad \text { with } \quad h=1 /(m+1),$</td><td>$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F} \quad \text { with } \quad h=1 /(m+1),$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0723</td><td>258</td><td>■ 0.999 • N|+0</td><td><code>\boldsymbol{D}_{2}=\operatorname{diag}\left(1, \omega_{4} \right)=\left[\begin{array}{rr} 1 & 0 \\ 0 & -i \end{array}\right] .</code></td><td>$\boldsymbol{D}_{2}=\operatorname{diag}\left(1, \omega_{4} \right)=\left[\begin{array}{rr} 1 & 0 \\ 0 & -i \end{array}\right] .$</td><td>$\boldsymbol{D}_{2}=\operatorname{diag}(1, \omega_{4})=\left[\begin{array}{rr} 1 & 0 \\ 0 & -i \end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0846</td><td>293</td><td>■ 0.999 • N|+0</td><td><code>\begin{aligned} & \boldsymbol{t}_{0}=3\left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_{0}=\frac{1}{2}, \quad \boldsymbol{x}_{1}=-4^{-1} \left[\begin{array}{l} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{1}=3 * 4^{-1}\left[\begin{array}{l} 0 \ \ 1 \end{array}\right] \\ & \boldsymbol{t}_{1}=3 * 4^{-1} \left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_{1}=\frac{1}{2}, \quad \boldsymbol{x}_{2}=-4^{-1} \left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{2}=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}</code></td><td>$\begin{aligned} & \boldsymbol{t}_{0}=3\left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_{0}=\frac{1}{2}, \quad \boldsymbol{x}_{1}=-4^{-1} \left[\begin{array}{l} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{1}=3 * 4^{-1}\left[\begin{array}{l} 0 \\ 1 \end{array}\right] \\ & \boldsymbol{t}_{1}=3 * 4^{-1} \left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_{1}=\frac{1}{2}, \quad \boldsymbol{x}_{2}=-4^{-1} \left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{2}=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}$</td><td>$\boldsymbol{t}_{0}=3\left[\begin{smallmatrix} 1 \\ -1/2 \end{smallmatrix}\right], \quad \alpha_{0}=\frac{1}{2}, \quad \boldsymbol{x}_{1}=-4^{-1}\left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right], \quad \boldsymbol{r}_{1}=3 * 4^{-1}\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$
$\boldsymbol{t}_{1}=3 * 4^{-1}\left[\begin{smallmatrix} -1 \\ 2 \end{smallmatrix}\right], \quad \alpha_{1}=\frac{1}{2}, \quad \boldsymbol{x}_{2}=-4^{-1}\left[\begin{smallmatrix} 1 \\ 1/2 \end{smallmatrix}\right], \quad \boldsymbol{r}_{2}=3 * 4^{-1}\left[\begin{smallmatrix} 1/2 \\ 0 \end{smallmatrix}\right],$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0937</td><td>317</td><td>■ 0.999 • K|+0</td><td><code>c_{0}\left(x^{T} \boldsymbol{A}_{p} x\right) \leq x^{T} \boldsymbol{A} \boldsymbol{x} \leq c_{1}\left(x^{T} \boldsymbol{A}_{p} x\right)</code></td><td>$c_{0}\left(x^{T} \boldsymbol{A}_{p} x\right) \leq x^{T} \boldsymbol{A} \boldsymbol{x} \leq c_{1}\left(x^{T} \boldsymbol{A}_{p} x\right)$</td><td>$c_{0}(x^{T} \boldsymbol{A}_{p} x) \leq x^{T} \boldsymbol{A} \boldsymbol{x} \leq c_{1}(x^{T} \boldsymbol{A}_{p} x)$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0945</td><td>320</td><td>■ 0.999 • K|+0</td><td><code>\boldsymbol{A} \boldsymbol{p}_{k-1} \in \operatorname{span}\left(\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}, \boldsymbol{p}_{k}\right) .</code></td><td>$\boldsymbol{A} \boldsymbol{p}_{k-1} \in \operatorname{span}\left(\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}, \boldsymbol{p}_{k}\right) .$</td><td>$\boldsymbol{A} \boldsymbol{p}_{k-1} \in \operatorname{span}(\boldsymbol{p}_{k-2}, \boldsymbol{p}_{k-1}, \boldsymbol{p}_{k}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0960 (13.69)</td><td>323</td><td>■ 0.999 • W|+0</td><td><code>\begin{aligned} & \boldsymbol{x}_{k+1}=\left(1-\omega_{k}\right) \boldsymbol{x}_{k-1}+\omega_{k} \left(\boldsymbol{x}_{k}+\boldsymbol{r}_{k}\right), \\ & \boldsymbol{r}_{k+1}=\left(1-\omega_{k}\right) \boldsymbol{r}_{k-1}+\omega_{k} \boldsymbol{B} \boldsymbol{r}_{k}, \end{aligned}</code></td><td>$\begin{aligned} & \boldsymbol{x}_{k+1}=\left(1-\omega_{k}\right) \boldsymbol{x}_{k-1}+\omega_{k} \left(\boldsymbol{x}_{k}+\boldsymbol{r}_{k}\right), \\ & \boldsymbol{r}_{k+1}=\left(1-\omega_{k}\right) \boldsymbol{r}_{k-1}+\omega_{k} \boldsymbol{B} \boldsymbol{r}_{k}, \end{aligned}$</td><td>$\begin{aligned} & \boldsymbol{x}_{k+1}=(1-\omega_{k}) \boldsymbol{x}_{k-1}+\omega_{k} (\boldsymbol{x}_{k}+\boldsymbol{r}_{k}), \\ & \boldsymbol{r}_{k+1}=(1-\omega_{k}) \boldsymbol{r}_{k-1}+\omega_{k} \boldsymbol{B} \boldsymbol{r}_{k}, \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0986</td><td>332</td><td>■ 0.999 • N|+0</td><td><code>\frac{1}{\lambda} \leq\left\|\boldsymbol{A}^{-1}\right \|_{p}=\frac{K_{p}(\boldsymbol{A})}{\|\boldsymbol{A} \|_{p}}</code></td><td>$\frac{1}{\lambda} \leq\left\|\boldsymbol{A}^{-1}\right\|_{p}=\frac{K_{p}(\boldsymbol{A})}{\|\boldsymbol{A}\|_{p}}$</td><td>$\frac{1}{\lambda} \leq\|\boldsymbol{A}^{-1}\|_{p}=\frac{K_{p}(\boldsymbol{A})}{\|\boldsymbol{A}\|_{p}}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1025</td><td>346</td><td>■ 0.999 • C|+4</td><td><code>\boldsymbol{x}_{k}=\frac{\boldsymbol{z}_{k}}{ \left\|\boldsymbol{z}_{k}\right\|}=\frac{c_{1} \lambda_{1}^{k}}{\left|c_{1} \lambda_{1}^{k}\right|} \frac{\boldsymbol{v}_{1}+\frac{c_{2}}{c_{1}}\left(\frac{ \lambda_{2}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+ \cdots+\frac{c_{n}}{c_{1}}\left(\frac{\lambda_{n}}{ \lambda_{1}}\right)^{k} \boldsymbol{v}_{n}}{\left\| \boldsymbol{v}_{1}+\frac{c_{2}}{c_{1}}\left(\frac{ \lambda_{2}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+ \cdots+\frac{c_{n}}{c_{1}}\left(\frac{\lambda_{n}}{ \lambda_{1}}\right)^{k} \boldsymbol{v}_{n}\right\|}, \quad k=0,1,2, \ldots,</code></td><td>$\boldsymbol{x}_{k}=\frac{\boldsymbol{z}_{k}}{\left\|\boldsymbol{z}_{k}\right\|}=\frac{c_{1} \lambda_{1}^{k}}{\left|c_{1} \lambda_{1}^{k}\right|} \frac{\boldsymbol{v}_{1}+\frac{c_{2}}{c_{1}}\left(\frac{\lambda_{2}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+\frac{c_{n}}{c_{1}}\left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}}{\left\|\boldsymbol{v}_{1}+\frac{c_{2}}{c_{1}}\left(\frac{\lambda_{2}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+\frac{c_{n}}{c_{1}}\left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}\right\|}, \quad k=0,1,2, \ldots,$</td><td>$\boldsymbol{x}_{k}=\frac{z_{k}}{\|z_{k}\|}=\frac{c_{1} \lambda_{1}^{k}}{|c_{1} \lambda_{1}^{k}|} \frac{\boldsymbol{v}_{1}+\frac{c_{2}}{c_{1}}\left(\frac{\lambda_{2}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+\frac{c_{n}}{c_{1}}\left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}}{\|\boldsymbol{v}_{1}+\frac{c_{2}}{c_{1}}\left(\frac{\lambda_{2}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{2}+\cdots+\frac{c_{n}}{c_{1}}\left(\frac{\lambda_{n}}{\lambda_{1}}\right)^{k} \boldsymbol{v}_{n}\|}, \quad k=0,1,2, \ldots,$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

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lyche-numerical-linear-algebra EQ1034	349	■ 0.999 • W|+0	`\begin{aligned} & s_{k}=\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{A} \boldsymbol{y}_{k}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}=s_{k-1}+\frac{\boldsymbol{y}_{k}^{*} \left(\boldsymbol{A}-s_{k-1} \boldsymbol{I} \right) \boldsymbol{y}_{k}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}=s_{k-1}+\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k-1}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}=s_{k-1}+\rho_{k}, \ \ & \boldsymbol{r}_{k}=\boldsymbol{A} \boldsymbol{x}_{k}-s_{k} \boldsymbol{x}_{k}=\frac{ \boldsymbol{A} \boldsymbol{y}_{k}-\left(s_{k-1}+\rho_{k} \right) \boldsymbol{y}_{k}}{\left|\boldsymbol{y}_{k} \right|_{2}}=\frac{\boldsymbol{x}_{k-1}-\rho_{k} \boldsymbol{y}_{k}}{\left|\boldsymbol{y}_{k}\right |_{2}}=\boldsymbol{w}_{k}-\rho_{k} \boldsymbol{x}_{k} . \end{aligned}`	$\begin{aligned} & s_{k}=\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{A} \boldsymbol{y}_{k}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}=s_{k-1}+\frac{\boldsymbol{y}_{k}^{*} \left(\boldsymbol{A}-s_{k-1} \boldsymbol{I} \right) \boldsymbol{y}_{k}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}=s_{k-1}+\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k-1}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}=s_{k-1}+\rho_{k}, \\ & \boldsymbol{r}_{k}=\boldsymbol{A} \boldsymbol{x}_{k}-s_{k} \boldsymbol{x}_{k}=\frac{ \boldsymbol{A} \boldsymbol{y}_{k}-\left(s_{k-1}+\rho_{k} \right) \boldsymbol{y}_{k}}{\left\|\boldsymbol{y}_{k} \right\|_{2}}=\frac{\boldsymbol{x}_{k-1}-\rho_{k} \boldsymbol{y}_{k}}{\left\|\boldsymbol{y}_{k}\right \|_{2}}=\boldsymbol{w}_{k}-\rho_{k} \boldsymbol{x}_{k} . \end{aligned}$	$s_k=\frac{\boldsymbol{y}_k^*\boldsymbol{A}\boldsymbol{y}_k}{\boldsymbol{y}_k^*\boldsymbol{y}_k}=s_{k-1}+\frac{\boldsymbol{y}_k^*(\boldsymbol{A}-s_{k-1}\boldsymbol{I})\boldsymbol{y}_k}{\boldsymbol{y}_k^*\boldsymbol{y}_k}=s_{k-1}+\frac{\boldsymbol{y}_k^*\boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^*\boldsymbol{y}_k}=s_{k-1}+\rho_k,$ $\boldsymbol{r}_k=\boldsymbol{A}\boldsymbol{x}_k-s_k\boldsymbol{x}_k=\frac{\boldsymbol{A}\boldsymbol{y}_k-(s_{k-1}+\rho_k)\boldsymbol{y}_k}{\|\boldsymbol{y}_k\|_2}=\frac{\boldsymbol{x}_{k-1}-\rho_k\boldsymbol{y}_k}{\|\boldsymbol{y}_k\|_2}=\boldsymbol{w}_k-\rho_k\boldsymbol{x}_k.$
lyche-numerical-linear-algebra EQ0097	054	■ 0.999 • W|+0	`\left[\begin{array}{lll} 1 & 0 & 0 \\ l_{1} & 1 & 0 \\ 0 & l_{2} & 1 \end{array}\right]\left[\begin{array}{l} z_{1} \\ z_{2} \\ z_{3} \end{array}\right]=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array} \right], \quad\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ 0 & u_{2} & c_{2} \\ 0 & 0 & u_{3} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{l} z_{1} \\ z_{2} \\ z_{3} \end{array}\right] .`	$\left[\begin{array}{lll} 1 & 0 & 0 \\ l_{1} & 1 & 0 \\ 0 & l_{2} & 1 \end{array}\right]\left[\begin{array}{l} z_{1} \\ z_{2} \\ z_{3} \end{array}\right]=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array} \right], \quad\left[\begin{array}{ccc} u_{1} & c_{1} & 0 \\ 0 & u_{2} & c_{2} \\ 0 & 0 & u_{3} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{l} z_{1} \\ z_{2} \\ z_{3} \end{array}\right] .$	$\begin{bmatrix}1&0&0\\l_1&1&0\\0&l_2&1\end{bmatrix}\begin{bmatrix}z_1\\z_2\\z_3\end{bmatrix}=\begin{bmatrix}b_1\\b_2\\b_3\end{bmatrix}, \quad \begin{bmatrix}u_1&c_1&0\\0&u_2&c_2\\0&0&u_3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}z_1\\z_2\\z_3\end{bmatrix}.$
lyche-numerical-linear-algebra EQ0380	154	■ 0.999 • K|+0	`\boldsymbol{B} \boldsymbol{B}^{*}=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{A}^{*} \boldsymbol{U}=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{A}^{*} \boldsymbol{U}= \boldsymbol{U}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{B}^{*} \boldsymbol{B} .`	$\boldsymbol{B} \boldsymbol{B}^{*}=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^{*} \boldsymbol{A}^{*} \boldsymbol{U}=\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{A}^{*} \boldsymbol{U}= \boldsymbol{U}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{U}=\boldsymbol{B}^{*} \boldsymbol{B} .$	$\boldsymbol{B}\boldsymbol{B}^*=\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}\boldsymbol{U}^*\boldsymbol{A}^*\boldsymbol{U}=\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{A}^*\boldsymbol{U}=\boldsymbol{U}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{U}=\boldsymbol{B}^*\boldsymbol{B}.$
lyche-numerical-linear-algebra EQ0394 (6.13)	158	■ 0.999 • W|-2	`\lambda_{k}=\min _{\operatorname{dim}(\mathcal{S})=n-k+1} \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x})=\max _{\operatorname{dim}(\mathcal{S})=k} \min _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k=1, \ldots, n .`	$\lambda_{k}=\min _{\operatorname{dim}(\mathcal{S})=n-k+1} \max _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x})=\max _{\operatorname{dim}(\mathcal{S})=k} \min _{\substack{ \boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k=1, \ldots, n .$	$\lambda_k=\min_{\dim(\mathcal{S})=n-k+1}\max_{\substack{\boldsymbol{x}\in\mathcal{S}\\ \boldsymbol{x}\neq\mathbf{0}}}R(\boldsymbol{x})=\max_{\dim(\mathcal{S})=k}\min_{\substack{\boldsymbol{x}\in\mathcal{S}\\ \boldsymbol{x}\neq\mathbf{0}}}R(\boldsymbol{x}), \quad k=1,\ldots,n.$
lyche-numerical-linear-algebra EQ0445	178	■ 0.999 • W|+1	`\left|\boldsymbol{A}-\boldsymbol{A}^{\prime} \right|_{F}=\min _{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \operatorname{rank}( \boldsymbol{B})=r}}|\boldsymbol{A}-\boldsymbol{B}|_{F}= \sqrt{\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}} .`	$\left\|\boldsymbol{A}-\boldsymbol{A}^{\prime} \right\|_{F}=\min _{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \operatorname{rank}(\boldsymbol{B})=r}}\|\boldsymbol{A}-\boldsymbol{B}\|_{F}= \sqrt{\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}} .$	$\|\boldsymbol{A}-\boldsymbol{A}'\|_F=\min_{\substack{\boldsymbol{B}\in\mathbb{R}^{m\times n}\\ rank(\boldsymbol{B})=r}}\|\boldsymbol{A}-\boldsymbol{B}\|_F=\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● N|+0</td><td>\left\|\boldsymbol{\epsilon}_{k}\right\|_{2}^{2}- \left\|\boldsymbol{\epsilon}_{k+1}\right\|_{2}^{2}= \frac{\left\|\boldsymbol{p}_{k}\right\|_{2}^{2}}{\left \|\boldsymbol{p}_{k}\right\|_{\boldsymbol{A}}^{2}} \left(\left\|\boldsymbol{\epsilon}_{k}\right\|_{ \boldsymbol{A}}^{2}+\left\|\boldsymbol{\epsilon}_{k+1} \right\|_{\boldsymbol{A}}^{2}\right)</td><td>$\left\|\boldsymbol{\epsilon}_{k}\right\|_{2}^{2}-\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{2}^{2}=\frac{\left\|\boldsymbol{p}_{k}\right\|_{2}^{2}}{\left\|\boldsymbol{p}_{k}\right\|_{\boldsymbol{A}}^{2}}\left(\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}}^{2}+\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{\boldsymbol{A}}^{2}\right)$</td><td>$\|\boldsymbol{\epsilon}_{k}\|_{2}^{2}-\|\boldsymbol{\epsilon}_{k+1}\|_{2}^{2}=\frac{\|\boldsymbol{p}_{k}\|_{2}^{2}}{\|\boldsymbol{p}_{k}\|_{\boldsymbol{A}}^{2}}(\|\boldsymbol{\epsilon}_{k}\|_{\boldsymbol{A}}^{2}+\|\boldsymbol{\epsilon}_{k+1}\|_{\boldsymbol{A}}^{2})$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0003</td><td>022</td><td>■ 1.000
● W|-2</td><td>\boldsymbol{A}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m 1} & a_{m 2} & \cdots & a_{m n} \end{array}\right] .</td><td>$\boldsymbol{A}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m 1} & a_{m 2} & \cdots & a_{m n} \end{array}\right] .$</td><td>$\boldsymbol{A}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m 1} & a_{m 2} & \cdots & a_{m n} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0008</td><td>023</td><td>■ 1.000
● K|+0</td><td>\boldsymbol{e}_{1}:=\left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array}\right], \quad \boldsymbol{e}_{2}:=\left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array}\right], \quad \boldsymbol{e}_{3}:=\left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array}\right], \quad \ldots, \quad \boldsymbol{e}_{n}:=\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array}\right],</td><td>$\boldsymbol{e}_{1}:=\left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array}\right], \quad \boldsymbol{e}_{2}:=\left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array}\right], \quad \boldsymbol{e}_{3}:=\left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array}\right], \quad \ldots, \quad \boldsymbol{e}_{n}:=\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array}\right],$</td><td>$\boldsymbol{e}_{1}:=\left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array}\right], \quad \boldsymbol{e}_{2}:=\left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array}\right], \quad \boldsymbol{e}_{3}:=\left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array}\right], \quad \ldots, \quad \boldsymbol{e}_{n}:=\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0025</td><td>030</td><td>■ 1.000
● W|+0</td><td>\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}=\boldsymbol{x}= \sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j} \stackrel{(1.9)}{=} \sum_{j=1}^{n} c_{j}\left(\sum_{i=1}^{m} a_{i j} \boldsymbol{v}_{i}\right)= \sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{i j} c_{j}\right) \boldsymbol{v}_{i} .</td><td>$\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}=\boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j} \stackrel{(1.9)}{=} \sum_{j=1}^{n} c_{j}\left(\sum_{i=1}^{m} a_{i j} \boldsymbol{v}_{i}\right)=\sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{i j} c_{j}\right) \boldsymbol{v}_{i} .$</td><td>$\sum_{i=1}^{m} b_{i} \boldsymbol{v}_{i}=\boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{s}_{j} \stackrel{(1.9)}{=} \sum_{j=1}^{n} c_{j}(\sum_{i=1}^{m} a_{i j} \boldsymbol{v}_{i})=\sum_{i=1}^{m}(\sum_{j=1}^{n} a_{i j} c_{j}) \boldsymbol{v}_{i} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0053</td><td>039</td><td>■ 1.000
● N|+0</td><td>\pi_{A}(\lambda)=(-1)^{n} \lambda^{n}+c_{n-1} \lambda^{n-1}+\cdots+c_{0},</td><td>$\pi_{A}(\lambda)=(-1)^{n} \lambda^{n}+c_{n-1} \lambda^{n-1}+\cdots+c_{0},$</td><td>$\pi_{\boldsymbol{A}}(\lambda)=(-1)^{n} \lambda^{n}+c_{n-1} \lambda^{n-1}+\cdots+c_{0},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0078 (2.2)</td><td>046</td><td>■ 1.000
● K|+0</td><td>g\left(x_{i}\right)=y_{i}, \text { for } i=1, \ldots, n+1 .</td><td>$g\left(x_{i}\right)=y_{i}, \text { for } i=1, \ldots, n+1 .$</td><td>$g(x_{i})=y_{i}, \text { for } i=1, \ldots, n+1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0087 (2.9)</td><td>051</td><td>■ 1.000
● N|+0</td><td>\mu_{i-1}+4 \mu_{i}+\mu_{i+1}=\frac{6}{h^{2}} \left(y_{i+1}-2 y_{i}+y_{i-1}\right), \quad i=2, \ldots, n,</td><td>$\mu_{i-1}+4 \mu_{i}+\mu_{i+1}=\frac{6}{h^{2}}\left(y_{i+1}-2 y_{i}+y_{i-1}\right), \quad i=2, \ldots, n,$</td><td>$\mu_{i-1}+4 \mu_{i}+\mu_{i+1}=\frac{6}{h^{2}}(y_{i+1}-2 y_{i}+y_{i-1}), \quad i=2, \ldots, n,$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0099 (2.16)</td><td>054</td><td>■ 1.000 ● N|+0</td><td>u_{1}=d_{1}, \quad l_{k}=a_{k} / u_{k}, \quad u_{k+1}=d_{k+1}-l_{k} c_{k}, \quad k=1,2, \ldots, n-1,</td><td>$u_{1}=d_{1}, \quad l_{k}=a_{k} / u_{k}, \quad u_{k+1}=d_{k+1}-l_{k} c_{k}, \quad k=1,2, \ldots, n-1,$</td><td>$u_1=d_1, \quad l_k=a_k/u_k, \quad u_{k+1}=d_{k+1}-l_kc_k, \quad k=1,2,\ldots,n-1,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0102 (2.19)</td><td>056</td><td>■ 1.000 ● W|+2</td><td>\left|l_{k}\right|=\frac{\left|a_{k}\right|}{\left|u_{k}\right|} \leq \frac{\left|a_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|},\left|u_{k+1}\right| \leq \left|d_{k+1}\right|+\frac{\left|a_{k}\right|\left|c_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|}, k=1, \ldots, n-1 .</td><td>$\left|l_{k}\right|=\frac{\left|a_{k}\right|}{\left|u_{k}\right|} \leq \frac{\left|a_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|},\left|u_{k+1}\right| \leq \left|d_{k+1}\right|+\frac{\left|a_{k}\right|\left|c_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|}, k=1, \ldots, n-1 .$</td><td>$|l_k|=\frac{|a_k|}{|u_k|} \leq \frac{|a_k|}{|d_k|-|a_{k-1}|}, \quad |u_{k+1}| \leq |d_{k+1}|+\frac{|a_k||c_k|}{|d_k|-|a_{k-1}|}, \quad k=1,\ldots,n-1.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0103</td><td>056</td><td>■ 1.000 ● W|+0</td><td>\left|l_{k}\right|=\frac{\left|a_{k}\right|}{\left|u_{k}\right|} \leq \frac{\left|a_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|},\left|u_{k+1}\right| \leq\left|d_{k+1}\right|+\left|l_{k}\right|\left|c_{k}\right| \leq\left|d_{k+1}\right|+\frac{\left|a_{k}\right|\left|c_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|} .</td><td>$\left|l_{k}\right|=\frac{\left|a_{k}\right|}{\left|u_{k}\right|} \leq \frac{\left|a_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|},\left|u_{k+1}\right| \leq\left|d_{k+1}\right|+\left|l_{k}\right|\left|c_{k}\right| \leq\left|d_{k+1}\right|+\frac{\left|a_{k}\right|\left|c_{k}\right|}{\left|d_{k}\right|-\left|a_{k-1}\right|} .$</td><td>$|l_k|=\frac{|a_k|}{|u_k|} \leq \frac{|a_k|}{|d_k|-|a_{k-1}|}, \quad |u_{k+1}| \leq |d_{k+1}|+|l_k||c_k| \leq |d_{k+1}|+\frac{|a_k||c_k|}{|d_k|-|a_{k-1}|}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0121 (2.31)</td><td>061</td><td>■ 1.000 ● W|+2</td><td>\boldsymbol{S}=\left[\boldsymbol{s}_{1}, \boldsymbol{s}_{2}, \boldsymbol{s}_{3}\right]=\left[\begin{array}{c} \sin \frac{\pi}{4} \sin \frac{2 \pi}{4} \sin \frac{3 \pi}{4} \\ \sin \frac{2 \pi}{4} \sin \frac{4 \pi}{4} \sin \frac{6 \pi}{4} \\ \sin \frac{3 \pi}{4} \sin \frac{6 \pi}{4} \sin \frac{9 \pi}{4} \end{array}\right]= \left[\begin{array}{ccc} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{array}\right], \quad t:=\frac{1}{\sqrt{2}} .</td><td>$\boldsymbol{S}=\left[\boldsymbol{s}_{1}, \boldsymbol{s}_{2}, \boldsymbol{s}_{3}\right]=\left[\begin{array}{c} \sin \frac{\pi}{4} \sin \frac{2 \pi}{4} \sin \frac{3 \pi}{4} \\ \sin \frac{2 \pi}{4} \sin \frac{4 \pi}{4} \sin \frac{6 \pi}{4} \\ \sin \frac{3 \pi}{4} \sin \frac{6 \pi}{4} \sin \frac{9 \pi}{4} \end{array}\right]= \left[\begin{array}{ccc} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{array}\right], \quad t:=\frac{1}{\sqrt{2}} .$</td><td>$\boldsymbol{S}=[\boldsymbol{s}_1,\boldsymbol{s}_2,\boldsymbol{s}_3]=\begin{bmatrix} \sin\frac{\pi}{4} & \sin\frac{2\pi}{4} & \sin\frac{3\pi}{4} \\ \sin\frac{2\pi}{4} & \sin\frac{4\pi}{4} & \sin\frac{6\pi}{4} \\ \sin\frac{3\pi}{4} & \sin\frac{6\pi}{4} & \sin\frac{9\pi}{4} \end{bmatrix}=\begin{bmatrix} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{bmatrix}, \quad t:=\frac{1}{\sqrt{2}}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0123 (2.32)</td><td>061</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{s}_{j}^{T} \boldsymbol{s}_{k}=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m .</td><td>$\boldsymbol{s}_{j}^{T} \boldsymbol{s}_{k}=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m .$</td><td>$\boldsymbol{s}_j^T\boldsymbol{s}_k=\frac{m+1}{2}\delta_{j,k}=\frac{1}{2h}\delta_{j,k}, \quad j,k=1,\ldots,m.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0128</td><td>062</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{A}=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]</td><td>$\boldsymbol{A}=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]$</td><td>$\boldsymbol{A}=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0186 (3.5)</td><td>079</td><td>■ 1.000 ● W|+0</td><td>(\boldsymbol{L} \boldsymbol{U})_{i j}=\sum_{k=1}^{i-1} l_{i k}^{(k)} a_{k j}^{(k)}+a_{i j}^{(i)}= \sum_{k=1}^{i-1}\left(a_{i j}^{(k)}-a_{i j}^{(k+1)}\right)+a_{i j}^{(i)}=a_{i j}^{(1)}=a_{i j},</td><td>$(\boldsymbol{L} \boldsymbol{U})_{i j}=\sum_{k=1}^{i-1} l_{i k}^{(k)} a_{k j}^{(k)}+a_{i j}^{(i)}= \sum_{k=1}^{i-1}\left(a_{i j}^{(k)}-a_{i j}^{(k+1)}\right)+a_{i j}^{(i)}=a_{i j}^{(1)}=a_{i j},$</td><td>$(\boldsymbol{LU})_{ij}=\sum_{k=1}^{i-1} l_{ik}^{(k)}a_{kj}^{(k)}+a_{ij}^{(i)}=\sum_{k=1}^{i-1}\left(a_{ij}^{(k)}-a_{ij}^{(k+1)}\right)+a_{ij}^{(i)}=a_{ij}^{(1)}=a_{ij},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0193</td><td>082</td><td>■ 1.000 ● C|-8</td><td>\left[\begin{array}{cccc} a_{k+1, k+1} & 0 & \cdots & 0 \\ a_{k+2, k+1} & a_{k+2, k+2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k+1} & & \cdots & a_{n, n} \end{array}\right]\left[\begin{array}{c} x_{k+1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{k+1} \\ \vdots \\ b_{n} \end{array}\right]-x_{k} \left[\begin{array}{c} a_{k+1, k} \\ \vdots \\ a_{n, k} \end{array}\right] .</td><td>$\left[\begin{array}{cccc} a_{k+1, k+1} & 0 & \cdots & 0 \\ a_{k+2, k+1} & a_{k+2, k+2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k+1} & & \cdots & a_{n, n} \end{array}\right]\left[\begin{array}{c} x_{k+1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{k+1} \\ \vdots \\ b_{n} \end{array}\right]-x_{k} \left[\begin{array}{c} a_{k+1, k} \\ \vdots \\ a_{n, k} \end{array}\right] .$</td><td>$\begin{bmatrix} a_{k+1,k+1} & 0 & \cdots & 0 \\ a_{k+2,k+1} & a_{k+2,k+2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n,k+1} & & \cdots & a_{n,n} \end{bmatrix}\begin{bmatrix} x_{k+1} \\ \vdots \\ x_n \end{bmatrix}=\begin{bmatrix} b_{k+1} \\ \vdots \\ b_n \end{bmatrix}-x_k\begin{bmatrix} a_{k+1,k} \\ \vdots \\ a_{n,k} \end{bmatrix}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0240 (3.32)	097	■ 1.000 ● N|+0	\boldsymbol{L}=\left[\begin{array}{cc} 1 & 0 \\ \boldsymbol{\ell}_{1} & \boldsymbol{L}_{2,2} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1} & \boldsymbol{u}_{1}^{T} \\ 0 & \boldsymbol{U}_{2,2} \end{array}\right],	$\boldsymbol{L}=\left[\begin{array}{cc} 1 & 0 \\ \boldsymbol{\ell}_{1} & \boldsymbol{L}_{2,2} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1} & \boldsymbol{u}_{1}^{T} \\ 0 & \boldsymbol{U}_{2,2} \end{array}\right],$	$\boldsymbol{L}=\left[\begin{array}{cc} 1 & 0 \\ \boldsymbol{\ell}_{1} & \boldsymbol{L}_{2,2} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1} & \boldsymbol{u}_{1}^{T} \\ 0 & \boldsymbol{U}_{2,2} \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0270	113	■ 1.000 ● N|+1	\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2 & 2-a \\ a & 1 \end{array}\right], \quad a \in \mathbb{R}	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2 & 2-a \\ a & 1 \end{array}\right], \quad a \in \mathbb{R}$	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2 & 2-a \\ a & 1 \end{array}\right], \quad a \in \mathbb{R}$
lyche-numerical-linear-algebra_ EQ0274	114	■ 1.000 ● K|+0	\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A} & \boldsymbol{a} \\ \boldsymbol{a}^{T} & \alpha \end{array}\right],	$\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A} & \boldsymbol{a} \\ \boldsymbol{a}^{T} & \alpha \end{array}\right],$	$\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A} & \boldsymbol{a} \\ \boldsymbol{a}^{T} & \alpha \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0283 (5.4)	118	■ 1.000 ● N|+0	\langle a \boldsymbol{y}, \boldsymbol{z}\rangle+\langle \boldsymbol{z}, a \boldsymbol{y}\rangle=a \overline{ \langle\boldsymbol{z}, \boldsymbol{y}\rangle}+\bar{a} \langle\boldsymbol{z}, \boldsymbol{y}\rangle=0 .	$\langle a \boldsymbol{y}, \boldsymbol{z}\rangle+\langle \boldsymbol{z}, a \boldsymbol{y}\rangle=a \overline{ \langle\boldsymbol{z}, \boldsymbol{y}\rangle}+\bar{a} \langle\boldsymbol{z}, \boldsymbol{y}\rangle=0 .$	$\langle a \boldsymbol{y}, \boldsymbol{z}\rangle+\langle \boldsymbol{z}, a \boldsymbol{y}\rangle=a \overline{ \langle\boldsymbol{z}, \boldsymbol{y}\rangle}+\bar{a} \langle\boldsymbol{z}, \boldsymbol{y}\rangle=0 .$
lyche-numerical-linear-algebra_ EQ0328	133	■ 1.000 ● N|+0	\boldsymbol{R}_{1}=\boldsymbol{R}=\left[\begin{array}{cc} \left|\boldsymbol{v}_{1}\right|_{2} & \boldsymbol{a}_{2}^{T} \boldsymbol{q}_{1} \\ 0 & \left|\boldsymbol{v}_{2}\right|_{2} \end{array}\right]= \frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]	$\boldsymbol{R}_{1}=\boldsymbol{R}=\left[\begin{array}{cc} \lVert\boldsymbol{v}_{1}\rVert_{2} & \boldsymbol{a}_{2}^{T} \boldsymbol{q}_{1} \\ 0 & \lVert\boldsymbol{v}_{2}\rVert_{2} \end{array}\right]= \frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]$	$\boldsymbol{R}_{1}=\boldsymbol{R}=\left[\begin{array}{cc} \lVert\boldsymbol{v}_{1}\rVert_{2} & \boldsymbol{a}_{2}^{T} \boldsymbol{q}_{1} \\ 0 & \lVert\boldsymbol{v}_{2}\rVert_{2} \end{array}\right]= \frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0341	137	■ 1.000 ● K|+0	\boldsymbol{B}:=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right],	$\boldsymbol{B}:=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right],$	$\boldsymbol{B}:=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0345	139	■ 1.000 ● N|-1	\boldsymbol{V}_{r}:=\boldsymbol{I}_{r}-2 \frac{ \boldsymbol{v}_{r} \boldsymbol{v}_{r}^{*}}{ \boldsymbol{v}_{r}^{*} \boldsymbol{v}_{r}}= \boldsymbol{I}_{r}-\boldsymbol{u}_{r} \boldsymbol{u}_{r}^{*}, \text { with } \boldsymbol{u}_{r}:=\sqrt{2} \frac{\boldsymbol{v}_{r}}{ \left|\boldsymbol{v}_{r}\right|_{2}} .	$\boldsymbol{V}_{r}:=\boldsymbol{I}_{r}-2 \frac{ \boldsymbol{v}_{r} \boldsymbol{v}_{r}^{*}}{ \boldsymbol{v}_{r}^{*} \boldsymbol{v}_{r}}= \boldsymbol{I}_{r}-\boldsymbol{u}_{r} \boldsymbol{u}_{r}^{*}, \text { with } \boldsymbol{u}_{r}:=\sqrt{2} \frac{\boldsymbol{v}_{r}}{ \lVert\boldsymbol{v}_{r}\rVert_{2}} .$	$\boldsymbol{V}_{r}:=\boldsymbol{I}_{r}-2 \frac{ \boldsymbol{v}_{r} \boldsymbol{v}_{r}^{*}}{ \boldsymbol{v}_{r}^{*} \boldsymbol{v}_{r}}= \boldsymbol{I}_{r}-\boldsymbol{u}_{r} \boldsymbol{u}_{r}^{*}, \text { with } \boldsymbol{u}_{r}:=\sqrt{2} \frac{\boldsymbol{v}_{r}}{ \lVert\boldsymbol{v}_{r}\rVert_{2}} .$
lyche-numerical-linear-algebra_ EQ0363	147	■ 1.000 ● K|+0	E:=\left[\begin{array}{cc} A C & 0 \\ C & 0 \end{array} \right], \quad F:=\left[\begin{array}{cc} 0 & 0 \\ C & C A \end{array}\right], \quad S=\left[\begin{array}{cc} I & A \\ 0 & I \end{array}\right] .	$E:=\left[\begin{array}{cc} A C & 0 \\ C & 0 \end{array} \right], \quad F:=\left[\begin{array}{cc} 0 & 0 \\ C & C A \end{array}\right], \quad S=\left[\begin{array}{cc} I & A \\ 0 & I \end{array}\right] .$	$\boldsymbol{E}:=\left[\begin{array}{cc} \boldsymbol{AC} & \boldsymbol{0} \\ \boldsymbol{C} & \boldsymbol{0} \end{array} \right], \quad \boldsymbol{F}:=\left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{C} & \boldsymbol{CA} \end{array}\right], \quad \boldsymbol{S}=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{A} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0364 (6.2)	148	■ 1.000 ● N|+0	\pi_{\boldsymbol{A}}(\lambda):=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\left(\lambda_{1}- \lambda\right)^{a_{1}} \cdots\left(\lambda_{k}-\lambda \right)^{a_{k}}, \quad \lambda_{i} \neq \lambda_{j}, i \neq j, \sum_{i=1}^{k} a_{i}=n .	$\pi_{\boldsymbol{A}}(\lambda):=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\left(\lambda_{1}- \lambda\right)^{a_{1}} \cdots\left(\lambda_{k}-\lambda \right)^{a_{k}}, \quad \lambda_{i} \neq \lambda_{j}, i \neq j, \sum_{i=1}^{k} a_{i}=n .$	$\pi_{\boldsymbol{A}}(\lambda):=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\left(\lambda_{1}- \lambda\right)^{a_{1}} \cdots\left(\lambda_{k}-\lambda \right)^{a_{k}}, \quad \lambda_{i} \neq \lambda_{j},\ i \neq j,\ \sum_{i=1}^{k} a_{i}=n .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra EQ0393 (6.12)</td><td>157</td><td>■ 1.000 ● N|-1</td><td>\lambda_{k} \geq \min _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),</td><td>$\lambda_{k} \geq \min _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$</td><td>$\lambda_{k} \geq \min _{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0413</td><td>165</td><td>■ 1.000 ● N|-1</td><td>\lambda_{\min }=\min _{\boldsymbol{x} \neq 0} R(\boldsymbol{x})=\min _{\|\boldsymbol{x}\|_{2}=1} R(\boldsymbol{x}),</td><td>$\lambda_{\min }=\min _{\boldsymbol{x} \neq 0} R(\boldsymbol{x})=\min _{\|\boldsymbol{x}\|_{2}=1} R(\boldsymbol{x}),$</td><td>$\lambda_{\min }=\min _{\boldsymbol{x} \neq 0} R(\boldsymbol{x})=\min _{\|\boldsymbol{x}\|_{2}=1} R(\boldsymbol{x}),$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0427 (2)</td><td>172</td><td>■ 1.000 ● K|+0</td><td>\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array} \right]=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & -1 \end{array}\right] .</td><td>$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array} \right]=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & -1 \end{array}\right] .$</td><td>$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array} \right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array} \right]=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & -1 \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0435</td><td>173</td><td>■ 1.000 ● W|+2</td><td>\boldsymbol{U}:=\left[\begin{array}{ccc} 1 / \sqrt{2} & -1 / \sqrt{2} & 0 \\ 1 / \sqrt{2} & 1 / \sqrt{2} & 0 \\ 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{3,3}, \quad \boldsymbol{\Sigma}:=\left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] \in \mathbb{R}^{3,2}, \quad \boldsymbol{V}:=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \in \mathbb{R}^{2,2} .</td><td>$\boldsymbol{U}:=\left[\begin{array}{ccc} 1 / \sqrt{2} & -1 / \sqrt{2} & 0 \\ 1 / \sqrt{2} & 1 / \sqrt{2} & 0 \\ 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{3,3}, \quad \boldsymbol{\Sigma}:=\left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] \in \mathbb{R}^{3,2}, \quad \boldsymbol{V}:=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \in \mathbb{R}^{2,2} .$</td><td>$\boldsymbol{U}:=\left[\begin{array}{ccc} 1 / \sqrt{2} & -1 / \sqrt{2} & 0 \\ 1 / \sqrt{2} & 1 / \sqrt{2} & 0 \\ 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{3,3}, \quad \boldsymbol{\Sigma}:=\left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] \in \mathbb{R}^{3,2}, \quad \boldsymbol{V}:=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \in \mathbb{R}^{2,2} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0439</td><td>175</td><td>■ 1.000 ● K|+0</td><td>A:=\frac{1}{25}\left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array}\right]</td><td>$A:=\frac{1}{25}\left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array}\right]$</td><td>$\boldsymbol{A}:=\frac{1}{25}\left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0442</td><td>177</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A} \boldsymbol{B}\|_{F}^{2}=\sum_{i=1}^{m} \sum_{j=1}^{k}\left|\boldsymbol{a}_{i:}^{*} \boldsymbol{b}_{: j}\right|^{2} \leq \sum_{i=1}^{m} \sum_{j=1}^{k}\left\|\boldsymbol{a}_{i:}\right \|_{2}^{2}\left\|\boldsymbol{b}_{: j}\right\|_{2}^{2}=\| \boldsymbol{A}\|_{F}^{2}\|\boldsymbol{B}\|_{F}^{2} .</td><td>$\|\boldsymbol{A} \boldsymbol{B}\|_{F}^{2}=\sum_{i=1}^{m} \sum_{j=1}^{k}\left|\boldsymbol{a}_{i:}^{*} \boldsymbol{b}_{: j}\right|^{2} \leq \sum_{i=1}^{m} \sum_{j=1}^{k}\left\|\boldsymbol{a}_{i:}\right \|_{2}^{2}\left\|\boldsymbol{b}_{: j}\right\|_{2}^{2}=\| \boldsymbol{A}\|_{F}^{2}\|\boldsymbol{B}\|_{F}^{2} .$</td><td>$\|\boldsymbol{A} \boldsymbol{B}\|_{F}^{2}=\sum_{i=1}^{m} \sum_{j=1}^{k}\left|\boldsymbol{a}_{i:}^{*} \boldsymbol{b}_{: j}\right|^{2} \leq \sum_{i=1}^{m} \sum_{j=1}^{k}\left\|\boldsymbol{a}_{i:}\right \|_{2}^{2}\left\|\boldsymbol{b}_{: j}\right\|_{2}^{2}=\| \boldsymbol{A}\|_{F}^{2}\|\boldsymbol{B}\|_{F}^{2} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0447 (7.11)</td><td>178</td><td>■ 1.000 ● N|+1</td><td>\boldsymbol{A}:=\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]=: \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{T}</td><td>$\boldsymbol{A}:=\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]=: \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{T}$</td><td>$\boldsymbol{A}:=\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right]=: \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{T}$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0457</td><td>181</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{A}=\left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right] .</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right] .$</td><td>$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0463</td><td>183</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{T}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .</td><td>$\boldsymbol{T}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .$</td><td>$\boldsymbol{T}=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra EQ00484 (8.16)</td><td>190</td><td>■ 1.000 ● K|+0</td><td>\|\boldsymbol{A}\|=\left\|\boldsymbol{A} \boldsymbol{x}^{*}\right\| \text { for some } \boldsymbol{x}^{*} \in \mathcal{S} .</td><td>$\|\boldsymbol{A}\|=\left\|\boldsymbol{A} \boldsymbol{x}^{*}\right\| \text { for some } \boldsymbol{x}^{*} \in \mathcal{S} .$</td><td>$\|\boldsymbol{A}\|=\|\boldsymbol{A} \boldsymbol{x}^{*}\| \text { for some } \boldsymbol{x}^{*} \in \mathcal{S} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00490</td><td>192</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{\infty}=\max _{1 \leq k \leq m}\left|\sum_{j=1}^{n} a_{k j} x_{j}\right| \leq \max _{1 \leq k \leq m} \sum_{j=1}^{n}\left|a_{k j} \right|\left|x_{j}\right| \leq K_{\infty} .</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{\infty}=\max _{1 \leq k \leq m}\left|\sum_{j=1}^{n} a_{k j} x_{j}\right| \leq \max _{1 \leq k \leq m} \sum_{j=1}^{n}\left|a_{k j} \right|\left|x_{j}\right| \leq K_{\infty} .$</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{\infty}=\max _{1 \leq k \leq m}|\sum_{j=1}^{n} a_{k j} x_{j}| \leq \max _{1 \leq k \leq m} \sum_{j=1}^{n}|a_{k j}||x_{j}| \leq K_{\infty} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00493 (8.21)</td><td>193</td><td>■ 1.000 ● K|+0</td><td>\|\boldsymbol{A}\|_{2}^{2}\|\boldsymbol{v}\|_{1}= \sigma^{2}\|\boldsymbol{v}\|_{1}=\left\|\sigma^{2} \boldsymbol{v}\right\|_{1}=\left\|\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}\right\|_{1} \leq\left\| \boldsymbol{A}^{*}\right\|_{1}\|\boldsymbol{A}\|_{1}\| \boldsymbol{v}\|_{1} .</td><td>$\|\boldsymbol{A}\|_{2}^{2}\|\boldsymbol{v}\|_{1}= \sigma^{2}\|\boldsymbol{v}\|_{1}=\left\|\sigma^{2} \boldsymbol{v}\right\|_{1}=\left\|\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}\right\|_{1} \leq\left\| \boldsymbol{A}^{*}\right\|_{1}\|\boldsymbol{A}\|_{1}\| \boldsymbol{v}\|_{1} .$</td><td>$\|\boldsymbol{A}\|_{2}^{2}\|\boldsymbol{v}\|_{1}= \sigma^{2}\|\boldsymbol{v}\|_{1}=\|\sigma^{2} \boldsymbol{v}\|_{1}=\|\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}\|_{1} \leq\| \boldsymbol{A}^{*}\|_{1}\|\boldsymbol{A}\|_{1}\| \boldsymbol{v}\|_{1} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00505 (8.27)</td><td>198</td><td>■ 1.000 ● N|+0</td><td>\frac{\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1} \right\|}{\left\|\boldsymbol{B}^{-1}\right\|} \leq \left\|\boldsymbol{A}^{-1} \boldsymbol{E}\right\| \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{E}\|}{\| \boldsymbol{A}\|},</td><td>$\frac{\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1} \right\|}{\left\|\boldsymbol{B}^{-1}\right\|} \leq \left\|\boldsymbol{A}^{-1} \boldsymbol{E}\right\| \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{E}\|}{\| \boldsymbol{A}\|},$</td><td>$\frac{\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\|}{\|\boldsymbol{B}^{-1}\|} \leq \|\boldsymbol{A}^{-1} \boldsymbol{E}\| \leq K(\boldsymbol{A}) \frac{\|\boldsymbol{E}\|}{\| \boldsymbol{A}\|},$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00509</td><td>198</td><td>■ 1.000 ● N|+0</td><td>\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right \| \leq\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \right\|\left\|\boldsymbol{B}^{-1}\right\| \leq K(A) \frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|}\left\| \boldsymbol{B}^{-1}\right\| .</td><td>$\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right \| \leq\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \right\|\left\|\boldsymbol{B}^{-1}\right\| \leq K(A) \frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|}\left\| \boldsymbol{B}^{-1}\right\| .$</td><td>$\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\| \leq\|\boldsymbol{A}^{-1} \boldsymbol{E}\|\|\boldsymbol{B}^{-1}\| \leq K(A) \frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|}\| \boldsymbol{B}^{-1}\| .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00521 (8.34)</td><td>201</td><td>■ 1.000 ● N|+1</td><td>a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a+ \frac{1}{q} b, \text { where } \frac{1}{p}+\frac{1}{q}=1 .</td><td>$a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a+ \frac{1}{q} b, \text { where } \frac{1}{p}+\frac{1}{q}=1 .$</td><td>$a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a+ \frac{1}{q} b, \text { where } \frac{1}{p}+\frac{1}{q}=1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00528 (8.36)</td><td>202</td><td>■ 1.000 ● N|+0</td><td>\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \frac{1}{4}\left(\|\boldsymbol{x}+\boldsymbol{y}\|^{2}- \|\boldsymbol{x}-\boldsymbol{y}\|^{2}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}</td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \frac{1}{4}\left(\|\boldsymbol{x}+\boldsymbol{y}\|^{2}- \|\boldsymbol{x}-\boldsymbol{y}\|^{2}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$</td><td>$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:= \frac{1}{4}(\|\boldsymbol{x}+\boldsymbol{y}\|^{2}- \|\boldsymbol{x}-\boldsymbol{y}\|^{2}), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00551</td><td>208</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{l} b_{1} \\ b_{2} \end{array}\right]=\left[\begin{array}{l} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{l} e_{1} \\ e_{2} \end{array}\right], \quad\|\boldsymbol{e}\|_{2}=0.1 .</td><td>$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{l} b_{1} \\ b_{2} \end{array}\right]=\left[\begin{array}{l} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{l} e_{1} \\ e_{2} \end{array}\right], \quad\|\boldsymbol{e}\|_{2}=0.1 .$</td><td>$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{l} b_{1} \\ b_{2} \end{array}\right]=\left[\begin{array}{l} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{l} e_{1} \\ e_{2} \end{array}\right], \quad\|\boldsymbol{e}\|_{2}=0.1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00598</td><td>220</td><td>■ 1.000 ● N|+1</td><td>\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{*} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_{1} \boldsymbol{x}=\boldsymbol{c}_{1}, \quad \boldsymbol{c}_{1}:=\boldsymbol{Q}_{1}^{*} \boldsymbol{b} .</td><td>$\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{*} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_{1} \boldsymbol{x}=\boldsymbol{c}_{1}, \quad \boldsymbol{c}_{1}:=\boldsymbol{Q}_{1}^{*} \boldsymbol{b} .$</td><td>$\boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{x}= \boldsymbol{A}^{*} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_{1} \boldsymbol{x}=\boldsymbol{c}_{1}, \quad \boldsymbol{c}_{1}:=\boldsymbol{Q}_{1}^{*} \boldsymbol{b} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00608 (9.11)</td><td>223</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*} .</td><td>$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*} .$</td><td>$\boldsymbol{A}^{\dagger}=(\boldsymbol{A}^{*} \boldsymbol{A})^{-1} \boldsymbol{A}^{*} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ00682 (10.15)</td><td>245</td><td>■ 1.000 ● N|+0</td><td>\lambda_{j}=d+2 a \cos (j \pi h), \quad h:=\frac{1}{m+1},</td><td>$\lambda_{j}=d+2 a \cos (j \pi h), \quad h:=\frac{1}{m+1},$</td><td>$\lambda_{j}=d+2 a \cos (j \pi h), \quad h:=\frac{1}{m+1},$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0692 (10.22)</td><td>248</td><td>■ 1.000 ● N|+1</td><td>\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^{2} \mu \boldsymbol{F}</td><td>$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^{2} \mu \boldsymbol{F}$</td><td>$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} - \frac{1}{6}\boldsymbol{T}\boldsymbol{V}\boldsymbol{T} = h^2\mu\boldsymbol{F}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0695 (10.21)</td><td>248</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U}+\boldsymbol{U} \boldsymbol{T}=h^{2} \boldsymbol{V} .</td><td>$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{2} \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U}+\boldsymbol{U} \boldsymbol{T}=h^{2} \boldsymbol{V} .$</td><td>$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}, \qquad \boldsymbol{T}\boldsymbol{U} + \boldsymbol{U}\boldsymbol{T} = h^2\boldsymbol{V}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0721</td><td>257</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A} \boldsymbol{P}_{N}=\left[\boldsymbol{a}_{1}, \boldsymbol{a}_{3}, \ldots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_{2}, \boldsymbol{a}_{4}, \ldots, \boldsymbol{a}_{N}\right],</td><td>$\boldsymbol{A} \boldsymbol{P}_{N}=\left[\boldsymbol{a}_{1}, \boldsymbol{a}_{3}, \ldots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_{2}, \boldsymbol{a}_{4}, \ldots, \boldsymbol{a}_{N}\right],$</td><td>$\boldsymbol{A}\boldsymbol{P}_N = [\boldsymbol{a}_1, \boldsymbol{a}_3, \ldots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \ldots, \boldsymbol{a}_N],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0754 (12.9)</td><td>270</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{M} \boldsymbol{x}_{k+1}=(\boldsymbol{M}-\boldsymbol{A}) \boldsymbol{x}_{k}+\boldsymbol{b}</td><td>$\boldsymbol{M} \boldsymbol{x}_{k+1}=(\boldsymbol{M}-\boldsymbol{A}) \boldsymbol{x}_{k}+\boldsymbol{b}$</td><td>$\boldsymbol{M}\boldsymbol{x}_{k+1} = (\boldsymbol{M} - \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0789 (12.29)</td><td>277</td><td>■ 1.000 ● K|+0</td><td>\rho\left(\boldsymbol{G}_{\omega}\right)>\rho\left(\boldsymbol{G}_{\omega^{*}}\right) \text { for } \omega \in(0,2) \backslash\left\{\omega^{*}\right\} .</td><td>$\rho\left(\boldsymbol{G}_{\omega}\right)>\rho\left(\boldsymbol{G}_{\omega^{*}}\right) \text { for } \omega \in(0,2) \backslash\left\{\omega^{*}\right\} .$</td><td>$\rho(\boldsymbol{G}_\omega) > \rho(\boldsymbol{G}_{\omega^*})\, for\ \omega \in (0,2) \setminus \{\omega^*\}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0791</td><td>278</td><td>■ 1.000 ● N|+0</td><td>\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right \| \leq\left\|\boldsymbol{G}^{k}\right\|\left\| \boldsymbol{x}_{0}-\boldsymbol{x}\right\| \approx \rho(\boldsymbol{G})^{k}\left\|\boldsymbol{x}_{0}- \boldsymbol{x}\right\| .</td><td>$\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right \| \leq\left\|\boldsymbol{G}^{k}\right\|\left\| \boldsymbol{x}_{0}-\boldsymbol{x}\right\| \approx \rho(\boldsymbol{G})^{k}\left\|\boldsymbol{x}_{0}- \boldsymbol{x}\right\| .$</td><td>$\|\boldsymbol{x}_k - \boldsymbol{x}\| \le \|\boldsymbol{G}^k\|\|\boldsymbol{x}_0 - \boldsymbol{x}\| \approx \rho(\boldsymbol{G})^k\|\boldsymbol{x}_0 - \boldsymbol{x}\|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0820</td><td>285</td><td>■ 1.000 ● K|+0</td><td>d^{\prime}(\omega)=2\left(\omega \mu^{2}-2\right)<2(\omega-2)<0 .</td><td>$d^{\prime}(\omega)=2\left(\omega \mu^{2}-2\right)<2(\omega-2)<0 .$</td><td>$d'(\omega) = 2(\omega\mu^2 - 2) < 2(\omega - 2) < 0.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0840 (13.6)</td><td>292</td><td>■ 1.000 ● N|+0</td><td>\begin{aligned} \frac{\partial}{\partial y_{i}} Q(\boldsymbol{y}) & :=\lim _{\varepsilon \rightarrow 0} \frac{1}{\varepsilon}\left(Q\left(\boldsymbol{y}+\varepsilon \boldsymbol{e}_{i} \right)-Q(\boldsymbol{y})\right) \\ & \left.=\lim _{ \varepsilon \rightarrow 0} \frac{1}{\varepsilon}\left(- \varepsilon \boldsymbol{e}_{i}^{T} \boldsymbol{r}(\boldsymbol{y})\right)+\frac{1}{2} \varepsilon^{2} \boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{i} \right)=-\boldsymbol{e}_{i}^{T} \boldsymbol{r}(\boldsymbol{y}), \quad i=1, \ldots, n, \end{aligned}</td><td>$\begin{aligned} \frac{\partial}{\partial y_{i}} Q(\boldsymbol{y}) & :=\lim _{\varepsilon \rightarrow 0} \frac{1}{\varepsilon}\left(Q\left(\boldsymbol{y}+\varepsilon \boldsymbol{e}_{i} \right)-Q(\boldsymbol{y})\right) \\ & \left.=\lim _{ \varepsilon \rightarrow 0} \frac{1}{\varepsilon}\left(- \varepsilon \boldsymbol{e}_{i}^{T} \boldsymbol{r}(\boldsymbol{y})\right)+\frac{1}{2} \varepsilon^{2} \boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{i} \right)=-\boldsymbol{e}_{i}^{T} \boldsymbol{r}(\boldsymbol{y}), \quad i=1, \ldots, n, \end{aligned}$</td><td>$\begin{aligned} \frac{\partial}{\partial y_i} Q(\boldsymbol{y}) &:= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}\left(Q(\boldsymbol{y} + \varepsilon\boldsymbol{e}_i) - Q(\boldsymbol{y})\right) \\ &= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}\left(-\varepsilon\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y})) + \frac{1}{2}\varepsilon^2\boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i\right) = -\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y}), \quad i = 1,\ldots,n, \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0866 (13.23)</td><td>301</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{r}_{k}^{T} \boldsymbol{w}= \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{w}=0, \quad \boldsymbol{w} \in \mathbb{W}_{k} .</td><td>$\boldsymbol{r}_{k}^{T} \boldsymbol{w}= \boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{w}=0, \quad \boldsymbol{w} \in \mathbb{W}_{k} .$</td><td>$\boldsymbol{r}_k^T\boldsymbol{w} = \boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{w} = 0, \quad \boldsymbol{w} \in \mathbb{W}_k.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0874 (13.29)</td><td>303</td><td>■ 1.000 ● N|-1</td><td>\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{ \boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0} \right\|_{\boldsymbol{A}}} \leq \min _{\substack{Q \in \Pi_{k} \\ Q(0)=1}} \max _{\lambda_{\min } \leq x \leq \lambda_{\max }}|Q(x)|=\frac{2}{a^{-k}+a^{k}}, \quad a:= \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},</td><td>$\frac{\left\|\boldsymbol{x}-\boldsymbol{x}_{k}\right\|_{ \boldsymbol{A}}}{\left\|\boldsymbol{x}-\boldsymbol{x}_{0} \right\|_{\boldsymbol{A}}} \leq \min _{\substack{Q \in \Pi_{k} \\ Q(0)=1}} \max _{\lambda_{\min } \leq x \leq \lambda_{\max }}|Q(x)|=\frac{2}{a^{-k}+a^{k}}, \quad a:= \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},$</td><td>$\frac{\|\boldsymbol{x} - \boldsymbol{x}_k\|_{\boldsymbol{A}}}{\|\boldsymbol{x} - \boldsymbol{x}_0\|_{\boldsymbol{A}}} \le \min_{\substack{Q \in \Pi_k \\ Q(0)=1}} \max_{\lambda_{min} \le x \le \lambda_{max}} |Q(x)| = \frac{2}{a^{-k} + a^k}, \quad a := \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1},$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0875 (13.30)	303	■1.000 ● N|+0	`\frac{x-1}{x+1}<e^{-2 / x} \quad \text { for } \quad x>1`	$\frac{x-1}{x+1}<e^{-2 / x} \quad \text { for } \quad x>1$	$\frac{x-1}{x+1}<e^{-2/x} \quad \text{for} \quad x>1$
lyche-numerical-linear-algebra EQ0929 (13.58)	315	■1.000 ● N|+0	`\langle\boldsymbol{V}, \boldsymbol{W}\rangle:=h^{2} \sum_{j, k=1}^{m} v_{j, k} w_{j, k},`	$\langle\boldsymbol{V}, \boldsymbol{W}\rangle:=h^{2} \sum_{j, k=1}^{m} v_{j, k} w_{j, k},$	$\langle\boldsymbol{V}, \boldsymbol{W}\rangle:=h^{2} \sum_{j, k=1}^{m} v_{j, k} w_{j, k},$
lyche-numerical-linear-algebra EQ0930 (13.59)	315	■1.000 ● W|-1	`\begin{aligned} \left\langle L_{h} v, \boldsymbol{W} \right\rangle & =\sum_{j=1}^{m} \sum_{k=0}^{m} c_{j, k+ \frac{1}{2}}\left(v_{j, k+1}-v_{j, k}\right)\left(w_{j, k+1}-w_{j, k}\right) \\ & +\sum_{j=0}^{m} \sum_{k=1}^{m} c_{j+\frac{1}{2}, k}\left(v_{j+1, k}-v_{j, k}\right) \left(w_{j+1, k}-w_{j, k}\right) . \end{aligned}`	$\begin{aligned} \left\langle L_{h} v, \boldsymbol{W} \right\rangle & =\sum_{j=1}^{m} \sum_{k=0}^{m} c_{j, k+ \frac{1}{2}}\left(v_{j, k+1}-v_{j, k}\right)\left(w_{j, k+1}-w_{j, k}\right) \\ & +\sum_{j=0}^{m} \sum_{k=1}^{m} c_{j+\frac{1}{2}, k}\left(v_{j+1, k}-v_{j, k}\right) \left(w_{j+1, k}-w_{j, k}\right) . \end{aligned}$	$\begin{aligned} \langle L_h v, \boldsymbol{W}\rangle &= \sum_{j=1}^{m}\sum_{k=0}^{m} c_{j,k+\frac{1}{2}}\big(v_{j,k+1}-v_{j,k}\big)\big(w_{j,k+1}-w_{j,k}\big) \\ &+ \sum_{j=0}^{m}\sum_{k=1}^{m} c_{j+\frac{1}{2},k}\big(v_{j+1,k}-v_{j,k}\big)\big(w_{j+1,k}-w_{j,k}\big). \end{aligned}$
lyche-numerical-linear-algebra EQ0946	320	■1.000 ● N|-1	`\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right], \quad \text { and } \quad \boldsymbol{b}=\left[\begin{array}{l} 4 \\ 0 \\ 0 \end{array}\right] .`	$\boldsymbol{A}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right], \quad \text { and } \quad \boldsymbol{b}=\left[\begin{array}{l} 4 \\ 0 \\ 0 \end{array}\right] .$	$\boldsymbol{A}=\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b}=\begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0957	322	■1.000 ● N|+0	`\boldsymbol{r}_{k}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}, \text { and } \rho_{k}:=\left|\boldsymbol{r}_{k}\right|_{2}^{2} .`	$\boldsymbol{r}_{k}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}, \text { and } \rho_{k}:=\left\|\boldsymbol{r}_{k}\right\|_{2}^{2} .$	$\boldsymbol{r}_k := \boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k, \text{ and } \rho_k := \\|\boldsymbol{r}_k\\|_2^2.$
lyche-numerical-linear-algebra EQ0967	326	■1.000 ● S|+0	`\begin{aligned} \boldsymbol{A} \boldsymbol{D} & = \operatorname{diag}\left(\boldsymbol{A}_{1}, \ldots, \boldsymbol{A}_{r}\right) \operatorname{diag}\left(\boldsymbol{X}_{1}, \ldots, \boldsymbol{X}_{r} \right)=\operatorname{diag}\left(\boldsymbol{A}_{1} \boldsymbol{X}_{1}, \ldots, \boldsymbol{A}_{r} \boldsymbol{X}_{r}\right) \\ & =\operatorname{diag} \left(\boldsymbol{X}_{1} \boldsymbol{D}_{1}, \ldots, \boldsymbol{X}_{r} \boldsymbol{D}_{r}\right)= \boldsymbol{X} \boldsymbol{D} . \end{aligned}`	$\begin{aligned} \boldsymbol{A} \boldsymbol{D} & = \operatorname{diag}\left(\boldsymbol{A}_{1}, \ldots, \boldsymbol{A}_{r}\right) \operatorname{diag}\left(\boldsymbol{X}_{1}, \ldots, \boldsymbol{X}_{r} \right)=\operatorname{diag}\left(\boldsymbol{A}_{1} \boldsymbol{X}_{1}, \ldots, \boldsymbol{A}_{r} \boldsymbol{X}_{r}\right) \\ & =\operatorname{diag} \left(\boldsymbol{X}_{1} \boldsymbol{D}_{1}, \ldots, \boldsymbol{X}_{r} \boldsymbol{D}_{r}\right)= \boldsymbol{X} \boldsymbol{D} . \end{aligned}$	$\begin{aligned} \boldsymbol{A}\boldsymbol{D} &= \operatorname{diag}(\boldsymbol{A}_1,\ldots,\boldsymbol{A}_r)\operatorname{diag}(\boldsymbol{X}_1,\ldots,\boldsymbol{X}_r) = \operatorname{diag}(\boldsymbol{A}_1\boldsymbol{X}_1,\ldots,\boldsymbol{A}_r\boldsymbol{X}_r) \\ &= \operatorname{diag}(\boldsymbol{X}_1\boldsymbol{D}_1,\ldots,\boldsymbol{X}_r\boldsymbol{D}_r) = \boldsymbol{X}\boldsymbol{D}. \end{aligned}$
lyche-numerical-linear-algebra EQ1030	348	■1.000 ● N|+0	`\mu_{1}(s)=\left(\lambda_{1}-s\right)^{-1}, \mu_{2}(s)= \left(\lambda_{2}-s\right)^{-1}, \ldots, \mu_{n}(s)= \left(\lambda_{n}-s\right)^{-1} .`	$\mu_{1}(s)=\left(\lambda_{1}-s\right)^{-1}, \mu_{2}(s)= \left(\lambda_{2}-s\right)^{-1}, \ldots, \mu_{n}(s)= \left(\lambda_{n}-s\right)^{-1} .$	$\mu_1(s)=(\lambda_1-s)^{-1}, \mu_2(s)=(\lambda_2-s)^{-1}, \ldots, \mu_n(s)=(\lambda_n-s)^{-1}.$
lyche-numerical-linear-algebra EQ1041	352	■1.000 ● W|+0	`A_{14}=\left[\begin{array}{r|rr|r} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ \hline-2.1 e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1 e-10 & -0.58622 & 0.052576 & -0.25774 \\ \hline 1.2 e-14 & 3.3 e-05 & -1.1 e-05 & 0.22746 \end{array}\right] .`	$A_{14}=\left[\begin{array}{r\|rr\|r} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ \hline-2.1 e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1 e-10 & -0.58622 & 0.052576 & -0.25774 \\ \hline 1.2 e-14 & 3.3 e-05 & -1.1 e-05 & 0.22746 \end{array}\right] .$	$A_{14}=\left[\begin{array}{r\|rr\|r} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ \hline -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ \hline 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{array}\right].$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1045</td><td>353</td><td>■ 1.000
● N|+0</td><td><code>\tilde{\boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k}= \tilde{\boldsymbol{Q}}_{k-1}\left(\boldsymbol{Q}_{k} \boldsymbol{R}_{k}\right) \tilde{\boldsymbol{R}}_{k-1}= \tilde{\boldsymbol{Q}}_{k-1} \boldsymbol{A}_{k} \tilde{\boldsymbol{R}}_{k-1}=\left(\tilde{ \boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{Q}}_{k-1}^{*} \right) \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^{k} .</code></td><td>$\tilde{\boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k}= \tilde{\boldsymbol{Q}}_{k-1}\left(\boldsymbol{Q}_{k} \boldsymbol{R}_{k}\right) \tilde{\boldsymbol{R}}_{k-1}= \tilde{\boldsymbol{Q}}_{k-1} \boldsymbol{A}_{k} \tilde{\boldsymbol{R}}_{k-1}=\left(\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{Q}}_{k-1}^{*}\right) \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^{k} .$</td><td>$\tilde{\boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k}= \tilde{\boldsymbol{Q}}_{k-1}(\boldsymbol{Q}_{k} \boldsymbol{R}_{k}) \tilde{\boldsymbol{R}}_{k-1}= \tilde{\boldsymbol{Q}}_{k-1} \boldsymbol{A}_{k} \tilde{\boldsymbol{R}}_{k-1}=(\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{Q}}_{k-1}^{*}) \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1}=\boldsymbol{A}^{k} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1047</td><td>354</td><td>■ 1.000
● N|+1</td><td><code>\hat{\boldsymbol{A}}_{k}=\tilde{ \boldsymbol{Q}}_{k-1}^{*}(\boldsymbol{A}+\boldsymbol{E}) \tilde{\boldsymbol{Q}}_{k-1}, \quad \boldsymbol{E}= \tilde{\boldsymbol{Q}}_{k-1}\left(a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}\right) \tilde{\boldsymbol{Q}}_{k-1}^{*} .</code></td><td>$\hat{\boldsymbol{A}}_{k}=\tilde{\boldsymbol{Q}}_{k-1}^{*}(\boldsymbol{A}+\boldsymbol{E}) \tilde{\boldsymbol{Q}}_{k-1}, \quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{k-1}\left(a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}\right) \tilde{\boldsymbol{Q}}_{k-1}^{*} .$</td><td>$\hat{\boldsymbol{A}}_{k}=\tilde{\boldsymbol{Q}}_{k-1}^{*}(\boldsymbol{A}+\boldsymbol{E}) \tilde{\boldsymbol{Q}}_{k-1}, \quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{k-1}(a_{i+1, i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_{i}^{T}) \tilde{\boldsymbol{Q}}_{k-1}^{*} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0002</td><td>022</td><td>■ 1.000
● N|+0</td><td><code>\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c} x_{1}+y_{1} \\ \vdots \\ x_{n}+y_{n} \end{array} \right], \quad a \boldsymbol{x}:=\left[\begin{array}{c} a x_{1} \\ \vdots \\ a x_{n} \end{array}\right], \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \quad a \in \mathbb{R} .</code></td><td>$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c} x_{1}+y_{1} \\ \vdots \\ x_{n}+y_{n} \end{array}\right], \quad a \boldsymbol{x}:=\left[\begin{array}{c} a x_{1} \\ \vdots \\ a x_{n} \end{array}\right], \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \quad a \in \mathbb{R} .$</td><td>$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c} x_{1}+y_{1} \\ \vdots \\ x_{n}+y_{n} \end{array}\right], \quad a \boldsymbol{x}:=\left[\begin{array}{c} a x_{1} \\ \vdots \\ a x_{n} \end{array}\right], \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^{n}, \quad a \in \mathbb{R} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0004</td><td>022</td><td>■ 1.000
● S|+1</td><td><code>\boldsymbol{a}_{: j}:=\left[\begin{array}{c} a_{1 j} \\ a_{2 j} \\ \vdots \\ a_{m j} \end{array}\right], \quad \boldsymbol{a}_{i:}^{T}:=\left[a_{i 1}, a_{i 2}, \ldots, a_{i n}\right], \quad \boldsymbol{A}=\left[\boldsymbol{a}_{: 1}, \boldsymbol{a}_{: 2}, \ldots \boldsymbol{a}_{: n}\right]=\left[\begin{array}{c} \boldsymbol{a}_{1:}^{T} \\ \boldsymbol{a}_{2:}^{T} \\ \vdots \\ \boldsymbol{a}_{m:}^{T} \end{array}\right]</code></td><td>$\boldsymbol{a}_{: j}:=\left[\begin{array}{c} a_{1 j} \\ a_{2 j} \\ \vdots \\ a_{m j} \end{array}\right], \quad \boldsymbol{a}_{i:}^{T}:=\left[a_{i 1}, a_{i 2}, \ldots, a_{i n}\right], \quad \boldsymbol{A}=\left[\boldsymbol{a}_{: 1}, \boldsymbol{a}_{: 2}, \ldots \boldsymbol{a}_{: n}\right]=\left[\begin{array}{c} \boldsymbol{a}_{1:}^{T} \\ \boldsymbol{a}_{2:}^{T} \\ \vdots \\ \boldsymbol{a}_{m:}^{T} \end{array}\right]$</td><td>$\boldsymbol{a}_{: j}:=\left[\begin{array}{c} a_{1 j} \\ a_{2 j} \\ \vdots \\ a_{m j} \end{array}\right], \quad \boldsymbol{a}_{i:}^{T}:=[a_{i 1}, a_{i 2}, \ldots, a_{i n}], \quad \boldsymbol{A}=[\boldsymbol{a}_{: 1}, \boldsymbol{a}_{: 2}, \ldots \boldsymbol{a}_{: n}]=\left[\begin{array}{c} \boldsymbol{a}_{1:}^{T} \\ \boldsymbol{a}_{2:}^{T} \\ \vdots \\ \boldsymbol{a}_{m:}^{T} \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0005</td><td>022</td><td>■ 1.000
● N|+0</td><td><code>e^{x}=e^{a+i b}:=e^{a}(\cos b+i \sin b) .</code></td><td>$e^{x}=e^{a+i b}:=e^{a}(\cos b+i \sin b) .$</td><td>$e^{x}=e^{a+i b}:=e^{a}(\cos b+i \sin b) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0006</td><td>022</td><td>■ 1.000
● N|+0</td><td><code>e^{i \pi / 2}=i, \quad e^{i \pi}=-1, \quad e^{2 i \pi}=1 .</code></td><td>$e^{i \pi / 2}=i, \quad e^{i \pi}=-1, \quad e^{2 i \pi}=1 .$</td><td>$e^{i \pi / 2}=i, \quad e^{i \pi}=-1, \quad e^{2 i \pi}=1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0007</td><td>023</td><td>■ 1.000
● N|+0</td><td><code>x=a+i b=r e^{i \theta}, \quad r=|x|=\sqrt{a^{2}+b^{2}}, \quad \cos \theta=\frac{a}{r}, \quad \sin \theta= \frac{b}{r} .</code></td><td>$x=a+i b=r e^{i \theta}, \quad r=|x|=\sqrt{a^{2}+b^{2}}, \quad \cos \theta=\frac{a}{r}, \quad \sin \theta=\frac{b}{r} .$</td><td>$x=a+i b=r e^{i \theta}, \quad r=|x|=\sqrt{a^{2}+b^{2}}, \quad \cos \theta=\frac{a}{r}, \quad \sin \theta=\frac{b}{r} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0014 (1.2)</td><td>025</td><td>■ 1.000
● N|+1</td><td><code>p(t):=a_{0}+a_{1} t+a_{2} t^{2}+\cdots+a_{n} t^{n},</code></td><td>$p(t):=a_{0}+a_{1} t+a_{2} t^{2}+\cdots+a_{n} t^{n},$</td><td>$p(t):=a_{0}+a_{1} t+a_{2} t^{2}+\cdots+a_{n} t^{n},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0017 (1.5)</td><td>028</td><td>■ 1.000
● N|+0</td><td><code>\mathcal{S} \cap \mathcal{T}:=\{\boldsymbol{x}: \boldsymbol{x} \in \mathcal{S} \text { and } \boldsymbol{x} \in \mathcal{T}\} .</code></td><td>$\mathcal{S} \cap \mathcal{T}:=\{\boldsymbol{x}: \boldsymbol{x} \in \mathcal{S} \text { and } \boldsymbol{x} \in \mathcal{T}\} .$</td><td>$\mathcal{S} \cap \mathcal{T}:=\{\boldsymbol{x}: \boldsymbol{x} \in \mathcal{S} \text { and } \boldsymbol{x} \in \mathcal{T}\} .$</td></tr>
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● N|-1</td><td><code>\mathcal{S} \cup \mathcal{T}:=\{\boldsymbol{x}: \boldsymbol{x} \in \mathcal{S} \text { or } \boldsymbol{x} \in \mathcal{T}\} .</code></td><td>$\mathcal{S} \cup \mathcal{T}:=\{\boldsymbol{x}: \boldsymbol{x} \in \mathcal{S} \text { or } \boldsymbol{x} \in \mathcal{T}\} .$</td><td>$\mathcal{S} \cup \mathcal{T}:=\{\boldsymbol{x}: \boldsymbol{x} \in \mathcal{S} \text { or } \boldsymbol{x} \in \mathcal{T}\} .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0022</td><td>029</td><td>■ 1.000 ● N|+0</td><td>\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}\right\}</td><td>$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}\right\}$</td><td>$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}\right\}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0023</td><td>029</td><td>■ 1.000 ● C|-4</td><td>\operatorname{dim}(\mathcal{S}+\mathcal{T})=p+q+r=(p+q)+(p+r)-p=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S} \cap \mathcal{T})</td><td>$\operatorname{dim}(\mathcal{S}+\mathcal{T})=p+q+r=(p+q)+(p+r)-p=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S} \cap \mathcal{T})$</td><td>$\operatorname{dim}(\mathcal{S}+\mathcal{T})=p+q+r=(p+q)+(p+r)-p=\operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T})-\operatorname{dim}(\mathcal{S} \cap \mathcal{T})$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0026</td><td>030</td><td>■ 1.000 ● N|+0</td><td>\mathcal{R}(\boldsymbol{X}):=\left\{\boldsymbol{X} \boldsymbol{c}: \boldsymbol{c} \in \mathbb{R}^{n}\right \}=\operatorname{span}(\mathcal{X}) .</td><td>$\mathcal{R}(\boldsymbol{X}):=\left\{\boldsymbol{X} \boldsymbol{c}: \boldsymbol{c} \in \mathbb{R}^{n}\right\}=\operatorname{span}(\mathcal{X}) .$</td><td>$\mathcal{R}(\boldsymbol{X}):=\left\{\boldsymbol{X} \boldsymbol{c}: \boldsymbol{c} \in \mathbb{R}^{n}\right\}=\operatorname{span}(\mathcal{X}) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0030</td><td>031</td><td>■ 1.000 ● N|+0</td><td>x_{1} \boldsymbol{a}_{1}+x_{2} \boldsymbol{a}_{2}+\cdots+x_{n} \boldsymbol{a}_{n}=\boldsymbol{b},</td><td>$x_{1} \boldsymbol{a}_{1}+x_{2} \boldsymbol{a}_{2}+\cdots+x_{n} \boldsymbol{a}_{n}=\boldsymbol{b},$</td><td>$x_{1} \boldsymbol{a}_{1}+x_{2} \boldsymbol{a}_{2}+\cdots+x_{n} \boldsymbol{a}_{n}=\boldsymbol{b},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0035 (1.12)</td><td>035</td><td>■ 1.000 ● N|+1</td><td>\operatorname{det}(\boldsymbol{A})=\sum_{\sigma \in S_{n}} \operatorname{sign}(\sigma) a_{\sigma(1), 1} a_{\sigma(2), 2} \cdots a_{\sigma(n), n}</td><td>$\operatorname{det}(\boldsymbol{A})=\sum_{\sigma \in S_{n}} \operatorname{sign}(\sigma) a_{\sigma(1), 1} a_{\sigma(2), 2} \cdots a_{\sigma(n), n}$</td><td>$\operatorname{det}(\boldsymbol{A})=\sum_{\sigma \in S_{n}} \operatorname{sign}(\sigma) a_{\sigma(1), 1} a_{\sigma(2), 2} \cdots a_{\sigma(n), n}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0036</td><td>035</td><td>■ 1.000 ● N|+0</td><td>\left|\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & & \vdots \\ a_{n 1} & a_{n 2} & \cdots & a_{n n} \end{array}\right| .</td><td>$\left|\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & & \vdots \\ a_{n 1} & a_{n 2} & \cdots & a_{n n} \end{array}\right| .$</td><td>$\left|\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1 n} \\ a_{21} & a_{22} & \cdots & a_{2 n} \\ \vdots & \vdots & & \vdots \\ a_{n 1} & a_{n 2} & \cdots & a_{n n} \end{array}\right| .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0037</td><td>035</td><td>■ 1.000 ● N|+0</td><td>\left|\begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array}\right|=a_{11} a_{22}-a_{21} a_{12} .</td><td>$\left|\begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array}\right|=a_{11} a_{22}-a_{21} a_{12} .$</td><td>$\left|\begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array}\right|=a_{11} a_{22}-a_{21} a_{12} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0039 (1.13)</td><td>036</td><td>■ 1.000 ● K|+0</td><td>\operatorname{det}(\boldsymbol{B})=\operatorname{det}(\boldsymbol{A}) .</td><td>$\operatorname{det}(\boldsymbol{B})=\operatorname{det}(\boldsymbol{A}) .$</td><td>$\operatorname{det}(\boldsymbol{B})=\operatorname{det}(\boldsymbol{A}) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0041</td><td>036</td><td>■ 1.000 ● N|+0</td><td>\operatorname{det}(\boldsymbol{A}):=\left| \begin{array}{ccc} 1 & x & y \\ 1 & x_{1} & y_{1} \\ 1 & x_{2} & y_{2} \end{array}\right|=0</td><td>$\operatorname{det}(\boldsymbol{A}):=\left|\begin{array}{ccc} 1 & x & y \\ 1 & x_{1} & y_{1} \\ 1 & x_{2} & y_{2} \end{array}\right|=0$</td><td>$\operatorname{det}(\boldsymbol{A}):=\left|\begin{array}{ccc} 1 & x & y \\ 1 & x_{1} & y_{1} \\ 1 & x_{2} & y_{2} \end{array}\right|=0$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0043</td><td>036</td><td>■ 1.000 ● N|+0</td><td>y-y_{1}=\frac{y_{2}-y_{1}}{x_{2}-x_{1}}\left(x-x_{1} \right)</td><td>$y-y_{1}=\frac{y_{2}-y_{1}}{x_{2}-x_{1}}\left(x-x_{1}\right)$</td><td>$y-y_{1}=\frac{y_{2}-y_{1}}{x_{2}-x_{1}}\left(x-x_{1}\right)$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0044</td><td>037</td><td>■ 1.000 ● N|+0</td><td>x_{j}=\frac{\operatorname{det}\left(\boldsymbol{A}_{j}(\boldsymbol{b})\right)}{\operatorname{det}(\boldsymbol{A})}, \quad j=1,2, \ldots, n,</td><td>$x_{j}=\frac{\operatorname{det}\left(\boldsymbol{A}_{j}(\boldsymbol{b})\right)}{\operatorname{det}(\boldsymbol{A})}, \quad j=1,2, \ldots, n,$</td><td>$x_{j}=\frac{\operatorname{det}\left(\boldsymbol{A}_{j}(\boldsymbol{b})\right)}{\operatorname{det}(\boldsymbol{A})}, \quad j=1,2, \ldots, n,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0045</td><td>037</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}^{-1}=\frac{1}{\operatorname{det}(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),</td><td>$\boldsymbol{A}^{-1}=\frac{1}{\operatorname{det}(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$</td><td>$\boldsymbol{A}^{-1}=\frac{1}{\operatorname{det}(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0048	038	■ 1.000 ● N|+0	`\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\left|\begin{array}{ccc} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{array}\right| .`	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} .$	$\det(\boldsymbol{A}-\lambda \boldsymbol{I})=\begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} .$
lyche-numerical-linear-algebra EQ0050 (1.16)	038	■ 1.000 ● N|+0	`\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\left(a_{11}-\lambda\right)\left(a_{22}-\lambda\right) \cdots\left(a_{n n}-\lambda\right)+r(\lambda),`	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\left(a_{11}-\lambda\right)\left(a_{22}-\lambda\right) \cdots\left(a_{n n}-\lambda\right)+r(\lambda),$	$\det(\boldsymbol{A}-\lambda \boldsymbol{I})=(a_{11}-\lambda)(a_{22}-\lambda) \cdots (a_{nn}-\lambda)+r(\lambda),$
lyche-numerical-linear-algebra EQ0051 (1.17)	039	■ 1.000 ● N|+0	`\operatorname{trace}(\boldsymbol{A})=\lambda_{1}+\lambda_{2}+\cdots+\lambda_{n}, \quad \operatorname{det}(\boldsymbol{A})=\lambda_{1} \lambda_{2} \cdots \lambda_{n},`	$\operatorname{trace}(\boldsymbol{A})=\lambda_{1}+\lambda_{2}+\cdots+\lambda_{n}, \quad \operatorname{det}(\boldsymbol{A})=\lambda_{1} \lambda_{2} \cdots \lambda_{n},$	$\operatorname{trace}(\boldsymbol{A})=\lambda_{1}+\lambda_{2}+\cdots+\lambda_{n}, \quad \det(\boldsymbol{A})=\lambda_{1} \lambda_{2} \cdots \lambda_{n},$
lyche-numerical-linear-algebra EQ0052 (1.18)	039	■ 1.000 ● N|-1	`\operatorname{trace}(\boldsymbol{A}):=a_{11}+a_{22}+\cdots+a_{n n} .`	$\operatorname{trace}(\boldsymbol{A}):=a_{11}+a_{22}+\cdots+a_{n n} .$	$\operatorname{trace}(\boldsymbol{A}):=a_{11}+a_{22}+\cdots+a_{nn}.$
lyche-numerical-linear-algebra EQ0054	039	■ 1.000 ● N|+0	`\pi_{A}(\lambda)=\left(\lambda_{1}-\lambda\right) \cdots\left(\lambda_{n}-\lambda\right)=(-1)^{n} \lambda^{n}+d_{n-1} \lambda^{n-1}+\cdots+d_{0},`	$\pi_{A}(\lambda)=\left(\lambda_{1}-\lambda\right) \cdots\left(\lambda_{n}-\lambda\right)=(-1)^{n} \lambda^{n}+d_{n-1} \lambda^{n-1}+\cdots+d_{0},$	$\pi_{\boldsymbol{A}}(\lambda)=(\lambda_{1}-\lambda) \cdots (\lambda_{n}-\lambda)=(-1)^{n} \lambda^{n}+d_{n-1} \lambda^{n-1}+\cdots+d_{0},$
lyche-numerical-linear-algebra EQ0055 (1.19)	039	■ 1.000 ● K|+0	`\lambda^{2}-\operatorname{trace}(\boldsymbol{A}) \lambda+\operatorname{det}(\boldsymbol{A})=0 .`	$\lambda^{2}-\operatorname{trace}(\boldsymbol{A}) \lambda+\operatorname{det}(\boldsymbol{A})=0 .$	$\lambda^{2}-\operatorname{trace}(\boldsymbol{A}) \lambda+\det(\boldsymbol{A})=0.$
lyche-numerical-linear-algebra EQ0056	040	■ 1.000 ● K|+0	`\left[\begin{array}{ll} W & X \\ Y & Z \end{array}\right]=\left[\begin{array}{ll} A & B \\ C & D \end{array}\right]\left[\begin{array}{ll} E & F \\ G & H \end{array}\right],`	$\left[\begin{array}{ll} W & X \\ Y & Z \end{array}\right]=\left[\begin{array}{ll} A & B \\ C & D \end{array}\right]\left[\begin{array}{ll} E & F \\ G & H \end{array}\right],$	$\begin{bmatrix} \boldsymbol{W} & \boldsymbol{X} \\ \boldsymbol{Y} & \boldsymbol{Z} \end{bmatrix}=\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}\begin{bmatrix} \boldsymbol{E} & \boldsymbol{F} \\ \boldsymbol{G} & \boldsymbol{H} \end{bmatrix},$
lyche-numerical-linear-algebra EQ0059	041	■ 1.000 ● S|+0	`A=\left[\begin{array}{cc} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{array}\right] .`	$A=\left[\begin{array}{cc} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{array}\right] .$	$A=\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0063	042	■ 1.000 ● K|+0	`\left[\begin{array}{ll} 1 & 2 \\ 2 & 1 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{l} 3 \\ 6 \end{array}\right]`	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 1 \end{array}\right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right]=\left[\begin{array}{l} 3 \\ 6 \end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x_{1} \\ x_{2} \end{bmatrix}=\begin{bmatrix} 3 \\ 6 \end{bmatrix}$
lyche-numerical-linear-algebra EQ0065	042	■ 1.000 ● N|+0	`\operatorname{adj}(\boldsymbol{A})=\left[ \begin{array}{ccc} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{array}\right] .`	$\operatorname{adj}(\boldsymbol{A})=\left[\begin{array}{ccc} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{array}\right] .$	$\operatorname{adj}(\boldsymbol{A})=\begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0066	042	■ 1.000 ● K|+0	`\operatorname{adj}(\boldsymbol{A}) \boldsymbol{A}=\left[ \begin{array}{ccc} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{array}\right]=\operatorname{det}(\boldsymbol{A}) \boldsymbol{I} .`	$\operatorname{adj}(\boldsymbol{A}) \boldsymbol{A}=\left[\begin{array}{ccc} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{array}\right]=\operatorname{det}(\boldsymbol{A}) \boldsymbol{I} .$	$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A}=\begin{bmatrix} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{bmatrix}=\det(\boldsymbol{A})\boldsymbol{I}.$
lyche-numerical-linear-algebra EQ0067	042	■ 1.000 ● W|+1	`\left|\begin{array}{llll} x & y & z & 1 \\ x_{1} & y_{1} & z_{1} & 1 \\ x_{2} & y_{2} & z_{2} & 1 \\ x_{3} & y_{3} & z_{3} & 1 \end{array}\right|=0`	$\begin{vmatrix} x & y & z & 1 \\ x_{1} & y_{1} & z_{1} & 1 \\ x_{2} & y_{2} & z_{2} & 1 \\ x_{3} & y_{3} & z_{3} & 1 \end{vmatrix}=0$	$\begin{vmatrix} x & y & z & 1 \\ x_{1} & y_{1} & z_{1} & 1 \\ x_{2} & y_{2} & z_{2} & 1 \\ x_{3} & y_{3} & z_{3} & 1 \end{vmatrix}=0.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0068	043	1.000 W|+0	A(T)=\frac{1}{2}\left|\begin{array}{ccc} 1 & 1 & 1 \\ x_{1} & x_{2} & x_{3} \\ y_{1} & y_{2} & y_{3} \end{array}\right| .	$A(T)=\frac{1}{2}\left\lvert\begin{array}{ccc} 1 & 1 & 1 \\ x_{1} & x_{2} & x_{3} \\ y_{1} & y_{2} & y_{3} \end{array}\right\rvert .$	$A(T)=\frac{1}{2}\left\lvert\begin{array}{ccc} 1 & 1 & 1 \\ x_{1} & x_{2} & x_{3} \\ y_{1} & y_{2} & y_{3} \end{array}\right\rvert .$
lyche-numerical-linear-algebra EQ0069	043	1.000 W|+0	\left|\begin{array}{ccccc} 1 & x_{1} & x_{1}^{2} & \cdots & x_{1}^{n-1} \\ 1 & x_{2} & x_{2}^{2} & \cdots & x_{2}^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n} & x_{n}^{2} & \cdots & x_{n}^{n-1} \end{array} \right|=\prod_{i>j}\left(x_{i}-x_{j}\right),	$\left\lvert\begin{array}{ccccc} 1 & x_{1} & x_{1}^{2} & \cdots & x_{1}^{n-1} \\ 1 & x_{2} & x_{2}^{2} & \cdots & x_{2}^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n} & x_{n}^{2} & \cdots & x_{n}^{n-1} \end{array} \right\rvert=\prod_{i>j}\left(x_{i}-x_{j}\right),$	$\left\lvert\begin{array}{ccccc} 1 & x_{1} & x_{1}^{2} & \cdots & x_{1}^{n-1} \\ 1 & x_{2} & x_{2}^{2} & \cdots & x_{2}^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n} & x_{n}^{2} & \cdots & x_{n}^{n-1} \end{array} \right\rvert=\prod_{i>j}(x_{i}-x_{j}),$
lyche-numerical-linear-algebra EQ0070	043	1.000 N|+0	\operatorname{det}(\boldsymbol{A})=P g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})	$\operatorname{det}(\boldsymbol{A})=P g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$	$\operatorname{det}(\boldsymbol{A})=P g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$
lyche-numerical-linear-algebra EQ0071	044	1.000 N|+0	g(\boldsymbol{\gamma})=\prod_{i=2}^{n}\left(\gamma_{i}-\gamma_{1}\right)\left(\gamma_{i}-\gamma_{2}\right) \cdots\left(\gamma_{i}-\gamma_{i-1}\right)	$g(\boldsymbol{\gamma})=\prod_{i=2}^{n}\left(\gamma_{i}-\gamma_{1}\right)\left(\gamma_{i}-\gamma_{2}\right) \cdots\left(\gamma_{i}-\gamma_{i-1}\right)$	$g(\boldsymbol{\gamma})=\prod_{i=2}^{n}(\gamma_{i}-\gamma_{1})(\gamma_{i}-\gamma_{2}) \cdots(\gamma_{i}-\gamma_{i-1})$
lyche-numerical-linear-algebra EQ0072	044	1.000 N|+0	q(\boldsymbol{\alpha}, \boldsymbol{\beta})=k g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})=k g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})=k g(\boldsymbol{\alpha}) g(\boldsymbol{\beta})$
lyche-numerical-linear-algebra EQ0074	044	1.000 N|+0	A_{k}(x)=\prod_{s \neq k}\left(\frac{\alpha_{s}-x}{\alpha_{s}-\alpha_{k}}\right), \quad B_{k}(x)=\prod_{s \neq k}\left(\frac{\beta_{s}-x}{\beta_{s}-\beta_{k}}\right) .	$A_{k}(x)=\prod_{s \neq k}\left(\frac{\alpha_{s}-x}{\alpha_{s}-\alpha_{k}}\right), \quad B_{k}(x)=\prod_{s \neq k}\left(\frac{\beta_{s}-x}{\beta_{s}-\beta_{k}}\right) .$	$A_{k}(x)=\prod_{s \neq k}\left(\frac{\alpha_{s}-x}{\alpha_{s}-\alpha_{k}}\right), \qquad B_{k}(x)=\prod_{s \neq k}\left(\frac{\beta_{s}-x}{\beta_{s}-\beta_{k}}\right) .$
lyche-numerical-linear-algebra EQ0075	044	1.000 N|+0	t_{i, j}^{n}=\frac{f(i) f(j)}{i+j-1},	$t_{i, j}^{n}=\frac{f(i) f(j)}{i+j-1},$	$t_{i, j}^{n}=\frac{f(i) f(j)}{i+j-1},$
lyche-numerical-linear-algebra EQ0076	044	1.000 N|+0	f(i+1)=\left(\frac{i^{2}-n^{2}}{i^{2}}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n .	$f(i+1)=\left(\frac{i^{2}-n^{2}}{i^{2}}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n .$	$f(i+1)=\left(\frac{i^{2}-n^{2}}{i^{2}}\right) f(i), \quad i=1,2, \ldots, \quad f(1)=-n .$
lyche-numerical-linear-algebra EQ0077 (2.1)	046	1.000 N|+0	x_{i}=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1	$x_{i}=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1$	$x_{i}=a+\frac{i-1}{n}(b-a), \quad i=1,2, \ldots, n+1$
lyche-numerical-linear-algebra EQ0079 (2.3)	047	1.000 N|+0	g(x)=y_{1}+\frac{y_{2}-y_{1}}{x_{2}-x_{1}}\left(x-x_{1}\right),	$g(x)=y_{1}+\frac{y_{2}-y_{1}}{x_{2}-x_{1}}\left(x-x_{1}\right),$	$g(x)=y_{1}+\frac{y_{2}-y_{1}}{x_{2}-x_{1}}(x-x_{1}),$
lyche-numerical-linear-algebra EQ0080	047	1.000 N|-2	f(x)=\arctan (10 x)+\pi / 2, \quad x \in[-1,1] .	$f(x)=\arctan (10 x)+\pi / 2, \quad x \in[-1,1] .$	$f(x)=\arctan (10 x)+\pi / 2, \quad x \in[-1,1] .$
lyche-numerical-linear-algebra EQ0082 (2.5)	049	1.000 N|+0	g(x):= \begin{cases}p_{1}(x)=-\frac{1}{2} x+\frac{3}{2} x^{3}, & \text { if } 0 \leq x<1, \\ p_{2}(x)=1+4(x-1)+\frac{9}{2}(x-1)^{2}+\frac{13}{2}(x-1)^{3}, & \text { if } 1 \leq x \leq 2,\end{cases}	$g(x):= \begin{cases}p_{1}(x)=-\frac{1}{2} x+\frac{3}{2} x^{3}, & \text { if } 0 \leq x<1, \\ p_{2}(x)=1+4(x-1)+\frac{9}{2}(x-1)^{2}+\frac{13}{2}(x-1)^{3}, & \text { if } 1 \leq x \leq 2,\end{cases}$	$g(x):= \begin{cases}p_{1}(x)=-\frac{1}{2} x+\frac{3}{2} x^{3}, & \text { if } 0 \leq x<1, \\ p_{2}(x)=1+4(x-1)+\frac{9}{2}(x-1)^{2}+\frac{13}{2}(x-1)^{3}, & \text { if } 1 \leq x \leq 2,\end{cases}$
lyche-numerical-linear-algebra EQ0083	049	1.000 N|+0	p_{i-1}^{(j)}\left(x_{i}\right)=p_{i}^{(j)}\left(x_{i}\right), \quad j=0,1,2, \quad i=2, \ldots, n .	$p_{i-1}^{(j)}\left(x_{i}\right)=p_{i}^{(j)}\left(x_{i}\right), \quad j=0,1,2, \quad i=2, \ldots, n .$	$p_{i-1}^{(j)}(x_{i})=p_{i}^{(j)}(x_{i}), \quad j=0,1,2, \quad i=2, \ldots, n .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0085 (2.7)</td><td>050</td><td>■ 1.000 ● N|+0</td><td><code>p_{i}(x)=c_{i, 1}+c_{i, 2}\left(x-x_{i}\right)+c_{i, 3}\left(x-x_{i}\right)^{2}+c_{i, 4}\left(x-x_{i}\right)^{3} \quad i=1, \ldots, n,</code></td><td>$p_{i}(x)=c_{i, 1}+c_{i, 2}\left(x-x_{i}\right)+c_{i, 3}\left(x-x_{i}\right)^{2}+c_{i, 4}\left(x-x_{i}\right)^{3} \quad i=1, \ldots, n,$</td><td>$p_i(x)=c_{i,1}+c_{i,2}(x-x_i)+c_{i,3}(x-x_i)^2+c_{i,4}(x-x_i)^3 \quad i=1,\ldots,n,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0086 (2.8)</td><td>050</td><td>■ 1.000 ● W|+2</td><td><code>c_{i 1}=y_{i}, c_{i 2}=\frac{y_{i+1}-y_{i}}{h}-\frac{h}{3} \mu_{i}-\frac{h}{6} \mu_{i+1}, c_{i, 3}=\frac{\mu_{i}}{2}, c_{i, 4}=\frac{\mu_{i+1}-\mu_{i}}{6 h} .</code></td><td>$c_{i 1}=y_{i}, c_{i 2}=\frac{y_{i+1}-y_{i}}{h}-\frac{h}{3} \mu_{i}-\frac{h}{6} \mu_{i+1}, c_{i, 3}=\frac{\mu_{i}}{2}, c_{i, 4}=\frac{\mu_{i+1}-\mu_{i}}{6 h} .$</td><td>$c_{i1}=y_i,\ c_{i2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1},\ c_{i,3}=\frac{\mu_i}{2},\ c_{i,4}=\frac{\mu_{i+1}-\mu_i}{6h}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0088 (2.10)</td><td>051</td><td>■ 1.000 ● W|+0</td><td><code>\begin{aligned} p_{i-1}^{\prime}\left(x_{i}\right) & =c_{i-1,2}+2 h c_{i-1,3}+3 h^{2} c_{i-1,4} \\ & =\frac{y_{i}-y_{i-1}}{h}-\frac{h}{3} \mu_{i-1}-\frac{h}{6} \mu_{i}+2 h \frac{\mu_{i-1}}{2}+3 h^{2} \frac{\mu_{i}-\mu_{i-1}}{6 h} \\ & =\frac{y_{i}-y_{i-1}}{h}+\frac{h}{6} \mu_{i-1}+\frac{h}{3} \mu_{i} \\ p_{i}^{\prime}\left(x_{i}\right) & =c_{i 2}=\frac{y_{i+1}-y_{i}}{h}-\frac{h}{3} \mu_{i}-\frac{h}{6} \mu_{i+1} . \end{aligned}</code></td><td>$\begin{aligned} p_{i-1}^{\prime}\left(x_{i}\right) & =c_{i-1,2}+2 h c_{i-1,3}+3 h^{2} c_{i-1,4} \\ & =\frac{y_{i}-y_{i-1}}{h}-\frac{h}{3} \mu_{i-1}-\frac{h}{6} \mu_{i}+2 h \frac{\mu_{i-1}}{2}+3 h^{2} \frac{\mu_{i}-\mu_{i-1}}{6 h} \\ & =\frac{y_{i}-y_{i-1}}{h}+\frac{h}{6} \mu_{i-1}+\frac{h}{3} \mu_{i} \\ p_{i}^{\prime}\left(x_{i}\right) & =c_{i 2}=\frac{y_{i+1}-y_{i}}{h}-\frac{h}{3} \mu_{i}-\frac{h}{6} \mu_{i+1} . \end{aligned}$</td><td>$\begin{aligned} p'_{i-1}(x_i) &= c_{i-1,2}+2hc_{i-1,3}+3h^2c_{i-1,4} \\ &= \frac{y_i-y_{i-1}}{h}-\frac{h}{3}\mu_{i-1}-\frac{h}{6}\mu_i+2h\frac{\mu_{i-1}}{2}+3h^2\frac{\mu_i-\mu_{i-1}}{6h} \\ &= \frac{y_i-y_{i-1}}{h}+\frac{h}{6}\mu_{i-1}+\frac{h}{3}\mu_i \\ p'_i(x_i) &= c_{i2}=\frac{y_{i+1}-y_i}{h}-\frac{h}{3}\mu_i-\frac{h}{6}\mu_{i+1}. \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0090 (2.12)</td><td>052</td><td>■ 1.000 ● N|+0</td><td><code>\left|a_{i i}\right|>\sum_{j \neq i}\left|a_{i j}\right|, i=1, \ldots, n .</code></td><td>$\left|a_{i i}\right|>\sum_{j \neq i}\left|a_{i j}\right|, i=1, \ldots, n .$</td><td>$|a_{ii}|>\sum_{j\neq i}|a_{ij}|,\ i=1,\ldots,n.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0091 (2.13)</td><td>052</td><td>■ 1.000 ● W|+0</td><td><code>\max _{1 \leq i \leq n}\left|x_{i}\right| \leq \max _{1 \leq i \leq n}\left(\frac{\left|b_{i}\right|}{\sigma_{i}}\right), \text { where } \sigma_{i}:=\left|a_{i i}\right|-\sum_{j \neq i}\left|a_{i j}\right| .</code></td><td>$\max _{1 \leq i \leq n}\left|x_{i}\right| \leq \max _{1 \leq i \leq n}\left(\frac{\left|b_{i}\right|}{\sigma_{i}}\right), \text { where } \sigma_{i}:=\left|a_{i i}\right|-\sum_{j \neq i}\left|a_{i j}\right| .$</td><td>$\max_{1\le i\le n}|x_i|\le\max_{1\le i\le n}\Bigl(\frac{|b_i|}{\sigma_i}\Bigr),\ \textit{where } \sigma_i:=|a_{ii}|-\sum_{j\neq i}|a_{ij}|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0092</td><td>052</td><td>■ 1.000 ● W|+0</td><td><code>\left|b_{k}\right|=\left|a_{k k} x_{k}+\sum_{j \neq k} a_{k j} x_{j}\right| \geq\left|a_{k k}\right|\left|x_{k}\right|-\sum_{j \neq k}\left|a_{k j}\right|\left|x_{j}\right| \geq\left|x_{k}\right|\left(\left|a_{k k}\right|-\sum_{j \neq k}\left|a_{k j}\right|\right),</code></td><td>$\left|b_{k}\right|=\left|a_{k k} x_{k}+\sum_{j \neq k} a_{k j} x_{j}\right| \geq\left|a_{k k}\right|\left|x_{k}\right|-\sum_{j \neq k}\left|a_{k j}\right|\left|x_{j}\right| \geq\left|x_{k}\right|\left(\left|a_{k k}\right|-\sum_{j \neq k}\left|a_{k j}\right|\right),$</td><td>$|b_k|=|a_{kk}x_k+\sum_{j\neq k}a_{kj}x_j|\ge|a_{kk}||x_k|-\sum_{j\neq k}|a_{kj}||x_j|\ge|x_k|\bigl(|a_{kk}|-\sum_{j\neq k}|a_{kj}|\bigr),$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0093</td><td>052</td><td>■ 1.000 ● K|+0</td><td><code>\left[\begin{array}{lll} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{array}\right]\left[\begin{array}{l} \mu_{2} \\ \mu_{3} \\ \mu_{4} \end{array}\right]=\left[\begin{array}{c} 2 \\ -6 \\ 2 \end{array}\right] .</code></td><td>$\left[\begin{array}{lll} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{array}\right]\left[\begin{array}{l} \mu_{2} \\ \mu_{3} \\ \mu_{4} \end{array}\right]=\left[\begin{array}{c} 2 \\ -6 \\ 2 \end{array}\right] .$</td><td>$\begin{bmatrix}4&1&0\\1&4&1\\0&1&4\end{bmatrix}\begin{bmatrix}\mu_2\\\mu_3\\\mu_4\end{bmatrix}=\begin{bmatrix}2\\-6\\2\end{bmatrix}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0094 (2.14)</td><td>053</td><td>1.000 W|+0</td><td>g(x):= \begin{cases}p_{1}(x)=\frac{1}{6} x^{3}, & \text { if } 0 \leq x<1, \\ p_{2}(x)= \frac{1}{6}+\frac{1}{2}(x-1)+\frac{1}{2}(x-1)^{2}- \frac{1}{2}(x-1)^{3}, & \text { if } 1 \leq x<2, \\ p_{3}(x)=\frac{2}{3}-(x-2)^{2}+\frac{1}{2}(x-2)^{3}, & \text { if } 2 \leq x<3, \\ p_{4}(x)= \frac{1}{6}-\frac{1}{2}(x-3)+\frac{1}{2}(x-3)^{2}- \frac{1}{6}(x-3)^{3}, & \text { if } 3 \leq x \leq 4, \end{cases}</td><td>$g(x):= \begin{cases}p_{1}(x)=\frac{1}{6} x^{3}, & \text { if } 0 \leq x<1, \\ p_{2}(x)= \frac{1}{6}+\frac{1}{2}(x-1)+\frac{1}{2}(x-1)^{2}- \frac{1}{2}(x-1)^{3}, & \text { if } 1 \leq x<2, \\ p_{3}(x)=\frac{2}{3}-(x-2)^{2}+\frac{1}{2}(x-2)^{3}, & \text { if } 2 \leq x<3, \\ p_{4}(x)= \frac{1}{6}-\frac{1}{2}(x-3)+\frac{1}{2}(x-3)^{2}- \frac{1}{6}(x-3)^{3}, & \text { if } 3 \leq x \leq 4, \end{cases}$</td><td>$g(x):= \begin{cases}p_{1}(x)=\frac{1}{6} x^{3}, & \text { if } 0 \leq x<1, \\ p_{2}(x)= \frac{1}{6}+\frac{1}{2}(x-1)+\frac{1}{2}(x-1)^{2}- \frac{1}{2}(x-1)^{3}, & \text { if } 1 \leq x<2, \\ p_{3}(x)=\frac{2}{3}-(x-2)^{2}+\frac{1}{2}(x-2)^{3}, & \text { if } 2 \leq x<3, \\ p_{4}(x)= \frac{1}{6}-\frac{1}{2}(x-3)+\frac{1}{2}(x-3)^{2}- \frac{1}{6}(x-3)^{3}, & \text { if } 3 \leq x \leq 4, \end{cases}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0095 (2.15)</td><td>054</td><td>1.000 C|-4</td><td>\left[\begin{array}{ccccc} d_{1} & c_{1} & & & \\ a_{1} & d_{2} & c_{2} & & \\ & \ddots & \ddots & \ddots & \\ & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_{n} \end{array}\right]=\left[\begin{array}{cccc} 1 & & & \\ l_{1} & 1 & & \\ & \ddots & \ddots & \\ & & l_{n-1} & 1 \end{array}\right]\left[\begin{array}{cccc} u_{1} & c_{1} & & \\ & \ddots & \ddots & \\ & & u_{n-1} & c_{n-1} \\ & & & u_{n} \end{array}\right]</td><td>$\left[\begin{array}{ccccc} d_{1} & c_{1} & & & \\ a_{1} & d_{2} & c_{2} & & \\ & \ddots & \ddots & \ddots & \\ & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_{n} \end{array}\right]=\left[\begin{array}{cccc} 1 & & & \\ l_{1} & 1 & & \\ & \ddots & \ddots & \\ & & l_{n-1} & 1 \end{array}\right]\left[\begin{array}{cccc} u_{1} & c_{1} & & \\ & \ddots & \ddots & \\ & & u_{n-1} & c_{n-1} \\ & & & u_{n} \end{array}\right]$</td><td>$\left[\begin{array}{ccccc} d_{1} & c_{1} & & & \\ a_{1} & d_{2} & c_{2} & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_{n} \end{array}\right]=\left[\begin{array}{cccc} 1 & & & \\ l_{1} & 1 & & \\ & \ddots & \ddots & \\ & & l_{n-1} & 1 \end{array}\right]\left[\begin{array}{cccc} u_{1} & c_{1} & & \\ & \ddots & \ddots & \\ & & u_{n-1} & c_{n-1} \\ & & & u_{n} \end{array}\right].$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0100 (2.17)</td><td>055</td><td>1.000 N|+0</td><td>\left|u_{k}\right| \geq \sigma_{k}+\left|c_{k}\right|, \quad \text { where, } \sigma_{k}:=\left|d_{k}\right|- \left|a_{k-1}\right|-\left|c_{k}\right|>0, k=1, \ldots, n,</td><td>$\left|u_{k}\right| \geq \sigma_{k}+\left|c_{k}\right|, \quad \text { where, } \sigma_{k}:=\left|d_{k}\right|- \left|a_{k-1}\right|-\left|c_{k}\right|>0, k=1, \ldots, n,$</td><td>$|u_k| \geq \sigma_k + |c_k|, \quad \text{where, } \sigma_k := |d_k| - |a_{k-1}| - |c_k| > 0,\ k = 1, \ldots, n,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0104 (2.20)</td><td>056</td><td>1.000 N|+0</td><td>-u^{\prime \prime}(x)=f(x), \quad x \in[0,1], \quad u(0)=0, u(1)=0,</td><td>$-u^{\prime \prime}(x)=f(x), \quad x \in[0,1], \quad u(0)=0, u(1)=0,$</td><td>$-u''(x) = f(x), \quad x \in [0,1], \quad u(0) = 0,\ u(1) = 0,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0105</td><td>056</td><td>1.000 W|+0</td><td>g^{\prime}(x)=\lim _{h \rightarrow 0} \frac{g\left(x+ \frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}</td><td>$g^{\prime}(x)=\lim _{h \rightarrow 0} \frac{g\left(x+ \frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$</td><td>$g'(x) = \lim_{h \to 0} \frac{g(x+\frac{h}{2}) - g(x-\frac{h}{2})}{h}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0107</td><td>057</td><td>1.000 N|+0</td><td>\frac{-v_{j-1}+2 v_{j}-v_{j+1}}{h^{2}}=f(j h), \quad j=1, \ldots, m, \quad v_{0}=v_{m+1}=0 .</td><td>$\frac{-v_{j-1}+2 v_{j}-v_{j+1}}{h^{2}}=f(j h), \quad j=1, \ldots, m, \quad v_{0}=v_{m+1}=0 .$</td><td>$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2} = f(jh), \quad j = 1, \ldots, m, \quad v_0 = v_{m+1} = 0.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0109 (2.22)</td><td>057</td><td>1.000 N|+0</td><td>\left|a_{i i}\right| \geq \sum_{j \neq i}\left|a_{i j} \right|, i=1, \ldots, n .</td><td>$\left|a_{i i}\right| \geq \sum_{j \neq i}\left|a_{i j} \right|, i=1, \ldots, n .$</td><td>$|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|, \ i = 1, \ldots, n.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0111 (2.23)</td><td>058</td><td>1.000 C|+3</td><td>\left|u_{k+1}\right|=\left|d_{k+1}-l_{k} c_{k}\right|= \left|d_{k+1}-\frac{a_{k} c_{k}}{u_{k}}\right| \geq \left|d_{k+1}\right|-\frac{\left|a_{k}\right|\left|c_{k} \right|}{\left|u_{k}\right|}>\left|d_{k+1}\right|- \left|a_{k}\right| .</td><td>$\left|u_{k+1}\right|=\left|d_{k+1}-l_{k} c_{k}\right|= \left|d_{k+1}-\frac{a_{k} c_{k}}{u_{k}}\right| \geq \left|d_{k+1}\right|-\frac{\left|a_{k}\right|\left|c_{k} \right|}{\left|u_{k}\right|}>\left|d_{k+1}\right|- \left|a_{k}\right| .$</td><td>$|u_{k+1}| = |d_{k+1} - l_k c_k| = |d_{k+1} - \frac{a_k c_k}{u_k}| \geq |d_{k+1}| - \frac{|a_k||c_k|}{|u_k|} > |d_{k+1}| - |a_k|.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0112 (2.24)</td><td>059</td><td>1.000 N|+0</td><td>R y^{\prime \prime}(x)=-F y(x), \quad y(0)=y(L)=0,</td><td>$R y^{\prime \prime}(x)=-F y(x), \quad y(0)=y(L)=0,$</td><td>$Ry''(x) = -Fy(x), \quad y(0) = y(L) = 0,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0113 (2.25)</td><td>059</td><td>1.000 N|+0</td><td>u^{\prime \prime}(t)=-K u(t), \quad u(0)=u(1)=0, \quad K:=\frac{F L^{2}}{R} .</td><td>$u^{\prime \prime}(t)=-K u(t), \quad u(0)=u(1)=0, \quad K:=\frac{F L^{2}}{R} .$</td><td>$u''(t) = -Ku(t), \quad u(0) = u(1) = 0, \quad K := \frac{FL^2}{R}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-a EQ0115	059	■ 1.000 ● N|+0	\frac{-v_{j-1}+2 v_{j}-v_{j+1}}{h^{2}}=K v_{j}, \quad j=1, \ldots, m, h=\frac{1}{m+1}, \quad v_{0}=v_{m+1}=0,	$\frac{-v_{j-1}+2 v_{j}-v_{j+1}}{h^{2}}=K v_{j}, \quad j=1, \ldots, m, h=\frac{1}{m+1}, \quad v_{0}=v_{m+1}=0,$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1,\ldots,m,\ h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$
lyche-numerical-linear-a EQ0116 (2.26)	060	■ 1.000 ● N|-1	\boldsymbol{T} \boldsymbol{v}=\lambda \boldsymbol{v}, \text { with } \boldsymbol{v}=\left[v_{1}, \ldots, v_{m} \right]^{T},	$\boldsymbol{T} \boldsymbol{v}=\lambda \boldsymbol{v}, \text { with } \boldsymbol{v}=\left[v_{1}, \ldots, v_{m}\right]^{T},$	$\boldsymbol{T}\boldsymbol{v}=\lambda\boldsymbol{v}, \text{ with } \boldsymbol{v}=[v_1,\ldots,v_m]^T,$
lyche-numerical-linear-a EQ0118	060	■ 1.000 ● N|+0	F=\frac{4 \sin ^{2}(\pi h / 2) R}{h^{2} L^{2}} .	$F=\frac{4 \sin ^{2}(\pi h / 2) R}{h^{2} L^{2}} .$	$F=\frac{4\sin^2(\pi h/2)R}{h^2L^2}.$
lyche-numerical-linear-a EQ0119 (2.28)	060	■ 1.000 ● N|-1	\boldsymbol{T}_{1}:=\operatorname{tridiag}(a, d, a)	$\boldsymbol{T}_{1}:=\operatorname{tridiag}(a, d, a)$	$\boldsymbol{T}_1:=\operatorname{tridiag}(a,d,a)$
lyche-numerical-linear-a EQ0120 (2.29)	060	■ 1.000 ● N|+0	\boldsymbol{S}=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^{m} \in \mathbb{R}^{m \times m} .	$\boldsymbol{S}=\left[\sin \frac{j k \pi}{m+1}\right]_{j, k=1}^{m} \in \mathbb{R}^{m \times m} .$	$\boldsymbol{S}=\left[\sin\frac{jk\pi}{m+1}\right]_{j,k=1}^{m} \in \mathbb{R}^{m\times m}.$
lyche-numerical-linear-a EQ0126	062	■ 1.000 ● W|+0	\sum_{k=0}^{m} \cos (2 k j \pi h)+i \sum_{k=0}^{m} \sin (2 k j \pi h)=\sum_{k=0}^{m} e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,	$\sum_{k=0}^{m} \cos (2 k j \pi h)+i \sum_{k=0}^{m} \sin (2 k j \pi h)=\sum_{k=0}^{m} e^{2 i k j \pi h}=\frac{e^{2 i(m+1) j \pi h}-1}{e^{2 i j \pi h}-1}=0,$	$\sum_{k=0}^{m}\cos(2kj\pi h)+i\sum_{k=0}^{m}\sin(2kj\pi h)=\sum_{k=0}^{m}e^{2ikj\pi h}=\frac{e^{2i(m+1)j\pi h}-1}{e^{2ij\pi h}-1}=0,$
lyche-numerical-linear-a EQ0130	063	■ 1.000 ● N|+1	\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{: 1}, \boldsymbol{A} \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \boldsymbol{b}_{: n}\right] .	$\boldsymbol{A} \boldsymbol{B}=\left[\boldsymbol{A} \boldsymbol{b}_{: 1}, \boldsymbol{A} \boldsymbol{b}_{: 2}, \ldots, \boldsymbol{A} \boldsymbol{b}_{: n}\right] .$	$\boldsymbol{AB}=\left[\boldsymbol{Ab}_{:1},\boldsymbol{Ab}_{:2},\ldots,\boldsymbol{Ab}_{:n}\right].$
lyche-numerical-linear-a EQ0131	063	■ 1.000 ● K|+0	\boldsymbol{A}=\boldsymbol{A} \boldsymbol{I}= \boldsymbol{A}\left[\boldsymbol{e}_{1}, \boldsymbol{e}_{2}, \ldots, \boldsymbol{e}_{p} \right]=\left[\boldsymbol{A} \boldsymbol{e}_{1}, \boldsymbol{A} \boldsymbol{e}_{2}, \ldots, \boldsymbol{A} \boldsymbol{e}_{p}\right]	$\boldsymbol{A}=\boldsymbol{A} \boldsymbol{I}= \boldsymbol{A}\left[\boldsymbol{e}_{1}, \boldsymbol{e}_{2}, \ldots, \boldsymbol{e}_{p} \right]=\left[\boldsymbol{A} \boldsymbol{e}_{1}, \boldsymbol{A} \boldsymbol{e}_{2}, \ldots, \boldsymbol{A} \boldsymbol{e}_{p}\right]$	$\boldsymbol{A}=\boldsymbol{AI}=\boldsymbol{A}\left[\boldsymbol{e}_1,\boldsymbol{e}_2,\ldots,\boldsymbol{e}_p\right]=\left[\boldsymbol{Ae}_1,\boldsymbol{Ae}_2,\ldots,\boldsymbol{Ae}_p\right]$
lyche-numerical-linear-a EQ0133	063	■ 1.000 ● N|+0	\boldsymbol{A} \boldsymbol{x}=x_{1} \boldsymbol{a}_{: 1}+x_{2} \boldsymbol{a}_{: 2}+\cdots+x_{p} \boldsymbol{a}_{: p} .	$\boldsymbol{A} \boldsymbol{x}=x_{1} \boldsymbol{a}_{: 1}+x_{2} \boldsymbol{a}_{: 2}+\cdots+x_{p} \boldsymbol{a}_{: p} .$	$\boldsymbol{Ax}=x_1\boldsymbol{a}_{:1}+x_2\boldsymbol{a}_{:2}+\cdots+x_p\boldsymbol{a}_{:p}.$
lyche-numerical-linear-a EQ0134	064	■ 1.000 ● K|+0	\boldsymbol{A}\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]=\left[\boldsymbol{A} \boldsymbol{B}_{1}, \boldsymbol{A} \boldsymbol{B}_{2} \right] .	$\boldsymbol{A}\left[\boldsymbol{B}_{1}, \boldsymbol{B}_{2}\right]=\left[\boldsymbol{A} \boldsymbol{B}_{1}, \boldsymbol{A} \boldsymbol{B}_{2} \right] .$	$\boldsymbol{A}\left[\boldsymbol{B}_1,\boldsymbol{B}_2\right]=\left[\boldsymbol{AB}_1,\boldsymbol{AB}_2\right].$
lyche-numerical-linear-a EQ0136	064	■ 1.000 ● K|+0	\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]\left[\begin{array}{l} \boldsymbol{B}_{1} \\ \boldsymbol{B}_{2} \end{array}\right]=\left[\boldsymbol{A}_{1} \boldsymbol{B}_{1}+\boldsymbol{A}_{2} \boldsymbol{B}_{2} \right] .	$\left[\boldsymbol{A}_{1}, \boldsymbol{A}_{2}\right]\left[\begin{array}{l} \boldsymbol{B}_{1} \\ \boldsymbol{B}_{2} \end{array}\right]=\left[\boldsymbol{A}_{1} \boldsymbol{B}_{1}+\boldsymbol{A}_{2} \boldsymbol{B}_{2} \right] .$	$\left[\boldsymbol{A}_1,\boldsymbol{A}_2\right]\begin{bmatrix}\boldsymbol{B}_1\\ \boldsymbol{B}_2\end{bmatrix}=\left[\boldsymbol{A}_1\boldsymbol{B}_1+\boldsymbol{A}_2\boldsymbol{B}_2\right].$
lyche-numerical-linear-a EQ0140	065	■ 1.000 ● N|+0	\boldsymbol{C}_{i j}=\sum_{k=1}^{s} \boldsymbol{A}_{i k} \boldsymbol{B}_{k j}, \quad i=1, \ldots, p, j=1, \ldots, q	$\boldsymbol{C}_{i j}=\sum_{k=1}^{s} \boldsymbol{A}_{i k} \boldsymbol{B}_{k j}, \quad i=1, \ldots, p, j=1, \ldots, q$	$\boldsymbol{C}_{ij}=\sum_{k=1}^{s}\boldsymbol{A}_{ik}\boldsymbol{B}_{kj}, \quad i=1,\ldots,p,\ j=1,\ldots,q$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0141	065	■ 1.000 ● N|+0	`\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p 1} & \cdots & \boldsymbol{A}_{p s} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1 q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s 1} & \cdots & \boldsymbol{B}_{s q} \end{array}\right]=\left[\begin{array}{ccc} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1 q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p 1} & \cdots & \boldsymbol{C}_{p q} \end{array}\right] .`	$\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p 1} & \cdots & \boldsymbol{A}_{p s} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1 q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s 1} & \cdots & \boldsymbol{B}_{s q} \end{array}\right]=\left[\begin{array}{ccc} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1 q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p 1} & \cdots & \boldsymbol{C}_{p q} \end{array}\right] .$	$\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p 1} & \cdots & \boldsymbol{A}_{p s} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1 q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s 1} & \cdots & \boldsymbol{B}_{s q} \end{array}\right]=\left[\begin{array}{ccc} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1 q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p 1} & \cdots & \boldsymbol{C}_{p q} \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0143 (2.33)	065	■ 1.000 ● K|+0	`\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \mathbf{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],`	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \mathbf{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \mathbf{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0146	066	■ 1.000 ● K|+0	`\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} .`	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} .$	$\boldsymbol{B}_{12}=\boldsymbol{C}=-\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} .$
lyche-numerical-linear-algebra_ EQ0148	066	■ 1.000 ● N|+0	`\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{k} & \boldsymbol{a}_{k} \\ \mathbf{0} & a_{k+1, k+1} \end{array}\right]`	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{k} & \boldsymbol{a}_{k} \\ \mathbf{0} & a_{k+1, k+1} \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{k} & \boldsymbol{a}_{k} \\ \mathbf{0} & a_{k+1, k+1} \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0149	066	■ 1.000 ● N|+0	`\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{k}^{-1} & \boldsymbol{c} \\ \mathbf{0} & a_{k+1, k+1}^{-1} \end{array}\right],`	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{k}^{-1} & \boldsymbol{c} \\ \mathbf{0} & a_{k+1, k+1}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{k}^{-1} & \boldsymbol{c} \\ \mathbf{0} & a_{k+1, k+1}^{-1} \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0150	067	■ 1.000 ● W|+0	`\max _{2 \leq j \leq n}\left|\mu_{j}\right| \leq \frac{3}{h^{2}} \max _{2 \leq i \leq n}\left|y_{i+1}-2 y_{i}+y_{i-1}\right| .`	$\max _{2 \leq j \leq n}\left\|\mu_{j}\right\| \leq \frac{3}{h^{2}} \max _{2 \leq i \leq n}\left\|y_{i+1}-2 y_{i}+y_{i-1}\right\| .$	$\max _{2 \leq j \leq n}\left\|\mu_{j}\right\| \leq \frac{3}{h^{2}} \max _{2 \leq i \leq n}\left\|y_{i+1}-2 y_{i}+y_{i-1}\right\| .$
lyche-numerical-linear-algebra_ EQ0151 (2.34)	067	■ 1.000 ● C|-8	`\left[\begin{array}{ccccc} 2 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 2 \end{array}\right]\left[\begin{array}{c} \mu_{1} \\ \mu_{2} \\ \vdots \\ \mu_{n} \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^{2}}\left[\begin{array}{c} y_{2}-y_{1}-h s_{1} \\ \delta^{2} y_{2} \\ \delta^{2} y_{3} \\ \vdots \\ \delta^{2} y_{n-1} \\ \delta^{2} y_{n} \\ h s_{n+1}-y_{n+1}+y_{n} \end{array}\right],`	$\left[\begin{array}{ccccc} 2 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 2 \end{array}\right]\left[\begin{array}{c} \mu_{1} \\ \mu_{2} \\ \vdots \\ \mu_{n} \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^{2}}\left[\begin{array}{c} y_{2}-y_{1}-h s_{1} \\ \delta^{2} y_{2} \\ \delta^{2} y_{3} \\ \vdots \\ \delta^{2} y_{n-1} \\ \delta^{2} y_{n} \\ h s_{n+1}-y_{n+1}+y_{n} \end{array}\right],$	$\left[\begin{array}{ccccc} 2 & 1 & & & \\ 1 & 4 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & 4 & 1 \\ & & & 1 & 2 \end{array}\right]\left[\begin{array}{c} \mu_{1} \\ \mu_{2} \\ \vdots \\ \mu_{n} \\ \mu_{n+1} \end{array}\right]=\frac{6}{h^{2}}\left[\begin{array}{c} y_{2}-y_{1}-h s_{1} \\ \delta^{2} y_{2} \\ \delta^{2} y_{3} \\ \vdots \\ \delta^{2} y_{n-1} \\ \delta^{2} y_{n} \\ h s_{n+1}-y_{n+1}+y_{n} \end{array}\right],$
lyche-numerical-linear-algebra_ EQ0152	068	■ 1.000 ● N|+0	`\int_{a}^{b}\left(g^{\prime \prime}(x)\right)^{2} d x \leq \int_{a}^{b}\left(h^{\prime \prime}(x)\right)^{2} d x`	$\int_{a}^{b}\left(g^{\prime \prime}(x)\right)^{2} d x \leq \int_{a}^{b}\left(h^{\prime \prime}(x)\right)^{2} d x$	$\int_{a}^{b}\left(g^{\prime \prime}(x)\right)^{2} d x \leq \int_{a}^{b}\left(h^{\prime \prime}(x)\right)^{2} d x$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0153 (2.35)	068	■ 1.000 ● N|+0	`\int_{a}^{b} g^{\prime \prime} e^{\prime \prime}=0, \quad e:=h-g .`	$\int_{a}^{b} g^{\prime \prime} e^{\prime \prime}=0, \quad e:=h-g .$	$\int_a^b g''e'' = 0, \quad e := h-g.$
lyche-numerical-linear-algebra_ EQ0154	068	■ 1.000 ● N|+0	`\int_{a}^{b} g^{\prime \prime \prime} e^{\prime}=\sum_{i=1}^{n} \int_{x_{i}}^{x_{i+1}} g^{\prime \prime} e^{\prime}=\sum_{i=1}^{n} v_{i} \int_{x_{i}}^{x_{i+1}} e^{\prime}=\sum_{i=1}^{n} v_{i} \left(e\left(x_{i+1}\right)-e\left(x_{i}\right)\right)=0 .`	$\int_{a}^{b} g^{\prime \prime \prime} e^{\prime}=\sum_{i=1}^{n} \int_{x_{i}}^{x_{i+1}} g^{\prime \prime \prime} e^{\prime}=\sum_{i=1}^{n} v_{i} \int_{x_{i}}^{x_{i+1}} e^{\prime}=\sum_{i=1}^{n} v_{i} \left(e\left(x_{i+1}\right)-e\left(x_{i}\right)\right)=0 .$	$\int_a^b g'''e' = \sum_{i=1}^n \int_{x_i}^{x_{i+1}} g'''e' = \sum_{i=1}^n v_i \int_{x_i}^{x_{i+1}} e' = \sum_{i=1}^n v_i\big(e(x_{i+1}) - e(x_i)\big) = 0.$
lyche-numerical-linear-algebra_ EQ0155	068	■ 1.000 ● W|+0	`\begin{aligned} \int_{a}^{b}\left(h^{\prime \prime} \right)^{2} & =\int_{a}^{b}\left(g^{\prime \prime}+e^{\prime \prime}\right)^{2} \\ & =\int_{a}^{b}\left(g^{\prime \prime}\right)^{2}+\int_{a}^{b}\left(e^{\prime \prime}\right)^{2}+2 \int_{a}^{b} g^{\prime \prime} e^{\prime \prime} \\ & =\int_{a}^{b}\left(g^{\prime \prime}\right)^{2}+\int_{a}^{b}\left(e^{\prime \prime} \right)^{2} \geq \int_{a}^{b}\left(g^{\prime \prime} \right)^{2} \end{aligned}`	$\begin{aligned} \int_{a}^{b}\left(h^{\prime \prime} \right)^{2} & =\int_{a}^{b}\left(g^{\prime \prime}+e^{\prime \prime}\right)^{2} \\ & =\int_{a}^{b}\left(g^{\prime \prime}\right)^{2}+\int_{a}^{b}\left(e^{\prime \prime}\right)^{2}+2 \int_{a}^{b} g^{\prime \prime} e^{\prime \prime} \\ & =\int_{a}^{b}\left(g^{\prime \prime}\right)^{2}+\int_{a}^{b}\left(e^{\prime \prime} \right)^{2} \geq \int_{a}^{b}\left(g^{\prime \prime} \right)^{2} \end{aligned}$	$\begin{aligned} \int_a^b (h'')^2 &= \int_a^b (g''+e'')^2 \\ &= \int_a^b (g'')^2 + \int_a^b (e'')^2 + 2\int_a^b g''e'' \\ &= \int_a^b (g'')^2 + \int_a^b (e'')^2 \geq \int_a^b (g'')^2 \end{aligned}$
lyche-numerical-linear-algebra_ EQ0157	071	■ 1.000 ● N|+0	`\delta^{2} f(x)=f^{\prime \prime}\left(\eta_{2}\right), \quad x-h<\eta_{2}<x+h .`	$\delta^{2} f(x)=f^{\prime \prime}\left(\eta_{2}\right), \quad x-h<\eta_{2}<x+h .$	$\delta^2 f(x) = f''(\eta_2), \quad x-h < \eta_2 < x+h.$
lyche-numerical-linear-algebra_ EQ0158	071	■ 1.000 ● N|+0	`\delta^{2} f(x)=f^{\prime \prime}(x)+\frac{h^{2}}{12} f^{(4)}\left(\eta_{4}\right), \quad x-h<\eta_{4}<x+h .`	$\delta^{2} f(x)=f^{\prime \prime}(x)+\frac{h^{2}}{12} f^{(4)}\left(\eta_{4}\right), \quad x-h<\eta_{4}<x+h .$	$\delta^2 f(x) = f''(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x-h < \eta_4 < x+h.$
lyche-numerical-linear-algebra_ EQ0159 (2.36)	071	■ 1.000 ● S|+0	`\begin{aligned} -u^{\prime \prime}(x)+r(x) u^{\prime}(x)+q(x) u(x) & =f(x), \text { for } x \in[a, b], \\ u(a) & =g_{0}, \quad u(b)=g_{1} . \end{aligned}`	$\begin{aligned} -u^{\prime \prime}(x)+r(x) u^{\prime}(x)+q(x) u(x) & =f(x), \text { for } x \in[a, b], \\ u(a) & =g_{0}, \quad u(b)=g_{1} . \end{aligned}$	$\begin{gathered} -u''(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a,b], \\ u(a) = g_0, \quad u(b) = g_1. \end{gathered}$
lyche-numerical-linear-algebra_ EQ0160 (2.37)	071	■ 1.000 ● N|+0	`\frac{-v_{j-1}+2 v_{j}-v_{j+1}}{h^{2}}+r\left(x_{j}\right) \frac{v_{j+1}-v_{j-1}}{2 h}+q\left(x_{j}\right) v_{j}=f\left(x_{j}\right), \quad j=1, \ldots, m,`	$\frac{-v_{j-1}+2 v_{j}-v_{j+1}}{h^{2}}+r\left(x_{j}\right) \frac{v_{j+1}-v_{j-1}}{2 h}+q\left(x_{j}\right) v_{j}=f\left(x_{j}\right), \quad j=1, \ldots, m,$	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j)\frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \ldots, m,$
lyche-numerical-linear-algebra_ EQ0162	072	■ 1.000 ● W|+0	`b_{j}= \begin{cases}h^{2} f\left(x_{1}\right)-a_{1} g_{0}, & \text { if } j=1, \\ h^{2} f\left(x_{j}\right), & \text { if } 2 \leq j \leq m-1, \\ h^{2} f\left(x_{m} \right)-c_{m} g_{1}, & \text { if } j=m .\end{cases}`	$b_{j}= \begin{cases}h^{2} f\left(x_{1}\right)-a_{1} g_{0}, & \text { if } j=1, \\ h^{2} f\left(x_{j}\right), & \text { if } 2 \leq j \leq m-1, \\ h^{2} f\left(x_{m} \right)-c_{m} g_{1}, & \text { if } j=m .\end{cases}$	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m-1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$
lyche-numerical-linear-algebra_ EQ0163	072	■ 1.000 ● N|+0	`F=\frac{4 \sin ^{2}(\pi h / 2) R}{h^{2} L^{2}}=\frac{\pi^{2} R}{L^{2}}+O\left(h^{2}\right) .`	$F=\frac{4 \sin ^{2}(\pi h / 2) R}{h^{2} L^{2}}=\frac{\pi^{2} R}{L^{2}}+O\left(h^{2}\right) .$	$F = \frac{4\sin^2(\pi h/2)R}{h^2L^2} = \frac{\pi^2 R}{L^2} + O(h^2).$
lyche-numerical-linear-algebra_ EQ0165 (2.39)	073	■ 1.000 ● N|+0	`s_{i, j}=s_{j, i}=\frac{1}{m+1} j(m+1-i), \quad 1 \leq j \leq i \leq m .`	$s_{i, j}=s_{j, i}=\frac{1}{m+1} j(m+1-i), \quad 1 \leq j \leq i \leq m .$	$s_{i,j} = s_{j,i} = \frac{1}{m+1} j\big(m+1-i\big), \quad 1 \leq j \leq i \leq m.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0167	073	■ 1.000 ● N|+0	`\boldsymbol{A} \boldsymbol{X}=\boldsymbol{B} \quad \Longleftrightarrow \quad \boldsymbol{A} \boldsymbol{x}_{: j}=\boldsymbol{b}_{: j}, j=1, \ldots, p .`	$\boldsymbol{A} \boldsymbol{X}=\boldsymbol{B} \quad \Longleftrightarrow \quad \boldsymbol{A} \boldsymbol{x}_{: j}=\boldsymbol{b}_{: j}, j=1, \ldots, p .$	$\boldsymbol{A} \boldsymbol{X}=\boldsymbol{B} \quad \Longleftrightarrow \quad \boldsymbol{A} \boldsymbol{x}_{: j}=\boldsymbol{b}_{: j},\ j=1, \ldots, p .$
lyche-numerical-linear-algebra_ EQ0173	076	■ 1.000 ● S|+0	`\begin{aligned} & x_{3}=b_{3}^{(3)} / a_{33}^{(3)} \\ & x_{2}=\left(b_{2}^{(2)}-a_{23}^{(2)} x_{3}\right) / a_{22}^{(2)} \\ & x_{1}=\left(b_{1}^{(1)}-a_{12}^{(1)} x_{2}-a_{13}^{(1)} x_{3}\right) / a_{11}^{(1)} . \end{aligned}`	$\begin{aligned} & x_{3}=b_{3}^{(3)} / a_{33}^{(3)} \\ & x_{2}=\left(b_{2}^{(2)}-a_{23}^{(2)} x_{3}\right) / a_{22}^{(2)} \\ & x_{1}=\left(b_{1}^{(1)}-a_{12}^{(1)} x_{2}-a_{13}^{(1)} x_{3}\right) / a_{11}^{(1)} . \end{aligned}$	$\begin{aligned} & x_{3}=b_{3}^{(3)} / a_{33}^{(3)} \\ & x_{2}=(b_{2}^{(2)}-a_{23}^{(2)} x_{3}) / a_{22}^{(2)} \\ & x_{1}=(b_{1}^{(1)}-a_{12}^{(1)} x_{2}-a_{13}^{(1)} x_{3}) / a_{11}^{(1)} . \end{aligned}$
lyche-numerical-linear-algebra_ EQ0177 (3.2)	077	■ 1.000 ● N|+0	`\text { for } j=k: n`	$\text { for } j=k: n$	for $j=k: n$
lyche-numerical-linear-algebra_ EQ0178	077	■ 1.000 ● W|+0	`a_{i j}^{(k+1)}=a_{i j}^{(k)}-l_{i k}^{(k)} a_{k j}^{(k)}`	$a_{i j}^{(k+1)}=a_{i j}^{(k)}-l_{i k}^{(k)} a_{k j}^{(k)}$	$a_{i j}^{(k+1)}=a_{i j}^{(k)}-l_{i k}^{(k)} a_{k j}^{(k)}$
lyche-numerical-linear-algebra_ EQ0179	078	■ 1.000 ● N|+0	`\boldsymbol{A}_{[k]}:=\boldsymbol{A}(1: k, 1: k)=\left[ \begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]`	$\boldsymbol{A}_{[k]}:=\boldsymbol{A}(1: k, 1: k)=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\boldsymbol{A}(1: k, 1: k)=\left[\begin{array}{cc} a_{11} \cdots & a_{k 1} \\ \vdots & \vdots \\ a_{k 1} \cdots & a_{k k} \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0180	078	■ 1.000 ● K|+0	`b_{i, j}=a_{r_{i}, r_{j}}, \quad i, j=1, \ldots, k .`	$b_{i, j}=a_{r_{i}, r_{j}}, \quad i, j=1, \ldots, k .$	$b_{i, j}=a_{r_{i}, r_{j}}, \quad i, j=1, \ldots, k .$
lyche-numerical-linear-algebra_ EQ0182	078	■ 1.000 ● K|+0	`[1],\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \boldsymbol{A} .`	$[1],\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right], \boldsymbol{A} .$	$[1],\ \left[\begin{smallmatrix} 1 & 2 \\ 4 & 5 \end{smallmatrix}\right],\ \boldsymbol{A} .$
lyche-numerical-linear-algebra_ EQ0183 (3.3)	078	■ 1.000 ● W|+0	`\operatorname{det}\left(\boldsymbol{A}_{[k]} \right)=a_{11}^{(1)} a_{22}^{(2)} \cdots a_{k k}^{(k)}, \quad k=1, \ldots, n .`	$\operatorname{det}\left(\boldsymbol{A}_{[k]} \right)=a_{11}^{(1)} a_{22}^{(2)} \cdots a_{k k}^{(k)}, \quad k=1, \ldots, n .$	$\operatorname{det}(\boldsymbol{A}_{[k]})=a_{11}^{(1)} a_{22}^{(2)} \cdots a_{k k}^{(k)}, \quad k=1, \ldots, n .$
lyche-numerical-linear-algebra_ EQ0187 (3.6)	079	■ 1.000 ● W|+0	`(\boldsymbol{L} \boldsymbol{U})_{i j}=\sum_{k=1}^{j-1} l_{i k}^{(k)} a_{k j}^{(k)}+l_{i j} a_{j j}^{(j)}= \sum_{k=1}^{j-1}\left(a_{i j}^{(k)}-a_{i j}^{(k+1)} \right)+a_{i j}^{(j)}=a_{i j} .`	$(\boldsymbol{L} \boldsymbol{U})_{i j}=\sum_{k=1}^{j-1} l_{i k}^{(k)} a_{k j}^{(k)}+l_{i j} a_{j j}^{(j)}= \sum_{k=1}^{j-1}\left(a_{i j}^{(k)}-a_{i j}^{(k+1)} \right)+a_{i j}^{(j)}=a_{i j} .$	$(\boldsymbol{L} \boldsymbol{U})_{i j}=\sum_{k=1}^{j-1} l_{i k}^{(k)} a_{k j}^{(k)}+l_{i j} a_{j j}^{(j)}= \sum_{k=1}^{j-1}\left(a_{i j}^{(k)}-a_{i j}^{(k+1)} \right)+a_{i j}^{(j)}=a_{i j} .$
lyche-numerical-linear-algebra_ EQ0188	080	■ 1.000 ● W|+0	`\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array}\right] .`	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array}\right] .$	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \\ x_{3} \end{array}\right]=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array}\right] .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0191</td><td>080</td><td>■ 1.000 ● W|+0</td><td><code>\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \ \ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]</code></td><td>$\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]$</td><td>$\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0192</td><td>081</td><td>■ 1.000 ● W|+0</td><td><code>\left[\begin{array}{cccc} a_{k, k} & 0 & \cdots & 0 \ \ a_{k+1, k} & a_{k+1, k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k} & & \cdots & a_{n \times n} \end{array}\right]\left[\begin{array}{c} x_{k} \\ x_{k+1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{k} \\ b_{k+1} \\ \vdots \\ b_{n} \end{array}\right] .</code></td><td>$\left[\begin{array}{cccc} a_{k, k} & 0 & \cdots & 0 \\ a_{k+1, k} & a_{k+1, k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k} & & \cdots & a_{n \times n} \end{array}\right]\left[\begin{array}{c} x_{k} \\ x_{k+1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{k} \\ b_{k+1} \\ \vdots \\ b_{n} \end{array}\right] .$</td><td>$\left[\begin{array}{cccc} a_{k, k} & 0 & \cdots & 0 \\ a_{k+1, k} & a_{k+1, k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k} & & \cdots & a_{n \times n} \end{array}\right]\left[\begin{array}{c} x_{k} \\ x_{k+1} \\ \vdots \\ x_{n} \end{array}\right]=\left[\begin{array}{c} b_{k} \\ b_{k+1} \\ \vdots \\ b_{n} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0194</td><td>082</td><td>■ 1.000 ● N|+0</td><td><code>b((k+1): n)=b((k+1): n)-x(k) * A((k+1): n, k) .</code></td><td>$b((k+1): n)=b((k+1): n)-x(k) * A((k+1): n, k) .$</td><td>$b((k+1): n)=b((k+1): n)-x(k) * A((k+1): n, k) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0195 (3.9)</td><td>082</td><td>■ 1.000 ● U|+10</td><td><code>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.3 cforwardsolve} \[N_{L U}:= \frac{2}{3} n^{3}-\frac{1}{2} n^{2}-\frac{1}{6} n \] \end{figure}</code></td><td>(not rendered)</td><td>$N_{LU} := \frac{2}{3} n^{3}-\frac{1}{2} n^{2}-\frac{1}{6} n$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0196</td><td>083</td><td>■ 1.000 ● N|+0</td><td><code>M+D+A+S=\frac{2}{3} n(n-1)\left(n-\frac{1}{2}\right)+ \frac{1}{2} n(n-1)=N_{L U}</code></td><td>$M+D+A+S=\frac{2}{3} n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2} n(n-1)=N_{L U}$</td><td>$M+D+A+S=\frac{2}{3} n(n-1)(n-\frac{1}{2})+\frac{1}{2} n(n-1)=N_{L U}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0197</td><td>083</td><td>■ 1.000 ● N|+0</td><td><code>M+S=2 \sum_{k=1}^{n-1}(n-k)^{2} \approx 2 \int_{1}^{n-1}(n-k)^{2} d k \approx 2 \int_{0}^{n}(n-k)^{2} d k=\frac{2}{3} n^{3}</code></td><td>$M+S=2 \sum_{k=1}^{n-1}(n-k)^{2} \approx 2 \int_{1}^{n-1}(n-k)^{2} d k \approx 2 \int_{0}^{n}(n-k)^{2} d k=\frac{2}{3} n^{3}$</td><td>$M+S=2 \sum_{k=1}^{n-1}(n-k)^{2} \approx 2 \int_{1}^{n-1}(n-k)^{2} d k \approx 2 \int_{0}^{n}(n-k)^{2} d k=\frac{2}{3} n^{3}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0198</td><td>083</td><td>■ 1.000 ● N|+1</td><td><code>N_{S}=2 \sum_{k=1}^{n}(2 k-1) \approx 2 \int_{1}^{n}(2 k-1) d k \approx 4 \int_{0}^{n} k d k=2 n^{2}</code></td><td>$N_{S}=2 \sum_{k=1}^{n}(2 k-1) \approx 2 \int_{1}^{n}(2 k-1) d k \approx 4 \int_{0}^{n} k d k=2 n^{2}$</td><td>$N_{S}=2 \sum_{k=1}^{n}(2 k-1) \approx 2 \int_{1}^{n}(2 k-1) d k \approx 4 \int_{0}^{n} k d k=2 n^{2}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0199</td><td>084</td><td>■ 1.000 ● N|+0</td><td><code>\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_{1}}, \boldsymbol{e}_{i_{2}}, \ldots, \boldsymbol{e}_{i_{n}}\right] \in \mathbb{R}^{n \times n},</code></td><td>$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_{1}}, \boldsymbol{e}_{i_{2}}, \ldots, \boldsymbol{e}_{i_{n}}\right] \in \mathbb{R}^{n \times n},$</td><td>$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_{1}}, \boldsymbol{e}_{i_{2}}, \ldots, \boldsymbol{e}_{i_{n}}\right] \in \mathbb{R}^{n \times n},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0200 (3.10)</td><td>084</td><td>■ 1.000 ● K|+0</td><td><code>\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^{T} \boldsymbol{A}= \boldsymbol{A}(\boldsymbol{p},:) .</code></td><td>$\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^{T} \boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:) .$</td><td>$\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^{T} \boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:) .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0206</td><td>086</td><td>■ 1.000 ● N|-1</td><td><code>a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}=a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)} \text { for } k<\min (i, j), \text { and } a_{p_{i}, j}^{(k)} a_{p_{j}, j}^{(j)}=a_{p_{i}, j}^{(j)} \text { for } i>j .</code></td><td>$a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}=a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)} \text { for } k<\min (i, j), \text { and } a_{p_{i}, j}^{(k)} a_{p_{j}, j}^{(j)}=a_{p_{i}, j}^{(j)} \text { for } i>j .$</td><td>$a_{p_{i}, k}^{(k)} a_{p_{k}, j}^{(k)}=a_{p_{i}, j}^{(k)}-a_{p_{i}, j}^{(k+1)} \text { for } k<\min (i, j), \text { and } a_{p_{i}, j}^{(k)} a_{p_{j}, j}^{(j)}=a_{p_{i}, j}^{(j)} \text { for } i>j .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0209</td><td>087</td><td>■ 1.000 ● N|+0</td><td><code>\left|a_{r_{k}, k}^{(k)}\right|:=\max \left\{\left|a_{i, k}^{(k)}\right|: k \leq i \leq n\right\}</code></td><td>$\left|a_{r_{k}, k}^{(k)}\right|:=\max \left\{\left|a_{i, k}^{(k)}\right|: k \leq i \leq n\right\}$</td><td>$|a_{r_{k}, k}^{(k)}|:=\max \{|a_{i, k}^{(k)}|: k \leq i \leq n\}$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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K|+0</td><td>\begin{aligned} x_{1}+x_{2} & =3 \\ 10^{-4} x_{1}+2 x_{2} & =4 \end{aligned}</td><td>$\begin{aligned} x_{1}+x_{2} & =3 \\ 10^{-4} x_{1}+2 x_{2} & =4 \end{aligned}$</td><td>$\begin{aligned} x_{1}+x_{2} & =3 \\ 10^{-4} x_{1}+2 x_{2} & =4 \end{aligned}$</td></tr>
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N|+0</td><td>\begin{aligned} x_{1}+x_{2} & =3 \\ \left(2-10^{-4}\right) x_{2} & =4-3 \times 10^{-4} \end{aligned}</td><td>$\begin{aligned} x_{1}+x_{2} & =3 \\ \left(2-10^{-4}\right) x_{2} & =4-3 \times 10^{-4} \end{aligned}$</td><td>$\begin{aligned} x_{1}+x_{2} & =3 \\ (2-10^{-4}) x_{2} & =4-3 \times 10^{-4} \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00217</td><td>088</td><td>1.000
W|-1</td><td>\frac{\left|a_{r_{k}, k}^{(k)}\right|}{s_{k}}:=\max \left\{\frac{\left|a_{i, k}^{(k)}\right|}{s_{k}}: k \leq i \leq n\right\}, \quad s_{k}:=\max _{1 \leq j \leq n}\left|a_{k j}\right| .</td><td>$\frac{\left|a_{r_{k}, k}^{(k)}\right|}{s_{k}}:=\max \left\{\frac{\left|a_{i, k}^{(k)}\right|}{s_{k}}: k \leq i \leq n\right\}, \quad s_{k}:=\max _{1 \leq j \leq n}\left|a_{k j}\right| .$</td><td>$\frac{|a_{r_{k}, k}^{(k)}|}{s_{k}}:=\max \left\{\frac{|a_{i, k}^{(k)}|}{s_{k}}: k \leq i \leq n\right\}, \quad s_{k}:=\max _{1 \leq j \leq n}|a_{k j}| .$</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ00219</td><td>088</td><td>1.000
W|+0</td><td>\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n, 1} & \cdots & l_{n, n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1, n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n, n} \end{array}\right] .</td><td>$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n, 1} & \cdots & l_{n, n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1, n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n, n} \end{array}\right] .$</td><td>$\boldsymbol{L}=\left[\begin{array}{ccc} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n, 1} & \cdots & l_{n, n} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{ccc} u_{1,1} & \cdots & u_{1, n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n, n} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00221 (3.15)</td><td>089</td><td>1.000
N|+0</td><td>u_{1}=a, \quad u_{2}=b, \quad a l_{1}=c, \quad b l_{1}+u_{3}=d .</td><td>$u_{1}=a, \quad u_{2}=b, \quad a l_{1}=c, \quad b l_{1}+u_{3}=d .$</td><td>$u_{1}=a, \quad u_{2}=b, \quad a l_{1}=c, \quad b l_{1}+u_{3}=d .$</td></tr>
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N|+0</td><td>\boldsymbol{A}_{1}:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_{2}:=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{3}:=\left[\begin{array}{ll} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{4}:=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right] .</td><td>$\boldsymbol{A}_{1}:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_{2}:=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{3}:=\left[\begin{array}{ll} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{4}:=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right] .$</td><td>$\boldsymbol{A}_{1}:=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right], \quad \boldsymbol{A}_{2}:=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{3}:=\left[\begin{array}{ll} 0 & 1 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{4}:=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00223</td><td>089</td><td>1.000
K|+0</td><td>\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]</td><td>$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$</td><td>$\boldsymbol{A}_{[k]}:=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k 1} \\ \vdots & & \vdots \\ a_{k 1} & \cdots & a_{k k} \end{array}\right]$</td></tr>
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Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-a EQ0225 (3.17)	090	■ 1.000 ● W|+0	`\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_{n} \\ \boldsymbol{r}_{n}^{T} & a_{n n} \end{array}\right]= \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_{n}^{T} & 1 \end{array} \right]\left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_{n} \\ 0 & u_{n n} \end{array} \right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1} \boldsymbol{u}_{n} \\ \boldsymbol{l}_{n}^{T} \boldsymbol{U}_{n-1} & \boldsymbol{l}_{n}^{T} \boldsymbol{u}_{n}+u_{n n} \end{array}\right],`	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_{n} \\ \boldsymbol{r}_{n}^{T} & a_{n n} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_{n}^{T} & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_{n} \\ 0 & u_{n n} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1} \boldsymbol{u}_{n} \\ \boldsymbol{l}_{n}^{T} \boldsymbol{U}_{n-1} & \boldsymbol{l}_{n}^{T} \boldsymbol{u}_{n}+u_{n n} \end{array}\right],$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_{n} \\ \boldsymbol{r}_{n}^{T} & a_{n n} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_{n}^{T} & 1 \end{array}\right]\left[\begin{array}{cc} \boldsymbol{U}_{n-1} & \boldsymbol{u}_{n} \\ 0 & u_{n n} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1} & \boldsymbol{L}_{n-1} \boldsymbol{u}_{n} \\ \boldsymbol{l}_{n}^{T} \boldsymbol{U}_{n-1} & \boldsymbol{l}_{n}^{T} \boldsymbol{u}_{n}+u_{n n} \end{array}\right],$
lyche-numerical-linear-a EQ0226 (3.18)	090	■ 1.000 ● S|+0	`\boldsymbol{U}_{n-1}^{T} \boldsymbol{l}_{n}= \boldsymbol{r}_{n}, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_{n}=\boldsymbol{c}_{n}, \quad u_{n n}=a_{n n}-\boldsymbol{l}_{n}^{T} \boldsymbol{u}_{n} .`	$\boldsymbol{U}_{n-1}^{T} \boldsymbol{l}_{n}=\boldsymbol{r}_{n}, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_{n}=\boldsymbol{c}_{n}, \quad u_{n n}=a_{n n}-\boldsymbol{l}_{n}^{T} \boldsymbol{u}_{n} .$	$\boldsymbol{U}_{n-1}^{T} \boldsymbol{l}_{n}=\boldsymbol{r}_{n}, \quad \boldsymbol{L}_{n-1} \boldsymbol{u}_{n}=\boldsymbol{c}_{n}, \quad u_{n n}=a_{n n}-\boldsymbol{l}_{n}^{T} \boldsymbol{u}_{n} .$
lyche-numerical-linear-a EQ0227	091	■ 1.000 ● W|+0	`\boldsymbol{U}_{k-1}^{T} \boldsymbol{l}_{k}= \boldsymbol{r}_{k}, \quad\left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \mathbf{0} \\ \boldsymbol{l}_{k}^{T} & 1 \end{array}\right] \left[\begin{array}{c} \boldsymbol{u}_{k} \\ u_{k k} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{c}_{k} \\ a_{k k} \end{array}\right] .`	$\boldsymbol{U}_{k-1}^{T} \boldsymbol{l}_{k}=\boldsymbol{r}_{k}, \quad\left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \mathbf{0} \\ \boldsymbol{l}_{k}^{T} & 1 \end{array}\right]\left[\begin{array}{c} \boldsymbol{u}_{k} \\ u_{k k} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{c}_{k} \\ a_{k k} \end{array}\right] .$	$\boldsymbol{U}_{k-1}^{T} \boldsymbol{l}_{k}=\boldsymbol{r}_{k}, \quad\left[\begin{array}{cc} \boldsymbol{L}_{k-1} & \mathbf{0} \\ \boldsymbol{l}_{k}^{T} & 1 \end{array}\right]\left[\begin{array}{c} \boldsymbol{u}_{k} \\ u_{k k} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{c}_{k} \\ a_{k k} \end{array}\right].$
lyche-numerical-linear-a EQ0230	092	■ 1.000 ● N|+0	`\boldsymbol{A}_{\{k\}}:=\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k 1} & \cdots & \boldsymbol{A}_{k k} \end{array}\right]`	$\boldsymbol{A}_{\{k\}}:=\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k 1} & \cdots & \boldsymbol{A}_{k k} \end{array}\right]$	$\boldsymbol{A}_{\{k\}}:=\left[\begin{array}{ccc} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1 k} \\ \vdots & & \vdots \\ \boldsymbol{A}_{k 1} & \cdots & \boldsymbol{A}_{k k} \end{array}\right]$
lyche-numerical-linear-a EQ0232 (3.22)	093	■ 1.000 ● U|+54	`\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.5 cbacksolve} \[ \boldsymbol{A}((k+1): n,(k+1): n) \boldsymbol{b}_{k}((k+1): n)=- \boldsymbol{A}((k+1): n, k) b_{k}(k), \quad k=1, \ldots, n-1 . \] \end{figure}`	*(not rendered)*	$\boldsymbol{A}((k+1):n,(k+1):n) \boldsymbol{b}_{k}((k+1):n)=-\boldsymbol{A}((k+1):n, k) b_{k}(k), \quad k=1, \ldots, n-1.$
lyche-numerical-linear-a EQ0233 (3.23)	094	■ 1.000 ● K|+0	`\boldsymbol{A}(1: k, 1: k) \boldsymbol{b}_{k}(1: k)= \boldsymbol{I}(1: k, k), \quad k=n, n-1, \ldots, 1,`	$\boldsymbol{A}(1: k, 1: k) \boldsymbol{b}_{k}(1: k)=\boldsymbol{I}(1: k, k), \quad k=n, n-1, \ldots, 1,$	$\boldsymbol{A}(1:k, 1:k) \boldsymbol{b}_{k}(1:k)=\boldsymbol{I}(1:k, k), \quad k=n, n-1, \ldots, 1,$
lyche-numerical-linear-a EQ0235 (3.28)	095	■ 1.000 ● K|+0	`\boldsymbol{C}:=\boldsymbol{U}+\boldsymbol{v} \boldsymbol{e}_{1}^{T},`	$\boldsymbol{C}:=\boldsymbol{U}+\boldsymbol{v} \boldsymbol{e}_{1}^{T},$	$\boldsymbol{C}:=\boldsymbol{U}+\boldsymbol{v} \boldsymbol{e}_{1}^{T},$
lyche-numerical-linear-a EQ0236 (3.29)	095	■ 1.000 ● N|+0	`\boldsymbol{P}:=\boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1, n},`	$\boldsymbol{P}:=\boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1, n},$	$\boldsymbol{P}:=\boldsymbol{I}_{1,2} \boldsymbol{I}_{2,3} \cdots \boldsymbol{I}_{n-1, n},$
lyche-numerical-linear-a EQ0237 (3.30)	096	■ 1.000 ● K|+0	`\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{w} \boldsymbol{e}_{1}^{T} .`	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{w} \boldsymbol{e}_{1}^{T} .$	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{w} \boldsymbol{e}_{1}^{T}.$
lyche-numerical-linear-a EQ0238	096	■ 1.000 ● N|+0	`\operatorname{det}\left(\boldsymbol{A}_{[k]} \right)=u_{11} u_{22} \cdots u_{k k} \text { for } k=1, \ldots, n .`	$\operatorname{det}\left(\boldsymbol{A}_{[k]}\right)=u_{11} u_{22} \cdots u_{k k} \text { for } k=1, \ldots, n .$	$\operatorname{det}(\boldsymbol{A}_{[k]})=u_{11} u_{22} \cdots u_{k k} \text { for } k=1, \ldots, n.$
lyche-numerical-linear-a EQ0239 (3.31)	097	■ 1.000 ● N|+0	`u_{11}=a_{11}, \quad u_{k k}=\frac{\operatorname{det} \left(\boldsymbol{A}_{[k]}\right)}{\operatorname{det} \left(\boldsymbol{A}_{[k-1]}\right)}, \text { for } k=2, \ldots, n .`	$u_{11}=a_{11}, \quad u_{k k}=\frac{\operatorname{det}\left(\boldsymbol{A}_{[k]}\right)}{\operatorname{det}\left(\boldsymbol{A}_{[k-1]}\right)}, \text { for } k=2, \ldots, n .$	$u_{11}=a_{11}, \quad u_{k k}=\frac{\operatorname{det}(\boldsymbol{A}_{[k]})}{\operatorname{det}(\boldsymbol{A}_{[k-1]})}, \text { for } k=2, \ldots, n.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0242	098	1.000 N|+0	\boldsymbol{B}:=\left[\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \ldots, \boldsymbol{a}_{n}, \boldsymbol{b}\right] \in \mathbb{R}^{n \times n} .	$\boldsymbol{B}:=\left[\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \ldots, \boldsymbol{a}_{n}, \boldsymbol{b}\right] \in \mathbb{R}^{n \times n} .$	$\boldsymbol{B} := [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_{p-1}, \boldsymbol{a}_{p+1}, \ldots, \boldsymbol{a}_n, \boldsymbol{b}] \in \mathbb{R}^{n\times n}.$
lyche-numerical-linear-algebra EQ0243	098	1.000 N|+0	A:=\left[\begin{array}{rr} -3 & -2 \\\\ 4 & 2 \end{array} \right] .	$A:=\left[\begin{array}{rr} -3 & -2 \\ 4 & 2 \end{array} \right] .$	$\boldsymbol{A} := \begin{bmatrix} -3 & -2 \\ 4 & 2 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0244	098	1.000 N|+0	\boldsymbol{P}:=\left[\boldsymbol{e}_{n}, \boldsymbol{e}_{n-1}, \ldots, \boldsymbol{e}_{1}\right] .	$\boldsymbol{P}:=\left[\boldsymbol{e}_{n}, \boldsymbol{e}_{n-1}, \ldots, \boldsymbol{e}_{1}\right] .$	$\boldsymbol{P} := [\boldsymbol{e}_n, \boldsymbol{e}_{n-1}, \ldots, \boldsymbol{e}_1].$
lyche-numerical-linear-algebra EQ0246 (4.1)	101	1.000 N|+0	d_{1}=a . \quad a l_{1}=b, \quad d_{2}=d-a\left|l_{1} \right|^{2} .	$d_{1}=a . \quad a l_{1}=b, \quad d_{2}=d-a\left\lvert l_{1} \right\rvert^{2} .$	$d_1 = a. \quad al_1 = b, \quad d_2 = d - a\lvert l_1\rvert^2.$
lyche-numerical-linear-algebra EQ0251	102	1.000 N|+0	f(\boldsymbol{x})=\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^{n} \sum_{j=1}^{n} a_{i j} \bar{x}_{i} x_{j}	$f(\boldsymbol{x})=\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^{n} \sum_{j=1}^{n} a_{i j} \bar{x}_{i} x_{j}$	$f(\boldsymbol{x}) = \boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}\overline{x}_i x_j$
lyche-numerical-linear-algebra EQ0255 (4.5)	104	1.000 N|+0	\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})= \boldsymbol{X}^{*} \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X}:=\left[\boldsymbol{e}_{r_{1}}, \ldots, \boldsymbol{e}_{r_{k}}\right] \in \mathbb{C}^{n \times k} .	$\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})= \boldsymbol{X}^{*} \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X}:=\left[\boldsymbol{e}_{r_{1}}, \ldots, \boldsymbol{e}_{r_{k}}\right] \in \mathbb{C}^{n \times k} .$	$\boldsymbol{A}(\boldsymbol{r},\boldsymbol{r}) = \boldsymbol{X}^*\boldsymbol{A}\boldsymbol{X}, \quad \boldsymbol{X} := [\boldsymbol{e}_{r_1}, \ldots, \boldsymbol{e}_{r_k}] \in \mathbb{C}^{n\times k}.$
lyche-numerical-linear-algebra EQ0256 (4.6)	104	1.000 W|+1	\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}= \boldsymbol{y}^{*} \boldsymbol{X}^{*} \boldsymbol{A} \boldsymbol{X} \boldsymbol{y}=\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^{k}, \quad \boldsymbol{x}:= \boldsymbol{X} \boldsymbol{y} .	$\boldsymbol{y}^{*} \boldsymbol{B} \boldsymbol{y}= \boldsymbol{y}^{*} \boldsymbol{X}^{*} \boldsymbol{A} \boldsymbol{X} \boldsymbol{y}=\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x} \geq 0, \quad \boldsymbol{y} \in \mathbb{C}^{k}, \quad \boldsymbol{x}:= \boldsymbol{X} \boldsymbol{y} .$	$\boldsymbol{y}^*\boldsymbol{B}\boldsymbol{y} = \boldsymbol{y}^*\boldsymbol{X}^*\boldsymbol{A}\boldsymbol{X}\boldsymbol{y} = \boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x} \ge 0, \quad \boldsymbol{y} \in \mathbb{C}^k, \quad \boldsymbol{x} := \boldsymbol{X}\boldsymbol{y}.$
lyche-numerical-linear-algebra EQ0257 (0)	105	1.000 W|-1	\left[\begin{array}{cc} 2 & -1 \\\\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1 & 0 \\\\ -\frac{1}{2} & 1 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\\\ 0 & \frac{3}{2} \end{array}\right]\left[\begin{array}{cc} 1 & -\frac{1}{2} \\\\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} \sqrt{2} & 0 \\\\ -1 / \sqrt{2} & \sqrt{3 / 2} \end{array}\right]\left[\begin{array}{cc} \sqrt{2} & -1 / \sqrt{2} \\\\ 0 & \sqrt{3 / 2} \end{array}\right] .	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]=\left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array}\right]\left[\begin{array}{ll} 2 & 0 \\ 0 & \frac{3}{2} \end{array}\right]\left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} \sqrt{2} & 0 \\ -1 / \sqrt{2} & \sqrt{3 / 2} \end{array}\right]\left[\begin{array}{cc} \sqrt{2} & -1 / \sqrt{2} \\ 0 & \sqrt{3 / 2} \end{array}\right] .$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & \frac{3}{2} \end{bmatrix}\begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 0 \\ -1/\sqrt{2} & \sqrt{3/2} \end{bmatrix}\begin{bmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0258 (4.7)	106	1.000 U|+54	\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 4.2 bandcholesky} \\[0<(\leq)\left(\alpha \boldsymbol{e}_{i}+\beta \boldsymbol{e}_{j}\right)^{*} \boldsymbol{A}\left(\alpha \boldsymbol{e}_{i}+\beta \boldsymbol{e}_{j} \right)=|\alpha|^{2} a_{i i}+|\beta|^{2} a_{j j}+2 \operatorname{Re}\left(\bar{\alpha} \beta a_{i j}\right) . \\] \end{figure}	*(not rendered)*	$0 < (\le)(\alpha\boldsymbol{e}_i + \beta\boldsymbol{e}_j)^*\boldsymbol{A}(\alpha\boldsymbol{e}_i + \beta\boldsymbol{e}_j) = \lvert\alpha\rvert^2 a_{ii} + \lvert\beta\rvert^2 a_{jj} + 2\mathrm{Re}\,(\overline{\alpha}\beta a_{ij}).$
lyche-numerical-linear-algebra EQ0259	106	1.000 N|+0	0<\left|a_{i j}\right|^{2} a_{i i}+a_{i i}^{2} a_{j j}-2 \left|a_{i j}\right|^{2} a_{i i}=a_{i i}\left(a_{i i} a_{j j}-\left|a_{i j}\right|^{2}\right) .	$0<\left\lvert a_{i j}\right\rvert^{2} a_{i i}+a_{i i}^{2} a_{j j}-2 \left\lvert a_{i j}\right\rvert^{2} a_{i i}=a_{i i}\left(a_{i i} a_{j j}-\left\lvert a_{i j}\right\rvert^{2}\right) .$	$0 < \lvert a_{ij}\rvert^2 a_{ii} + a_{ii}^2 a_{jj} - 2\lvert a_{ij}\rvert^2 a_{ii} = a_{ii}(a_{ii}a_{jj} - \lvert a_{ij}\rvert^2).$
lyche-numerical-linear-algebra EQ0260	106	1.000 N|+0	\left|a_{i j}\right|=\left|b_{i j}\right|<\sqrt{b_{i i} b_{j j}}=\sqrt{\left(a_{i i}+\varepsilon\right)\left(a_{j j}+\varepsilon\right)}, \quad i \neq j .	$\left\lvert a_{i j}\right\rvert=\left\lvert b_{i j}\right\rvert<\sqrt{b_{i i} b_{j j}}=\sqrt{\left(a_{i i}+\varepsilon\right)\left(a_{j j}+\varepsilon\right)}, \quad i \neq j .$	$\lvert a_{ij}\rvert = \lvert b_{ij}\rvert < \sqrt{b_{ii}b_{jj}} = \sqrt{(a_{ii}+\varepsilon)(a_{jj}+\varepsilon)}, \quad i \ne j.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0261</td><td>107</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{1}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{2}=\left[\begin{array}{ll} 1 & 2 \\ 2 & 2 \end{array}\right], \quad \boldsymbol{A}_{3}=\left[\begin{array}{rr} -2 & 1 \\ 1 & 2 \end{array}\right] .</td><td>$\boldsymbol{A}_{1}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{2}=\left[\begin{array}{ll} 1 & 2 \\ 2 & 2 \end{array}\right], \quad \boldsymbol{A}_{3}=\left[\begin{array}{rr} -2 & 1 \\ 1 & 2 \end{array}\right] .$</td><td>$\boldsymbol{A}_{1}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{2}=\left[\begin{array}{ll} 1 & 2 \\ 2 & 2 \end{array}\right], \quad \boldsymbol{A}_{3}=\left[\begin{array}{rr} -2 & 1 \\ 1 & 2 \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0266 (4.10)</td><td>109</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{L}^{*}:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^{*} / \beta \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right], \quad \beta:= \sqrt{\alpha},</td><td>$\boldsymbol{L}^{*}:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^{*} / \beta \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right], \quad \beta:= \sqrt{\alpha},$</td><td>$\boldsymbol{L}^{*}:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^{*} / \beta \\ \mathbf{0} & \boldsymbol{L}_{1}^{*} \end{array}\right], \quad \beta:= \sqrt{\alpha},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0271</td><td>113</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}=2 x_{1}^{2}+(2-a) x_{1} x_{2}+a x_{2} x_{1}+x_{2}^{2}=x_{1}^{2}+\left(x_{1}+x_{2}\right)^{2}>0 .</td><td>$\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}=2 x_{1}^{2}+(2-a) x_{1} x_{2}+a x_{2} x_{1}+x_{2}^{2}=x_{1}^{2}+\left(x_{1}+x_{2}\right)^{2}>0 .$</td><td>$\boldsymbol{x}^{T} \boldsymbol{A} \boldsymbol{x}=2 x_{1}^{2}+(2-a) x_{1} x_{2}+a x_{2} x_{1}+x_{2}^{2}=x_{1}^{2}+\left(x_{1}+x_{2}\right)^{2}>0 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0273</td><td>114</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+ \boldsymbol{v} \boldsymbol{v}^{T}\right) \boldsymbol{L}^{T},</td><td>$\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+ \boldsymbol{v} \boldsymbol{v}^{T}\right) \boldsymbol{L}^{T},$</td><td>$\boldsymbol{A}=\boldsymbol{L}(\boldsymbol{I}+ \boldsymbol{v} \boldsymbol{v}^{T}) \boldsymbol{L}^{T},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0275</td><td>114</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{L}_{+}=\left[\begin{array}{cc} \boldsymbol{L} & 0 \\ \boldsymbol{y}^{T} & \lambda \end{array}\right],</td><td>$\boldsymbol{L}_{+}=\left[\begin{array}{cc} \boldsymbol{L} & 0 \\ \boldsymbol{y}^{T} & \lambda \end{array}\right],$</td><td>$\boldsymbol{L}_{+}=\left[\begin{array}{cc} \boldsymbol{L} & 0 \\ \boldsymbol{y}^{T} & \lambda \end{array}\right],$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0276</td><td>117</td><td>■ 1.000 ● N|+0</td><td>\langle\boldsymbol{x}, a \boldsymbol{y}+b \boldsymbol{z} \rangle=\overline{\langle a \boldsymbol{y}+b \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{a \langle\boldsymbol{y}, \boldsymbol{x}\rangle+b\langle \boldsymbol{z}, \boldsymbol{x}\rangle}=\bar{a} \overline{ \langle\boldsymbol{y}, \boldsymbol{x}\rangle}+\overline{b \langle\boldsymbol{z}, \boldsymbol{x}\rangle}</td><td>$\langle\boldsymbol{x}, a \boldsymbol{y}+b \boldsymbol{z} \rangle=\overline{\langle a \boldsymbol{y}+b \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{a \langle\boldsymbol{y}, \boldsymbol{x}\rangle+b\langle \boldsymbol{z}, \boldsymbol{x}\rangle}=\bar{a} \overline{ \langle\boldsymbol{y}, \boldsymbol{x}\rangle}+\overline{b \langle\boldsymbol{z}, \boldsymbol{x}\rangle}$</td><td>$\langle\boldsymbol{x}, a \boldsymbol{y}+b \boldsymbol{z} \rangle=\overline{\langle a \boldsymbol{y}+b \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{a \langle\boldsymbol{y}, \boldsymbol{x}\rangle+b\langle \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{a} \overline{ \langle\boldsymbol{y}, \boldsymbol{x}\rangle}+\overline{b \langle\boldsymbol{z}, \boldsymbol{x}\rangle}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0277 (5.1)</td><td>117</td><td>■ 1.000 ● N|+1</td><td>\langle\boldsymbol{x}, a \boldsymbol{y}+b \boldsymbol{z} \rangle=\bar{a}\langle\boldsymbol{x}, \boldsymbol{y} \rangle+\bar{b}\langle\boldsymbol{x}, \boldsymbol{z} \rangle, \quad\langle a \boldsymbol{x}, a \boldsymbol{y} \rangle=|a|^{2}\langle\boldsymbol{x}, \boldsymbol{y} \rangle .</td><td>$\langle\boldsymbol{x}, a \boldsymbol{y}+b \boldsymbol{z} \rangle=\bar{a}\langle\boldsymbol{x}, \boldsymbol{y} \rangle+\bar{b}\langle\boldsymbol{x}, \boldsymbol{z} \rangle, \quad\langle a \boldsymbol{x}, a \boldsymbol{y} \rangle=|a|^{2}\langle\boldsymbol{x}, \boldsymbol{y} \rangle .$</td><td>$\langle\boldsymbol{x}, a \boldsymbol{y}+b \boldsymbol{z} \rangle=\overline{a}\langle\boldsymbol{x}, \boldsymbol{y} \rangle+\overline{b}\langle\boldsymbol{x}, \boldsymbol{z} \rangle, \quad\langle a \boldsymbol{x}, a \boldsymbol{y} \rangle=|a|^{2}\langle\boldsymbol{x}, \boldsymbol{y} \rangle .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0280 (5.2)</td><td>117</td><td>■ 1.000 ● N|+0</td><td>\|\cdot\|: \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \longmapsto\|\boldsymbol{x}\|:=\sqrt{ \langle\boldsymbol{x}, \boldsymbol{x}\rangle}</td><td>$\|\cdot\|: \mathcal{V} \rightarrow \mathbb{R}, \quad \boldsymbol{x} \longmapsto\|\boldsymbol{x}\|:=\sqrt{ \langle\boldsymbol{x}, \boldsymbol{x}\rangle}$</td><td>$\|\cdot\|: \mathcal{V} \rightarrow \mathbb{R}, \qquad \boldsymbol{x} \longmapsto\|\boldsymbol{x}\|:=\sqrt{ \langle\boldsymbol{x}, \boldsymbol{x}\rangle}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0281 (5.3)</td><td>118</td><td>■ 1.000 ● N|+0</td><td>|\langle\boldsymbol{x}, \boldsymbol{y}\rangle| \leq\| \boldsymbol{x}\|\|\boldsymbol{y}\|,</td><td>$|\langle\boldsymbol{x}, \boldsymbol{y}\rangle| \leq\| \boldsymbol{x}\|\|\boldsymbol{y}\|,$</td><td>$|\langle\boldsymbol{x}, \boldsymbol{y}\rangle| \leq\| \boldsymbol{x}\|\|\boldsymbol{y}\|,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0285 (5.5)</td><td>119</td><td>■ 1.000 ● W|+0</td><td>\|\boldsymbol{x}+a \boldsymbol{y}\|^{2}=\|\boldsymbol{x} \|^{2}+a\langle\boldsymbol{y}, \boldsymbol{x} \rangle+\bar{a}\langle\boldsymbol{x}, \boldsymbol{y} \rangle+|a|^{2}\|\boldsymbol{y}\|^{2}, \quad a \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V} .</td><td>$\|\boldsymbol{x}+a \boldsymbol{y}\|^{2}=\|\boldsymbol{x} \|^{2}+a\langle\boldsymbol{y}, \boldsymbol{x} \rangle+\bar{a}\langle\boldsymbol{x}, \boldsymbol{y} \rangle+|a|^{2}\|\boldsymbol{y}\|^{2}, \quad a \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V} .$</td><td>$\|\boldsymbol{x}+a \boldsymbol{y}\|^{2}=\|\boldsymbol{x} \|^{2}+a\langle\boldsymbol{y}, \boldsymbol{x} \rangle+\overline{a}\langle\boldsymbol{x}, \boldsymbol{y} \rangle+|a|^{2}\|\boldsymbol{y}\|^{2}, \quad a \in \mathbb{C}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0287 (5.6)</td><td>119</td><td>■ 1.000 ● N|+0</td><td>\cos \theta=\frac{\langle\boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} .</td><td>$\cos \theta=\frac{\langle\boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} .$</td><td>$\cos \theta=\frac{\langle\boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0288 (5.7)</td><td>119</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{x}+\boldsymbol{y}\|^{2}=\|\boldsymbol{x} \|^{2}+\|\boldsymbol{y}\|^{2}, \quad \text { if } \quad \boldsymbol{x} \perp \boldsymbol{y} .</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2}=\|\boldsymbol{x} \|^{2}+\|\boldsymbol{y}\|^{2}, \quad \text { if } \quad \boldsymbol{x} \perp \boldsymbol{y} .$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2}=\|\boldsymbol{x} \|^{2}+\|\boldsymbol{y}\|^{2}, \quad \textit{if} \quad \boldsymbol{x} \perp \boldsymbol{y} .$</td></tr>
</table>

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<tr><td>lyche-numerical-linear-algebra_ EQ00290</td><td>120</td><td>■ 1.000 ● N|+0</td><td>\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k} \right\}:=\left\{\frac{\boldsymbol{v}_{1}}{\left \|\boldsymbol{v}_{1}\right\|}, \ldots, \frac{ \boldsymbol{v}_{k}}{\left\|\boldsymbol{v}_{k}\right \|}\right\}</td><td>$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{k} \right\}:=\left\{\frac{\boldsymbol{v}_{1}}{\left\|\boldsymbol{v}_{1}\right\|}, \ldots, \frac{\boldsymbol{v}_{k}}{\left\|\boldsymbol{v}_{k}\right\|}\right\}$</td><td>$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\} := \left\{\frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}, \ldots, \frac{\boldsymbol{v}_k}{\|\boldsymbol{v}_k\|}\right\}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00291</td><td>120</td><td>■ 1.000 ● C|+3</td><td>\left\langle\boldsymbol{v}_{j}, \boldsymbol{v}_{l} \right\rangle=\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{l}\right\rangle-\sum_{i=1}^{j-1} \frac{ \left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{i} \right\rangle}{\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle}\left\langle \boldsymbol{v}_{i}, \boldsymbol{v}_{l}\right\rangle= \left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{l} \right\rangle-\frac{\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{l}\right\rangle}{\left\langle \boldsymbol{v}_{l}, \boldsymbol{v}_{l}\right\rangle} \left\langle\boldsymbol{v}_{l}, \boldsymbol{v}_{l}\right \rangle=0</td><td>$\left\langle\boldsymbol{v}_{j}, \boldsymbol{v}_{l} \right\rangle=\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{l}\right\rangle-\sum_{i=1}^{j-1} \frac{\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{i} \right\rangle}{\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle}\left\langle \boldsymbol{v}_{i}, \boldsymbol{v}_{l}\right\rangle= \left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{l} \right\rangle-\frac{\left\langle\boldsymbol{s}_{j}, \boldsymbol{v}_{l}\right\rangle}{\left\langle \boldsymbol{v}_{l}, \boldsymbol{v}_{l}\right\rangle} \left\langle\boldsymbol{v}_{l}, \boldsymbol{v}_{l}\right\rangle=0$</td><td>$\langle\boldsymbol{v}_j, \boldsymbol{v}_l\rangle = \langle\boldsymbol{s}_j, \boldsymbol{v}_l\rangle - \sum_{i=1}^{j-1}\frac{\langle\boldsymbol{s}_j, \boldsymbol{v}_i\rangle}{\langle\boldsymbol{v}_i, \boldsymbol{v}_i\rangle}\langle\boldsymbol{v}_i, \boldsymbol{v}_l\rangle = \langle\boldsymbol{s}_j, \boldsymbol{v}_l\rangle - \frac{\langle\boldsymbol{s}_j, \boldsymbol{v}_l\rangle}{\langle\boldsymbol{v}_l, \boldsymbol{v}_l\rangle}\langle\boldsymbol{v}_l, \boldsymbol{v}_l\rangle = 0$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00292</td><td>121</td><td>■ 1.000 ● N|+0</td><td>\operatorname{dim}(\mathcal{S} \oplus \mathcal{T})= \operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T}) .</td><td>$\operatorname{dim}(\mathcal{S} \oplus \mathcal{T})= \operatorname{dim}(\mathcal{S})+\operatorname{dim}(\mathcal{T}) .$</td><td>$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00295</td><td>122</td><td>■ 1.000 ● W|+0</td><td>\left\langle\boldsymbol{s}_{0}, \boldsymbol{v}_{j}\right \rangle=\left\langle\sum_{i=1}^{k} \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_{i}\right\rangle}{\left \langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right \rangle} \boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right \rangle=\sum_{i=1}^{k} \frac{\left\langle\boldsymbol{v}, \boldsymbol{v}_{i}\right\rangle}{\left\langle \boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle} \left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right \rangle=\left\langle\boldsymbol{v}, \boldsymbol{v}_{j} \right\rangle .</td><td>$\left\langle\boldsymbol{s}_{0}, \boldsymbol{v}_{j}\right\rangle=\left\langle\sum_{i=1}^{k} \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_{i}\right\rangle}{\left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle} \boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle=\sum_{i=1}^{k} \frac{\left\langle\boldsymbol{v}, \boldsymbol{v}_{i}\right\rangle}{\left\langle \boldsymbol{v}_{i}, \boldsymbol{v}_{i}\right\rangle} \left\langle\boldsymbol{v}_{i}, \boldsymbol{v}_{j}\right\rangle=\left\langle\boldsymbol{v}, \boldsymbol{v}_{j} \right\rangle .$</td><td>$\langle\boldsymbol{s}_0, \boldsymbol{v}_j\rangle = \langle\sum_{i=1}^{k}\frac{\langle\boldsymbol{v}, \boldsymbol{v}_i\rangle}{\langle\boldsymbol{v}_i, \boldsymbol{v}_i\rangle}\boldsymbol{v}_i, \boldsymbol{v}_j\rangle = \sum_{i=1}^{k}\frac{\langle\boldsymbol{v}, \boldsymbol{v}_i\rangle}{\langle\boldsymbol{v}_i, \boldsymbol{v}_i\rangle}\langle\boldsymbol{v}_i, \boldsymbol{v}_j\rangle = \langle\boldsymbol{v}, \boldsymbol{v}_j\rangle.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00297</td><td>123</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{v}-\boldsymbol{s}\|^{2}=\left\| \boldsymbol{v}-\boldsymbol{s}_{0}+\boldsymbol{u}\right \|^{2}=\left\|\boldsymbol{v}-\boldsymbol{s}_{0}\right \|^{2}+\|\boldsymbol{u}\|^{2}>\left\|\boldsymbol{v}- \boldsymbol{s}_{0}\right\|^{2} .</td><td>$\|\boldsymbol{v}-\boldsymbol{s}\|^{2}=\left\| \boldsymbol{v}-\boldsymbol{s}_{0}+\boldsymbol{u}\right\|^{2}=\left\|\boldsymbol{v}-\boldsymbol{s}_{0}\right\|^{2}+\|\boldsymbol{u}\|^{2}>\left\|\boldsymbol{v}- \boldsymbol{s}_{0}\right\|^{2} .$</td><td>$\|\boldsymbol{v} - \boldsymbol{s}\|^2 = \|\boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u}\|^2 = \|\boldsymbol{v} - \boldsymbol{s}_0\|^2 + \|\boldsymbol{u}\|^2 > \|\boldsymbol{v} - \boldsymbol{s}_0\|^2.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00299</td><td>123</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{j}= \left\langle\boldsymbol{A} \boldsymbol{e}_{j}, \boldsymbol{e}_{i}\right\rangle=\left\langle \boldsymbol{e}_{j}, \overline{\boldsymbol{A}} \boldsymbol{e}_{i}\right\rangle=\boldsymbol{e}_{i}^{T} \boldsymbol{A}^{T} \boldsymbol{e}_{j} .</td><td>$\boldsymbol{e}_{i}^{T} \boldsymbol{A} \boldsymbol{e}_{j}= \left\langle\boldsymbol{A} \boldsymbol{e}_{j}, \boldsymbol{e}_{i}\right\rangle=\left\langle \boldsymbol{e}_{j}, \overline{\boldsymbol{A}} \boldsymbol{e}_{i}\right\rangle=\boldsymbol{e}_{i}^{T} \boldsymbol{A}^{T} \boldsymbol{e}_{j} .$</td><td>$\boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_j = \langle\boldsymbol{A}\boldsymbol{e}_j, \boldsymbol{e}_i\rangle = \langle\boldsymbol{e}_j, \overline{\boldsymbol{A}}\boldsymbol{e}_i\rangle = \boldsymbol{e}_i^T\boldsymbol{A}^T\boldsymbol{e}_j.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ00302</td><td>124</td><td>■ 1.000 ● N|-1</td><td>\boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}, \text { where } \boldsymbol{u} \in \mathbb{C}^{n} \text { and } \boldsymbol{u}^{*} \boldsymbol{u}=2</td><td>$\boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}, \text { where } \boldsymbol{u} \in \mathbb{C}^{n} \text { and } \boldsymbol{u}^{*} \boldsymbol{u}=2$</td><td>$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^*\boldsymbol{u} = 2$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0304	124	■ 1.000 ● N|+0	\boldsymbol{H}^{*} \boldsymbol{H}=\boldsymbol{H}^{2}= \left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*} \right)\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}\right)=\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^{*}+\boldsymbol{u}\left(\boldsymbol{u}^{*} \boldsymbol{u}\right) \boldsymbol{u}^{*}=\boldsymbol{I} .	$\boldsymbol{H}^{*} \boldsymbol{H}=\boldsymbol{H}^{2}=\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}\right)\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*}\right)=\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^{*}+\boldsymbol{u}\left(\boldsymbol{u}^{*} \boldsymbol{u}\right) \boldsymbol{u}^{*}=\boldsymbol{I} .$	$\boldsymbol{H}^{*} \boldsymbol{H}=\boldsymbol{H}^{2}=(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*})(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^{*})=\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^{*}+\boldsymbol{u}(\boldsymbol{u}^{*} \boldsymbol{u}) \boldsymbol{u}^{*}=\boldsymbol{I} .$
lyche-numerical-linear-algebra EQ0305 (5.12)	124	■ 1.000 ● K|+0	\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}	$\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}$	$\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}$
lyche-numerical-linear-algebra EQ0307	125	■ 1.000 ● K|+0	\boldsymbol{H}=\boldsymbol{I}-\frac{2 \boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}= \boldsymbol{P}-\frac{\boldsymbol{v} \boldsymbol{v}^{*}}{ \boldsymbol{v}^{*} \boldsymbol{v}}, \text { where } \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}},	$\boldsymbol{H}=\boldsymbol{I}-\frac{2 \boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}=\boldsymbol{P}-\frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}, \text { where } \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}},$	$\boldsymbol{H}=\boldsymbol{I}-\frac{2 \boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}=\boldsymbol{P}-\frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}}, \text { where } \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v} \boldsymbol{v}^{*}}{\boldsymbol{v}^{*} \boldsymbol{v}},$
lyche-numerical-linear-algebra EQ0315	128	■ 1.000 ● K|+0	\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array}\right], \quad \left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{array}\right] .	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array}\right], \quad\left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array}\right], \quad\left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{array}\right] .$
lyche-numerical-linear-algebra EQ0316	128	■ 1.000 ● N|+0	\boldsymbol{A}_{n+1}:=\boldsymbol{H}_{n} \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}=\left[\begin{array}{c} \boldsymbol{R}_{1} \\ \mathbf{0} \end{array}\right]=\boldsymbol{R},	$\boldsymbol{A}_{n+1}:=\boldsymbol{H}_{n} \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}=\left[\begin{array}{c} \boldsymbol{R}_{1} \\ \mathbf{0} \end{array}\right]=\boldsymbol{R},$	$\boldsymbol{A}_{n+1}:=\boldsymbol{H}_{n} \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_{1} \boldsymbol{A}=\left[\begin{array}{c} \boldsymbol{R}_{1} \\ \mathbf{0} \end{array}\right]=\boldsymbol{R},$
lyche-numerical-linear-algebra EQ0317	128	■ 1.000 ● N|+0	\boldsymbol{A}_{1}:=\boldsymbol{A}, \quad \boldsymbol{A}_{k+1}=\boldsymbol{H}_{k} \boldsymbol{A}_{k}, \quad k=1,2, \ldots, n .	$\boldsymbol{A}_{1}:=\boldsymbol{A}, \quad \boldsymbol{A}_{k+1}=\boldsymbol{H}_{k} \boldsymbol{A}_{k}, \quad k=1,2, \ldots, n .$	$\boldsymbol{A}_{1}:=\boldsymbol{A}, \quad \boldsymbol{A}_{k+1}=\boldsymbol{H}_{k} \boldsymbol{A}_{k}, \quad k=1,2, \ldots, n .$
lyche-numerical-linear-algebra EQ0319	129	■ 1.000 ● N|+0	\boldsymbol{A}_{k+1}:=\boldsymbol{H}_{k} \boldsymbol{A}_{k}=\left[\begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \mathbf{0} & \hat{\boldsymbol{H}}_{k} \boldsymbol{D}_{k} \end{array} \right]=\left[\begin{array}{cc} \boldsymbol{B}_{k+1} & \boldsymbol{C}_{k+1} \\ \mathbf{0} & \boldsymbol{D}_{k+1} \end{array}\right],	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_{k} \boldsymbol{A}_{k}=\left[\begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \mathbf{0} & \hat{\boldsymbol{H}}_{k} \boldsymbol{D}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{B}_{k+1} & \boldsymbol{C}_{k+1} \\ \mathbf{0} & \boldsymbol{D}_{k+1} \end{array}\right],$	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_{k} \boldsymbol{A}_{k}=\left[\begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \mathbf{0} & \hat{\boldsymbol{H}}_{k} \boldsymbol{D}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{B}_{k+1} & \boldsymbol{C}_{k+1} \\ \mathbf{0} & \boldsymbol{D}_{k+1} \end{array}\right],$
lyche-numerical-linear-algebra EQ0322	131	■ 1.000 ● N|+0	\boldsymbol{R}=\left[\begin{array}{c} \boldsymbol{R}_{1} \\ \mathbf{0}_{m-n, n} \end{array}\right]	$\boldsymbol{R}=\left[\begin{array}{c} \boldsymbol{R}_{1} \\ \mathbf{0}_{m-n, n} \end{array}\right]$	$\boldsymbol{R}=\left[\begin{array}{c} \boldsymbol{R}_{1} \\ \mathbf{0}_{m-n, n} \end{array}\right]$
lyche-numerical-linear-algebra EQ0325 (5.17)	133	■ 1.000 ● N|+0	\boldsymbol{v}_{1}=\boldsymbol{a}_{1}, \quad \boldsymbol{v}_{j}=\boldsymbol{a}_{j}-\sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{j}^{T} \boldsymbol{v}_{i}}{ \boldsymbol{v}_{i}^{T} \boldsymbol{v}_{i}} \boldsymbol{v}_{i}, \quad \text { for } j=2, \ldots, n .	$\boldsymbol{v}_{1}=\boldsymbol{a}_{1}, \quad \boldsymbol{v}_{j}=\boldsymbol{a}_{j}-\sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{j}^{T} \boldsymbol{v}_{i}}{\boldsymbol{v}_{i}^{T} \boldsymbol{v}_{i}} \boldsymbol{v}_{i}, \quad \text { for } j=2, \ldots, n .$	$\boldsymbol{v}_{1}=\boldsymbol{a}_{1}, \quad \boldsymbol{v}_{j}=\boldsymbol{a}_{j}-\sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{j}^{T} \boldsymbol{v}_{i}}{\boldsymbol{v}_{i}^{T} \boldsymbol{v}_{i}} \boldsymbol{v}_{i}, \quad \textit{for } j=2, \ldots, n .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
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EQ0327</td><td>133</td><td>1.000
S|+0</td><td><code>\boldsymbol{a}_{j}=\boldsymbol{v}_{j}+\sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{j}^{T} \boldsymbol{v}_{i}}{ \boldsymbol{v}_{i}^{T} \boldsymbol{v}_{i}} \boldsymbol{v}_{i}=r_{j j} \boldsymbol{q}_{j}+ \sum_{i=1}^{j-1} \boldsymbol{q}_{i} r_{i j}= \boldsymbol{Q}_{1} \boldsymbol{R}_{1} \boldsymbol{e}_{j}, j=1, \ldots, n .</code></td><td>$\boldsymbol{a}_{j}=\boldsymbol{v}_{j}+\sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{j}^{T} \boldsymbol{v}_{i}}{ \boldsymbol{v}_{i}^{T} \boldsymbol{v}_{i}} \boldsymbol{v}_{i}=r_{j j} \boldsymbol{q}_{j}+ \sum_{i=1}^{j-1} \boldsymbol{q}_{i} r_{i j}= \boldsymbol{Q}_{1} \boldsymbol{R}_{1} \boldsymbol{e}_{j}, j=1, \ldots, n .$</td><td>$\boldsymbol{a}_{j}=\boldsymbol{v}_{j}+\sum_{i=1}^{j-1} \frac{\boldsymbol{a}_{j}^{T} \boldsymbol{v}_{i}}{ \boldsymbol{v}_{i}^{T} \boldsymbol{v}_{i}} \boldsymbol{v}_{i}=r_{j j} \boldsymbol{q}_{j}+ \sum_{i=1}^{j-1} \boldsymbol{q}_{i} r_{i j}= \boldsymbol{Q}_{1} \boldsymbol{R}_{1} \boldsymbol{e}_{j},\ j=1, \ldots, n .$</td></tr>
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N|+0</td><td><code>\left[\begin{array}{cc} p_{i i} & p_{i j} \\ p_{j i} & p_{j j} \end{array}\right]=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^{2}+s^{2}=1 .</code></td><td>$\left[\begin{array}{cc} p_{i i} & p_{i j} \\ p_{j i} & p_{j j} \end{array}\right]=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^{2}+s^{2}=1 .$</td><td>$\left[\begin{array}{cc} p_{i i} & p_{i j} \\ p_{j i} & p_{j j} \end{array}\right]=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^{2}+s^{2}=1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
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C|-6</td><td><code>\boldsymbol{P}_{12}=\left[\begin{array}{cccc} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{13}=\left[\begin{array}{cccc} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{23}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] .</code></td><td>$\boldsymbol{P}_{12}=\left[\begin{array}{cccc} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{13}=\left[\begin{array}{cccc} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{23}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] .$</td><td>$\boldsymbol{P}_{12}=\left[\begin{array}{cccc} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{13}=\left[\begin{array}{cccc} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{23}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] .$</td></tr>
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EQ0337 (5.20)</td><td>136</td><td>1.000
N|+0</td><td><code>\cos \theta=\frac{|\langle\boldsymbol{x}, \boldsymbol{y} \rangle|}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} .</code></td><td>$\cos \theta=\frac{|\langle\boldsymbol{x}, \boldsymbol{y} \rangle|}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} .$</td><td>$\cos \theta=\frac{|\langle\boldsymbol{x}, \boldsymbol{y} \rangle|}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0339</td><td>137</td><td>1.000
W|+0</td><td><code>\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right] .</code></td><td>$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right] .$</td><td>$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0340</td><td>137</td><td>1.000
N|+1</td><td><code>\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}, \quad \text { where } \quad \boldsymbol{H}:=\boldsymbol{I}-2 \frac{ \boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}} .</code></td><td>$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}, \quad \text { where } \quad \boldsymbol{H}:=\boldsymbol{I}-2 \frac{ \boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}} .$</td><td>$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}, \quad \text { where } \quad \boldsymbol{H}:=\boldsymbol{I}-2 \frac{ \boldsymbol{v} \boldsymbol{v}^{T}}{\boldsymbol{v}^{T} \boldsymbol{v}} .$</td></tr>
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K|+0</td><td><code>A:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right] .</code></td><td>$A:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right] .$</td><td>$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right] .$</td></tr>
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EQ0344 (5.21)</td><td>138</td><td>1.000
N|+0</td><td><code>|\operatorname{det}(\boldsymbol{A})| \leq \prod_{j=1}^{n} \left\|\boldsymbol{a}_{j}\right\|_{2} .</code></td><td>$|\operatorname{det}(\boldsymbol{A})| \leq \prod_{j=1}^{n} \left\|\boldsymbol{a}_{j}\right\|_{2} .$</td><td>$|\operatorname{det}(\boldsymbol{A})| \leq \prod_{j=1}^{n} \|\boldsymbol{a}_{j}\|_{2} .$</td></tr>
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EQ0346</td><td>139</td><td>1.000
K|+0</td><td><code>A:=\left[\begin{array}{rr} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]</code></td><td>$A:=\left[\begin{array}{rr} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$</td><td>$\boldsymbol{A}:=\left[\begin{array}{rr} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$</td></tr>
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K|+0</td><td><code>\boldsymbol{G}_{3} \boldsymbol{G}_{2} \boldsymbol{G}_{1} \boldsymbol{H}=\boldsymbol{R}</code></td><td>$\boldsymbol{G}_{3} \boldsymbol{G}_{2} \boldsymbol{G}_{1} \boldsymbol{H}=\boldsymbol{R}$</td><td>$\boldsymbol{G}_{3} \boldsymbol{G}_{2} \boldsymbol{G}_{1} \boldsymbol{H}=\boldsymbol{R}$</td></tr>
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Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0349	141	1.000 W|+0	`\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]`	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array}\right]$
lyche-numerical-linear-algebra EQ0351	141	1.000 K|+0	`\left(a_{1,1}, a_{2,2}, \ldots, a_{n, n}\right)`	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n, n}\right)$	$(a_{1,1}, a_{2,2}, \ldots, a_{n,n})$
lyche-numerical-linear-algebra EQ0352	141	1.000 N|+0	`\left(a_{2,1}, a_{3,2}, \ldots, a_{n, n-1}\right) .`	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n, n-1}\right) .$	$(a_{2,1}, a_{3,2}, \ldots, a_{n,n-1}).$
lyche-numerical-linear-algebra EQ0353	142	1.000 N|+0	`\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}`	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1,m}\boldsymbol{P}_{m-2,m-1} \cdots \boldsymbol{P}_{2,3}\boldsymbol{P}_{1,2}\boldsymbol{H}$
lyche-numerical-linear-algebra EQ0356	145	1.000 K|+0	`\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) .`	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) .$	$\pi_{\boldsymbol{A}}(\lambda) = \det(\boldsymbol{A}-\lambda\boldsymbol{I}).$
lyche-numerical-linear-algebra EQ0357	146	1.000 N|+0	`\sum_{j=1}^{m} c_{j} \boldsymbol{x}_{j}=\mathbf{0} \Rightarrow \sum_{j=1}^{m} c_{j} \boldsymbol{A} \boldsymbol{x}_{j}=\sum_{j=1}^{m} c_{j} \lambda_{j} \boldsymbol{x}_{j}=\mathbf{0} .`	$\sum_{j=1}^{m} c_{j} \boldsymbol{x}_{j}=\mathbf{0} \Rightarrow \sum_{j=1}^{m} c_{j} \boldsymbol{A} \boldsymbol{x}_{j}=\sum_{j=1}^{m} c_{j} \lambda_{j} \boldsymbol{x}_{j}=\mathbf{0} .$	$\sum_{j=1}^{m} c_j\boldsymbol{x}_j = \mathbf{0} \Rightarrow \sum_{j=1}^{m} c_j\boldsymbol{A}\boldsymbol{x}_j = \sum_{j=1}^{m} c_j\lambda_j\boldsymbol{x}_j = \mathbf{0}.$
lyche-numerical-linear-algebra EQ0358	146	1.000 K|+0	`\boldsymbol{I}:=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \boldsymbol{J}:=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right] .`	$\boldsymbol{I}:=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \boldsymbol{J}:=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right] .$	$\boldsymbol{I} := \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}, \quad \boldsymbol{J} := \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}.$
lyche-numerical-linear-algebra EQ0359	146	1.000 N|+0	`\boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{x}_{j} \text { for some scalars } c_{1}, \ldots, c_{n}`	$\boldsymbol{x}=\sum_{j=1}^{n} c_{j} \boldsymbol{x}_{j} \text { for some scalars } c_{1}, \ldots, c_{n}$	$\boldsymbol{x} = \sum_{j=1}^{n} c_j\boldsymbol{x}_j \text{ for some scalars } c_1, \ldots, c_n.$
lyche-numerical-linear-algebra EQ0360	146	1.000 K|+0	`\boldsymbol{x}=\frac{x_{1}+x_{2}}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]+\frac{x_{1}-x_{2}}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right] .`	$\boldsymbol{x}=\frac{x_{1}+x_{2}}{2}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]+\frac{x_{1}-x_{2}}{2}\left[\begin{array}{c} 1 \\ -1 \end{array}\right] .$	$\boldsymbol{x} = \frac{x_1+x_2}{2}\begin{bmatrix}1\\1\end{bmatrix} + \frac{x_1-x_2}{2}\begin{bmatrix}1\\-1\end{bmatrix}.$
lyche-numerical-linear-algebra EQ0365 (6.3)	148	1.000 N|+0	`\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^{n}:(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}=\mathbf{0}\right\}`	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^{n}:(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}=\mathbf{0}\right\}$	$\mathcal{N}(\boldsymbol{A}-\lambda\boldsymbol{I}) := \{\boldsymbol{x} \in \mathbb{C}^n : (\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{x} = \mathbf{0}\}$
lyche-numerical-linear-algebra EQ0368 (6.6)	150	1.000 W|+0	`\boldsymbol{U}_{i}=\operatorname{diag}\left(\boldsymbol{J}_{m_{i, 1}}\left(\lambda_{i}\right), \ldots, \boldsymbol{J}_{m_{i, g_{i}}}\left(\lambda_{i}\right)\right) .`	$\boldsymbol{U}_{i}=\operatorname{diag}\left(\boldsymbol{J}_{m_{i, 1}}\left(\lambda_{i}\right), \ldots, \boldsymbol{J}_{m_{i, g_{i}}}\left(\lambda_{i}\right)\right) .$	$\boldsymbol{U}_i = \operatorname{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i), \ldots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)).$
lyche-numerical-linear-algebra EQ0372	152	1.000 N|+0	`\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_{1}=\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{v}_{1}=\lambda_{1} \boldsymbol{V}^{*} \boldsymbol{v}_{1}=\lambda_{1} \boldsymbol{e}_{1} .`	$\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_{1}=\boldsymbol{V}^{*} \boldsymbol{A} \boldsymbol{v}_{1}=\lambda_{1} \boldsymbol{V}^{*} \boldsymbol{v}_{1}=\lambda_{1} \boldsymbol{e}_{1} .$	$\boldsymbol{V}^*\boldsymbol{A}\boldsymbol{V}\boldsymbol{e}_1 = \boldsymbol{V}^*\boldsymbol{A}\boldsymbol{v}_1 = \lambda_1\boldsymbol{V}^*\boldsymbol{v}_1 = \lambda_1\boldsymbol{e}_1.$
lyche-numerical-linear-algebra EQ0376	153	1.000 N|+0	`\boldsymbol{V}=\left[\begin{array}{ccc} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{array}\right] .`	$\boldsymbol{V}=\left[\begin{array}{ccc} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{array}\right] .$	$\boldsymbol{V} = \begin{bmatrix}-\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\\0 & 1 & 0\\\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\end{bmatrix}.$
lyche-numerical-linear-algebra EQ0377	153	1.000 N|+0	`\boldsymbol{M}=\left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{array}\right] .`	$\boldsymbol{M}=\left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{array}\right] .$	$\boldsymbol{M} = \begin{bmatrix}2 & -\sqrt{2}\\-\sqrt{2} & 2\end{bmatrix}.$

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value

Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0378</td><td>153</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}^{*} \boldsymbol{A}=\operatorname{diag} \left(\overline{d_{1}} d_{1}, \ldots, \overline{d_{n}} d_{n}\right)=\operatorname{diag}\left(\left|d_{1} \right|^{2}, \ldots,\left|d_{n}\right|^{2}\right)= \boldsymbol{A} \boldsymbol{A}^{*},</td><td>$\boldsymbol{A}^{*} \boldsymbol{A}=\operatorname{diag} \left(\overline{d_{1}} d_{1}, \ldots, \overline{d_{n}} d_{n}\right)=\operatorname{diag}\left(\left|d_{1} \right|^{2}, \ldots,\left|d_{n}\right|^{2}\right)= \boldsymbol{A} \boldsymbol{A}^{*},$</td><td>$\boldsymbol{A}^{*} \boldsymbol{A}=\operatorname{diag}(\overline{d_{1}} d_{1}, \ldots, \overline{d_{n}} d_{n})=\operatorname{diag}(|d_{1}|^{2}, \ldots,|d_{n}|^{2})= \boldsymbol{A} \boldsymbol{A}^{*},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0382</td><td>154</td><td>■ 1.000 ● N|+0</td><td>e_{i i}=\sum_{k=1}^{i}\left|b_{k i}\right|^{2}=\left|b_{i i}\right|^{2}=\sum_{k=i}^{n}\left|b_{i k}\right|^{2}=f_{i i}</td><td>$e_{i i}=\sum_{k=1}^{i}\left|b_{k i}\right|^{2}=\left|b_{i i}\right|^{2}=\sum_{k=i}^{n}\left|b_{i k}\right|^{2}=f_{i i}$</td><td>$e_{i i}=\sum_{k=1}^{i}|b_{k i}|^{2}=|b_{i i}|^{2}=\sum_{k=i}^{n}|b_{i k}|^{2}=f_{i i}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0385 (6.10)</td><td>155</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{M}=\left[\begin{array}{cc} \mu & v \\ -v & \mu \end{array}\right]</td><td>$\boldsymbol{M}=\left[\begin{array}{cc} \mu & v \\ -v & \mu \end{array}\right]$</td><td>$\boldsymbol{M}=\left[\begin{array}{cc} \mu & \nu \\ -\nu & \mu \end{array}\right]$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0388</td><td>156</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}= \operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n} \right) .</td><td>$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}= \operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n} \right) .$</td><td>$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U}= \operatorname{diag}(\lambda_{1}, \ldots, \lambda_{n}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0389</td><td>156</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}= \operatorname{diag}\left(\lambda_{1}, \lambda_{2}, \ldots, \lambda_{n}\right) .</td><td>$\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}= \operatorname{diag}\left(\lambda_{1}, \lambda_{2}, \ldots, \lambda_{n}\right) .$</td><td>$\boldsymbol{U}^{T} \boldsymbol{A} \boldsymbol{U}= \operatorname{diag}(\lambda_{1}, \lambda_{2}, \ldots, \lambda_{n}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0391</td><td>157</td><td>■ 1.000 ● N|+0</td><td>\operatorname{dim}\left(\mathcal{S} \cap \mathcal{S}^{ \prime}\right)=\operatorname{dim}(\mathcal{S})+ \operatorname{dim}\left(\mathcal{S}^{\prime}\right)- \operatorname{dim}\left(\mathcal{S}+\mathcal{S}^{\prime} \right) \geq(n-k+1)+k-n=1 .</td><td>$\operatorname{dim}\left(\mathcal{S} \cap \mathcal{S}^{ \prime}\right)=\operatorname{dim}(\mathcal{S})+ \operatorname{dim}\left(\mathcal{S}^{\prime}\right)- \operatorname{dim}\left(\mathcal{S}+\mathcal{S}^{\prime} \right) \geq(n-k+1)+k-n=1 .$</td><td>$\operatorname{dim}(\mathcal{S} \cap \mathcal{S}^{\prime})=\operatorname{dim}(\mathcal{S})+ \operatorname{dim}(\mathcal{S}^{\prime})- \operatorname{dim}(\mathcal{S}+\mathcal{S}^{\prime}) \geq(n-k+1)+k-n=1.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0395 (6.14)</td><td>158</td><td>■ 1.000 ● N|+0</td><td>\alpha_{k}+\varepsilon_{n} \leq \beta_{k} \leq \alpha_{k}+\varepsilon_{1}, \text { for } k=1, \ldots, n,</td><td>$\alpha_{k}+\varepsilon_{n} \leq \beta_{k} \leq \alpha_{k}+\varepsilon_{1}, \text { for } k=1, \ldots, n,$</td><td>$\alpha_{k}+\varepsilon_{n} \leq \beta_{k} \leq \alpha_{k}+\varepsilon_{1}, \textit{ for } k=1, \ldots, n,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0398 (6.16)</td><td>159</td><td>■ 1.000 ● W|+1</td><td>\boldsymbol{v}=\sum_{j=1}^{n} \frac{ \boldsymbol{y}_{j}^{*} \boldsymbol{v}}{ \boldsymbol{y}_{j}^{*} \boldsymbol{x}_{j}} \boldsymbol{x}_{j}=\sum_{k=1}^{n} \frac{ \boldsymbol{x}_{k}^{*} \boldsymbol{v}}{ \boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k}} \boldsymbol{y}_{k}, \quad \boldsymbol{v} \in \mathbb{C}^{n}</td><td>$\boldsymbol{v}=\sum_{j=1}^{n} \frac{ \boldsymbol{y}_{j}^{*} \boldsymbol{v}}{ \boldsymbol{y}_{j}^{*} \boldsymbol{x}_{j}} \boldsymbol{x}_{j}=\sum_{k=1}^{n} \frac{ \boldsymbol{x}_{k}^{*} \boldsymbol{v}}{ \boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k}} \boldsymbol{y}_{k}, \quad \boldsymbol{v} \in \mathbb{C}^{n}$</td><td>$\boldsymbol{v}=\sum_{j=1}^{n} \frac{ \boldsymbol{y}_{j}^{*} \boldsymbol{v}}{ \boldsymbol{y}_{j}^{*} \boldsymbol{x}_{j}} \boldsymbol{x}_{j}=\sum_{k=1}^{n} \frac{ \boldsymbol{x}_{k}^{*} \boldsymbol{v}}{ \boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k}} \boldsymbol{y}_{k}, \quad \boldsymbol{v} \in \mathbb{C}^{n}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0399</td><td>160</td><td>■ 1.000 ● N|-1</td><td>\boldsymbol{X}:=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right], \quad \boldsymbol{Y}:= \left[\boldsymbol{y}_{1}, \ldots, \boldsymbol{y}_{n} \right], \quad \boldsymbol{D}:=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right) .</td><td>$\boldsymbol{X}:=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right], \quad \boldsymbol{Y}:= \left[\boldsymbol{y}_{1}, \ldots, \boldsymbol{y}_{n} \right], \quad \boldsymbol{D}:=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right) .$</td><td>$\boldsymbol{X}:=[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}], \quad \boldsymbol{Y}:= [\boldsymbol{y}_{1}, \ldots, \boldsymbol{y}_{n}], \quad \boldsymbol{D}:=\operatorname{diag}(\lambda_{1}, \ldots, \lambda_{n}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0400 (6.17)</td><td>160</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{v}=\sum_{j=1}^{n} \frac{ \boldsymbol{x}_{j}^{*} \boldsymbol{v}}{ \boldsymbol{x}_{j}^{*} \boldsymbol{x}_{j}} \boldsymbol{x}_{j} .</td><td>$\boldsymbol{v}=\sum_{j=1}^{n} \frac{ \boldsymbol{x}_{j}^{*} \boldsymbol{v}}{ \boldsymbol{x}_{j}^{*} \boldsymbol{x}_{j}} \boldsymbol{x}_{j} .$</td><td>$\boldsymbol{v}=\sum_{j=1}^{n} \frac{ \boldsymbol{x}_{j}^{*} \boldsymbol{v}}{ \boldsymbol{x}_{j}^{*} \boldsymbol{x}_{j}} \boldsymbol{x}_{j}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0402</td><td>161</td><td>■ 1.000 ● N|+0</td><td>\left(\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V}\right) \boldsymbol{u}=\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{V}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{u},</td><td>$\left(\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V}\right) \boldsymbol{u}=\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{V}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{u},$</td><td>$(\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{V}) \boldsymbol{u}=\boldsymbol{V}^{*} \boldsymbol{A}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{V}^{*} \boldsymbol{y}=\bar{\lambda} \boldsymbol{u},$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0405 (6.18)	162	■ 1.000 ● N|+0	`\boldsymbol{A}^{k} \boldsymbol{x}=\sum_{j=1}^{n} c_{j} \lambda_{j}^{k} \boldsymbol{x}_{j} \text { for some scalars } c_{1}, \ldots, c_{n}`	$\boldsymbol{A}^{k} \boldsymbol{x}=\sum_{j=1}^{n} c_{j} \lambda_{j}^{k} \boldsymbol{x}_{j} \text { for some scalars } c_{1}, \ldots, c_{n}$	$\boldsymbol{A}^{k} \boldsymbol{x}=\sum_{j=1}^{n} c_{j} \lambda_{j}^{k} \boldsymbol{x}_{j}$ for some scalars $c_{1}, \ldots, c_{n}$.
lyche-numerical-linear-algebra_ EQ0411	164	■ 1.000 ● N|+0	`p(\boldsymbol{A}):=\sum_{j=0}^{r} b_{j} \boldsymbol{A}^{j},`	$p(\boldsymbol{A}):=\sum_{j=0}^{r} b_{j} \boldsymbol{A}^{j},$	$p(\boldsymbol{A}):=\sum_{j=0}^{r} b_{j} \boldsymbol{A}^{j},$
lyche-numerical-linear-algebra_ EQ0417 (7.1)	168	■ 1.000 ● K|+0	`\boldsymbol{A}=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{*}, \text { where } \boldsymbol{U}^{*} \boldsymbol{U}=\boldsymbol{I} .`	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{*}, \text { where } \boldsymbol{U}^{*} \boldsymbol{U}=\boldsymbol{I} .$	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{*}$, where $\boldsymbol{U}^{*} \boldsymbol{U}=\boldsymbol{I}$.
lyche-numerical-linear-algebra_ EQ0419	169	■ 1.000 ● N|+0	`\lambda_{1} \geq \cdots \geq \lambda_{r}>0=\lambda_{r+1}= \cdots=\lambda_{n} .`	$\lambda_{1} \geq \cdots \geq \lambda_{r}>0=\lambda_{r+1}= \cdots=\lambda_{n} .$	$\lambda_{1} \geq \cdots \geq \lambda_{r}>0=\lambda_{r+1}= \cdots=\lambda_{n} .$
lyche-numerical-linear-algebra_ EQ0421 (7.3)	170	■ 1.000 ● N|+0	`\lambda=\frac{\boldsymbol{v}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}}{\boldsymbol{v}^{*} \boldsymbol{v}}=\frac{|\boldsymbol{A} \boldsymbol{v}|_{2}^{2}}{|\boldsymbol{v}|_{2}^{2}} \geq 0 .`	$\lambda=\frac{\boldsymbol{v}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}}{\boldsymbol{v}^{*} \boldsymbol{v}}=\frac{\lVert\boldsymbol{A} \boldsymbol{v}\rVert_{2}^{2}}{\lVert\boldsymbol{v}\rVert_{2}^{2}} \geq 0 .$	$\lambda=\frac{\boldsymbol{v}^{*} \boldsymbol{A}^{*} \boldsymbol{A} \boldsymbol{v}}{\boldsymbol{v}^{*} \boldsymbol{v}}=\frac{\lVert\boldsymbol{A} \boldsymbol{v}\rVert_{2}^{2}}{\lVert\boldsymbol{v}\rVert_{2}^{2}} \geq 0 .$
lyche-numerical-linear-algebra_ EQ0423	171	■ 1.000 ● K|+0	`\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}= \frac{1}{25}\left[\begin{array}{ll} 52 & 36 \\ 36 & 73 \end{array}\right]`	$\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}= \frac{1}{25}\left[\begin{array}{ll} 52 & 36 \\ 36 & 73 \end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}= \frac{1}{25}\left[\begin{array}{ll} 52 & 36 \\ 36 & 73 \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0424	171	■ 1.000 ● N|+0	`\boldsymbol{B}\left[\begin{array}{l} 3 \\ 4 \end{array} \right]=4\left[\begin{array}{l} 3 \\ 4 \end{array} \right], \quad \boldsymbol{B}\left[\begin{array}{r} 4 \\ -3 \end{array}\right]=1\left[\begin{array}{r} 4 \\ -3 \end{array}\right] .`	$\boldsymbol{B}\left[\begin{array}{l} 3 \\ 4 \end{array} \right]=4\left[\begin{array}{l} 3 \\ 4 \end{array} \right], \quad \boldsymbol{B}\left[\begin{array}{r} 4 \\ -3 \end{array}\right]=1\left[\begin{array}{r} 4 \\ -3 \end{array}\right] .$	$\boldsymbol{B}\left[\begin{array}{l} 3 \\ 4 \end{array} \right]=4\left[\begin{array}{l} 3 \\ 4 \end{array} \right], \quad \boldsymbol{B}\left[\begin{array}{r} 4 \\ -3 \end{array}\right]=1\left[\begin{array}{r} 4 \\ -3 \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0428 (7.6)	172	■ 1.000 ● N|+0	`\boldsymbol{A}=\sum_{j=1}^{r} \sigma_{j} \boldsymbol{u}_{j} \boldsymbol{v}_{j}^{*} .`	$\boldsymbol{A}=\sum_{j=1}^{r} \sigma_{j} \boldsymbol{u}_{j} \boldsymbol{v}_{j}^{*} .$	$\boldsymbol{A}=\sum_{j=1}^{r} \sigma_{j} \boldsymbol{u}_{j} \boldsymbol{v}_{j}^{*} .$
lyche-numerical-linear-algebra_ EQ0430	172	■ 1.000 ● K|+0	`\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}=\left[ \begin{array}{ll} 2 & 2 \\ 2 & 2 \end{array}\right]`	$\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}=\left[\begin{array}{ll} 2 & 2 \\ 2 & 2 \end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^{T} \boldsymbol{A}=\left[\begin{array}{ll} 2 & 2 \\ 2 & 2 \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0433	173	■ 1.000 ● K|+0	`\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[ \begin{array}{ll} 1 & 1 \end{array}\right] .`	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right] .$	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0434	173	■ 1.000 ● N|+1	`\boldsymbol{s}_{1}=\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_{2}=\left[ \begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_{3}=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],`	$\boldsymbol{s}_{1}=\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_{2}=\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_{3}=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],$	$\boldsymbol{s}_{1}=\left[\begin{smallmatrix} 1 \\ 1 \\ 0 \end{smallmatrix}\right], \quad \boldsymbol{s}_{2}=\left[\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}\right], \quad \boldsymbol{s}_{3}=\left[\begin{smallmatrix} 0 \\ 0 \\ 1 \end{smallmatrix}\right],$
lyche-numerical-linear-algebra_ EQ0437	174	■ 1.000 ● W|+0	`\mathcal{E}=\boldsymbol{U}_{1} \tilde{\mathcal{E}} \text { where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}= \left[y_{1}, \ldots, y_{n}\right]^{T} \in \mathbb{R}^{n}: \frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+\frac{y_{n}^{2}}{ \sigma_{n}^{2}}=1\right\} .`	$\mathcal{E}=\boldsymbol{U}_{1} \tilde{\mathcal{E}} \text { where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}= \left[y_{1}, \ldots, y_{n}\right]^{T} \in \mathbb{R}^{n}: \frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+\frac{y_{n}^{2}}{ \sigma_{n}^{2}}=1\right\} .$	$\mathcal{E}=\boldsymbol{U}_{1} \tilde{\mathcal{E}}$ *where* $\tilde{\mathcal{E}}:=\{\boldsymbol{y}=[y_{1}, \ldots, y_{n}]^{T} \in \mathbb{R}^{n}: \frac{y_{1}^{2}}{\sigma_{1}^{2}}+\cdots+\frac{y_{n}^{2}}{\sigma_{n}^{2}}=1\}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0441 (7.9)</td><td>176</td><td>■ 1.000 ● K|+0</td><td>\|\boldsymbol{A}\|_{F}:=\left(\sum_{i=1}^{m} \sum_{j=1}^{n}\left|a_{i j}\right|^{2}\right)^{1 / 2} .</td><td>$\|\boldsymbol{A}\|_{F}:=\left(\sum_{i=1}^{m} \sum_{j=1}^{n}\left|a_{i j}\right|^{2}\right)^{1 / 2} .$</td><td>$\|\boldsymbol{A}\|_F := \big(\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2\big)^{1/2}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0444</td><td>177</td><td>■ 1.000 ● N|+0</td><td>\left\|\boldsymbol{A}-\boldsymbol{A}^{\prime}\right \|_{F}=\left\|\boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^{*}\right\|_{F}=\left \|\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{ \prime} \\ \mathbf{0} \end{array}\right]\right\|_{F}= \sqrt{\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}}<\epsilon .</td><td>$\left\|\boldsymbol{A}-\boldsymbol{A}^{\prime}\right\|_{F}=\left\|\boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^{*}\right\|_{F}=\left\|\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right]\right\|_{F}=\sqrt{\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2}}<\epsilon .$</td><td>$\|\boldsymbol{A}-\boldsymbol{A}'\|_F = \|\boldsymbol{U}\left[\begin{smallmatrix}\boldsymbol{D}-\boldsymbol{D}'\\ \mathbf{0}\end{smallmatrix}\right]\boldsymbol{V}^*\|_F = \|\left[\begin{smallmatrix}\boldsymbol{D}-\boldsymbol{D}'\\ \mathbf{0}\end{smallmatrix}\right]\|_F = \sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2} < \epsilon.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0449</td><td>180</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right] \in \mathbb{R}^{(m+n) \times(m+n)}</td><td>$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^{*} & \mathbf{0} \end{array}\right] \in \mathbb{R}^{(m+n) \times(m+n)}$</td><td>$\boldsymbol{C} := \begin{bmatrix}\mathbf{0} & \boldsymbol{A}\\ \boldsymbol{A}^* & \mathbf{0}\end{bmatrix} \in \mathbb{R}^{(m+n)\times(m+n)}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0450</td><td>180</td><td>■ 1.000 ● N|+0</td><td>\left\{\left(\sigma_{1}, \boldsymbol{p}_{1}\right), \ldots,\left(\sigma_{n}, \boldsymbol{p}_{n}\right), \left(-\sigma_{1}, \boldsymbol{q}_{1}\right), \ldots,\left(-\sigma_{n}, \boldsymbol{q}_{n}\right), \left(0, \boldsymbol{r}_{n+1}\right), \ldots,\left(0, \boldsymbol{r}_{m}\right)\right\},</td><td>$\left\{\left(\sigma_{1}, \boldsymbol{p}_{1}\right), \ldots,\left(\sigma_{n}, \boldsymbol{p}_{n}\right), \left(-\sigma_{1}, \boldsymbol{q}_{1}\right), \ldots,\left(-\sigma_{n}, \boldsymbol{q}_{n}\right), \left(0, \boldsymbol{r}_{n+1}\right), \ldots,\left(0, \boldsymbol{r}_{m}\right)\right\},$</td><td>$\{(\sigma_1, \boldsymbol{p}_1), \ldots, (\sigma_n, \boldsymbol{p}_n), (-\sigma_1, \boldsymbol{q}_1), \ldots, (-\sigma_n, \boldsymbol{q}_n), (0, \boldsymbol{r}_{n+1}), \ldots, (0, \boldsymbol{r}_m)\},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0452 (7.12)</td><td>180</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{Q}:=\boldsymbol{U} \boldsymbol{V}^{T}, \quad \boldsymbol{P}:=\boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^{T}</td><td>$\boldsymbol{Q}:=\boldsymbol{U} \boldsymbol{V}^{T}, \quad \boldsymbol{P}:=\boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^{T}$</td><td>$\boldsymbol{Q} := \boldsymbol{U}\boldsymbol{V}^T, \quad \boldsymbol{P} := \boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{V}^T$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0453 (7.13)</td><td>180</td><td>■ 1.000 ● N|+0</td><td>A=Q P</td><td>$A=Q P$</td><td>$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{P}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0455 (7.15)</td><td>181</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2} \boldsymbol{X}_{k}^{-T}\left(\boldsymbol{X}_{k}^{T}- \boldsymbol{Q}^{T}\right)\left(\boldsymbol{X}_{k}- \boldsymbol{Q}\right)</td><td>$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2} \boldsymbol{X}_{k}^{-T}\left(\boldsymbol{X}_{k}^{T}-\boldsymbol{Q}^{T}\right)\left(\boldsymbol{X}_{k}-\boldsymbol{Q}\right)$</td><td>$\boldsymbol{X}_{k+1} - \boldsymbol{Q} = \frac{1}{2}\boldsymbol{X}_k^{-T}(\boldsymbol{X}_k^T - \boldsymbol{Q}^T)(\boldsymbol{X}_k - \boldsymbol{Q})$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0456 (7.16)</td><td>181</td><td>■ 1.000 ● N|+0</td><td>\left\|\boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\|_{F} \leq \frac{1}{2}\left\|\boldsymbol{X}_{k}^{-1}\right \|_{F}\left\|\boldsymbol{X}_{k}-\boldsymbol{Q}\right \|_{F}^{2} .</td><td>$\left\|\boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\|_{F} \leq \frac{1}{2}\left\|\boldsymbol{X}_{k}^{-1}\right\|_{F}\left\|\boldsymbol{X}_{k}-\boldsymbol{Q}\right\|_{F}^{2} .$</td><td>$\|\boldsymbol{X}_{k+1} - \boldsymbol{Q}\|_F \le \frac{1}{2}\|\boldsymbol{X}_k^{-1}\|_F\|\boldsymbol{X}_k - \boldsymbol{Q}\|_F^2.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0458</td><td>181</td><td>■ 1.000 ● K|+0</td><td>\boldsymbol{B}=\left[\begin{array}{ccc} 1 & 0 & -1 \\ 1 & 1 & 1 \end{array}\right] .</td><td>$\boldsymbol{B}=\left[\begin{array}{ccc} 1 & 0 & -1 \\ 1 & 1 & 1 \end{array}\right] .$</td><td>$\boldsymbol{B} = \begin{bmatrix}1 & 0 & -1\\ 1 & 1 & 1\end{bmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0461</td><td>182</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}=\left[\begin{array}{rrrr} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right] .</td><td>$\boldsymbol{A}=\left[\begin{array}{rrrr} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right] .$</td><td>$\boldsymbol{A} = \begin{bmatrix}1 & -1 & -1 & -1\\ 0 & 1 & -1 & -1\\ 0 & 0 & 1 & -1\\ 0 & 0 & 0 & 1\end{bmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0464</td><td>183</td><td>■ 1.000 ● N|+0</td><td>\left\|\boldsymbol{A}-\boldsymbol{A}^{\prime}\right \|_{F}^{2}=\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2} .</td><td>$\left\|\boldsymbol{A}-\boldsymbol{A}^{\prime}\right\|_{F}^{2}=\sigma_{r+1}^{2}+\cdots+\sigma_{n}^{2} .$</td><td>$\|\boldsymbol{A} - \boldsymbol{A}'\|_F^2 = \sigma_{r+1}^2 + \cdots + \sigma_n^2.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0472</td><td>187</td><td>■ 1.000 ● N|+0</td><td>\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}:\| \boldsymbol{y}\|^{\prime}=1\right\} .</td><td>$\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}:\|\boldsymbol{y}\|^{\prime}=1\right\} .$</td><td>$\mathcal{S} := \{\boldsymbol{y} \in \mathcal{V} : \|\boldsymbol{y}\|' = 1\}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0473 (8.8)</td><td>187</td><td>■ 1.000 ● N|+0</td><td>m \leq\|\boldsymbol{y}\| \leq M, \quad \boldsymbol{y} \in \mathcal{S} .</td><td>$m \leq\|\boldsymbol{y}\| \leq M, \quad \boldsymbol{y} \in \mathcal{S} .$</td><td>$m \le \|\boldsymbol{y}\| \le M, \quad \boldsymbol{y} \in \mathcal{S}.$</td></tr>
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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra EQ0475</td><td>188</td><td>■ 1.000 ● K|+0</td><td>\mu\|\boldsymbol{A}\| \leq\|\boldsymbol{A}\|^{\prime} \leq M\|\boldsymbol{A}\|</td><td>$\mu\|\boldsymbol{A}\| \leq\|\boldsymbol{A}\|^{\prime} \leq M\|\boldsymbol{A}\|$</td><td>$\mu\|\boldsymbol{A}\| \leq\|\boldsymbol{A}\|^{\prime} \leq M\|\boldsymbol{A}\|$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0476 (8.9)</td><td>188</td><td>■ 1.000 ● S|-1</td><td>\|\boldsymbol{A}\|_{S}:=\sum_{i=1}^{m} \sum_{j=1}^{n} \left|a_{i j}\right|, \quad\|\boldsymbol{A}\|_{F}:=\left(\sum_{i=1}^{m} \sum_{j=1}^{n}\left|a_{i j}\right|^{2} \right)^{1 / 2}, \quad\|\boldsymbol{A}\|_{M}:=\max _{i, j}\left|a_{i j}\right| .</td><td>$\|\boldsymbol{A}\|_{S}:=\sum_{i=1}^{m} \sum_{j=1}^{n} \left|a_{i j}\right|, \quad\|\boldsymbol{A}\|_{F}:=\left(\sum_{i=1}^{m} \sum_{j=1}^{n}\left|a_{i j}\right|^{2} \right)^{1 / 2}, \quad\|\boldsymbol{A}\|_{M}:=\max _{i, j}\left|a_{i j}\right| .$</td><td>$\|\boldsymbol{A}\|_{S}:=\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{i j}|, \quad\|\boldsymbol{A}\|_{F}:=\big(\sum_{i=1}^{m} \sum_{j=1}^{n}|a_{i j}|^{2} \big)^{1 / 2}, \quad\|\boldsymbol{A}\|_{M}:=\max _{i, j}|a_{i j}| .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0478 (8.11)</td><td>189</td><td>■ 1.000 ● K|+0</td><td>\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\|,</td><td>$\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\|,$</td><td>$\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \, \|\boldsymbol{x}\|,$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0479 (8.12)</td><td>190</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A}\|:=\max _{\boldsymbol{x} \neq 0} \frac{ \|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\|} .</td><td>$\|\boldsymbol{A}\|:=\max _{\boldsymbol{x} \neq 0} \frac{ \|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\|} .$</td><td>$\|\boldsymbol{A}\|:=\max _{\boldsymbol{x} \neq 0} \frac{ \|\boldsymbol{A} \boldsymbol{x}\|}{\|\boldsymbol{x}\|} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0480 (8.13)</td><td>190</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A}\|=\max _{\|\boldsymbol{x}\|=1}\| \boldsymbol{A} \boldsymbol{x}\| .</td><td>$\|\boldsymbol{A}\|=\max _{\|\boldsymbol{x}\|=1}\| \boldsymbol{A} \boldsymbol{x}\| .$</td><td>$\|\boldsymbol{A}\|=\max _{\|\boldsymbol{x}\|=1}\| \boldsymbol{A} \boldsymbol{x}\| .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0483 (8.15)</td><td>190</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\| \text { for all } \boldsymbol{A} \in \mathbb{C}^{m \times n} \text { and } \boldsymbol{x} \in \mathbb{C}^{n} .</td><td>$\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\| \text { for all } \boldsymbol{A} \in \mathbb{C}^{m \times n} \text { and } \boldsymbol{x} \in \mathbb{C}^{n} .$</td><td>$\|\boldsymbol{A} \boldsymbol{x}\| \leq\|\boldsymbol{A}\| \|\boldsymbol{x}\| \text { for all } \boldsymbol{A} \in \mathbb{C}^{m \times n} \text { and } \boldsymbol{x} \in \mathbb{C}^{n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0489</td><td>192</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{1}=\sum_{k=1}^{m} \left|\sum_{j=1}^{n} a_{k j} x_{j}\right| \leq \sum_{k=1}^{m} \sum_{j=1}^{n}\left|a_{k j}\right| \left|x_{j}\right|=\sum_{j=1}^{n}\left(\sum_{k=1}^{m} \left|a_{k j}\right|\right)\left|x_{j}\right| \leq K_{1} .</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{1}=\sum_{k=1}^{m} \left|\sum_{j=1}^{n} a_{k j} x_{j}\right| \leq \sum_{k=1}^{m} \sum_{j=1}^{n}\left|a_{k j}\right| \left|x_{j}\right|=\sum_{j=1}^{n}\left(\sum_{k=1}^{m} \left|a_{k j}\right|\right)\left|x_{j}\right| \leq K_{1} .$</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{1}=\sum_{k=1}^{m} \big|\sum_{j=1}^{n} a_{k j} x_{j}\big| \leq \sum_{k=1}^{m} \sum_{j=1}^{n}|a_{k j}| |x_{j}|=\sum_{j=1}^{n}\big(\sum_{k=1}^{m} |a_{k j}|\big)|x_{j}| \leq K_{1} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra EQ0491</td><td>192</td><td>■ 1.000 ● K|+0</td><td>\|\boldsymbol{A}\|_{1}=\frac{29}{15}, \quad\| \boldsymbol{A}\|_{2}=2, \quad\|\boldsymbol{A}\|_{\infty}= \frac{37}{15}, \quad\|\boldsymbol{A}\|_{F}=\sqrt{5} .</td><td>$\|\boldsymbol{A}\|_{1}=\frac{29}{15}, \quad\| \boldsymbol{A}\|_{2}=2, \quad\|\boldsymbol{A}\|_{\infty}= \frac{37}{15}, \quad\|\boldsymbol{A}\|_{F}=\sqrt{5} .$</td><td>$\|\boldsymbol{A}\|_{1}=\frac{29}{15}, \quad\| \boldsymbol{A}\|_{2}=2, \quad\|\boldsymbol{A}\|_{\infty}= \frac{37}{15}, \quad\|\boldsymbol{A}\|_{F}=\sqrt{5} .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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● K|+0</td><td>x=\frac{x_{2}-x}{x_{2}-x_{1}} x_{1}+\frac{x-x_{1}}{x_{2}-x_{1}} x_{2}=(1-\lambda) x_{1}+\lambda x_{2}</td><td>$x=\frac{x_{2}-x}{x_{2}-x_{1}} x_{1}+\frac{x-x_{1}}{x_{2}-x_{1}} x_{2}=(1-\lambda) x_{1}+\lambda x_{2}$</td><td>$x=\frac{x_{2}-x}{x_{2}-x_{1}} x_{1}+\frac{x-x_{1}}{x_{2}-x_{1}} x_{2}=(1-\lambda) x_{1}+\lambda x_{2}$</td></tr>
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● N|+0</td><td>f\left(\sum_{j=1}^{n} \lambda_{j} z_{j}\right) \leq \sum_{j=1}^{n} \lambda_{j} f\left(z_{j}\right) .</td><td>$f\left(\sum_{j=1}^{n} \lambda_{j} z_{j}\right) \leq \sum_{j=1}^{n} \lambda_{j} f\left(z_{j}\right) .$</td><td>$f(\sum_{j=1}^{n} \lambda_{j} z_{j}) \leq \sum_{j=1}^{n} \lambda_{j} f(z_{j}) .$</td></tr>
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● N|+0</td><td>f\left(\sum_{j=1}^{n} \lambda_{j} z_{j}\right)=f\left(\lambda_{1} z_{1}+\left(1-\lambda_{1}\right) u\right) \leq \lambda_{1} f\left(z_{1}\right)+\left(1-\lambda_{1} \right) f(u) \leq \sum_{j=1}^{n} \lambda_{j} f\left(z_{j} \right)</td><td>$f\left(\sum_{j=1}^{n} \lambda_{j} z_{j}\right)=f\left(\lambda_{1} z_{1}+\left(1-\lambda_{1}\right) u\right) \leq \lambda_{1} f\left(z_{1}\right)+\left(1-\lambda_{1} \right) f(u) \leq \sum_{j=1}^{n} \lambda_{j} f\left(z_{j} \right)$</td><td>$f\Big(\sum_{j=1}^{n} \lambda_{j} z_{j}\Big)=f(\lambda_{1} z_{1}+(1-\lambda_{1}) u) \leq \lambda_{1} f(z_{1})+(1-\lambda_{1}) f(u) \leq \sum_{j=1}^{n} \lambda_{j} f(z_{j})$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0518</td><td>201</td><td>■ 1.000
● S|+0</td><td>-\log \left(\sum_{j=1}^{n} \lambda_{j} a_{j}\right) \leq-\sum_{j=1}^{n} \lambda_{j} \log \left(a_{j}\right)=-\log \left(a_{1}^{\lambda_{1}} \cdots a_{n}^{\lambda_{n}} \right)</td><td>$-\log \left(\sum_{j=1}^{n} \lambda_{j} a_{j}\right) \leq-\sum_{j=1}^{n} \lambda_{j} \log \left(a_{j}\right)=-\log \left(a_{1}^{\lambda_{1}} \cdots a_{n}^{\lambda_{n}} \right)$</td><td>$-\log (\sum_{j=1}^{n} \lambda_{j} a_{j}) \leq-\sum_{j=1}^{n} \lambda_{j} \log (a_{j})=-\log (a_{1}^{\lambda_{1}} \cdots a_{n}^{\lambda_{n}})$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0519 (8.33)</td><td>201</td><td>■ 1.000
● N|+0</td><td>\left(a_{1} a_{2} \cdots a_{n}\right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^{n} a_{j} .</td><td>$\left(a_{1} a_{2} \cdots a_{n}\right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^{n} a_{j} .$</td><td>$(a_{1} a_{2} \cdots a_{n})^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^{n} a_{j} .$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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EQ0520</td><td>201</td><td>■ 1.000
● N|+1</td><td>\sum_{j=1}^{n}\left|x_{j} y_{j}\right| \leq\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}, \text { where } \frac{1}{p}+\frac{1}{q}=1</td><td>$\sum_{j=1}^{n}\left|x_{j} y_{j}\right| \leq\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}, \text { where } \frac{1}{p}+\frac{1}{q}=1$</td><td>$\sum_{j=1}^{n}|x_{j} y_{j}| \leq\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q},\ \textit{where}\ \frac{1}{p}+\frac{1}{q}=1.$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0522</td><td>201</td><td>■ 1.000
● W|+0</td><td>\frac{1}{\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}} \sum_{j=1}^{n}\left|x_{j} y_{j}\right|=\sum_{j=1}^{n} \left(\frac{\left|x_{j}\right|^{p}}{\|\boldsymbol{x} \|_{p}^{p}}\right)^{\frac{1}{p}}\left(\frac{\left|y_{j} \right|^{q}}{\|\boldsymbol{y}\|_{q}^{q}}\right)^{ \frac{1}{q}} \leq \sum_{j=1}^{n}\left(\frac{1}{p} \frac{\left|x_{j}\right|^{p}}{\|\boldsymbol{x} \|_{p}^{p}}+\frac{1}{q} \frac{\left|y_{j}\right|^{q}}{\| \boldsymbol{y}\|_{q}^{q}}\right)=1</td><td>$\frac{1}{\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}} \sum_{j=1}^{n}\left|x_{j} y_{j}\right|=\sum_{j=1}^{n} \left(\frac{\left|x_{j}\right|^{p}}{\|\boldsymbol{x} \|_{p}^{p}}\right)^{\frac{1}{p}}\left(\frac{\left|y_{j} \right|^{q}}{\|\boldsymbol{y}\|_{q}^{q}}\right)^{ \frac{1}{q}} \leq \sum_{j=1}^{n}\left(\frac{1}{p} \frac{\left|x_{j}\right|^{p}}{\|\boldsymbol{x} \|_{p}^{p}}+\frac{1}{q} \frac{\left|y_{j}\right|^{q}}{\| \boldsymbol{y}\|_{q}^{q}}\right)=1$</td><td>$\frac{1}{\|\boldsymbol{x}\|_{p}\|\boldsymbol{y}\|_{q}} \sum_{j=1}^{n}|x_{j} y_{j}|=\sum_{j=1}^{n} \left(\frac{|x_{j}|^{p}}{\|\boldsymbol{x} \|_{p}^{p}}\right)^{\frac{1}{p}}\left(\frac{|y_{j}|^{q}}{\|\boldsymbol{y}\|_{q}^{q}}\right)^{ \frac{1}{q}} \leq \sum_{j=1}^{n}\left(\frac{1}{p} \frac{|x_{j}|^{p}}{\|\boldsymbol{x} \|_{p}^{p}}+\frac{1}{q} \frac{|y_{j}|^{q}}{\| \boldsymbol{y}\|_{q}^{q}}\right)=1$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0523</td><td>202</td><td>■ 1.000
● N|+0</td><td>\|\boldsymbol{x}+\boldsymbol{y}\|_{p} \leq\|\boldsymbol{x}\|_{p}+\|\boldsymbol{y}\|_{p} .</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p} \leq\|\boldsymbol{x}\|_{p}+\|\boldsymbol{y}\|_{p} .$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p} \leq\|\boldsymbol{x}\|_{p}+\|\boldsymbol{y}\|_{p} .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0524</td><td>202</td><td>■ 1.000
● N|+0</td><td>\|\boldsymbol{x}+\boldsymbol{y}\|_{p}^{p}=\sum_{j=1}^{n} \left|x_{j}+y_{j}\right|^{p} \leq \sum_{j=1}^{n} \left|x_{j}\right|\left|x_{j}+y_{j}\right|^{p-1}+ \sum_{j=1}^{n}\left|y_{j}\right|\left|x_{j}+y_{j} \right|^{p-1} .</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p}^{p}=\sum_{j=1}^{n} \left|x_{j}+y_{j}\right|^{p} \leq \sum_{j=1}^{n} \left|x_{j}\right|\left|x_{j}+y_{j}\right|^{p-1}+ \sum_{j=1}^{n}\left|y_{j}\right|\left|x_{j}+y_{j} \right|^{p-1} .$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p}^{p}=\sum_{j=1}^{n} |x_{j}+y_{j}|^{p} \leq \sum_{j=1}^{n} |x_{j}||x_{j}+y_{j}|^{p-1}+ \sum_{j=1}^{n}|y_{j}||x_{j}+y_{j}|^{p-1} .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0525</td><td>202</td><td>■ 1.000
● N|+0</td><td>\|\boldsymbol{x}+\boldsymbol{y}\|_{p}^{p} \leq\| \boldsymbol{x}\|_{p}\|\boldsymbol{x}+\boldsymbol{y} \|_{p}^{p / q}+\|\boldsymbol{y}\|_{p}\|\boldsymbol{x}+ \boldsymbol{y}\|_{p}^{p / q}=\left(\|\boldsymbol{x} \|_{p}+\|\boldsymbol{y}\|_{p}\right)\|\boldsymbol{x}+ \boldsymbol{y}\|_{p}^{p-1},</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p}^{p} \leq\| \boldsymbol{x}\|_{p}\|\boldsymbol{x}+\boldsymbol{y} \|_{p}^{p / q}+\|\boldsymbol{y}\|_{p}\|\boldsymbol{x}+ \boldsymbol{y}\|_{p}^{p / q}=\left(\|\boldsymbol{x} \|_{p}+\|\boldsymbol{y}\|_{p}\right)\|\boldsymbol{x}+ \boldsymbol{y}\|_{p}^{p-1},$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|_{p}^{p} \leq\| \boldsymbol{x}\|_{p}\|\boldsymbol{x}+\boldsymbol{y} \|_{p}^{p / q}+\|\boldsymbol{y}\|_{p}\|\boldsymbol{x}+ \boldsymbol{y}\|_{p}^{p / q}=(\|\boldsymbol{x} \|_{p}+\|\boldsymbol{y}\|_{p})\|\boldsymbol{x}+ \boldsymbol{y}\|_{p}^{p-1},$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0526 (8.35)</td><td>202</td><td>■ 1.000
● N|+0</td><td>\|\boldsymbol{x}+\boldsymbol{y}\|^{2}+\|\boldsymbol{x}- \boldsymbol{y}\|^{2}=2\|\boldsymbol{x}\|^{2}+2\| \boldsymbol{y}\|^{2},</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2}+\|\boldsymbol{x}- \boldsymbol{y}\|^{2}=2\|\boldsymbol{x}\|^{2}+2\| \boldsymbol{y}\|^{2},$</td><td>$\|\boldsymbol{x}+\boldsymbol{y}\|^{2}+\|\boldsymbol{x}- \boldsymbol{y}\|^{2}=2\|\boldsymbol{x}\|^{2}+2\| \boldsymbol{y}\|^{2},$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0534</td><td>203</td><td>■ 1.000
● N|+0</td><td>\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y} \right\rangle=\frac{n}{m}\langle\boldsymbol{x}, \boldsymbol{y}\rangle .</td><td>$\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y} \right\rangle=\frac{n}{m}\langle\boldsymbol{x}, \boldsymbol{y}\rangle .$</td><td>$\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y} \rangle=\frac{n}{m}\langle\boldsymbol{x}, \boldsymbol{y}\rangle .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0539</td><td>204</td><td>■ 1.000
● N|+0</td><td>\tilde{\boldsymbol{b}}_{k}:=\boldsymbol{b}_{k}- \boldsymbol{B}_{k-1}\left(\boldsymbol{B}_{k-1}^{T} \boldsymbol{A} \boldsymbol{B}_{k-1}\right)^{-1} \boldsymbol{B}_{k-1}^{T} \boldsymbol{A} \boldsymbol{b}_{k}, \quad k=2, \ldots, n .</td><td>$\tilde{\boldsymbol{b}}_{k}:=\boldsymbol{b}_{k}- \boldsymbol{B}_{k-1}\left(\boldsymbol{B}_{k-1}^{T} \boldsymbol{A} \boldsymbol{B}_{k-1}\right)^{-1} \boldsymbol{B}_{k-1}^{T} \boldsymbol{A} \boldsymbol{b}_{k}, \quad k=2, \ldots, n .$</td><td>$\tilde{\boldsymbol{b}}_{k}:=\boldsymbol{b}_{k}- \boldsymbol{B}_{k-1}\big(\boldsymbol{B}_{k-1}^{T} \boldsymbol{A} \boldsymbol{B}_{k-1}\big)^{-1} \boldsymbol{B}_{k-1}^{T} \boldsymbol{A} \boldsymbol{b}_{k}, \quad k=2, \ldots, n .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0540</td><td>205</td><td>■ 1.000
● N|+0</td><td>\left\|\boldsymbol{b}_{1}\right\|_{\boldsymbol{A}}\left \|\boldsymbol{b}_{2}\right\|_{\boldsymbol{A}} \cdots \left\|\boldsymbol{b}_{n}\right\|_{\boldsymbol{A}} \leq 2^{n(n-1) / 4} \sqrt{\operatorname{det}(\boldsymbol{A})} .</td><td>$\left\|\boldsymbol{b}_{1}\right\|_{\boldsymbol{A}}\left \|\boldsymbol{b}_{2}\right\|_{\boldsymbol{A}} \cdots \left\|\boldsymbol{b}_{n}\right\|_{\boldsymbol{A}} \leq 2^{n(n-1) / 4} \sqrt{\operatorname{det}(\boldsymbol{A})} .$</td><td>$\|\boldsymbol{b}_{1}\|_{\boldsymbol{A}}\|\boldsymbol{b}_{2}\|_{\boldsymbol{A}} \cdots \|\boldsymbol{b}_{n}\|_{\boldsymbol{A}} \leq 2^{n(n-1) / 4} \sqrt{\operatorname{det}(\boldsymbol{A})} .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0541</td><td>205</td><td>■ 1.000
● N|+0</td><td>\|\boldsymbol{A}\|:=\sqrt{m n}\|\boldsymbol{A}\|_{M}, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}</td><td>$\|\boldsymbol{A}\|:=\sqrt{m n}\|\boldsymbol{A}\|_{M}, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$</td><td>$\|\boldsymbol{A}\|:=\sqrt{m n}\|\boldsymbol{A}\|_{M}, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0542</td><td>205</td><td>■ 1.000
● N|-1</td><td>\|\boldsymbol{A}\|_{2}=\max _{\|\boldsymbol{x}\|_{2}= \|\boldsymbol{y}\|_{2}=1}\left|\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}\right| .</td><td>$\|\boldsymbol{A}\|_{2}=\max _{\|\boldsymbol{x}\|_{2}= \|\boldsymbol{y}\|_{2}=1}\left|\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}\right| .$</td><td>$\|\boldsymbol{A}\|_{2}=\max _{\|\boldsymbol{x}\|_{2}= \|\boldsymbol{y}\|_{2}=1}|\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{x}| .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0546</td><td>207</td><td>1.000 N|+0</td><td>\boldsymbol{r}_{k}=\sum_{i=1}^{n} c_{i k} \boldsymbol{u}_{i},</td><td>$\boldsymbol{r}_{k}=\sum_{i=1}^{n} c_{i k} \boldsymbol{u}_{i},$</td><td>$\boldsymbol{r}_{k}=\sum_{i=1}^{n} c_{i k} \boldsymbol{u}_{i},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0548</td><td>207</td><td>1.000 N|+0</td><td>\lambda_{j}=d+2 c \cos \left(\frac{j \pi}{n+1}\right), \quad j=1, \ldots, n .</td><td>$\lambda_{j}=d+2 c \cos \left(\frac{j \pi}{n+1}\right), \quad j=1, \ldots, n .$</td><td>$\lambda_{j}=d+2 c \cos \left(\frac{j \pi}{n+1}\right), \quad j=1, \ldots, n .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0549 (8.40)</td><td>207</td><td>1.000 K|+0</td><td>\boldsymbol{T} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B} \boldsymbol{x}_{k} .</td><td>$\boldsymbol{T} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B} \boldsymbol{x}_{k} .$</td><td>$\boldsymbol{T} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{B} \boldsymbol{x}_{k} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0550</td><td>207</td><td>1.000 N|-2</td><td>\min \left\{\max _{i} \sum_{j=1}^{n}\left|b_{i j}\right|, \max _{j} \sum_{i=1}^{n}\left|b_{i j}\right|\right\}<d-2 c .</td><td>$\min \left\{\max _{i} \sum_{j=1}^{n}\left|b_{i j}\right|, \max _{j} \sum_{i=1}^{n}\left|b_{i j}\right|\right\}<d-2 c .$</td><td>$\min \left\{\max _{i} \sum_{j=1}^{n}\left|b_{i j}\right|, \max _{j} \sum_{i=1}^{n}\left|b_{i j}\right|\right\}<d-2 c .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0552 (8.41)</td><td>208</td><td>1.000 N|+0</td><td>\frac{4}{\pi^{2}}(m+1)^{2}-2 / 3<\operatorname{cond}_{p}(\boldsymbol{T}) \leq \frac{1}{2}(m+1)^{2}, \quad p=1,2, \infty,</td><td>$\frac{4}{\pi^{2}}(m+1)^{2}-2 / 3<\operatorname{cond}_{p}(\boldsymbol{T}) \leq \frac{1}{2}(m+1)^{2}, \quad p=1,2, \infty,$</td><td>$\frac{4}{\pi^{2}}(m+1)^{2}-2 / 3<\operatorname{cond}_{p}(\boldsymbol{T}) \leq \frac{1}{2}(m+1)^{2}, \quad p=1,2, \infty,$</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0555 (8.43)</td><td>209</td><td>1.000 N|+0</td><td>\frac{4}{\pi^{2}} h^{-2}-\frac{2}{3}< \operatorname{cond}_{2}(\boldsymbol{T})<\frac{4}{\pi^{2}} h^{-2} .</td><td>$\frac{4}{\pi^{2}} h^{-2}-\frac{2}{3}<\operatorname{cond}_{2}(\boldsymbol{T})<\frac{4}{\pi^{2}} h^{-2} .$</td><td>$\frac{4}{\pi^{2}} h^{-2}-\frac{2}{3}<\operatorname{cond}_{2}(\boldsymbol{T})<\frac{4}{\pi^{2}} h^{-2} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0556</td><td>209</td><td>1.000 N|+0</td><td>\frac{\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\right\|}{\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}\right\|} \leq\|\boldsymbol{E}\|</td><td>$\frac{\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\right\|}{\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}\right\|} \leq\|\boldsymbol{E}\|$</td><td>$\frac{\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}-\boldsymbol{I}\right\|}{\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}\right\|} \leq\|\boldsymbol{E}\|$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0557</td><td>209</td><td>1.000 K|+0</td><td>\frac{1}{1+\|\boldsymbol{E}\|} \leq\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}\right\| \leq \frac{1}{1-\|\boldsymbol{E}\|}</td><td>$\frac{1}{1+\|\boldsymbol{E}\|} \leq\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}\right\| \leq \frac{1}{1-\|\boldsymbol{E}\|}$</td><td>$\frac{1}{1+\|\boldsymbol{E}\|} \leq\left\|(\boldsymbol{I}-\boldsymbol{E})^{-1}\right\| \leq \frac{1}{1-\|\boldsymbol{E}\|}$</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0560</td><td>209</td><td>1.000 N|+0</td><td>\frac{K(\boldsymbol{B})^{-1}}{1+r} \frac{\| \boldsymbol{E}\|}{\|\boldsymbol{A}\|} \leq \frac{\left\| \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\|}{\left\| \boldsymbol{A}^{-1}\right\|} .</td><td>$\frac{K(\boldsymbol{B})^{-1}}{1+r} \frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|} \leq \frac{\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\|}{\left\|\boldsymbol{A}^{-1}\right\|} .$</td><td>$\frac{K(\boldsymbol{B})^{-1}}{1+r} \frac{\|\boldsymbol{E}\|}{\|\boldsymbol{A}\|} \leq \frac{\left\|\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\|}{\left\|\boldsymbol{A}^{-1}\right\|} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0561</td><td>209</td><td>1.000 K|+0</td><td>a=x_{0}<x_{1}<\cdots<x_{n}=b,</td><td>$a=x_{0}<x_{1}<\cdots<x_{n}=b,$</td><td>$a=x_{0}<x_{1}<\cdots<x_{n}=b,$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0562	210	1.000 W|+0	`\begin{array}{ll} g\left(x_{i}\right)=y_{i}, & i=0,1, \ldots, n, \\ g^{\prime}(a)=g^{\prime}(b), & g^{\prime \prime}(a)=g^{\prime \prime}(b) . \end{array}`	$\begin{array}{ll} g\left(x_{i}\right)=y_{i}, & i=0,1, \ldots, n, \\ g^{\prime}(a)=g^{\prime}(b), & g^{\prime \prime}(a)=g^{\prime \prime}(b) . \end{array}$	$g(x_i)=y_i, \quad i=0,1,\ldots,n,$ $g'(a)=g'(b), \quad g''(a)=g''(b).$
lyche-numerical-linear-algebra_ EQ0565	210	1.000 S|+0	`\lambda_{i}:=\frac{h_{i}}{h_{i-1}+h_{i}}, \quad \mu_{i}:=\frac{h_{i-1}}{h_{i-1}+h_{i}}, \quad, i=1, \ldots, n,`	$\lambda_{i}:=\frac{h_{i}}{h_{i-1}+h_{i}}, \quad \mu_{i}:=\frac{h_{i-1}}{h_{i-1}+h_{i}}, \quad, i=1, \ldots, n,$	$\lambda_i:=\frac{h_i}{h_{i-1}+h_i}, \quad \mu_i:=\frac{h_{i-1}}{h_{i-1}+h_i}, \quad, i=1,\ldots,n,$
lyche-numerical-linear-algebra_ EQ0566	210	1.000 N|-1	`h_{i}=x_{i+1}-x_{i}, \quad i=0, \ldots, n-1, \text { and } h_{n}=h_{0} .`	$h_{i}=x_{i+1}-x_{i}, \quad i=0, \ldots, n-1, \text { and } h_{n}=h_{0} .$	$h_i=x_{i+1}-x_i, \quad i=0,\ldots,n-1, \text{ and } h_n=h_0.$
lyche-numerical-linear-algebra_ EQ0567	210	1.000 N|+0	`\frac{1}{2} \leq \frac{h_{i}}{h_{i-1}} \leq 2, \quad i=1, \ldots, n .`	$\frac{1}{2} \leq \frac{h_{i}}{h_{i-1}} \leq 2, \quad i=1, \ldots, n .$	$\frac{1}{2}\leq\frac{h_i}{h_{i-1}}\leq 2, \quad i=1,\ldots,n.$
lyche-numerical-linear-algebra_ EQ0568	211	1.000 N|-1	`|\boldsymbol{A}|_{\infty}=3 \quad\text { and that } \quad|\boldsymbol{A}|_{1} \leq \frac{10}{3} .`	$\Vert\boldsymbol{A}\Vert_{\infty}=3 \quad\text { and that } \quad\Vert\boldsymbol{A}\Vert_{1} \leq \frac{10}{3} .$	$\Vert\boldsymbol{A}\Vert_\infty=3 \quad \text{and that} \quad \Vert\boldsymbol{A}\Vert_1\leq\frac{10}{3}.$
lyche-numerical-linear-algebra_ EQ0571 (8.45)	211	1.000 N|+0	`\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\frac{1}{4}\left(|\boldsymbol{x}+\boldsymbol{y}|^{2}-|\boldsymbol{x}-\boldsymbol{y}|^{2}+i|\boldsymbol{x}+i \boldsymbol{y}|^{2}-i|\boldsymbol{x}-i \boldsymbol{y}|^{2}\right),`	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\frac{1}{4}\left(\Vert\boldsymbol{x}+\boldsymbol{y}\Vert^{2}-\Vert\boldsymbol{x}-\boldsymbol{y}\Vert^{2}+i\Vert\boldsymbol{x}+i \boldsymbol{y}\Vert^{2}-i\Vert\boldsymbol{x}-i \boldsymbol{y}\Vert^{2}\right),$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle:=\frac{1}{4}\big(\Vert\boldsymbol{x}+\boldsymbol{y}\Vert^2-\Vert\boldsymbol{x}-\boldsymbol{y}\Vert^2+i\Vert\boldsymbol{x}+i\boldsymbol{y}\Vert^2-i\Vert\boldsymbol{x}-i\boldsymbol{y}\Vert^2\big),$
lyche-numerical-linear-algebra_ EQ0574	213	1.000 K|+0	`E(\boldsymbol{x}):=|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}|_{2}^{2},`	$E(\boldsymbol{x}):=\Vert\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\Vert_{2}^{2},$	$E(\boldsymbol{x}):=\Vert\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\Vert_2^2,$
lyche-numerical-linear-algebra_ EQ0576	214	1.000 N|+1	`|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}|_{2}^{2}=\left(x_{1}-1\right)^{2}+\left(x_{1}-1\right)^{2}+\left(x_{1}-2\right)^{2}=3 x_{1}^{2}-8 x_{1}+6 .`	$\Vert\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\Vert_{2}^{2}=\left(x_{1}-1\right)^{2}+\left(x_{1}-1\right)^{2}+\left(x_{1}-2\right)^{2}=3 x_{1}^{2}-8 x_{1}+6 .$	$\Vert\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\Vert_2^2=(x_1-1)^2+(x_1-1)^2+(x_1-2)^2=3x_1^2-8x_1+6.$
lyche-numerical-linear-algebra_ EQ0581	215	1.000 N|+0	`y=\boldsymbol{u}^{*} \boldsymbol{x}=\sum_{i=1}^{n} u_{i} x_{i},`	$y=\boldsymbol{u}^{*} \boldsymbol{x}=\sum_{i=1}^{n} u_{i} x_{i},$	$y=\boldsymbol{u}^*\boldsymbol{x}=\sum_{i=1}^{n}u_ix_i,$
lyche-numerical-linear-algebra_ EQ0585 (9.2)	216	1.000 N|+0	`E(\boldsymbol{x}):=|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}|_{2}^{2}=\sum_{k=1}^{m}\left(\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right)-y_{k}\right)^{2} .`	$E(\boldsymbol{x}):=\Vert\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\Vert_{2}^{2}=\sum_{k=1}^{m}\left(\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right)-y_{k}\right)^{2} .$	$E(\boldsymbol{x}):=\Vert\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\Vert_2^2=\sum_{k=1}^{m}\big(\sum_{j=1}^{n}x_j\phi_j(t_k)-y_k\big)^2.$
lyche-numerical-linear-algebra_ EQ0586	217	1.000 N|+0	`E(\boldsymbol{x}):=\sum_{k=1}^{m} w_{k}\left(\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right)-y_{k}\right)^{2} .`	$E(\boldsymbol{x}):=\sum_{k=1}^{m} w_{k}\left(\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right)-y_{k}\right)^{2} .$	$E(\boldsymbol{x}):=\sum_{k=1}^{m}w_k\big(\sum_{j=1}^{n}x_j\phi_j(t_k)-y_k\big)^2.$
lyche-numerical-linear-algebra_ EQ0587 (9.3)	217	1.000 K|+0	`p\left(t_{k}\right):=\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right)=0, \quad k=1, \ldots, m \Rightarrow x_{1}=\cdots=x_{n}=0 .`	$p\left(t_{k}\right):=\sum_{j=1}^{n} x_{j} \phi_{j}\left(t_{k}\right)=0, \quad k=1, \ldots, m \Rightarrow x_{1}=\cdots=x_{n}=0 .$	$p(t_k):=\sum_{j=1}^{n}x_j\phi_j(t_k)=0, \quad k=1,\ldots,m \;\Rightarrow\; x_1=\cdots=x_n=0.$
lyche-numerical-linear-algebra_ EQ0595	219	1.000 W|+2	`\left(\boldsymbol{A}^{*} \boldsymbol{b}\right)_{i}=\sum_{k=1}^{m} \bar{a}_{k, i} b_{k}, i=1, \ldots, n,`	$\left(\boldsymbol{A}^{*} \boldsymbol{b}\right)_{i}=\sum_{k=1}^{m} \bar{a}_{k, i} b_{k}, i=1, \ldots, n,$	$(\boldsymbol{A}^*\boldsymbol{b})_i=\sum_{k=1}^{m}\bar{a}_{k,i}b_k, i=1,\ldots,n,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>lyche-numerical-linear-al… EQ0596 (9.6)</td><td>220</td><td>■ 1.000 ● N|+1</td><td><code>K_{2}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right):= \left\|\boldsymbol{A}^{*} \boldsymbol{A}\right \|_{2}\left\|\left(\boldsymbol{A}^{*} \boldsymbol{A} \right)^{-1}\right\|_{2}=\frac{\lambda_{1}}{ \lambda_{n}}=\frac{\sigma_{1}^{2}}{\sigma_{n}^{2}}=K_{2}(\boldsymbol{A})^{2},</code></td><td>$K_{2}\left(\boldsymbol{A}^{*} \boldsymbol{A}\right):= \left\|\boldsymbol{A}^{*} \boldsymbol{A}\right\|_{2}\left\|\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1}\right\|_{2}=\frac{\lambda_{1}}{\lambda_{n}}=\frac{\sigma_{1}^{2}}{\sigma_{n}^{2}}=K_{2}(\boldsymbol{A})^{2},$</td><td>$K_2(\boldsymbol{A}^*\boldsymbol{A}) := \|\boldsymbol{A}^*\boldsymbol{A}\|_2\|(\boldsymbol{A}^*\boldsymbol{A})^{-1}\|_2 = \frac{\lambda_1}{\lambda_n} = \frac{\sigma_1^2}{\sigma_n^2} = K_2(\boldsymbol{A})^2,$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0597</td><td>220</td><td>■ 1.000 ● N|+0</td><td><code>\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{R}_{1}^{*} \boldsymbol{Q}_{1}^{*} \boldsymbol{Q}_{1} \boldsymbol{R}_{1}=\boldsymbol{R}_{1}^{*} \boldsymbol{R}_{1}, \quad \boldsymbol{A}^{*} \boldsymbol{b}=\boldsymbol{R}_{1}^{*} \boldsymbol{Q}_{1}^{*} \boldsymbol{b} .</code></td><td>$\boldsymbol{A}^{*} \boldsymbol{A}=\boldsymbol{R}_{1}^{*} \boldsymbol{Q}_{1}^{*} \boldsymbol{Q}_{1} \boldsymbol{R}_{1}=\boldsymbol{R}_{1}^{*} \boldsymbol{R}_{1}, \quad \boldsymbol{A}^{*} \boldsymbol{b}=\boldsymbol{R}_{1}^{*} \boldsymbol{Q}_{1}^{*} \boldsymbol{b} .$</td><td>$\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{R}_1^*\boldsymbol{Q}_1^*\boldsymbol{Q}_1\boldsymbol{R}_1 = \boldsymbol{R}_1^*\boldsymbol{R}_1, \quad \boldsymbol{A}^*\boldsymbol{b} = \boldsymbol{R}_1^*\boldsymbol{Q}_1^*\boldsymbol{b}.$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0605</td><td>222</td><td>■ 1.000 ● N|+0</td><td><code>A=U_{1} \Sigma_{1} V_{1}^{*}, \quad A^{\dagger}:=V_{1} \Sigma_{1}^{-1} U_{1}^{*}</code></td><td>$A=U_{1} \Sigma_{1} V_{1}^{*}, \quad A^{\dagger}:=V_{1} \Sigma_{1}^{-1} U_{1}^{*}$</td><td>$\boldsymbol{A} = \boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*, \quad \boldsymbol{A}^\dagger := \boldsymbol{V}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{U}_1^*$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0612</td><td>224</td><td>■ 1.000 ● K|+0</td><td><code>\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}= \boldsymbol{A}^{\dagger} \boldsymbol{b}_{1}, \quad \boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*},</code></td><td>$\boldsymbol{x}=\boldsymbol{A}^{\dagger} \boldsymbol{b}=\boldsymbol{A}^{\dagger} \boldsymbol{b}_{1}, \quad \boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)^{-1} \boldsymbol{A}^{*},$</td><td>$\boldsymbol{x} = \boldsymbol{A}^\dagger\boldsymbol{b} = \boldsymbol{A}^\dagger\boldsymbol{b}_1, \quad \boldsymbol{A}^\dagger = (\boldsymbol{A}^*\boldsymbol{A})^{-1}\boldsymbol{A}^*,$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0616</td><td>226</td><td>■ 1.000 ● N|-1</td><td><code>\boldsymbol{b}_{1}=\boldsymbol{A} \boldsymbol{A}^{ \dagger} \boldsymbol{b}=\left[10^{-4}, 0,0\right]^{*}, \quad \text { and } \quad \boldsymbol{e}_{1}= \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{e}= \left[10^{-6}, 0,0\right]^{*} .</code></td><td>$\boldsymbol{b}_{1}=\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}=\left[10^{-4}, 0,0\right]^{*}, \quad \text { and } \quad \boldsymbol{e}_{1}=\boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{e}=\left[10^{-6}, 0,0\right]^{*} .$</td><td>$\boldsymbol{b}_1 = \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{b} = [10^{-4}, 0, 0]^*, \quad \text{and} \quad \boldsymbol{e}_1 = \boldsymbol{A}\boldsymbol{A}^\dagger\boldsymbol{e} = [10^{-6}, 0, 0]^*.$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0618</td><td>226</td><td>■ 1.000 ● N|+1</td><td><code>\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_{ \infty}}{\|\boldsymbol{x}\|_{\infty}}= \frac{10^{-6}}{10^{-4}}=10^{-2}</code></td><td>$\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_{\infty}}{\|\boldsymbol{x}\|_{\infty}}=\frac{10^{-6}}{10^{-4}}=10^{-2}$</td><td>$\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_\infty}{\|\boldsymbol{x}\|_\infty} = \frac{10^{-6}}{10^{-4}} = 10^{-2},$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0619 (9.15)</td><td>226</td><td>■ 1.000 ● S|+2</td><td><code>\rho=\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_{2}}{\| \boldsymbol{x}\|_{2}} \leq \frac{1}{\alpha} K(1+\beta K) \frac{\|\boldsymbol{E}\|_{2}}{\|\boldsymbol{A}\|_{2}}, \quad \beta=\frac{\left\|\boldsymbol{b}_{2}\right\|_{2}}{ \left\|\boldsymbol{b}_{1}\right\|_{2}}, \quad K=\| \boldsymbol{A}\|_{2}\left\|\boldsymbol{A}^{\dagger} \right\|_{2} .</code></td><td>$\rho=\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_{2}}{\|\boldsymbol{x}\|_{2}} \leq \frac{1}{\alpha} K(1+\beta K) \frac{\|\boldsymbol{E}\|_{2}}{\|\boldsymbol{A}\|_{2}}, \quad \beta=\frac{\left\|\boldsymbol{b}_{2}\right\|_{2}}{\left\|\boldsymbol{b}_{1}\right\|_{2}}, \quad K=\|\boldsymbol{A}\|_{2}\left\|\boldsymbol{A}^{\dagger}\right\|_{2} .$</td><td>$\rho = \frac{\|\boldsymbol{x}-\boldsymbol{y}\|_2}{\|\boldsymbol{x}\|_2} \le \frac{1}{\alpha}K(1+\beta K)\frac{\|\boldsymbol{E}\|_2}{\|\boldsymbol{A}\|_2}, \quad \beta = \frac{\|\boldsymbol{b}_2\|_2}{\|\boldsymbol{b}_1\|_2}, \quad K = \|\boldsymbol{A}\|_2\|\boldsymbol{A}^\dagger\|_2.$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0623 (9.18)</td><td>228</td><td>■ 1.000 ● N|+0</td><td><code>\left|\alpha_{j}-\beta_{j}\right| \leq\|\boldsymbol{A}- \boldsymbol{B}\|_{2}, \text { for } j=1,2, \ldots, n .</code></td><td>$\left|\alpha_{j}-\beta_{j}\right| \leq\|\boldsymbol{A}-\boldsymbol{B}\|_{2}, \text { for } j=1,2, \ldots, n .$</td><td>$|\alpha_j - \beta_j| \le \|\boldsymbol{A}-\boldsymbol{B}\|_2, \textit{ for } j = 1, 2, \ldots, n.$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0625</td><td>228</td><td>■ 1.000 ● W|+0</td><td><code>\left\|(\boldsymbol{A}+\boldsymbol{E})^{\dagger}\right \|_{2}=\frac{1}{\beta_{r}} \leq \frac{1}{\alpha_{r}- \epsilon_{1}}=\frac{1 / \alpha_{r}}{1-\epsilon_{1} / \alpha_{r}}=\frac{\left\|\boldsymbol{A}^{\dagger}\right \|_{2}}{1-\left\|\boldsymbol{A}^{\dagger}\right\|_{2}\| \boldsymbol{E}\|_{2}} .</code></td><td>$\left\|(\boldsymbol{A}+\boldsymbol{E})^{\dagger}\right\|_{2}=\frac{1}{\beta_{r}} \leq \frac{1}{\alpha_{r}-\epsilon_{1}}=\frac{1 / \alpha_{r}}{1-\epsilon_{1} / \alpha_{r}}=\frac{\left\|\boldsymbol{A}^{\dagger}\right\|_{2}}{1-\left\|\boldsymbol{A}^{\dagger}\right\|_{2}\|\boldsymbol{E}\|_{2}} .$</td><td>$\|(\boldsymbol{A}+\boldsymbol{E})^\dagger\|_2 = \frac{1}{\beta_r} \le \frac{1}{\alpha_r - \epsilon_1} = \frac{1/\alpha_r}{1-\epsilon_1/\alpha_r} = \frac{\|\boldsymbol{A}^\dagger\|_2}{1-\|\boldsymbol{A}^\dagger\|_2\|\boldsymbol{E}\|_2}.$</td></tr>
<tr><td>lyche-numerical-linear-al… EQ0626 (9.19)</td><td>228</td><td>■ 1.000 ● N|+0</td><td><code>\sum_{j=1}^{n}\left|\mu_{j}-\lambda_{j}\right|^{2} \leq\| \boldsymbol{A}-\boldsymbol{B}\|_{F}^{2}:=\sum_{i=1}^{n} \sum_{j=1}^{n}\left|a_{i j}-b_{i j}\right|^{2},</code></td><td>$\sum_{j=1}^{n}\left|\mu_{j}-\lambda_{j}\right|^{2} \leq\|\boldsymbol{A}-\boldsymbol{B}\|_{F}^{2}:=\sum_{i=1}^{n} \sum_{j=1}^{n}\left|a_{i j}-b_{i j}\right|^{2},$</td><td>$\sum_{j=1}^n |\mu_j - \lambda_j|^2 \le \|\boldsymbol{A}-\boldsymbol{B}\|_F^2 := \sum_{i=1}^n\sum_{j=1}^n |a_{ij} - b_{ij}|^2,$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0627 (9.20)	229	■ 1.000 ● N|+0	`\sum_{j=1}^{n}\left|\beta_{j}-\alpha_{j}\right|^{2} \leq |\boldsymbol{A}-\boldsymbol{B}|_{F}^{2} .`	$\sum_{j=1}^{n}\left\|\beta_{j}-\alpha_{j}\right\|^{2} \leq \|\boldsymbol{A}-\boldsymbol{B}\|_{F}^{2} .$	$\sum_{j=1}^{n}\|\beta_j-\alpha_j\|^2 \leq \|\boldsymbol{A}-\boldsymbol{B}\|_F^2.$
lyche-numerical-linear-algebra EQ0629 (9.21)	229	■ 1.000 ● N|+0	`\sum_{j=1}^{m+n}\left|\lambda_{j}-\mu_{j}\right|^{2} \leq |\boldsymbol{C}-\boldsymbol{D}|_{F}^{2} .`	$\sum_{j=1}^{m+n}\left\|\lambda_{j}-\mu_{j}\right\|^{2} \leq \|\boldsymbol{C}-\boldsymbol{D}\|_{F}^{2} .$	$\sum_{j=1}^{m+n}\|\lambda_j-\mu_j\|^2 \leq \|\boldsymbol{C}-\boldsymbol{D}\|_F^2.$
lyche-numerical-linear-algebra EQ0631	229	■ 1.000 ● N|-1	`t:=\max (r, s) .`	$t:=\max (r, s) .$	$t := \max(r,s).$
lyche-numerical-linear-algebra EQ0633	230	■ 1.000 ● W|+0	`\sum_{j=1}^{m+n}\left|\lambda_{j}-\mu_{j}\right|^{2}= \sum_{i=1}^{t}\left|\alpha_{i}-\beta_{i}\right|^{2}+ \sum_{i=1}^{t}\left|-\alpha_{i}+\beta_{i}\right|^{2}=2 \sum_{i=1}^{t}\left|\alpha_{i}-\beta_{i}\right|^{2}`	$\sum_{j=1}^{m+n}\left\|\lambda_{j}-\mu_{j}\right\|^{2}=\sum_{i=1}^{t}\left\|\alpha_{i}-\beta_{i}\right\|^{2}+\sum_{i=1}^{t}\left\|-\alpha_{i}+\beta_{i}\right\|^{2}=2 \sum_{i=1}^{t}\left\|\alpha_{i}-\beta_{i}\right\|^{2}$	$\sum_{j=1}^{m+n}\|\lambda_j-\mu_j\|^2=\sum_{i=1}^{t}\|\alpha_i-\beta_i\|^2+\sum_{i=1}^{t}\|-\alpha_i+\beta_i\|^2=2\sum_{i=1}^{t}\|\alpha_i-\beta_i\|^2$
lyche-numerical-linear-algebra EQ0635 (9.22)	230	■ 1.000 ● K|+0	`\left(t_{i}-c_{1}\right)^{2}+\left(y_{i}-c_{2} \right)^{2}=r^{2}, i=1, \ldots, m,`	$\left(t_{i}-c_{1}\right)^{2}+\left(y_{i}-c_{2}\right)^{2}=r^{2}, i=1, \ldots, m,$	$(t_i-c_1)^2+(y_i-c_2)^2=r^2,\ i=1,\ldots,m,$
lyche-numerical-linear-algebra EQ0636 (9.23)	230	■ 1.000 ● W|+1	`t_{i} x_{1}+y_{i} x_{2}+x_{3}=t_{i}^{2}+y_{i}^{2}, i=1, \ldots, m,`	$t_{i} x_{1}+y_{i} x_{2}+x_{3}=t_{i}^{2}+y_{i}^{2}, i=1, \ldots, m,$	$t_ix_1+y_ix_2+x_3=t_i^2+y_i^2,\ i=1,\ldots,m,$
lyche-numerical-linear-algebra EQ0637	231	■ 1.000 ● N|+0	`\sum_{i=1}^{3}\left\\|p\left(x_{i}\right)-y_{i}\right \\|^{2},`	$\sum_{i=1}^{3}\left\\|p\left(x_{i}\right)-y_{i}\right\\|^{2},$	$\sum_{i=1}^{3}\\|p(x_i)-y_i\\|^2,$
lyche-numerical-linear-algebra EQ0638	231	■ 1.000 ● N|+0	`F(\boldsymbol{x}):=|\boldsymbol{A} \boldsymbol{x}- \boldsymbol{b}|_{2}^{2}+|\boldsymbol{x}|_{2}^{2} .`	$F(\boldsymbol{x}):=\|\boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\|_{2}^{2}+\|\boldsymbol{x}\|_{2}^{2} .$	$F(\boldsymbol{x}) := \|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\|_2^2+\|\boldsymbol{x}\|_2^2.$
lyche-numerical-linear-algebra EQ0639	231	■ 1.000 ● N|+1	`F(\boldsymbol{x})=\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}-2 \boldsymbol{c}^{T} \boldsymbol{x}+ \boldsymbol{b}^{T} \boldsymbol{b}, \quad \text { where } \quad \boldsymbol{c}=\boldsymbol{A}^{T} \boldsymbol{b} .`	$F(\boldsymbol{x})=\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}-2 \boldsymbol{c}^{T} \boldsymbol{x}+\boldsymbol{b}^{T} \boldsymbol{b}, \quad \text { where } \quad \boldsymbol{c}=\boldsymbol{A}^{T} \boldsymbol{b} .$	$F(\boldsymbol{x})=\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}-2\boldsymbol{c}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{b}, \quad \text{where} \quad \boldsymbol{c}=\boldsymbol{A}^T\boldsymbol{b}.$
lyche-numerical-linear-algebra EQ0640 (9.24)	231	■ 1.000 ● N|+1	`|\boldsymbol{r}(\boldsymbol{x})|_{D}^{2}:= \sum_{i=1}^{m} r_{i}(x)^{2} d_{i}, \quad \boldsymbol{x} \in \mathbb{R}^{n}`	$\|\boldsymbol{r}(\boldsymbol{x})\|_{D}^{2}:=\sum_{i=1}^{m} r_{i}(x)^{2} d_{i}, \quad \boldsymbol{x} \in \mathbb{R}^{n}$	$\|\boldsymbol{r}(\boldsymbol{x})\|_D^2 := \sum_{i=1}^{m} r_i(x)^2 d_i, \quad \boldsymbol{x}\in\mathbb{R}^n,$
lyche-numerical-linear-algebra EQ0642	232	■ 1.000 ● K|+0	`\boldsymbol{A}^{T} \boldsymbol{D} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{A}^{T} \boldsymbol{D} \boldsymbol{b} .`	$\boldsymbol{A}^{T} \boldsymbol{D} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{A}^{T} \boldsymbol{D} \boldsymbol{b} .$	$\boldsymbol{A}^T\boldsymbol{D}\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^T\boldsymbol{D}\boldsymbol{b}.$
lyche-numerical-linear-algebra EQ0643	232	■ 1.000 ● N|+0	`K_{2}\left(\boldsymbol{A}^{T} \boldsymbol{D} \boldsymbol{A}\right) \leq K_{2}\left(\boldsymbol{A}^{T} \boldsymbol{A}\right) K_{2}(\boldsymbol{D}),`	$K_{2}\left(\boldsymbol{A}^{T} \boldsymbol{D} \boldsymbol{A}\right) \leq K_{2}\left(\boldsymbol{A}^{T} \boldsymbol{A}\right) K_{2}(\boldsymbol{D}),$	$K_2(\boldsymbol{A}^T\boldsymbol{D}\boldsymbol{A}) \leq K_2(\boldsymbol{A}^T\boldsymbol{A})K_2(\boldsymbol{D}),$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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$$\boldsymbol{P}_{\mathcal{N}(\boldsymbol{A})}=\boldsymbol{I}-\boldsymbol{A}^{\dagger} \boldsymbol{A}=\boldsymbol{V}_{2} \boldsymbol{V}_{2}^{*}=\sum_{j=r+1}^{n} \boldsymbol{v}_{j} \boldsymbol{v}_{j}^{*} \in \mathbb{C}^{n \times n}.$$</td></tr>
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● K|+0</td><td>A=\left[\begin{array}{l} B \\ C \end{array}\right]</td><td>$$A=\left[\begin{array}{l} B \\ C \end{array}\right]$$</td><td>$$\boldsymbol{A}=\left[\begin{array}{l} \boldsymbol{B} \\ \boldsymbol{C} \end{array}\right]$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
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● N|+0</td><td>\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array} \right] .</td><td>$$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array}\right] .$$</td><td>$$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{l} b_{1} \\ b_{2} \\ b_{3} \end{array}\right].$$</td></tr>
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● K|+0</td><td>\left|\sigma_{i}(\epsilon)-\sigma_{i}(0)\right| \leq| \epsilon|, \quad i=1, \ldots, n,</td><td>$$\left|\sigma_{i}(\epsilon)-\sigma_{i}(0)\right| \leq|\epsilon|, \quad i=1, \ldots, n,$$</td><td>$$|\sigma_{i}(\epsilon)-\sigma_{i}(0)| \leq|\epsilon|, \quad i=1, \ldots, n,$$</td></tr>
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● N|+0</td><td>\bar{\Omega}_{h}:=\{(j h, k h): j, k=0,1, \ldots, m+1\}, \quad \text { where } \quad h=1 /(m+1) .</td><td>$$\bar{\Omega}_{h}:=\{(j h, k h): j, k=0,1, \ldots, m+1\}, \quad \text { where } \quad h=1 /(m+1) .$$</td><td>$$\overline{\Omega}_{h}:=\{(jh, kh):\ j, k=0,1, \ldots, m+1\}, \quad \text { where } \quad h=1/(m+1).$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0658 (10.2)</td><td>239</td><td>■ 1.000
● N|+0</td><td>\frac{\partial^{2} u(j h, k h)}{\partial x^{2}} \approx \frac{v_{j-1, k}-2 v_{j, k}+v_{j+1, k}}{h^{2}}, \frac{ \partial^{2} u(j h, k h)}{\partial y^{2}} \approx \frac{v_{j, k-1}-2 v_{j, k}+v_{j, k+1}}{h^{2}} .</td><td>$$\frac{\partial^{2} u(j h, k h)}{\partial x^{2}} \approx \frac{v_{j-1, k}-2 v_{j, k}+v_{j+1, k}}{h^{2}}, \frac{\partial^{2} u(j h, k h)}{\partial y^{2}} \approx \frac{v_{j, k-1}-2 v_{j, k}+v_{j, k+1}}{h^{2}} .$$</td><td>$$\frac{\partial^{2} u(jh, kh)}{\partial x^{2}} \approx \frac{v_{j-1, k}-2 v_{j, k}+v_{j+1, k}}{h^{2}}, \quad \frac{\partial^{2} u(jh, kh)}{\partial y^{2}} \approx \frac{v_{j, k-1}-2 v_{j, k}+v_{j, k+1}}{h^{2}}.$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0660 (10.4)</td><td>239</td><td>■ 1.000
● N|+0</td><td>-\Delta_{h} v_{j, k}:=\frac{-v_{j-1, k}+2 v_{j, k}-v_{j+1, k}}{h^{2}}+\frac{-v_{j, k-1}+2 v_{j, k}-v_{j, k+1}}{h^{2}} .</td><td>$$-\Delta_{h} v_{j, k}:=\frac{-v_{j-1, k}+2 v_{j, k}-v_{j+1, k}}{h^{2}}+\frac{-v_{j, k-1}+2 v_{j, k}-v_{j, k+1}}{h^{2}} .$$</td><td>$$-\Delta_{h} v_{j, k}:=\frac{-v_{j-1, k}+2 v_{j, k}-v_{j+1, k}}{h^{2}}+\frac{-v_{j, k-1}+2 v_{j, k}-v_{j, k+1}}{h^{2}}.$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0662 (10.6)</td><td>239</td><td>■ 1.000
● W|+0</td><td>\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1, m} \\ \vdots & & \vdots \\ v_{m, 1} & \cdots & v_{m, m} \end{array}\right] \in \mathbb{R}^{m \times m}, \quad \boldsymbol{F}:=\left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \cdots & f_{m, m} \end{array}\right] \in \mathbb{R}^{m \times m} .</td><td>$$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1, m} \\ \vdots & & \vdots \\ v_{m, 1} & \cdots & v_{m, m} \end{array}\right] \in \mathbb{R}^{m \times m}, \quad \boldsymbol{F}:=\left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \cdots & f_{m, m} \end{array}\right] \in \mathbb{R}^{m \times m} .$$</td><td>$$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \cdots & v_{1, m} \\ \vdots & & \vdots \\ v_{m, 1} & \cdots & v_{m, m} \end{array}\right] \in \mathbb{R}^{m \times m}, \quad \boldsymbol{F}:=\left[\begin{array}{ccc} f_{1,1} & \cdots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \cdots & f_{m, m} \end{array}\right] \in \mathbb{R}^{m \times m}.$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0663</td><td>239</td><td>■ 1.000
● W|+0</td><td>\sum_{i=1}^{m} \boldsymbol{T}_{j, i} v_{i, k}+ \sum_{i=1}^{m} v_{j, i} \boldsymbol{T}_{i, k}=-v_{j-1, k}+2 v_{j, k}-v_{j+1, k}-v_{j, k-1}+2 v_{j, k}-v_{j, k+1},</td><td>$$\sum_{i=1}^{m} \boldsymbol{T}_{j, i} v_{i, k}+\sum_{i=1}^{m} v_{j, i} \boldsymbol{T}_{i, k}=-v_{j-1, k}+2 v_{j, k}-v_{j+1, k}-v_{j, k-1}+2 v_{j, k}-v_{j, k+1},$$</td><td>$$\sum_{i=1}^{m} \boldsymbol{T}_{j, i} v_{i, k}+\sum_{i=1}^{m} v_{j, i} \boldsymbol{T}_{i, k}=-v_{j-1, k}+2 v_{j, k}-v_{j+1, k}-v_{j, k-1}+2 v_{j, k}-v_{j, k+1},$$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0664	239	■ 1.000 • S|+0	`\operatorname{vec}(\boldsymbol{B}):=\left[b_{11}, \ldots, b_{m 1}, b_{12}, \ldots, b_{m 2}, \ldots, b_{1 n}, \ldots, b_{m n}\right]^{T} \in \mathbb{R}^{m n}`	$\operatorname{vec}(\boldsymbol{B}):=\left[b_{11}, \ldots, b_{m 1}, b_{12}, \ldots, b_{m 2}, \ldots, b_{1 n}, \ldots, b_{m n}\right]^{T} \in \mathbb{R}^{m n}$	$vec(\boldsymbol{B}) := [b_{11}, \ldots, b_{m1}, b_{12}, \ldots, b_{m2}, \ldots, b_{1n}, \ldots, b_{mn}]^T \in \mathbb{R}^{mn}$
lyche-numerical-linear-algebra EQ0665	240	■ 1.000 • N|+0	`4 x_{i}-x_{i-1}-x_{i+1}-x_{i-m}-x_{i+m}=b_{i},`	$4 x_{i}-x_{i-1}-x_{i+1}-x_{i-m}-x_{i+m}=b_{i},$	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$
lyche-numerical-linear-algebra EQ0669 (10.9)	241	■ 1.000 • N|-2	`\boldsymbol{T}_{1}:=\operatorname{tridiag}(a, d, a) .`	$\boldsymbol{T}_{1}:=\operatorname{tridiag}(a, d, a) .$	$\boldsymbol{T}_1 := \operatorname{tridiag}(a, d, a).$
lyche-numerical-linear-algebra EQ0672	242	■ 1.000 • N|+0	`\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A} b_{1,1} & \boldsymbol{A} b_{1,2} & \cdots & \boldsymbol{A} b_{1, s} \\ \boldsymbol{A} b_{2,1} & \boldsymbol{A} b_{2,2} & \cdots & \boldsymbol{A} b_{2, s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} b_{r, 1} & \boldsymbol{A} b_{r, 2} & \cdots & \boldsymbol{A} b_{r, s} \end{array}\right] .`	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A} b_{1,1} & \boldsymbol{A} b_{1,2} & \cdots & \boldsymbol{A} b_{1, s} \\ \boldsymbol{A} b_{2,1} & \boldsymbol{A} b_{2,2} & \cdots & \boldsymbol{A} b_{2, s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} b_{r, 1} & \boldsymbol{A} b_{r, 2} & \cdots & \boldsymbol{A} b_{r, s} \end{array}\right] .$	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A} b_{1,1} & \boldsymbol{A} b_{1,2} & \cdots & \boldsymbol{A} b_{1, s} \\ \boldsymbol{A} b_{2,1} & \boldsymbol{A} b_{2,2} & \cdots & \boldsymbol{A} b_{2, s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} b_{r, 1} & \boldsymbol{A} b_{r, 2} & \cdots & \boldsymbol{A} b_{r, s} \end{array}\right] .$
lyche-numerical-linear-algebra EQ0673	242	■ 1.000 • N|+1	`\boldsymbol{T}_{1}=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right], \quad \boldsymbol{I}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]`	$\boldsymbol{T}_{1}=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right], \quad \boldsymbol{I}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\boldsymbol{T}_{1}=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right], \quad \boldsymbol{I}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$
lyche-numerical-linear-algebra EQ0675 (10.12)	243	■ 1.000 • N|+1	`\boldsymbol{T}_{2}=\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}, \quad \boldsymbol{T}_{1}, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_{2} \in \mathbb{R}^{\left(m^{2}\right) \times\left(m^{2}\right)} .`	$\boldsymbol{T}_{2}=\boldsymbol{T}_{1} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}_{1}, \quad \boldsymbol{T}_{1}, \boldsymbol{I} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{T}_{2} \in \mathbb{R}^{\left(m^{2}\right) \times\left(m^{2}\right)} .$	$\boldsymbol{T}_2 = \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1, \quad \boldsymbol{T}_1, \boldsymbol{I} \in \mathbb{R}^{m\times m}, \quad \boldsymbol{T}_2 \in \mathbb{R}^{(m^2)\times(m^2)}.$
lyche-numerical-linear-algebra EQ0678	243	■ 1.000 • N|+0	`(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})=\left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1, t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r, 1} & \cdots & \boldsymbol{A} b_{r, t} \end{array}\right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1, s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t, 1} & \cdots & \boldsymbol{C} d_{t, s} \end{array}\right] .`	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})=\left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1, t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r, 1} & \cdots & \boldsymbol{A} b_{r, t} \end{array}\right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1, s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t, 1} & \cdots & \boldsymbol{C} d_{t, s} \end{array}\right] .$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})=\left[\begin{array}{ccc} \boldsymbol{A} b_{1,1} & \cdots & \boldsymbol{A} b_{1, t} \\ \vdots & & \vdots \\ \boldsymbol{A} b_{r, 1} & \cdots & \boldsymbol{A} b_{r, t} \end{array}\right] \left[\begin{array}{ccc} \boldsymbol{C} d_{1,1} & \cdots & \boldsymbol{C} d_{1, s} \\ \vdots & & \vdots \\ \boldsymbol{C} d_{t, 1} & \cdots & \boldsymbol{C} d_{t, s} \end{array}\right] .$
lyche-numerical-linear-algebra EQ0681	245	■ 1.000 • S|+0	`\begin{aligned} \left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+\boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V}) & =\operatorname{vec}(\boldsymbol{F}) \\ \Leftrightarrow \quad\left(\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_{s}^{T}+\boldsymbol{I}_{r} \boldsymbol{V} \boldsymbol{B}^{T}\right) & =\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V}+ \boldsymbol{V} \boldsymbol{B}^{T}=\boldsymbol{F} . \end{aligned}`	$\begin{aligned} \left(\boldsymbol{A} \otimes \boldsymbol{I}_{s}+\boldsymbol{I}_{r} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V}) & =\operatorname{vec}(\boldsymbol{F}) \\ \Leftrightarrow \quad\left(\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_{s}^{T}+\boldsymbol{I}_{r} \boldsymbol{V} \boldsymbol{B}^{T}\right) & =\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V}+ \boldsymbol{V} \boldsymbol{B}^{T}=\boldsymbol{F} . \end{aligned}$	$\begin{aligned} &(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ &\Leftrightarrow \quad (\boldsymbol{A}\boldsymbol{V}\boldsymbol{I}_s^T + \boldsymbol{I}_r\boldsymbol{V}\boldsymbol{B}^T) = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-al
EQ0683 (10.16)</td><td>246</td><td>■ 1.000
● N|+0</td><td><code>\boldsymbol{s}_{j}=[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T} .</code></td><td>$\boldsymbol{s}_{j}=[\sin (j \pi h), \sin (2 j \pi h), \ldots, \sin (m j \pi h)]^{T} .$</td><td>$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \ldots, \sin(mj\pi h)]^T.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0685 (10.18)</td><td>246</td><td>■ 1.000
● N|+0</td><td><code>\boldsymbol{T}_{2}\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k}\right)=\left(\lambda_{j}+ \lambda_{k}\right)\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k}\right) \quad j, k=1, \ldots, m,</code></td><td>$\boldsymbol{T}_{2}\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k}\right)=\left(\lambda_{j}+ \lambda_{k}\right)\left(\boldsymbol{s}_{j} \otimes \boldsymbol{s}_{k}\right) \quad j, k=1, \ldots, m,$</td><td>$\boldsymbol{T}_2(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) = (\lambda_j + \lambda_k)(\boldsymbol{s}_j \otimes \boldsymbol{s}_k) \quad j,k = 1,\ldots,m,$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0688 (10.20)</td><td>246</td><td>■ 1.000
● N|+1</td><td><code>\|\boldsymbol{A}\|_{2}\left\|\boldsymbol{A}^{-1}\right \|_{2}=\frac{\cos ^{2} w}{\sin ^{2} w}, \quad w:=\frac{ \pi}{2(m+1)}</code></td><td>$\|\boldsymbol{A}\|_{2}\left\|\boldsymbol{A}^{-1}\right \|_{2}=\frac{\cos ^{2} w}{\sin ^{2} w}, \quad w:=\frac{ \pi}{2(m+1)}$</td><td>$\|\boldsymbol{A}\|_2 \|\boldsymbol{A}^{-1}\|_2 = \frac{\cos^2 w}{\sin^2 w}, \quad w := \frac{\pi}{2(m+1)}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0689</td><td>246</td><td>■ 1.000
● N|+0</td><td><code>\lambda_{j, k}=4-2 \cos (2 j w)-2 \cos (2 k w)=4 \sin ^{2}(j w)+4 \sin ^{2}(k w), \quad j, k=1, \ldots, m .</code></td><td>$\lambda_{j, k}=4-2 \cos (2 j w)-2 \cos (2 k w)=4 \sin ^{2}(j w)+4 \sin ^{2}(k w), \quad j, k=1, \ldots, m .$</td><td>$\lambda_{j,k} = 4-2\cos(2jw)-2\cos(2kw) = 4\sin^2(jw)+4\sin^2(kw), \quad j,k=1,\ldots,m.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0690</td><td>246</td><td>■ 1.000
● N|+0</td><td><code>\lambda_{\max }=8 \cos ^{2} w, \quad \lambda_{\min }=8 \sin ^{2} w .</code></td><td>$\lambda_{\max }=8 \cos ^{2} w, \quad \lambda_{\min }=8 \sin ^{2} w .$</td><td>$\lambda_{max} = 8\cos^2 w, \quad \lambda_{min} = 8\sin^2 w.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0696</td><td>249</td><td>■ 1.000
● N|+0</td><td><code>(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V})=h^{2} \operatorname{vec}(\boldsymbol{F}), \quad(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U})=h^{2} \operatorname{vec}(\boldsymbol{V}) .</code></td><td>$(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{V})=h^{2} \operatorname{vec}(\boldsymbol{F}), \quad(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}) \operatorname{vec}(\boldsymbol{U})=h^{2} \operatorname{vec}(\boldsymbol{V}) .$</td><td>$(\boldsymbol{T}\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T})\mathrm{vec}(\boldsymbol{V}) = h^2\mathrm{vec}(\boldsymbol{F}), \quad (\boldsymbol{T}\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T})\mathrm{vec}(\boldsymbol{U}) = h^2\mathrm{vec}(\boldsymbol{V}).$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0697 (10.25)</td><td>249</td><td>■ 1.000
● N|-1</td><td><code>\boldsymbol{T}^{2} \boldsymbol{U}+2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T}+\boldsymbol{U} \boldsymbol{T}^{2}=h^{4} \boldsymbol{F} \quad \text { and } \quad \boldsymbol{A} \boldsymbol{x}=\boldsymbol{b},</code></td><td>$\boldsymbol{T}^{2} \boldsymbol{U}+2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T}+\boldsymbol{U} \boldsymbol{T}^{2}=h^{4} \boldsymbol{F} \quad \text { and } \quad \boldsymbol{A} \boldsymbol{x}=\boldsymbol{b},$</td><td>$\boldsymbol{T}^2\boldsymbol{U} + 2\boldsymbol{TUT} + \boldsymbol{UT}^2 = h^4\boldsymbol{F} \quad \text{and} \quad \boldsymbol{Ax} = \boldsymbol{b},$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0700 (11.2)</td><td>252</td><td>■ 1.000
● N|+0</td><td><code>\boldsymbol{U}_{1}=\boldsymbol{D}_{1}, \quad \boldsymbol{L}_{k}=\boldsymbol{A}_{k} \boldsymbol{U}_{k}^{-1}, \quad \boldsymbol{U}_{k+1}= \boldsymbol{D}_{k+1}-\boldsymbol{L}_{k} \boldsymbol{C}_{k}, \quad k=1,2, \ldots, m-1 .</code></td><td>$\boldsymbol{U}_{1}=\boldsymbol{D}_{1}, \quad \boldsymbol{L}_{k}=\boldsymbol{A}_{k} \boldsymbol{U}_{k}^{-1}, \quad \boldsymbol{U}_{k+1}= \boldsymbol{D}_{k+1}-\boldsymbol{L}_{k} \boldsymbol{C}_{k}, \quad k=1,2, \ldots, m-1 .$</td><td>$\boldsymbol{U}_1 = \boldsymbol{D}_1, \quad \boldsymbol{L}_k = \boldsymbol{A}_k\boldsymbol{U}_k^{-1}, \quad \boldsymbol{U}_{k+1} = \boldsymbol{D}_{k+1} - \boldsymbol{L}_k\boldsymbol{C}_k, \quad k = 1,2,\ldots,m-1.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0704 (11.4)</td><td>253</td><td>■ 1.000
● N|+0</td><td><code>\boldsymbol{S}:=\left[\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{m}\right]=[\sin (j k \pi h)]_{j, k=1}^{m} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D}= \operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{m} \right) .</code></td><td>$\boldsymbol{S}:=\left[\boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{m}\right]=[\sin (j k \pi h)]_{j, k=1}^{m} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{D}= \operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{m} \right) .$</td><td>$\boldsymbol{S} := [\boldsymbol{s}_1,\ldots,\boldsymbol{s}_m] = \left[\sin(jk\pi h)\right]_{j,k=1}^m \in \mathbb{R}^{m\times m}, \quad \boldsymbol{D} = \mathrm{diag}(\lambda_1,\ldots,\lambda_m).$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0705 (2h)</td><td>253</td><td>■ 1.000
● N|+0</td><td><code>\boldsymbol{T} \boldsymbol{S}=\left[\boldsymbol{T} \boldsymbol{s}_{1}, \ldots, \boldsymbol{T} \boldsymbol{s}_{m}\right]=\left[\lambda_{1} \boldsymbol{s}_{1}, \ldots, \lambda_{m} \boldsymbol{s}_{m}\right]=\boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^{2}=\boldsymbol{S}^{T} \boldsymbol{S}=\frac{1}{2 h} \boldsymbol{I} .</code></td><td>$\boldsymbol{T} \boldsymbol{S}=\left[\boldsymbol{T} \boldsymbol{s}_{1}, \ldots, \boldsymbol{T} \boldsymbol{s}_{m}\right]=\left[\lambda_{1} \boldsymbol{s}_{1}, \ldots, \lambda_{m} \boldsymbol{s}_{m}\right]=\boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^{2}=\boldsymbol{S}^{T} \boldsymbol{S}=\frac{1}{2 h} \boldsymbol{I} .$</td><td>$\boldsymbol{TS} = [\boldsymbol{Ts}_1,\ldots,\boldsymbol{Ts}_m] = [\lambda_1\boldsymbol{s}_1,\ldots,\lambda_m\boldsymbol{s}_m] = \boldsymbol{SD}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T\boldsymbol{S} = \frac{1}{2h}\boldsymbol{I}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0708</td><td>253</td><td>■ 1.000
● K|+0</td><td><code>x_{j, k}=4 h^{4} g_{j, k} /\left(\lambda_{j}+\lambda_{k} \right)=h^{4} g_{j, k} /\left(\sigma_{j}+\sigma_{k} \right), \quad \sigma_{j}:=\lambda_{j} / 4=\sin ^{2}(j \pi h / 2) .</code></td><td>$x_{j, k}=4 h^{4} g_{j, k} /\left(\lambda_{j}+\lambda_{k} \right)=h^{4} g_{j, k} /\left(\sigma_{j}+\sigma_{k} \right), \quad \sigma_{j}:=\lambda_{j} / 4=\sin ^{2}(j \pi h / 2) .$</td><td>$x_{j,k} = 4h^4 g_{j,k}/(\lambda_j+\lambda_k) = h^4 g_{j,k}/(\sigma_j+\sigma_k), \quad \sigma_j := \lambda_j/4 = \sin^2(j\pi h/2).$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0710</td><td>255</td><td>■ 1.000
● N|+0</td><td><code>w_{j}=\sum_{k=1}^{m} \sin \left(\frac{j k \pi}{m+1} \right) v_{k}, \quad j=1, \ldots, m</code></td><td>$w_{j}=\sum_{k=1}^{m} \sin \left(\frac{j k \pi}{m+1} \right) v_{k}, \quad j=1, \ldots, m$</td><td>$w_j = \sum_{k=1}^m \sin\left(\frac{jk\pi}{m+1}\right)v_k, \quad j = 1,\ldots,m$</td></tr>
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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
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lyche-numerical-linear-algebra_EQ0712	256	■ 1.000 ● N|+0	`z_{j+1}=\sum_{k=0}^{N-1} \omega_{N}^{j k} y_{k+1}, \quad j=0, \ldots, N-1,`	$z_{j+1}=\sum_{k=0}^{N-1} \omega_{N}^{j k} y_{k+1}, \quad j=0, \ldots, N-1,$	$z_{j+1}=\sum_{k=0}^{N-1} \omega_{N}^{j k} y_{k+1}, \quad j=0, \ldots, N-1,$
lyche-numerical-linear-algebra_EQ0713	256	■ 1.000 ● K|+0	`\omega_{4}=\exp ^{-2 \pi i / 4}=\cos (\pi / 2)-i \sin (\pi / 2)=-i`	$\omega_{4}=\exp ^{-2 \pi i / 4}=\cos (\pi / 2)-i \sin (\pi / 2)=-i$	$\omega_{4}=\exp ^{-2 \pi i / 4}=\cos (\pi / 2)-i \sin (\pi / 2)=-i$
lyche-numerical-linear-algebra_EQ0715	256	■ 1.000 ● K|+0	`\sin w=\frac{e^{i w}-e^{-i w}}{2 i} .`	$\sin w=\frac{e^{i w}-e^{-i w}}{2 i} .$	$\sin w=\frac{e^{i w}-e^{-i w}}{2 i} .$
lyche-numerical-linear-algebra_EQ0716	256	■ 1.000 ● N|+0	`\boldsymbol{z}^{T}=\left[0, \boldsymbol{x}^{T}, 0,-\boldsymbol{x}_{B}^{T}\right] \in \mathbb{R}^{2 m+2}, \quad \boldsymbol{x}_{B}^{T}:=\left[x_{m}, \ldots, x_{2}, x_{1}\right] .`	$\boldsymbol{z}^{T}=\left[0, \boldsymbol{x}^{T}, 0,-\boldsymbol{x}_{B}^{T}\right] \in \mathbb{R}^{2 m+2}, \quad \boldsymbol{x}_{B}^{T}:=\left[x_{m}, \ldots, x_{2}, x_{1}\right] .$	$\boldsymbol{z}^{T}=[0, \boldsymbol{x}^{T}, 0,-\boldsymbol{x}_{B}^{T}] \in \mathbb{R}^{2 m+2}, \quad \boldsymbol{x}_{B}^{T}:=[x_{m}, \ldots, x_{2}, x_{1}].$
lyche-numerical-linear-algebra_EQ0718	256	■ 1.000 ● N|+0	`\omega^{j k}=e^{-\pi i j k /(m+1)}, \quad \omega^{(2 m+2-j) k}=e^{-2 \pi i} e^{\pi i j k /(m+1)}=e^{\pi i j k /(m+1)} .`	$\omega^{j k}=e^{-\pi i j k /(m+1)}, \quad \omega^{(2 m+2-j) k}=e^{-2 \pi i} e^{\pi i j k /(m+1)}=e^{\pi i j k /(m+1)} .$	$\omega^{j k}=e^{-\pi i j k /(m+1)}, \quad \omega^{(2 m+2-j) k}=e^{-2 \pi i} e^{\pi i j k /(m+1)}=e^{\pi i j k /(m+1)} .$
lyche-numerical-linear-algebra_EQ0720	257	■ 1.000 ● K|+0	`\boldsymbol{P}_{N}=\left[\boldsymbol{e}_{1}, \boldsymbol{e}_{3}, \ldots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_{2}, \boldsymbol{e}_{4}, \ldots, \boldsymbol{e}_{N}\right],`	$\boldsymbol{P}_{N}=\left[\boldsymbol{e}_{1}, \boldsymbol{e}_{3}, \ldots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_{2}, \boldsymbol{e}_{4}, \ldots, \boldsymbol{e}_{N}\right],$	$\boldsymbol{P}_{N}=[\boldsymbol{e}_{1}, \boldsymbol{e}_{3}, \ldots, \boldsymbol{e}_{N-1}, \boldsymbol{e}_{2}, \boldsymbol{e}_{4}, \ldots, \boldsymbol{e}_{N}],$
lyche-numerical-linear-algebra_EQ0724	258	■ 1.000 ● K|+0	`\boldsymbol{F}_{2}=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right], \quad \boldsymbol{D}_{2} \boldsymbol{F}_{2}=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],`	$\boldsymbol{F}_{2}=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right], \quad \boldsymbol{D}_{2} \boldsymbol{F}_{2}=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],$	$\boldsymbol{F}_{2}=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right], \quad \boldsymbol{D}_{2} \boldsymbol{F}_{2}=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],$
lyche-numerical-linear-algebra_EQ0727 (11.8)	258	■ 1.000 ● N|+1	`\boldsymbol{D}_{m}=\operatorname{diag}\left(1, \omega_{N}, \omega_{N}^{2}, \ldots, \omega_{N}^{m-1}\right) .`	$\boldsymbol{D}_{m}=\operatorname{diag}\left(1, \omega_{N}, \omega_{N}^{2}, \ldots, \omega_{N}^{m-1}\right) .$	$\boldsymbol{D}_{m}=\operatorname{diag}(1, \omega_{N}, \omega_{N}^{2}, \ldots, \omega_{N}^{m-1}).$
lyche-numerical-linear-algebra_EQ0728	258	■ 1.000 ● W|+0	`\omega_{m}^{m}=1, \omega_{2 m}^{2 k}=\omega_{m}^{k}, \omega_{2 m}^{m}=-1,\left(\boldsymbol{F}_{m}\right)_{p, q}=\omega_{m}^{j k},\left(\boldsymbol{D}_{m} \boldsymbol{F}_{m}\right)_{p, q}=\omega_{2 m}^{j} \omega_{m}^{j k},`	$\omega_{m}^{m}=1, \omega_{2 m}^{2 k}=\omega_{m}^{k}, \omega_{2 m}^{m}=-1,\left(\boldsymbol{F}_{m}\right)_{p, q}=\omega_{m}^{j k},\left(\boldsymbol{D}_{m} \boldsymbol{F}_{m}\right)_{p, q}=\omega_{2 m}^{j} \omega_{m}^{j k},$	$\omega_{m}^{m}=1,\ \omega_{2 m}^{2 k}=\omega_{m}^{k},\ \omega_{2 m}^{m}=-1,\ (\boldsymbol{F}_{m})_{p, q}=\omega_{m}^{j k},\ (\boldsymbol{D}_{m} \boldsymbol{F}_{m})_{p, q}=\omega_{2 m}^{j} \omega_{m}^{j k},$
lyche-numerical-linear-algebra_EQ0730	258	■ 1.000 ● N|+0	`\boldsymbol{w}_{1}^{T}=\left[y_{1}, y_{3}, \ldots, y_{2 m-1}\right], \quad \boldsymbol{w}_{2}^{T}=\left[y_{2}, y_{4}, \ldots, y_{2 m}\right] .`	$\boldsymbol{w}_{1}^{T}=\left[y_{1}, y_{3}, \ldots, y_{2 m-1}\right], \quad \boldsymbol{w}_{2}^{T}=\left[y_{2}, y_{4}, \ldots, y_{2 m}\right] .$	$\boldsymbol{w}_{1}^{T}=[y_{1}, y_{3}, \ldots, y_{2 m-1}], \quad \boldsymbol{w}_{2}^{T}=[y_{2}, y_{4}, \ldots, y_{2 m}].$
lyche-numerical-linear-algebra_EQ0732	259	■ 1.000 ● N|+0	`\boldsymbol{q}_{1}=\boldsymbol{F}_{m} \boldsymbol{w}_{1}, \quad \text { and } \quad \boldsymbol{q}_{2}=\boldsymbol{D}_{m}\left(\boldsymbol{F}_{m} \boldsymbol{w}_{2}\right) .`	$\boldsymbol{q}_{1}=\boldsymbol{F}_{m} \boldsymbol{w}_{1}, \quad \text { and } \quad \boldsymbol{q}_{2}=\boldsymbol{D}_{m}\left(\boldsymbol{F}_{m} \boldsymbol{w}_{2}\right) .$	$\boldsymbol{q}_{1}=\boldsymbol{F}_{m} \boldsymbol{w}_{1}, \quad \text { and } \quad \boldsymbol{q}_{2}=\boldsymbol{D}_{m}(\boldsymbol{F}_{m} \boldsymbol{w}_{2}).$
lyche-numerical-linear-algebra_EQ0733	260	■ 1.000 ● N|+0	`\frac{8 N^{2}}{5 N \log _{2} N} \approx 84000 .`	$\frac{8 N^{2}}{5 N \log _{2} N} \approx 84000 .$	$\frac{8 N^{2}}{5 N \log _{2} N} \approx 84000 .$
lyche-numerical-linear-algebra_EQ0734	260	■ 1.000 ● S|+0	`8 \gamma m(m+1) \log _{2}(2 m+2) \approx 8 \gamma m^{2} \log _{2} m=4 \gamma n \log _{2} n,`	$8 \gamma m(m+1) \log _{2}(2 m+2) \approx 8 \gamma m^{2} \log _{2} m=4 \gamma n \log _{2} n,$	$8 \gamma m(m+1) \log _{2}(2 m+2) \approx 8 \gamma m^{2} \log _{2} m=4 \gamma n \log _{2} n,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-al… EQ0735	261	■ 1.000 ● W|+0	`v_{i j}=\frac{1}{(m+1)^{4}} \sum_{p=1}^{m} \sum_{r=1}^{m} \sum_{k=1}^{m} \sum_{l=1}^{m} \frac{\sin \left(\frac{i p \pi}{m+1}\right) \sin \left(\frac{j r \pi}{m+1}\right) \sin \left(\frac{k p \pi}{m+1}\right) \sin \left(\frac{l r \pi}{m+1}\right)}{\left[\sin \left(\frac{p \pi}{2(m+1)}\right)\right]^{2}+\left[\sin \left(\frac{r \pi}{2(m+1)}\right)\right]^{2}} f_{k, l} .`	$v_{i j}=\frac{1}{(m+1)^{4}} \sum_{p=1}^{m} \sum_{r=1}^{m} \sum_{k=1}^{m} \sum_{l=1}^{m} \frac{\sin \left(\frac{i p \pi}{m+1}\right) \sin \left(\frac{j r \pi}{m+1}\right) \sin \left(\frac{k p \pi}{m+1}\right) \sin \left(\frac{l r \pi}{m+1}\right)}{\left[\sin \left(\frac{p \pi}{2(m+1)}\right)\right]^{2}+\left[\sin \left(\frac{r \pi}{2(m+1)}\right)\right]^{2}} f_{k, l} .$	$v_{ij}=\frac{1}{(m+1)^4}\sum_{p=1}^{m}\sum_{r=1}^{m}\sum_{k=1}^{m}\sum_{l=1}^{m}\frac{\sin\left(\frac{ip\pi}{m+1}\right)\sin\left(\frac{jr\pi}{m+1}\right)\sin\left(\frac{kp\pi}{m+1}\right)\sin\left(\frac{lr\pi}{m+1}\right)}{\left[\sin\left(\frac{p\pi}{2(m+1)}\right)\right]^2+\left[\sin\left(\frac{r\pi}{2(m+1)}\right)\right]^2} f_{k,l}.$
lyche-numerical-linear-al… EQ0737 (11.9)	261	■ 1.000 ● N|+0	`\left(\boldsymbol{T}+\lambda_{j} \boldsymbol{I}\right) \boldsymbol{x}_{j}=\boldsymbol{c}_{j} \quad j=1, \ldots, m,`	$\left(\boldsymbol{T}+\lambda_{j} \boldsymbol{I}\right) \boldsymbol{x}_{j}=\boldsymbol{c}_{j} \quad j=1, \ldots, m,$	$(\boldsymbol{T}+\lambda_j\boldsymbol{I})\boldsymbol{x}_j=\boldsymbol{c}_j \quad j=1,\ldots,m,$
lyche-numerical-linear-al… EQ0738	261	■ 1.000 ● N|+0	`\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^{2} \mu \boldsymbol{F},`	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T}=h^{2} \mu \boldsymbol{F},$	$\boldsymbol{TV}+\boldsymbol{VT}-\frac{1}{6}\boldsymbol{TVT}=h^2\mu\boldsymbol{F},$
lyche-numerical-linear-al… EQ0741	262	■ 1.000 ● K|+0	`\boldsymbol{T}^{2} \boldsymbol{U}+2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T}+\boldsymbol{U} \boldsymbol{T}^{2}=h^{4} \boldsymbol{F} .`	$\boldsymbol{T}^{2} \boldsymbol{U}+2 \boldsymbol{T} \boldsymbol{U} \boldsymbol{T}+\boldsymbol{U} \boldsymbol{T}^{2}=h^{4} \boldsymbol{F} .$	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{TUT}+\boldsymbol{UT}^2=h^4\boldsymbol{F}.$
lyche-numerical-linear-al… EQ0742	262	■ 1.000 ● N|+0	`\boldsymbol{D}^{2} \boldsymbol{X}+2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}+\boldsymbol{X} \boldsymbol{D}^{2}=4 h^{6} \boldsymbol{G}, \text { where } \boldsymbol{G}=\boldsymbol{S} \boldsymbol{F} \boldsymbol{S},`	$\boldsymbol{D}^{2} \boldsymbol{X}+2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}+\boldsymbol{X} \boldsymbol{D}^{2}=4 h^{6} \boldsymbol{G}, \text { where } \boldsymbol{G}=\boldsymbol{S} \boldsymbol{F} \boldsymbol{S},$	$\boldsymbol{D}^2\boldsymbol{X}+2\boldsymbol{DXD}+\boldsymbol{XD}^2=4h^6\boldsymbol{G}, \text{ where } \boldsymbol{G}=\boldsymbol{SFS},$
lyche-numerical-linear-al… EQ0743	262	■ 1.000 ● W|+0	`x_{j, k}=\frac{h^{6} g_{j, k}}{4\left(\sigma_{j}+\sigma_{k}\right)^{2}}, \text { where } \sigma_{j}=\sin ^{2}((j \pi h) / 2) \text { for } j, k=1,2, \ldots, m .`	$x_{j, k}=\frac{h^{6} g_{j, k}}{4\left(\sigma_{j}+\sigma_{k}\right)^{2}}, \text { where } \sigma_{j}=\sin ^{2}((j \pi h) / 2) \text { for } j, k=1,2, \ldots, m .$	$x_{j,k}=\frac{h^6 g_{j,k}}{4(\sigma_j+\sigma_k)^2}, \text{ where } \sigma_j=\sin^2((j\pi h)/2) \text{ for } j,k=1,2,\ldots,m.$
lyche-numerical-linear-al… EQ0744 (12.1)	266	■ 1.000 ● S|+0	`\boldsymbol{x}(i)=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}(j)+b_{i}\right) / a_{i i}, \quad i=1,2, \ldots, n .`	$\boldsymbol{x}(i)=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}(j)+b_{i}\right) / a_{i i}, \quad i=1,2, \ldots, n .$	$\boldsymbol{x}(i)=\big(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}(j)-\sum_{j=i+1}^{n}a_{ij}\boldsymbol{x}(j)+b_i\big)/a_{ii}, \quad i=1,2,\ldots,n.$
lyche-numerical-linear-al… EQ0745 (12.2)	266	■ 1.000 ● N|+0	`\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k}(j)+b_{i}\right) / a_{i i}, \text { for } i=1,2, \ldots, n .`	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k}(j)+b_{i}\right) / a_{i i}, \text { for } i=1,2, \ldots, n .$	$\boldsymbol{x}_{k+1}(i)=\big(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^{n}a_{ij}\boldsymbol{x}_k(j)+b_i\big)/a_{ii}, \text{ for } i=1,2,\ldots,n.$
lyche-numerical-linear-al… EQ0746 (12.3)	266	■ 1.000 ● W|+0	`\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k}(j)+b_{i}\right) / a_{i i}, \text { for } i=1,2, \ldots, n .`	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k}(j)+b_{i}\right) / a_{i i}, \text { for } i=1,2, \ldots, n .$	$\boldsymbol{x}_{k+1}(i)=\big(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^{n}a_{ij}\boldsymbol{x}_k(j)+b_i\big)/a_{ii}, \text{ for } i=1,2,\ldots,n.$
lyche-numerical-linear-al… EQ0747 (12.4)	266	■ 1.000 ● N|+0	`\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k}(j)+b_{i}\right) / a_{i i}+(1-\omega) \boldsymbol{x}_{k}(i) .`	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k}(j)+b_{i}\right) / a_{i i}+(1-\omega) \boldsymbol{x}_{k}(i) .$	$\boldsymbol{x}_{k+1}(i)=\omega\big(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^{n}a_{ij}\boldsymbol{x}_k(j)+b_i\big)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0748 (12.5)	267	■ 1.000 ● N|+0	`\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k+1 / 2}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k+1}(j)+b_{i}\right) / a_{i i}+(1-\omega) \boldsymbol{x}_{k+1 / 2}(i)`	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1} a_{i j} \boldsymbol{x}_{k+1 / 2}(j)-\sum_{j=i+1}^{n} a_{i j} \boldsymbol{x}_{k+1}(j)+b_{i}\right) / a_{i i}+(1-\omega) \boldsymbol{x}_{k+1 / 2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\big(-\sum_{j=1}^{i-1} a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^{n} a_{ij}\boldsymbol{x}_{k+1}(j)+b_i\big)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$
lyche-numerical-linear-algebra_ EQ0749	267	■ 1.000 ● N|+0	`4 \boldsymbol{v}(i, j)-\boldsymbol{v}(i-1, j)-\boldsymbol{v}(i+1, j)-\boldsymbol{v}(i, j-1)-\boldsymbol{v}(i, j+1)=h^{2} f_{i, j}, \quad i, j=1, \ldots, m,`	$4 \boldsymbol{v}(i, j)-\boldsymbol{v}(i-1, j)-\boldsymbol{v}(i+1, j)-\boldsymbol{v}(i, j-1)-\boldsymbol{v}(i, j+1)=h^{2} f_{i, j}, \quad i, j=1, \ldots, m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2 f_{i,j},\quad i,j=1,\ldots,m,$
lyche-numerical-linear-algebra_ EQ0750 (12.6)	267	■ 1.000 ● N|+0	`\boldsymbol{v}(i, j)=\left(\boldsymbol{v}(i-1, j)+\boldsymbol{v}(i+1, j)+\boldsymbol{v}(i, j-1)+v(i, j+1)+e_{i, j}\right) / 4,`	$\boldsymbol{v}(i, j)=\left(\boldsymbol{v}(i-1, j)+\boldsymbol{v}(i+1, j)+\boldsymbol{v}(i, j-1)+v(i, j+1)+e_{i, j}\right) / 4,$	$\boldsymbol{v}(i,j)=\big(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+v(i,j+1)+e_{i,j}\big)/4,$
lyche-numerical-linear-algebra_ EQ0755 (12.10)	270	■ 1.000 ● N|+0	`\boldsymbol{x}_{k+1}:=\boldsymbol{G} \boldsymbol{x}_{k}+\boldsymbol{c}, \quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1} \boldsymbol{A}, \quad, \boldsymbol{c}=\boldsymbol{M}^{-1} b .`	$\boldsymbol{x}_{k+1}:=\boldsymbol{G} \boldsymbol{x}_{k}+\boldsymbol{c}, \quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1} \boldsymbol{A}, \quad, \boldsymbol{c}=\boldsymbol{M}^{-1} b .$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad ,\boldsymbol{c}=\boldsymbol{M}^{-1}b.$
lyche-numerical-linear-algebra_ EQ0756	270	■ 1.000 ● W|+0	`\boldsymbol{x}=\lim _{k \rightarrow \infty} \boldsymbol{x}_{k+1}=\lim _{k \rightarrow \infty} \left(\boldsymbol{G} \boldsymbol{x}_{k}+\boldsymbol{c} \right)=\boldsymbol{G} \lim _{k \rightarrow \infty} \boldsymbol{x}_{k}+\boldsymbol{c}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c} .`	$\boldsymbol{x}=\lim _{k \rightarrow \infty} \boldsymbol{x}_{k+1}=\lim _{k \rightarrow \infty} \left(\boldsymbol{G} \boldsymbol{x}_{k}+\boldsymbol{c} \right)=\boldsymbol{G} \lim _{k \rightarrow \infty} \boldsymbol{x}_{k}+\boldsymbol{c}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c} .$	$\boldsymbol{x}=\lim_{k\to\infty}\boldsymbol{x}_{k+1}=\lim_{k\to\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\to\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$
lyche-numerical-linear-algebra_ EQ0758 (12.12)	271	■ 1.000 ● N|+0	`\boldsymbol{M}_{J}=\boldsymbol{D}, \quad \boldsymbol{M}_{1}=\boldsymbol{D}-\boldsymbol{A}_{L}, \quad \boldsymbol{M}_{\omega}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L} .`	$\boldsymbol{M}_{J}=\boldsymbol{D}, \quad \boldsymbol{M}_{1}=\boldsymbol{D}-\boldsymbol{A}_{L}, \quad \boldsymbol{M}_{\omega}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L} .$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$
lyche-numerical-linear-algebra_ EQ0759 (12.13)	271	■ 1.000 ● N|+0	`\boldsymbol{D} \boldsymbol{x}_{k+1}=\omega \left(\boldsymbol{A}_{L} \boldsymbol{x}_{k+1}+\boldsymbol{A}_{R} \boldsymbol{x}_{k}+\boldsymbol{b} \right)+(1-\omega) \boldsymbol{D} \boldsymbol{x}_{k} .`	$\boldsymbol{D} \boldsymbol{x}_{k+1}=\omega \left(\boldsymbol{A}_{L} \boldsymbol{x}_{k+1}+\boldsymbol{A}_{R} \boldsymbol{x}_{k}+\boldsymbol{b} \right)+(1-\omega) \boldsymbol{D} \boldsymbol{x}_{k} .$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega\big(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b}\big)+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$
lyche-numerical-linear-algebra_ EQ0760	271	■ 1.000 ● N|+0	`\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array} \right]=\left[\begin{array}{l} 1 \\ 1 \end{array}\right]`	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]\left[\begin{array}{l} x_{1} \\ x_{2} \end{array} \right]=\left[\begin{array}{l} 1 \\ 1 \end{array}\right]$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
lyche-numerical-linear-algebra_ EQ0761	271	■ 1.000 ● N|+0	`\boldsymbol{A}_{L}=\left[\begin{array}{ll} 0 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{D}=\left[ \begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{R}=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right],`	$\boldsymbol{A}_{L}=\left[\begin{array}{ll} 0 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_{R}=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right],$	$\boldsymbol{A}_L=\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},\quad \boldsymbol{D}=\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},\quad \boldsymbol{A}_R=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$
lyche-numerical-linear-algebra_ EQ0762	271	■ 1.000 ● K|+0	`\boldsymbol{M}_{J}=\boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{M}_{ \omega}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L}= \left[\begin{array}{cc} 2 \omega^{-1} & 0 \\ -1 & 2 \omega^{-1} \end{array}\right] .`	$\boldsymbol{M}_{J}=\boldsymbol{D}=\left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{M}_{\omega}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{L}= \left[\begin{array}{cc} 2 \omega^{-1} & 0 \\ -1 & 2 \omega^{-1} \end{array}\right] .$	$\boldsymbol{M}_J=\boldsymbol{D}=\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L=\begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0764 (12.15)	271	■ 1.000 ● K|+0	`\boldsymbol{G}_{J}=\boldsymbol{I}-\boldsymbol{D}^{-1} \boldsymbol{A}=\left[\begin{array}{cc} 0 & 1 / 2 \\ 1 / 2 & 0 \end{array}\right], \quad \boldsymbol{G}_{1}=\left[ \begin{array}{ll} 0 & 1 / 2 \\ 0 & 1 / 4 \end{array} \right] .`	$\boldsymbol{G}_{J}=\boldsymbol{I}-\boldsymbol{D}^{-1} \boldsymbol{A}=\left[\begin{array}{cc} 0 & 1 / 2 \\ 1 / 2 & 0 \end{array}\right], \quad \boldsymbol{G}_{1}=\left[\begin{array}{ll} 0 & 1 / 2 \\ 0 & 1 / 4 \end{array} \right] .$	$\boldsymbol{G}_{J}=\boldsymbol{I}-\boldsymbol{D}^{-1} \boldsymbol{A}=\begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_{1}=\begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0771 (12.18)	273	■ 1.000 ● N|+0	`\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha\left( \boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}\right) .`	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha\left(\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k}\right) .$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha\big(\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_{k}\big).$
lyche-numerical-linear-algebra EQ0773	274	■ 1.000 ● N|-1	`\rho_{\alpha}:=\rho(\boldsymbol{I}-\alpha \boldsymbol{A}):=\max _{j}\left|1-\alpha \lambda_{j} \right|= \begin{cases}1-\alpha \lambda_{1}, & \text { if } \alpha \leq 0 \\ 1-\alpha \lambda_{n}, & \text { if } 0<\alpha \leq \alpha_{o} \\ \alpha \lambda_{1}-1, & \text { if } \alpha>\alpha_{o},\end{cases}`	$\rho_{\alpha}:=\rho(\boldsymbol{I}-\alpha \boldsymbol{A}):=\max _{j}\left\|1-\alpha \lambda_{j} \right\|= \begin{cases}1-\alpha \lambda_{1}, & \text { if } \alpha \leq 0 \\ 1-\alpha \lambda_{n}, & \text { if } 0<\alpha \leq \alpha_{o} \\ \alpha \lambda_{1}-1, & \text { if } \alpha>\alpha_{o},\end{cases}$	$\rho_{\alpha}:=\rho(\boldsymbol{I}-\alpha \boldsymbol{A}):=\max_{j}\|1-\alpha \lambda_{j}\|= \begin{cases}1-\alpha \lambda_{1}, & \text{if } \alpha \leq 0 \\ 1-\alpha \lambda_{n}, & \text{if } 0<\alpha \leq \alpha_{o} \\ \alpha \lambda_{1}-1, & \text{if } \alpha>\alpha_{o},\end{cases}$
lyche-numerical-linear-algebra EQ0774	274	■ 1.000 ● N|+1	`\rho_{\alpha_{o}}=\alpha_{o} \lambda_{1}-1=\frac{ \lambda_{1}-\lambda_{n}}{\lambda_{1}+\lambda_{n}}=\frac{ \kappa-1}{\kappa+1}<1 .`	$\rho_{\alpha_{o}}=\alpha_{o} \lambda_{1}-1=\frac{ \lambda_{1}-\lambda_{n}}{\lambda_{1}+\lambda_{n}}=\frac{ \kappa-1}{\kappa+1}<1 .$	$\rho_{\alpha_{o}}=\alpha_{o} \lambda_{1}-1=\frac{\lambda_{1}-\lambda_{n}}{\lambda_{1}+\lambda_{n}}=\frac{\kappa-1}{\kappa+1}<1.$
lyche-numerical-linear-algebra EQ0776 (12.21)	275	■ 1.000 ● K|+0	`\boldsymbol{G}_{\omega}=(\boldsymbol{I}-\omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R}+(1-\omega) \boldsymbol{I}) .`	$\boldsymbol{G}_{\omega}=(\boldsymbol{I}-\omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R}+(1-\omega) \boldsymbol{I}) .$	$\boldsymbol{G}_{\omega}=(\boldsymbol{I}-\omega \boldsymbol{L})^{-1}\big(\omega \boldsymbol{R}+(1-\omega) \boldsymbol{I}\big).$
lyche-numerical-linear-algebra EQ0777 (12.22)	275	■ 1.000 ● K|+0	`\omega^{-1}(2-\omega)|1-\lambda|^{2} \boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x}=\left(1-|\lambda|^{2} \right) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x},`	$\omega^{-1}(2-\omega)\|1-\lambda\|^{2} \boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x}=\left(1-\|\lambda\|^{2} \right) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x},$	$\omega^{-1}(2-\omega)\|1-\lambda\|^{2} \boldsymbol{x}^{*} \boldsymbol{D} \boldsymbol{x}=(1-\|\lambda\|^{2}) \boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x},$
lyche-numerical-linear-algebra EQ0778	275	■ 1.000 ● N|+0	`\boldsymbol{G}_{\omega} \boldsymbol{x}=\left( \boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}\right) \boldsymbol{x}=\boldsymbol{x}- \left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{ \boldsymbol{L}}\right)^{-1} \boldsymbol{A} \boldsymbol{x}`	$\boldsymbol{G}_{\omega} \boldsymbol{x}=\left(\boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}\right) \boldsymbol{x}=\boldsymbol{x}- \left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{ \boldsymbol{L}}\right)^{-1} \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{G}_{\omega} \boldsymbol{x}=(\boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}) \boldsymbol{x}=\boldsymbol{x}-(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}})^{-1} \boldsymbol{A} \boldsymbol{x}$
lyche-numerical-linear-algebra EQ0779 (12.23)	276	■ 1.000 ● N|+0	`\boldsymbol{A} \boldsymbol{x}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right) \boldsymbol{y}, \quad \boldsymbol{y}:=(1-\lambda) \boldsymbol{x} .`	$\boldsymbol{A} \boldsymbol{x}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right) \boldsymbol{y}, \quad \boldsymbol{y}:=(1-\lambda) \boldsymbol{x} .$	$\boldsymbol{A} \boldsymbol{x}=(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}) \boldsymbol{y}, \quad \boldsymbol{y}:=(1-\lambda) \boldsymbol{x}.$
lyche-numerical-linear-algebra EQ0780 (12.24)	276	■ 1.000 ● N|+0	`\boldsymbol{E} \boldsymbol{y}=\lambda \boldsymbol{A} \boldsymbol{x}, \quad \boldsymbol{E}:=\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}- \boldsymbol{D}=\omega^{-1} \boldsymbol{D}- \boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A} .`	$\boldsymbol{E} \boldsymbol{y}=\lambda \boldsymbol{A} \boldsymbol{x}, \quad \boldsymbol{E}:=\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}- \boldsymbol{D}=\omega^{-1} \boldsymbol{D}- \boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A} .$	$\boldsymbol{E} \boldsymbol{y}=\lambda \boldsymbol{A} \boldsymbol{x}, \quad \boldsymbol{E}:=\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_{\boldsymbol{R}}-\boldsymbol{D}=\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}.$
lyche-numerical-linear-algebra EQ0781	276	■ 1.000 ● K|+0	`\boldsymbol{E} \boldsymbol{y}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}- \boldsymbol{A}\right) \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{y}= \boldsymbol{A} \boldsymbol{x}-(1-\lambda) \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A} \boldsymbol{x} .`	$\boldsymbol{E} \boldsymbol{y}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}- \boldsymbol{A}\right) \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{y}= \boldsymbol{A} \boldsymbol{x}-(1-\lambda) \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A} \boldsymbol{x} .$	$\boldsymbol{E} \boldsymbol{y}=(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}) \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-\boldsymbol{A} \boldsymbol{y}=\boldsymbol{A} \boldsymbol{x}-(1-\lambda) \boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{A} \boldsymbol{x}.$
lyche-numerical-linear-algebra EQ0784	276	■ 1.000 ● K|+0	`\lambda_{j, k}=4-2 \cos (j \pi h)-2 \cos (k \pi h), \quad j, k=1, \ldots, m, h=1 /(m+1) .`	$\lambda_{j, k}=4-2 \cos (j \pi h)-2 \cos (k \pi h), \quad j, k=1, \ldots, m, h=1 /(m+1) .$	$\lambda_{j,k}=4-2\cos(j\pi h)-2\cos(k\pi h), \quad j,k=1,\ldots,m, h=1/(m+1).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ0785 (12.25)	276	1.000 N|+0	`\mu_{j, k}=1-\frac{1}{4} \lambda_{j, k}=\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h), \quad j, k=1, \ldots, m .`	$\mu_{j, k}=1-\frac{1}{4} \lambda_{j, k}=\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h), \quad j, k=1, \ldots, m .$	$\mu_{j, k}=1-\frac{1}{4} \lambda_{j, k}=\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h), \quad j, k=1, \ldots, m .$
lyche-numerical-linear-algebra_ EQ0786 (12.26)	277	1.000 N|+0	`\left|\boldsymbol{x}_{k}-\boldsymbol{x}\right|_{2} \leq\left|\boldsymbol{G}_{J}\right|_{2}^{k}\left|\boldsymbol{x}_{0}-\boldsymbol{x}\right|_{2}=\cos ^{k}(\pi h)\left|\boldsymbol{x}_{0}-\boldsymbol{x}\right|_{2}, \quad k=0,1,2, \ldots`	$\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|_{2} \leq\left\|\boldsymbol{G}_{J}\right\|_{2}^{k}\left\|\boldsymbol{x}_{0}-\boldsymbol{x}\right\|_{2}=\cos ^{k}(\pi h)\left\|\boldsymbol{x}_{0}-\boldsymbol{x}\right\|_{2}, \quad k=0,1,2, \ldots$	$\\|\boldsymbol{x}_{k}-\boldsymbol{x}\\|_{2} \leq\\|\boldsymbol{G}_{J}\\|_{2}^{k}\\|\boldsymbol{x}_{0}-\boldsymbol{x}\\|_{2}=\cos ^{k}(\pi h)\\|\boldsymbol{x}_{0}-\boldsymbol{x}\\|_{2}, \quad k=0,1,2, \ldots$
lyche-numerical-linear-algebra_ EQ0788 (12.28)	277	1.000 K|+0	`\omega^{*}:=\frac{2}{1+\sqrt{1-\beta^{2}}}>1 .`	$\omega^{*}:=\frac{2}{1+\sqrt{1-\beta^{2}}}>1 .$	$\omega^{*}:=\frac{2}{1+\sqrt{1-\beta^{2}}}>1 .$
lyche-numerical-linear-algebra_ EQ0790 (12.30)	277	1.000 W|+2	`\omega^{*}=\frac{2}{1+\sin (\pi h)}, \quad \rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\omega^{*}-1=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}, \quad h=\frac{1}{m+1} .`	$\omega^{*}=\frac{2}{1+\sin (\pi h)}, \quad \rho\left(\boldsymbol{G}_{\omega^{*}}\right)=\omega^{*}-1=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}, \quad h=\frac{1}{m+1} .$	$\omega^{*}=\frac{2}{1+\sin (\pi h)}, \quad \rho(\boldsymbol{G}_{\omega^{*}})=\omega^{*}-1=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}, \quad h=\frac{1}{m+1} .$
lyche-numerical-linear-algebra_ EQ0792 (12.31)	278	1.000 N|+0	`\tilde{k}:=\frac{s \log (10)}{\eta}`	$\tilde{k}:=\frac{s \log (10)}{\eta}$	$\tilde{k}:=\frac{s \log (10)}{\eta}$
lyche-numerical-linear-algebra_ EQ0794	279	1.000 N|+0	`\tilde{k}_{n}=\frac{2 \log (10) s}{\pi^{2}} n+O\left(n^{-1}\right)=O(n) .`	$\tilde{k}_{n}=\frac{2 \log (10) s}{\pi^{2}} n+O\left(n^{-1}\right)=O(n) .$	$\tilde{k}_{n}=\frac{2 \log (10) s}{\pi^{2}} n+O(n^{-1})=O(n) .$
lyche-numerical-linear-algebra_ EQ0795	279	1.000 N|+0	`\tilde{k}_{n}=\frac{\log (10) s}{\pi^{2}} n+O\left(n^{-1}\right)=O(n) .`	$\tilde{k}_{n}=\frac{\log (10) s}{\pi^{2}} n+O\left(n^{-1}\right)=O(n) .$	$\tilde{k}_{n}=\frac{\log (10) s}{\pi^{2}} n+O(n^{-1})=O(n) .$
lyche-numerical-linear-algebra_ EQ0796	279	1.000 N|+0	`\tilde{k}_{n}=\frac{\log (10) s}{2 \pi} \sqrt{n}+O\left(n^{-1 / 2}\right)=O(\sqrt{n}) .`	$\tilde{k}_{n}=\frac{\log (10) s}{2 \pi} \sqrt{n}+O\left(n^{-1 / 2}\right)=O(\sqrt{n}) .$	$\tilde{k}_{n}=\frac{\log (10) s}{2 \pi} \sqrt{n}+O(n^{-1 / 2})=O(\sqrt{n}) .$
lyche-numerical-linear-algebra_ EQ0797 (12.32)	280	1.000 K|+0	`\left|\boldsymbol{x}_{k}-\boldsymbol{x}_{k-1}\right| \geq \frac{1-|\boldsymbol{G}|}{|\boldsymbol{G}|}\left|\boldsymbol{x}_{k}-\boldsymbol{x}\right|, \quad k \geq 1 .`	$\left\|\boldsymbol{x}_{k}-\boldsymbol{x}_{k-1}\right\| \geq \frac{1-\\|\boldsymbol{G}\\|}{\\|\boldsymbol{G}\\|}\left\|\boldsymbol{x}_{k}-\boldsymbol{x}\right\|, \quad k \geq 1 .$	$\\|\boldsymbol{x}_{k}-\boldsymbol{x}_{k-1}\\| \geq \frac{1-\\|\boldsymbol{G}\\|}{\\|\boldsymbol{G}\\|}\\|\boldsymbol{x}_{k}-\boldsymbol{x}\\|, \quad k \geq 1 .$
lyche-numerical-linear-algebra_ EQ0799 (12.33)	280	1.000 N|+0	`\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\boldsymbol{M}^{-1} \boldsymbol{r}_{k}, \quad \boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k} .`	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\boldsymbol{M}^{-1} \boldsymbol{r}_{k}, \quad \boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k} .$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\boldsymbol{M}^{-1} \boldsymbol{r}_{k}, \quad \boldsymbol{r}_{k}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k} .$
lyche-numerical-linear-algebra_ EQ0800	281	1.000 N|+0	`\lim _{k \rightarrow \infty} \boldsymbol{A}^{k}=\mathbf{0} \Longleftrightarrow \rho(\boldsymbol{A})<1,`	$\lim _{k \rightarrow \infty} \boldsymbol{A}^{k}=\mathbf{0} \Longleftrightarrow \rho(\boldsymbol{A})<1,$	$\lim _{k \rightarrow \infty} \boldsymbol{A}^{k}=\mathbf{0} \Longleftrightarrow \rho(\boldsymbol{A})<1,$
lyche-numerical-linear-algebra_ EQ0801	281	1.000 N|+0	`\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}=\left[\begin{array}{ccc} \lambda_{1} & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_{2} & t^{-1} r_{23} \\ 0 & 0 & \lambda_{3} \end{array}\right] .`	$\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}=\left[\begin{array}{ccc} \lambda_{1} & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_{2} & t^{-1} r_{23} \\ 0 & 0 & \lambda_{3} \end{array}\right] .$	$\boldsymbol{D}_{t} \boldsymbol{R} \boldsymbol{D}_{t}^{-1}=\left[\begin{array}{ccc} \lambda_{1} & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_{2} & t^{-1} r_{23} \\ 0 & 0 & \lambda_{3} \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ0803 (12.34)	282	1.000 N|+0	`\lim _{k \rightarrow \infty}\left|\boldsymbol{A}^{k}\right|^{1 / k}=\rho(\boldsymbol{A}) .`	$\lim _{k \rightarrow \infty}\left\|\boldsymbol{A}^{k}\right\|^{1 / k}=\rho(\boldsymbol{A}) .$	$\lim _{k \rightarrow \infty}\\|\boldsymbol{A}^{k}\\|^{1 / k}=\rho(\boldsymbol{A}) .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
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<tr><td>lyche-numerical-linear-al
EQ0804</td><td>282</td><td>■ 1.000
● N|+0</td><td>\left\|\boldsymbol{A}^{k}\right\|=\left\|(\rho(\boldsymbol{A})+\epsilon)^{k} \boldsymbol{B}^{k}\right\|=(\rho(\boldsymbol{A})+\epsilon)^{k}\left\|\boldsymbol{B}^{k}\right\|<(\rho(\boldsymbol{A})+\epsilon)^{k} .</td><td>$\left\|\boldsymbol{A}^{k}\right\|=\left\|(\rho(\boldsymbol{A})+\epsilon)^{k} \boldsymbol{B}^{k}\right\|=(\rho(\boldsymbol{A})+\epsilon)^{k}\left\|\boldsymbol{B}^{k}\right\|<(\rho(\boldsymbol{A})+\epsilon)^{k} .$</td><td>$\|\boldsymbol{A}^{k}\| = \|(\rho(\boldsymbol{A})+\epsilon)^{k}\boldsymbol{B}^{k}\| = \big(\rho(\boldsymbol{A})+\epsilon\big)^{k}\|\boldsymbol{B}^{k}\| < \big(\rho(\boldsymbol{A})+\epsilon\big)^{k}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0805</td><td>282</td><td>■ 1.000
● N|+0</td><td>\sum_{k=0}^{\infty} \boldsymbol{B}^{k}</td><td>$\sum_{k=0}^{\infty} \boldsymbol{B}^{k}$</td><td>$\sum_{k=0}^{\infty} \boldsymbol{B}^{k}$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0807</td><td>283</td><td>■ 1.000
● S|+0</td><td>\left\|\boldsymbol{S}_{l}-\boldsymbol{S}_{m}\right\|=\left\|\sum_{k=m+1}^{l} \boldsymbol{B}^{k}\right\| \leq \sum_{k=m+1}^{l}\|\boldsymbol{B}\|^{k} \leq\|\boldsymbol{B}\|^{m+1} \sum_{k=0}^{\infty}\|\boldsymbol{B}\|^{k}=\frac{\|\boldsymbol{B}\|^{m+1}}{1-\|\boldsymbol{B}\|} .</td><td>$\left\|\boldsymbol{S}_{l}-\boldsymbol{S}_{m}\right\|=\left\|\sum_{k=m+1}^{l} \boldsymbol{B}^{k}\right\| \leq \sum_{k=m+1}^{l}\|\boldsymbol{B}\|^{k} \leq\|\boldsymbol{B}\|^{m+1} \sum_{k=0}^{\infty}\|\boldsymbol{B}\|^{k}=\frac{\|\boldsymbol{B}\|^{m+1}}{1-\|\boldsymbol{B}\|} .$</td><td>$\|\boldsymbol{S}_{l}-\boldsymbol{S}_{m}\| = \|\sum_{k=m+1}^{l}\boldsymbol{B}^{k}\| \leq \sum_{k=m+1}^{l}\|\boldsymbol{B}\|^{k} \leq \|\boldsymbol{B}\|^{m+1}\sum_{k=0}^{\infty}\|\boldsymbol{B}\|^{k} = \frac{\|\boldsymbol{B}\|^{m+1}}{1-\|\boldsymbol{B}\|}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0808 (12.36)</td><td>283</td><td>■ 1.000
● N|+0</td><td>\left(\sum_{k=0}^{m} \boldsymbol{B}^{k}\right)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}+\boldsymbol{B}+\cdots+\boldsymbol{B}^{m}-\left(\boldsymbol{B}+\cdots+\boldsymbol{B}^{m+1}\right)=\boldsymbol{I}-\boldsymbol{B}^{m+1} .</td><td>$\left(\sum_{k=0}^{m} \boldsymbol{B}^{k}\right)(\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}+\boldsymbol{B}+\cdots+\boldsymbol{B}^{m}-\left(\boldsymbol{B}+\cdots+\boldsymbol{B}^{m+1}\right)=\boldsymbol{I}-\boldsymbol{B}^{m+1} .$</td><td>$(\sum_{k=0}^{m}\boldsymbol{B}^{k})(\boldsymbol{I}-\boldsymbol{B}) = \boldsymbol{I}+\boldsymbol{B}+\cdots+\boldsymbol{B}^{m}-(\boldsymbol{B}+\cdots+\boldsymbol{B}^{m+1}) = \boldsymbol{I}-\boldsymbol{B}^{m+1}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0809 (12.37)</td><td>283</td><td>■ 1.000
● N|+0</td><td>\frac{1}{4}\left(v_{i-1, j}+v_{i, j-1}+v_{i+1, j}+v_{i, j+1}\right)=\mu v_{i, j}, \quad i, j=1, \ldots, m,</td><td>$\frac{1}{4}\left(v_{i-1, j}+v_{i, j-1}+v_{i+1, j}+v_{i, j+1}\right)=\mu v_{i, j}, \quad i, j=1, \ldots, m,$</td><td>$\frac{1}{4}(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \ldots, m,$</td></tr>
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EQ0810 (12.38)</td><td>283</td><td>■ 1.000
● K|+0</td><td>(\omega \boldsymbol{R}+\lambda \omega \boldsymbol{L}) \boldsymbol{w}=(\lambda+\omega-1) \boldsymbol{w} .</td><td>$(\omega \boldsymbol{R}+\lambda \omega \boldsymbol{L}) \boldsymbol{w}=(\lambda+\omega-1) \boldsymbol{w} .$</td><td>$(\omega\boldsymbol{R}+\lambda\omega\boldsymbol{L})\boldsymbol{w} = (\lambda+\omega-1)\boldsymbol{w}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0811 (12.39)</td><td>284</td><td>■ 1.000
● N|+1</td><td>\frac{\omega}{4}\left(\lambda w_{i-1, j}+\lambda w_{i, j-1}+w_{i+1, j}+w_{i, j+1}\right)=(\lambda+\omega-1) w_{i, j}</td><td>$\frac{\omega}{4}\left(\lambda w_{i-1, j}+\lambda w_{i, j-1}+w_{i+1, j}+w_{i, j+1}\right)=(\lambda+\omega-1) w_{i, j}$</td><td>$\frac{\omega}{4}(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1}) = (\lambda+\omega-1)w_{i,j},$</td></tr>
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EQ0812 (12.40)</td><td>284</td><td>■ 1.000
● K|+0</td><td>\mu:=\frac{\lambda+\omega-1}{\omega \lambda^{1 / 2}}</td><td>$\mu:=\frac{\lambda+\omega-1}{\omega \lambda^{1 / 2}}$</td><td>$\mu := \frac{\lambda+\omega-1}{\omega\lambda^{1/2}}$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0813 (12.41)</td><td>284</td><td>■ 1.000
● K|+0</td><td>\mu \omega \lambda^{1 / 2}=\lambda+\omega-1</td><td>$\mu \omega \lambda^{1 / 2}=\lambda+\omega-1$</td><td>$\mu\omega\lambda^{1/2} = \lambda+\omega-1$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0814</td><td>284</td><td>■ 1.000
● K|+0</td><td>\frac{\omega}{4}\left(v_{i-1, j}+v_{i, j-1}+v_{i+1, j}+v_{i, j+1}\right)=\lambda^{-1 / 2}(\lambda+\omega-1) v_{i, j} .</td><td>$\frac{\omega}{4}\left(v_{i-1, j}+v_{i, j-1}+v_{i+1, j}+v_{i, j+1}\right)=\lambda^{-1 / 2}(\lambda+\omega-1) v_{i, j} .$</td><td>$\frac{\omega}{4}(v_{i-1,j}+v_{i,j-1}+v_{i+1,j}+v_{i,j+1}) = \lambda^{-1/2}(\lambda+\omega-1)v_{i,j}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0817</td><td>284</td><td>■ 1.000
● N|+1</td><td>\frac{\omega}{4}\left(\lambda w_{i-1, j}+\lambda w_{i, j-1}+w_{i+1, j}+w_{i, j+1}\right)=\omega \mu \lambda^{1 / 2} w_{i, j}</td><td>$\frac{\omega}{4}\left(\lambda w_{i-1, j}+\lambda w_{i, j-1}+w_{i+1, j}+w_{i, j+1}\right)=\omega \mu \lambda^{1 / 2} w_{i, j}$</td><td>$\frac{\omega}{4}(\lambda w_{i-1,j}+\lambda w_{i,j-1}+w_{i+1,j}+w_{i,j+1}) = \omega\mu\lambda^{1/2}w_{i,j},$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0818 (12.42)</td><td>285</td><td>■ 1.000
● K|+0</td><td>\lambda(\mu):=\frac{1}{4}\left(\omega \mu \pm \sqrt{(\omega \mu)^{2}-4(\omega-1)}\right)^{2} .</td><td>$\lambda(\mu):=\frac{1}{4}\left(\omega \mu \pm \sqrt{(\omega \mu)^{2}-4(\omega-1)}\right)^{2} .$</td><td>$\lambda(\mu) := \frac{1}{4}\Big(\omega\mu \pm \sqrt{(\omega\mu)^{2}-4(\omega-1)}\Big)^{2}.$</td></tr>
<tr><td>lyche-numerical-linear-al
EQ0821 (12.43)</td><td>285</td><td>■ 1.000
● N|+0</td><td>\omega=\tilde{\omega}(\mu):=2 \frac{1-\sqrt{1-\mu^{2}}}{\mu^{2}}=\frac{2}{1+\sqrt{1-\mu^{2}}} .</td><td>$\omega=\tilde{\omega}(\mu):=2 \frac{1-\sqrt{1-\mu^{2}}}{\mu^{2}}=\frac{2}{1+\sqrt{1-\mu^{2}}} .$</td><td>$\omega = \tilde{\omega}(\mu) := 2\frac{1-\sqrt{1-\mu^{2}}}{\mu^{2}} = \frac{2}{1+\sqrt{1-\mu^{2}}}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

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<tr><td>lyche-numerical-linear-a EQ0822</td><td>285</td><td>■ 1.000 ● K|+0</td><td><code>|\lambda(\mu)|=\frac{1}{4}\left((\omega \mu)^{2}+4(\omega-1)-(\omega \mu)^{2}\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2 .</code></td><td>$|\lambda(\mu)|=\frac{1}{4}\left((\omega \mu)^{2}+4(\omega-1)-(\omega \mu)^{2}\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2 .$</td><td>$|\lambda(\mu)|=\frac{1}{4}\Big((\omega\mu)^{2}+4(\omega-1)-(\omega\mu)^{2}\Big)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0823 (12.44)</td><td>285</td><td>■ 1.000 ● K|+0</td><td><code>\lambda(\mu)=\frac{1}{4}\left(\omega \mu+\sqrt{(\omega \mu)^{2}-4(\omega-1)}\right)^{2}, \quad 0<\omega \leq \tilde{\omega}(\mu) .</code></td><td>$\lambda(\mu)=\frac{1}{4}\left(\omega \mu+\sqrt{(\omega \mu)^{2}-4(\omega-1)}\right)^{2}, \quad 0<\omega \leq \tilde{\omega}(\mu) .$</td><td>$\lambda(\mu)=\frac{1}{4}\Big(\omega\mu+\sqrt{(\omega\mu)^{2}-4(\omega-1)}\Big)^{2}, \quad 0<\omega\leq\tilde{\omega}(\mu).$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0824</td><td>285</td><td>■ 1.000 ● N|+0</td><td><code>\frac{d}{d \omega}\left(\omega \beta+\sqrt{(\omega \beta)^{2}-4(\omega-1)}\right)=\frac{\beta \sqrt{(\omega \beta)^{2}-4(\omega-1)}+\omega \beta^{2}-2}{\sqrt{(\omega \beta)^{2}-4(\omega-1)}} .</code></td><td>$\frac{d}{d \omega}\left(\omega \beta+\sqrt{(\omega \beta)^{2}-4(\omega-1)}\right)=\frac{\beta \sqrt{(\omega \beta)^{2}-4(\omega-1)}+\omega \beta^{2}-2}{\sqrt{(\omega \beta)^{2}-4(\omega-1)}} .$</td><td>$\frac{d}{d\omega}\Big(\omega\beta+\sqrt{(\omega\beta)^{2}-4(\omega-1)}\Big)=\frac{\beta\sqrt{(\omega\beta)^{2}-4(\omega-1)}+\omega\beta^{2}-2}{\sqrt{(\omega\beta)^{2}-4(\omega-1)}}.$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0826</td><td>286</td><td>■ 1.000 ● N|+0</td><td><code>\rho\left(\boldsymbol{G}_{1}\right)=\left|a_{21} a_{12} / a_{11} a_{22}\right| .</code></td><td>$\rho\left(\boldsymbol{G}_{1}\right)=\left|a_{21} a_{12} / a_{11} a_{22}\right| .$</td><td>$\rho(\boldsymbol{G}_{1})=|a_{21}a_{12}/a_{11}a_{22}|.$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0829 (12.45)</td><td>287</td><td>■ 1.000 ● N|+0</td><td><code>\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{N} \boldsymbol{x}_{k}+\boldsymbol{b},</code></td><td>$\boldsymbol{M} \boldsymbol{x}_{k+1}=\boldsymbol{N} \boldsymbol{x}_{k}+\boldsymbol{b},$</td><td>$\boldsymbol{M}\boldsymbol{x}_{k+1}=\boldsymbol{N}\boldsymbol{x}_{k}+\boldsymbol{b},$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0832 (12.48)</td><td>288</td><td>■ 1.000 ● N|+0</td><td><code>x(i)=b(i)-\sum_{j \neq i} a(i, j) x(j)</code></td><td>$x(i)=b(i)-\sum_{j \neq i} a(i, j) x(j)$</td><td>$x(i)=b(i)-\sum_{j\neq i}a(i,j)x(j)$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0838 (13.4)</td><td>291</td><td>■ 1.000 ● W|+0</td><td><code>Q(x, y):=\frac{1}{2}\left[\begin{array}{ll} x & y \end{array}\right]\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=x^{2}-x y+y^{2}</code></td><td>$Q(x, y):=\frac{1}{2}\left[\begin{array}{ll} x & y \end{array}\right]\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=x^{2}-x y+y^{2}$</td><td>$Q(x,y):=\frac{1}{2}[x\ y]\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=x^{2}-xy+y^{2}$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0839 (13.5)</td><td>291</td><td>■ 1.000 ● N|+1</td><td><code>Q(\boldsymbol{y}+\varepsilon \boldsymbol{h})=Q(\boldsymbol{y})-\varepsilon \boldsymbol{h}^{T} r(\boldsymbol{y})+\frac{1}{2} \varepsilon^{2} \boldsymbol{h}^{T} \boldsymbol{A} \boldsymbol{h}, \text { where } r(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y} .</code></td><td>$Q(\boldsymbol{y}+\varepsilon \boldsymbol{h})=Q(\boldsymbol{y})-\varepsilon \boldsymbol{h}^{T} r(\boldsymbol{y})+\frac{1}{2} \varepsilon^{2} \boldsymbol{h}^{T} \boldsymbol{A} \boldsymbol{h}, \text { where } r(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{y} .$</td><td>$Q(\boldsymbol{y}+\varepsilon\boldsymbol{h})=Q(\boldsymbol{y})-\varepsilon\boldsymbol{h}^{T}r(\boldsymbol{y})+\frac{1}{2}\varepsilon^{2}\boldsymbol{h}^{T}\boldsymbol{A}\boldsymbol{h}, \text{ where } r(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{y}.$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0841</td><td>293</td><td>■ 1.000 ● N|+0</td><td><code>Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_{k}+\alpha \boldsymbol{p}_{k}\right) .</code></td><td>$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_{k}+\alpha \boldsymbol{p}_{k}\right) .$</td><td>$Q(\boldsymbol{x}_{k+1})=Q(\boldsymbol{x}_{k}+\alpha_{k}\boldsymbol{p}_{k})=\min_{\alpha\in\mathbb{R}}Q(\boldsymbol{x}_{k}+\alpha\boldsymbol{p}_{k}).$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0842 (13.7)</td><td>293</td><td>■ 1.000 ● W|+0</td><td><code>\alpha_{k}:=\frac{\boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} .</code></td><td>$\alpha_{k}:=\frac{\boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}} .$</td><td>$\alpha_{k}:=\frac{\boldsymbol{p}_{k}^{T}\boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T}\boldsymbol{A}\boldsymbol{p}_{k}}.$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0843 (13.8)</td><td>293</td><td>■ 1.000 ● W|+0</td><td><code>\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\left(\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}}\right) \boldsymbol{r}_{k} .</code></td><td>$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\left(\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}}\right) \boldsymbol{r}_{k} .$</td><td>$\boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+(\frac{\boldsymbol{r}_{k}^{T}\boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T}\boldsymbol{A}\boldsymbol{r}_{k}})\boldsymbol{r}_{k}.$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0844 (13.9)</td><td>293</td><td>■ 1.000 ● W|+0</td><td><code>\begin{aligned} \boldsymbol{p}_{k} & =\boldsymbol{r}_{k}, \boldsymbol{t}_{k}=\boldsymbol{A} \boldsymbol{p}_{k}, \\ \alpha_{k} & =\left(\boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k}\right) /\left(\boldsymbol{p}_{k}^{T} \boldsymbol{t}_{k}\right), \\ \boldsymbol{x}_{k+1} & = \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \\ \boldsymbol{r}_{k+1} & =\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{t}_{k} . \end{aligned}</code></td><td>$\begin{aligned} \boldsymbol{p}_{k} & =\boldsymbol{r}_{k}, \boldsymbol{t}_{k}=\boldsymbol{A} \boldsymbol{p}_{k}, \\ \alpha_{k} & =\left(\boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k}\right) /\left(\boldsymbol{p}_{k}^{T} \boldsymbol{t}_{k}\right), \\ \boldsymbol{x}_{k+1} & = \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \\ \boldsymbol{r}_{k+1} & =\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{t}_{k} . \end{aligned}$</td><td>$\begin{aligned} \boldsymbol{p}_{k} & =\boldsymbol{r}_{k},\ \boldsymbol{t}_{k}=\boldsymbol{A} \boldsymbol{p}_{k}, \\ \alpha_{k} & =(\boldsymbol{p}_{k}^{T} \boldsymbol{r}_{k})/(\boldsymbol{p}_{k}^{T} \boldsymbol{t}_{k}), \\ \boldsymbol{x}_{k+1} & = \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \\ \boldsymbol{r}_{k+1} & =\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{t}_{k} . \end{aligned}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0845 (13.10)</td><td>293</td><td>■ 1.000 ● N|+0</td><td><code>\boldsymbol{r}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \left(\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k} \right)=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k} .</code></td><td>$\boldsymbol{r}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \left(\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k} \right)=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k} .$</td><td>$\boldsymbol{r}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{A}(\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k})=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0853 (13.17)</td><td>296</td><td>■ 1.000 ● W|+0</td><td><code>\boldsymbol{p}_{k+1}:=\boldsymbol{r}_{k+1}+ \beta_{k} \boldsymbol{p}_{k}, \quad \beta_{k}:= \frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} .</code></td><td>$\boldsymbol{p}_{k+1}:=\boldsymbol{r}_{k+1}+ \beta_{k} \boldsymbol{p}_{k}, \quad \beta_{k}:= \frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} .$</td><td>$\boldsymbol{p}_{k+1}:=\boldsymbol{r}_{k+1}+ \beta_{k} \boldsymbol{p}_{k}, \quad \beta_{k}:= \frac{\boldsymbol{r}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0856</td><td>298</td><td>■ 1.000 ● W|-2</td><td><code>\boldsymbol{A}=\operatorname{tridiag}_{m}(a, d, a) \otimes \boldsymbol{I}_{m}+\boldsymbol{I}_{m} \otimes \operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{ \left(m^{2}\right) \times\left(m^{2}\right)}</code></td><td>$\boldsymbol{A}=\operatorname{tridiag}_{m}(a, d, a) \otimes \boldsymbol{I}_{m}+\boldsymbol{I}_{m} \otimes \operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{ \left(m^{2}\right) \times\left(m^{2}\right)}$</td><td>$\boldsymbol{A}=\operatorname{tridiag}_{m}(a, d, a) \otimes \boldsymbol{I}_{m}+\boldsymbol{I}_{m} \otimes \operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{(m^{2}) \times (m^{2})}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0859</td><td>300</td><td>■ 1.000 ● C|-3</td><td><code>\boldsymbol{T}_{2}:=\operatorname{tridiag}_{m}(a, d, a) \otimes \boldsymbol{I}_{m}+\boldsymbol{I}_{m} \otimes \operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{ \left(m^{2}\right) \times\left(m^{2}\right)} .</code></td><td>$\boldsymbol{T}_{2}:=\operatorname{tridiag}_{m}(a, d, a) \otimes \boldsymbol{I}_{m}+\boldsymbol{I}_{m} \otimes \operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{ \left(m^{2}\right) \times\left(m^{2}\right)} .$</td><td>$\boldsymbol{T}_{2}:=\operatorname{tridiag}_{m}(a, d, a) \otimes \boldsymbol{I}_{m}+\boldsymbol{I}_{m} \otimes \operatorname{tridiag}_{m}(a, d, a) \in \mathbb{R}^{(m^{2}) \times (m^{2})} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0860 (13.21)</td><td>300</td><td>■ 1.000 ● N|+0</td><td><code>\lambda_{j, k}=2 d+2 a \cos (j \pi h)+2 a \cos (k \pi h), \quad j, k=1, \ldots, m .</code></td><td>$\lambda_{j, k}=2 d+2 a \cos (j \pi h)+2 a \cos (k \pi h), \quad j, k=1, \ldots, m .$</td><td>$\lambda_{j, k}=2 d+2 a \cos (j \pi h)+2 a \cos (k \pi h), \quad j, k=1, \ldots, m .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0864</td><td>301</td><td>■ 1.000 ● S|+0</td><td><code>\mathbb{W}_{0} \subset \mathbb{W}_{1} \subset \mathbb{W}_{2} \subset \cdots \subset \mathbb{W}_{n} \subset \mathbb{R}^{n}</code></td><td>$\mathbb{W}_{0} \subset \mathbb{W}_{1} \subset \mathbb{W}_{2} \subset \cdots \subset \mathbb{W}_{n} \subset \mathbb{R}^{n}$</td><td>$\mathbb{W}_{0} \subset \mathbb{W}_{1} \subset \mathbb{W}_{2} \subset \cdots \subset \mathbb{W}_{n} \subset \mathbb{R}^{n}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0865 (13.22)</td><td>301</td><td>■ 1.000 ● N|+0</td><td><code>\boldsymbol{x}_{k}-\boldsymbol{x}_{0} \in \mathbb{W}_{k}, \quad \boldsymbol{r}_{k}, \boldsymbol{p}_{k} \in \mathbb{W}_{k+1}, \quad k=0,1, \ldots,</code></td><td>$\boldsymbol{x}_{k}-\boldsymbol{x}_{0} \in \mathbb{W}_{k}, \quad \boldsymbol{r}_{k}, \boldsymbol{p}_{k} \in \mathbb{W}_{k+1}, \quad k=0,1, \ldots,$</td><td>$\boldsymbol{x}_{k}-\boldsymbol{x}_{0} \in \mathbb{W}_{k}, \quad \boldsymbol{r}_{k}, \boldsymbol{p}_{k} \in \mathbb{W}_{k+1}, \quad k=0,1, \ldots,$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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● N|-1</td><td>\|f\|_{\infty}=\max _{a \leq x \leq b}|f(x)| .</td><td>$\|f\|_{\infty}=\max _{a \leq x \leq b}|f(x)| .$</td><td>$\|f\|_{\infty}=\max _{a \leq x \leq b}|f(x)| .$</td></tr>
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EQ0882</td><td>305</td><td>■ 1.000
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EQ0883 (13.32)</td><td>305</td><td>■ 1.000
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<tr><td>lyche-numerical-linear-a
EQ0886</td><td>305</td><td>■ 1.000
● N|+0</td><td>-Q\left(\mu_{j-1}\right) \leq\|Q\|_{\infty} \leq\left \|Q^{*}\right\|_{\infty}=-Q^{*}\left(\mu_{j-1}\right) .</td><td>$-Q\left(\mu_{j-1}\right) \leq\|Q\|_{\infty} \leq\left\|Q^{*}\right\|_{\infty}=-Q^{*}\left(\mu_{j-1}\right) .$</td><td>$-Q(\mu_{j-1}) \leq\|Q\|_{\infty} \leq\|Q^{*}\|_{\infty}=-Q^{*}(\mu_{j-1}) .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0887 (13.33)</td><td>306</td><td>■ 1.000
● K|+0</td><td>Q^{*}(x)=T_{k}\left(\frac{b+a-2 x}{b-a}\right) / T_{k} \left(\frac{b+a}{b-a}\right) .</td><td>$Q^{*}(x)=T_{k}\left(\frac{b+a-2 x}{b-a}\right) / T_{k}\left(\frac{b+a}{b-a}\right) .$</td><td>$Q^{*}(x)=T_{k}\left(\frac{b+a-2 x}{b-a}\right) / T_{k}\left(\frac{b+a}{b-a}\right) .$</td></tr>
<tr><td>lyche-numerical-linear-a
EQ0888 (13.34)</td><td>306</td><td>■ 1.000
● N|-1</td><td>\max _{a \leq x \leq b}\left|T_{k}\left(\frac{b+a-2 x}{b-a}\right)\right|=\max _{-1 \leq t \leq 1} \left|T_{k}(t)\right|=1 .</td><td>$\max _{a \leq x \leq b}\left|T_{k}\left(\frac{b+a-2 x}{b-a}\right)\right|=\max _{-1 \leq t \leq 1}\left|T_{k}(t)\right|=1 .$</td><td>$\max _{a \leq x \leq b}\left|T_{k}\left(\frac{b+a-2 x}{b-a}\right)\right|=\max _{-1 \leq t \leq 1}\left|T_{k}(t)\right|=1 .$</td></tr>
<tr><td>lyche-numerical-linear-a
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● N|+1</td><td>t+\sqrt{t^{2}-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}, \quad \kappa=b / a .</td><td>$t+\sqrt{t^{2}-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}, \quad \kappa=b / a .$</td><td>$t+\sqrt{t^{2}-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}, \quad \kappa=b / a .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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EQ0890 (13.35)</td><td>306</td><td>■ 1.000
● W|+0</td><td><code>T_{k}\left(\frac{b+a}{b-a}\right)=T_{k}\left(\frac{ \kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{ \sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^{k}+\left(\frac{ \sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}\right]</code></td><td>$T_{k}\left(\frac{b+a}{b-a}\right)=T_{k}\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^{k}+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}\right]$</td><td>$T_{k}\left(\frac{b+a}{b-a}\right)=T_{k}\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^{k}+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}\right]$</td></tr>
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EQ0893 (13.38)</td><td>307</td><td>■ 1.000
● N|+0</td><td><code>t_{i}=\frac{c_{i}^{2}}{\sum_{j=1}^{n} c_{j}^{2}} \geq 0, \quad i=1, \ldots, n \text { and } \sum_{i=1}^{n} t_{i}=1 .</code></td><td>$t_{i}=\frac{c_{i}^{2}}{\sum_{j=1}^{n} c_{j}^{2}} \geq 0, \quad i=1, \ldots, n \text { and } \sum_{i=1}^{n} t_{i}=1 .$</td><td>$t_{i}=\frac{c_{i}^{2}}{\sum_{j=1}^{n} c_{j}^{2}} \geq 0, \quad i=1, \ldots, n \text { and } \sum_{i=1}^{n} t_{i}=1 .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
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● N|+0</td><td><code>\sqrt{a b}=\sqrt{(a c)(b / c)} \leq(a c+b / c) / 2= \frac{1}{2} \sum_{i=1}^{n} t_{i}\left(\lambda_{i} c+1 /\left(\lambda_{i} c\right)\right)=\frac{1}{2} \sum_{i=1}^{n} t_{i} f\left(\lambda_{i} c\right),</code></td><td>$\sqrt{a b}=\sqrt{(a c)(b / c)} \leq(a c+b / c) / 2= \frac{1}{2} \sum_{i=1}^{n} t_{i}\left(\lambda_{i} c+1 /\left(\lambda_{i} c\right)\right)=\frac{1}{2} \sum_{i=1}^{n} t_{i} f\left(\lambda_{i} c\right),$</td><td>$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq(ac+b/c)/2=\frac{1}{2}\sum_{i=1}^{n}t_{i}\big(\lambda_{i}c+1/(\lambda_{i}c)\big)=\frac{1}{2}\sum_{i=1}^{n}t_{i}f(\lambda_{i}c),$</td></tr>
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EQ0895</td><td>307</td><td>■ 1.000
● N|+0</td><td><code>\sqrt{a b} \leq \frac{1}{2} \max _{m c \leq x \leq M c} f(x) .</code></td><td>$\sqrt{a b} \leq \frac{1}{2} \max _{m c \leq x \leq M c} f(x) .$</td><td>$\sqrt{ab}\leq\frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$</td></tr>
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● N|-1</td><td><code>\sqrt{a b} \leq \frac{1}{2} \max \{f(m c), f(M c)\} .</code></td><td>$\sqrt{a b} \leq \frac{1}{2} \max \{f(m c), f(M c)\} .$</td><td>$\sqrt{ab}\leq\frac{1}{2}\max\{f(mc),f(Mc)\}.$</td></tr>
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● W|+0</td><td><code>1=\left(\sum_{i=1}^{n} t_{i}\right)^{2}=\left(\sum_{i=1}^{n}\left(t_{i} \lambda_{i}\right)^{1 / 2} \left(t_{i} / \lambda_{i}\right)^{1 / 2}\right)^{2} \leq\left(\sum_{i=1}^{n} t_{i} \lambda_{i}\right)\left(\sum_{i=1}^{n} t_{i} / \lambda_{i}\right)=a b .</code></td><td>$1=\left(\sum_{i=1}^{n} t_{i}\right)^{2}=\left(\sum_{i=1}^{n}\left(t_{i} \lambda_{i}\right)^{1 / 2}\left(t_{i} / \lambda_{i}\right)^{1 / 2}\right)^{2} \leq\left(\sum_{i=1}^{n} t_{i} \lambda_{i}\right)\left(\sum_{i=1}^{n} t_{i} / \lambda_{i}\right)=a b .$</td><td>$1=\big(\sum_{i=1}^{n}t_{i}\big)^{2}=\left(\sum_{i=1}^{n}(t_{i}\lambda_{i})^{1/2}(t_{i}/\lambda_{i})^{1/2}\right)^{2}\leq\big(\sum_{i=1}^{n}t_{i}\lambda_{i}\big)\big(\sum_{i=1}^{n}t_{i}/\lambda_{i}\big)=ab.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
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● W|+0</td><td><code>\boldsymbol{\epsilon}_{k+1}=\boldsymbol{\epsilon}_{k}- \alpha_{k} \boldsymbol{r}_{k}, \quad \alpha_{k}:= \frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}} .</code></td><td>$\boldsymbol{\epsilon}_{k+1}=\boldsymbol{\epsilon}_{k}-\alpha_{k} \boldsymbol{r}_{k}, \quad \alpha_{k}:=\frac{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}} .$</td><td>$\boldsymbol{\epsilon}_{k+1}=\boldsymbol{\epsilon}_{k}-\alpha_{k}\boldsymbol{r}_{k},\quad\alpha_{k}:=\frac{\boldsymbol{r}_{k}^{T}\boldsymbol{r}_{k}}{\boldsymbol{r}_{k}^{T}\boldsymbol{A}\boldsymbol{r}_{k}}.$</td></tr>
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● W|+0</td><td><code>\frac{\left\|\boldsymbol{\epsilon}_{k+1}\right \|_{\boldsymbol{A}}^{2}}{\left\|\boldsymbol{ \epsilon}_{k}\right\|_{\boldsymbol{A}}^{2}}=1-\frac{ \left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k} \right)^{2}}{\left(\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}\right)\left(\boldsymbol{r}_{k}^{T} \boldsymbol{A}^{-1} \boldsymbol{r}_{k}\right)} \leq 1- \frac{4 \lambda_{\min } \lambda_{\max }}{\left(\lambda_{ \min }+\lambda_{\max }\right)^{2}}=\left(\frac{\kappa-1}{ \kappa+1}\right)^{2}</code></td><td>$\frac{\left\|\boldsymbol{\epsilon}_{k+1}\right\|_{\boldsymbol{A}}^{2}}{\left\|\boldsymbol{\epsilon}_{k}\right\|_{\boldsymbol{A}}^{2}}=1-\frac{\left(\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}\right)^{2}}{\left(\boldsymbol{r}_{k}^{T} \boldsymbol{A} \boldsymbol{r}_{k}\right)\left(\boldsymbol{r}_{k}^{T} \boldsymbol{A}^{-1} \boldsymbol{r}_{k}\right)} \leq 1-\frac{4 \lambda_{\min } \lambda_{\max }}{\left(\lambda_{\min }+\lambda_{\max }\right)^{2}}=\left(\frac{\kappa-1}{\kappa+1}\right)^{2}$</td><td>$\frac{\|\boldsymbol{\epsilon}_{k+1}\|_{\boldsymbol{A}}^{2}}{\|\boldsymbol{\epsilon}_{k}\|_{\boldsymbol{A}}^{2}}=1-\frac{(\boldsymbol{r}_{k}^{T}\boldsymbol{r}_{k})^{2}}{(\boldsymbol{r}_{k}^{T}\boldsymbol{A}\boldsymbol{r}_{k})(\boldsymbol{r}_{k}^{T}\boldsymbol{A}^{-1}\boldsymbol{r}_{k})}\leq1-\frac{4\lambda_{min}\lambda_{max}}{(\lambda_{min}+\lambda_{max})^{2}}=\left(\frac{\kappa-1}{\kappa+1}\right)^{2}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_
EQ0905 (13.41)</td><td>309</td><td>■ 1.000
● S|+0</td><td><code>\boldsymbol{\epsilon}_{j}=\sum_{i=j}^{m} \alpha_{i} \boldsymbol{p}_{i}, \quad \alpha_{i}=\frac{ \boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{ \boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}} .</code></td><td>$\boldsymbol{\epsilon}_{j}=\sum_{i=j}^{m} \alpha_{i} \boldsymbol{p}_{i}, \quad \alpha_{i}=\frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}} .$</td><td>$\boldsymbol{\epsilon}_{j}=\sum_{i=j}^{m}\alpha_{i}\boldsymbol{p}_{i},\quad\alpha_{i}=\frac{\boldsymbol{r}_{i}^{T}\boldsymbol{r}_{i}}{\boldsymbol{p}_{i}^{T}\boldsymbol{A}\boldsymbol{p}_{i}}.$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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● W|+0</td><td>\left\|\boldsymbol{\epsilon}_{j}\right\|_{ \boldsymbol{A}}^{2}=\boldsymbol{\epsilon}_{j} \boldsymbol{A} \boldsymbol{\epsilon}_{j}=\sum_{i=j}^{m} \alpha_{i}^{2} \boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=\sum_{i=j}^{m} \frac{\left(\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}\right)^{2}}{ \boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}} .</td><td>$\left\|\boldsymbol{\epsilon}_{j}\right\|_{\boldsymbol{A}}^{2}=\boldsymbol{\epsilon}_{j} \boldsymbol{A} \boldsymbol{\epsilon}_{j}=\sum_{i=j}^{m} \alpha_{i}^{2} \boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=\sum_{i=j}^{m} \frac{\left(\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}\right)^{2}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}.$</td><td>$\|\boldsymbol{\epsilon}_{j}\|_{A}^{2}=\boldsymbol{\epsilon}_{j} \boldsymbol{A} \boldsymbol{\epsilon}_{j}=\sum_{i=j}^{m} \alpha_{i}^{2} \boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}=\sum_{i=j}^{m} \frac{(\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i})^{2}}{\boldsymbol{p}_{i}^{T} \boldsymbol{A} \boldsymbol{p}_{i}}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_EQ0907</td><td>309</td><td>■ 1.000
● S|+0</td><td>\boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}=\left(\boldsymbol{r}_{i}+\beta_{i-1} \boldsymbol{p}_{i-1} \right)^{T} \boldsymbol{p}_{k}=\beta_{i-1} \boldsymbol{p}_{i-1}^{T} \boldsymbol{p}_{k}=\cdots= \beta_{i-1} \cdots \beta_{k}\left(\boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\right),</td><td>$\boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}=\left(\boldsymbol{r}_{i}+\beta_{i-1} \boldsymbol{p}_{i-1}\right)^{T} \boldsymbol{p}_{k}=\beta_{i-1} \boldsymbol{p}_{i-1}^{T} \boldsymbol{p}_{k}=\cdots=\beta_{i-1} \cdots \beta_{k}\left(\boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}\right),$</td><td>$\boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}=(\boldsymbol{r}_{i}+\beta_{i-1} \boldsymbol{p}_{i-1})^{T} \boldsymbol{p}_{k}=\beta_{i-1} \boldsymbol{p}_{i-1}^{T} \boldsymbol{p}_{k}=\cdots=\beta_{i-1} \cdots \beta_{k}(\boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}),$</td></tr>
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● W|+1</td><td>\boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}=\frac{ \boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{ \boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}, \quad i \geq k</td><td>$\boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}=\frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}, \quad i \geq k$</td><td>$\boldsymbol{p}_{i}^{T} \boldsymbol{p}_{k}=\frac{\boldsymbol{r}_{i}^{T} \boldsymbol{r}_{i}}{\boldsymbol{r}_{k}^{T} \boldsymbol{r}_{k}} \boldsymbol{p}_{k}^{T} \boldsymbol{p}_{k}, \quad i \geq k.$</td></tr>
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● K|+0</td><td>\left\|\boldsymbol{\epsilon}_{k}\right\|_{2}^{2}=\left \|\boldsymbol{\epsilon}_{k+1}+\boldsymbol{x}_{k+1}- \boldsymbol{x}_{k}\right\|_{2}^{2}=\left\|\boldsymbol{ \epsilon}_{k+1}+\alpha_{k} \boldsymbol{p}_{k}\right \|_{2}^{2},</td><td>$\left\|\boldsymbol{\epsilon}_{k}\right\|_{2}^{2}=\left\|\boldsymbol{\epsilon}_{k+1}+\boldsymbol{x}_{k+1}-\boldsymbol{x}_{k}\right\|_{2}^{2}=\left\|\boldsymbol{\epsilon}_{k+1}+\alpha_{k} \boldsymbol{p}_{k}\right\|_{2}^{2},$</td><td>$\|\boldsymbol{\epsilon}_{k}\|_{2}^{2}=\|\boldsymbol{\epsilon}_{k+1}+\boldsymbol{x}_{k+1}-\boldsymbol{x}_{k}\|_{2}^{2}=\|\boldsymbol{\epsilon}_{k+1}+\alpha_{k} \boldsymbol{p}_{k}\|_{2}^{2},$</td></tr>
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● K|+0</td><td>\boldsymbol{C}^{T}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{C}^{-T}= \boldsymbol{B} \boldsymbol{A} .</td><td>$\boldsymbol{C}^{T}\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}\right) \boldsymbol{C}^{-T}=\boldsymbol{B} \boldsymbol{A}.$</td><td>$\boldsymbol{C}^{T}(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^{T}) \boldsymbol{C}^{-T}=\boldsymbol{B} \boldsymbol{A}.$</td></tr>
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● W|+0</td><td>\begin{aligned} & \boldsymbol{x}_{k+1}= \boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \quad \alpha_{k}=\frac{\boldsymbol{s}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}, \\ & \boldsymbol{r}_{k+1}= \boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}, \\ & \boldsymbol{s}_{k+1}= \boldsymbol{s}_{k}-\alpha_{k} \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_{k}, \\ & \boldsymbol{p}_{k+1}=\boldsymbol{s}_{k+1}+ \beta_{k} \boldsymbol{p}_{k}, \quad \beta_{k}=\frac{ \boldsymbol{s}_{k+1}^{T} \boldsymbol{r}_{k+1}}{ \boldsymbol{s}_{k}^{T} \boldsymbol{r}_{k}} . \end{aligned}</td><td>$\begin{aligned} & \boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \quad \alpha_{k}=\frac{\boldsymbol{s}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}, \\ & \boldsymbol{r}_{k+1}=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}, \\ & \boldsymbol{s}_{k+1}=\boldsymbol{s}_{k}-\alpha_{k} \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_{k}, \\ & \boldsymbol{p}_{k+1}=\boldsymbol{s}_{k+1}+\beta_{k} \boldsymbol{p}_{k}, \quad \beta_{k}=\frac{\boldsymbol{s}_{k+1}^{T} \boldsymbol{r}_{k+1}}{\boldsymbol{s}_{k}^{T} \boldsymbol{r}_{k}}. \end{aligned}$</td><td>$\begin{aligned} & \boldsymbol{x}_{k+1}=\boldsymbol{x}_{k}+\alpha_{k} \boldsymbol{p}_{k}, \quad \alpha_{k}=\frac{\boldsymbol{s}_{k}^{T} \boldsymbol{r}_{k}}{\boldsymbol{p}_{k}^{T} \boldsymbol{A} \boldsymbol{p}_{k}}, \\ & \boldsymbol{r}_{k+1}=\boldsymbol{r}_{k}-\alpha_{k} \boldsymbol{A} \boldsymbol{p}_{k}, \\ & \boldsymbol{s}_{k+1}=\boldsymbol{s}_{k}-\alpha_{k} \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_{k}, \\ & \boldsymbol{p}_{k+1}=\boldsymbol{s}_{k+1}+\beta_{k} \boldsymbol{p}_{k}, \quad \beta_{k}=\frac{\boldsymbol{s}_{k+1}^{T} \boldsymbol{r}_{k+1}}{\boldsymbol{s}_{k}^{T} \boldsymbol{r}_{k}}. \end{aligned}$</td></tr>
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Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
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lyche-numerical-linear-algebra_ EQ0922	313	■ 1.000 • N|+0	`\left\{\left(x_{j}, y_{k}\right)=(j h, k h):(j, k) \in \bar{I}_{m}\right\}`	$\left\{\left(x_{j}, y_{k}\right)=(j h, k h):(j, k) \in \bar{I}_{m}\right\}$	$\{(x_{j}, y_{k})=(j h, k h):(j, k) \in \bar{I}_{m}\}$
lyche-numerical-linear-algebra_ EQ0923 (13.54)	314	■ 1.000 • W|+2	`\begin{aligned} \frac{d}{d t}\left(f(t) \frac{d}{d t} g(t)\right) & \approx\left(f\left(t+\frac{h}{2}\right) \frac{d}{d t} g(t+h / 2)-f\left(t-\frac{h}{2}\right) \frac{d}{d t} g\left(t-\frac{h}{2}\right)\right) / h \\ & \approx\left(f\left(t+\frac{h}{2}\right)(g(t+h)-g(t))-f \left(t-\frac{h}{2}\right)(g(t)-g(t-h))\right) / h^{2} \end{aligned}`	$\begin{aligned} \frac{d}{d t}\left(f(t) \frac{d}{d t} g(t)\right) & \approx\left(f\left(t+\frac{h}{2}\right) \frac{d}{d t} g(t+h / 2)-f\left(t-\frac{h}{2}\right) \frac{d}{d t} g\left(t-\frac{h}{2}\right)\right) / h \\ & \approx\left(f\left(t+\frac{h}{2}\right)(g(t+h)-g(t))-f \left(t-\frac{h}{2}\right)(g(t)-g(t-h))\right) / h^{2} \end{aligned}$	$\begin{aligned} \frac{d}{d t}\left(f(t) \frac{d}{d t} g(t)\right) & \approx\left(f(t+\frac{h}{2}) \frac{d}{d t} g(t+h / 2)-f(t-\frac{h}{2}) \frac{d}{d t} g(t-\frac{h}{2})\right) / h \\ & \approx\left(f(t+\frac{h}{2})(g(t+h)-g(t))-f(t-\frac{h}{2})(g(t)-g(t-h))\right) / h^{2} \end{aligned}$
lyche-numerical-linear-algebra_ EQ0927	314	■ 1.000 • N|+0	`\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \ldots & v_{1, m} \\ \vdots & & \vdots \\ v_{m, 1} & \ldots & v_{m, m} \end{array}\right]`	$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \ldots & v_{1, m} \\ \vdots & & \vdots \\ v_{m, 1} & \ldots & v_{m, m} \end{array}\right]$	$\boldsymbol{V}:=\left[\begin{array}{ccc} v_{1,1} & \ldots & v_{1, m} \\ \vdots & & \vdots \\ v_{m, 1} & \ldots & v_{m, m} \end{array}\right]$
lyche-numerical-linear-algebra_ EQ0928 (13.57)	314	■ 1.000 • N|+0	`\boldsymbol{A} \boldsymbol{x}=\boldsymbol{A} \operatorname{vec}(\boldsymbol{V}):=h^{2} \operatorname{vec}\left(L_{h} \boldsymbol{v}\right) .`	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{A} \operatorname{vec}(\boldsymbol{V}):=h^{2} \operatorname{vec}\left(L_{h} \boldsymbol{v}\right) .$	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{A} \operatorname{vec}(\boldsymbol{V}):=h^{2} \operatorname{vec}(L_{h} v).$
lyche-numerical-linear-algebra_ EQ0931 (13.60)	315	■ 1.000 • W|+0	`\sum_{i=1}^{m}\left(a_{i-1}\left(b_{i}-b_{i-1} \right)-a_{i}\left(b_{i+1}-b_{i}\right)\right) c_{i}=\sum_{i=0}^{m} a_{i}\left(b_{i+1}-b_{i}\right) \left(c_{i+1}-c_{i}\right) .`	$\sum_{i=1}^{m}\left(a_{i-1}\left(b_{i}-b_{i-1} \right)-a_{i}\left(b_{i+1}-b_{i}\right)\right) c_{i}=\sum_{i=0}^{m} a_{i}\left(b_{i+1}-b_{i}\right) \left(c_{i+1}-c_{i}\right) .$	$\sum_{i=1}^{m}\left(a_{i-1}(b_{i}-b_{i-1})-a_{i}(b_{i+1}-b_{i})\right) c_{i}=\sum_{i=0}^{m} a_{i}(b_{i+1}-b_{i})(c_{i+1}-c_{i}).$
lyche-numerical-linear-algebra_ EQ0932	315	■ 1.000 • W|+0	`\sum_{i=0}^{m} a_{i}\left(b_{i+1}-b_{i}\right) c_{i+1}- \sum_{i=0}^{m} a_{i}\left(b_{i+1}-b_{i}\right) c_{i},`	$\sum_{i=0}^{m} a_{i}\left(b_{i+1}-b_{i}\right) c_{i+1}- \sum_{i=0}^{m} a_{i}\left(b_{i+1}-b_{i}\right) c_{i},$	$\sum_{i=0}^{m} a_{i}(b_{i+1}-b_{i}) c_{i+1}-\sum_{i=0}^{m} a_{i}(b_{i+1}-b_{i}) c_{i},$
lyche-numerical-linear-algebra_ EQ0933	316	■ 1.000 • N|+0	`c(x, y)=e^{-x+y} \text { and } f(x, y)=1 .`	$c(x, y)=e^{-x+y} \text { and } f(x, y)=1 .$	$c(x, y)=e^{-x+y} \text { and } f(x, y)=1.$
lyche-numerical-linear-algebra_ EQ0934	317	■ 1.000 • N|+0	`\kappa=\frac{\lambda_{\max }}{\lambda_{\min }} \leq \frac{c_{1}}{c_{0}} .`	$\kappa=\frac{\lambda_{\max }}{\lambda_{\min }} \leq \frac{c_{1}}{c_{0}} .$	$\kappa=\frac{\lambda_{max}}{\lambda_{min}} \leq \frac{c_{1}}{c_{0}}.$
lyche-numerical-linear-algebra_ EQ0936	317	■ 1.000 • N|+0	`\boldsymbol{x}^{T} \boldsymbol{A}_{p} x=\sum_{i=1}^{m} \sum_{j=0}^{m}\left(v_{i, j+1}-v_{i, j}\right)^{2}+ \sum_{j=1}^{m} \sum_{i=0}^{m}\left(v_{i+1, j}-v_{i, j} \right)^{2} .`	$\boldsymbol{x}^{T} \boldsymbol{A}_{p} x=\sum_{i=1}^{m} \sum_{j=0}^{m}\left(v_{i, j+1}-v_{i, j}\right)^{2}+ \sum_{j=1}^{m} \sum_{i=0}^{m}\left(v_{i+1, j}-v_{i, j} \right)^{2} .$	$\boldsymbol{x}^{T} \boldsymbol{A}_{p} x=\sum_{i=1}^{m} \sum_{j=0}^{m}\left(v_{i, j+1}-v_{i, j}\right)^{2}+\sum_{j=1}^{m} \sum_{i=0}^{m}\left(v_{i+1, j}-v_{i, j}\right)^{2}.$
lyche-numerical-linear-algebra_ EQ0938	318	■ 1.000 • N|+0	`Q(\boldsymbol{y})=\frac{1}{2} \sum_{j=1}^{n} \lambda_{j} v_{j}^{2}-\sum_{j=1}^{n} c_{j} v_{j} .`	$Q(\boldsymbol{y})=\frac{1}{2} \sum_{j=1}^{n} \lambda_{j} v_{j}^{2}-\sum_{j=1}^{n} c_{j} v_{j} .$	$Q(\boldsymbol{y})=\frac{1}{2} \sum_{j=1}^{n} \lambda_{j} v_{j}^{2}-\sum_{j=1}^{n} c_{j} v_{j}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0940</td><td>319</td><td>1.000 N|-1</td><td>\boldsymbol{w}^{T} \boldsymbol{A}^{T} \boldsymbol{A} \hat{\boldsymbol{x}}=\boldsymbol{w}^{T} \boldsymbol{A}^{T} \boldsymbol{b}, \quad \text { for all } \boldsymbol{w} \in \mathbb{W},</td><td>$\boldsymbol{w}^{T} \boldsymbol{A}^{T} \boldsymbol{A} \hat{\boldsymbol{x}}=\boldsymbol{w}^{T} \boldsymbol{A}^{T} \boldsymbol{b}, \quad \text { for all } \boldsymbol{w} \in \mathbb{W},$</td><td>$\boldsymbol{w}^{T} \boldsymbol{A}^{T} \boldsymbol{A} \hat{\boldsymbol{x}}=\boldsymbol{w}^{T} \boldsymbol{A}^{T} \boldsymbol{b}, \quad \text { for all } \boldsymbol{w} \in \mathbb{W},$</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0943</td><td>319</td><td>1.000 N|+0</td><td>\langle\boldsymbol{v}, \boldsymbol{w}\rangle_{ \boldsymbol{A}}:=\boldsymbol{v}^{T} \boldsymbol{A}^{T} \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^{n} .</td><td>$\langle\boldsymbol{v}, \boldsymbol{w}\rangle_{ \boldsymbol{A}}:=\boldsymbol{v}^{T} \boldsymbol{A}^{T} \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^{n} .$</td><td>$\langle\boldsymbol{v}, \boldsymbol{w}\rangle_{ \boldsymbol{A}}:=\boldsymbol{v}^{T} \boldsymbol{A}^{T} \boldsymbol{A} \boldsymbol{w}, \qquad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^{n} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0944</td><td>320</td><td>1.000 K|+0</td><td>\left\langle\boldsymbol{p}_{k}, \boldsymbol{w}\right \rangle_{\boldsymbol{A}}=0, \quad \text { for all } \boldsymbol{w} \in \mathbb{W}_{k} .</td><td>$\left\langle\boldsymbol{p}_{k}, \boldsymbol{w}\right \rangle_{\boldsymbol{A}}=0, \quad \text { for all } \boldsymbol{w} \in \mathbb{W}_{k} .$</td><td>$\langle\boldsymbol{p}_{k}, \boldsymbol{w} \rangle_{\boldsymbol{A}}=0, \quad \text { for all } \boldsymbol{w} \in \mathbb{W}_{k} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0952 (13.65)</td><td>321</td><td>1.000 K|+0</td><td>\langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle=\langle\boldsymbol{b}, \boldsymbol{w}\rangle \text { for all } \boldsymbol{w} \in \mathcal{W},</td><td>$\langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle=\langle\boldsymbol{b}, \boldsymbol{w}\rangle \text { for all } \boldsymbol{w} \in \mathcal{W},$</td><td>$\langle\boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle=\langle\boldsymbol{b}, \boldsymbol{w}\rangle \text { for all } \boldsymbol{w} \in \mathcal{W},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0953 (13.66)</td><td>322</td><td>1.000 N|+1</td><td>\sum_{j=1}^{k} x_{j}\left\langle\boldsymbol{A} \boldsymbol{w}_{j}, \boldsymbol{w}_{i}\right\rangle= \left\langle\boldsymbol{b}, \boldsymbol{w}_{i}\right \rangle, \quad i=1, \ldots, k</td><td>$\sum_{j=1}^{k} x_{j}\left\langle\boldsymbol{A} \boldsymbol{w}_{j}, \boldsymbol{w}_{i}\right\rangle= \left\langle\boldsymbol{b}, \boldsymbol{w}_{i}\right \rangle, \quad i=1, \ldots, k$</td><td>$\sum_{j=1}^{k} x_{j}\langle\boldsymbol{A} \boldsymbol{w}_{j}, \boldsymbol{w}_{i}\rangle= \langle\boldsymbol{b}, \boldsymbol{w}_{i}\rangle, \quad i=1, \ldots, k.$</td></tr>
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<tr><td>lyche-numerical-linear-algebra_ EQ0956</td><td>322</td><td>1.000 N|+0</td><td>\left\langle\boldsymbol{A} \boldsymbol{x}_{k}, \boldsymbol{w}\right\rangle=\langle\boldsymbol{b}, \boldsymbol{w}\rangle, \text { for all } \boldsymbol{w} \in \mathcal{W}_{k} .</td><td>$\left\langle\boldsymbol{A} \boldsymbol{x}_{k}, \boldsymbol{w}\right\rangle=\langle\boldsymbol{b}, \boldsymbol{w}\rangle, \text { for all } \boldsymbol{w} \in \mathcal{W}_{k} .$</td><td>$\langle\boldsymbol{A} \boldsymbol{x}_{k}, \boldsymbol{w}\rangle=\langle\boldsymbol{b}, \boldsymbol{w}\rangle, \text { for all } \boldsymbol{w} \in \mathcal{W}_{k} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0958</td><td>322</td><td>1.000 N|+0</td><td>\rho_{k} \neq 0, \quad k=0, \ldots, m .</td><td>$\rho_{k} \neq 0, \quad k=0, \ldots, m .$</td><td>$\rho_{k} \neq 0, \quad k=0, \ldots, m .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0962 (13.72)</td><td>323</td><td>1.000 N|+0</td><td>\left\langle\boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1}\right\rangle=\left\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{j}\right\rangle, \quad j=0,1, \ldots, k .</td><td>$\left\langle\boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1}\right\rangle=\left\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{j}\right\rangle, \quad j=0,1, \ldots, k .$</td><td>$\langle\boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1}\rangle=\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{j}\rangle, \quad j=0,1, \ldots, k .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0963 (13.70)</td><td>323</td><td>1.000 N|-1</td><td>T_{n}(t)=\cosh (n \operatorname{arccosh} t) \text { for } t \geq 1,</td><td>$T_{n}(t)=\cosh (n \operatorname{arccosh} t) \text { for } t \geq 1,$</td><td>$T_{n}(t)=\cosh (n \operatorname{arccosh} t) \text { for } t \geq 1,$</td></tr>
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Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-a EQ0965</td><td>324</td><td>■ 1.000 ● C|-8</td><td><code>\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array}\right]=\left[\begin{array}{l} c_{\frac{1}{2}, 1}+c_{1, \frac{1}{2}}+c_{1, \frac{3}{2}}+c_{\frac{3}{2}, 1} \\ c_{\frac{3}{2}, 1}+c_{2, \frac{1}{2}}+c_{2, \frac{3}{2}}+c_{\frac{5}{2}, 1} \\ c_{\frac{1}{2}, 2}+c_{1, \frac{3}{2}}+c_{1, \frac{5}{2}}+c_{\frac{3}{2}, 2} \\ c_{\frac{3}{2}, 2}+c_{2, \frac{3}{2}}+c_{2, \frac{5}{2}}+c_{\frac{5}{2}, 2} \end{array}\right] .</code></td><td>$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array}\right]=\left[\begin{array}{l} c_{\frac{1}{2}, 1}+c_{1, \frac{1}{2}}+c_{1, \frac{3}{2}}+c_{\frac{3}{2}, 1} \\ c_{\frac{3}{2}, 1}+c_{2, \frac{1}{2}}+c_{2, \frac{3}{2}}+c_{\frac{5}{2}, 1} \\ c_{\frac{1}{2}, 2}+c_{1, \frac{3}{2}}+c_{1, \frac{5}{2}}+c_{\frac{3}{2}, 2} \\ c_{\frac{3}{2}, 2}+c_{2, \frac{3}{2}}+c_{2, \frac{5}{2}}+c_{\frac{5}{2}, 2} \end{array}\right] .$</td><td>$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array}\right]=\left[\begin{array}{l} c_{\frac{1}{2}, 1}+c_{1, \frac{1}{2}}+c_{1, \frac{3}{2}}+c_{\frac{3}{2}, 1} \\ c_{\frac{3}{2}, 1}+c_{2, \frac{1}{2}}+c_{2, \frac{3}{2}}+c_{\frac{5}{2}, 1} \\ c_{\frac{1}{2}, 2}+c_{1, \frac{3}{2}}+c_{1, \frac{5}{2}}+c_{\frac{3}{2}, 2} \\ c_{\frac{3}{2}, 2}+c_{2, \frac{3}{2}}+c_{2, \frac{5}{2}}+c_{\frac{5}{2}, 2} \end{array}\right] .$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0966</td><td>326</td><td>■ 1.000 ● N|-1</td><td><code>\boldsymbol{A}=\operatorname{diag}\left(\boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \ldots, \boldsymbol{A}_{r}\right), \quad \boldsymbol{A}_{i} \in \mathbb{C}^{m_{i} \times m_{i}} .</code></td><td>$\boldsymbol{A}=\operatorname{diag}\left(\boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \ldots, \boldsymbol{A}_{r}\right), \quad \boldsymbol{A}_{i} \in \mathbb{C}^{m_{i} \times m_{i}} .$</td><td>$\boldsymbol{A}=\operatorname{diag}(\boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \ldots, \boldsymbol{A}_{r}), \quad \boldsymbol{A}_{i} \in \mathbb{C}^{m_{i} \times m_{i}} .$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0968</td><td>327</td><td>■ 1.000 ● N|+0</td><td><code>\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\prod_{i=1}^{r} \operatorname{det}\left(\boldsymbol{A}_{i i}-\lambda \boldsymbol{I}\right)</code></td><td>$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\prod_{i=1}^{r} \operatorname{det}\left(\boldsymbol{A}_{i i}-\lambda \boldsymbol{I}\right)$</td><td>$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\prod_{i=1}^{r} \operatorname{det}(\boldsymbol{A}_{i i}-\lambda \boldsymbol{I})$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0970</td><td>328</td><td>■ 1.000 ● N|+0</td><td><code>\left|\lambda-a_{i i}\right|=\left|\sum_{j \neq i} a_{i j} x_{j} / x_{i}\right| \leq \sum_{j \neq i}\left|a_{i j}\right|\left|x_{j} / x_{i}\right| \leq r_{i}</code></td><td>$\left|\lambda-a_{i i}\right|=\left|\sum_{j \neq i} a_{i j} x_{j} / x_{i}\right| \leq \sum_{j \neq i}\left|a_{i j}\right|\left|x_{j} / x_{i}\right| \leq r_{i}$</td><td>$|\lambda-a_{i i}|=|\sum_{j \neq i} a_{i j} x_{j} / x_{i}| \leq \sum_{j \neq i}|a_{i j}||x_{j} / x_{i}| \leq r_{i}$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0971</td><td>328</td><td>■ 1.000 ● N|+0</td><td><code>R_{i}=\{z \in \mathbb{R}:|z-2| \leq 2\}, \quad i=2,3, \ldots, m-1 .</code></td><td>$R_{i}=\{z \in \mathbb{R}:|z-2| \leq 2\}, \quad i=2,3, \ldots, m-1 .$</td><td>$R_{i}=\{z \in \mathbb{R}:|z-2| \leq 2\}, \quad i=2,3, \ldots, m-1 .$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0972</td><td>328</td><td>■ 1.000 ● N|+0</td><td><code>\lambda_{j}=4\left[\sin \frac{j \pi}{2(m+1)}\right]^{2}, \quad j=1,2, \ldots, m .</code></td><td>$\lambda_{j}=4\left[\sin \frac{j \pi}{2(m+1)}\right]^{2}, \quad j=1,2, \ldots, m .$</td><td>$\lambda_{j}=4\left[\sin \frac{j \pi}{2(m+1)}\right]^{2}, \quad j=1,2, \ldots, m .$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0973</td><td>329</td><td>■ 1.000 ● N|+0</td><td><code>\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:= \operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in[0,1] .</code></td><td>$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:= \operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in[0,1] .$</td><td>$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:= \operatorname{diag}(a_{11}, \ldots, a_{n n}), \quad t \in[0,1] .$</td></tr>
<tr><td>lyche-numerical-linear-a EQ0974</td><td>329</td><td>■ 1.000 ● N|+0</td><td><code>R_{i}(t)=\left\{z \in \mathbb{C}:\left|z-a_{i i}\right| \leq t r_{i}\right\}, \quad r_{i}:=\sum_{\substack{j=1 \\ j \neq i}}^{n}\left|a_{i j}\right| .</code></td><td>$R_{i}(t)=\left\{z \in \mathbb{C}:\left|z-a_{i i}\right| \leq t r_{i}\right\}, \quad r_{i}:=\sum_{\substack{j=1 \\ j \neq i}}^{n}\left|a_{i j}\right| .$</td><td>$R_{i}(t)=\{z \in \mathbb{C}:|z-a_{i i}| \leq t r_{i}\}, \quad r_{i}:=\sum_{\substack{j=1 \\ j \neq i}}^{n}|a_{i j}| .$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0978</td><td>330</td><td>■ 1.000 ● N|+1</td><td>\begin{aligned} & \left|\mu-\lambda_{1}\right|^{n} \leq \prod_{j=1}^{n}\left|\mu-\lambda_{j}\right|=|\operatorname{det}(\mu \boldsymbol{I}-\boldsymbol{A})|= \left|\operatorname{det}\left((\mu \boldsymbol{I}-\boldsymbol{A})\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right]\right)\right| \\ & \leq\left\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{1}\right\|_{2} \prod_{j=2}^{n}\left\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{j}\right\|_{2} \leq\|\boldsymbol{E}\|_{2}\left(\|(\boldsymbol{A}+\boldsymbol{E})\|_{2}+\|\boldsymbol{A}\|_{2}\right)^{n-1} . \end{aligned}</td><td>$$\begin{aligned} & \left|\mu-\lambda_{1}\right|^{n} \leq \prod_{j=1}^{n}\left|\mu-\lambda_{j}\right|=|\operatorname{det}(\mu \boldsymbol{I}-\boldsymbol{A})|= \left|\operatorname{det}\left((\mu \boldsymbol{I}-\boldsymbol{A})\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right]\right)\right| \\ & \leq\left\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{1}\right\|_{2} \prod_{j=2}^{n}\left\|(\mu \boldsymbol{I}-\boldsymbol{A}) \boldsymbol{u}_{j}\right\|_{2} \leq\|\boldsymbol{E}\|_{2}\left(\|(\boldsymbol{A}+\boldsymbol{E})\|_{2}+\|\boldsymbol{A}\|_{2}\right)^{n-1} . \end{aligned}$$</td><td>$$\begin{aligned} & |\mu-\lambda_1|^n \le \prod_{j=1}^{n}|\mu-\lambda_j| = |\det(\mu\boldsymbol{I}-\boldsymbol{A})| = \big|\det\big((\mu\boldsymbol{I}-\boldsymbol{A})[\boldsymbol{u}_1,\ldots,\boldsymbol{u}_n]\big)\big| \\ & \le \|(\mu\boldsymbol{I}-\boldsymbol{A})\boldsymbol{u}_1\|_2 \prod_{j=2}^{n}\|(\mu\boldsymbol{I}-\boldsymbol{A})\boldsymbol{u}_j\|_2 \le \|\boldsymbol{E}\|_2\big(\|(\boldsymbol{A}+\boldsymbol{E})\|_2+\|\boldsymbol{A}\|_2\big)^{n-1}. \end{aligned}$$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0979 (14.2)</td><td>331</td><td>■ 1.000 ● N|+0</td><td>|\lambda-\mu| \leq K_{p}(\boldsymbol{X})\|\boldsymbol{r}\|_{p}, \quad 1 \leq p \leq \infty,</td><td>$|\lambda-\mu| \leq K_{p}(\boldsymbol{X})\|\boldsymbol{r}\|_{p}, \quad 1 \leq p \leq \infty,$</td><td>$|\lambda-\mu| \le K_p(\boldsymbol{X})\|\boldsymbol{r}\|_p, \quad 1 \le p \le \infty,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0980 (14.3)</td><td>331</td><td>■ 1.000 ● N|+1</td><td>|\lambda-\mu| \leq K_{p}(\boldsymbol{X})\|\boldsymbol{E}\|_{p}, \quad 1 \leq p \leq \infty,</td><td>$|\lambda-\mu| \leq K_{p}(\boldsymbol{X})\|\boldsymbol{E}\|_{p}, \quad 1 \leq p \leq \infty,$</td><td>$|\lambda-\mu| \le K_p(\boldsymbol{X})\|\boldsymbol{E}\|_p, \quad 1 \le p \le \infty,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0983</td><td>332</td><td>■ 1.000 ● N|+0</td><td>\|\boldsymbol{A} \boldsymbol{x}\|_{p}=\left\|\left[\lambda_{1} x_{1}, \ldots, \lambda_{n} x_{n}\right]^{T} \right\|_{p}=\left(\sum_{j=1}^{n}\left|\lambda_{j} \right|^{p}\left|x_{j}\right|^{p}\right)^{1 / p} \leq \rho(\boldsymbol{A})\|\boldsymbol{x}\|_{p} .</td><td>$\|\boldsymbol{A} \boldsymbol{x}\|_{p}=\left\|\left[\lambda_{1} x_{1}, \ldots, \lambda_{n} x_{n}\right]^{T} \right\|_{p}=\left(\sum_{j=1}^{n}\left|\lambda_{j} \right|^{p}\left|x_{j}\right|^{p}\right)^{1 / p} \leq \rho(\boldsymbol{A})\|\boldsymbol{x}\|_{p} .$</td><td>$\|\boldsymbol{A}\boldsymbol{x}\|_p = \|[\lambda_1x_1,\ldots,\lambda_nx_n]^T\|_p = \big(\sum_{j=1}^{n}|\lambda_j|^p|x_j|^p\big)^{1/p} \le \rho(\boldsymbol{A})\|\boldsymbol{x}\|_p.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0984 (14.4)</td><td>332</td><td>■ 1.000 ● N|+0</td><td>\frac{|\lambda-\mu|}{|\lambda|} \leq K_{p}(\boldsymbol{X}) K_{p}(\boldsymbol{A}) \frac{\| \boldsymbol{r}\|_{p}}{\|\boldsymbol{A}\|_{p}}, \quad 1 \leq p \leq \infty,</td><td>$\frac{|\lambda-\mu|}{|\lambda|} \leq K_{p}(\boldsymbol{X}) K_{p}(\boldsymbol{A}) \frac{\| \boldsymbol{r}\|_{p}}{\|\boldsymbol{A}\|_{p}}, \quad 1 \leq p \leq \infty,$</td><td>$\frac{|\lambda-\mu|}{|\lambda|} \le K_p(\boldsymbol{X})K_p(\boldsymbol{A})\frac{\|\boldsymbol{r}\|_p}{\|\boldsymbol{A}\|_p}, \quad 1 \le p \le \infty,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0985 (14.5)</td><td>332</td><td>■ 1.000 ● N|+0</td><td>\frac{|\lambda-\mu|}{|\lambda|} \leq K_{p}(\boldsymbol{X})\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \right\|_{p} \leq K_{p}(\boldsymbol{X}) K_{p}(\boldsymbol{A}) \frac{\|\boldsymbol{E}\|_{p}}{\| \boldsymbol{A}\|_{p}}, \quad 1 \leq p \leq \infty,</td><td>$\frac{|\lambda-\mu|}{|\lambda|} \leq K_{p}(\boldsymbol{X})\left\|\boldsymbol{A}^{-1} \boldsymbol{E} \right\|_{p} \leq K_{p}(\boldsymbol{X}) K_{p}(\boldsymbol{A}) \frac{\|\boldsymbol{E}\|_{p}}{\| \boldsymbol{A}\|_{p}}, \quad 1 \leq p \leq \infty,$</td><td>$\frac{|\lambda-\mu|}{|\lambda|} \le K_p(\boldsymbol{X})\|\boldsymbol{A}^{-1}\boldsymbol{E}\|_p \le K_p(\boldsymbol{X})K_p(\boldsymbol{A})\frac{\|\boldsymbol{E}\|_p}{\|\boldsymbol{A}\|_p}, \quad 1 \le p \le \infty,$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0987</td><td>333</td><td>■ 1.000 ● K|+0</td><td>(\boldsymbol{B}+\boldsymbol{F}-\boldsymbol{I}) \boldsymbol{x}=\left(\mu \boldsymbol{A}^{-1}- \boldsymbol{A}^{-1} \boldsymbol{E}-\boldsymbol{I}\right) \boldsymbol{x}=\boldsymbol{A}^{-1}(\mu \boldsymbol{I}-(\boldsymbol{E}+\boldsymbol{A})) \boldsymbol{x}=\mathbf{0} .</td><td>$(\boldsymbol{B}+\boldsymbol{F}-\boldsymbol{I}) \boldsymbol{x}=\left(\mu \boldsymbol{A}^{-1}- \boldsymbol{A}^{-1} \boldsymbol{E}-\boldsymbol{I}\right) \boldsymbol{x}=\boldsymbol{A}^{-1}(\mu \boldsymbol{I}-(\boldsymbol{E}+\boldsymbol{A})) \boldsymbol{x}=\mathbf{0} .$</td><td>$(\boldsymbol{B}+\boldsymbol{F}-\boldsymbol{I})\boldsymbol{x} = (\mu\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1}\boldsymbol{E}-\boldsymbol{I})\boldsymbol{x} = \boldsymbol{A}^{-1}\big(\mu\boldsymbol{I}-(\boldsymbol{E}+\boldsymbol{A})\big)\boldsymbol{x} = \mathbf{0}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0988</td><td>333</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{k+1}^{*}=\left(\boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}\right)^{*}= \boldsymbol{H}_{k} \boldsymbol{A}_{k}^{*} \boldsymbol{H}_{k}=\boldsymbol{A}_{k+1} .</td><td>$\boldsymbol{A}_{k+1}^{*}=\left(\boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}\right)^{*}= \boldsymbol{H}_{k} \boldsymbol{A}_{k}^{*} \boldsymbol{H}_{k}=\boldsymbol{A}_{k+1} .$</td><td>$\boldsymbol{A}_{k+1}^* = (\boldsymbol{H}_k\boldsymbol{A}_k\boldsymbol{H}_k)^* = \boldsymbol{H}_k\boldsymbol{A}_k^*\boldsymbol{H}_k = \boldsymbol{A}_{k+1}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0990</td><td>334</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{H}_{k}=\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{array}\right] \in \mathbb{C}^{n \times n} .</td><td>$\boldsymbol{H}_{k}=\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_{k} \end{array}\right] \in \mathbb{C}^{n \times n} .$</td><td>$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{V}_k \end{bmatrix} \in \mathbb{C}^{n\times n}.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ0993</td><td>335</td><td>1.000 W|+0</td><td>\boldsymbol{Q}_{k}=\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{T} \end{array}\right] *\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \ \ \mathbf{0} & \boldsymbol{U}_{k}-\boldsymbol{v}_{k} \left(\boldsymbol{v}_{k}^{T} \boldsymbol{U}_{k}\right) \end{array}\right] .</td><td>$\boldsymbol{Q}_{k}=\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{T} \end{array}\right] *\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k} \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k}-\boldsymbol{v}_{k} \left(\boldsymbol{v}_{k}^{T} \boldsymbol{U}_{k}\right) \end{array}\right] .$</td><td>$\boldsymbol{Q}_{k}=\begin{bmatrix} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{T} \end{bmatrix} *\begin{bmatrix} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k} \end{bmatrix}=\begin{bmatrix} \boldsymbol{I}_{k} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{k}-\boldsymbol{v}_{k}(\boldsymbol{v}_{k}^{T} \boldsymbol{U}_{k}) \end{bmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ0996 (14.7)</td><td>336</td><td>1.000 N|+0</td><td>p_{k+1}(x)=\left(x-d_{k+1}\right) p_{k}(x)-c_{k}^{2} p_{k-1}(x) .</td><td>$p_{k+1}(x)=\left(x-d_{k+1}\right) p_{k}(x)-c_{k}^{2} p_{k-1}(x).$</td><td>$p_{k+1}(x)=(x-d_{k+1}) p_{k}(x)-c_{k}^{2} p_{k-1}(x).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1001</td><td>337</td><td>1.000 K|+0</td><td>p_{k+1}\left(y_{j}\right)=(-1)^{k+1-j}\left|p_{k+1} \left(y_{j}\right)\right| \neq 0, \text { for } j=0,1, \ldots, k+1 .</td><td>$p_{k+1}\left(y_{j}\right)=(-1)^{k+1-j}\left|p_{k+1} \left(y_{j}\right)\right| \neq 0, \text { for } j=0,1, \ldots, k+1.$</td><td>$p_{k+1}(y_{j})=(-1)^{k+1-j}|p_{k+1}(y_{j})| \neq 0, \text{ for } j=0,1, \ldots, k+1.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1002</td><td>337</td><td>1.000 N|+0</td><td>p_{k+1}\left(y_{j}\right)=-c_{k}^{2} p_{k-1}\left(y_{j} \right)=-c_{k}^{2}\left(y_{j}-z_{1}\right) \cdots \left(y_{j}-z_{k-1}\right) .</td><td>$p_{k+1}\left(y_{j}\right)=-c_{k}^{2} p_{k-1}\left(y_{j} \right)=-c_{k}^{2}\left(y_{j}-z_{1}\right) \cdots \left(y_{j}-z_{k-1}\right).$</td><td>$p_{k+1}(y_{j})=-c_{k}^{2} p_{k-1}(y_{j})=-c_{k}^{2}(y_{j}-z_{1}) \cdots (y_{j}-z_{k-1}).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1003</td><td>337</td><td>1.000 N|+0</td><td>\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{y}= \sum_{i=1}^{n} \lambda_{i}\left|y_{i}\right|^{2}= \sum_{i=1}^{k} \lambda_{i}\left|x_{i}\right|^{2}>0 .</td><td>$\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{y}= \sum_{i=1}^{n} \lambda_{i}\left|y_{i}\right|^{2}= \sum_{i=1}^{k} \lambda_{i}\left|x_{i}\right|^{2}>0.$</td><td>$\boldsymbol{y}^{*} \boldsymbol{A} \boldsymbol{y}= \sum_{i=1}^{n} \lambda_{i}|y_{i}|^{2}= \sum_{i=1}^{k} \lambda_{i}|x_{i}|^{2}>0.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1005 (14.8)</td><td>338</td><td>1.000 N|+0</td><td>d_{1}(\alpha)=d_{1}-\alpha, d_{k}(\alpha)=d_{k}-\alpha-c_{k-1}^{2} / d_{k-1}(\alpha), k=2,3, \ldots, n .</td><td>$d_{1}(\alpha)=d_{1}-\alpha, d_{k}(\alpha)=d_{k}-\alpha-c_{k-1}^{2} / d_{k-1}(\alpha), k=2,3, \ldots, n.$</td><td>$d_{1}(\alpha)=d_{1}-\alpha,\ d_{k}(\alpha)=d_{k}-\alpha-c_{k-1}^{2} / d_{k-1}(\alpha),\ k=2,3, \ldots, n.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1006</td><td>338</td><td>1.000 K|+0</td><td>\rho(x):=\#\left\{k: d_{k}(x)>0 \text { for } k=1, \ldots, n\right\}</td><td>$\rho(x):=\#\left\{k: d_{k}(x)>0 \text { for } k=1, \ldots, n\right\}$</td><td>$\rho(x):=\#\{k: d_{k}(x)>0 \text{ for } k=1, \ldots, n\}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1007</td><td>340</td><td>1.000 K|+0</td><td>\boldsymbol{A}=\left(\begin{array}{llll} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right) .</td><td>$\boldsymbol{A}=\left(\begin{array}{llll} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right).$</td><td>$\boldsymbol{A}=\begin{pmatrix} 4&1&0&0 \\ 1&4&1&0 \\ 0&1&4&1 \\ 0&0&1&4 \end{pmatrix}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1008</td><td>340</td><td>1.000 N|-1</td><td>\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:= \operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in \mathbb{R} .</td><td>$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:= \operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in \mathbb{R}.$</td><td>$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \quad \boldsymbol{D}:=\operatorname{diag}(a_{11}, \ldots, a_{nn}), \quad t \in \mathbb{R}.$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1009</td><td>340</td><td>1.000 N|+0</td><td>|\lambda-\mu| \leq C\left(t_{2}-t_{1}\right)^{1 / n}, \text { where } C \leq 2\left(\|\boldsymbol{D}\|_{2}+\| \boldsymbol{A}-\boldsymbol{D}\|_{2}\right) .</td><td>$|\lambda-\mu| \leq C\left(t_{2}-t_{1}\right)^{1 / n}, \text { where } C \leq 2\left(\|\boldsymbol{D}\|_{2}+\| \boldsymbol{A}-\boldsymbol{D}\|_{2}\right).$</td><td>$|\lambda-\mu| \leq C(t_{2}-t_{1})^{1/n}, \text{ where } C \leq 2\big(\|\boldsymbol{D}\|_{2}+\|\boldsymbol{A}-\boldsymbol{D}\|_{2}\big).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1013</td><td>342</td><td>1.000 N|+0</td><td>\boldsymbol{D}^{-1 / 2}:=\operatorname{diag} \left(d_{1}^{-1 / 2}, \ldots, d_{n}^{-1 / 2}\right) .</td><td>$\boldsymbol{D}^{-1 / 2}:=\operatorname{diag} \left(d_{1}^{-1 / 2}, \ldots, d_{n}^{-1 / 2}\right).$</td><td>$\boldsymbol{D}^{-1/2}:=\operatorname{diag}\big(d_{1}^{-1/2}, \ldots, d_{n}^{-1/2}\big).$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1027</td><td>347</td><td>1.000 N|+0</td><td>\rho(\lambda)^{2}=\left|\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda\right|^{2}+\left\| \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u}\right \|_{2}^{2},</td><td>$\rho(\lambda)^{2}=\left|\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda\right|^{2}+\left\| \boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u}\right \|_{2}^{2},$</td><td>$\rho(\lambda)^{2}=|\boldsymbol{u}^{*} \boldsymbol{A} \boldsymbol{u}-\lambda|^{2}+\|\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{u}\|_{2}^{2},$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

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<tr><td>lyche-numerical-linear-algebra_ EQ1028</td><td>347</td><td>■ 1.000 ● N|+0</td><td>\frac{|\lambda-\mu|}{|\lambda|} \leq K_{2}(\boldsymbol{X}) K_{2}(\boldsymbol{A}) \frac{\| \boldsymbol{A} \boldsymbol{u}-\mu \boldsymbol{u}\|_{2}}{ \|\boldsymbol{A}\|_{2}} .</td><td>$\frac{|\lambda-\mu|}{|\lambda|} \leq K_{2}(\boldsymbol{X}) K_{2}(\boldsymbol{A}) \frac{\|\boldsymbol{A} \boldsymbol{u}-\mu \boldsymbol{u}\|_{2}}{\|\boldsymbol{A}\|_{2}} .$</td><td>$\frac{|\lambda-\mu|}{|\lambda|} \leq K_{2}(\boldsymbol{X}) K_{2}(\boldsymbol{A}) \frac{\|\boldsymbol{A} \boldsymbol{u}-\mu \boldsymbol{u}\|_{2}}{\|\boldsymbol{A}\|_{2}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1033</td><td>349</td><td>■ 1.000 ● N|+0</td><td>\rho_{k}:=\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k-1}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}, \quad \boldsymbol{w}_{k}:=\frac{ \boldsymbol{x}_{k-1}}{\left\|\boldsymbol{y}_{k}\right \|_{2}} .</td><td>$\rho_{k}:=\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k-1}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}, \quad \boldsymbol{w}_{k}:=\frac{\boldsymbol{x}_{k-1}}{\left\|\boldsymbol{y}_{k}\right\|_{2}} .$</td><td>$\rho_{k}:=\frac{\boldsymbol{y}_{k}^{*} \boldsymbol{x}_{k-1}}{\boldsymbol{y}_{k}^{*} \boldsymbol{y}_{k}}, \quad \boldsymbol{w}_{k}:=\frac{\boldsymbol{x}_{k-1}}{\|\boldsymbol{y}_{k}\|_{2}} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1036 (15.9)</td><td>351</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{k+1}=\boldsymbol{R}_{k} \boldsymbol{Q}_{k}=\boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k} \boldsymbol{Q}_{k},</td><td>$\boldsymbol{A}_{k+1}=\boldsymbol{R}_{k} \boldsymbol{Q}_{k}=\boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k} \boldsymbol{Q}_{k},$</td><td>$\boldsymbol{A}_{k+1}=\boldsymbol{R}_{k} \boldsymbol{Q}_{k}=\boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k} \boldsymbol{Q}_{k},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1037</td><td>351</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{1}=\boldsymbol{A}=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array}\right]=\left(\frac{1}{ \sqrt{5}}\left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array}\right]\right) *\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array}\right] \right)=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}</td><td>$\boldsymbol{A}_{1}=\boldsymbol{A}=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array}\right]=\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array}\right]\right) *\left(\frac{1}{\sqrt{5}}\left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array}\right]\right)=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}$</td><td>$\boldsymbol{A}_{1}=\boldsymbol{A}=\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}=(\frac{1}{\sqrt{5}}\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}) *(\frac{1}{\sqrt{5}}\begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix})=\boldsymbol{Q}_{1} \boldsymbol{R}_{1}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1038</td><td>351</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{2}=\boldsymbol{R}_{1} \boldsymbol{Q}_{1}= \frac{1}{5}\left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array}\right] *\left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array}\right]=\frac{1}{5}\left[\begin{array}{cc} 14 & -3 \\ -3 & 6 \end{array}\right]= \left[\begin{array}{cc} 2.8 & -0.6 \\ -0.6 & 1.2 \end{array}\right] .</td><td>$\boldsymbol{A}_{2}=\boldsymbol{R}_{1} \boldsymbol{Q}_{1}=\frac{1}{5}\left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array}\right] *\left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array}\right]=\frac{1}{5}\left[\begin{array}{cc} 14 & -3 \\ -3 & 6 \end{array}\right]=\left[\begin{array}{cc} 2.8 & -0.6 \\ -0.6 & 1.2 \end{array}\right] .$</td><td>$\boldsymbol{A}_{2}=\boldsymbol{R}_{1} \boldsymbol{Q}_{1}=\frac{1}{5}\begin{bmatrix} -5 & -4 \\ 0 & 3 \end{bmatrix} *\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}=\frac{1}{5}\begin{bmatrix} 14 & -3 \\ -3 & 6 \end{bmatrix}=\begin{bmatrix} 2.8 & -0.6 \\ -0.6 & 1.2 \end{bmatrix} .$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1039</td><td>351</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{4} \approx\left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array}\right], \quad \boldsymbol{A}_{10} \approx\left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array}\right]</td><td>$\boldsymbol{A}_{4} \approx\left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array}\right], \quad \boldsymbol{A}_{10} \approx\left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array}\right]$</td><td>$\boldsymbol{A}_{4} \approx\begin{bmatrix} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{bmatrix}, \quad \boldsymbol{A}_{10} \approx\begin{bmatrix} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{bmatrix}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1040</td><td>352</td><td>■ 1.000 ● N|+0</td><td>A_{1}=A=\left[\begin{array}{llll} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{array}\right]</td><td>$A_{1}=A=\left[\begin{array}{llll} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{array}\right]$</td><td>$A_{1}=A=\begin{bmatrix} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{bmatrix}$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1042 (15.10)</td><td>352</td><td>■ 1.000 ● K|+0</td><td>\tilde{\boldsymbol{Q}}_{k}:=\boldsymbol{Q}_{1} \cdots \boldsymbol{Q}_{k} \text { and } \tilde{ \boldsymbol{R}}_{k}:=\boldsymbol{R}_{k} \cdots \boldsymbol{R}_{1},</td><td>$\tilde{\boldsymbol{Q}}_{k}:=\boldsymbol{Q}_{1} \cdots \boldsymbol{Q}_{k} \text { and } \tilde{\boldsymbol{R}}_{k}:=\boldsymbol{R}_{k} \cdots \boldsymbol{R}_{1},$</td><td>$\tilde{\boldsymbol{Q}}_{k}:=\boldsymbol{Q}_{1} \cdots \boldsymbol{Q}_{k} \textit{ and } \tilde{\boldsymbol{R}}_{k}:=\boldsymbol{R}_{k} \cdots \boldsymbol{R}_{1},$</td></tr>
<tr><td>lyche-numerical-linear-algebra_ EQ1044 (15.11)</td><td>353</td><td>■ 1.000 ● N|+0</td><td>\boldsymbol{A}_{k}=\boldsymbol{Q}_{k-1}^{*} \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1}= \boldsymbol{Q}_{k-1}^{*} \boldsymbol{Q}_{k-2}^{*} \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1}=\cdots=\tilde{ \boldsymbol{Q}}_{k-1}^{*} \boldsymbol{A} \tilde{ \boldsymbol{Q}}_{k-1} .</td><td>$\boldsymbol{A}_{k}=\boldsymbol{Q}_{k-1}^{*} \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1}=\boldsymbol{Q}_{k-1}^{*} \boldsymbol{Q}_{k-2}^{*} \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1}=\cdots=\tilde{\boldsymbol{Q}}_{k-1}^{*} \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} .$</td><td>$\boldsymbol{A}_{k}=\boldsymbol{Q}_{k-1}^{*} \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1}=\boldsymbol{Q}_{k-1}^{*} \boldsymbol{Q}_{k-2}^{*} \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1}=\cdots=\tilde{\boldsymbol{Q}}_{k-1}^{*} \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1} .$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra_ EQ1046	353	■ 1.000 ● W|+0	`\boldsymbol{A}^{k} \boldsymbol{e}_{1}=\tilde{ \boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k} \boldsymbol{e}_{1}=\tilde{r}_{11}^{(k)} \tilde{ \boldsymbol{Q}}_{k} \boldsymbol{e}_{1} \text { or } \tilde{\boldsymbol{q}}_{1}^{(k)}:=\tilde{ \boldsymbol{Q}}_{k} \boldsymbol{e}_{1}=\frac{1}{ \tilde{r}_{11}^{(k)}} \boldsymbol{A}^{k} \boldsymbol{e}_{1} .`	$\boldsymbol{A}^{k} \boldsymbol{e}_{1}=\tilde{\boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k} \boldsymbol{e}_{1}=\tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_{k} \boldsymbol{e}_{1} \text { or } \tilde{\boldsymbol{q}}_{1}^{(k)}:=\tilde{\boldsymbol{Q}}_{k} \boldsymbol{e}_{1}=\frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^{k} \boldsymbol{e}_{1} .$	$\boldsymbol{A}^{k} \boldsymbol{e}_{1}=\tilde{\boldsymbol{Q}}_{k} \tilde{\boldsymbol{R}}_{k} \boldsymbol{e}_{1}=\tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_{k} \boldsymbol{e}_{1} \text { or } \tilde{\boldsymbol{q}}_{1}^{(k)}:=\tilde{\boldsymbol{Q}}_{k} \boldsymbol{e}_{1}=\frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^{k} \boldsymbol{e}_{1} .$
lyche-numerical-linear-algebra_ EQ1048	355	■ 1.000 ● N|+0	`\boldsymbol{A}_{k+1}=\boldsymbol{Q}_{k}^{*} \left(\boldsymbol{A}_{k}-s_{k} \boldsymbol{I} \right) \boldsymbol{Q}_{k}+s_{k} \boldsymbol{I}= \boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k} \boldsymbol{Q}_{k}`	$\boldsymbol{A}_{k+1}=\boldsymbol{Q}_{k}^{*} \left(\boldsymbol{A}_{k}-s_{k} \boldsymbol{I}\right) \boldsymbol{Q}_{k}+s_{k} \boldsymbol{I}=\boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k} \boldsymbol{Q}_{k}$	$\boldsymbol{A}_{k+1}=\boldsymbol{Q}_{k}^{*}(\boldsymbol{A}_{k}-s_{k} \boldsymbol{I}) \boldsymbol{Q}_{k}+s_{k} \boldsymbol{I}=\boldsymbol{Q}_{k}^{*} \boldsymbol{A}_{k} \boldsymbol{Q}_{k}$
lyche-numerical-linear-algebra_ EQ1050 (16.1)	358	■ 1.000 ● N|+0	`\nabla f:=\left[\begin{array}{c} D_{1} f \\ \vdots \ \ D_{n} f \end{array}\right], \quad \boldsymbol{H} f:= \nabla \nabla^{T} f:=\left[\begin{array}{ccc} D_{1} D_{1} f & \cdots & D_{1} D_{n} f \\ \vdots & & \vdots \\ D_{n} D_{1} & \cdots & D_{n} D_{n} f \end{array}\right],`	$\nabla f:=\left[\begin{array}{c} D_{1} f \\ \vdots \\ D_{n} f \end{array}\right], \quad \boldsymbol{H} f:=\nabla \nabla^{T} f:=\left[\begin{array}{ccc} D_{1} D_{1} f & \cdots & D_{1} D_{n} f \\ \vdots & & \vdots \\ D_{n} D_{1} & \cdots & D_{n} D_{n} f \end{array}\right],$	$\nabla f:=\left[\begin{array}{c} D_{1} f \\ \vdots \\ D_{n} f \end{array}\right], \quad \boldsymbol{H} f:=\nabla \nabla^{T} f:=\left[\begin{array}{ccc} D_{1} D_{1} f & \cdots & D_{1} D_{n} f \\ \vdots & & \vdots \\ D_{n} D_{1} & \cdots & D_{n} D_{n} f \end{array}\right],$
lyche-numerical-linear-algebra_ EQ1051	359	■ 1.000 ● N|+0	`\nabla^{T} \boldsymbol{f}:=\left[\begin{array}{ccc} D_{1} f_{1} & \cdots & D_{n} f_{1} \\ \vdots & & \vdots \\ D_{1} f_{m} & \cdots & D_{n} f_{m} \end{array}\right] .`	$\nabla^{T} \boldsymbol{f}:=\left[\begin{array}{ccc} D_{1} f_{1} & \cdots & D_{n} f_{1} \\ \vdots & & \vdots \\ D_{1} f_{m} & \cdots & D_{n} f_{m} \end{array}\right] .$	$\nabla^{T} \boldsymbol{f}:=\left[\begin{array}{ccc} D_{1} f_{1} & \cdots & D_{n} f_{1} \\ \vdots & & \vdots \\ D_{1} f_{m} & \cdots & D_{n} f_{m} \end{array}\right] .$
lyche-numerical-linear-algebra_ EQ1053 (16.2)	359	■ 1.000 ● N|+0	`f(\boldsymbol{x}+\boldsymbol{h})=f(\boldsymbol{x})+ \boldsymbol{h}^{T} \nabla f(\boldsymbol{x})+\frac{1}{2} \boldsymbol{h}^{T} \nabla \nabla^{T} f(\boldsymbol{c}) \boldsymbol{h}, \text { for some } \boldsymbol{c} \in L .`	$f(\boldsymbol{x}+\boldsymbol{h})=f(\boldsymbol{x})+\boldsymbol{h}^{T} \nabla f(\boldsymbol{x})+\frac{1}{2} \boldsymbol{h}^{T} \nabla \nabla^{T} f(\boldsymbol{c}) \boldsymbol{h}, \text { for some } \boldsymbol{c} \in L .$	$f(\boldsymbol{x}+\boldsymbol{h})=f(\boldsymbol{x})+\boldsymbol{h}^{T} \nabla f(\boldsymbol{x})+\frac{1}{2} \boldsymbol{h}^{T} \nabla \nabla^{T} f(\boldsymbol{c}) \boldsymbol{h}, \textit{ for some } \boldsymbol{c} \in L .$
lyche-numerical-linear-algebra_ EQ1055	359	■ 1.000 ● N|+0	`g(1)=g(0)+g^{\prime}(0)+\frac{1}{2} g^{\prime \prime}(u), \text { for some } u \in(0,1),`	$g(1)=g(0)+g^{\prime}(0)+\frac{1}{2} g^{\prime \prime}(u), \text { for some } u \in(0,1),$	$g(1)=g(0)+g^{\prime}(0)+\frac{1}{2} g^{\prime \prime}(u), \text { for some } u \in(0,1),$
lyche-numerical-linear-algebra_ EQ1056	360	■ 1.000 ● W|+1	`\begin{aligned} D_{i}\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right) & =\lim _{h \rightarrow 0} \frac{1}{h}\left(\left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)^{T} \boldsymbol{B} \left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)- \boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x} \right) \\ & =\lim _{h \rightarrow 0}\left(\boldsymbol{e}_{i}^{T} \boldsymbol{B} \boldsymbol{x}+ \boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{e}_{i}+h \boldsymbol{e}_{i}^{T} \boldsymbol{e}_{i}\right)= \boldsymbol{e}_{i}^{T}\left(\boldsymbol{B}+ \boldsymbol{B}^{T}\right) \boldsymbol{x}, \end{aligned}`	$\begin{aligned} D_{i}\left(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right) & =\lim _{h \rightarrow 0} \frac{1}{h}\left(\left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)^{T} \boldsymbol{B}\left(\boldsymbol{x}+h \boldsymbol{e}_{i}\right)-\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\right) \\ & =\lim _{h \rightarrow 0}\left(\boldsymbol{e}_{i}^{T} \boldsymbol{B} \boldsymbol{x}+\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{e}_{i}+h \boldsymbol{e}_{i}^{T} \boldsymbol{e}_{i}\right)=\boldsymbol{e}_{i}^{T}\left(\boldsymbol{B}+\boldsymbol{B}^{T}\right) \boldsymbol{x}, \end{aligned}$	$\begin{aligned} D_{i}(\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}) & =\lim _{h \rightarrow 0} \frac{1}{h}\big((\boldsymbol{x}+h \boldsymbol{e}_{i})^{T} \boldsymbol{B}(\boldsymbol{x}+h \boldsymbol{e}_{i})-\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{x}\big) \\ & =\lim _{h \rightarrow 0}\big(\boldsymbol{e}_{i}^{T} \boldsymbol{B} \boldsymbol{x}+\boldsymbol{x}^{T} \boldsymbol{B} \boldsymbol{e}_{i}+h \boldsymbol{e}_{i}^{T} \boldsymbol{e}_{i}\big)=\boldsymbol{e}_{i}^{T}(\boldsymbol{B}+\boldsymbol{B}^{T}) \boldsymbol{x}, \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Tables

<table>
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<tr><td>LU and QR factorizations</td><td>2.4, 2.5, 3.2, 3.3, 3.5, 4.1, 4.2, 5.1 − 5.4, 5.6</td></tr>
<tr><td>SVD, norms and LSQ</td><td>6.1, 6.3, 7.1 − 7.4, 8.1 − 8.3, 9.1 − 9.3, 9.4.1</td></tr>
<tr><td>Kronecker products</td><td>10.1, 10.2, 10.3, 11.1, 11.2, 11.3</td></tr>
<tr><td>Iterative methods</td><td>12.1 − 12.4, 13.1 − 13.3, 13.5</td></tr>
<tr><td>Eigenpairs</td><td>14.1 − 14.5, 15.1 − 15.3</td></tr>
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<tr><th></th><th>k_{100}</th><th>k_{2500}</th><th>$k_{10\,000}$</th><th>$k_{40\,000}$</th><th>$k_{160\,000}$</th></tr>
<tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr>
<tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr>
<tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr>
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<tr><td>lyche-numerical-linear-algebra_TAB_003</td><td>278</td><td>—</td><td>(no LaTeX source; 560 × 173 px region)</td><td>—</td><td><table>
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<tr><td>GS</td><td>0.920627</td><td>0.99621</td><td>223</td><td>4852</td></tr>
<tr><td>SOR</td><td>0.56039</td><td>0.88402</td><td>32</td><td>150</td></tr>
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<tr><td>K</td><td>19</td><td>18</td><td>18</td><td>16</td><td>15</td></tr>
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<tr><td>K</td><td>94</td><td>188</td><td>370</td><td>735</td></tr>
<tr><td>$K/\sqrt{n}$</td><td>1.88</td><td>1.88</td><td>1.85</td><td>1.84</td></tr>
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<tr><td>K</td><td>222</td><td>472</td><td>728</td><td>986</td><td>1246</td></tr>
<tr><td>$K/\sqrt{n}$</td><td>4.44</td><td>4.72</td><td>4.85</td><td>4.93</td><td>4.98</td></tr>
<tr><td>K_{pre}</td><td>22</td><td>23</td><td>23</td><td>23</td><td>23</td></tr>
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<tr><td>k</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>$\|\boldsymbol{r}\|_2$</td><td>1.0e+000</td><td>7.7e−002</td><td>1.6e−004</td><td>8.2e−010</td><td>2.0e−020</td></tr>
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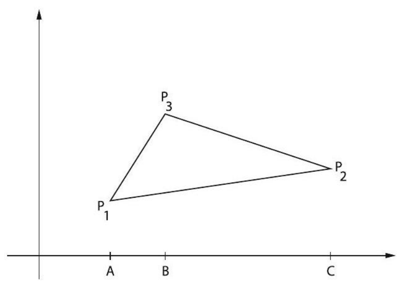
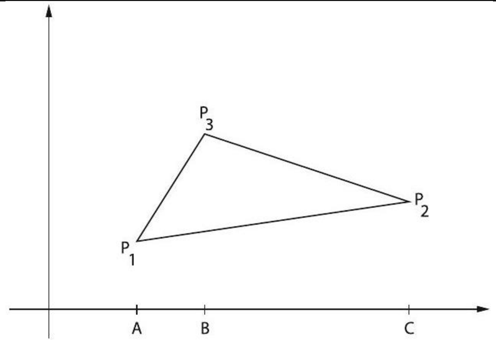
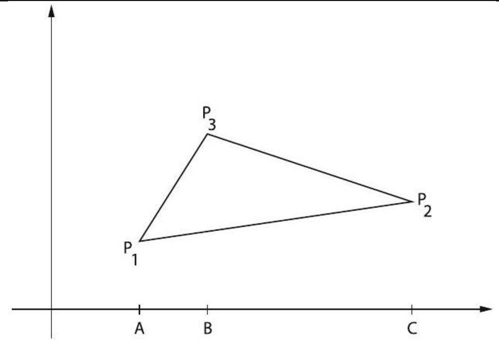
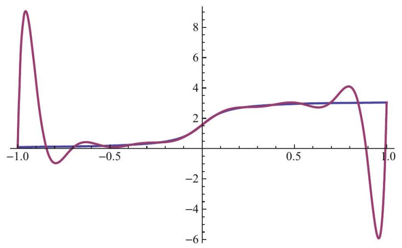
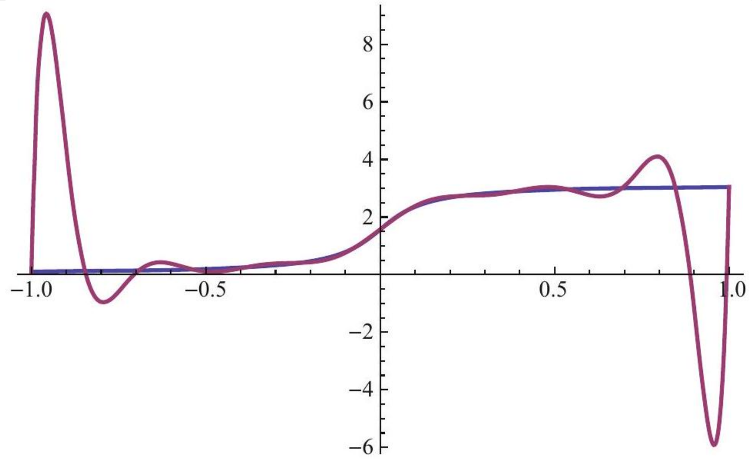
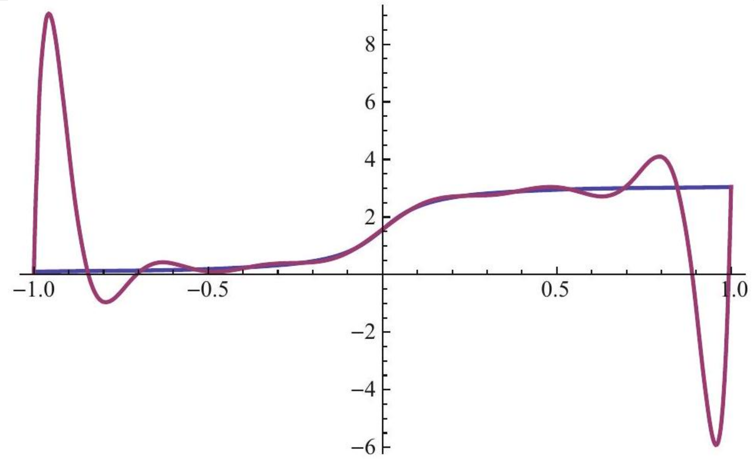
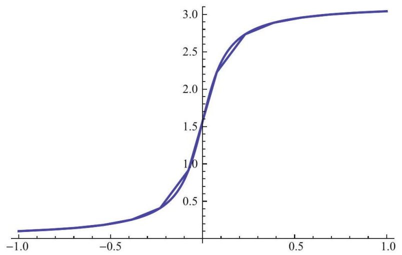
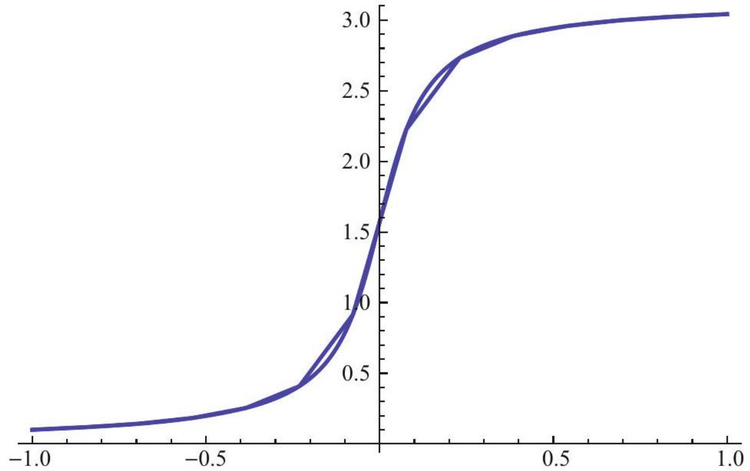
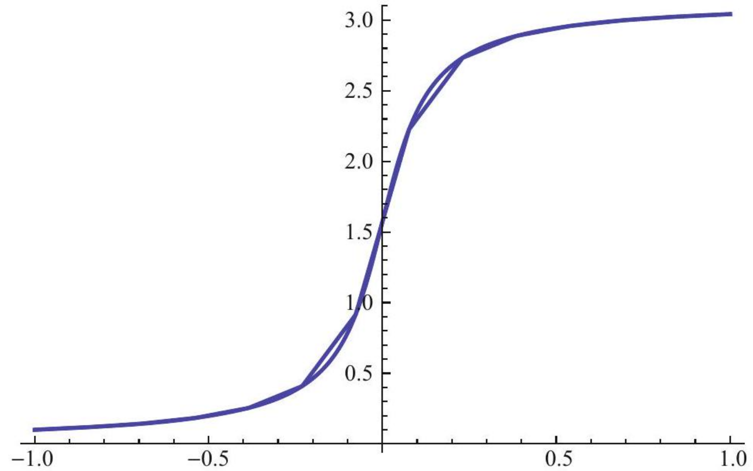
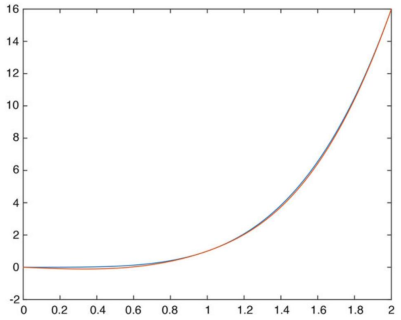
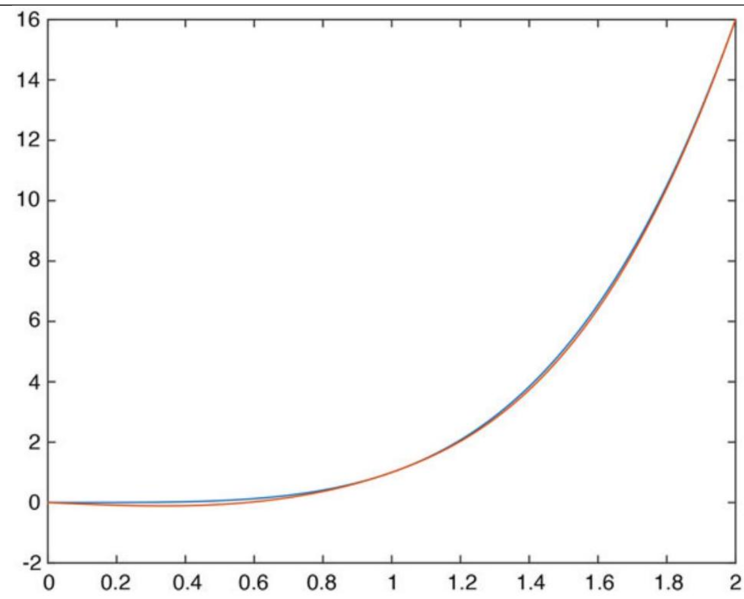
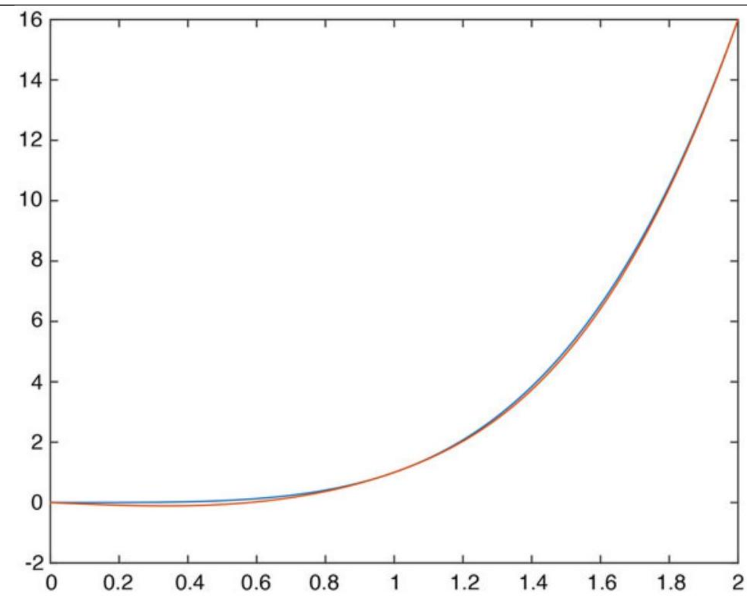
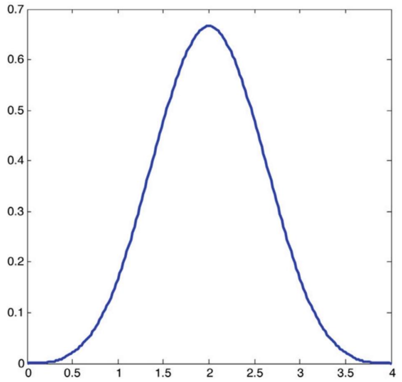
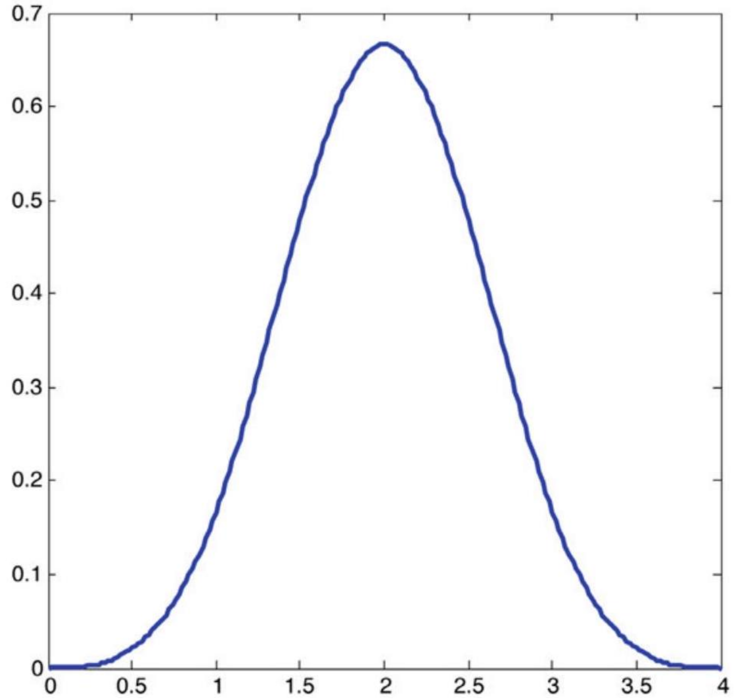
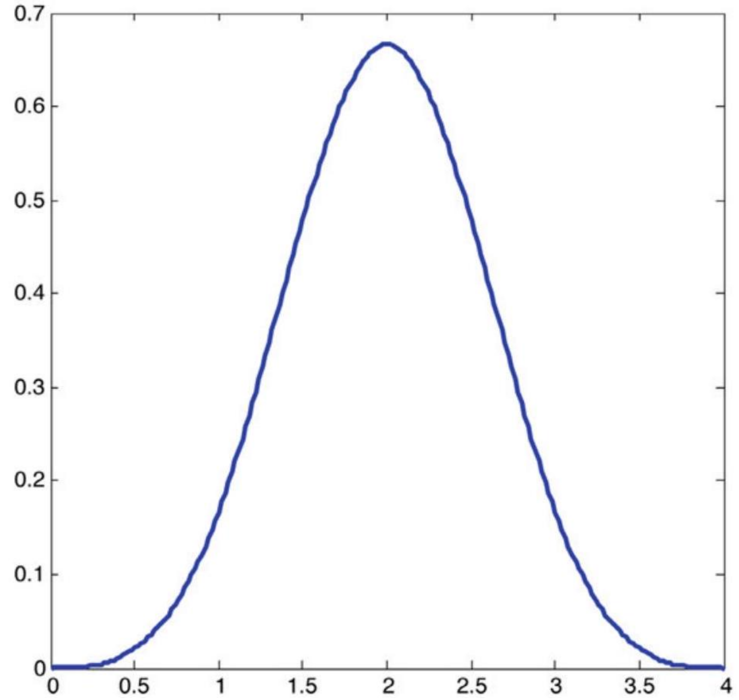
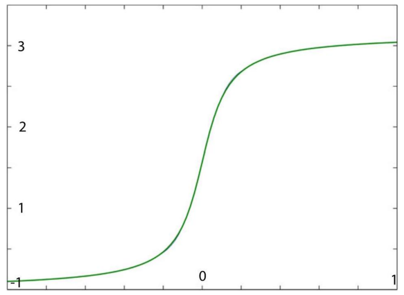
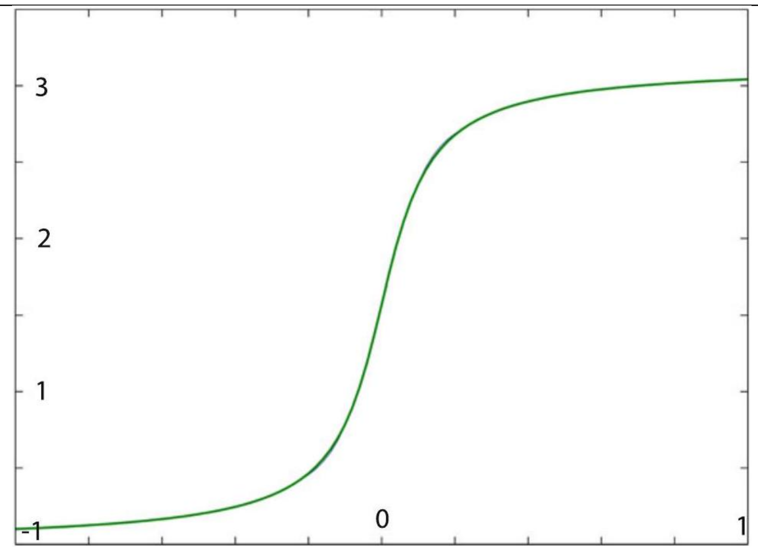
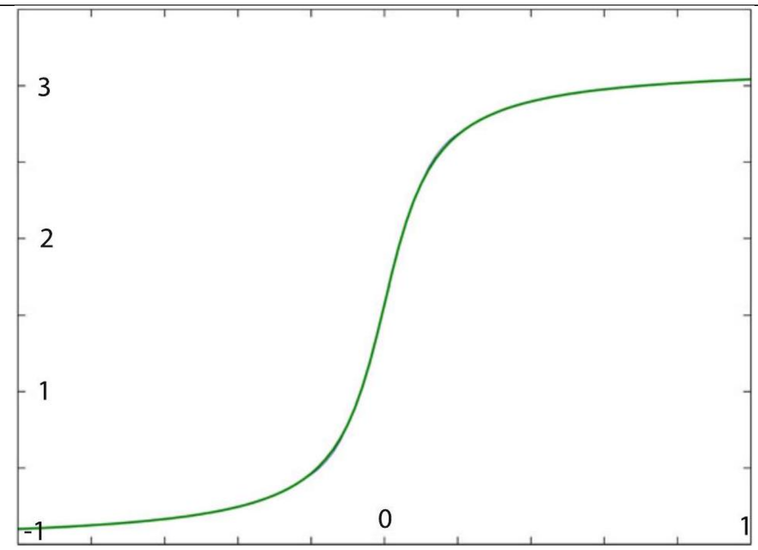
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

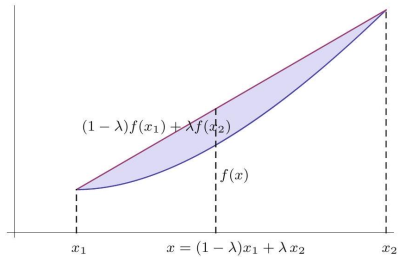
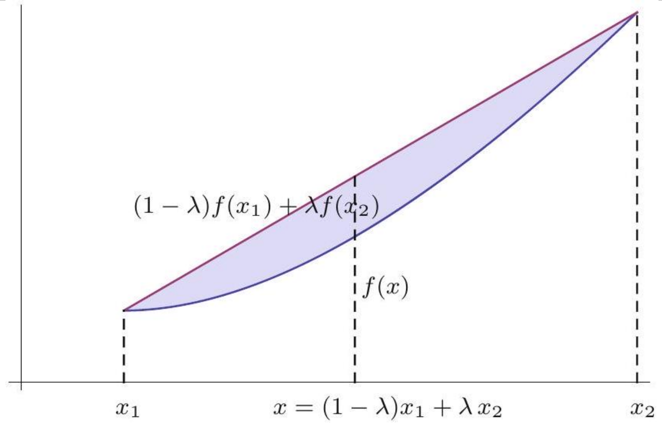
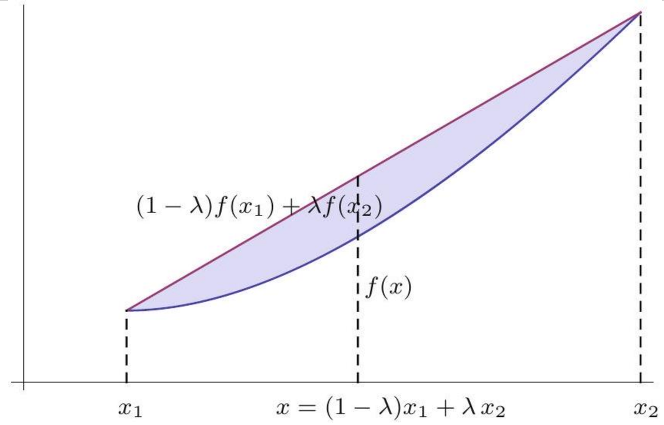
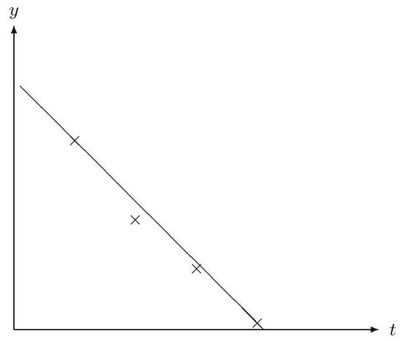
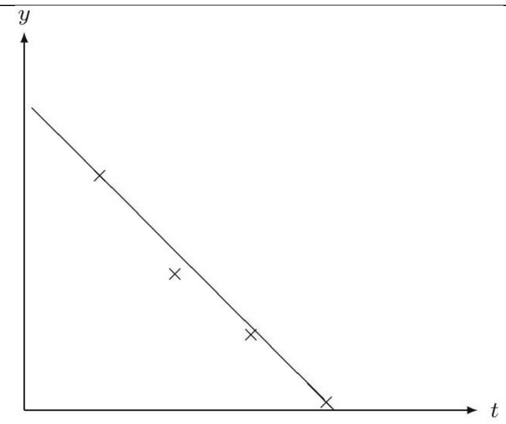
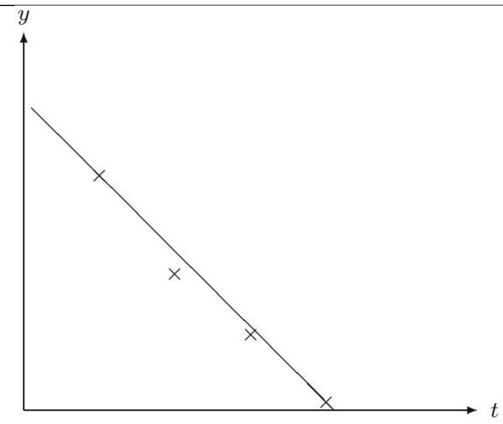
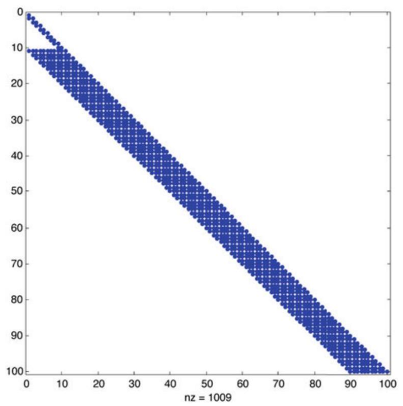
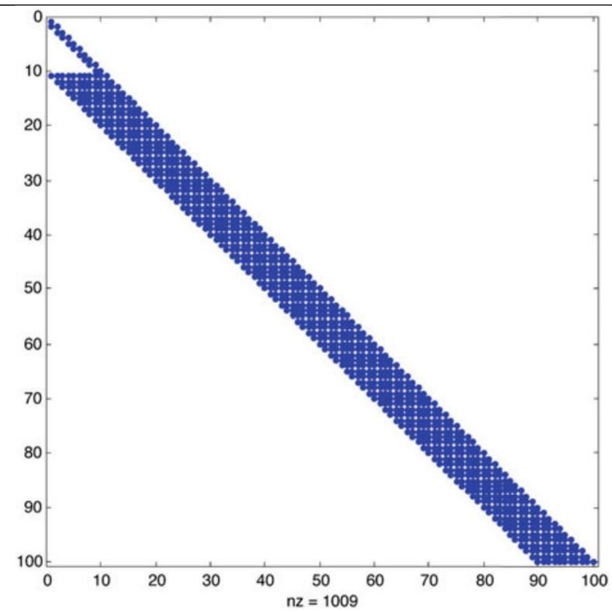
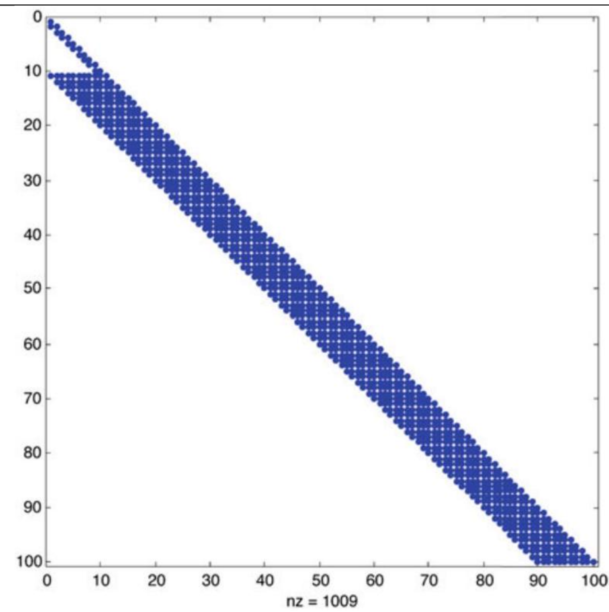
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

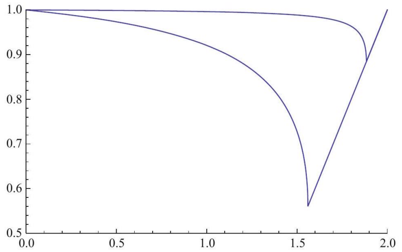
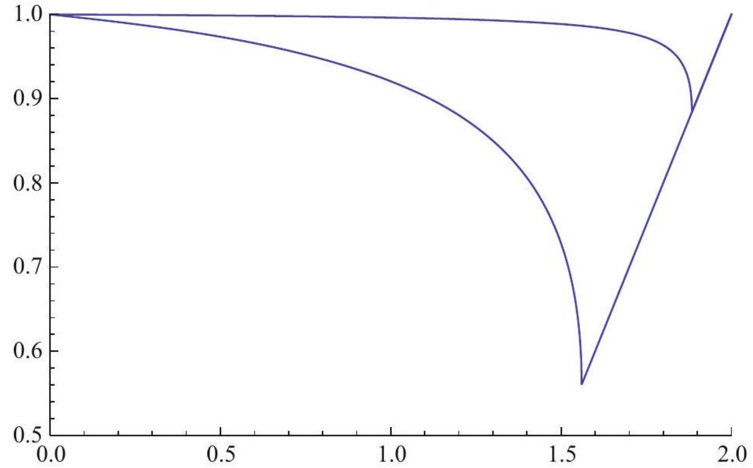
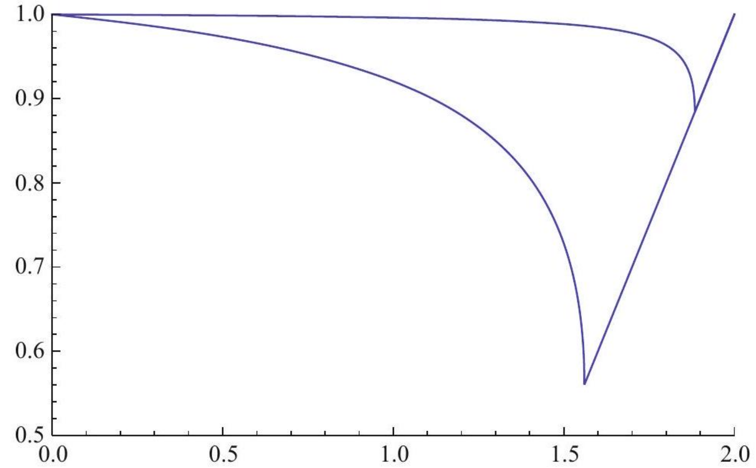
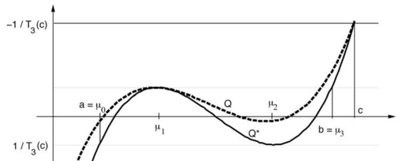
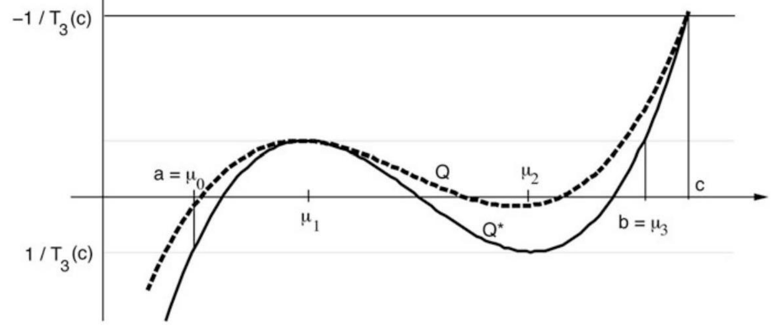
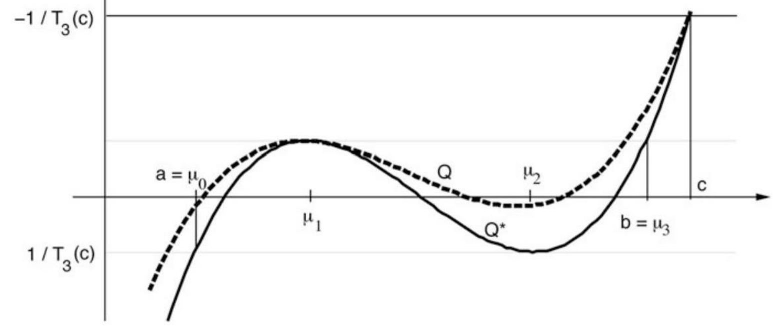
Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region's own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the `tex.zip` image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix's confidence, as the equation rows do. Setting a region's LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with `pdfdrill vision`; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
lyche-numerical-linear-algebra_PIC_0001 *(dup)*	043		750c5f9d-6269-44fe-81e3-a3e4f5704d07-043_441_635_213_705.jpg			
lyche-numerical-linear-algebra_PIC_0002 *(dup)*	048		750c5f9d-6269-44fe-81e3-a3e4f5704d07-048_598_971_213_276.jpg			
lyche-numerical-linear-algebra_PIC_0003 *(dup)*	048		750c5f9d-6269-44fe-81e3-a3e4f5704d07-048_615_969_946_278.jpg			

lyche-numerical-linear-algebra_PIC_~~0004~~ *(dup)*	050		750c5f9d-6269-44fe-81e3-a3e4f5704d07-050_769_958_213_284.jpg			
lyche-numerical-linear-algebra_PIC_~~0005~~ *(dup)*	053		750c5f9d-6269-44fe-81e3-a3e4f5704d07-053_910_948_217_289.jpg			
lyche-numerical-linear-algebra_PIC_~~0006~~ *(dup)*	070		750c5f9d-6269-44fe-81e3-a3e4f5704d07-070_704_963_213_282.jpg			

lyche-numerical-linear-algebra_PIC_0007	199	*(dup)*	750c5f9d-6269-44fe-81e3-a3e4f5704d07-199_550_851_1359_337.jpg			
lyche-numerical-linear-algebra_PIC_0008	215	*(dup)*	750c5f9d-6269-44fe-81e3-a3e4f5704d07-215_543_636_1546_706.jpg			
lyche-numerical-linear-algebra_PIC_0009	251	*(dup)*	750c5f9d-6269-44fe-81e3-a3e4f5704d07-251_788_777_1215_376.jpg			

lyche-numerical-linear-algebra_PIC_0010	278	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-278_609_969_215_278.jpg			
lyche-numerical-linear-algebra_PIC_0011	306	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-306_527_1156_1449_185.jpg			
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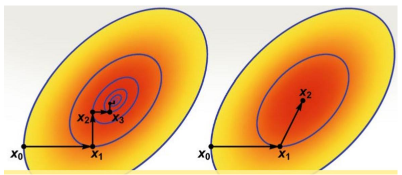
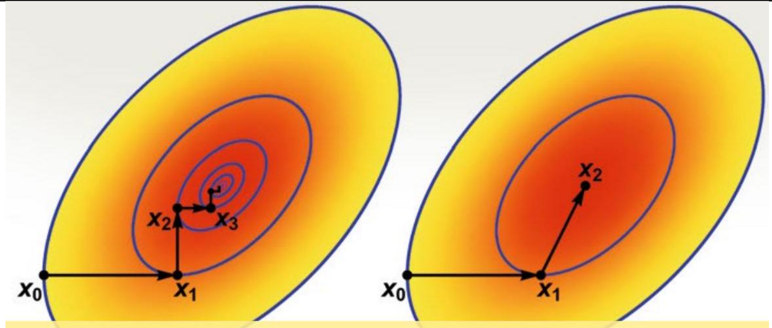
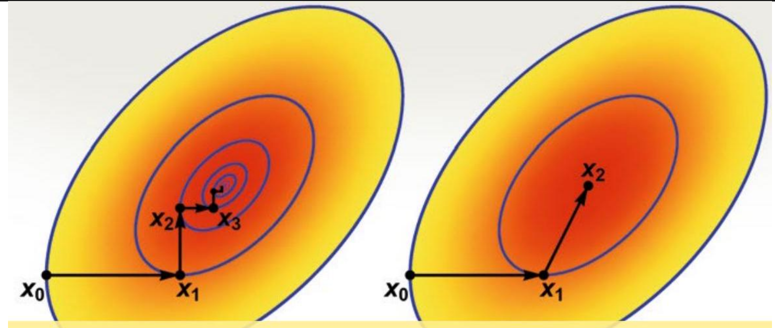
lyche-numerical-linear-algebra_DIA_0001	001	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-001_500_1166_59_176.jpg			
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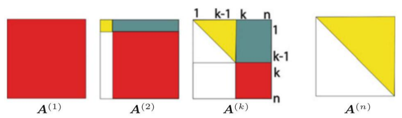
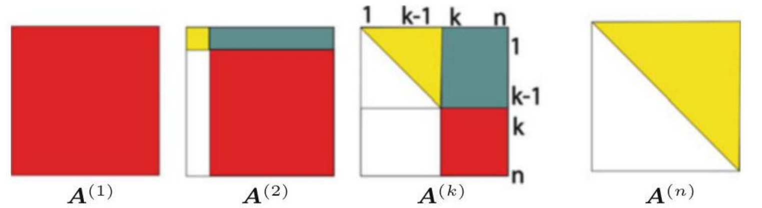
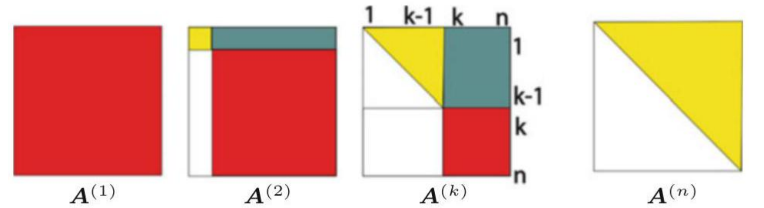
lyche-numerical-linear-algebra_DIA_0007	077	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-077_274_965_1748_280.jpg			
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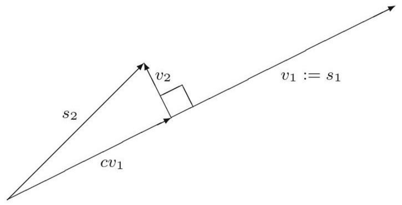
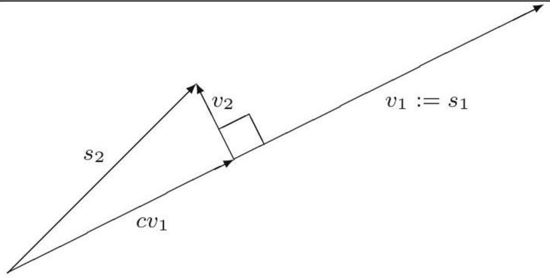
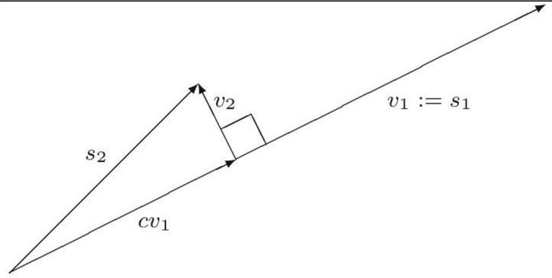
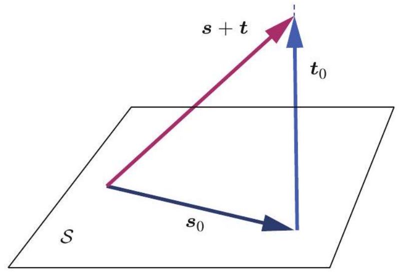
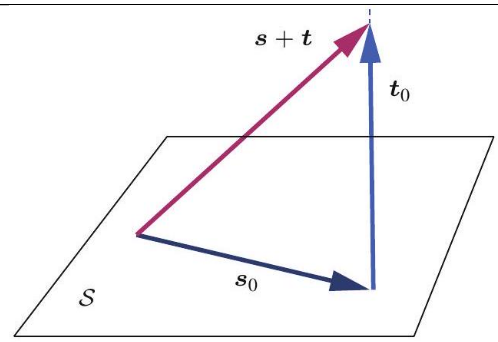
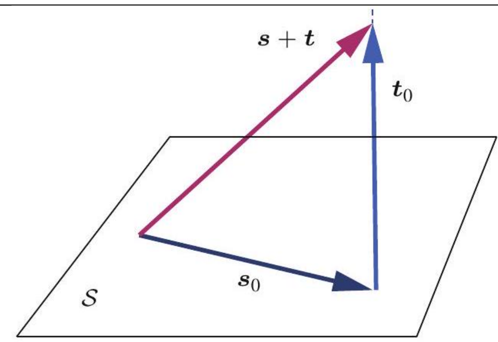
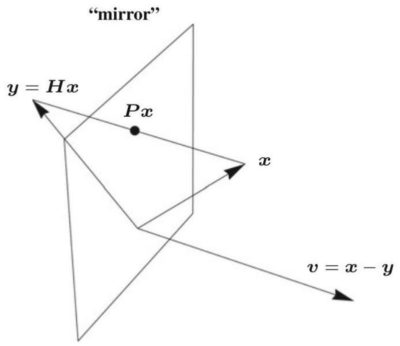
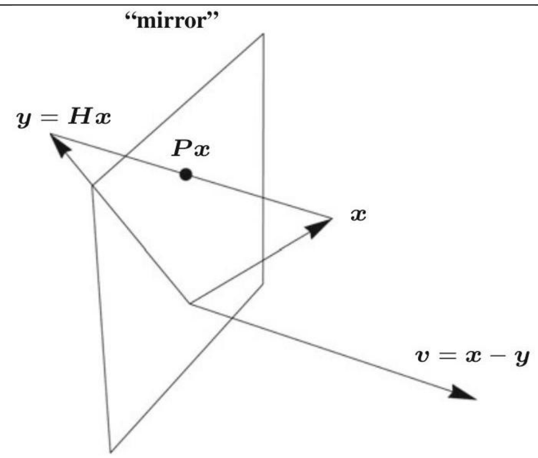
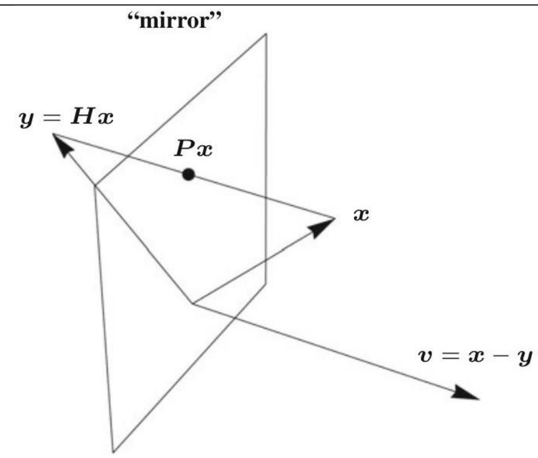
lyche-numerical-linear-algebra_DIA_0016	120	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-120_360_708_227_632.jpg			
lyche-numerical-linear-algebra_DIA_0017	122	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-122_438_638_213_704.jpg			
lyche-numerical-linear-algebra_DIA_0018	125	(dup)	750c5f9d-6269-44fe-81e3-a3e4f5704d07-125_586_680_1348_664.jpg			
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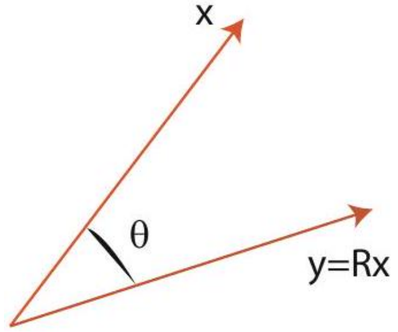
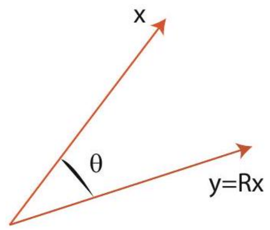
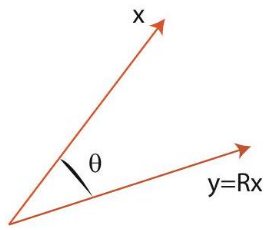
<tr><td>lyche-numerical-linear-algebra_DIA_0021 (dup)</td><td>134</td><td></td><td>750c5f9d-6269-44fe-81e3-a3e4f5704d07-134_291_343_1660_1001.jpg</td><td>
</td><td>
</td><td>
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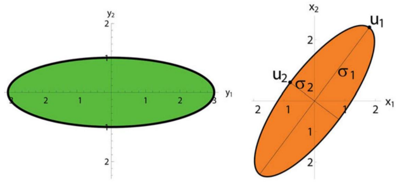
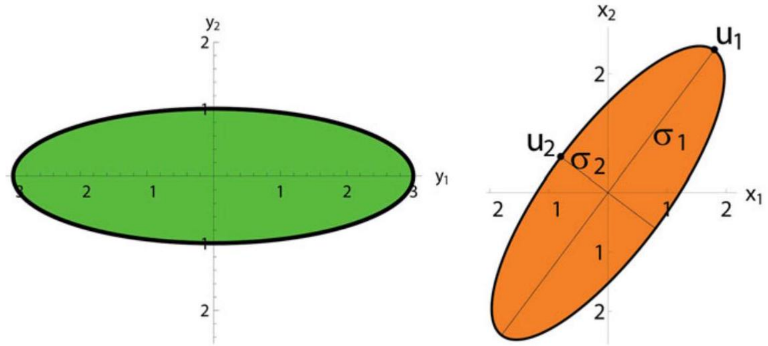
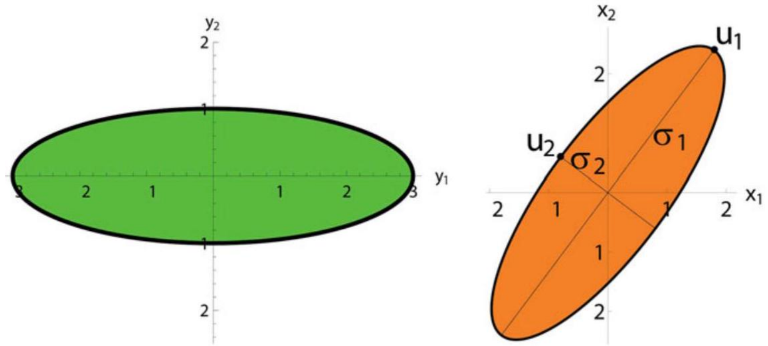
<tr><td>lyche-numerical-linear-algebra_DIA_0023 (dup)</td><td>175</td><td></td><td>750c5f9d-6269-44fe-81e3-a3e4f5704d07-175_524_1142_1371_196.jpg</td><td></td><td>
</td><td>
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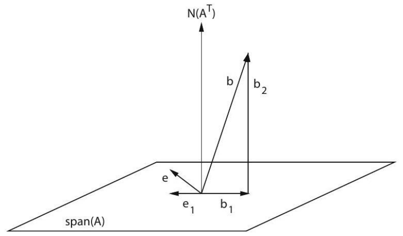
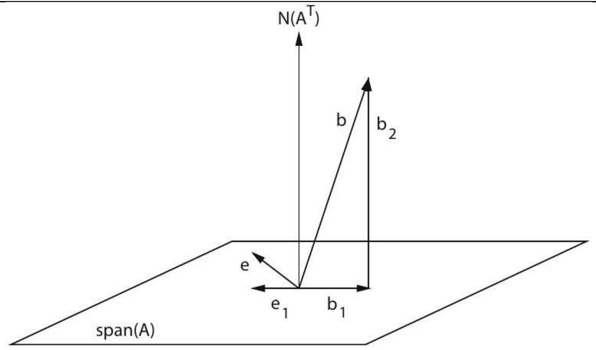
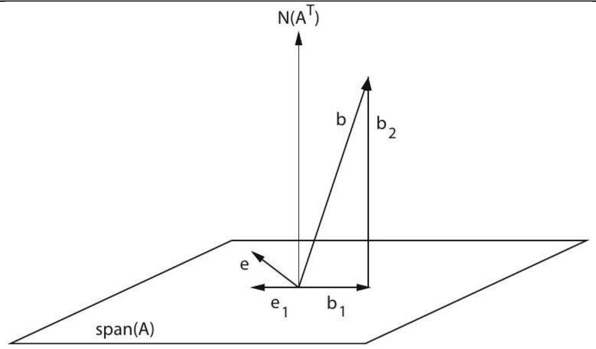
<tr><td>lyche-numerical-linear-algebra_DIA_0024 (dup)</td><td>225</td><td></td><td>750c5f9d-6269-44fe-81e3-a3e4f5704d07-225_449_765_1559_575.jpg</td><td></td><td>
</td><td>
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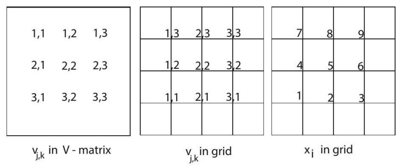
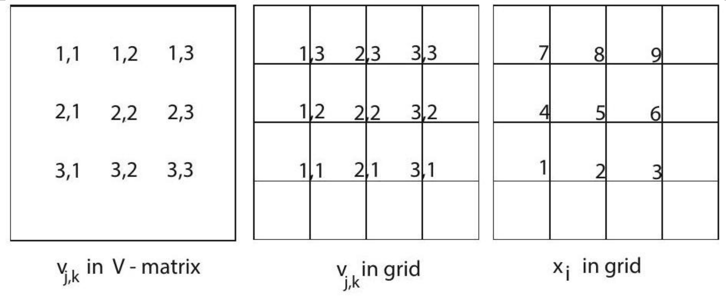
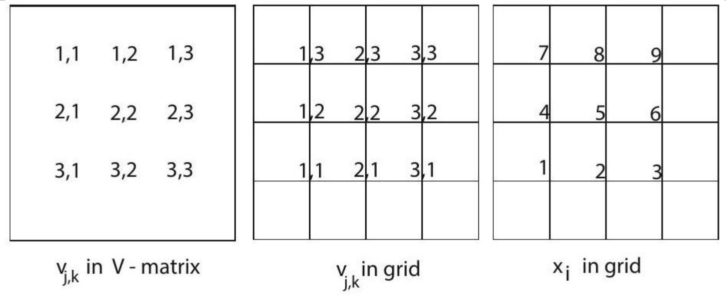
<tr><td>lyche-numerical-linear-algebra_DIA_0025 (dup)</td><td>240</td><td></td><td>750c5f9d-6269-44fe-81e3-a3e4f5704d07-240_385_935_211_295.jpg</td><td></td><td>
</td><td>
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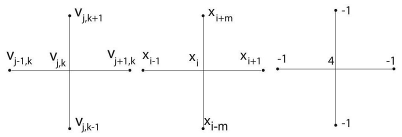
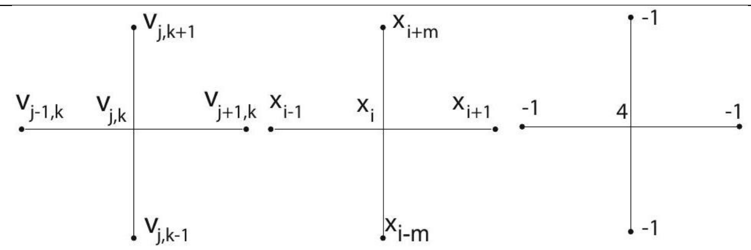
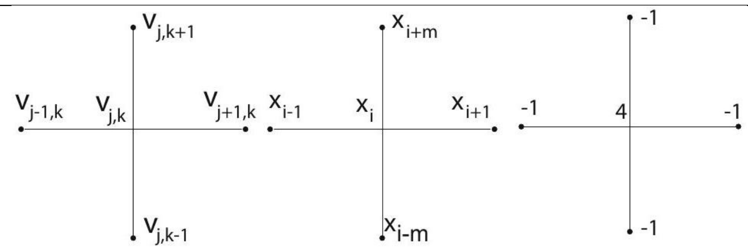
<tr><td>lyche-numerical-linear-algebra_DIA_0026 (dup)</td><td>240</td><td></td><td>750c5f9d-6269-44fe-81e3-a3e4f5704d07-240_316_958_704_284.jpg</td><td></td><td>
</td><td>
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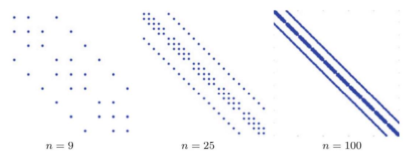
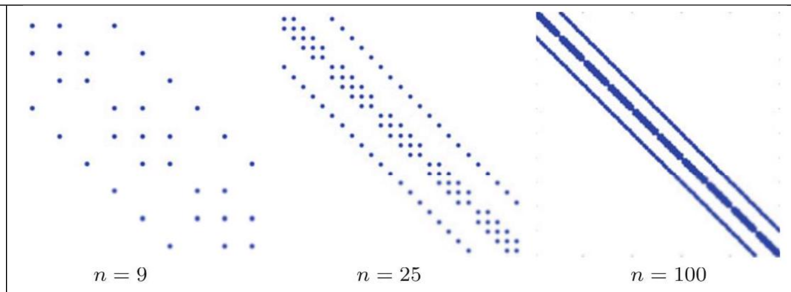
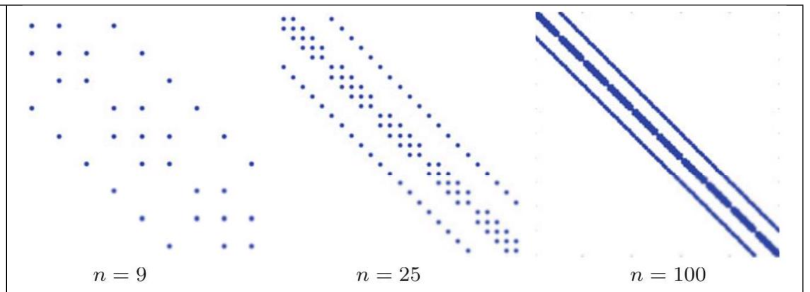
lyche-numerical-linear-algebra_DIA_0027 *(dup)*	241		750c5f9d-6269-44fe-81e3-a3e4f5704d07-241_398_1079_209_230.jpg			
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lyche-numerical-linear-algebra_DIA_0029 *(dup)*	259		*no image for this region (28 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*
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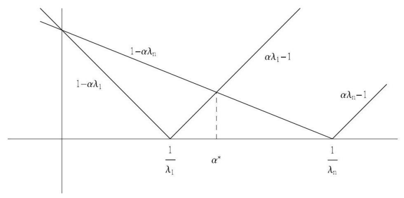
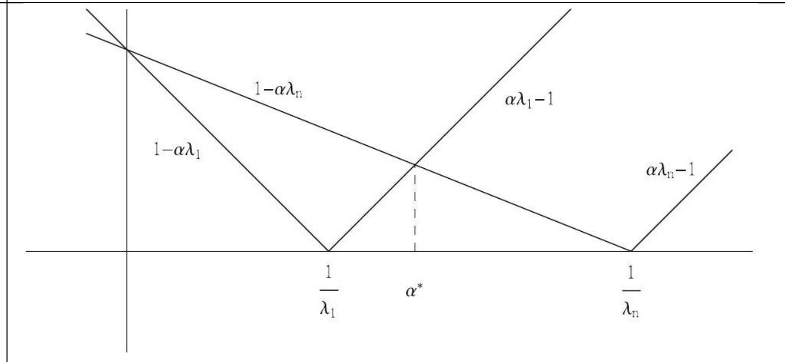
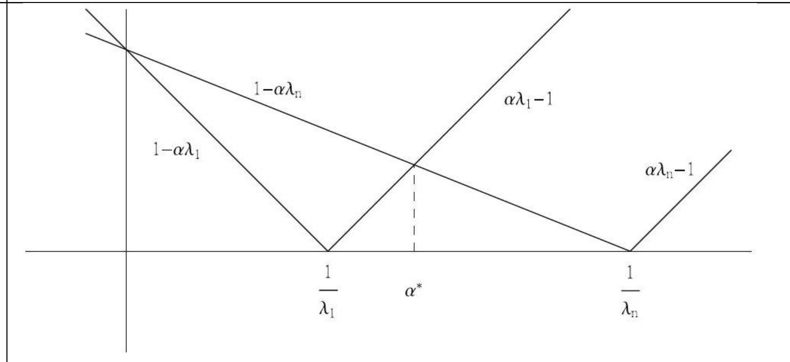
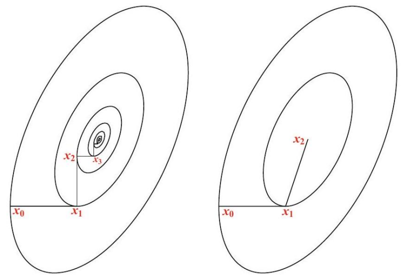
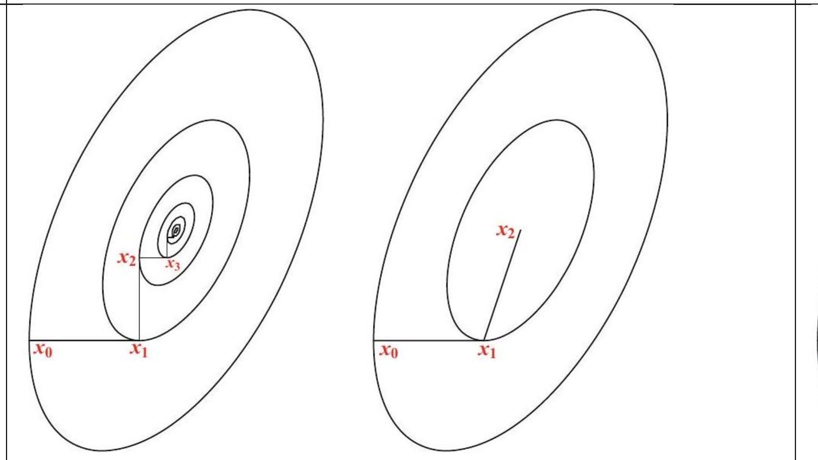
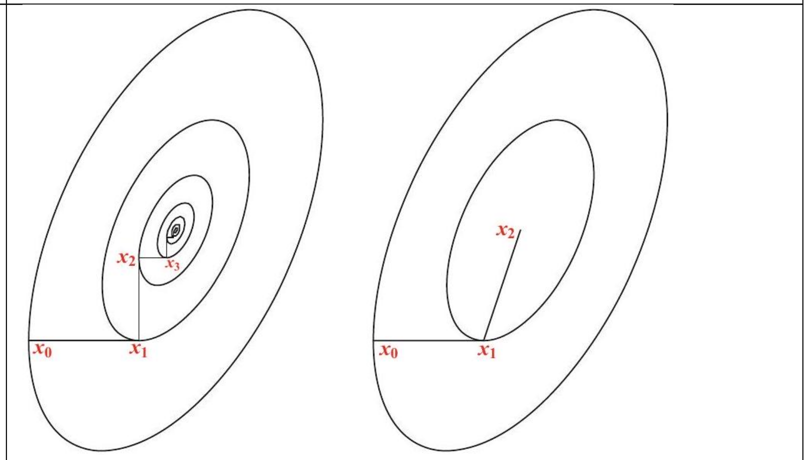
lyche-numerical-linear-algebra_DIA_0031 *(dup)*	268		*no image for this region (28 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*
lyche-numerical-linear-algebra_DIA_0032 *(dup)*	269		*no image for this region (28 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*
lyche-numerical-linear-algebra_DIA_0033 *(dup)*	274		750c5f9d-6269-44fe-81e3-a3e4f5704d07-274_459_956_209_286.jpg			
lyche-numerical-linear-algebra_DIA_0034 *(dup)*	292		750c5f9d-6269-44fe-81e3-a3e4f5704d07-292_586_847_211_339.jpg			
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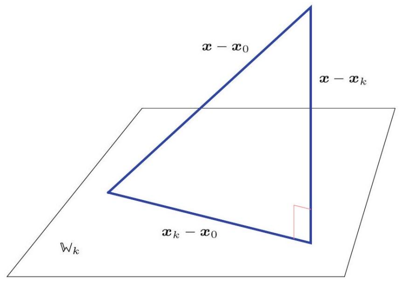
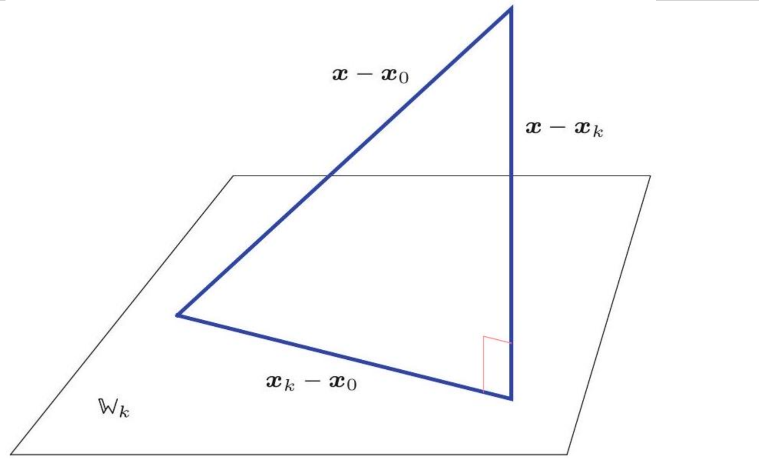
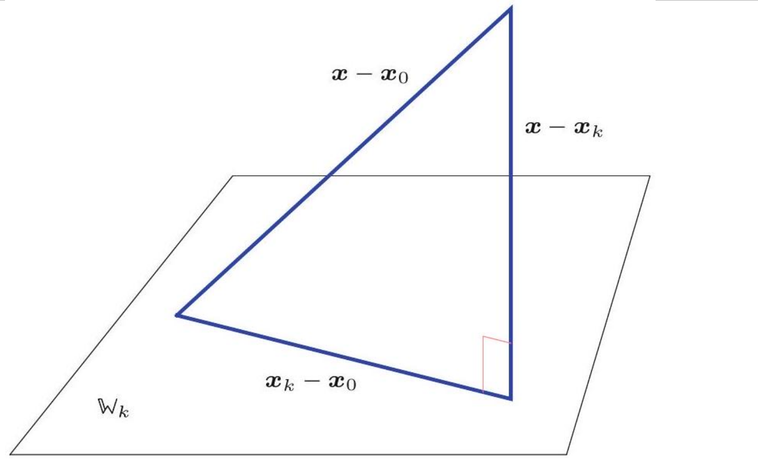
lyche-numerical-linear-algebra_DIA_0037 (dup)	302		750c5f9d-6269-44fe-81e3-a3e4f5704d07-302_596_847_211_339.jpg			
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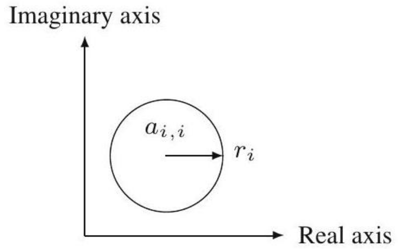
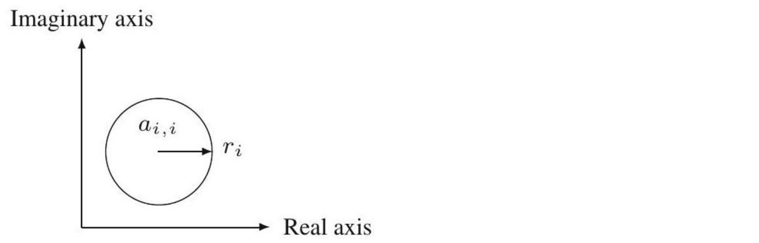
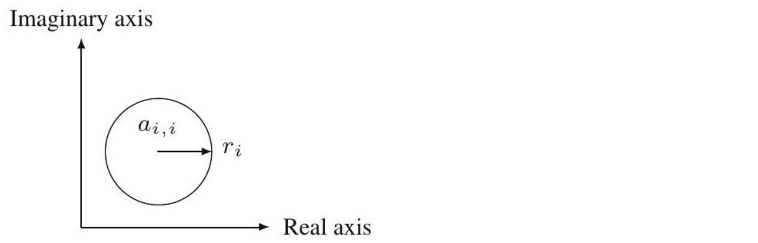
lyche-numerical-linear-algebra_DIA_0039 (dup)	328		750c5f9d-6269-44fe-81e3-a3e4f5704d07-328_308_484_213_853.jpg			
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lyche-numerical-linear-algebra_DIA_0045 (dup)	354		*no image for this region (28 in zip)*	*(no tex.zip crop for this row)*	*(image not on disk — pass –texzip or download the CDN crop)*	*(image not on disk — pass –texzip or download the CDN crop)*